THE PRACTICE OF HARMONY

THIRD EDITION

PETER SPENCER

THE PRACTICE OF HARMONY

third edition

Peter Spencer

The Florida State University



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Preface

Almost all music majors in colleges and universities in the United States are required to study music theory. A large portion of music theory involves the study of tertian harmony, that is, the general harmonic practice of composers from about 1700 to 1900. A thorough grasp of the basic principles of this practice is a prerequisite to the full appreciation and comprehension of the works of every composer from Bach to Brahms. One should not forget, however, that almost a hundred years have elapsed since Brahm's death and that in that time a number of harmonic practices have grown that demand attention, albeit at an elementary level. The object of the third edition of *The Practice of Harmony* is, therefore, to give the music student, regardless of his or her major, a thorough understanding of the basic materials of harmony.

The book is divided into four parts:

Part One: Foundations–Designed to ensure that the student has a solid grasp of fundamentals before advancing to subject matter in which these fundamentals are used as the bases for further development. These early chapters establish the pedagogical strategy that is employed throughout most of the remainder of the book. The materials are presented in an additive manner, so that the student uses what was learned in one chapter to comprehend the materials in the next. Such an approach allows coverage of not only the rudiments of music theory, but also the principles behind the rudiments. Because the author is convinced that understanding is most effectively reached by doing, each chapter contains a large number of exercises, most of which have time goals designed to force the student to operate quickly–indeed automatically.*

Upon successfully completing this portion of the book, the student will be in a position to make automatic responses to questions related to reading in G, F, and C clefs, major and minor scales, key signatures and scale degrees, all intervals, triads, and principles of rhythmic notation.

Parts Two and Three: Harmony in Common Practice–Designed to develop a complete understanding of the principles of tertian harmony as they

*Upon completion of each group of exercises, the preceding text will be torn out and, along with the exercises, inserted in a loose-leaf binding. The student can insert additional staff paper as needed.

pertain to common practice. Parts Two and Three, "The Diatonic Vocabulary" and "The Chromatic Vocabulary" respectively, continue to stress learning by doing. After each new concept has been introduced, several pages of carefully graduated exercises follow to ensure that the student completely understands that concept before approaching a new one. This method favors writing over analysis, for understanding involves more than the ability merely to analyze; it involves also a working knowledge of the problems that are inevitably encountered when musical pitches are committed to paper. One cannot begin to reach a true understanding of a Beethoven sonata, for instance, unless one has wrestled, even at a very low level, with the same kind of harmonic problems that confronted Beethoven himself. Analysis can be illuminating only to the student who understands some of the reasons for the notes in the first place. Not that analysis has been ignored; in addition to examples for analysis in the text, more elaborate *exercises* for analysis are included. These exercises call for the use of Charles Burkhart's *Anthology for Musical Analysis,* fifth edition (Fort Worth, TX: Harcourt Brace, 1994); Ralph Turek's *Analytical Anthology of Music,* second edition (New York: McGraw-Hill, 1992); or Mary Wennerstrom's *Anthology of Musical Structure and Style* (Englewood Cliffs, NJ: Prentice Hall, 1983).

Part Four: Post-Common Practice Harmony—Designed to introduce the student to some of the more important harmonic procedures that have either evolved from or developed as a reaction to common practice. In keeping with the philosophy embraced in the previous parts, the materials in Part Four are presented concisely and reinforced by a wide variety of exercises. The absence of a continuing "common practice" in relation to several of these materials, however, precludes the possibility for the same kind of step-by-step development of concepts that students encounter in the earlier parts of the text. Nonetheless, in the spirit of the rest of the book, the emphasis in Part Four is also on the exposition of theoretical procedures rather than on individual composers' interpretations of them. The pursuit of *stylistic* considerations is beyond the scope or intention of *The Practice of Harmony,* for such considerations fall more naturally into the purview of advanced theoretical studies.

In summation, *The Practice of Harmony* is designed for students of basic theory. The author has given considerable thought to the kind of people for whom this book would be valuable and consequently has endeavored to present the material in an entirely systematic manner, simply and logically, so that the subject matter is not only comprehensible to the student, but also easily taught by the instructor.

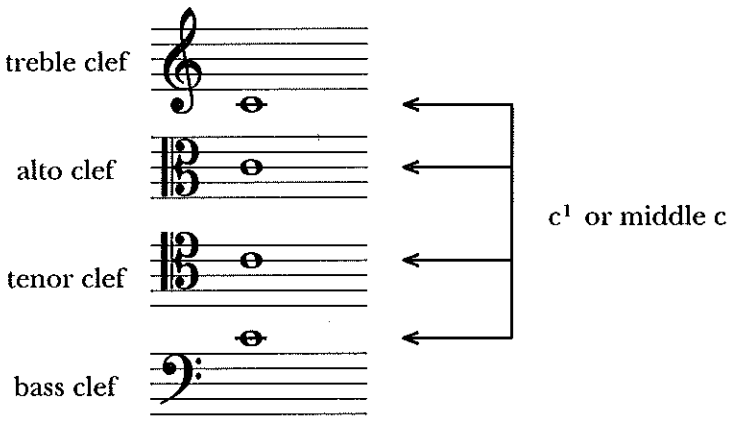
PETER SPENCER

PART ONE

FOUNDATIONS

1 Clefs and Basic Pitch Notation*

Four clefs are in common use:



Note:

a. The treble clef is called a *G clef* because the symbol is a corruption of the letter G, the "center" of which encircles g^1, the second line of the staff.

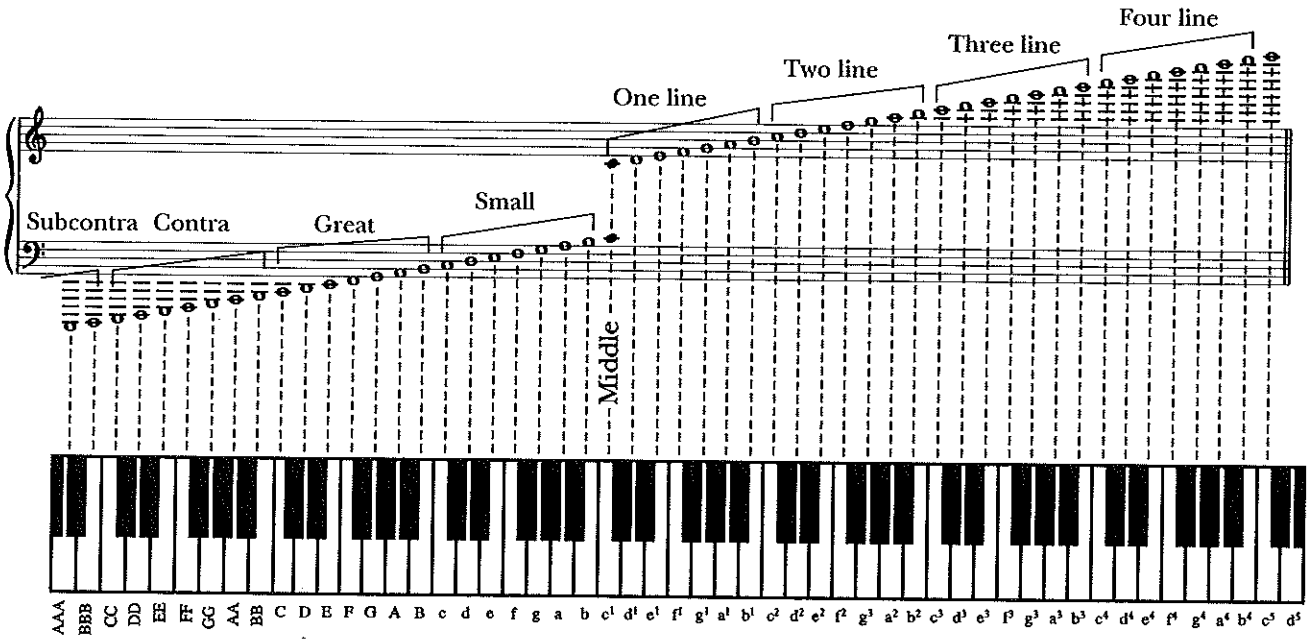
b. The alto and tenor clefs are called *C clefs* because the symbol (the same for each) is a corruption of the letter C, the center of which encircles c^1, the third line of the staff for the alto clef and the fourth line for the tenor clef.

c. The bass clef is called an *F clef* because the symbol is a corruption of the letter F, the "center" of which encircles f, the fourth line of the staff.

* Because Chapter 6, *The Notation of Rhythm*, is not dependent upon the contents of the first five chapters, this chapter may be studied concurrently with Chapters 1 through 5.

d. The term *pitch class* is used to describe a class of pitches with the same letter name. Thus, all C's (C, c, c^1, etc.) are members of the same pitch class. In addition, all pitches and their *enharmonic equivalents* belong to the same pitch class (for example, F♯ and G♭, or E♯ and F).

Because there are only seven letter names for pitches (A B C D E F G), but at least seven audible octave transpositions of any given pitch, each pitch has its own special designation.



Raymond Elliott, FUNDAMENTALS OF MUSIC, 3rd ed., © 1971, p. 15. Reprinted by permission of Prentice-Hall, Inc., Englewood Cliffs, N.J.

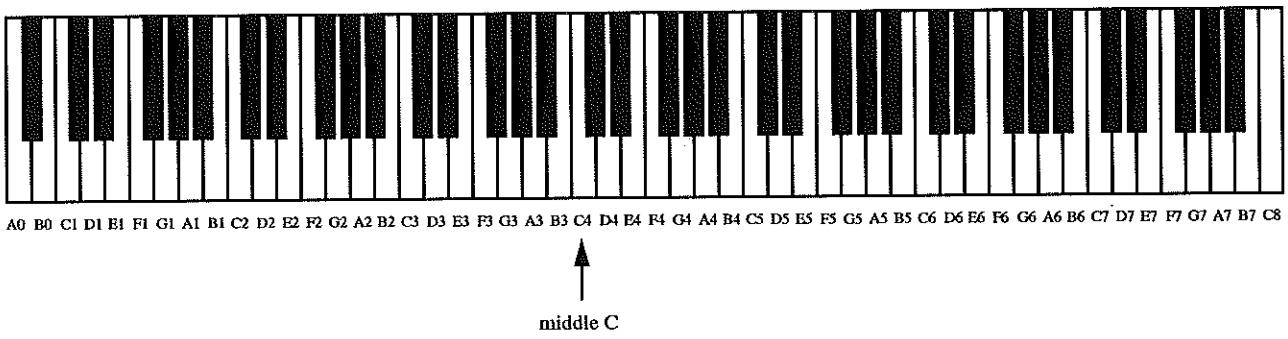
Note:

a. The shortest movement from one pitch to another, called a half step or semitone, sometimes involves a change in letter name and sometimes does not.

b. When a change of name is not involved, the pitch is raised or lowered a half step by the use of either a sharp (♯) or a flat (♭). The pitch may also be changed by two half steps without altering the letter name by the use of a double sharp (𝄪) or a double flat (𝄫).

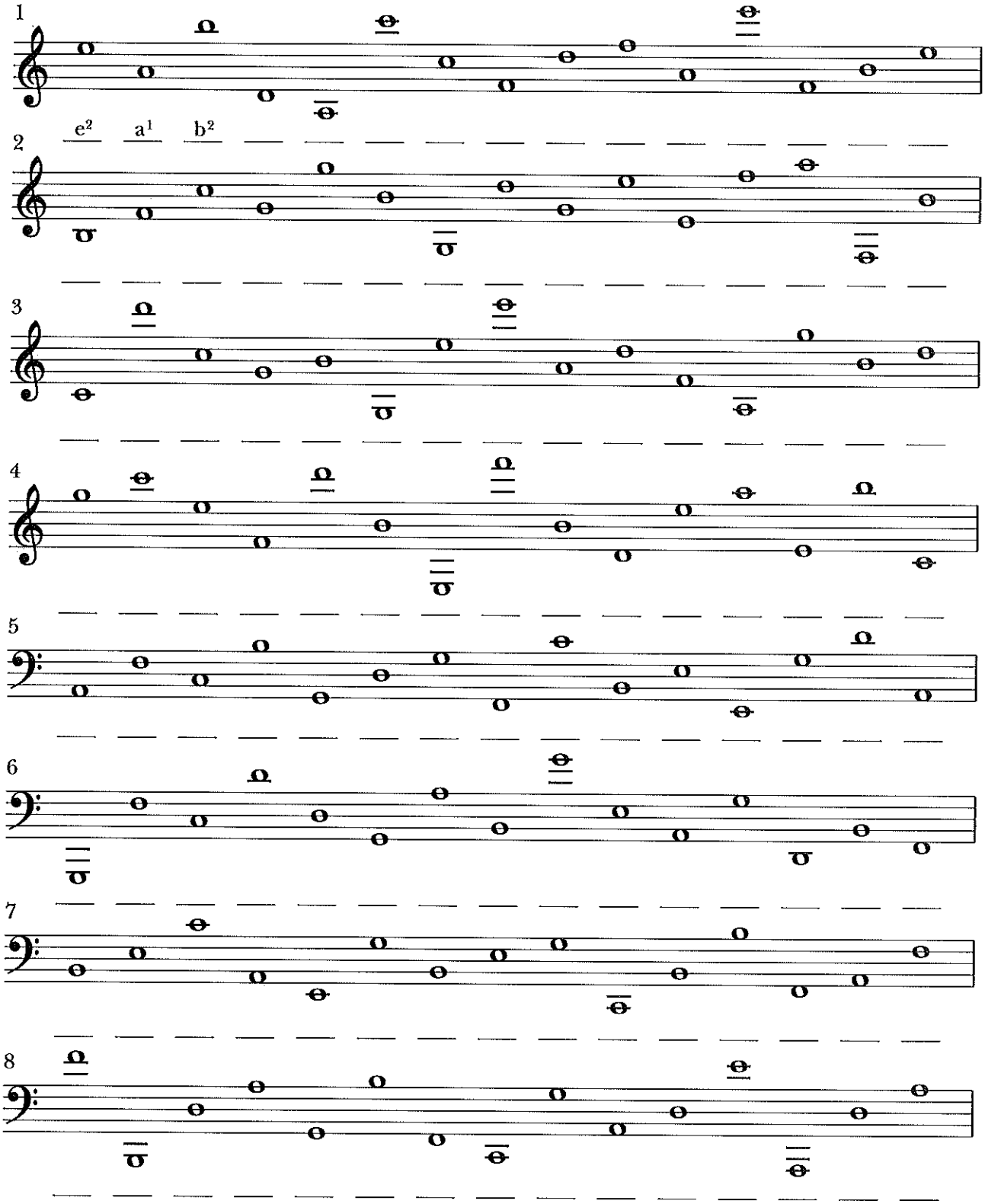
c. E to F and B to C are the only unaltered pitches that are one half step apart. C to D, D to E, F to G, G to A, and A to B are two half steps apart; this space or interval is called a whole step or a whole tone.

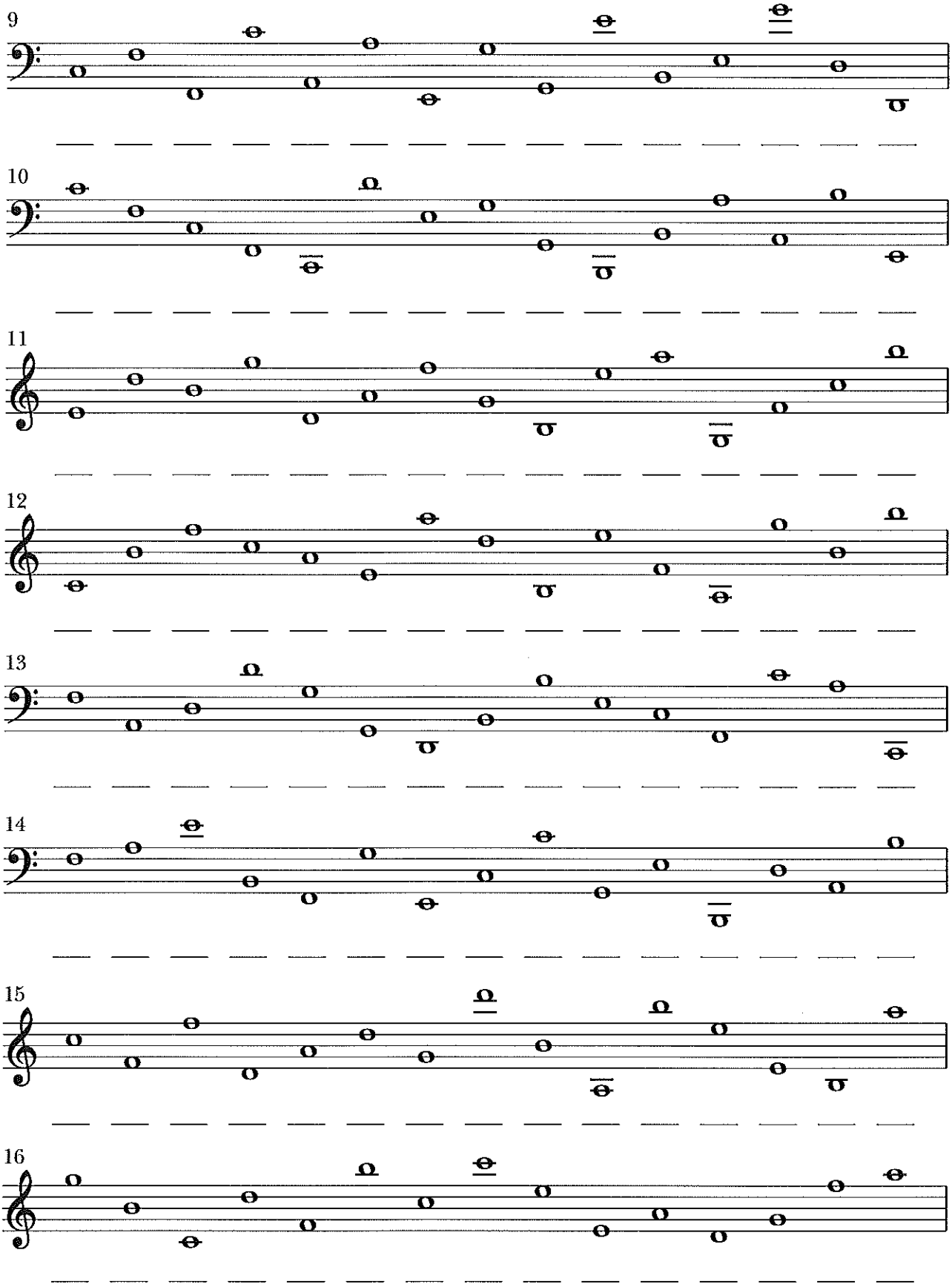
In a more recent system for pitch designation, introduced by the International Acoustic Society, the lowest C on the piano is C1. Under this designation, middle C is C4, and the piano's highest C is C8. This system, though still not embraced by the majority of writers of music texts, is universally used in electronic and computer-generated music.

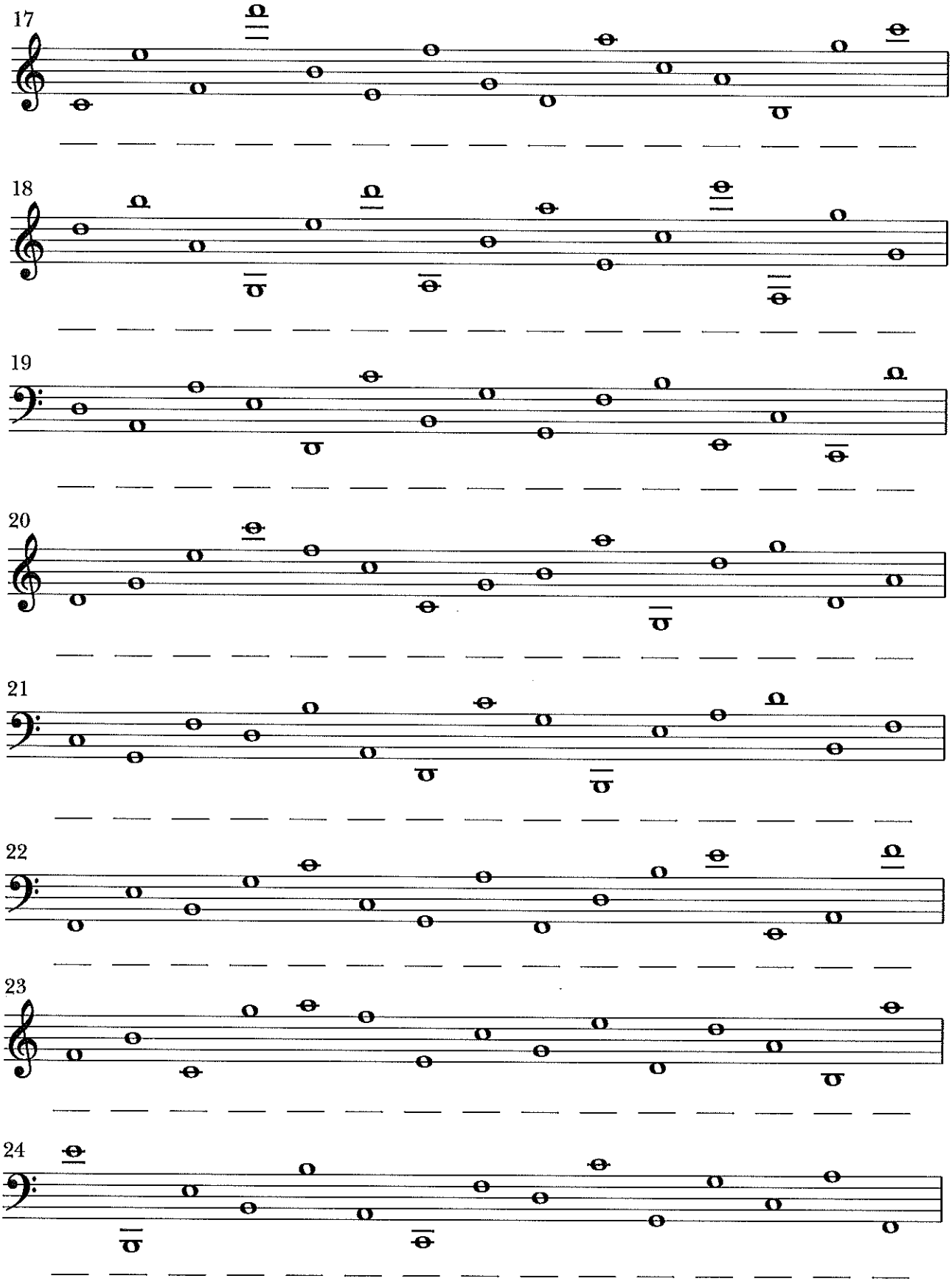


EXERCISES

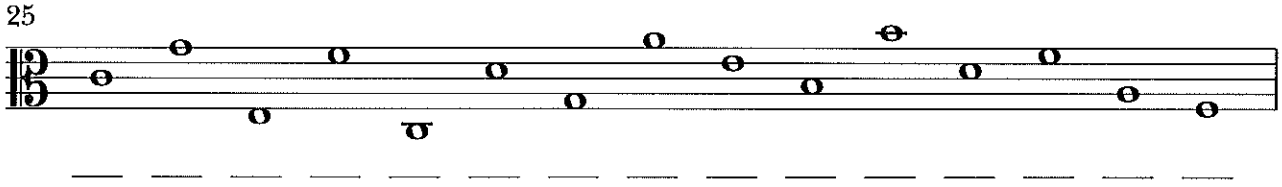
A. Write the letter name for each note in the manner indicated. Time goal—45 seconds per line (middle C = c^1).



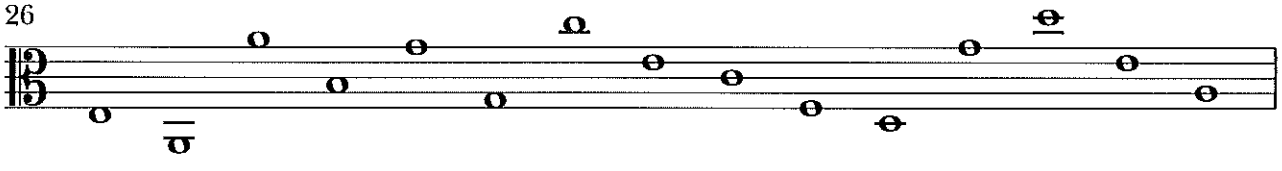

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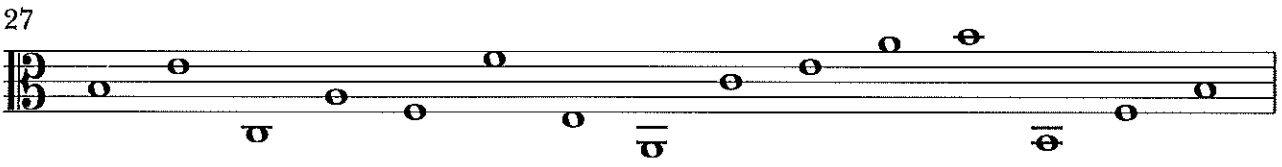
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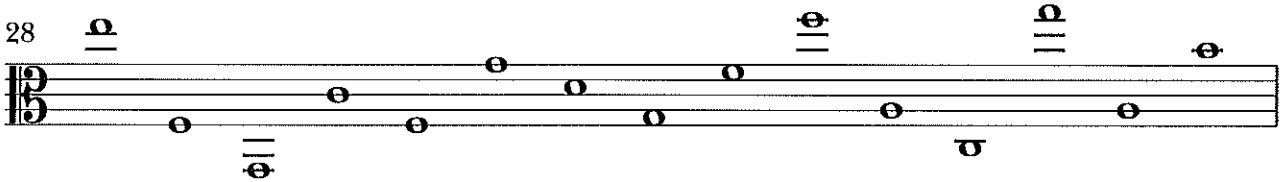
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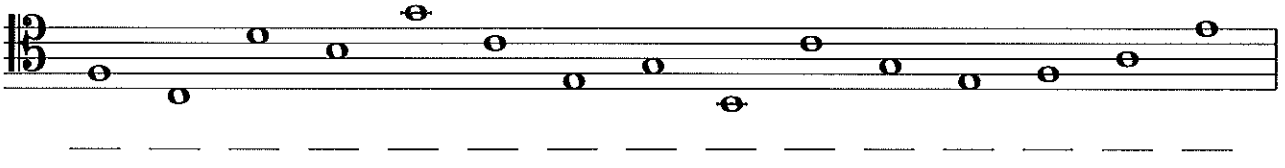
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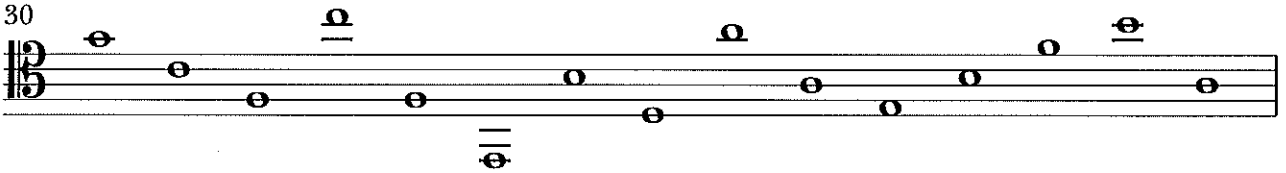
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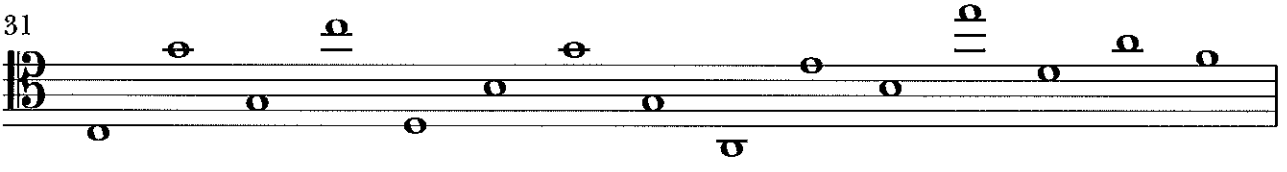
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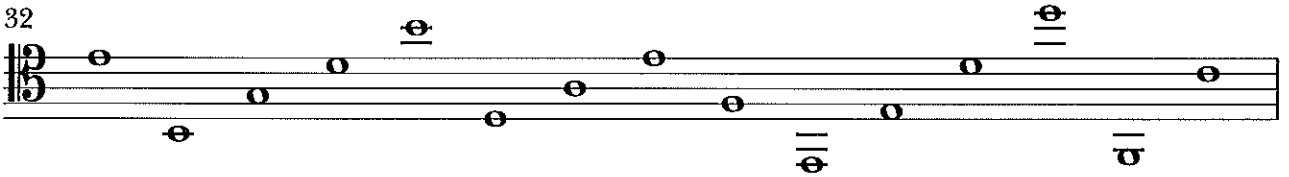
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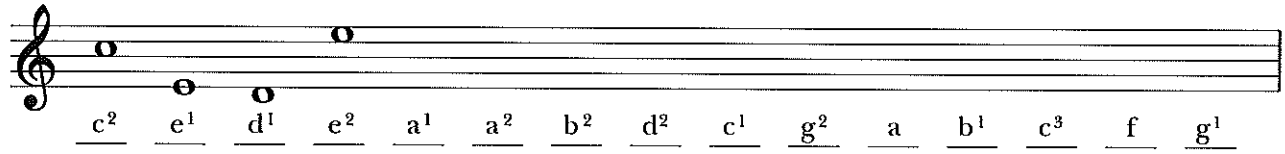


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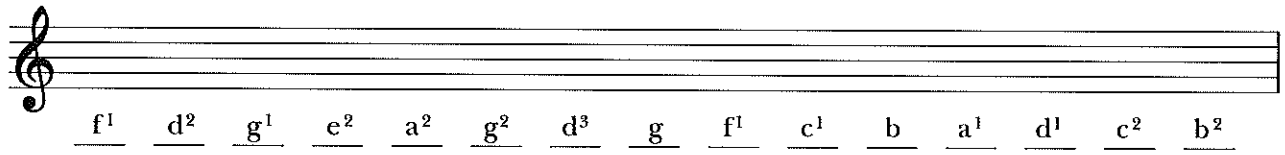
B. Notate the specified pitches. Time goal—45 seconds per line (middle C = c^1).

1



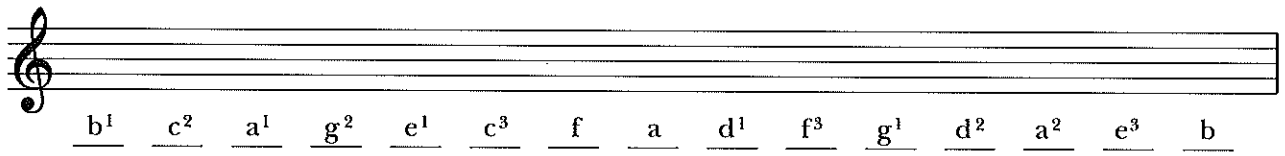
c^2 e^1 d^1 e^2 a^1 a^2 b^2 d^2 c^1 g^2 a b^1 c^3 f g^1

2



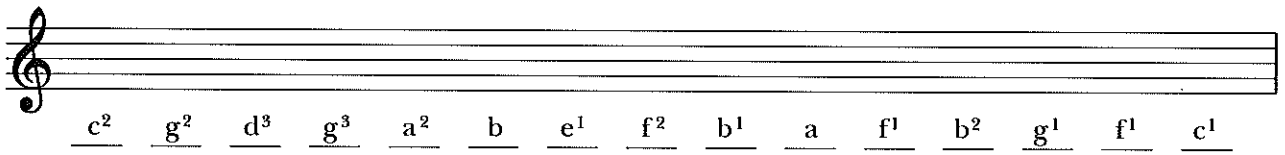
f^1 d^2 g^1 e^2 a^2 g^2 d^3 g f^1 c^1 b a^1 d^1 c^2 b^2

3



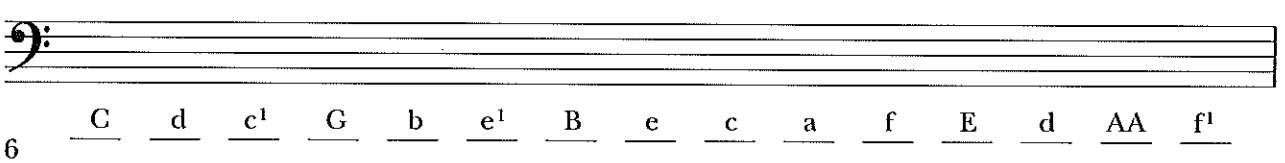
b^1 c^2 a^1 g^2 e^1 c^3 f a d^1 f^3 g^1 d^2 a^2 e^3 b

4



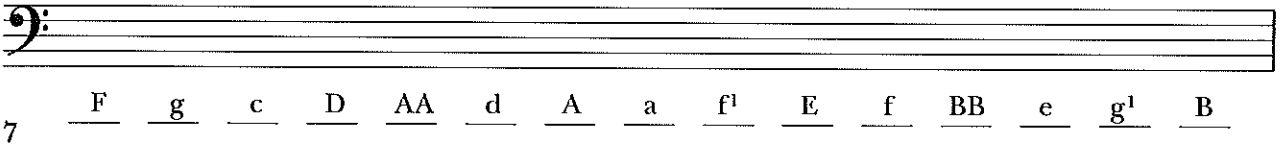
c^2 g^2 d^3 g^3 a^2 b e^1 f^2 b^1 a f^1 b^2 g^1 f^1 c^1

5



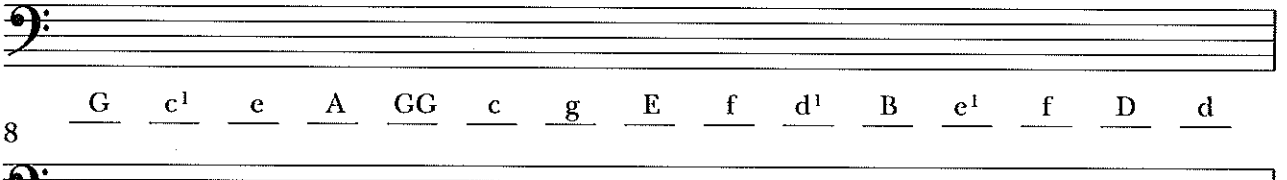
C d c^1 G b e^1 B e c a f E d AA f^1

6



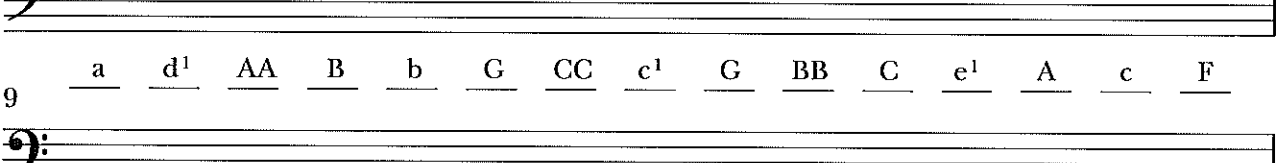
F g c D AA d A a f^1 E f BB e g^1 B

7



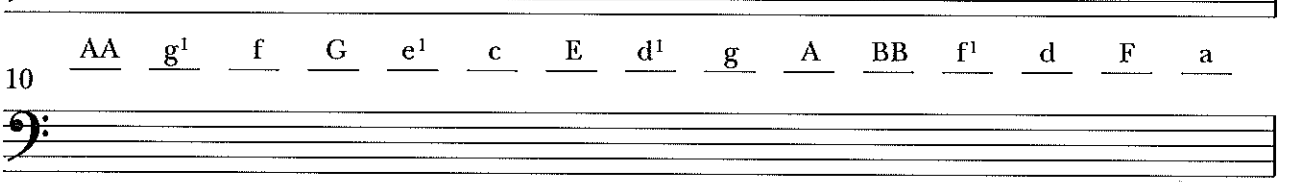
G c^1 e A GG c g E f d^1 B e^1 f D d

8



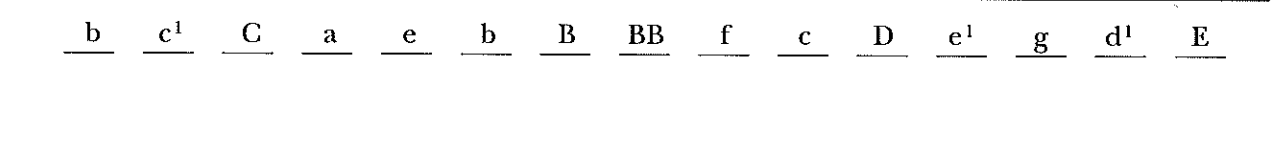
a d^1 AA B b G CC c^1 G BB C e^1 A c F

9



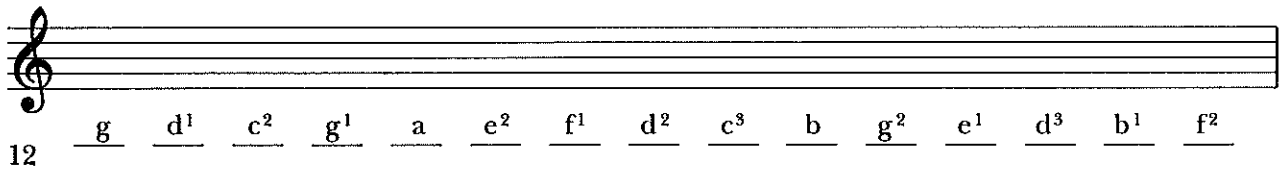
AA g^1 f G e^1 c E d^1 g A BB f^1 d F a

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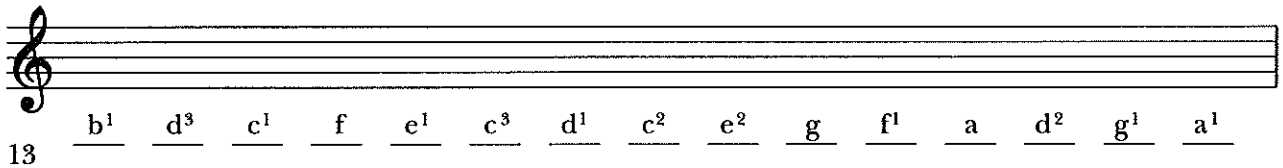
b c^1 C a e b B BB f c D e^1 g d^1 E

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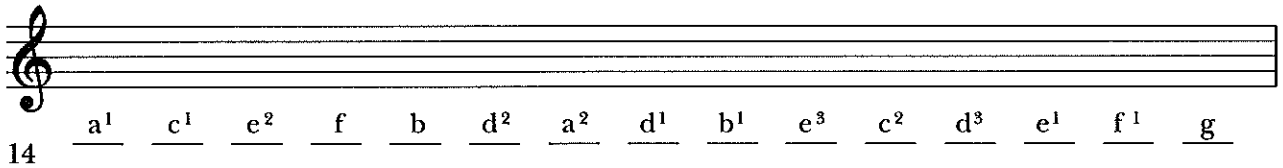
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12



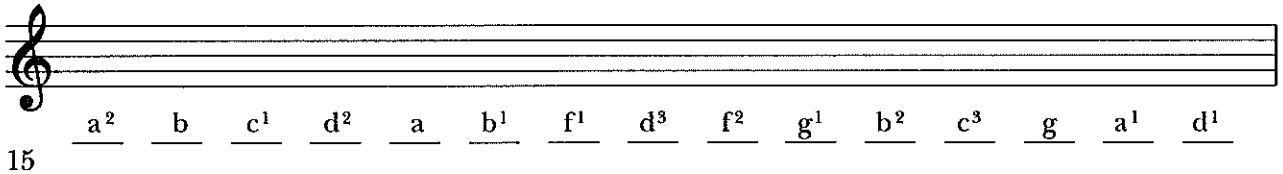
b¹ d³ c¹ f e¹ c³ d¹ c² e² g f¹ a d² g¹ a¹

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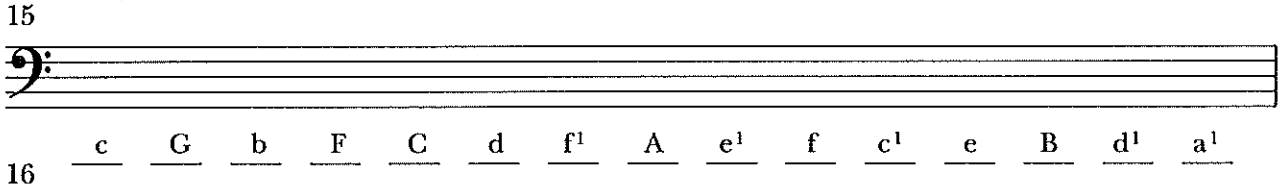
a¹ c¹ e² f b d² a² d¹ b¹ e³ c² d³ e¹ f¹ g

14



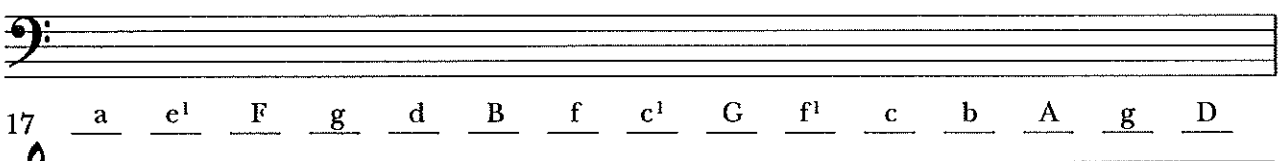
a² b c¹ d² a b¹ f¹ d³ f² g¹ b² c³ g a¹ d¹

15



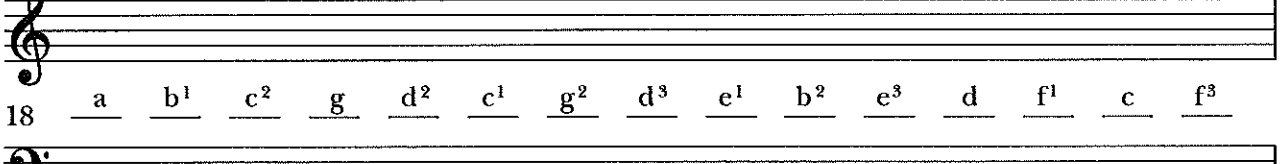
c G b F C d f¹ A e¹ f c¹ e B d¹ a¹

16



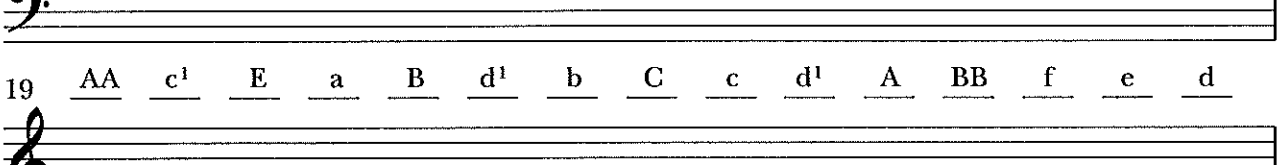
a e¹ F g d B f c¹ G f¹ c b A g D

17



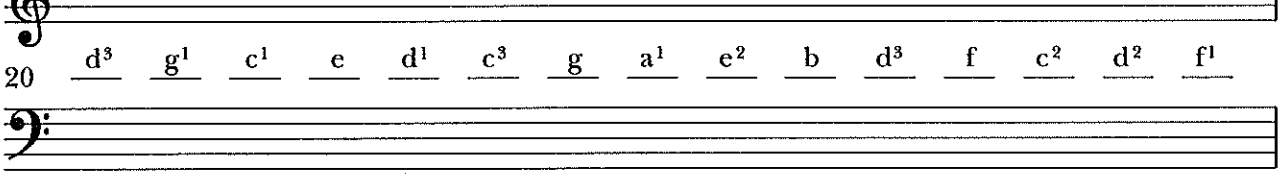
a b¹ c² g d² c¹ g² d³ e¹ b² e³ d f¹ c f³

18



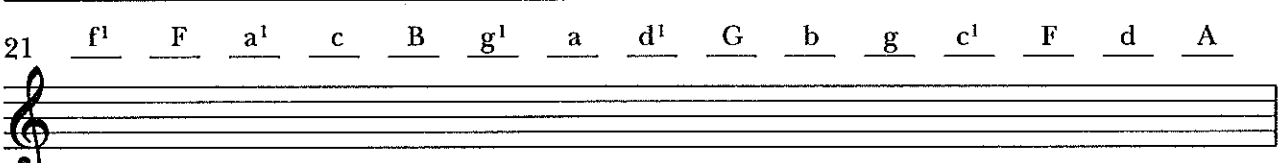
AA c¹ E a B d¹ b C c d¹ A BB f e d

19



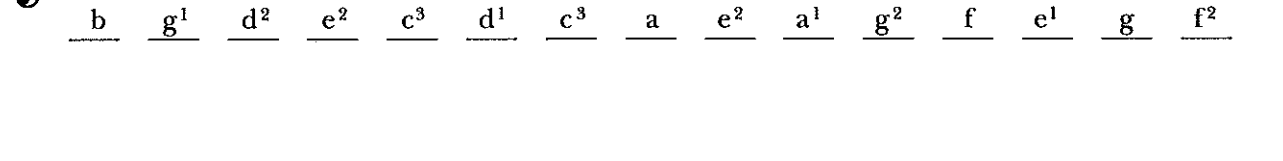
d³ g¹ c¹ e d¹ c³ g a¹ e² b d³ f c² d² f¹

20



f¹ F a¹ c B g¹ a d¹ G b g c¹ F d A

21



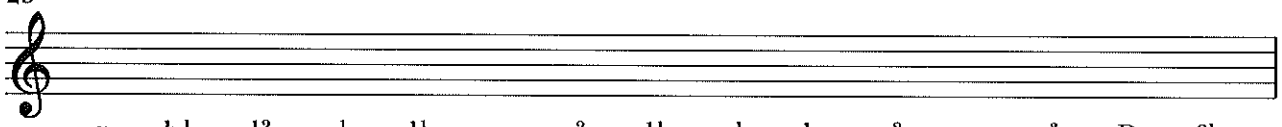
b g¹ d² e² c³ d¹ c³ a e² a¹ g² f e¹ g f²

22



e C c^1 AA G f e^1 D g F d^1 g BB E f^1

23



g b^1 d^2 c^1 d^1 a c^2 d^1 e^1 b e^2 e c^3 D f^1

24



c^1 B d f^1 e b c d^1 f C E BB g e^1 A

25



g a c^2 f g^1 B d b^1 e^1 b a^1 e c^1 d^2 c

26



d^1 f B e^1 a b^1 c^1 e g b a^1 g^1 A e^1 f^1

27



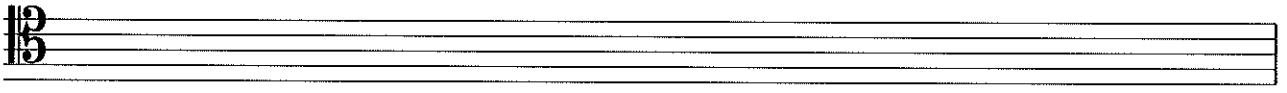
g a^1 g^1 d c^1 f B d^2 e e^1 f^1 c a d^1 b

28



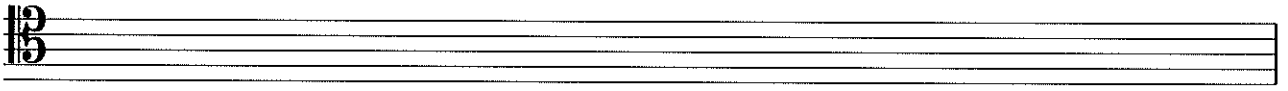
f g^1 g d^1 b A d b^1 c a c^2 f^1 a^1 e^1 c^1

29



F a^1 d^1 c B g^1 c^2 c^1 g A e^1 f d e f^1

30



e^1 d f^1 G b^1 d^1 c g^1 c^2 a^1 A a c^1 f F

31



F a e^1 b G G f^1 e c^1 A c^2 d^1 B c b^1

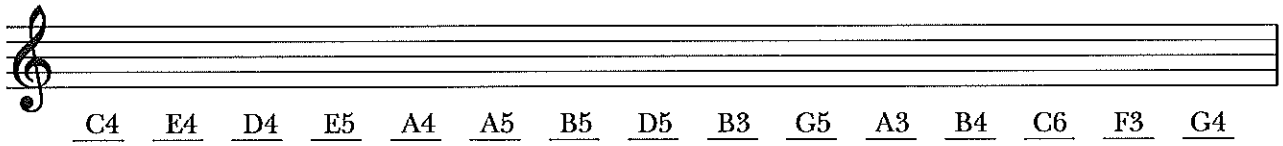
32



a^1 f d^1 g f^1 A c^1 B c^1 c^2 b g^1 a e^1 b^1

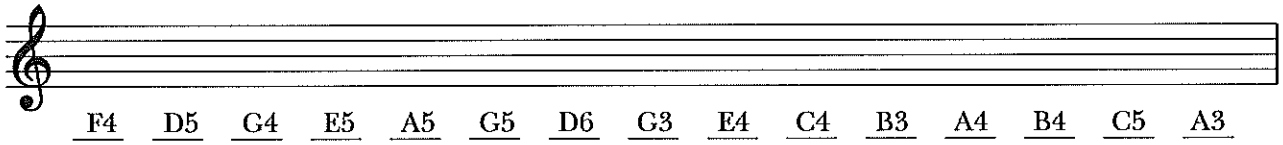
C. Notate the specified pitches. Time goal—45 seconds per line (middle C = c^4).

1



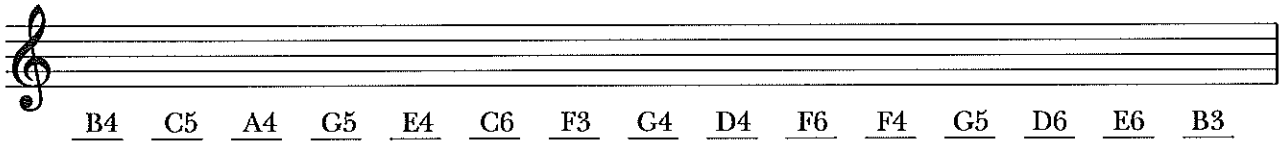
C4 E4 D4 E5 A4 A5 B5 D5 B3 G5 A3 B4 C6 F3 G4

2



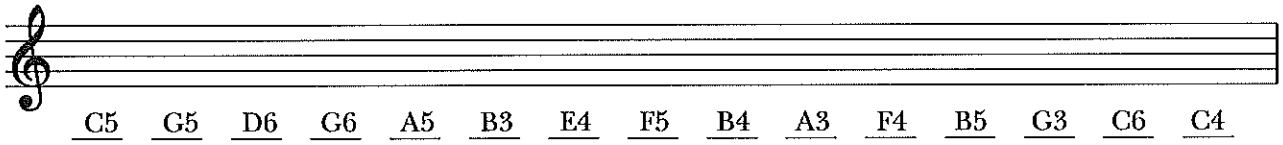
F4 D5 G4 E5 A5 G5 D6 G3 E4 C4 B3 A4 B4 C5 A3

3



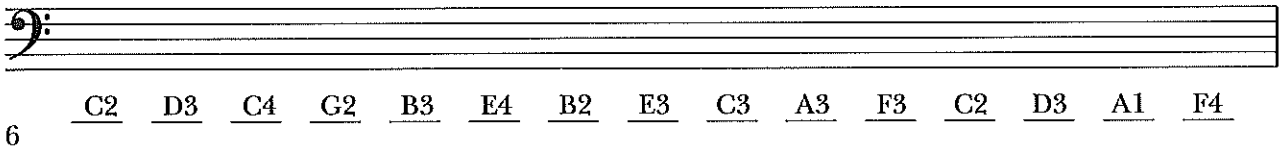
B4 C5 A4 G5 E4 C6 F3 G4 D4 F6 F4 G5 D6 E6 B3

4



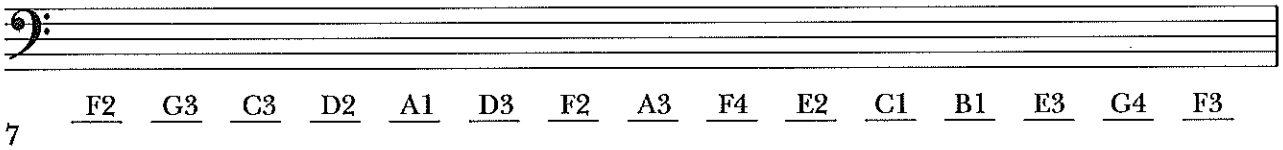
C5 G5 D6 G6 A5 B3 E4 F5 B4 A3 F4 B5 G3 C6 C4

5



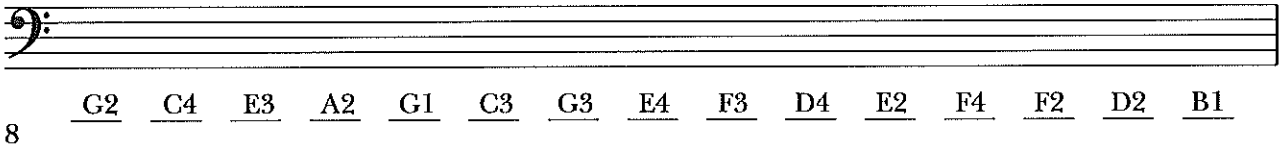
C2 D3 C4 G2 B3 E4 B2 E3 C3 A3 F3 C2 D3 A1 F4

6



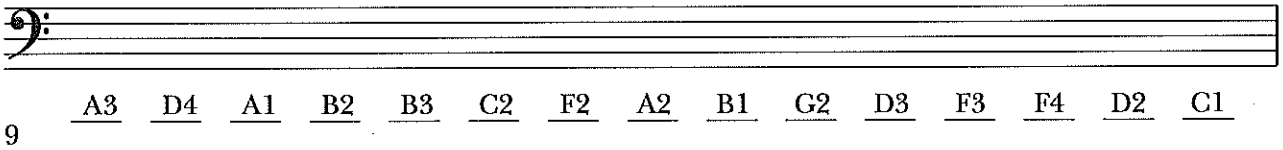
F2 G3 C3 D2 A1 D3 F2 A3 F4 E2 C1 B1 E3 G4 F3

7



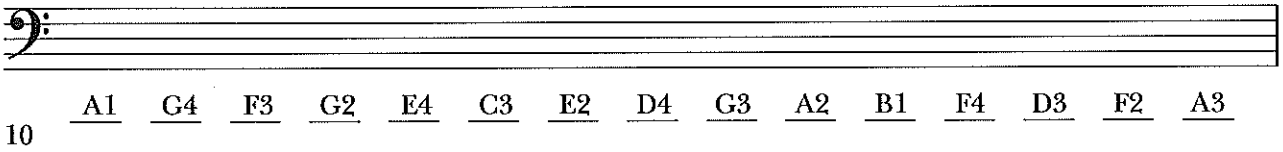
G2 C4 E3 A2 G1 C3 G3 E4 F3 D4 E2 F4 F2 D2 B1

8



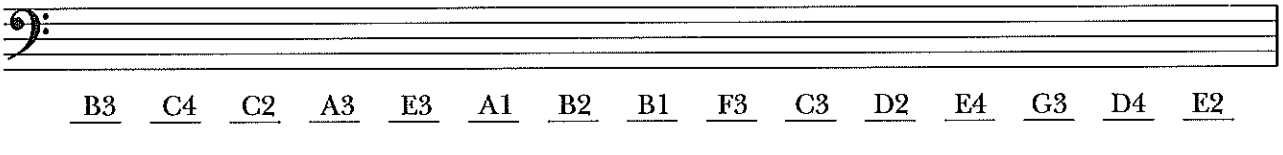
A3 D4 A1 B2 B3 C2 F2 A2 B1 G2 D3 F3 F4 D2 C1

9



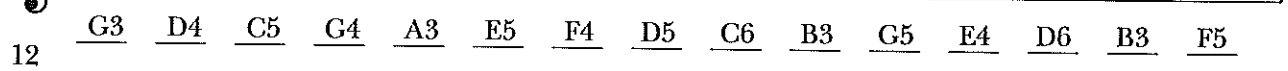
A1 G4 F3 G2 E4 C3 E2 D4 G3 A2 B1 F4 D3 F2 A3

10



B3 C4 C2 A3 E3 A1 B2 B1 F3 C3 D2 E4 G3 D4 E2

11



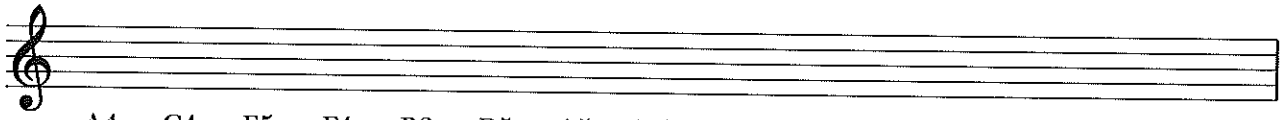
G3 D4 C5 G4 A3 E5 F4 D5 C6 B3 G5 E4 D6 B3 F5

12



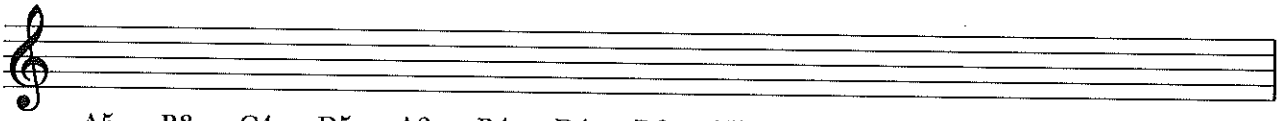
B4 D6 C4 F3 E4 C6 D4 C5 E3 G3 F4 A3 D4 G4 A5

13



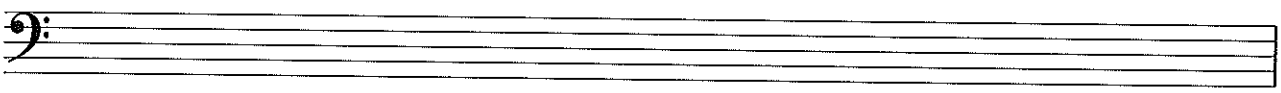
A4 C4 E5 F4 B3 D5 A5 D4 B4 E6 C3 D6 E4 F3 G3

14



A5 B3 C4 D5 A3 B4 D4 D6 F5 G4 B5 C6 G3 A4 G5

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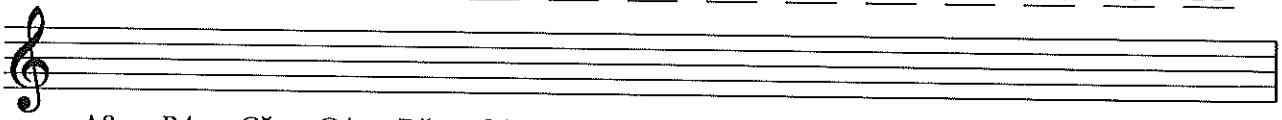


C3 G2 B3 G3 B3 D3 F4 A2 E4 D3 C4 E3 B2 D4 A3

16



17 A3 E4 F2 G3 A1 B2 F3 C4 G2 F4 C3 B3 A2 E3 B1



18 A3 B4 C5 G4 D5 C4 G5 E4 B5 B3 E6 D6 F4 C6 F6



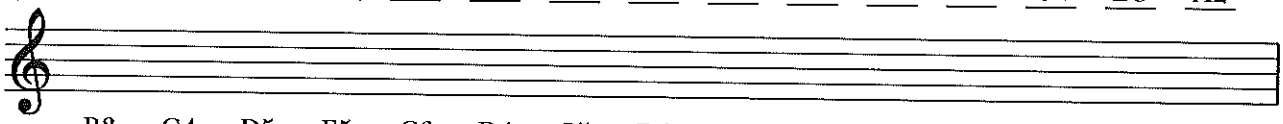
19 A1 C4 E2 A3 B1 D4 B3 C2 C3 D2 A2 E4 F3 B2 D3



20 D6 C4 G4 E4 D4 C6 G3 A4 E5 B3 E6 F3 C5 D5 F4



21 F4 F2 A3 C3 B2 G4 G3 F3 G2 B3 B1 A1 C4 D3 A2



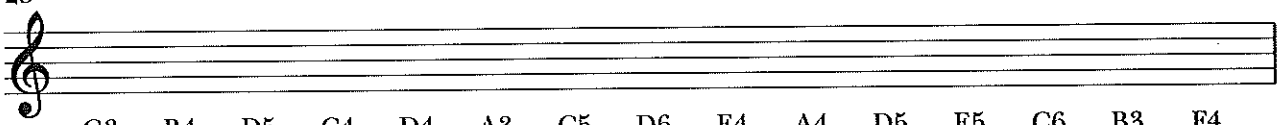
B3 G4 D5 E5 C6 D4 C5 D4 D6 A3 E5 B4 E4 G3 F5

22



E3 C2 C4 A1 G2 F3 E4 D2 G3 F2 C3 G2 B1 E2 D4

23



G3 B4 D5 C4 D4 A3 C5 D6 E4 A4 D5 E5 C6 B3 F4

24



C4 B2 D3 F4 E3 B3 C3 D4 F3 C2 E2 B1 G3 E4 D2

25



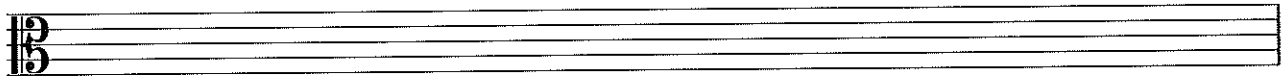
G3 A3 C5 F3 G4 B2 D3 B4 E4 B3 A4 B3 E5 D5 C3

26



D4 F3 B2 E4 A3 B4 C4 E3 B3 F4 A2 G4 C3 A4 D3

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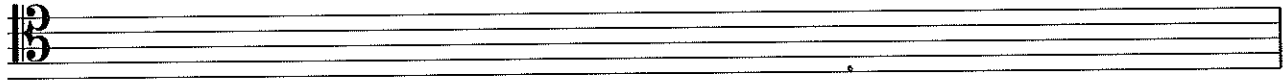
G3 A4 G4 D3 C4 F3 B2 D5 E3 C5 F4 C3 A3 D4 D5

28



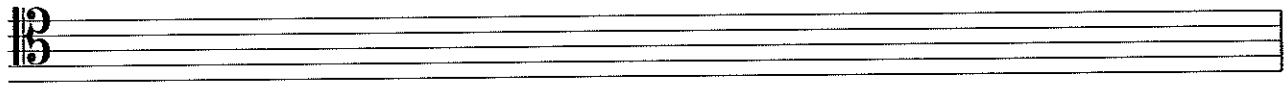
F3 A3 G4 D4 B3 A4 D3 B4 C3 E3 F4 C5 B4 E4 C4

29



F4 A4 C3 D3 B3 G3 A2 B2 E4 G4 D4 C5 F3 A3 E3

30



E4 D3 F4 G2 B3 D4 D3 G4 C5 A4 C3 A3 C4 A3 F2

31



F2 A3 E4 B3 G2 D3 F4 E3 C4 A2 C5 D4 F3 C3 B4

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A4 F3 D5 G3 F4 A2 C4 B2 C3 C5 B3 G4 B1 B2 D4

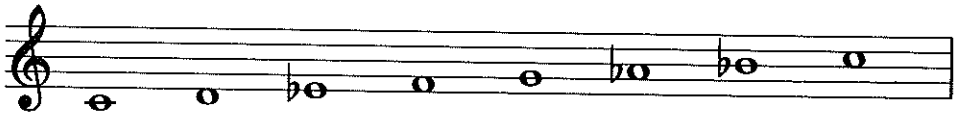
2 Scales

Strictly speaking, a scale is an ascending, ordered arrangement of pitches. In the tertian harmonic system, which is basic in the study of music theory, two scale types occur: the *major scale* and the *minor scale.* The minor scale has three variants: the *natural minor,* the *melodic minor,* and the *harmonic minor.*

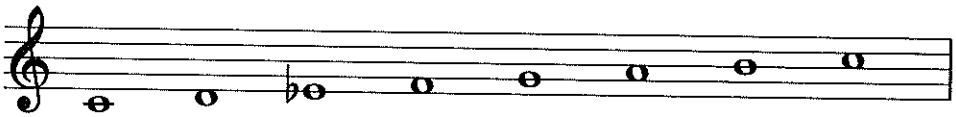
major scale



natural minor scale



melodic minor scale



harmonic minor scale



Note:

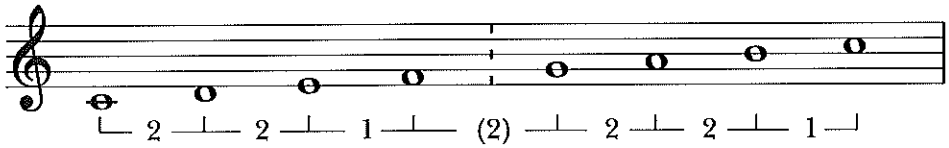
a. Each scale consists of eight pitches, the first and last being an octave apart.

b. Because any one of the scales may be built upon any given pitch, each scale has its own characteristic organization. In each case, this organization is most clearly seen when one examines not the pitches themselves, but the intervals between the pitches.

c. In all the scales, the interval between adjacent pitches is called a second. It is a *major second* if the interval between the pitches is two half steps and a *minor second* if the interval is one half step. (In the harmonic minor scale, there is one interval of three half steps. This interval is called an *augmented second.*)

THE MAJOR SCALE

The major scale may be viewed as consisting of two tetrachords (four-note groups) separated by two half steps. Each tetrachord contains five half steps.



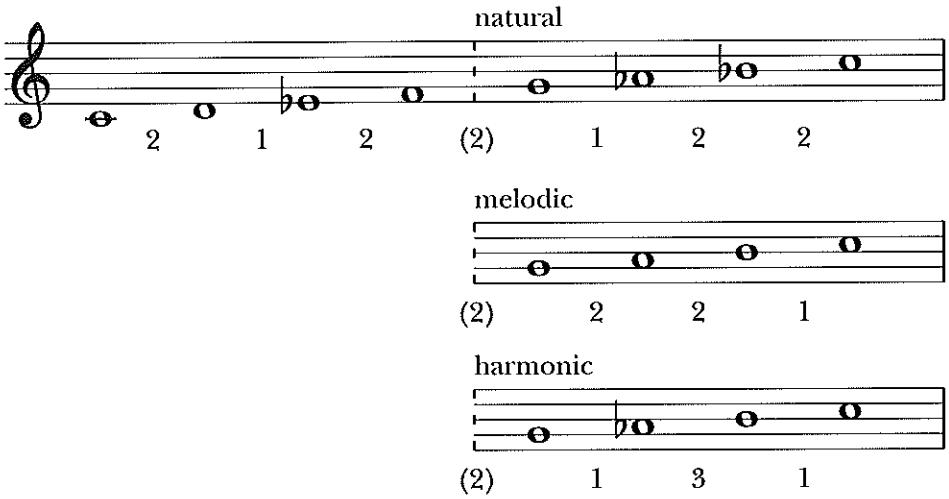
Note:

Each tetrachord has the same intervallic properties—two half steps, followed by two half steps, followed by one half step. Thus, the major scale may be expressed as

2–2–1–(2)–2–2–1

THE MINOR SCALE

The minor scale may also be viewed as consisting of two tetrachords separated by two half steps. The upper tetrachord, however, differs for each variant. Like the major scale, each tetrachord contains five half steps.



Note:

a. All three variants have the same lower tetrachord:

2–1–2

b. The upper tetrachord of the *natural* minor scale is the retrograde of the major tetrachord (i.e., the major tetrachord written backward). The scale, therefore, may be expressed as

2–1–2–(2)–1–2–2

c. The upper tetrachord of the *melodic* minor scale is the same as that of the major tetrachord. The scale, therefore, may be expressed as

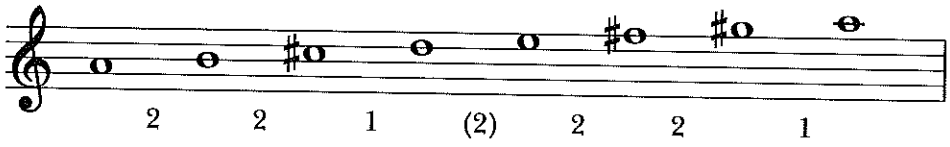
2–1–2–(2)–2–2–1

d. The upper tetrachord of the *harmonic* minor scale is palindromic (i.e., it reads the same forward and backward). The scale, therefore, may be expressed as

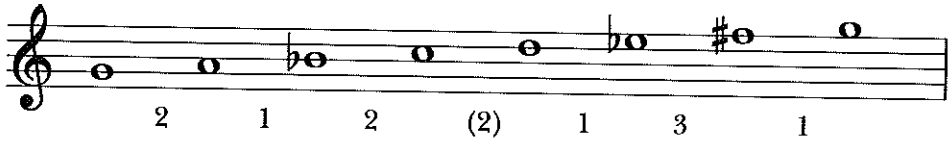
2–1–2–(2)–1–3–1

NAMING SCALES

A scale is named by identifying its lowest note and by describing its type, that is, major or variety of minor. Thus the following scale



is A major and



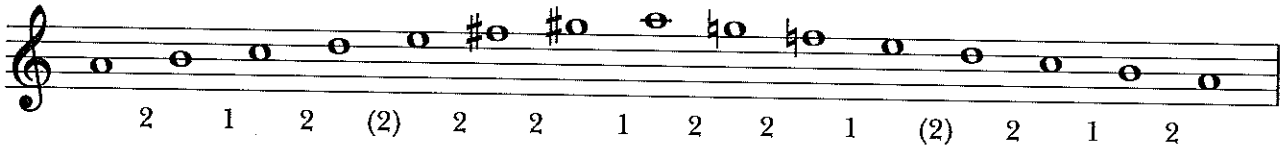
is g harmonic minor.

The major scale, by convention, is assigned an uppercase letter (e.g., A), and the minor a lowercase letter (e.g., g).

SCALES IN DESCENT

In this chapter each scale has been presented as an ascending series of pitches. In addition, a scale may be written as a descending series without altering its intervallic organization, except in the case of the melodic minor.

The descending melodic minor scale has exactly the same intervallic organization as the ascending (and descending) *natural minor.*

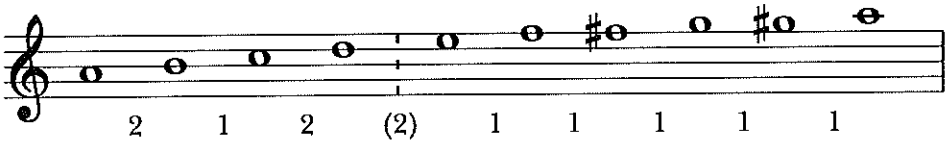


a melodic minor (ascending and descending)

THE SYNTHETIC MINOR SCALE

No effort has been made in this part of the text to explain when and why the various forms of the minor scale are used. In Chapter 10, the text shows

that the minor scale may, for practical purposes, be considered a synthesis of all three forms, containing ten pitches:



Note:

a. The upper part of the scale, now a hexachord (six-note group), has an intervallic organization entirely of half steps.

b. Considerations of voice leading and melodic activity will determine which pitches from the hexachord are appropriately employed in a given musical situation. These issues are discussed in Part Two.

MODES

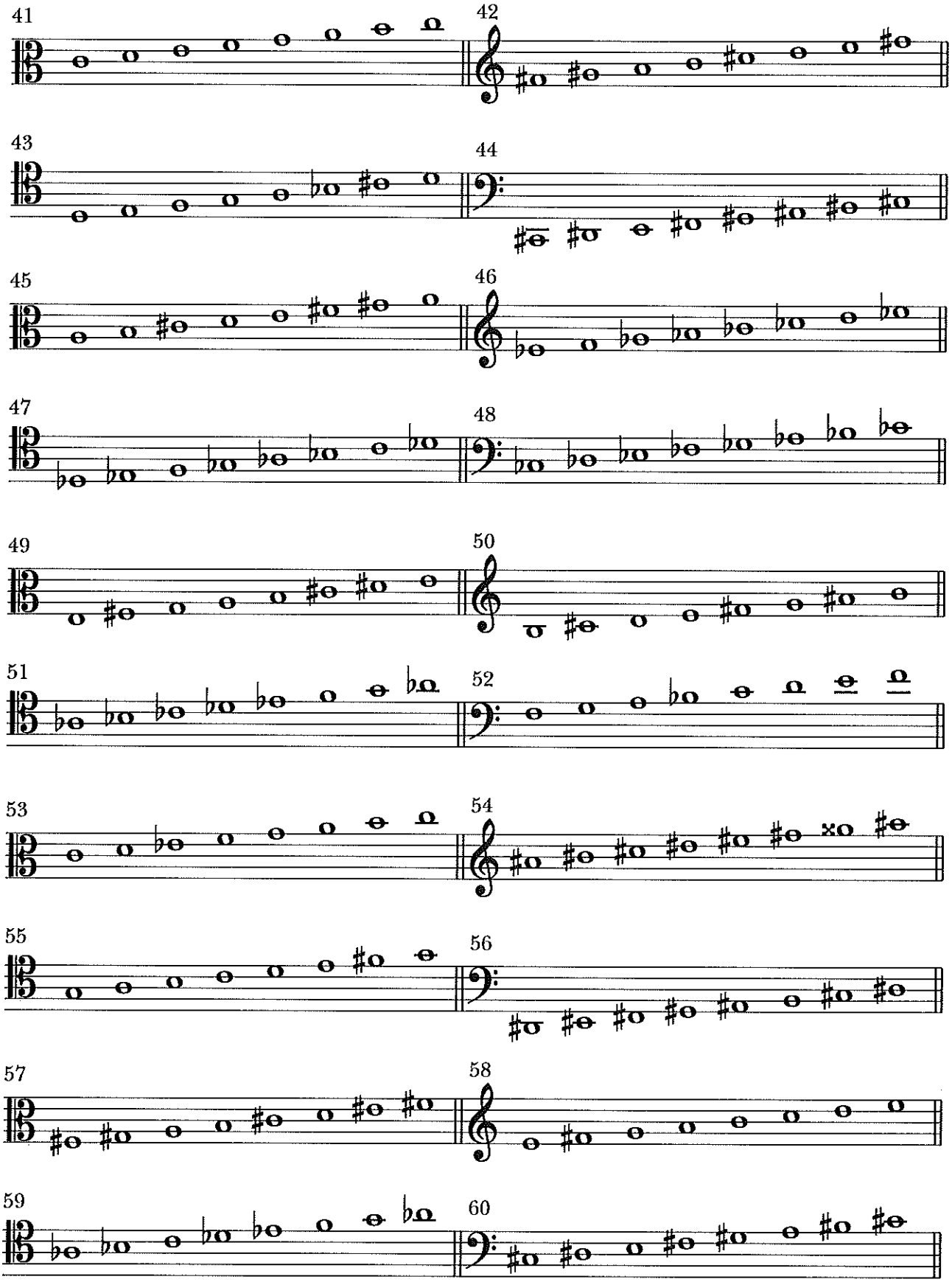
The study of the scalar structure of modes is not a necessary foundation for the majority of the chapters in this book; however, the fact that modes may be derived from the major scale makes an introductory study of them possible at this point. The reader may wish to refer, therefore, to the beginning of Chapter 26, in which the derivation of the seven modes from the major scale is shown, followed by each mode's essential characteristics.

EXERCISES

A. Identify each scale by analyzing its tetrachord structure. Time goal—30 seconds per scale.

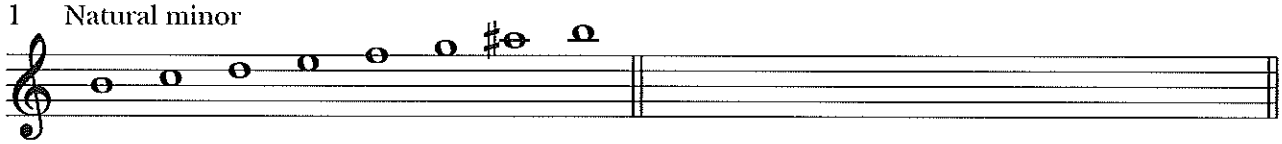



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49
50
51
52
53
54
55
56
57
58
59
60

B. Identify the error(s) in each scale and write the scale correctly. Time goal—45 seconds per scale.

1 Natural minor



2 Harmonic minor



3 Melodic minor



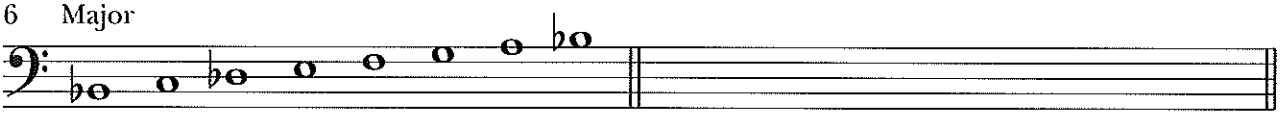
4 Harmonic minor



5 Major



6 Major



7 Natural minor



8 Melodic minor



9 Harmonic minor



10 Natural minor



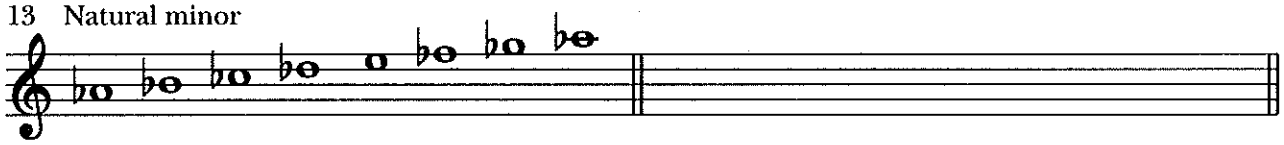
11 Major



12 Harmonic minor



13 Natural minor



14 Harmonic minor



15 Natural minor



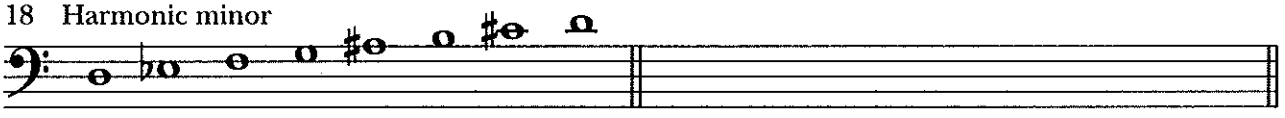
16 Harmonic minor



17 Melodic minor



18 Harmonic minor



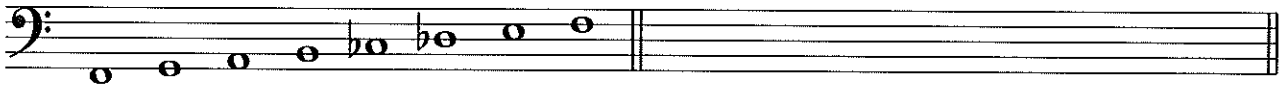
19 Melodic minor



20 Major



21 Harmonic minor



22 Major



23 Natural minor



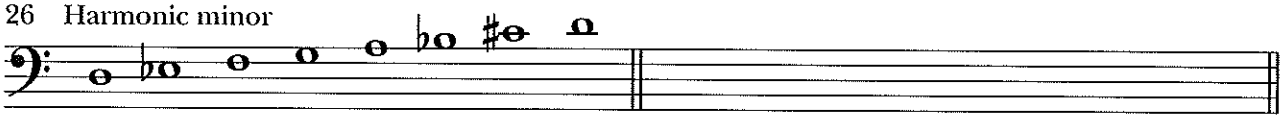
24 Melodic minor



25 Natural minor



26 Harmonic minor



27 Major



28 Melodic minor



29 Natural minor



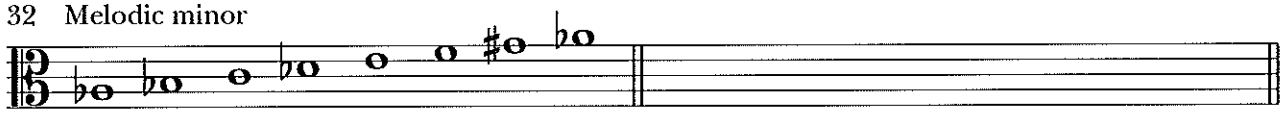
30 Harmonic minor



31 Melodic minor



32 Melodic minor



33 Major



34 Harmonic minor



35 Harmonic minor



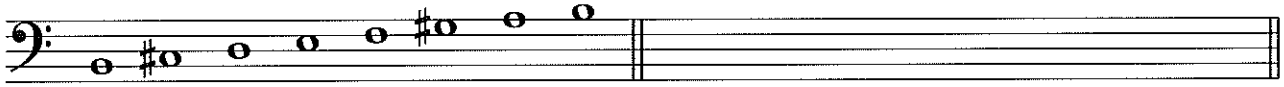
36 Melodic minor



37 Major



38 Natural minor



39 Melodic minor



40 Natural minor



C. Write the specified scales. Time goal–20 seconds per scale.

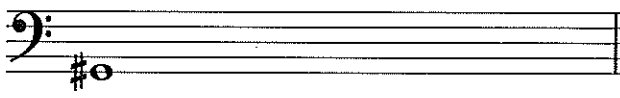
1 Major



2 Melodic minor



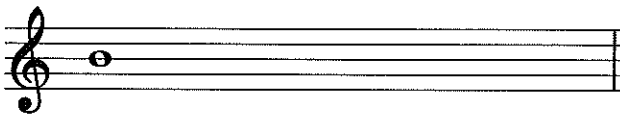
3 Natural minor



4 Harmonic minor



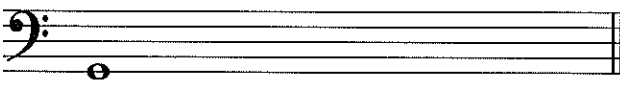
5 Melodic minor



6 Harmonic minor



7 Melodic minor



8 Natural minor



9 Natural minor



10 Harmonic minor



11 Major



12 Harmonic minor



13 Harmonic minor



14 Melodic minor



15 Natural minor



16 Major



17 Natural minor



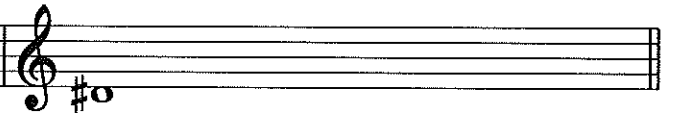
18 Melodic minor



19 Harmonic minor



20 Harmonic minor



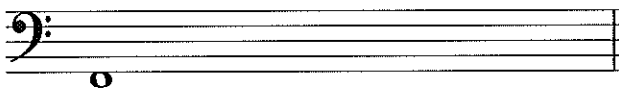
21 Melodic minor



22 Major



23 Natural minor



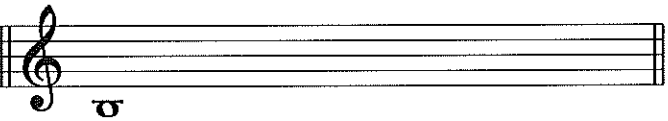
24 Harmonic minor



25 Major



26 Natural minor



27 Major



28 Natural minor



29 Melodic minor



30 Harmonic minor



31 Major



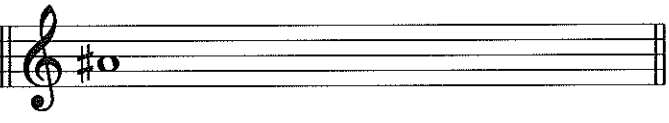
32 Natural minor



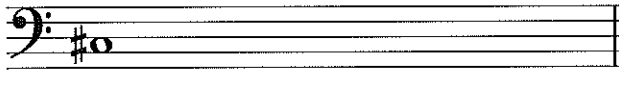
33 Major



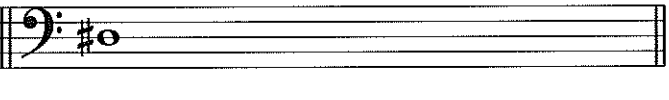
34 Melodic minor



35 Melodic minor



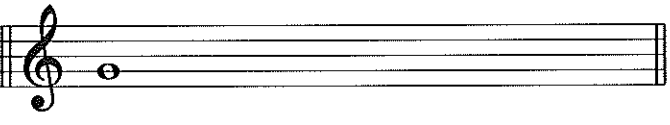
36 Natural minor



37 Major



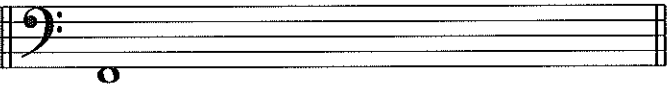
38 Harmonic minor



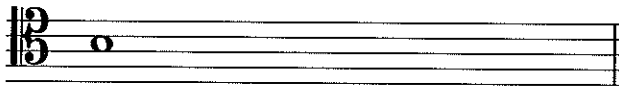
39 Melodic minor



40 Major



41 Major



42 Major



43 Major



44 Major



45 Harmonic minor



46 Natural minor



47 Melodic minor



48 Major



49 Harmonic minor



50 Harmonic minor



51 Natural minor



52 Major



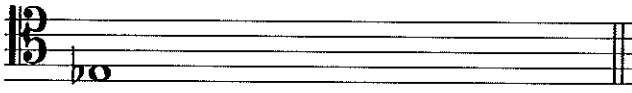
53 Melodic minor



54 Natural minor



55 Major



56 Major



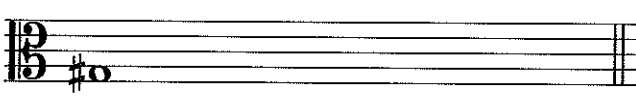
57 Melodic minor



58 Natural minor



59 Harmonic minor



60 Major

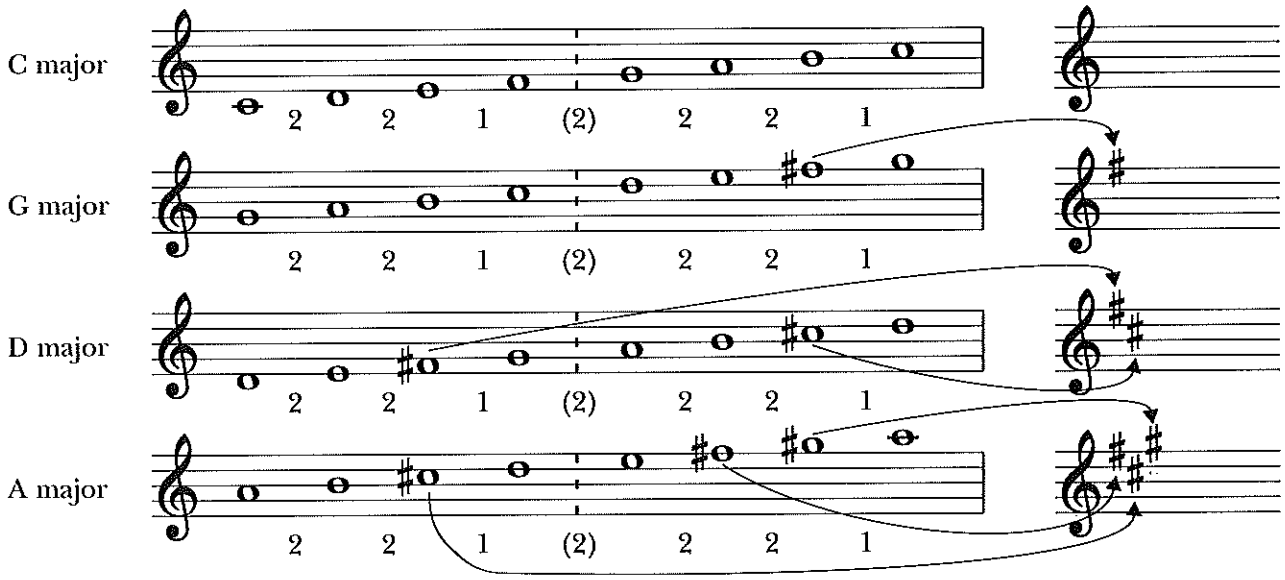


3 Key Signatures and Scale Degrees

A *key signature*, which appears to the right of the clef sign as an arrangement of sharps or flats, gives two essential pieces of information to the musician.

1. It tells the performer which pitches are to be raised by a sharp, or lowered by a flat.
2. It indicates (by implication) that there is a central pitch around which the other pitches are organized. This pitch is the lowest member of the scale (the scale may be major or minor), is called the *tonic*, and is the pitch after which the key is named. For example, if the tonic is A, the key is either A major or a minor, depending on the number of accidentals.

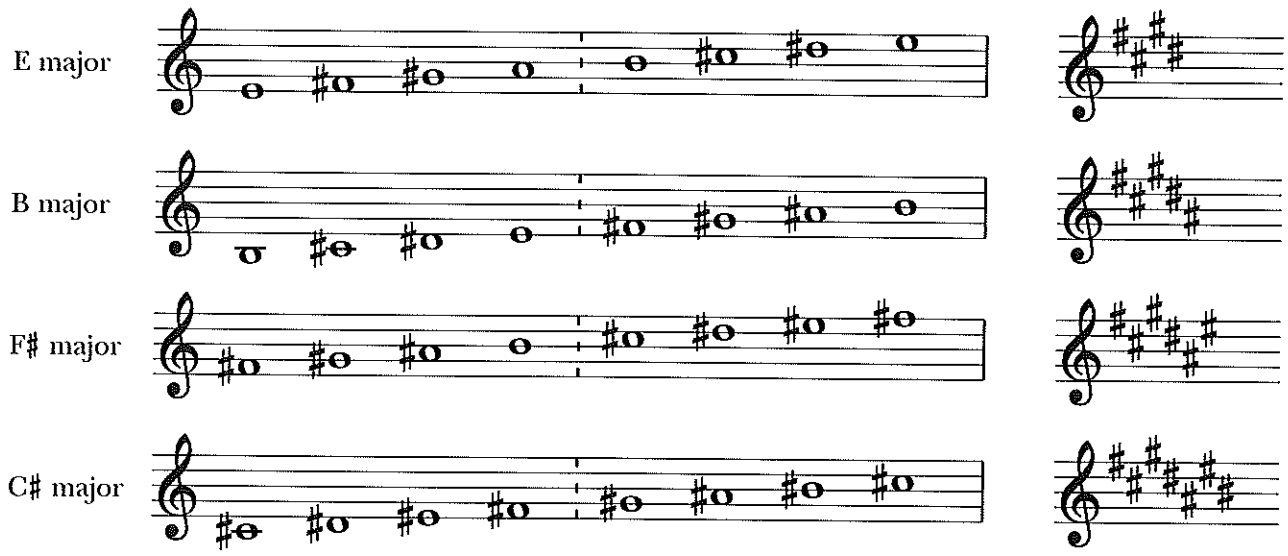
Key signatures are all derived from the major scale in a completely logical manner. Let us examine key signatures made up of sharps first.



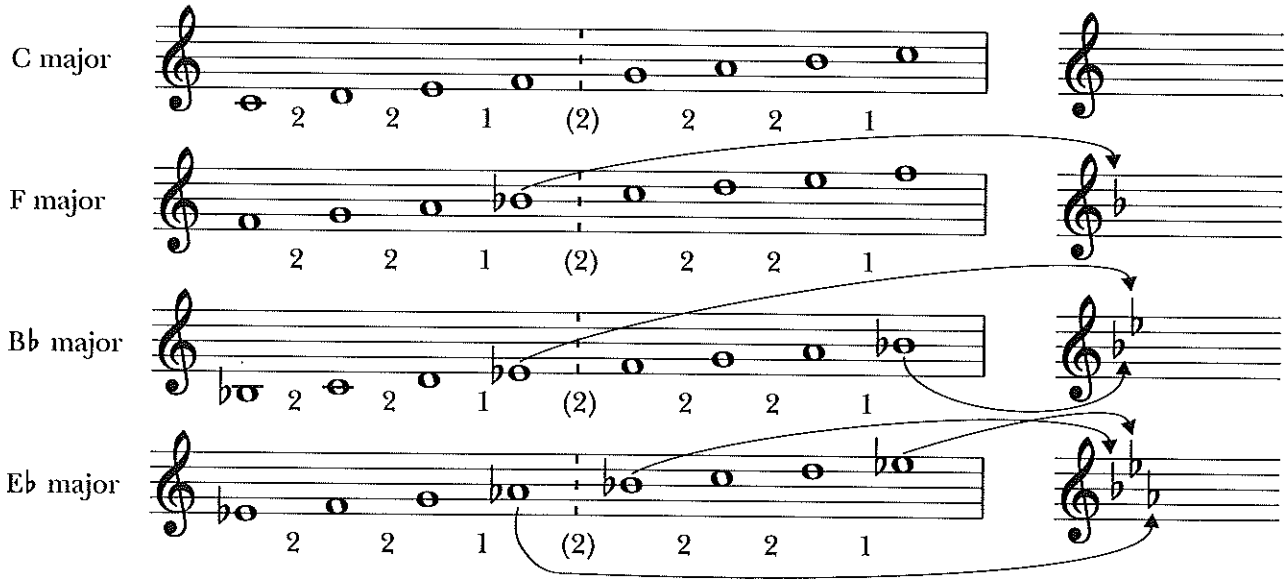
Note:

a. Each scale begins with the second tetrachord of the previous scale.
b. In each case a new accidental is introduced to the scale as a result of having to sharp the third pitch of the second tetrachord.
c. The key signatures show an "accumulation" of accidentals, F♯ always appearing first, C♯ second, and so on.

The remainder of the major scales with sharps and their derived key signatures are



The key signatures with flats are derived from the major scale in a manner similar to the derivation of key signatures with sharps.



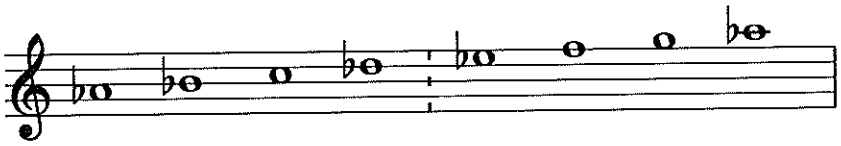
Note:

a. Each scale ends with the first tetrachord of the previous scale, and the last pitch of that tetrachord gives the new tonic.

b. In each case a new accidental is introduced to the scale as a result of having to flat the fourth pitch of the first tetrachord.

c. As with the sharps, the flat key signatures show an accumulation of accidentals, B♭ always appearing first, E♭ second, and so on.

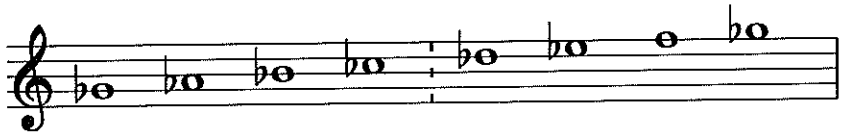
The remainder of the major scales with flats and their derived key signatures are

A♭ major 



D♭ major 



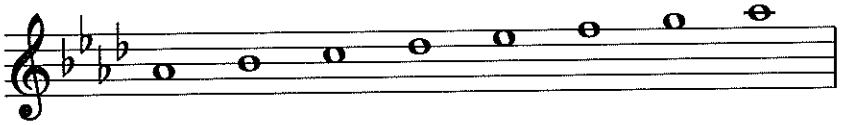
G♭ major 



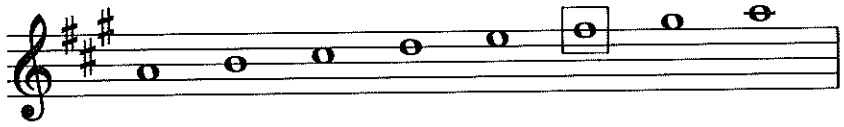
C♭ major 



The writing of a key signature renders it unnecessary to put an accidental beside each altered pitch in the scale. Thus, with a key signature, the A♭ major scale appears as



Each major key has a companion minor key with the same signature. The companion key is known as the *relative minor*. The tonic of the relative minor is the *sixth* pitch of the major key's scale. If, therefore, the major key is A, the major scale in that key is



The sixth pitch (or scale step, or degree) is F♯, and so the relative minor key of A major is f♯ minor. The key signatures



or



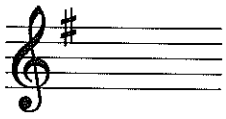
indicate a tonic of A or F♯. Examination of the organization of the music quickly reveals which one of the two is the tonic.

The key signatures for the major scales and their relative minors are shown below:

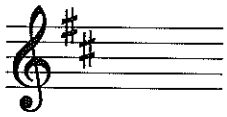
C major
a minor



G major
e minor



D major
b minor



A major
f♯ minor



E major
c♯ minor



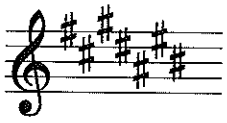
B major
g♯ minor



F♯ major
d♯ minor



C♯ major
a♯ minor



F major
d minor



B♭ major
g minor



E♭ major
c minor



A♭ major
f minor



D♭ major
b♭ minor



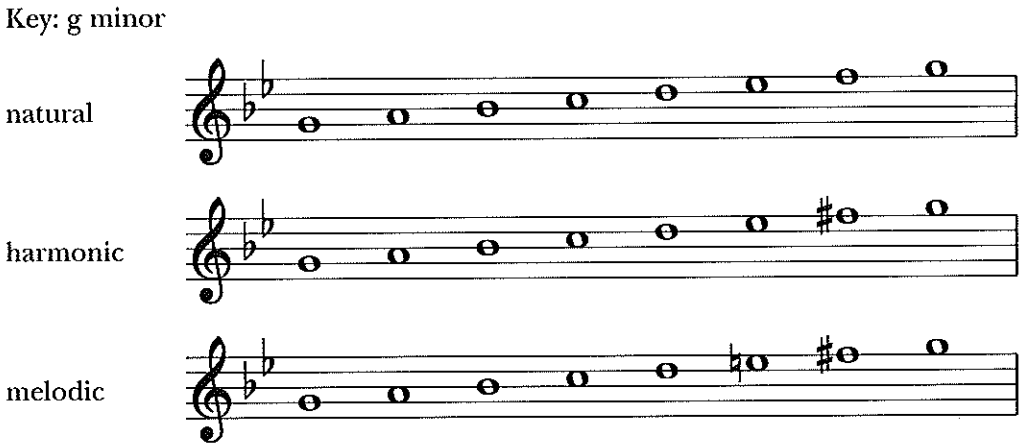
G♭ major
e♭ minor



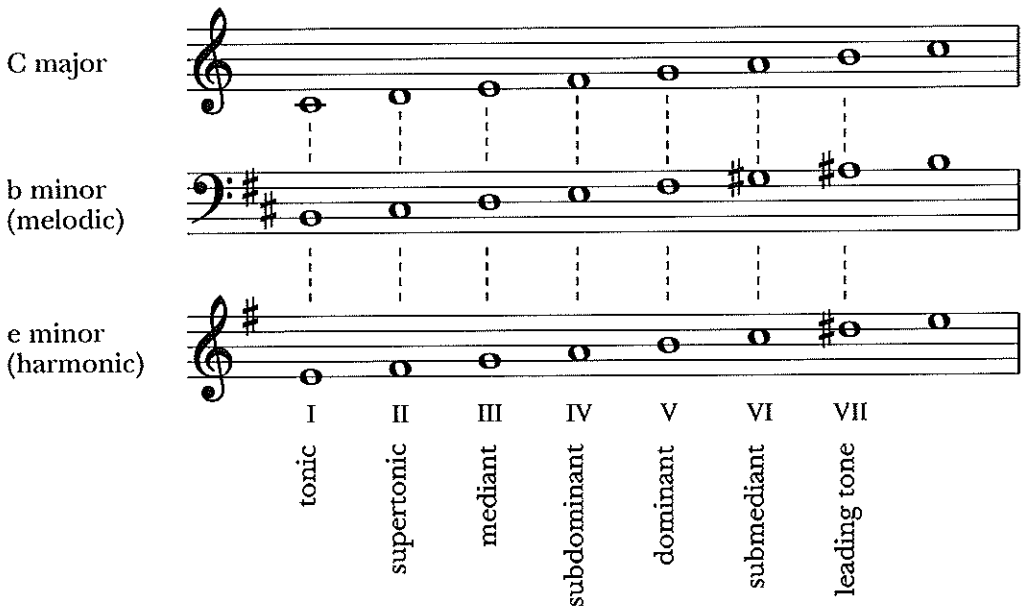
C♭ major
a♭ minor



For each tonic there are, of course, three minor scales; however, the key signature is not affected by this variation. If the natural minor scale is being used the accidentals in the key signature ensure the correct formation of the scale. In the case of the harmonic minor, the seventh degree of the scale has to be raised a half step by an accidental placed beside it; and both the sixth and seventh degrees have to be raised similarly if the melodic minor is being used.



Each scale degree, irrespective of key, has both a name and a number (written in Roman) by which to identify it. (Roman numerals are sometimes written in lowercase letters; see Chapter 5).



The natural minor has no leading tone because the leading tone must *always* be a half step below the tonic. The seventh scale degree in the natural minor is called the *subtonic.*

EXERCISES

A. Identify the *major* key signatures. Time goal–20 seconds per line.



B. Identify the *minor* key signatures. Time goal–20 seconds per line.

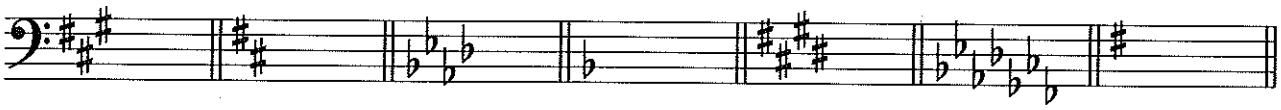
1



2



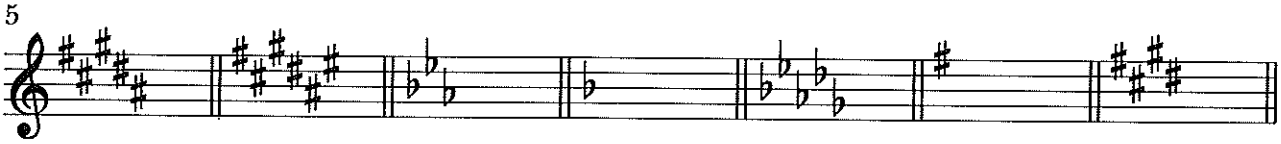
3



4



5



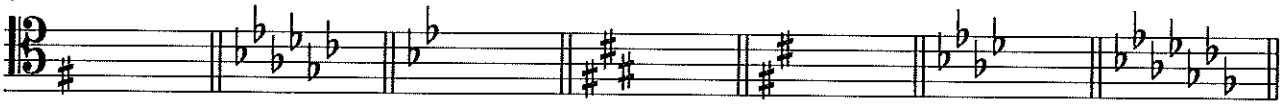
6



7



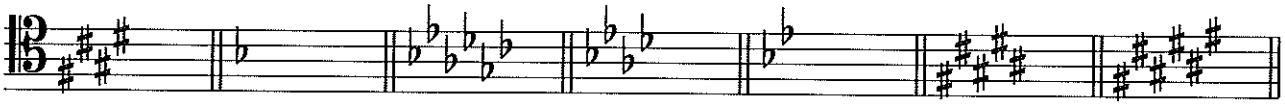
8



9

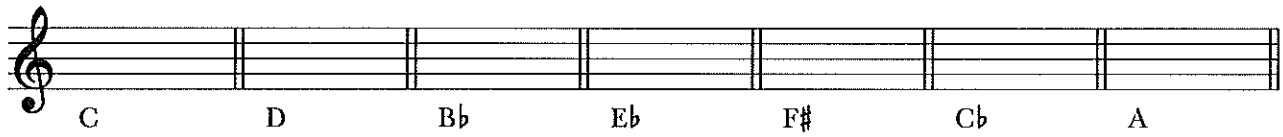


10



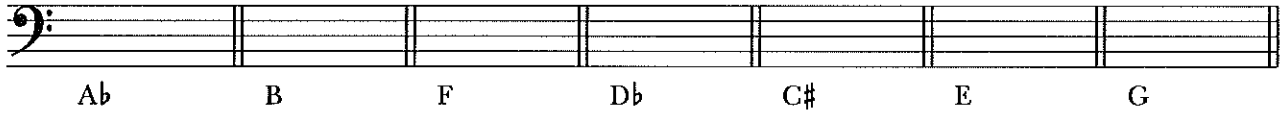
C. Supply the specified *major* key signatures. Time goal--60 seconds per line.

1



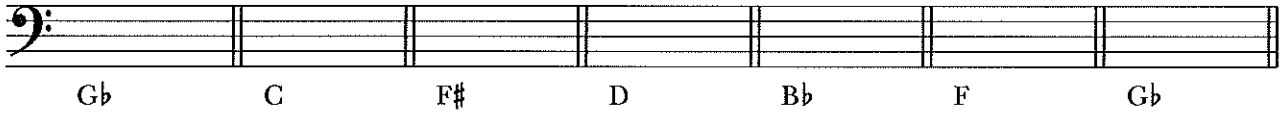
C D B♭ E♭ F♯ C♭ A

2



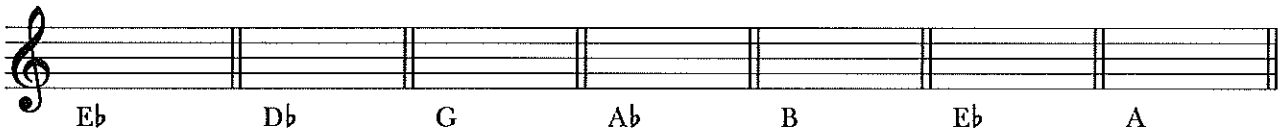
A♭ B F D♭ C♯ E G

3



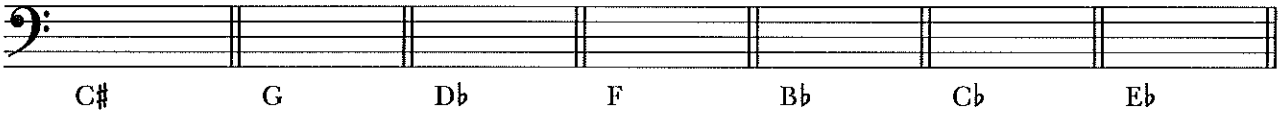
G♭ C F♯ D B♭ F G♭

4



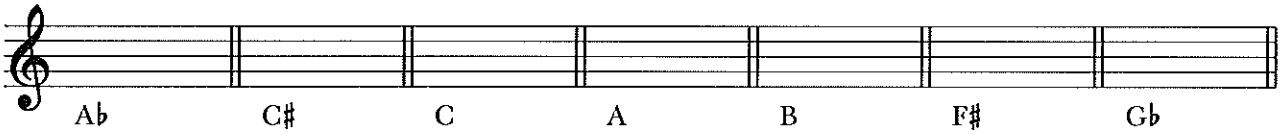
E♭ D♭ G A♭ B E♭ A

5



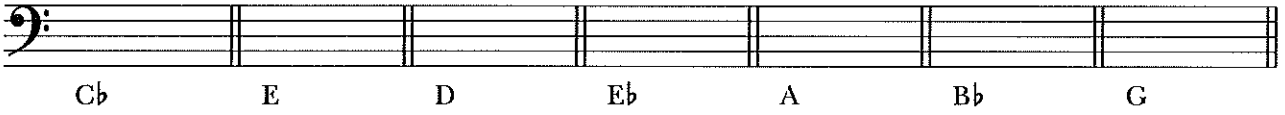
C♯ G D♭ F B♭ C♭ E♭

6



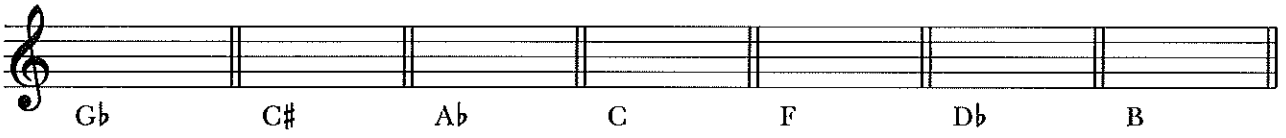
A♭ C♯ C A B F♯ G♭

7



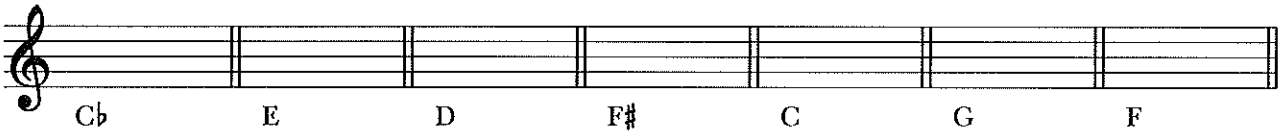
C♭ E D E♭ A B♭ G

8



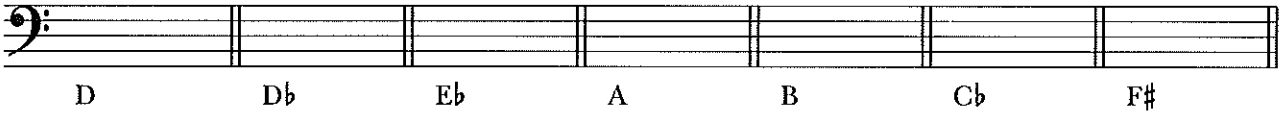
G♭ C♯ A♭ C F D♭ B

9



C♭ E D F♯ C G F

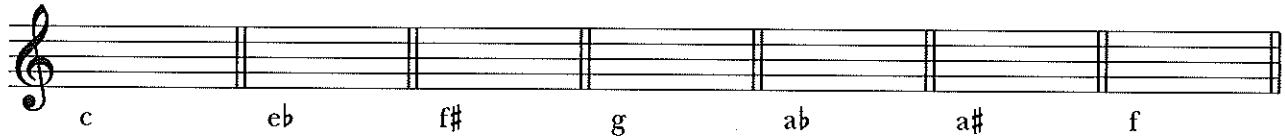
10



D D♭ E♭ A B C♭ F♯

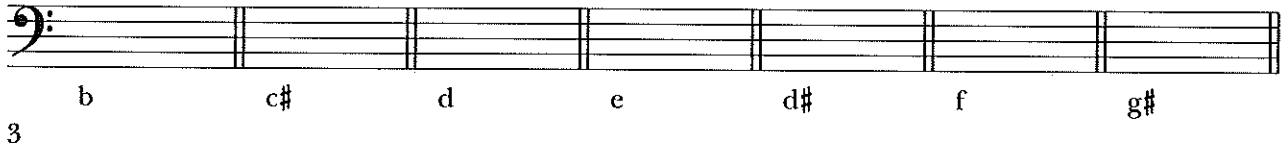
D. Supply the specified *minor* key signatures. Time goal–60 seconds per line.

1



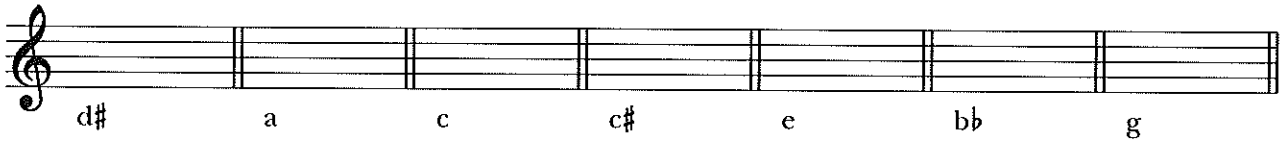
c eb f# g ab a# f

2



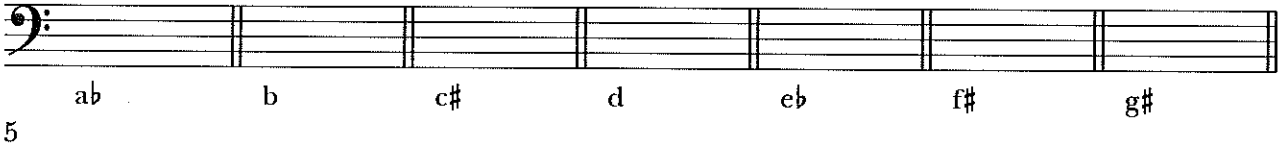
b c# d e d# f g#

3



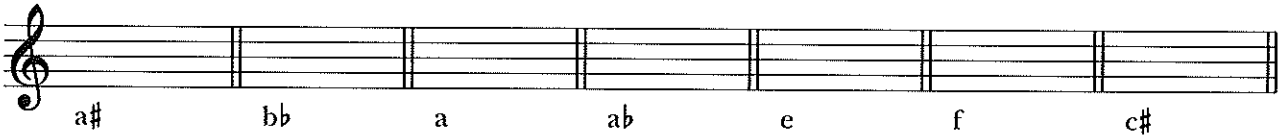
d# a c c# e bb g

4



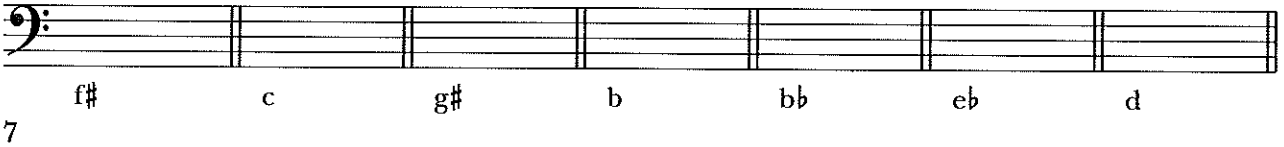
ab b c# d eb f# g#

5



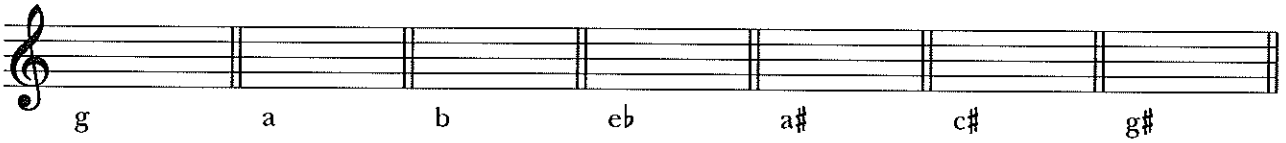
a# bb a ab e f c#

6



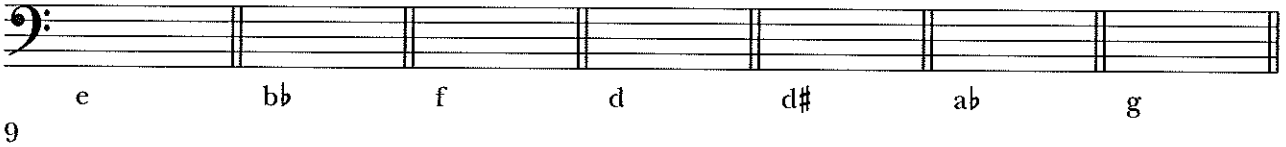
f# c g# b bb eb d

7



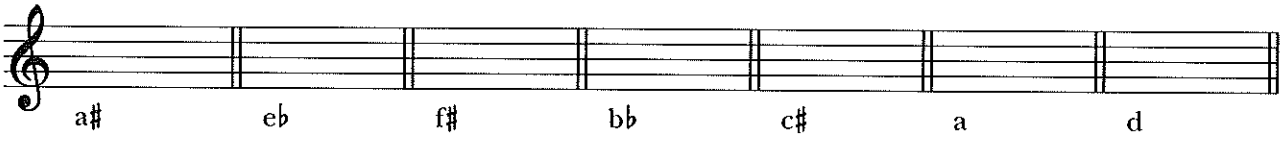
g a b eb a# c# g#

8



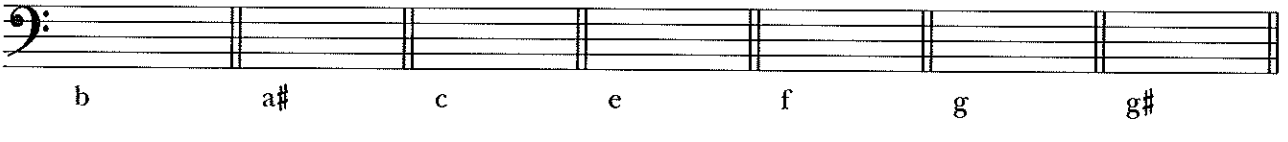
e bb f d d# ab g

9



a# eb f# bb c# a d

10



b a# c e f g g#

E. Using accidentals where necessary, write the specified scales *without* key signatures. Then write the key signature, as shown. Time goal–30 seconds per scale.

1 d melodic minor



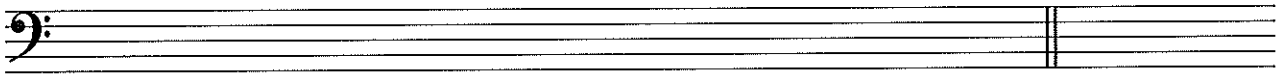
2 e harmonic minor



3 g♯ melodic minor



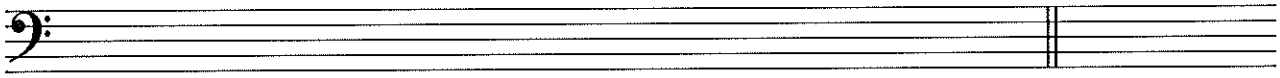
4 b natural minor



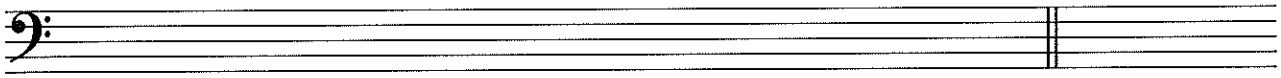
5 C major



6 f♯ harmonic minor



7 b harmonic minor



8 a♭ melodic minor



9 d♯ harmonic minor



10 a natural minor



11 f harmonic minor



12 b♭ melodic minor



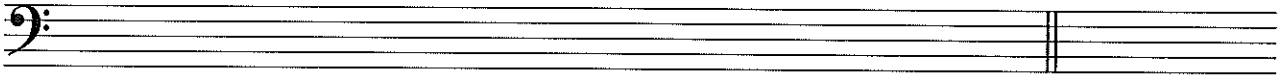
13 C♭ major



14 g harmonic minor



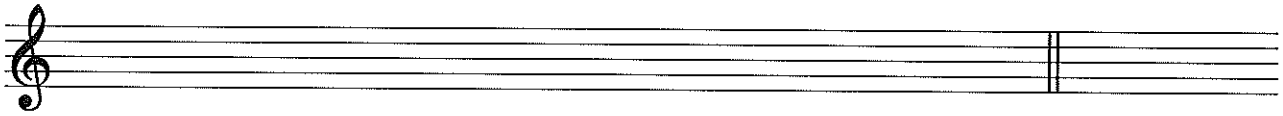
15 a harmonic minor



16 e melodic minor



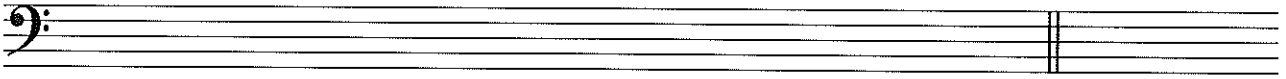
17 e♭ melodic minor



18 B major



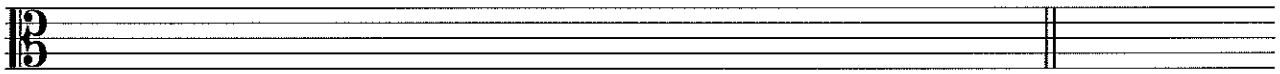
19 c♯ harmonic minor



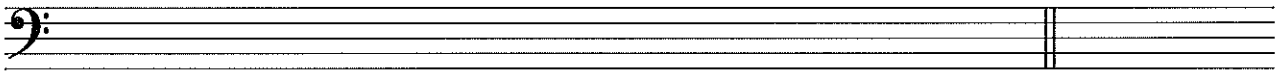
20 d♯ melodic minor



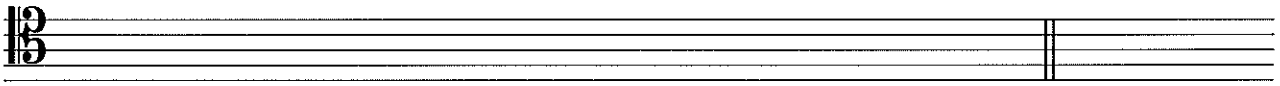
21 g natural minor



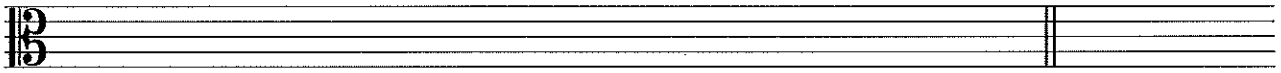
22 a♯ harmonic minor



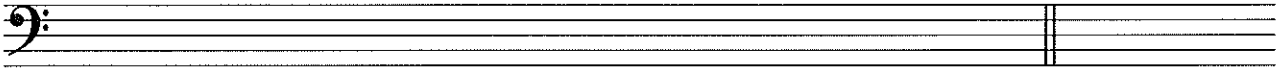
23 D♭ major



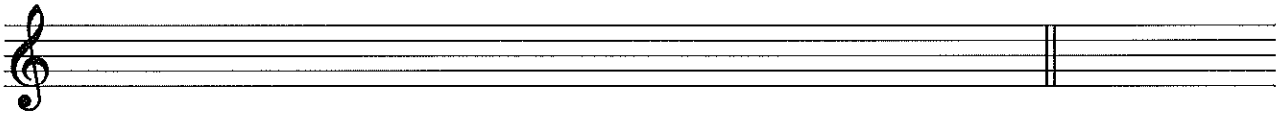
24 a melodic minor



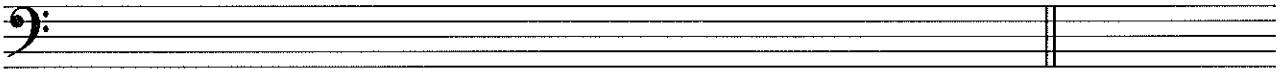
25 F♯ major



26 e♭ natural minor



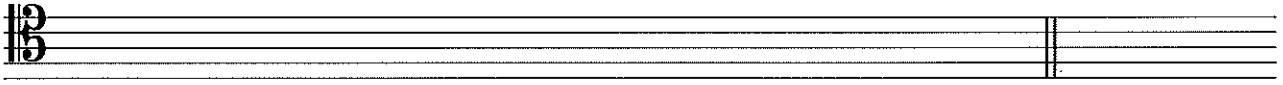
27 d harmonic minor



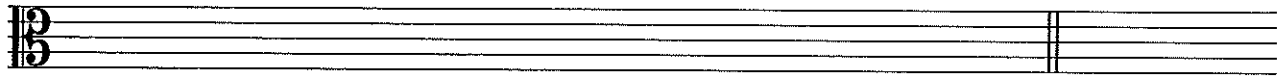
28 c melodic minor



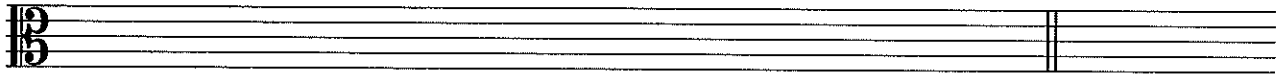
29 b♭ harmonic minor



30 f♯ natural minor



31 a♯ melodic minor



32 b harmonic minor



33 C♯ major



34 c♯ melodic minor



35 d natural minor



36 G♭ major



37 f♯ melodic minor



38 f natural minor



39 c melodic minor

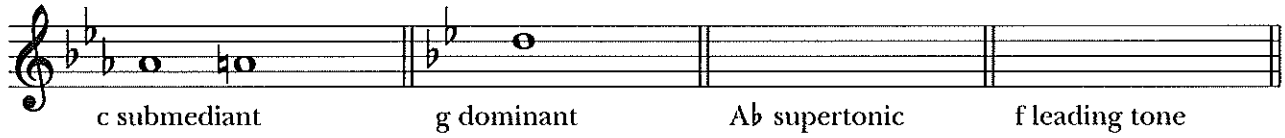


40 b melodic minor



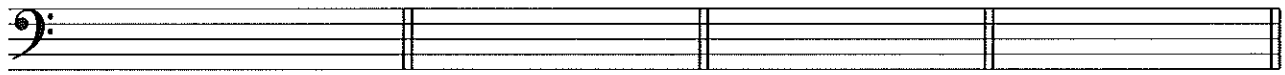
F. Write the signatures of the given keys and the pitch(es) for the specified scale degrees. Time goal—40 seconds per line.

1



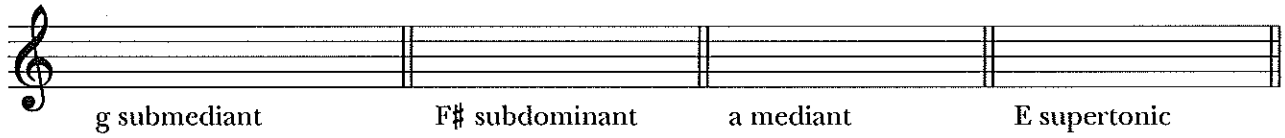
c submediant | g dominant | A♭ supertonic | f leading tone

2



e♭ subdominant | b♭ submediant | D mediant | c subtonic

3



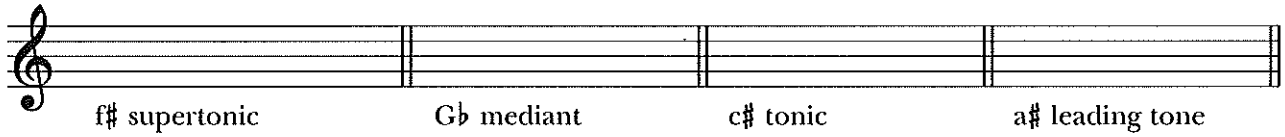
g submediant | F♯ subdominant | a mediant | E supertonic

4



b tonic | a♭ leading tone | g♯ subdominant | C♯ dominant

5



f♯ supertonic | G♭ mediant | c♯ tonic | a♯ leading tone

6



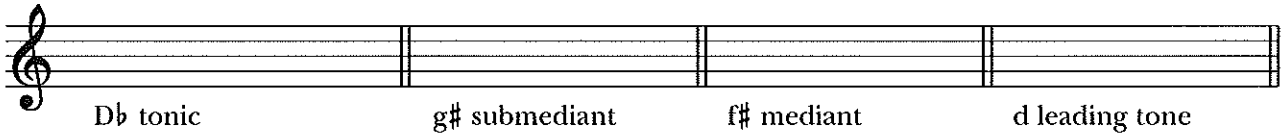
B♭ supertonic | c tonic | e mediant | C♭ dominant

7



B subdominant | d supertonic | E♭ submediant | G leading tone

8



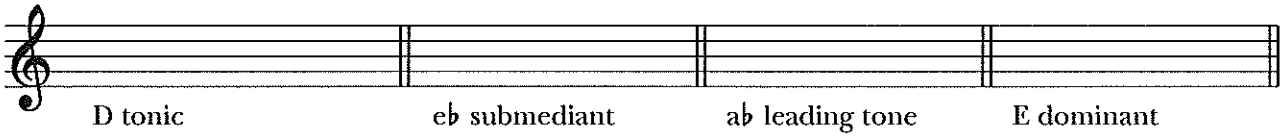
D♭ tonic | g♯ submediant | f♯ mediant | d leading tone

9



a♯ dominant | d subtonic | f subdominant | b supertonic

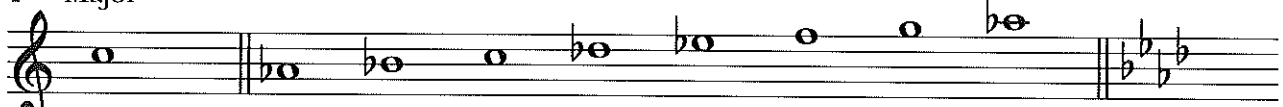
10



D tonic | e♭ submediant | a♭ leading tone | E dominant

G. Using accidentals where necessary, write the specified scales around the given scale degrees *without* key signatures. Then write the key signature, as shown. Time goal–30 seconds per scale.

1 Major



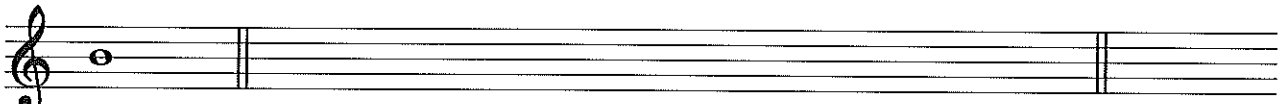
mediant

2 Harmonic minor



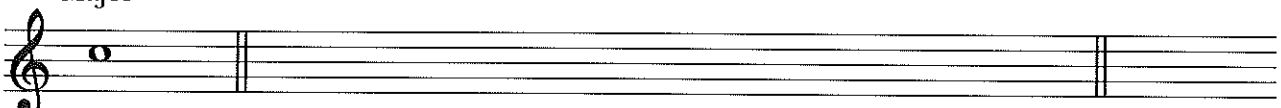
submediant

3 Melodic minor



supertonic

4 Major



subdominant

5 Natural minor



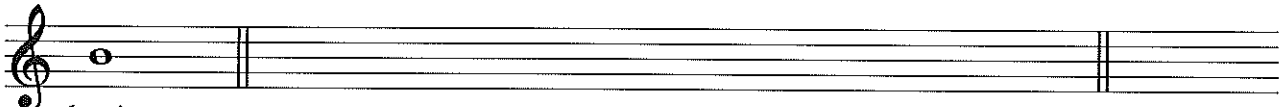
mediant

6 Harmonic minor



leading tone

7 Harmonic minor



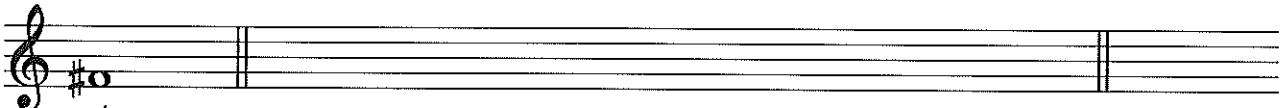
dominant

8 Melodic minor



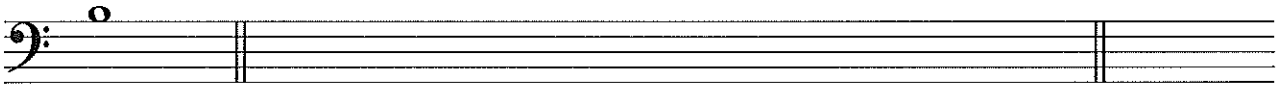
dominant

9 Natural minor



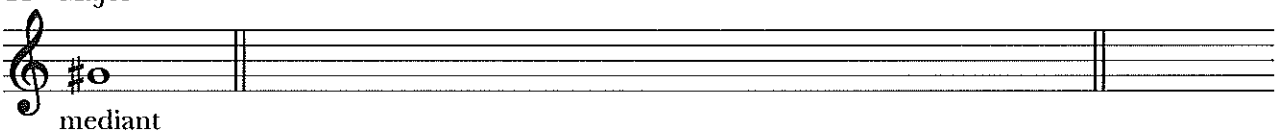
tonic

10 Major



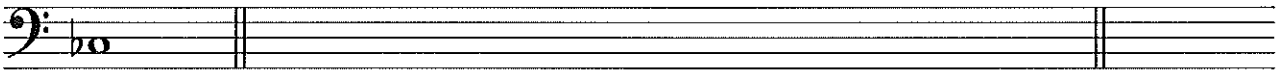
leading tone

11 Major



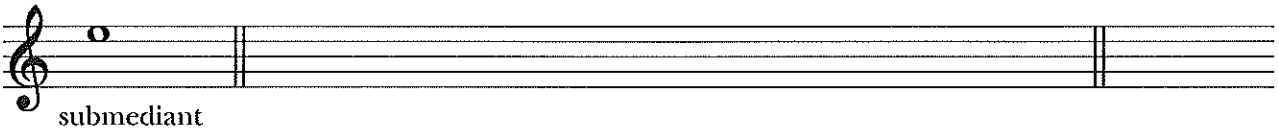
mediant

12 Harmonic minor



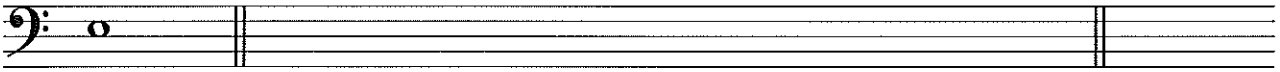
submediant

13 Melodic minor



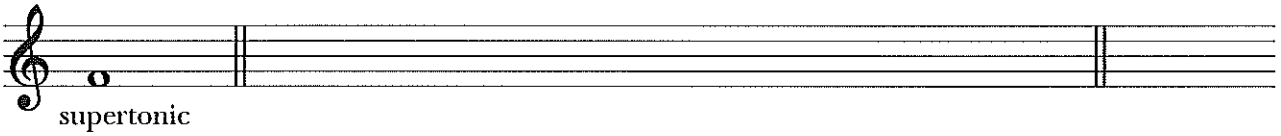
submediant

14 Natural minor



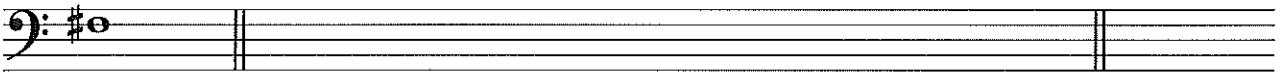
tonic

15 Harmonic minor



supertonic

16 Harmonic minor



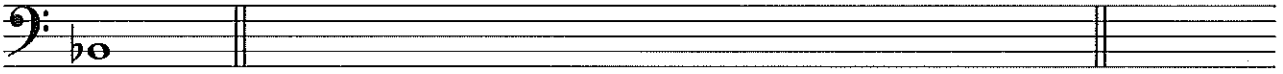
leading tone

17 Melodic minor



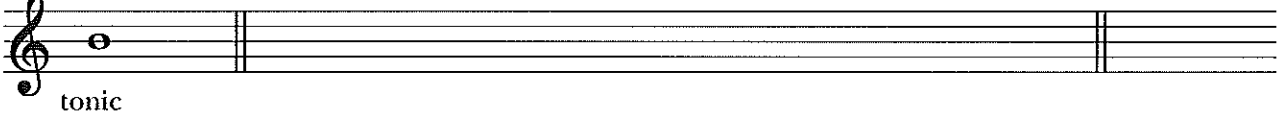
dominant

18 Major



subdominant

19 Major



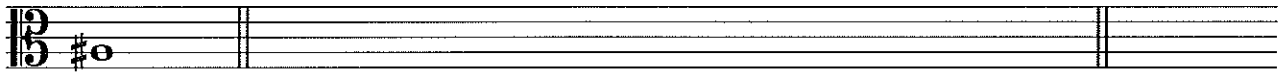
tonic

20 Natural minor



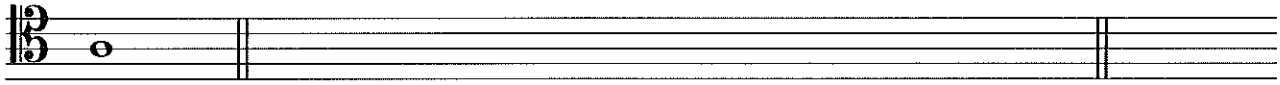
submediant

21 Melodic minor



submediant

22 Major



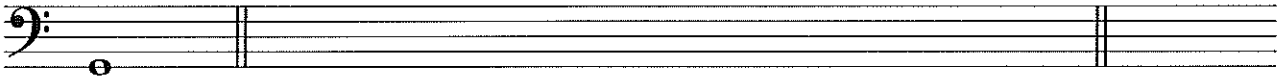
dominant

23 Natural minor



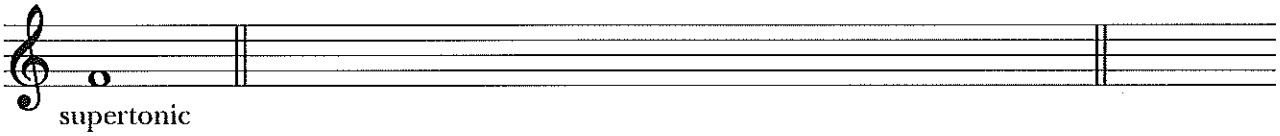
subdominant

24 Major



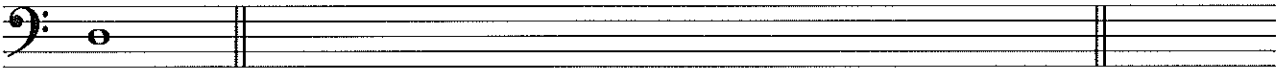
mediant

25 Natural minor



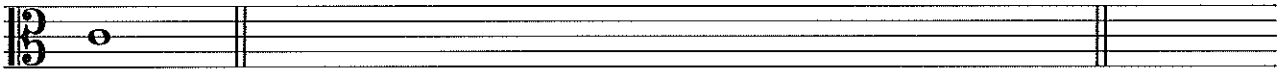
supertonic

26 Melodic minor



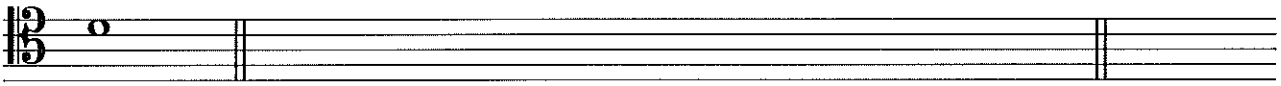
submediant

27 Natural minor



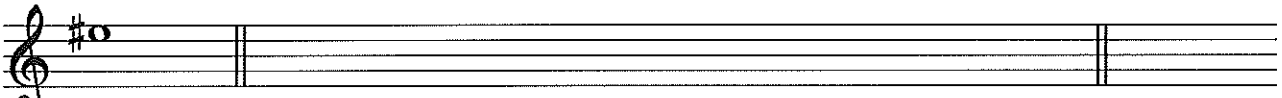
dominant

28 Melodic minor



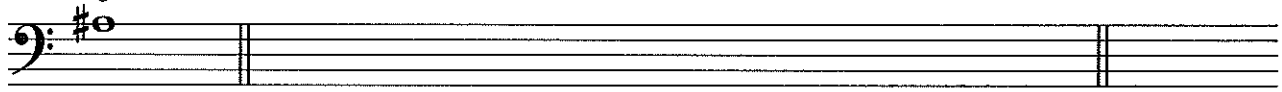
leading tone

29 Melodic minor



leading tone

30 Major



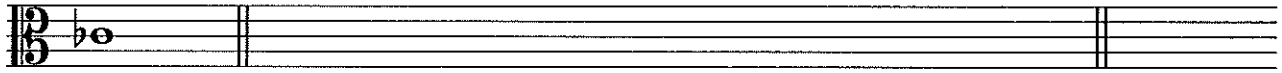
mediant

31 Melodic minor



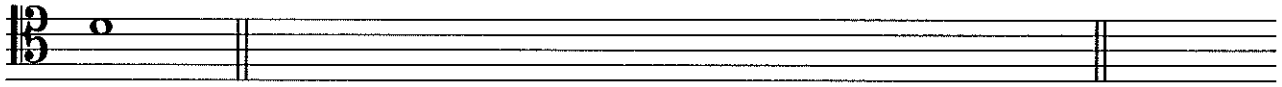
mediant

32 Major



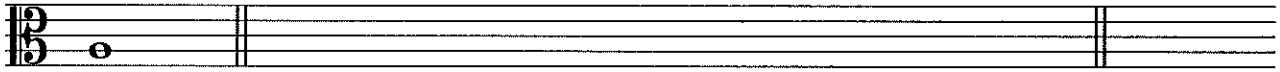
tonic

33 Melodic minor



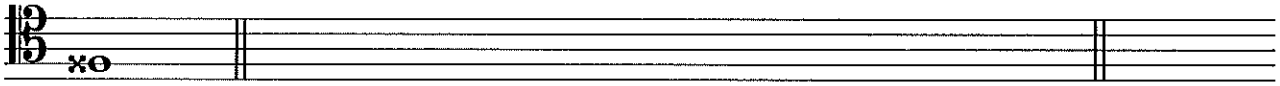
dominant

34 Harmonic minor



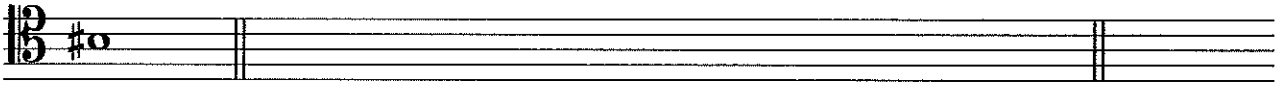
mediant

35 Melodic minor



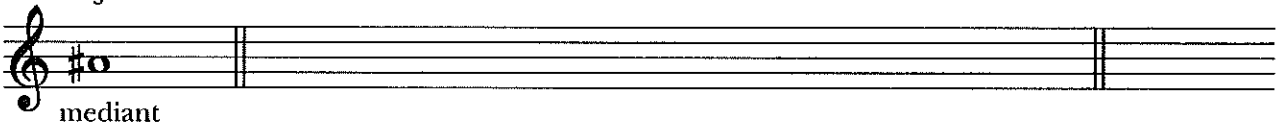
submediant

36 Harmonic minor



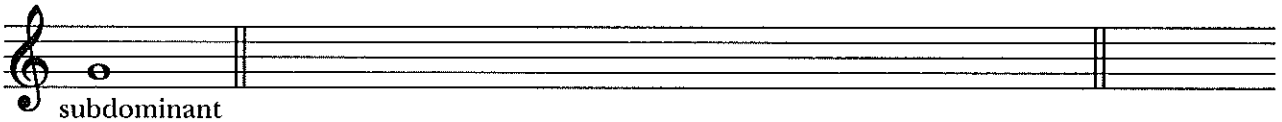
leading tone

37 Major



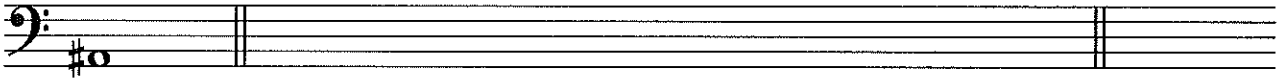
mediant

38 Natural minor



subdominant

39 Melodic minor



supertonic

40 Major



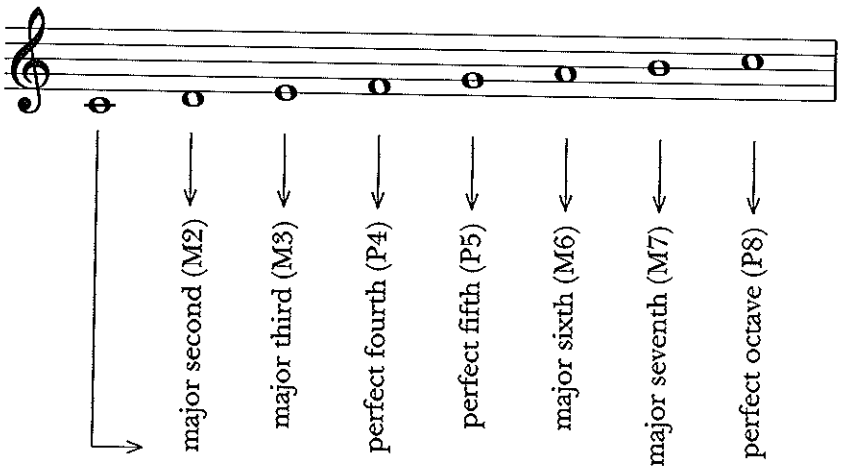
tonic

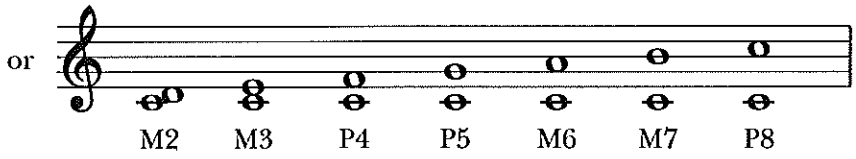
4 Intervals

The spatial relationship between two pitches is called an interval. In Chapter 2, to enable the notation of scales, the interval of the second was introduced. To notate an interval with accuracy one needs to know the following:

1. The precise names of the intervals.
2. The correct spelling of the intervals.
3. The number of half steps the intervals contain.

The major scale provides an excellent starting point for a formal study of intervals.





Note:

a. Each interval has a qualitative and a quantitative component in the description of its name. Take, for example, the major sixth. *Major* describes quality (type); *sixth* describes quantity (numerical value).

b. The quantity of each interval (e.g., third, sixth, etc.) coincides precisely with the number of note names it contains. For example, the perfect fifth, C up to G, is called a fifth because there are five note names in the interval–C D E F G. It is important to understand that the interval C up to G is *always* a fifth in quantity even if an accidental is placed beside either or both of the notes, thus altering its quality.

c. Each interval with a precise name (M2, P4, M7, etc.) spans a characteristic number of half steps. The major scale above shows that

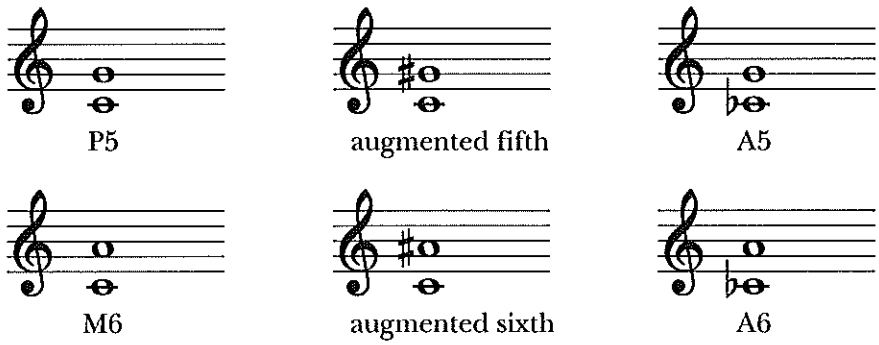
M2 spans 2 half steps
M3 spans 4 half steps
P4 spans 5 half steps
P5 spans 7 half steps
M6 spans 9 half steps
M7 spans 11 half steps
P8 spans 12 half steps

An understanding of these three points is most valuable, since it enables one to write or identify all of the intervals above without reference to the major scale.

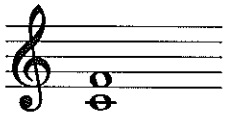
THE QUALITY OF INTERVALS

The major scale contains all the perfect and major intervals within the span of an octave. The qualities, *but not the quantities*, of these intervals may be altered by the appropriate placing of accidentals in the following manner.

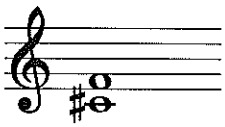
1. A *perfect* or *major* interval may be made larger by a half step. This interval is then said to be *augmented.*



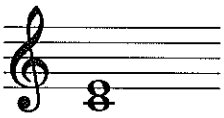
2. A *perfect* interval may be made smaller by a half step. This interval is then said to be *diminished.*


P4

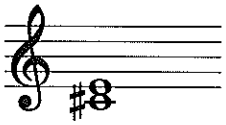

diminished fourth


d4

3. A *major* interval may be made smaller by a half step. This interval is then said to be *minor.*


M3


minor third


m3

4. A *minor* interval may be made smaller by a half step. This interval is then said to be *diminished.*


m7


diminished seventh


d7

Note:

a. The quality of a perfect interval can never be altered to produce a major or minor interval. Nor can a major or minor interval be so altered to produce a perfect one.

b. Perfect fourths, fifths, and octaves are traditionally called *perfect consonances.*

c. Major and minor thirds are called *imperfect consonances.*

d. Major and minor seconds and sevenths, and all augmented and diminished intervals, are called *dissonances.*

e. The augmented fourth and diminished fifth, both intervals with the same number of half steps (six), are sometimes referred to as *tritones,* because six half steps equal three (tri-) whole steps (tones).

f. The standard abbreviations for interval quality are:

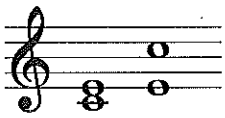
P for Perfect
M for Major
m for minor
A or + for Augmented
d or ° for diminished

The number of half steps in the most commonly encountered intervals up to an octave is as follows:

INTERVAL	HALF STEPS
m2	1
M2	2
m3	3
M3	4
P4	5
A4	6
d5	6
P5	7
m6	8
M6	9
d7	9
A6	10
m7	10
M7	11
P8	12

THE INVERSION OF INTERVALS

An interval is inverted either by raising its lower pitch an octave or by lowering its upper pitch an octave.



M3 m6



P5 P4



A4 d5

Note:

a. An interval and its inversion always span an octave. Such intervals are said to be *complementary*.

b. The sum of the intervals' quantity is always nine. (It is not eight, as one might suppose, because one of the pitch names is counted twice.)

c. With the exception of the perfect interval, the quality of an interval changes when it is inverted.

A major interval becomes minor.

A minor interval becomes major.

An augmented interval becomes diminished.

A diminished interval becomes augmented.

d. The sum of the half steps in complementary intervals is always twelve. For example, a major sixth (nine half steps) added to its inversion, a minor third (three half steps), produces a perfect octave (twelve half steps).

Or:

M6 + m3 = P8

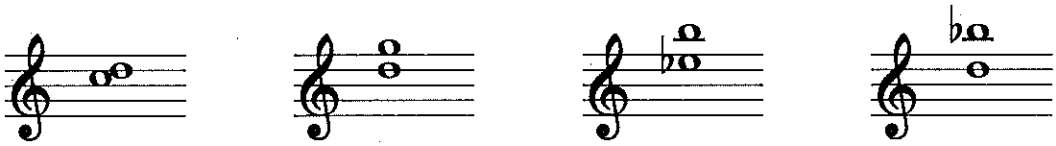
9 + 3 = 12

COMPOUND INTERVALS

All intervals greater than an octave are said to be *compound.* For example:



If one raises the lower pitch an octave in each interval above, the results are as follows:



Thus,

M9 = M2 + 7
P11 = P4 + 7
A12 = A5 + 7
m13 = m6 + 7

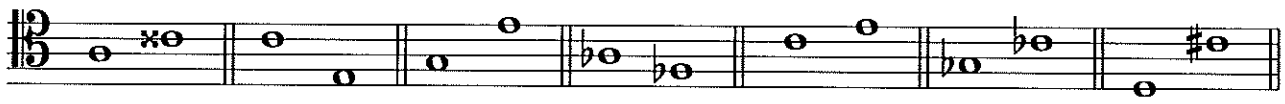
A compound interval, therefore, between one and two octaves in span is created by expanding the simple interval by an octave. The quality of the compound interval is the same as that of the simple one. Its quantity is determined by adding seven to the quantity of the simple interval.

EXERCISES

A. Identify the intervals. Time goal–20 seconds per line.



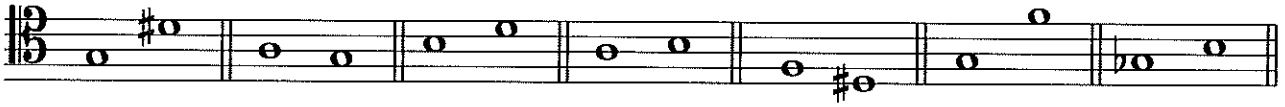
11



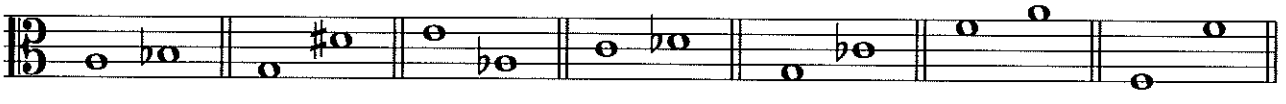
12



13



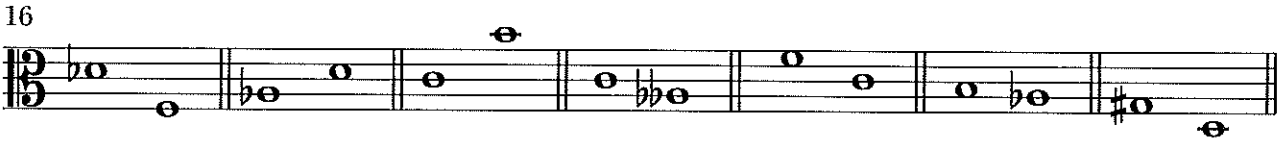
14



15



16



17



18



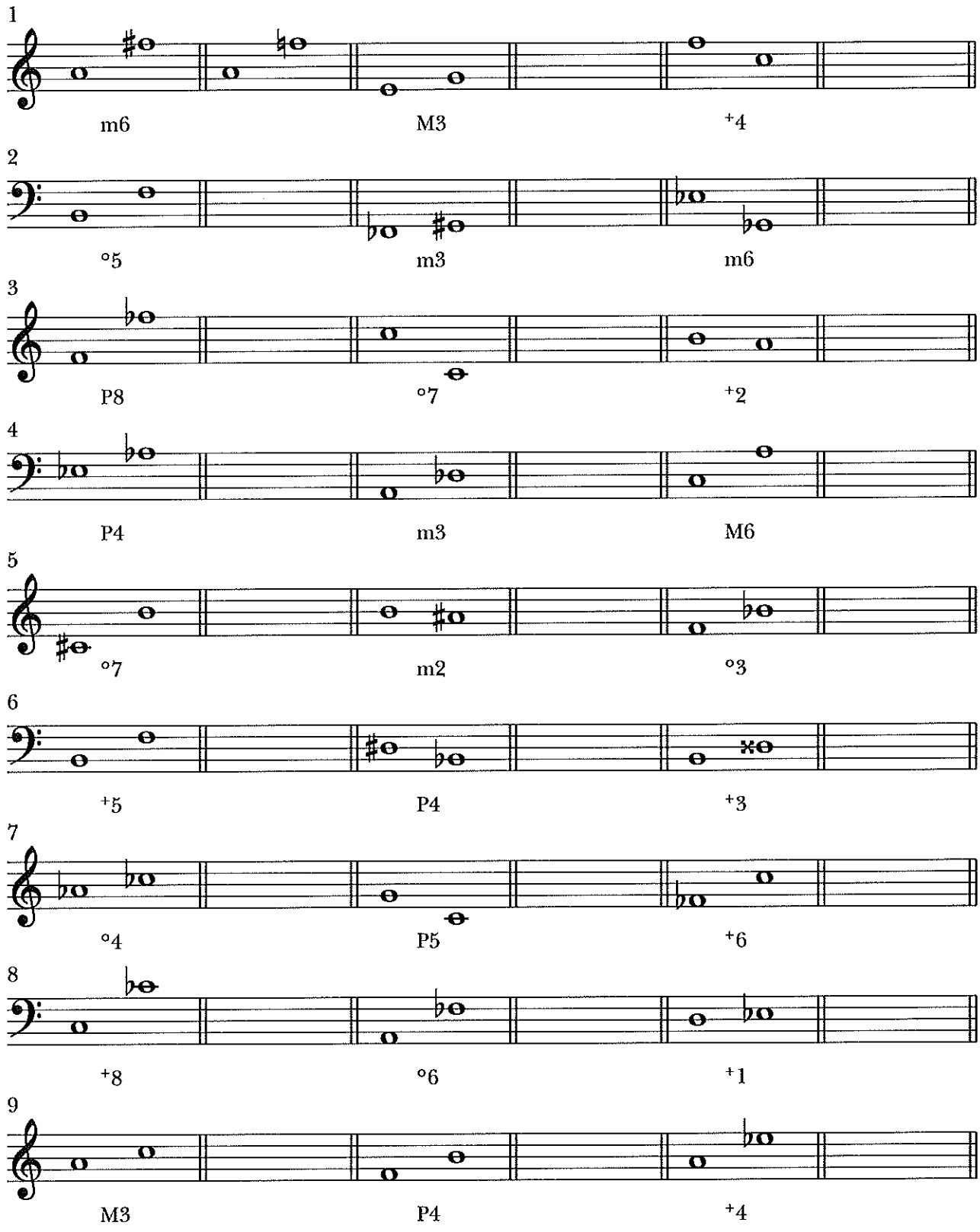
19



20



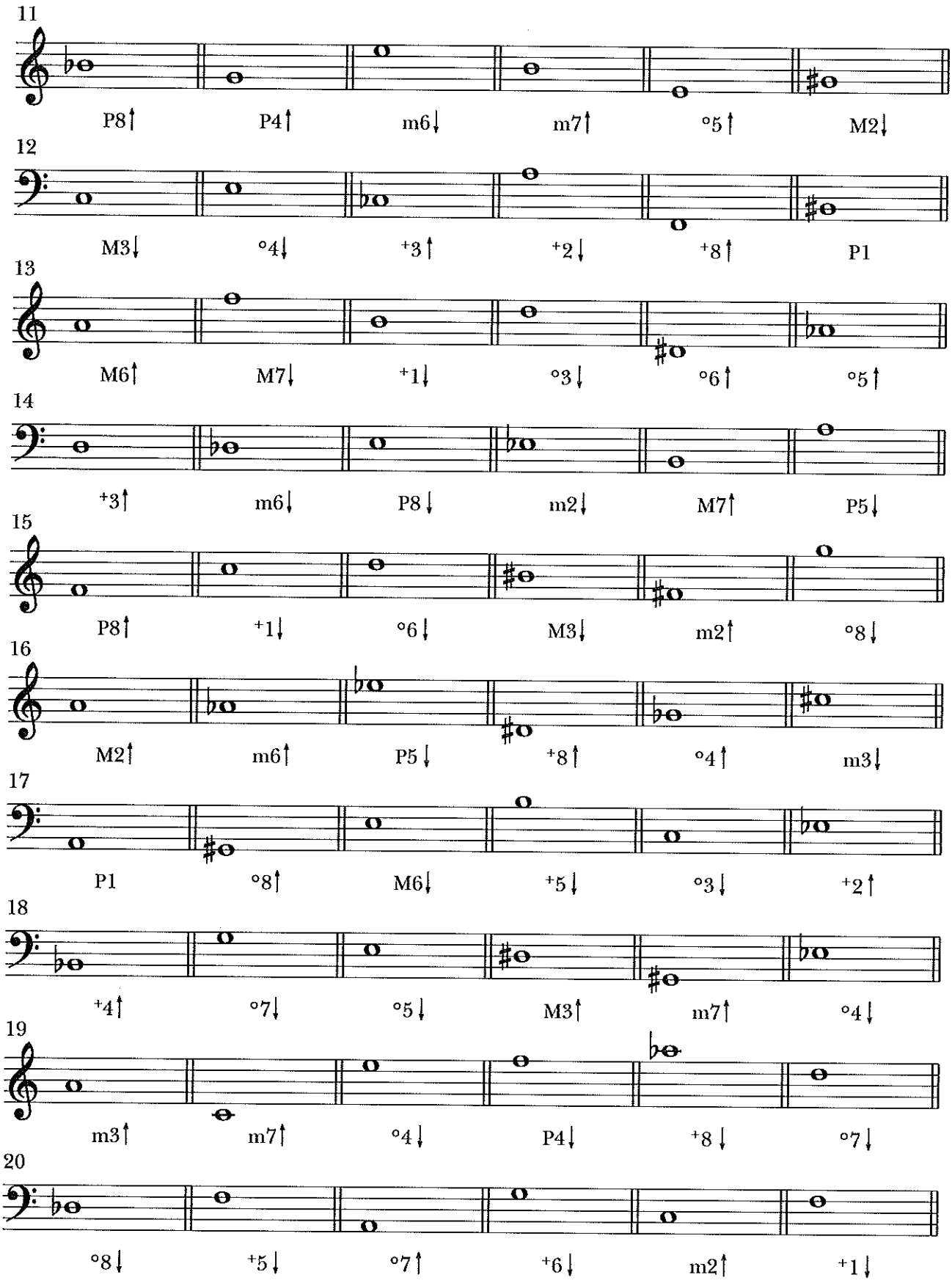
B. Most of the following intervals are spelled incorrectly. Rewrite the wrong intervals correctly without changing the first of the two pitches. Time goal–20 seconds per line.

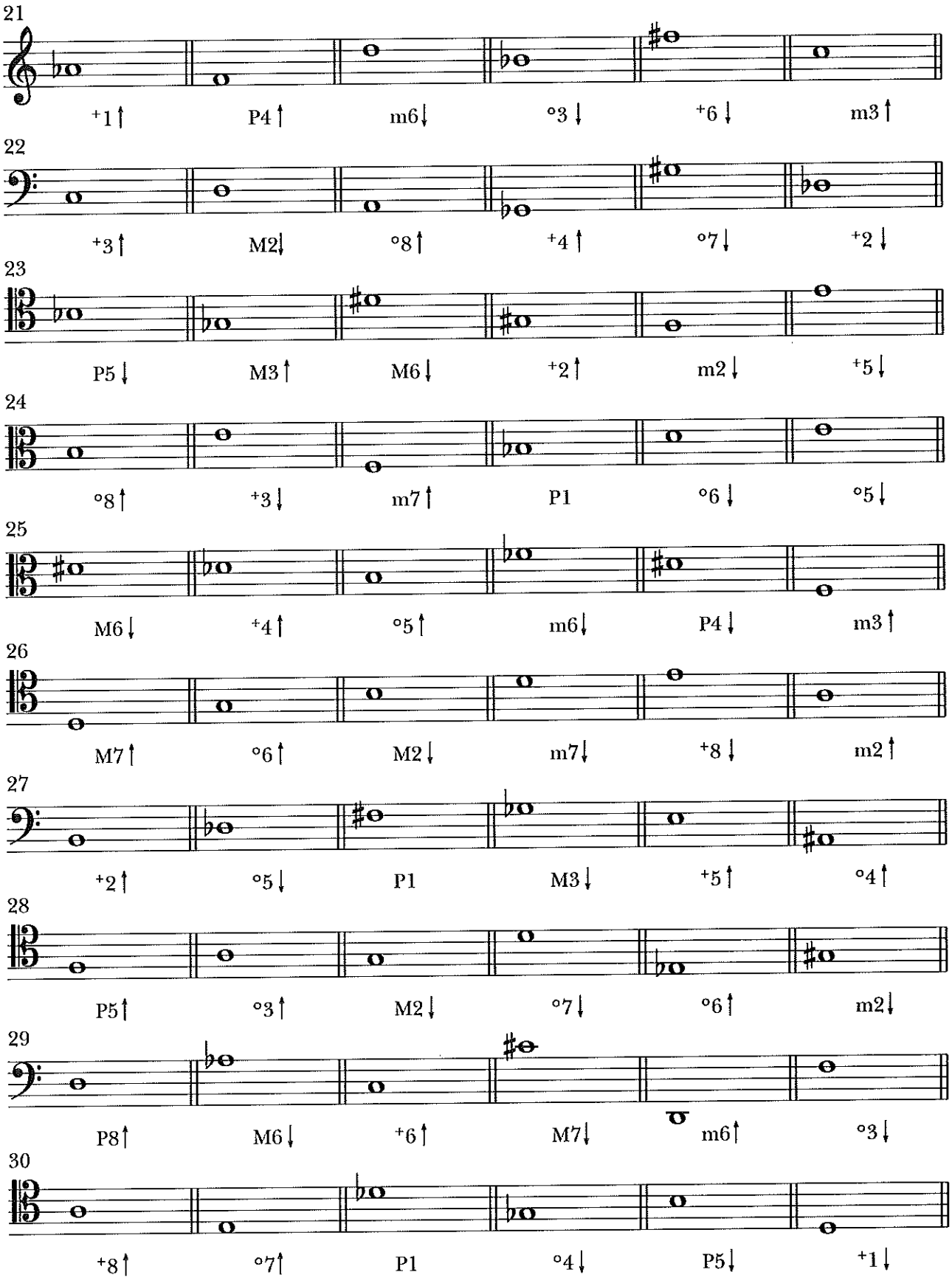



10
M2
M7
°4
11
M6
°3
°6
12
+3
+1
+2
13
P4
M3
+6
14
M7
M7
°7
15
°5
M6
M3
16
+2
°4
m6
17
°7
+3
P5
18
P5
M2
P8
19
°8
°4
°7
20
°3
+6
+3

C. Write the specified intervals above or below the given pitches as indicated. Time goal—30 seconds per line.




11
P8↑ P4↑ m6↓ m7↑ °5↑ M2↓
12
M3↓ °4↓ +3↑ +2↓ +8↑ P1
13
M6↑ M7↓ +1↓ °3↓ °6↑ °5↑
14
+3↑ m6↓ P8↓ m2↓ M7↑ P5↓
15
P8↑ +1↓ °6↓ M3↓ m2↑ °8↓
16
M2↑ m6↑ P5↓ +8↑ °4↑ m3↓
17
P1 °8↑ M6↓ +5↓ °3↓ +2↑
18
+4↑ °7↓ °5↓ M3↑ m7↑ °4↓
19
m3↑ m7↑ °4↓ P4↓ +8↓ °7↓
20
°8↓ +5↓ °7↑ +6↓ m2↑ +1↓


21
+1↑ P4↑ m6↓ °3↓ +6↓ m3↑
22
+3↑ M2↓ °8↑ +4↑ °7↓ +2↓
23
P5↓ M3↑ M6↓ +2↑ m2↓ +5↓
24
°8↑ +3↓ m7↑ P1 °6↓ °5↓
25
M6↓ +4↑ °5↑ m6↓ P4↓ m3↑
26
M7↑ °6↑ M2↓ m7↓ +8↓ m2↑
27
+2↑ °5↓ P1 M3↓ +5↑ °4↑
28
P5↑ °3↑ M2↓ °7↓ °6↑ m2↓
29
P8↑ M6↓ +6↑ M7↓ m6↑ °3↓
30
+8↑ °7↑ P1 °4↓ P5↓ +1↓

D. Write and identify the inversions of the given intervals. Time goal–60 seconds per line.

1



2



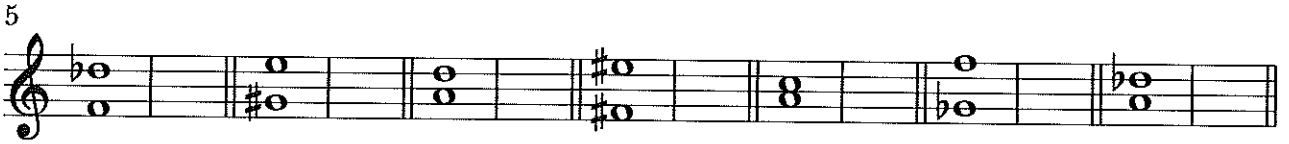
3



4



5



6



7



8



9



10



E. Rewrite the given intervals as compound intervals. In each case identify the compound interval. Time goal—75 seconds per line.



5 Triads

A triad is a three-note chord. In traditional harmony, a triad is a three-note chord built in intervals of thirds. As seen in the previous chapter, for two pitches to be separated by a third, the pitches must be either on adjacent lines or in adjacent spaces. There is no other way of spelling the interval correctly.

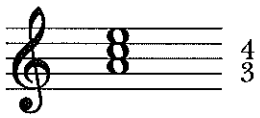
TYPES OF TRIADS

Triads contain two kinds of thirds: the major third (four half steps) and the minor third (three half steps). These thirds in combination make possible the formation of four triad types.

1. The *major triad:* constructed by writing a major third and a minor third upon it.



2. The *minor triad:* constructed by writing a minor third and a major third upon it.



3. The *augmented triad:* constructed by writing one major third upon another.



4. The *diminished triad:* constructed by writing one minor third upon another.



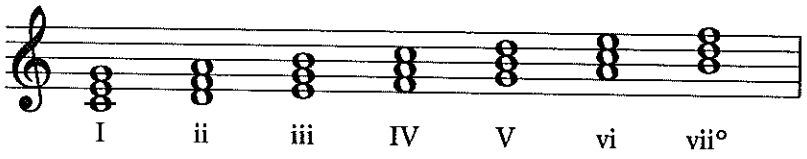
Note:

a. The lowest note of the triad is called the *root.*
b. The middle note is called the *third,* because it is an interval of a third above the root.
c. The highest note is called the *fifth,* because it is an interval of a fifth above the root.
d. In the major and minor triads, the fifth is *perfect.*
e. In the augmented triad, the fifth is *augmented.*
f. In the diminished triad, the fifth is *diminished.*

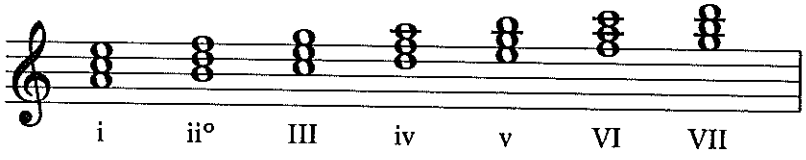
TRIADS IN MAJOR AND MINOR KEYS

Triads are named after the scale degree upon which they are built. Thus, the triad built on the first scale degree is called the tonic triad, on the second scale degree the supertonic, and so on. In the key of C major and its relative minor, a minor, the triads on all the scale degrees are as follows:

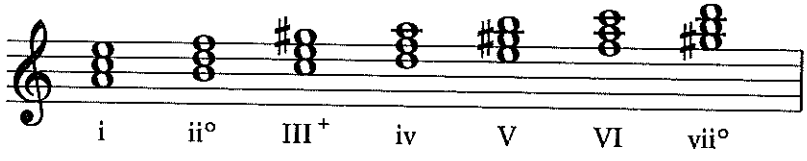
C major



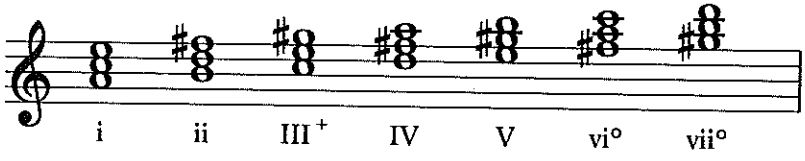
a minor (natural)



a minor (harmonic)



a minor (melodic)



Note:

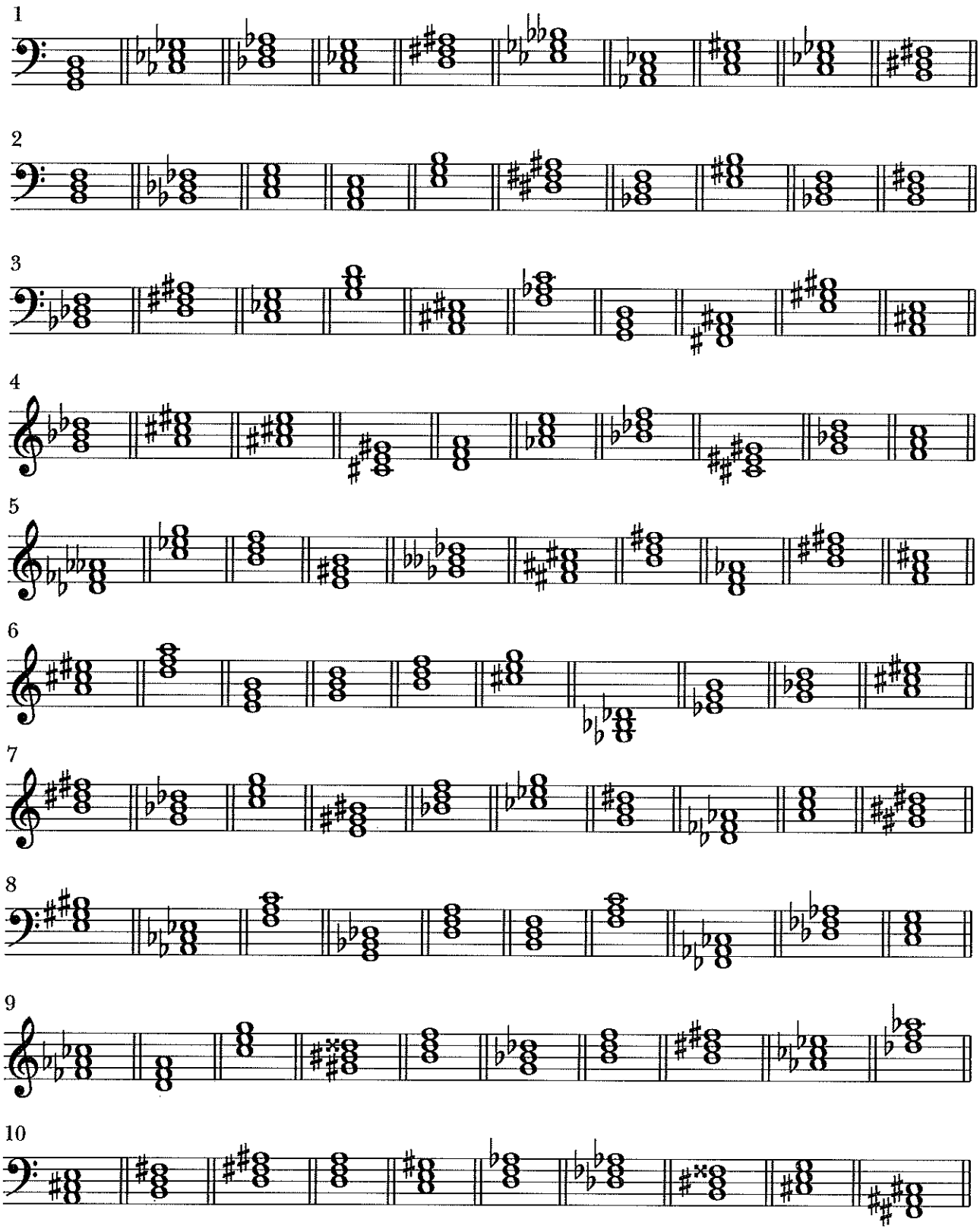
Capital Roman numerals are used for major triads (I), small Roman numerals for minor triads (ii), capital Roman numerals followed by plus signs for augmented triads (III+), and small Roman numerals followed by circles for diminished triads (vii°).

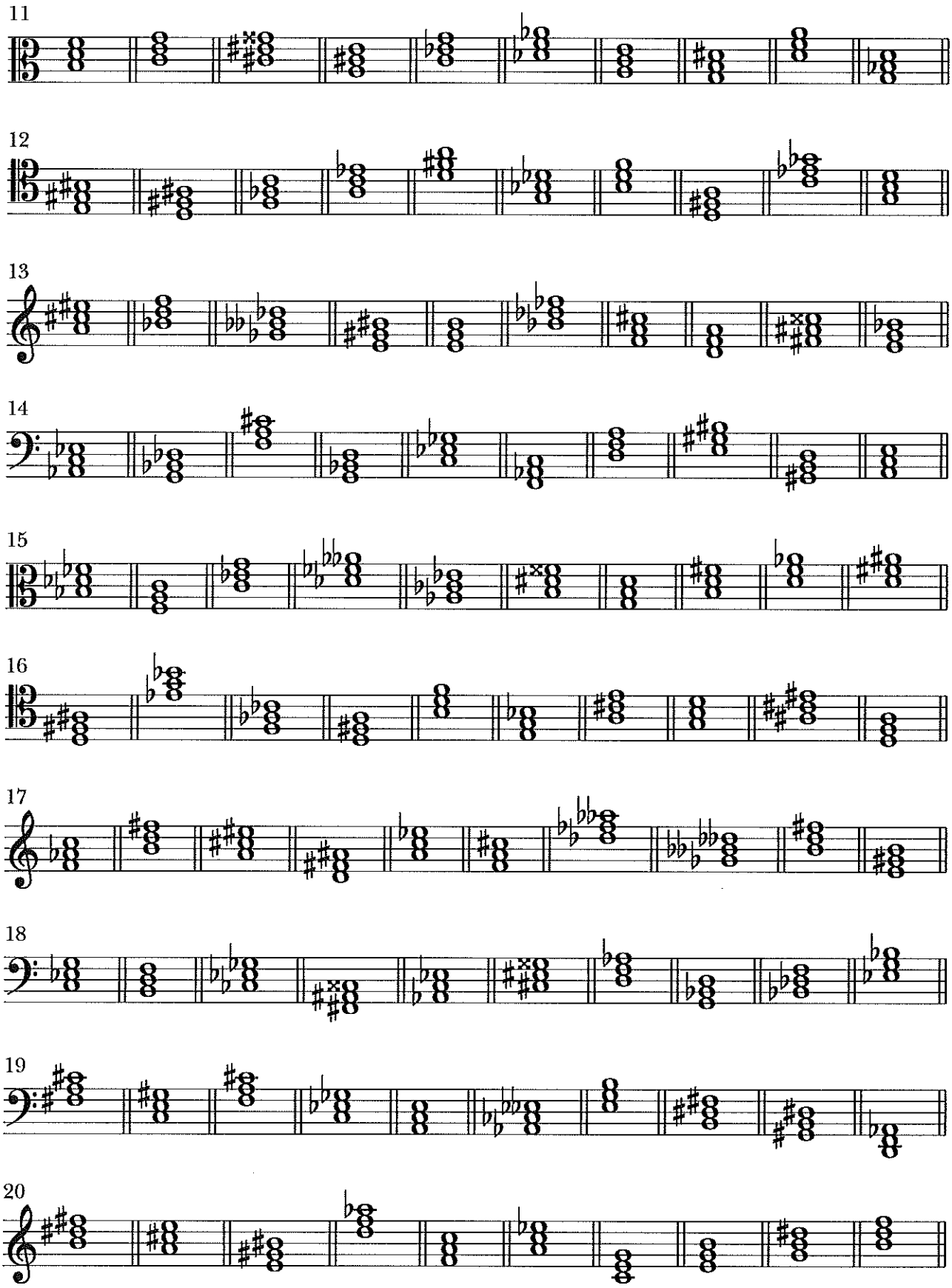
It is important to appreciate that the qualities of the triads in the major and minor keys are governed by the intervallic organizations of the major and minor scales. In all major keys, therefore, the tonic triad will be major, the supertonic will be minor, and so on. The chart below shows the distribution of the four types of triads in the major and minor keys.

	MAJOR	**MINOR**		
		NATURAL	HARMONIC	MELODIC
MAJOR TRIADS	I, IV, V	III, VI, VII	V, VI	IV, V
MINOR TRIADS	ii, iii, vi	i, iv, v	i, iv	i, ii
AUGMENTED TRIADS			III+	III+
DIMINISHED TRIADS	vii°	ii°	ii°, vii°	vi°, vii°

EXERCISES

A. Identify the quality of each triad. Time goal—45 seconds per line.




11
12
13
14
15
16
17
18
19
20

B. Using the given note as either the root (r), third (3), or fifth (5), write the specified triad. Time goal—75 seconds per line.




11
12
13
14
15
16
17
18
19
20

C. Supply the signature of the given key, and write the triad on the specified scale degree. Time goal—60 seconds per line.

1



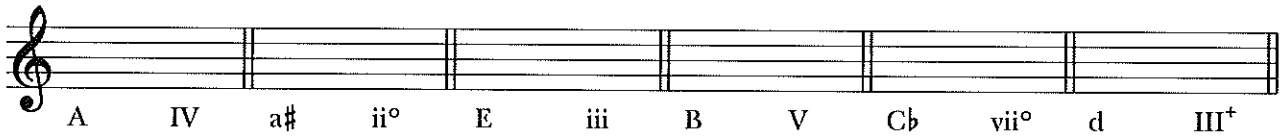
e i F iii b♭ III+ F♯ IV G♭ V a v

2



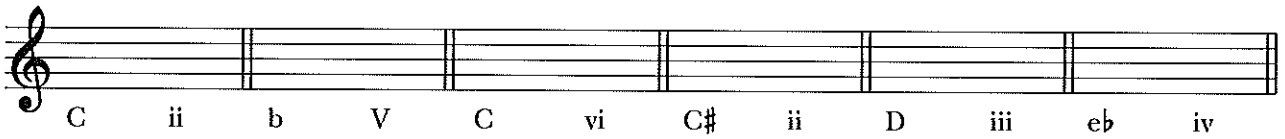
f♯ VII G vii° A♭ I D♭ ii d♯ vi° g♯ i

3



A IV a♯ ii° E iii B V C♭ vii° d III+

4



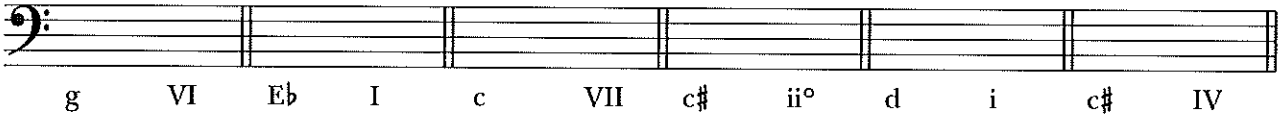
C ii b V C vi C♯ ii D iii e♭ iv

5



a III f VI G♭ vi A♭ iii a♭ vi° a♯ VII

6



g VI E♭ I c VII c♯ ii° d i c♯ IV

7



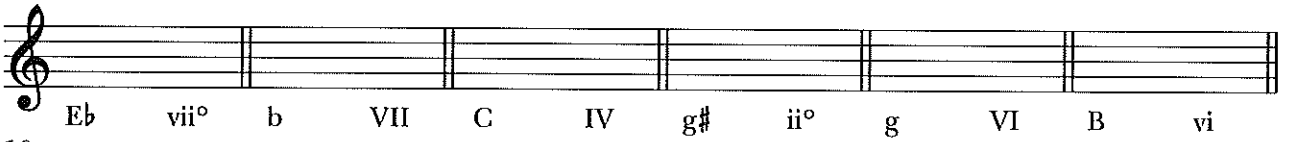
B♭ vii° f♯ iv G V D ii a iv a♭ vi°

8



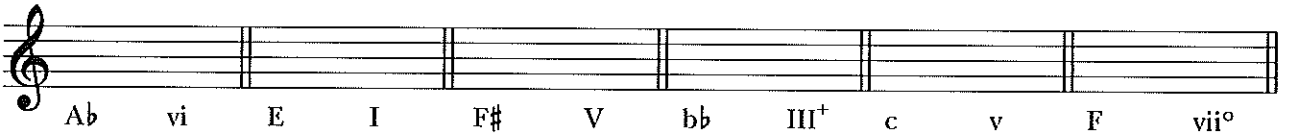
D iii E♭ iii A♭ ii C♭ V e♭ ii° f III+

9



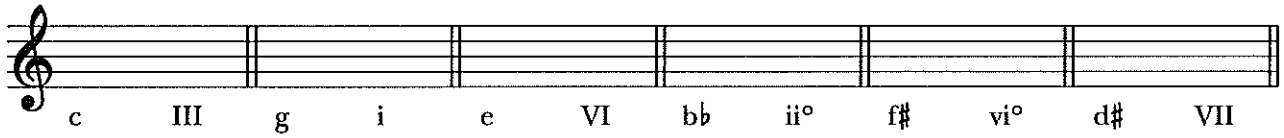
E♭ vii° b VII C IV g♯ ii° g VI B vi

10



A♭ vi E I F♯ V b♭ III+ c v F vii°

11



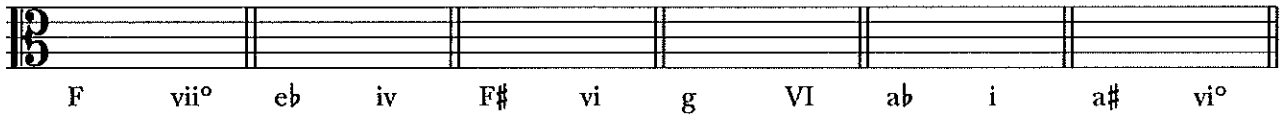
c III g i e VI b♭ ii° f♯ vi° d♯ VII

12



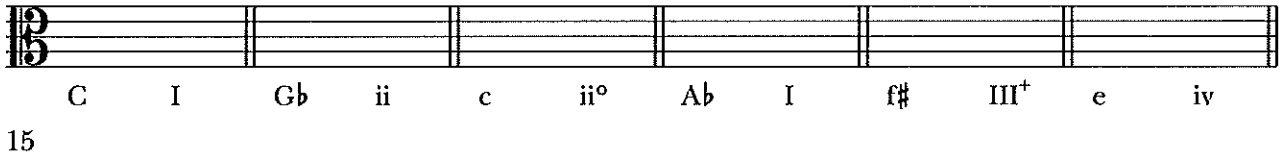
G I g♯ IV b v f ii A vi c♯ v

13



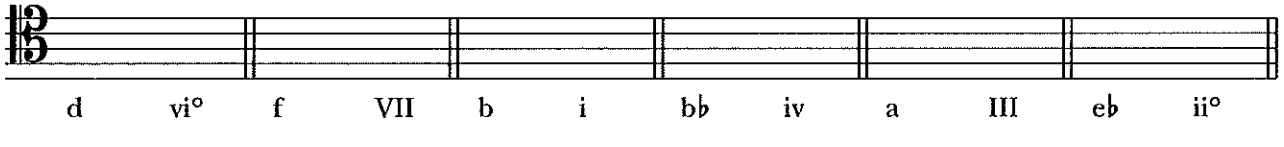
F vii° e♭ iv F♯ vi g VI a♭ i a♯ vi°

14



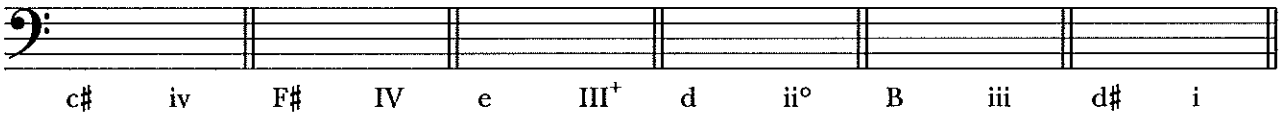
C I G♭ ii c ii° A♭ I f♯ III^{+} e iv

15



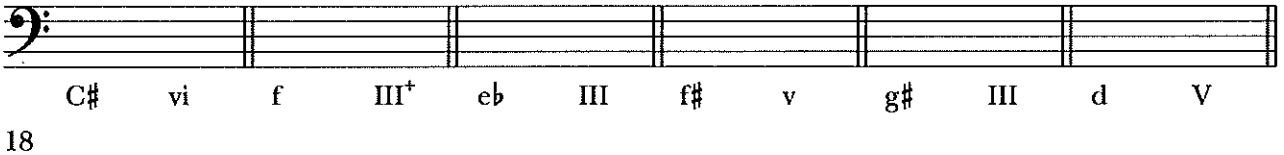
d vi° f VII b i b♭ iv a III e♭ ii°

16



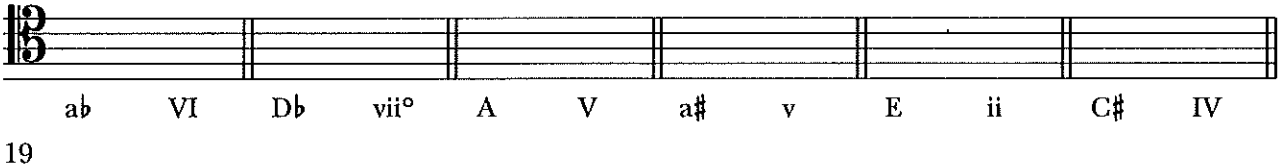
c♯ iv F♯ IV e III^{+} d ii° B iii d♯ i

17



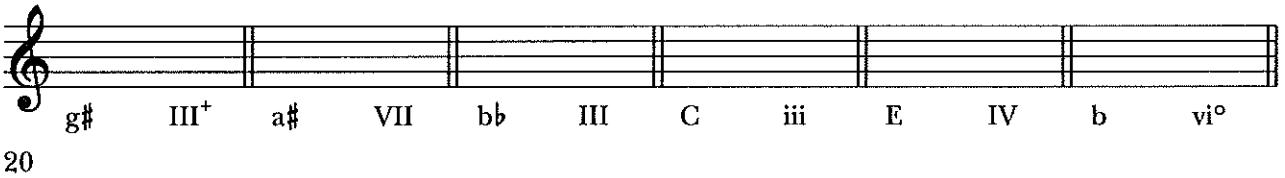
C♯ vi f III^{+} e♭ III f♯ v g♯ III d V

18



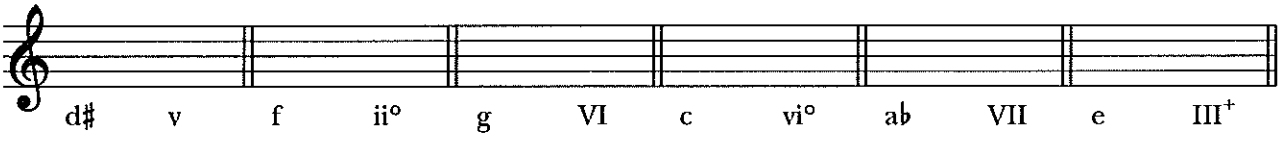
a♭ VI D♭ vii° A V a♯ v E ii C♯ IV

19



g♯ III^{+} a♯ VII b♭ III C iii E IV b vi°

20



d♯ v f ii° g VI c vi° a♭ VII e III^{+}

D. Write the specified key signature, the triad, and the triad's quality in the manner indicated. Time goal–100 seconds per line.

1 Submediant min. maj. dim. maj.



C maj. a har. min. g♯ mel. min. b nat. min. a♭ nat. min. c har. min.

2 Mediant



D maj. e nat. min. a har. min. f mel. min. G maj. g♯ har. min.

3 Dominant



a nat. min. b mel. min. C♭ maj. A maj. d har. min. e♭ nat. min.

4 Leading tone



c mel. min. b♭ har. min. a mel. min. E maj. g har. min. d♯ har. min.

5 Submediant



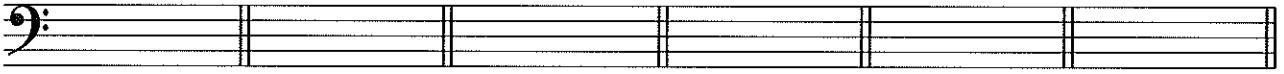
b har. min. g mel. min. f nat. min. C maj. f♯ mel. min. a♭ har. min.

6 Tonic



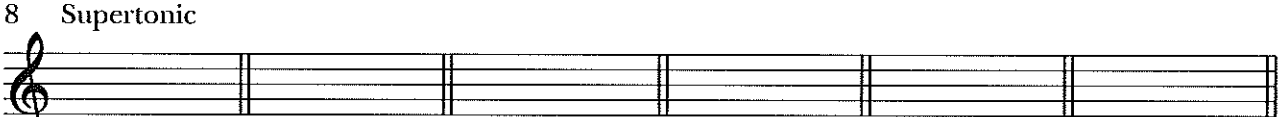
A♭ maj. b♭ nat. min. e♭ mel. min. d nat. min. c♯ har. min. B♭ maj.

7 Subdominant



E♭ maj. d♯ mel. min. f♯ nat. min. c har. min. c nat. min. b♭ mel. min.

8 Supertonic



G maj. g nat. min. f♯ har. min. c mel. min. F maj. a mel. min.

9 Dominant



d mel. min. c har min. c♯ nat. min. D maj. e♭ har. min. f har. min.

10 Mediant



d♯ har. min. c♯ mel. min. g♯ mel. min. B maj b nat. min. a♭ mel. min.

11 Subdominant



c♯ mel. min. F maj. d har. min. f♯ mel. min. G♭ maj. a nat. min.

12 Leading tone



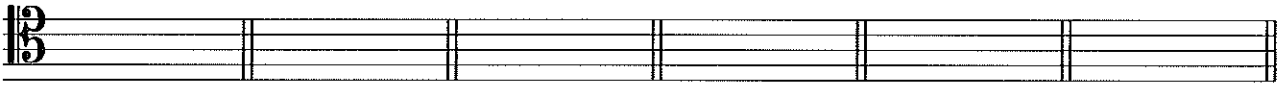
f har. min. e har. min. a♯ mel. min. B♭ maj. g mel. min. e♭ har. min.

13 Mediant



f nat. min. a♭ mel. min. c nat. min. b mel. min. c♯ nat. min. d♯ nat. min.

14 Tonic



E maj. c har. min. A maj. f♯ nat. min. a♯ nat. min. b♭ har. min.

15 Subdominant



G maj. b har. min. d mel. min. e♭ nat. min. c♯ nat. min. f har. min.

16 Supertonic



e nat. min a har. min. C♯ maj. g mel. min. a♯ har. min. g♯ har. min.

17 Submediant



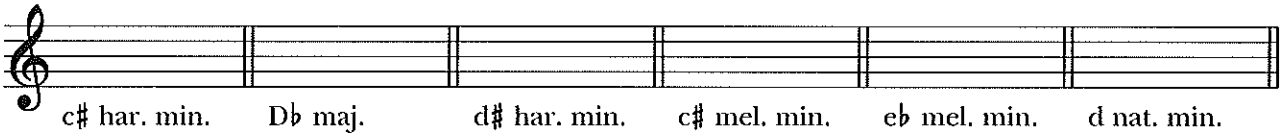
e♭ nat. min. d♯ mel. min. c mel. min. f♯ har. min. b♭ nat. min. B maj.

18 Dominant



b nat. min. a mel. min. A♭ maj. f mel. min. a♭ nat. min. e mel. min.

19 Submediant



c♯ har. min. D♭ maj. d♯ har. min. c♯ mel. min. e♭ mel. min. d nat. min.

20 Leading tone

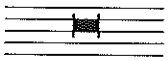
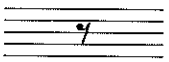


a♭ har. min. F maj. D maj. g♯ mel. min. b♭ mel. min. C♯ maj.

6 The Notation of Rhythm

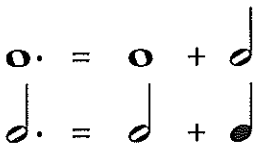
NOTE VALUES

The note with the longest duration is the breve. The duration of each note in the table below is half that of the note immediately above it.

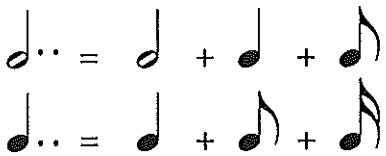
SYMBOL	NAME	EQUIVALENT REST
𝅜	breve (rarely used)	
𝅝	whole note	
𝅗𝅥	half note	
♩	quarter note	
♪	eighth note	
𝅘𝅥𝅯	sixteenth note	
𝅘𝅥𝅰	thirty-second note	
𝅘𝅥𝅱	sixty-fourth note	

A rest is used to account for the time during which a note is not sounding.

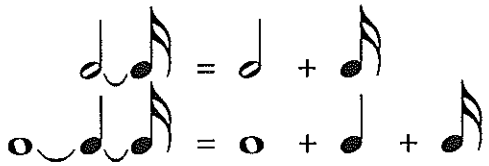
A note's duration may be increased by one half by placing a dot after the note, thus:



A second dot increases the duration by half the value of the first dot.



Because dots cannot always produce the desired duration, notes may be tied together:



METER SIGNATURES

A meter (or time) signature (e.g., $\frac{3}{4}, \frac{6}{8}, \frac{4}{4}$) gives the performer three pieces of information:

1. The lower number tells which note value is used as the basic unit of organization within the measure.
2. The upper number tells how many notes of a given value are in the measure.
3. By implication, the complete signature tells how many beats (pulsations) are in the measure.

For example, $\frac{3}{4}$ implies that the measure is organized in quarter notes, that there are three quarter notes in the measure, and that there are three beats in the measure.

There are two types of meter—*simple* and *compound*. In simple meter the note value that represents the beat may be divided into durations that are related to the beat by multiples of two. For example:

beat = ♩
simple division = ♪ ♪ or ♪ 𝅘𝅥𝅯 𝅘𝅥𝅯

In compound meter the note value that represents the beat may be divided into durations that are related to the beat by multiples of three. For example:

beat = ♩.
compound division = ♪ ♪ ♪

The vast majority of music of the seventeenth, eighteenth, and nineteenth centuries is constructed in measures that have either two (duple), three (triple), or four (quadruple) beats to the measure. The chart below shows the most commonly found meter signatures.

	SIMPLE		COMPOUND	
	METER SIG.	DURATION OF BEAT	METER SIG.	DURATION OF BEAT
DUPLE	2/2 (¢)	♩	6/4	♩.
	2/4	♩	6/8	♩.
	2/8	♪	6/16	♪.
TRIPLE	3/2	𝅗𝅥	9/4	♩.
	3/4	♩	9/8	♩.
	3/8	♪	9/16	♪.
QUADRUPLE	4/2	𝅗𝅥	12/4	♩.
	4/4 (𝐂)	♩	12/8	♩.
	4/8	♪	12/16	♪.

PRINCIPLES OF NOTATION

The procedure for the notation of rhythm in music is very difficult to codify because a given rule frequently has exceptions. One reason for these frequent exceptions is that a common practice has evolved that is not necessarily the outcome of a strict logic; another, that visual considerations overcome, in some cases, procedures that may be theoretically more correct.

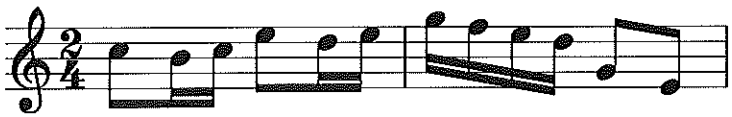
What follows, therefore, is an explanation of a number of principles that constitute the basis for the notation of rhythm. One should remember that, generally speaking, music is notated in such a way that the organization of the beat and its subdivisions is clear to the performer.

Flags and Beams

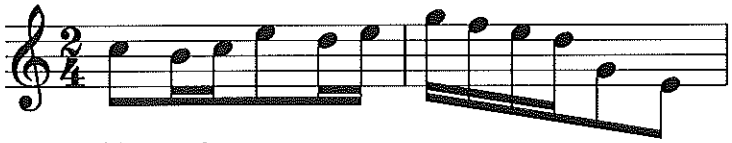
All pitches of a duration less than a quarter note are made up of note heads (𝅗𝅥), stems (|), and flags (𝅘). The stem is attached to the right of the note head and points up if the note head is placed below the middle line of the staff, and to the left, pointing down, if the note head is on or above the middle line. The flag is always placed on the right of the end of the stem.

In instrumental music (and in some editions of vocal music), flags are replaced by beams. For example:



becomes 

grouped in single beats

or sometimes 

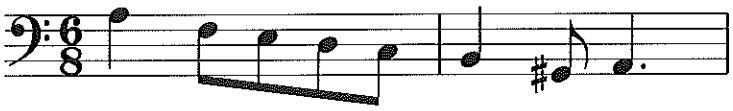
grouped in two beats

Notice that when groups of notes are embraced by a single beam, the stems always go in the same direction, the direction being determined by the placement of the note heads.

A beam is used for the duration of a fraction of a beat only if the remainder of the beat is occupied by a rest.

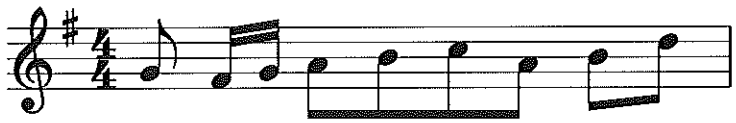


If a beam crosses a beat, either it must span the measure completely or it must span a whole number of beats completely. For example:



is undesirable because the beaming in the first measure obscures the fact that e is the first note of the second beat. The correct notation is



Similarily, 

is unsatisfactory for two reasons:

1. The first eighth note should be beamed to the sixteenth notes.
2. Beats two and three should not be beamed together without beaming the whole measure together.

Of the three possible solutions that follow, the first is the most common and the last the least common.



Since triple-meter measures cannot be divided in half without breaking into the beat, the first and second, or second and third, beats are occasionally beamed together.

Dots and Ties

Dots and ties are used to extend the duration of given note values. Dotted notes may be used anywhere within a measure provided that the beat is not obscured. For example:



is acceptable because the performer has no difficulty in understanding the temporal organization, but



is unacceptable because the performer would have difficulty in seeing where the second and third beats fall.

Ties are used within a measure when the use of dots would create confusion. The example directly above, for instance, can be rewritten with ties so that the beats may be clearly seen.



Ties are also used to extend a note value across a bar line.



Notice that the tie symbol ( ) always connects *note heads.*

Borrowed Divisions

Simple meter is used when the beat is divided into multiples of two. It is possible, however, to insert compound divisions of the beat into music written in simple meter, and simple divisions into music written in compound meter. For example:



In the example above, three eighth notes occupy the duration of two in the third beat. This is a *borrowed division.*

Notice that the three eighth notes are connected by a single beam and that a small "3" is written below the beam for the sake of clarification. This group of three is called a *triplet.*

Study the borrowed divisions in the following examples.



Augmentation and Diminution

The duration of a pitch is augmented by doubling its value and diminished by halving its value.

Thus, a piece of music in $\frac{2}{4}$ may be rewritten in $\frac{4}{4}$ by doubling the value of each of the original notes and rests. For example:



may be rewritten in $\frac{4}{4}$ as



Similarly,



becomes, in diminution,



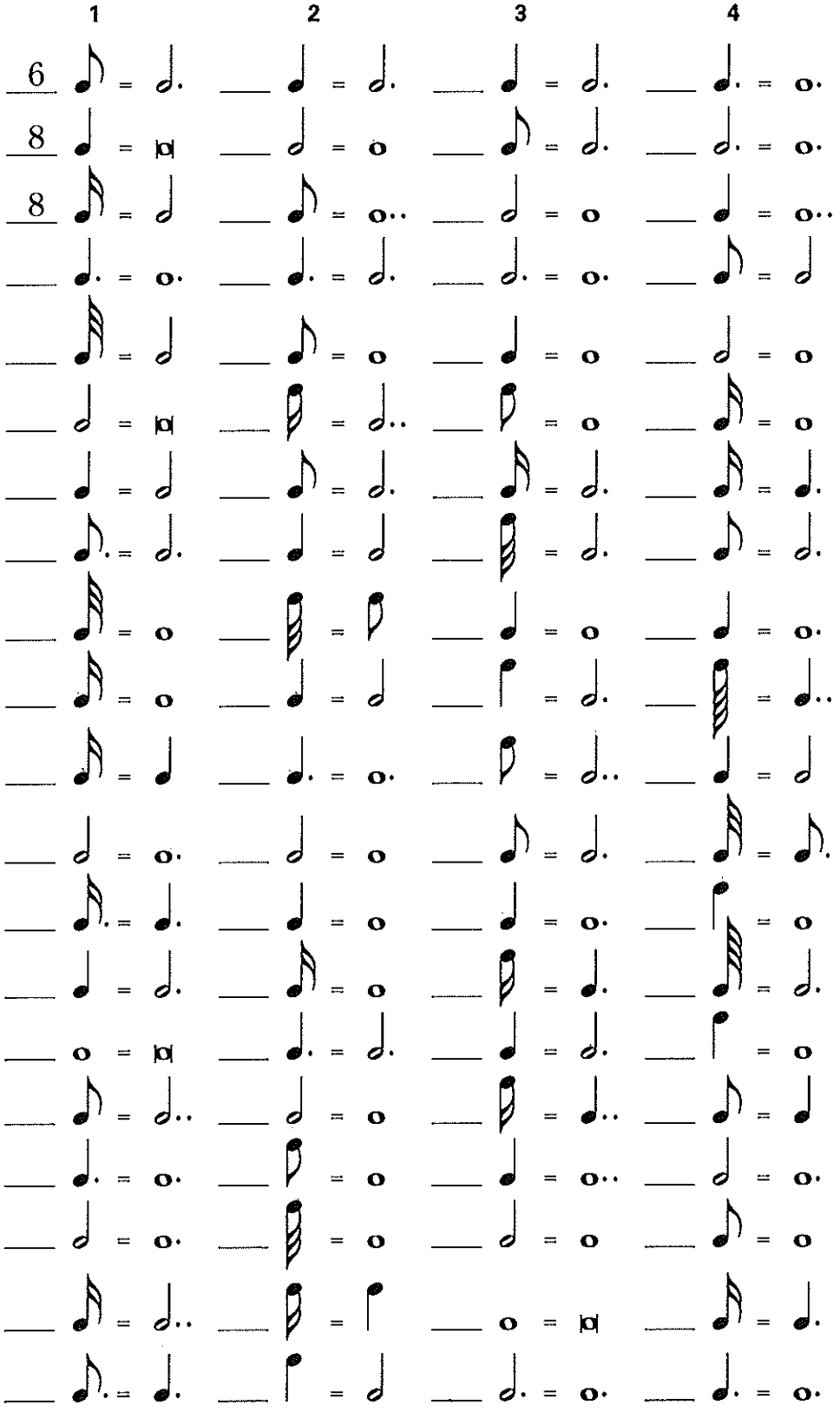
Rests

In any measure each beat, or part of each beat, must be accounted for by a note or a rest. The principles for writing rests, therefore, are very similar to those for writing notes, with some exceptions.

1. Rests may not be tied.
2. It is not the common practice to follow a rest with a dot. For example, 𝄽 𝅘𝅥𝅯 is preferable to 𝄽·.
3. A whole rest is used to indicate an empty measure, *regardless of the meter.*

EXERCISES

A. Write the numerical solution for each rhythmic equation. Time goal—90 seconds per *column.*



B. Rewrite the given melodies in the specified meters, using rhythmic augmentation or diminution.

1



2



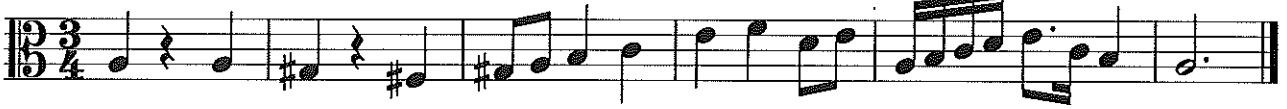


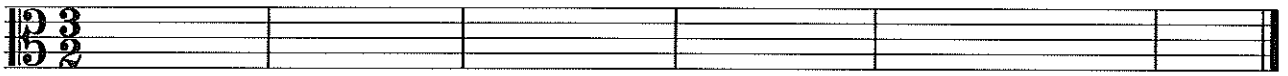
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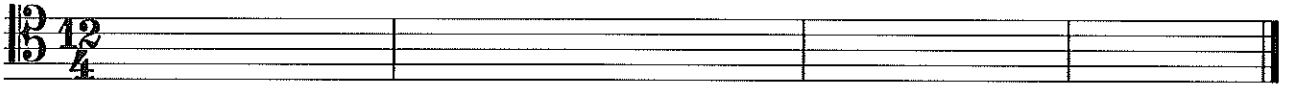
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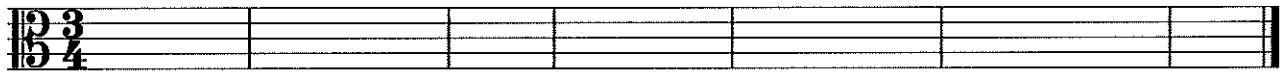


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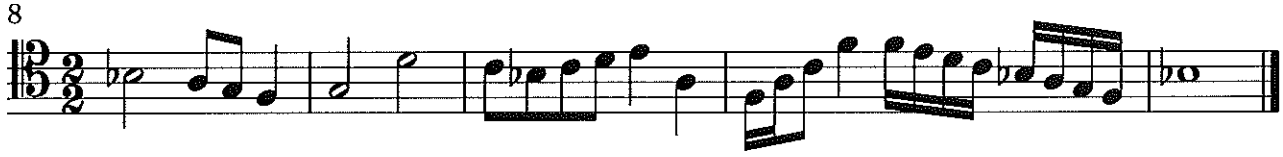


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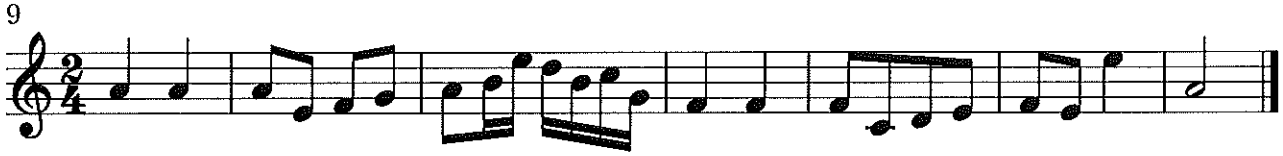


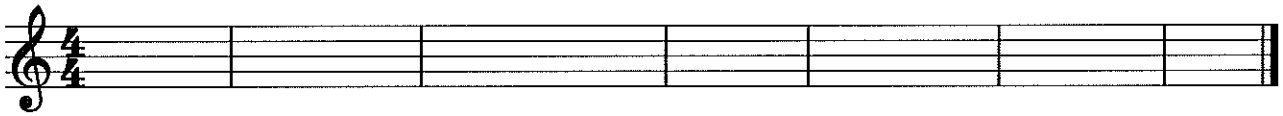
8





9





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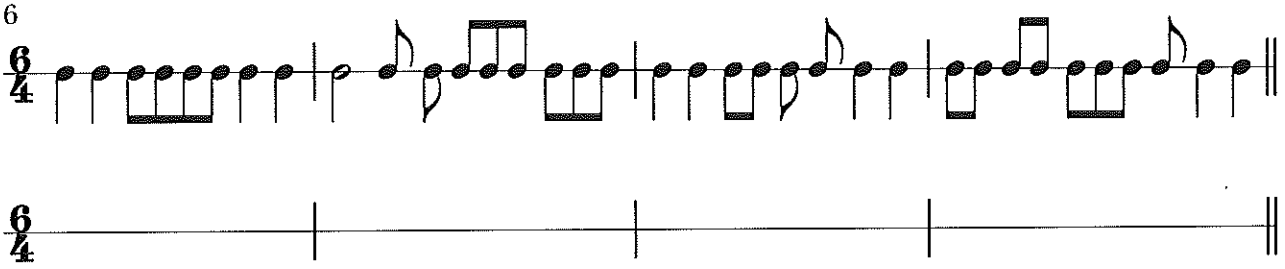




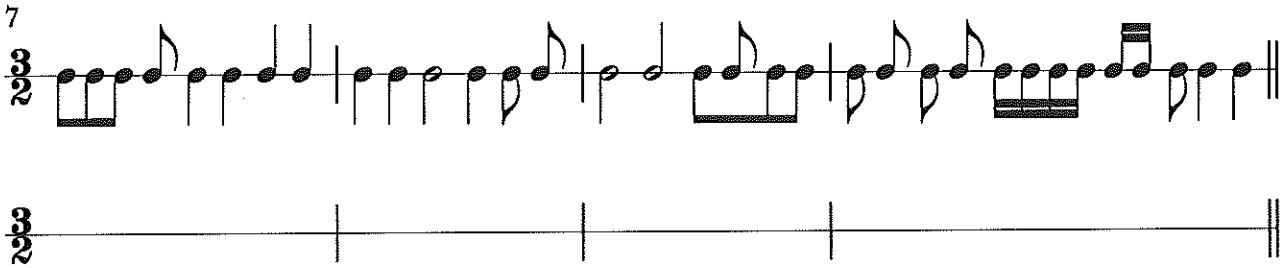
C. Rewrite the given rhythms using correct beaming procedures.



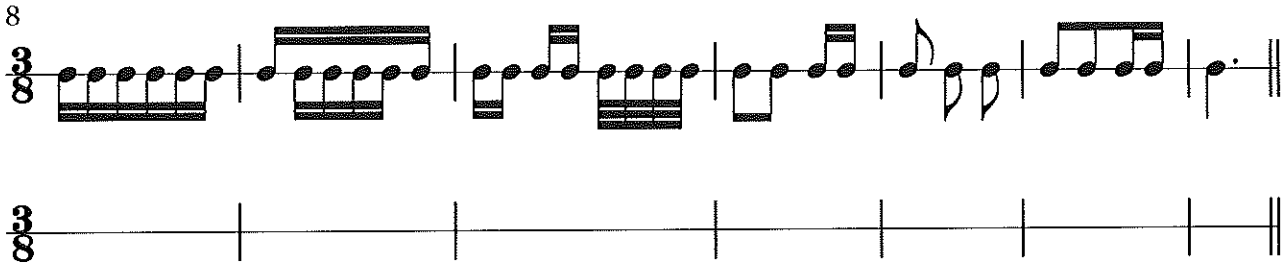
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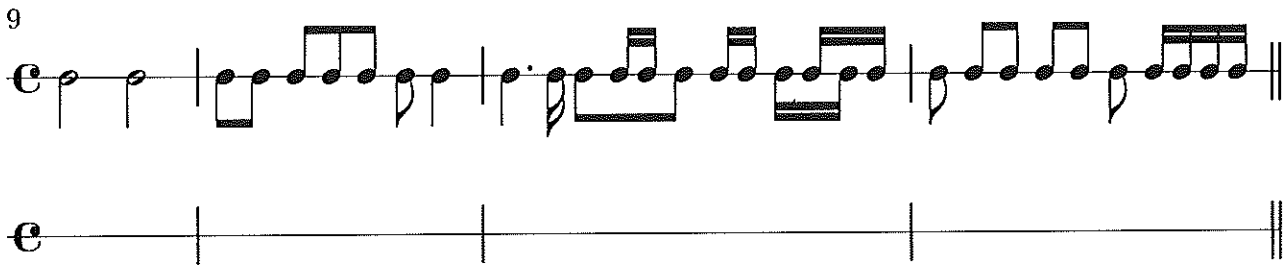
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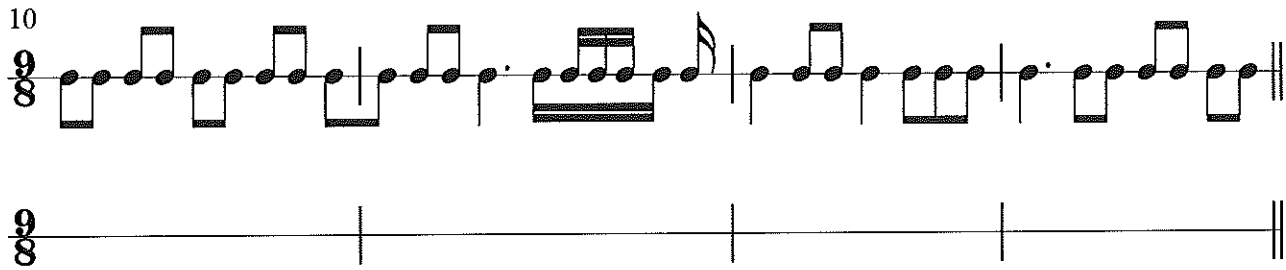
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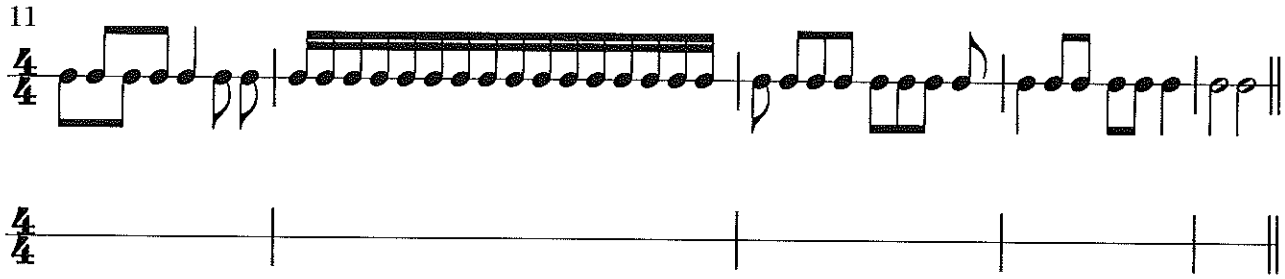
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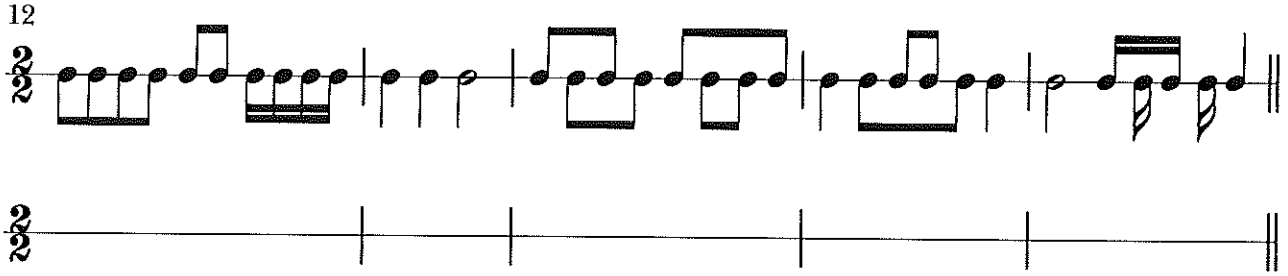
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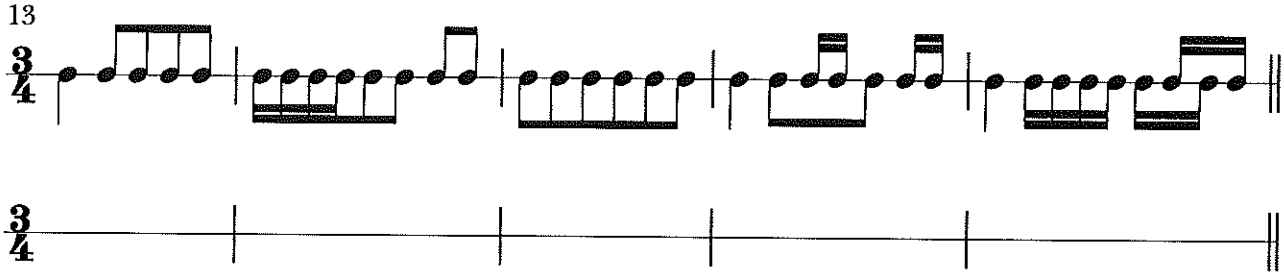
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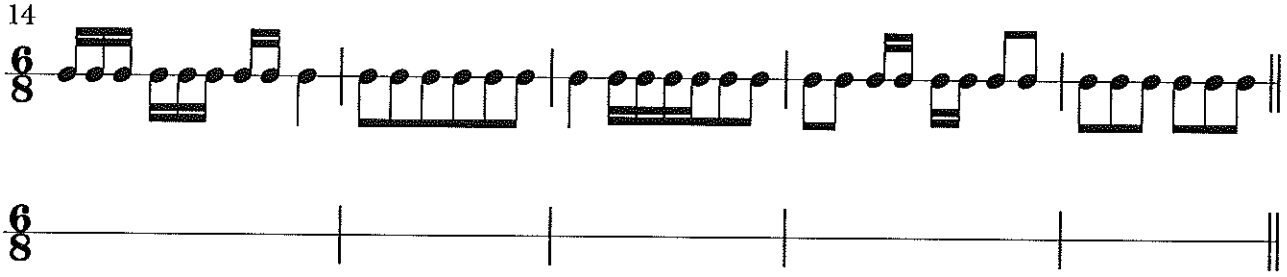
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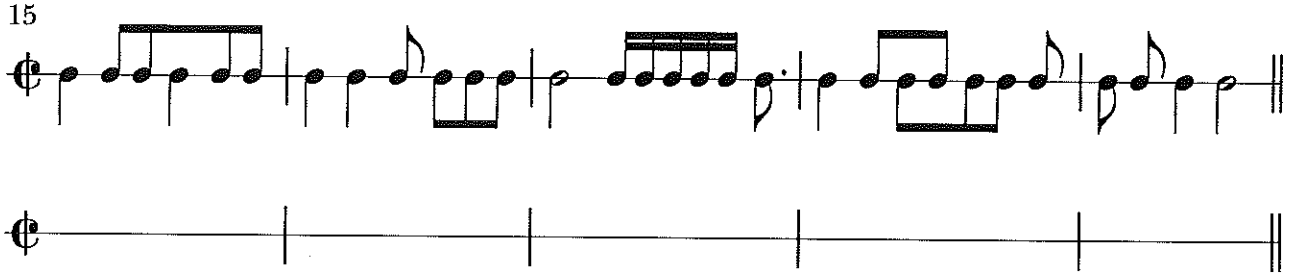
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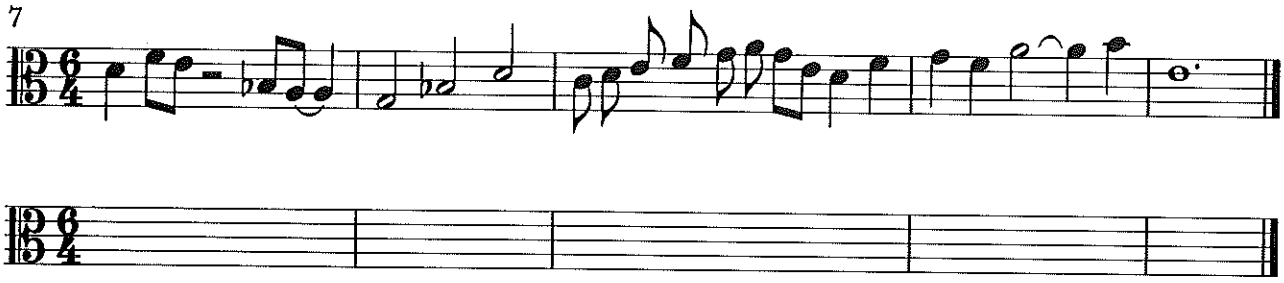
D. Correct the errors of notation in the following melodies.



6



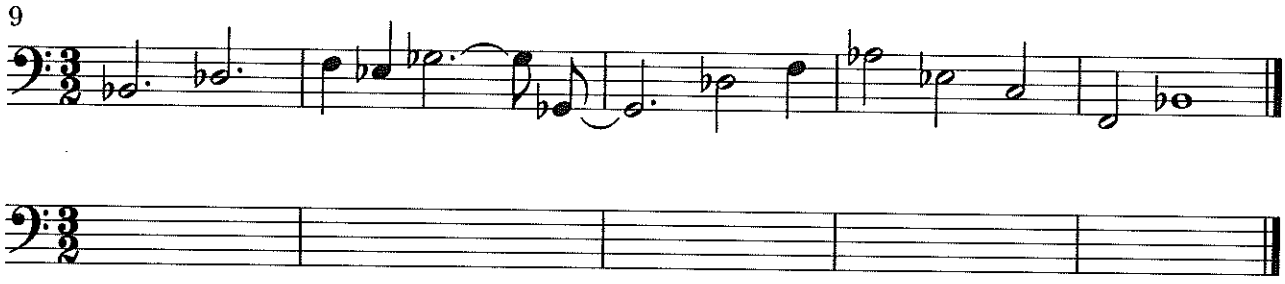
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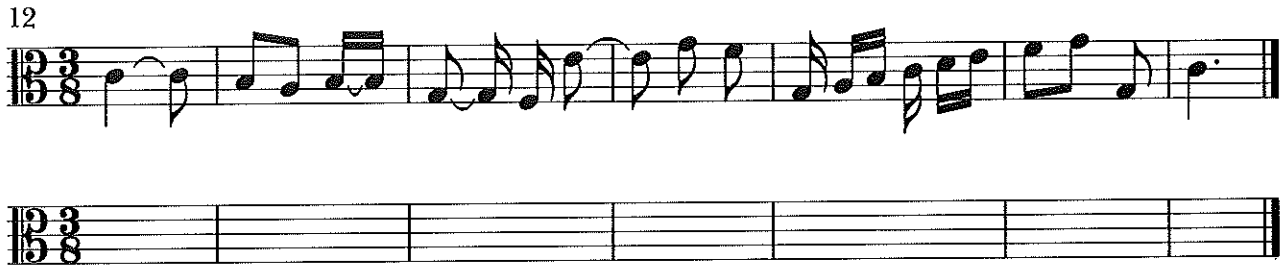
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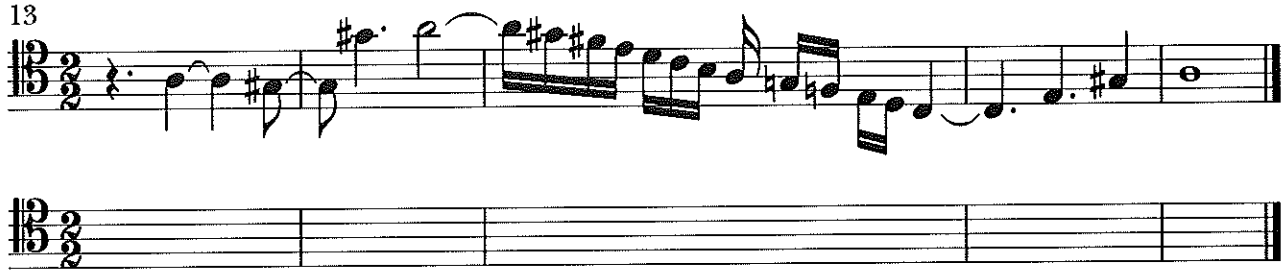
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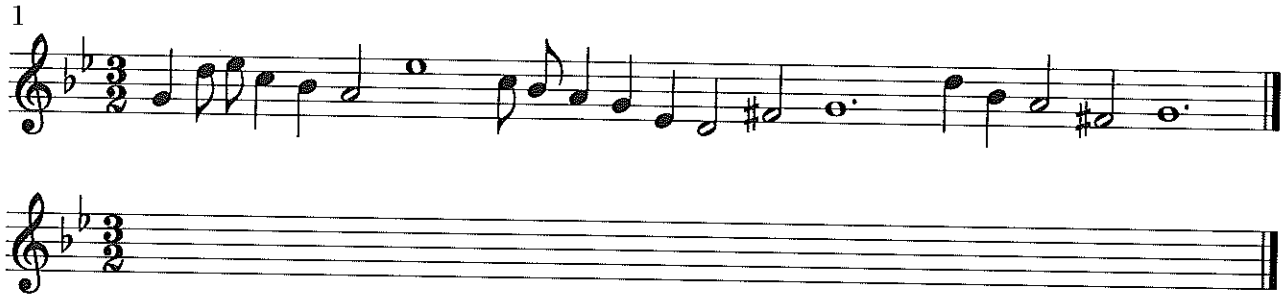


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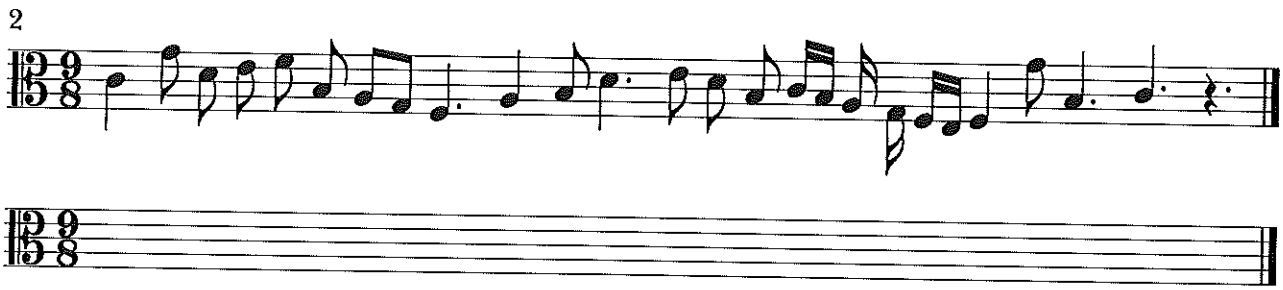


E. Rewrite the following melodies with bar lines, and correct errors of notation.

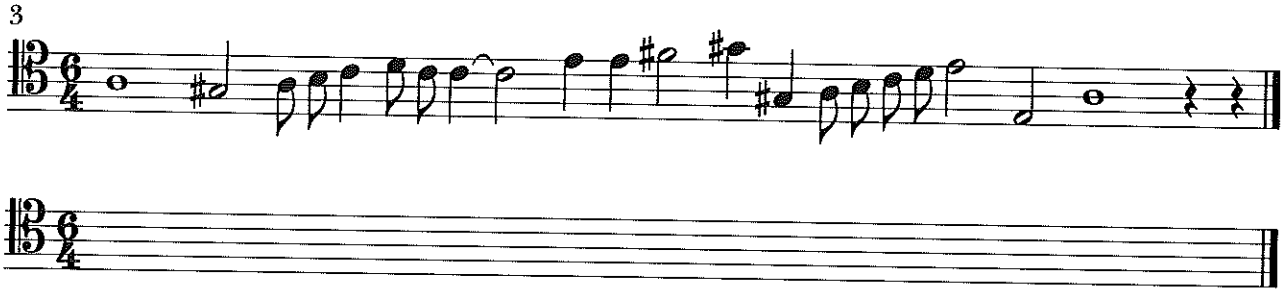
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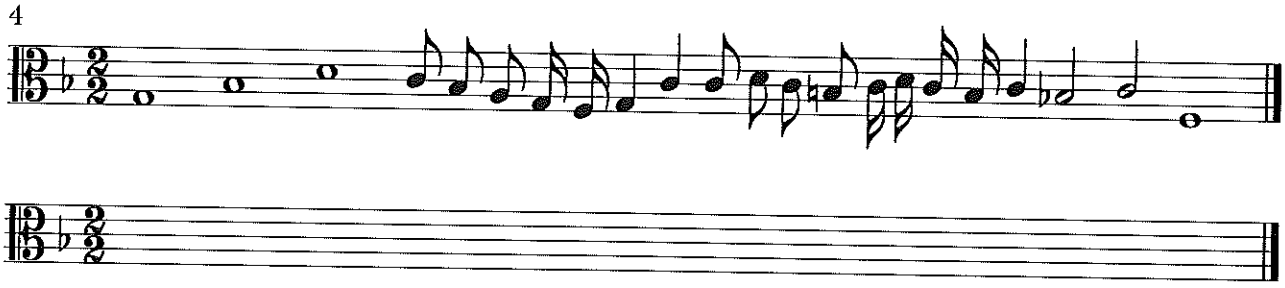
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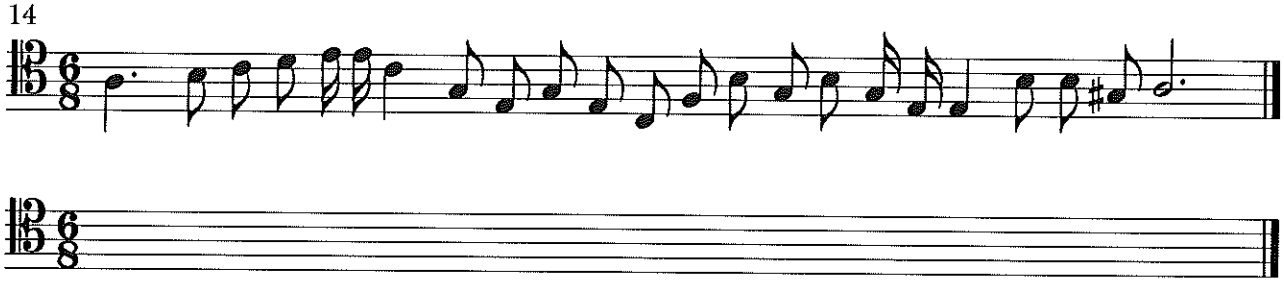
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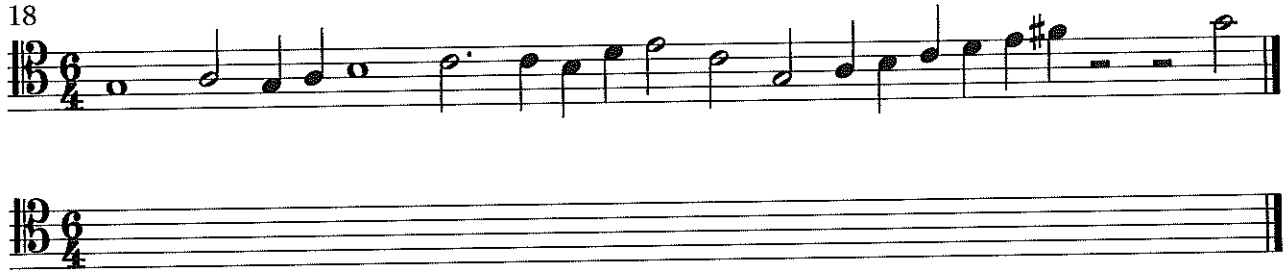
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PART TWO

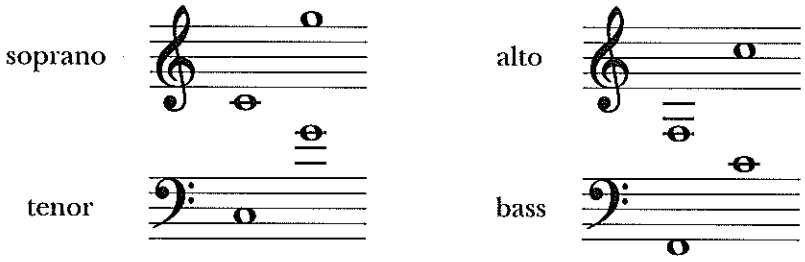
HARMONY IN COMMON PRACTICE

The Diatonic Vocabulary

7 Four-Part Vocal Writing

Although composers write harmonic music for voice or instruments, or both, in a wide variety of parts, students customarily begin the study of harmony by writing for voices in four parts. There are three good reasons for this approach. First, by writing for voices rather than for instruments, the student learns to be sensitive to linear problems that arise in individual parts, for singers cannot depress a key or cover a hole to make their pitch; they must create their own. Second, the four voices—soprano, alto, tenor, and bass (SATB)—embrace a three-octave range over which the ear has relatively little difficulty in distinguishing between one pitch and another. Third, the majority of chords in common usage have either three or four different pitches.

The ranges of the four voices are



Notice that the soprano and alto ranges are duplicated an octave lower by the tenor and bass.

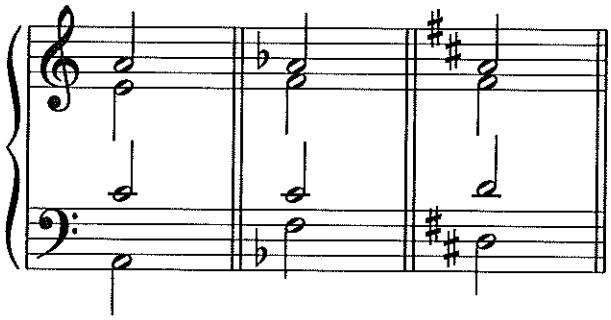
Because triads have only three different pitches, in a four-part setting one of those pitches has to be doubled. The problem of doubling is one that has plagued writers of music theory books for the better part of a century. Rules for the doubling of certain members of the triad vary from book to book, and some are so complex that the student is literally terrified to commit a pitch to paper.

For over three centuries composers have treated the problem of doubling with such considerable freedom that a series of elaborate rules is at best unrealistic, at worst stultifying. The guidelines below will help the student

develop an approach to doubling that is not only simple and practical, but also as representative of composers' practices as any other approach.

1. In major and minor triads, the most commonly doubled pitch is the root.
2. If the root cannot be doubled, the third or the fifth is doubled, depending on which produces the more desirable voice leading.
3. Pitches that have strong resolution tendencies are not doubled, for example, the leading tone. Such pitches are called *sensitive* pitches.

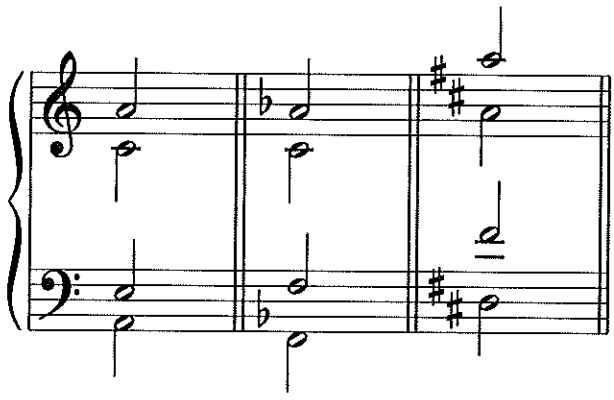
Triads may be written in *close* or *open structure.* In close structure in four parts, the soprano, alto, and tenor voices are placed as close together as the chord tones will allow.



Note:

a. The stems are written above the note head for the soprano and tenor and below the note head for the alto and bass.
b. There is no restriction upon the distance between the tenor and bass.
c. The triads are said to be in *root position* because the roots are in the bass.

In open structure, the soprano, alto, and tenor voices are not placed as close together as possible.



Note:

a. There is a space of an octave between the soprano and alto in the third example. Except for the tenor and bass, adjacent voices should not exceed this interval.
b. In the same example, the fifth is doubled. Although this is not incorrect, it is not the most common doubling.
c. In the same example, the a^2 in the soprano is extremely high for that voice.

EXERCISES

A. The key, triad, soprano part, and structure (close or open) are given. Add alto, tenor, and bass parts, and then rewrite each chord in the structure that differs from the one given, being careful to pay attention to voice ranges.



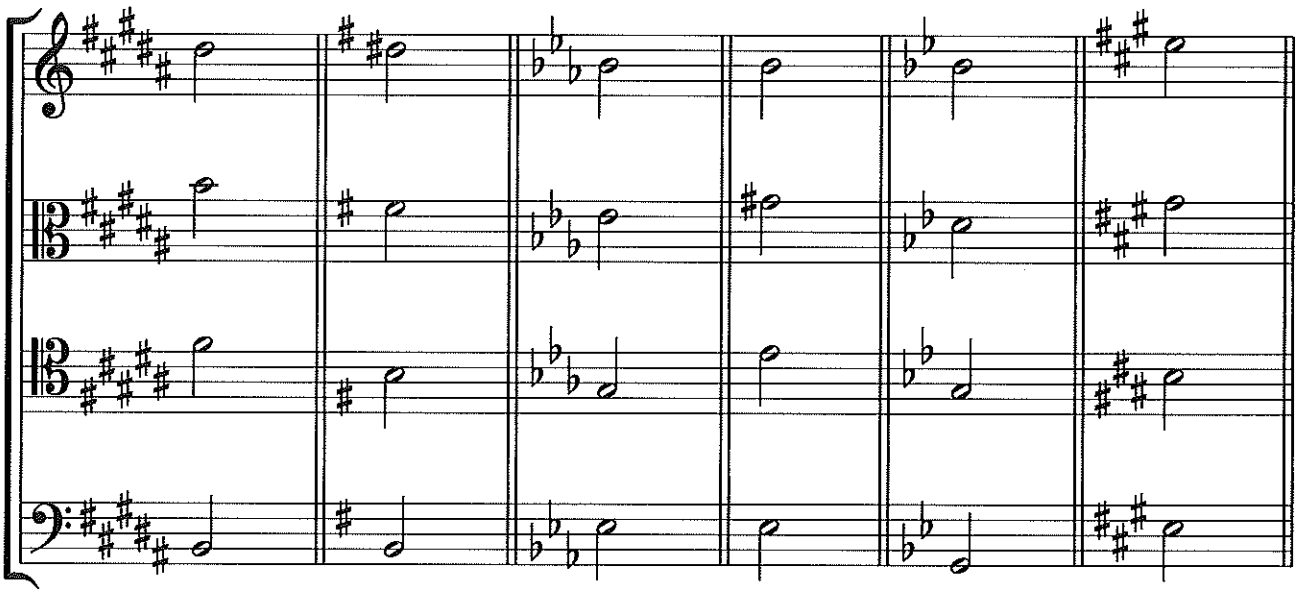
B. All of the chords below are tonic, dominant, or subdominant triads. For each one identify the key, triad, and structure.

1



F: V

2



8 Primary Triads in Root Position

The tonic, subdominant, and dominant triads are called the *primary triads.* In the major keys, the primary triads are all major (I, IV, V), whereas in the minor keys, only the dominant triad is major (i, iv, V).

CADENCES

The word *cadence* is derived from the Latin verb *cadere,* which means "to fall." Cadences in music are analogous to those punctuation marks in English that subdivide sentences into phrases and clauses. Thus, cadences occur at points of relative repose. A complete musical sentence is referred to as a *period,* and each major subdivision as a *phrase.*

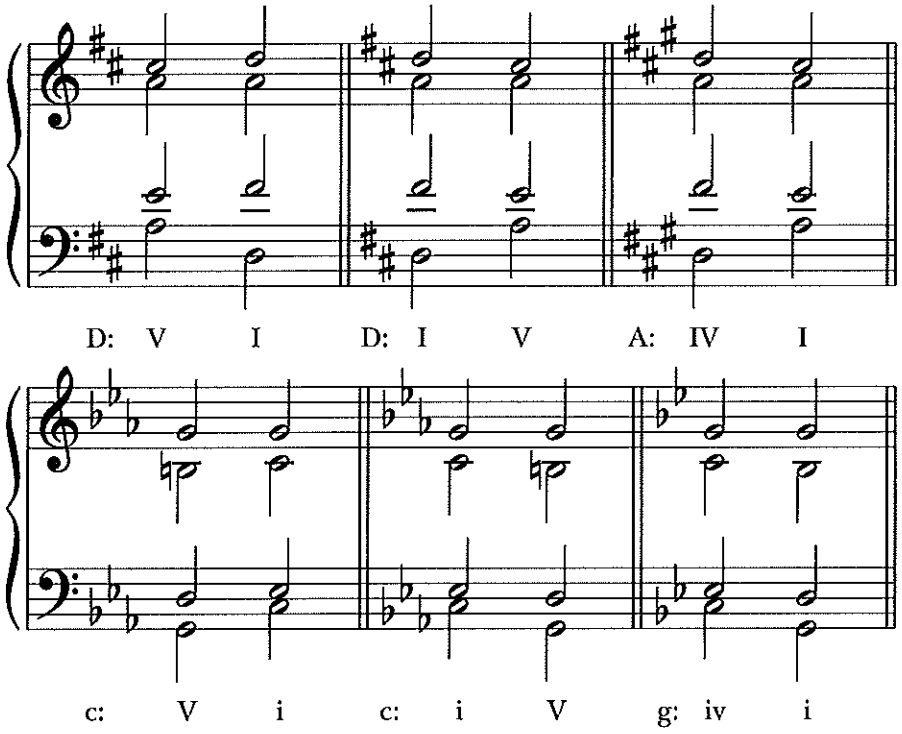
1. The *authentic cadence* is the cadence that most decisively terminates the last phrase of a period. It is represented by the harmonic progression V–I.
2. The *plagal cadence,* IV–I, is also used at the end of a period but more rarely than the authentic cadence because it is harmonically less decisive. It is sometimes described as the "amen cadence" due to its use at the end of hymns.
3. The *half cadence,* analogous to a comma, occurs in the middle of a musical period and typically terminates an internal phrase. This cadence is represented by several harmonic progressions such as I–V, IV–V, ii–V, and vi–V. The essence of the half cadence is arrival at the *dominant.*
4. The *deceptive cadence* is most commonly represented by the progression V–vi, although V may be followed by other chords. Since this cadence involves the movement to a chord that is not a primary triad, it will be discussed later.

A cadence does not occur every time one of the progressions described above is encountered. For a listener to perceive a cadence, the music must create a feeling of repose. Such a feeling is frequently achieved by a coincidental relaxation of harmonic, melodic, and rhythmic motion.

ROOTS A FOURTH AND FIFTH APART

Three procedures are common to all of the following progressions:

1. The root is doubled in both chords.
2. Two pitches in each progression move in parallel stepwise motion.
3. There is one common tone (a pitch common to both chords), and it remains in the same voice.



Note:

a. In the V–I and V–i progressions, the *leading tone* moves up to the *tonic.* This movement is its natural tendency.

b. Each progression is provided with a harmonic analysis that includes the key of the excerpt and the scale degree and quality of each triad. Students are encouraged to use this standard format to check the accuracy of their answers to the exercises.

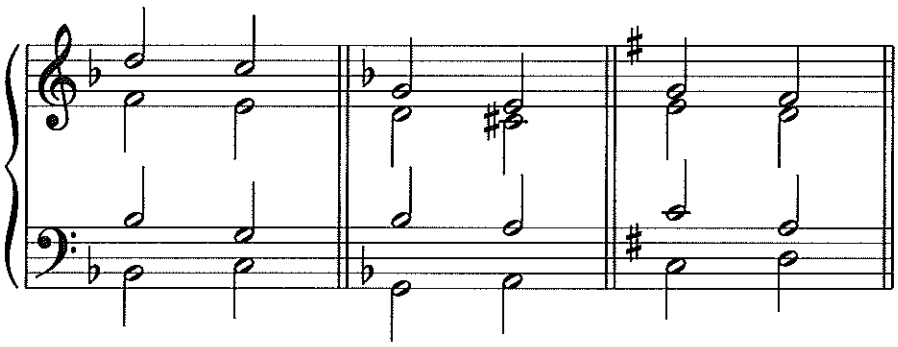
If, in the soprano voice in a V–I progression, the fifth of the dominant triad moves to the root of the tonic triad, the tonic triad will be incomplete, having three roots and a third. This voicing is acceptable.



ROOTS A SECOND APART

Two procedures are common to the following progressions:

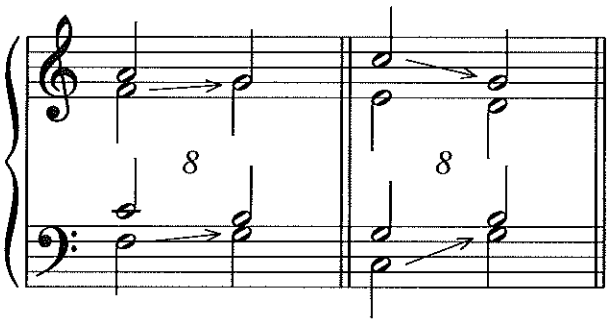
1. The root is doubled in both chords.
2. The top three voices move in contrary motion to the bass.



RESTRICTIONS IN VOICE LEADING

The contents of Part Two of this book represent a synthesis of harmonic techniques commonly practiced by composers over a period of about three hundred years, from 1600 through 1900. Composers throughout this period were usually careful, particularly in choral writing, to avoid the following movements of the voices:

1. Parallel or consecutive movement between any two voices in octaves.

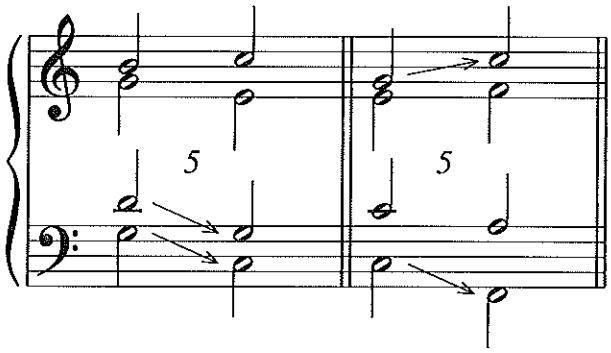


Note:

a. In the first example, *parallel octaves* are caused by the stepwise motion of the alto and bass voices.

b. In the second example, *consecutive octaves* are caused by the soprano and bass voices, which leap in opposite directions from c^1 to g^1 and c to g, respectively.

2. Parallel or consecutive movement between two voices in perfect fifths.

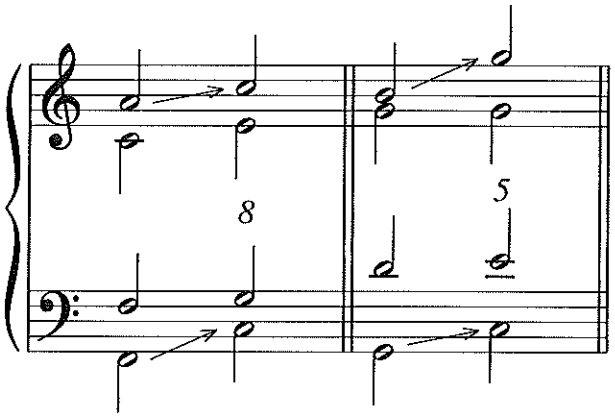


Note:

a. In the first example, *parallel fifths* are caused by tenor and bass voices, which, after a leap in the same direction, remain a perfect fifth apart.

b. In the second example, *consecutive fifths* are caused by the soprano and bass voices, which, after a leap in opposite directions, remain a (compound) perfect fifth apart.

3. Hidden octaves or fifths—the leap in the same direction by outer voices to a perfect octave or fifth.

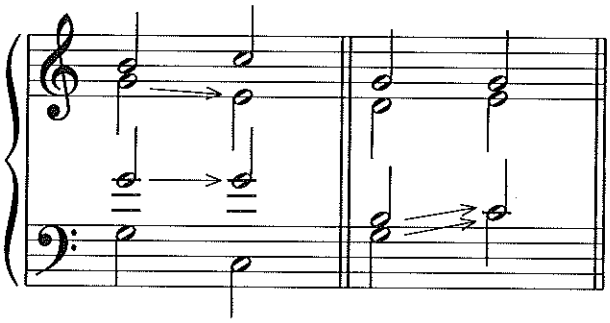


Note:

a. In both examples, *hidden octaves* and *fifths* are caused by the soprano and bass voices, which leap to (compound) perfect octaves and fifths.

b. The condition for hidden octaves or fifths exists only if both outer voices leap to a (compound) octave or fifth.

4. The overlapping or crossing of voices.



Note:

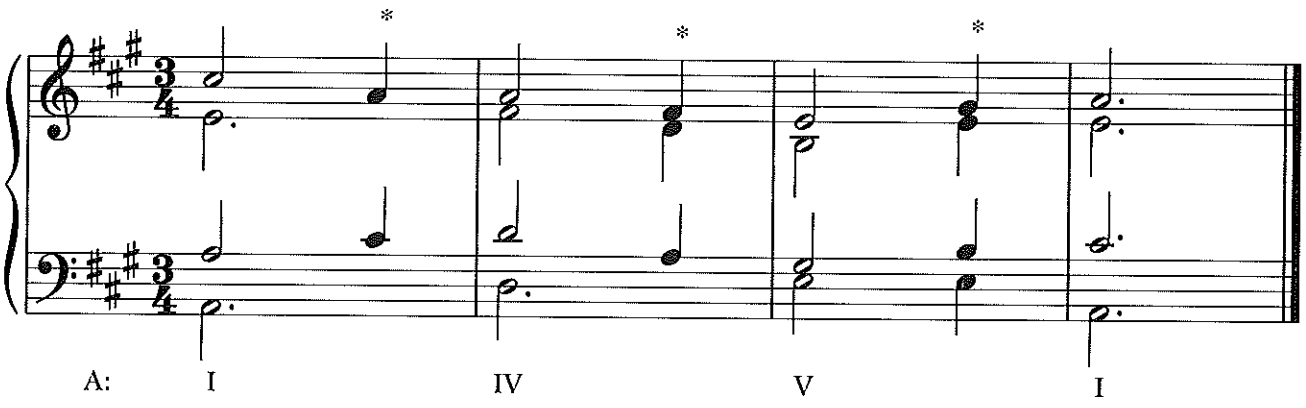
a. In the first example, the alto and tenor voices quite properly double g^1. The motion of the alto to c^1 while the tenor sustains the higher g^1 causes the overlapping (or crossing) of voices.

b. In the second example, the tenor and bass voices move in the same direction to middle c. The bass, however, "passes through" the tenor's b to arrive at middle c and in doing so causes the overlapping of voices.

As a general principle, motion in the inner parts should be as smooth as possible. Large leaps and diminished and augmented intervals should be treated with great care, because they are difficult to sing. In fact, good harmony is very often the result of combining good harmonic motion with good linear movement of the individual voices.

PRIMARY TRIADS IN COMBINATION

Study the following examples. They show how primary triads in root position are typically used in a simple four-voice texture.





Note:

a. In both examples, the principle of *revoicing* over a stationary bass note (see asterisks) is exemplified. Two or three of the upper voices move in such a way that the doubling is not altered.

b. The subdominant triad follows the tonic in both examples. Rarely does the subdominant follow the dominant triad.

c. The final chord of the cadence falls on a strong beat.

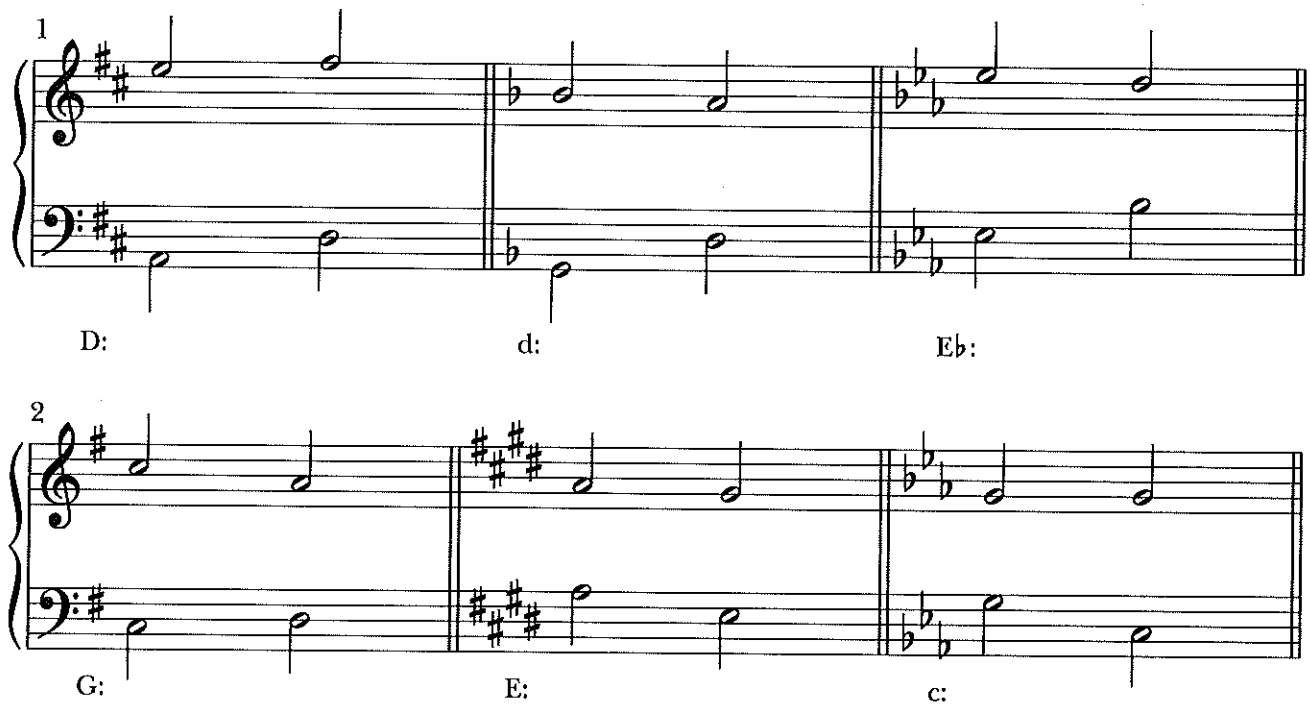
d. Each of the examples terminates with a *perfect authentic cadence.* For an authentic cadence to be classified as perfect, the penultimate chord must be a dominant in root position and the final chord must be a tonic in root position with the tonic note in the soprano voice. All other dominant to tonic cadences are called *imperfect authentic cadences.*

THE FUNCTION OF PRIMARY TRIADS

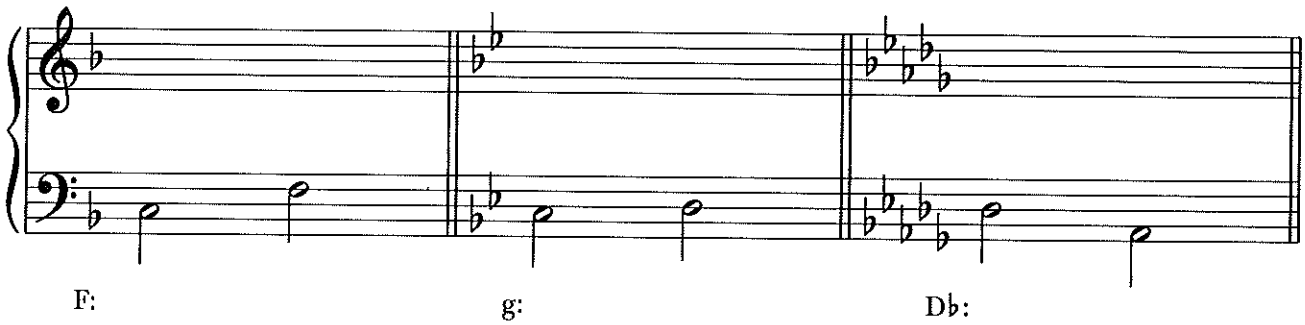
The main function of the primary triads is to establish the key of a piece of music. The composer realizes this goal by judiciously using the tonic, dominant, and subdominant triads at the beginning of a composition and by exploiting one of the four cadence types at the end of phrases. Music in which these two conditions are not realized is typically nebulous from a tonal standpoint.

EXERCISES

A. Add parts for alto and tenor. (The student should begin by making a Roman numeral analysis of each progression.)



B. Add parts for soprano, alto, and tenor.

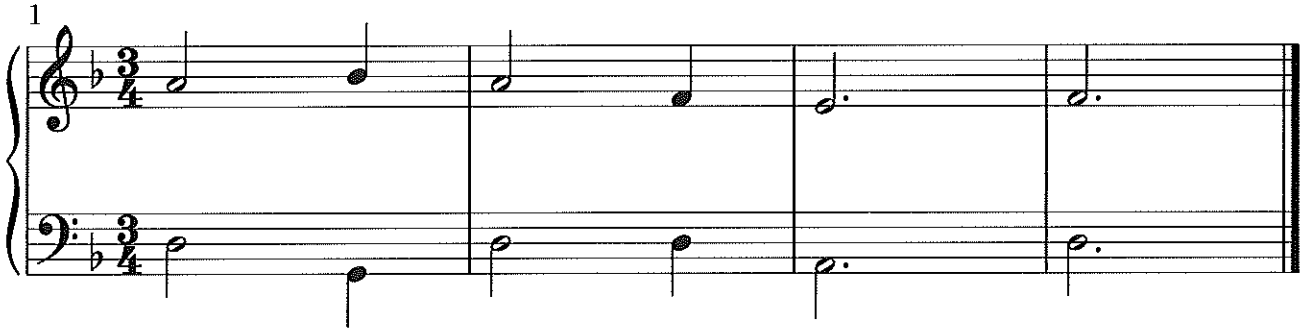


C. Add parts for alto, tenor, and bass.

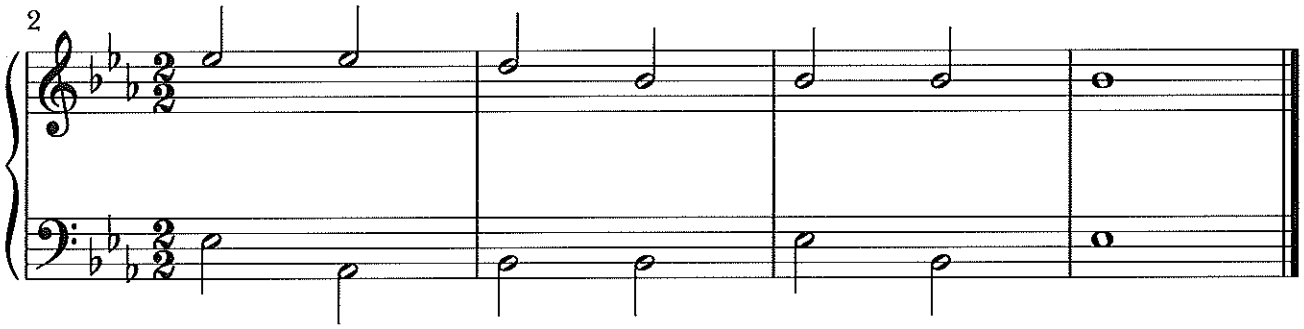


D. Add parts for alto and tenor. (The student is advised to determine the key in each case before proceeding.)

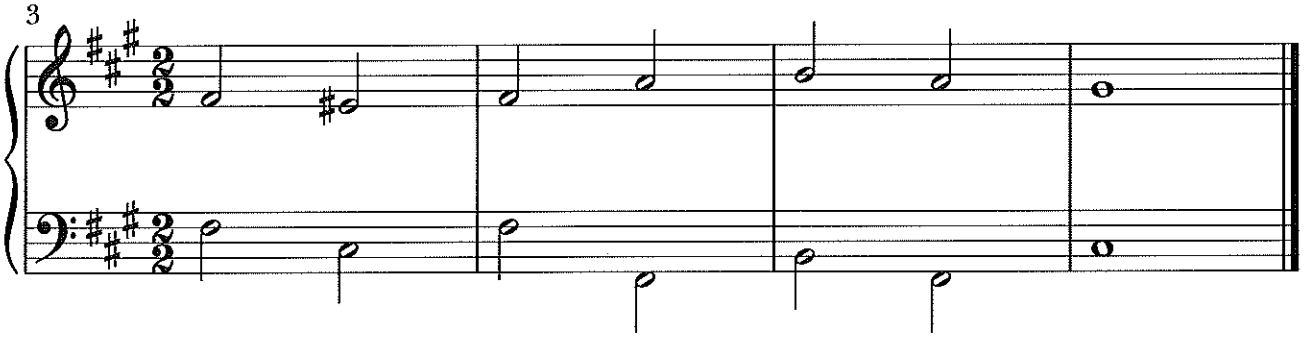
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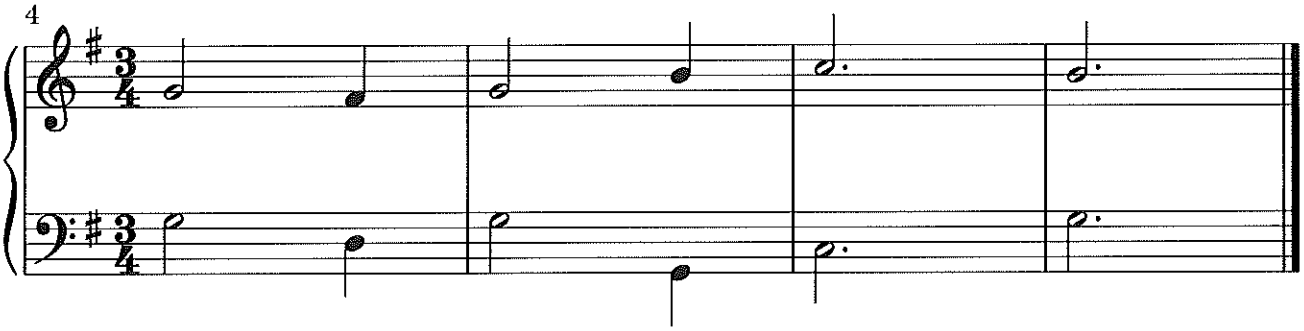
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E. Add parts for soprano, alto, and tenor.



F. Add parts for alto, tenor, and bass.



G. Make a harmonic analysis of the following phrases (see note b on page 106).

1



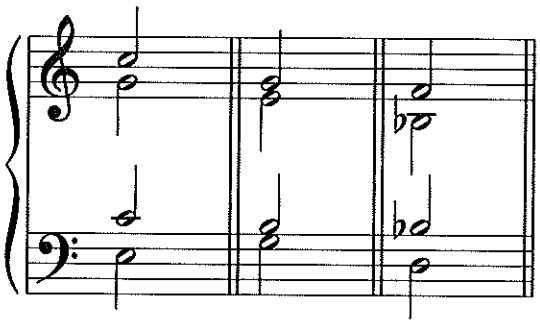
2



9 Inversions of the Primary Triads

A triad is said to be inverted when a pitch other than its root (the pitch on which the triad is constructed) is in the bass voice.

When the *third* is in the bass the triad is in *first inversion.*



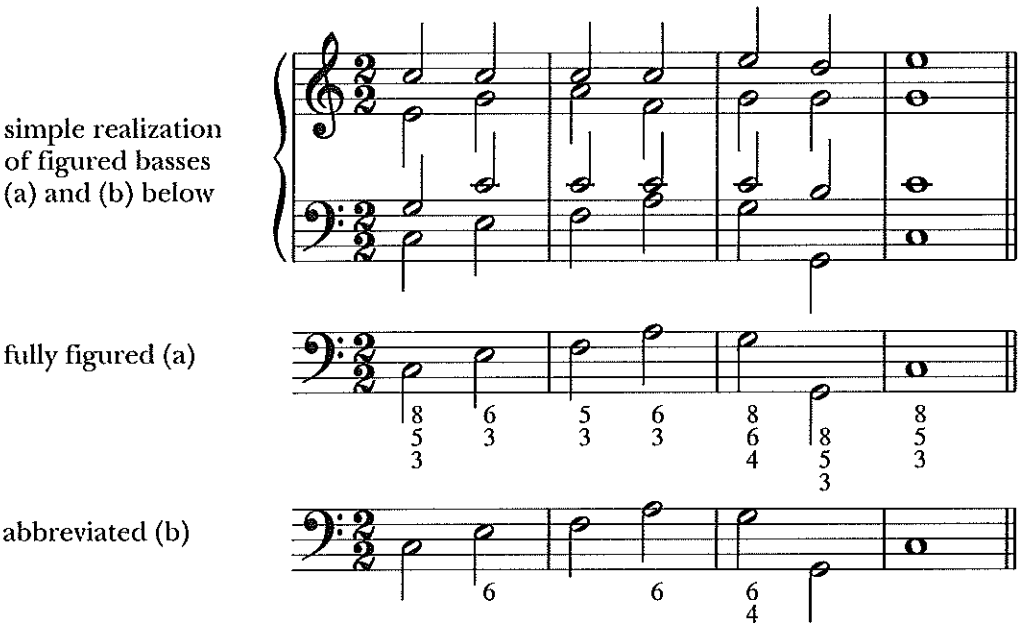
When the *fifth* is in the bass the triad is in *second inversion.*



CHORD SYMBOLIZATION: FIGURED BASS

Figured bass is a musical shorthand that originated in the sixteenth century. To save time, a composer, instead of writing a complete score, would provide a keyboard player in an ensemble with a bass line underneath which various Arabic numerals were written. The numeral (or numerals) under each bass note indicated to the performer which intervals above the bass were to be played. Usually, the numeral was expressed as a simple interval, although the performer was at liberty to perform that interval in any octave he wished.

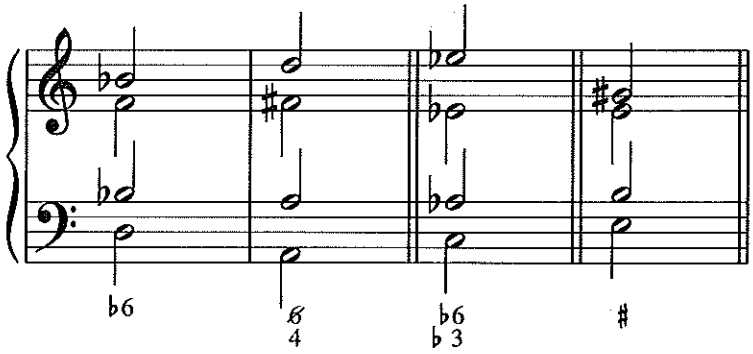
Examine the following example:



The figures under bass line (a) show all of the intervals that appear in the realization. But in practice this would be an extremely cumbersome form of shorthand. Thus, (b) is written, from which the performer may deduce that

1. The absence of any figures indicates a triad in root position.
2. 6 indicates a chord in first inversion. This chord is commonly referred to as a 6 chord.
3. $\frac{6}{4}$ indicates a chord in second inversion. This chord is commonly referred to as a $\frac{6}{4}$ chord.

Each interval is considered to be diatonic unless altered by placing an accidental in front of the figure. Occasionally, a figure has a diagonal line through it, which indicates that the interval is raised by a half step. An isolated accidental indicates that the third above the bass is raised or lowered.



PRIMARY TRIADS IN FIRST INVERSION

Primary triads in first inversion follow the same principles regarding doubling and voice leading as primary triads in root position.

Study the following examples:



Note:

a. The introduction of the primary triad in first inversion allows notes to be written on the third, sixth, and seventh scale degrees, in addition to the first, fourth, and fifth. Thus, a bass line that is more melodic than was previously possible can be created.

b. The figured bass and the harmonic analysis indicate when a triad is inverted.

c. At least one of the upper voices tends to move in contrary motion to the bass.

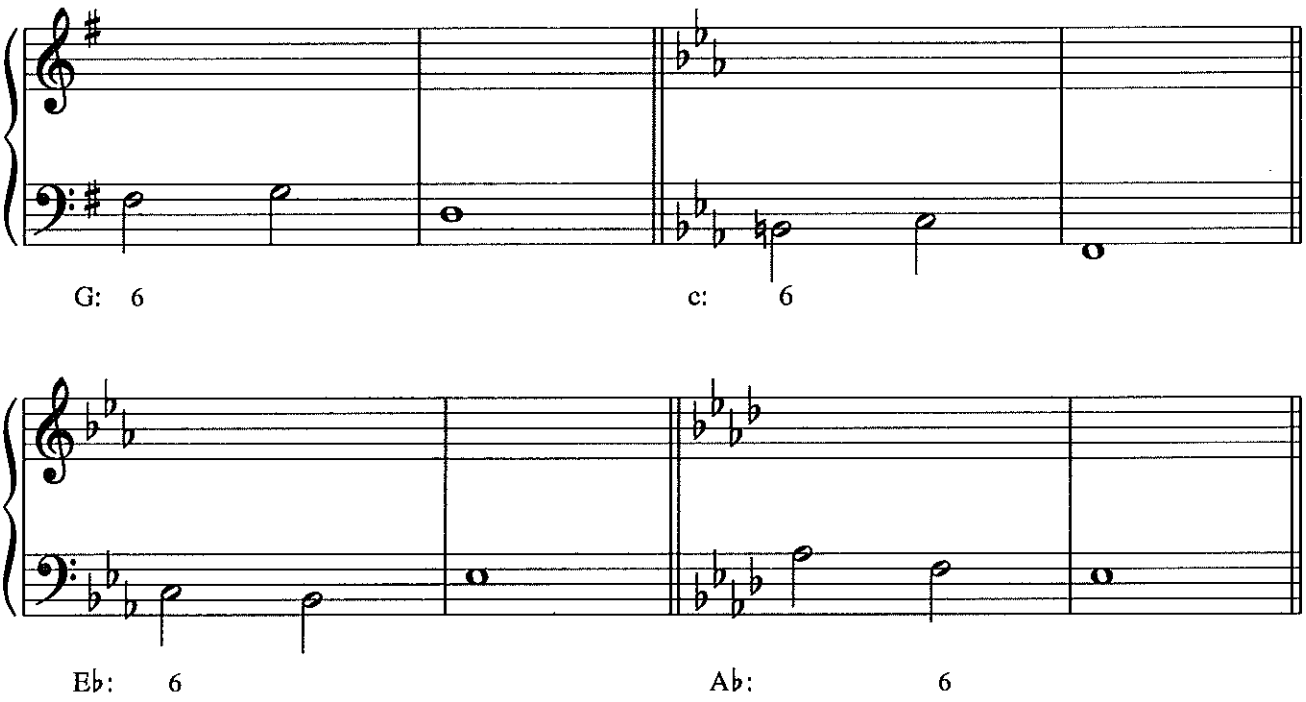
The exercises that follow are associated with triads in root position and first inversion. The discussion of triads in second inversion begins on page 125.

EXERCISES

A. Add parts for alto and tenor.

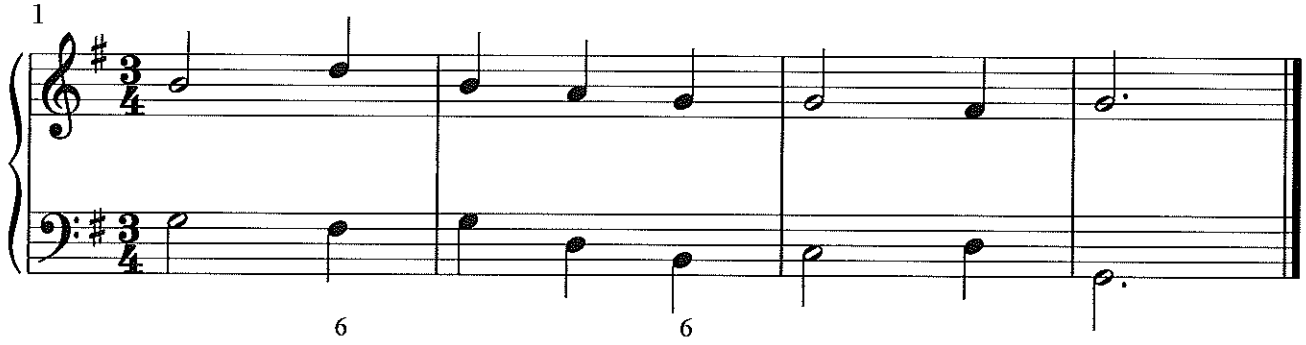


B. Add parts for soprano, alto, and tenor.



C. Add parts for alto and tenor.

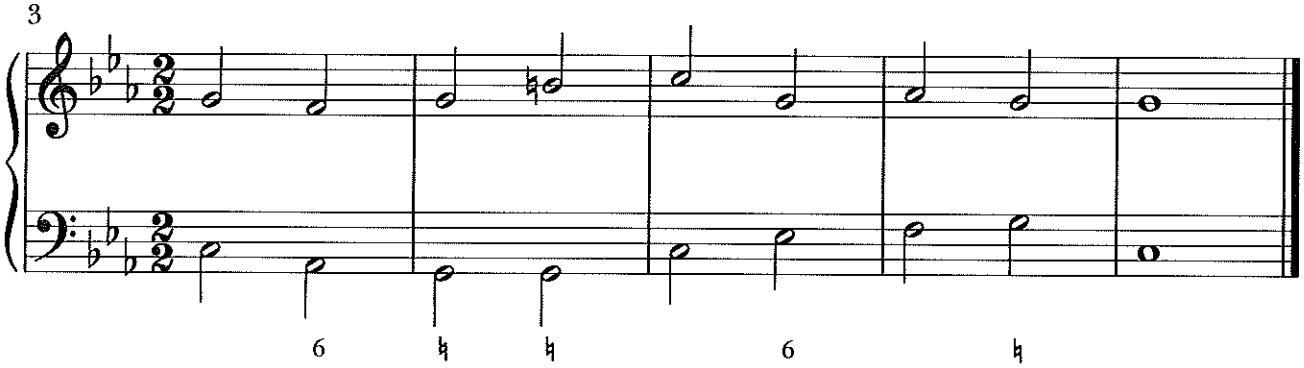
1



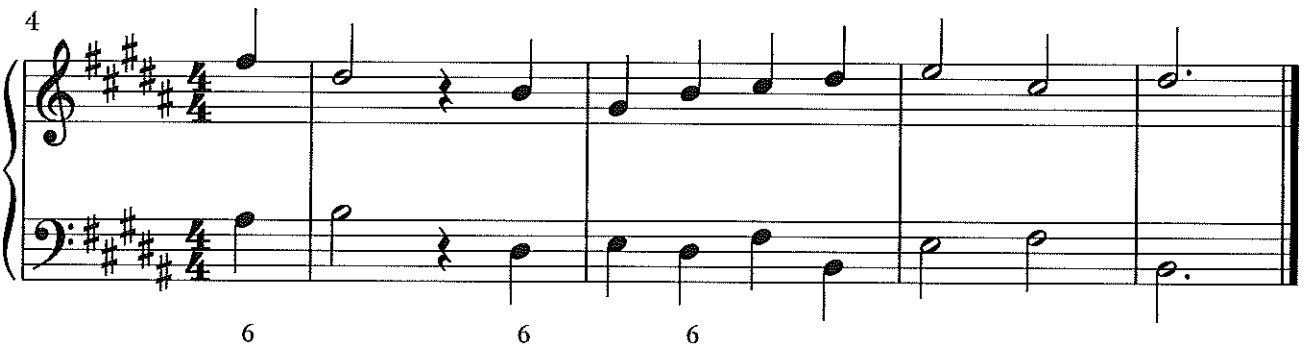
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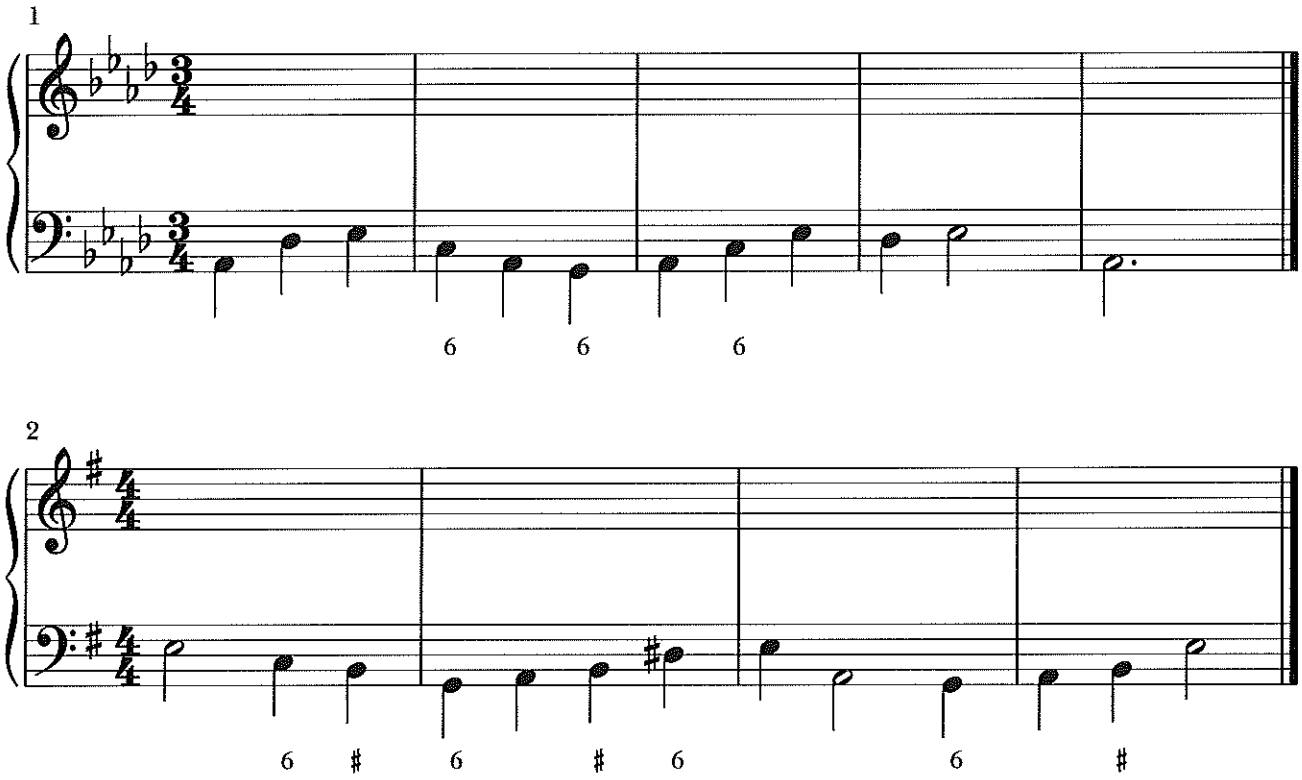
3



4



D. Add parts for soprano, alto, and tenor.



E. Add parts for alto, tenor, and bass, using suitable first inversions.



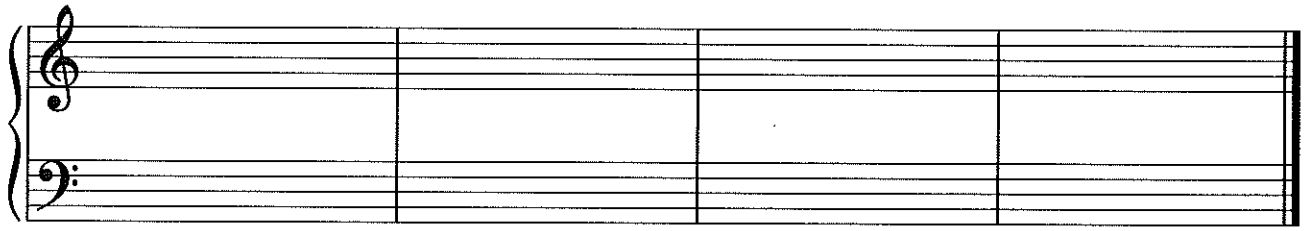
F. Using the given progressions, invent four-measure phrases for soprano, alto, tenor, and bass. The chords should be distributed evenly and logically, with particular attention to the rhythmic organization of the cadence (see model).

1. G: $\frac{2}{4}$ I–V–I–I6–IV–V–I (worked below as a model)
2. d: $\frac{3}{4}$ i–V6–i–iv6–iv–V–i
3. A: $\frac{2}{2}$ V6–I–I6–IV–IV6–V6–I
4. e: $\frac{4}{4}$ i–V–i–iv–iv6–V–V6–i

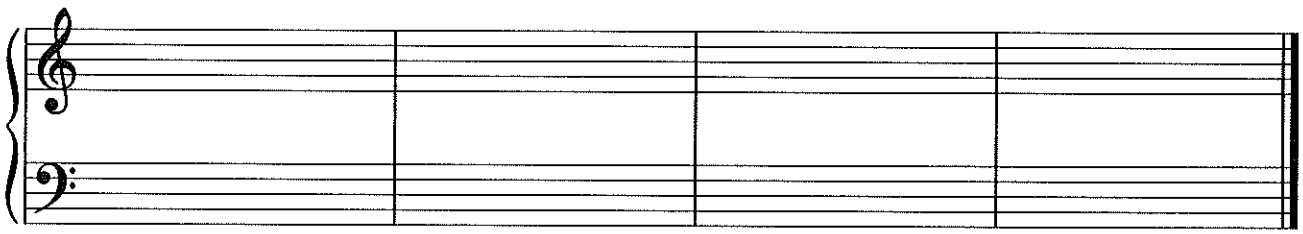
1



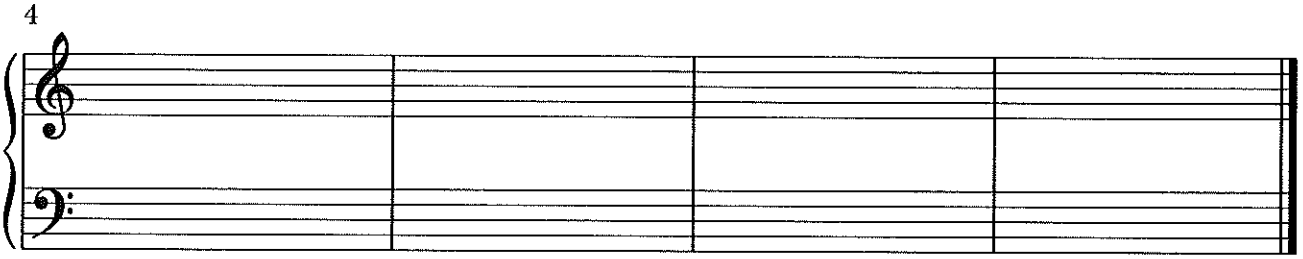
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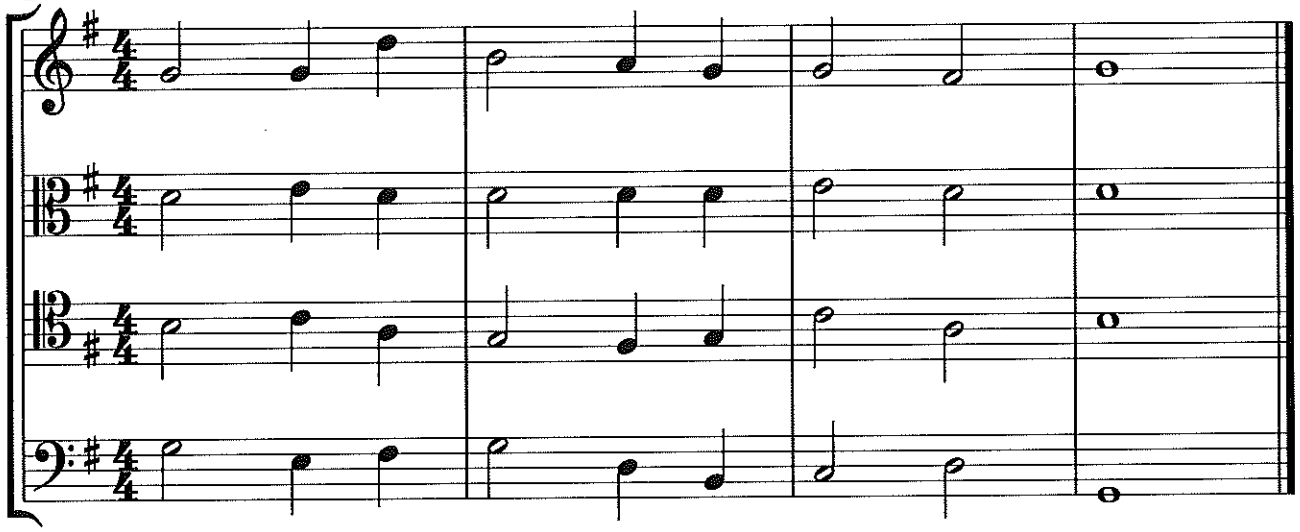


4



G. Make a harmonic analysis of the following phrases. Indicate first inversions with the appropriate symbol.

1



2



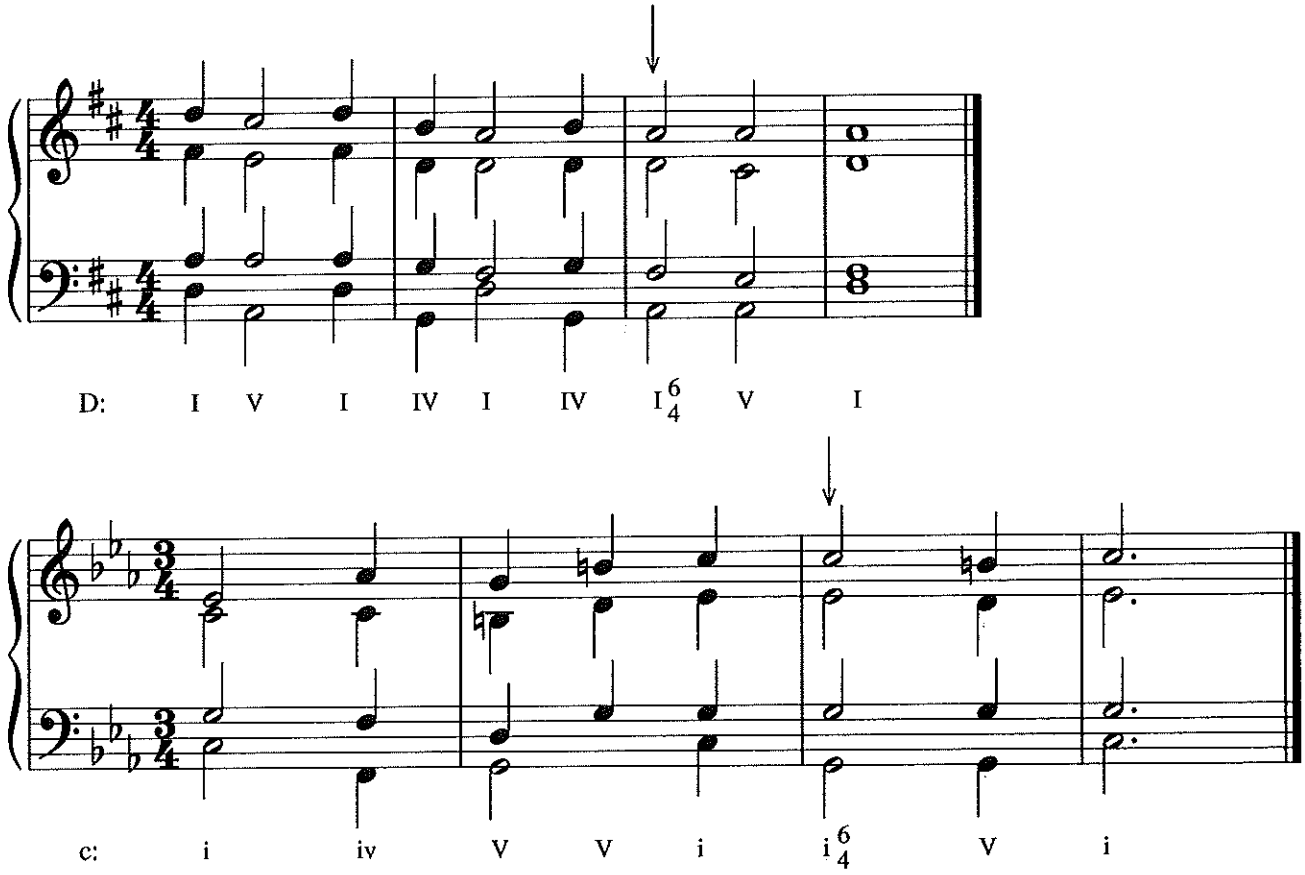
PRIMARY TRIADS IN SECOND INVERSION

Primary triads in second inversion are used far less frequently than primary triads in root position and first inversion, because in second inversion the interval of the fourth occurs between the bass and an upper voice. This interval, although technically a consonance, is considered to be sensitive in this inversion, and thus, complete freedom in the use of the chord is prohibited.

The $\frac{6}{4}$ chord has three common uses.

The Cadential $\frac{6}{4}$ Chord

As the title suggests, the cadential $\frac{6}{4}$ chord occurs at cadence points. Study the following examples:



Note:

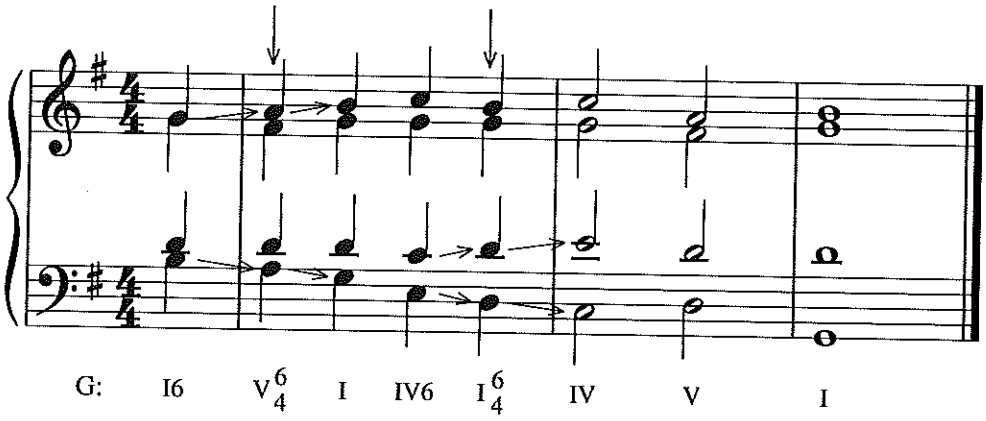
a. The bass is doubled.

b. The chord occurs in a metrically strong position.

c. The chord's root and third resolve by step.

The Passing $\frac{6}{4}$ Chord

The passing $\frac{6}{4}$ chord is used when the bass line moves stepwise. It most commonly separates a triad in root position from its first inversion, as the example shows:

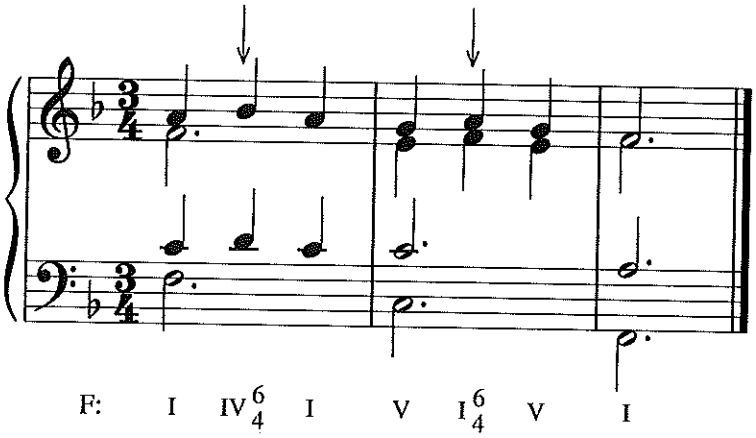


Note:

a. The bass is doubled.
b. The chord occurs both accented and unaccented, although the latter is more common.
c. One of the upper voices proceeds in contrary motion to the bass.

The Auxiliary $\frac{6}{4}$ Chord

The auxiliary $\frac{6}{4}$ chord is purely decorative and occurs when the bass is stationary, thus:

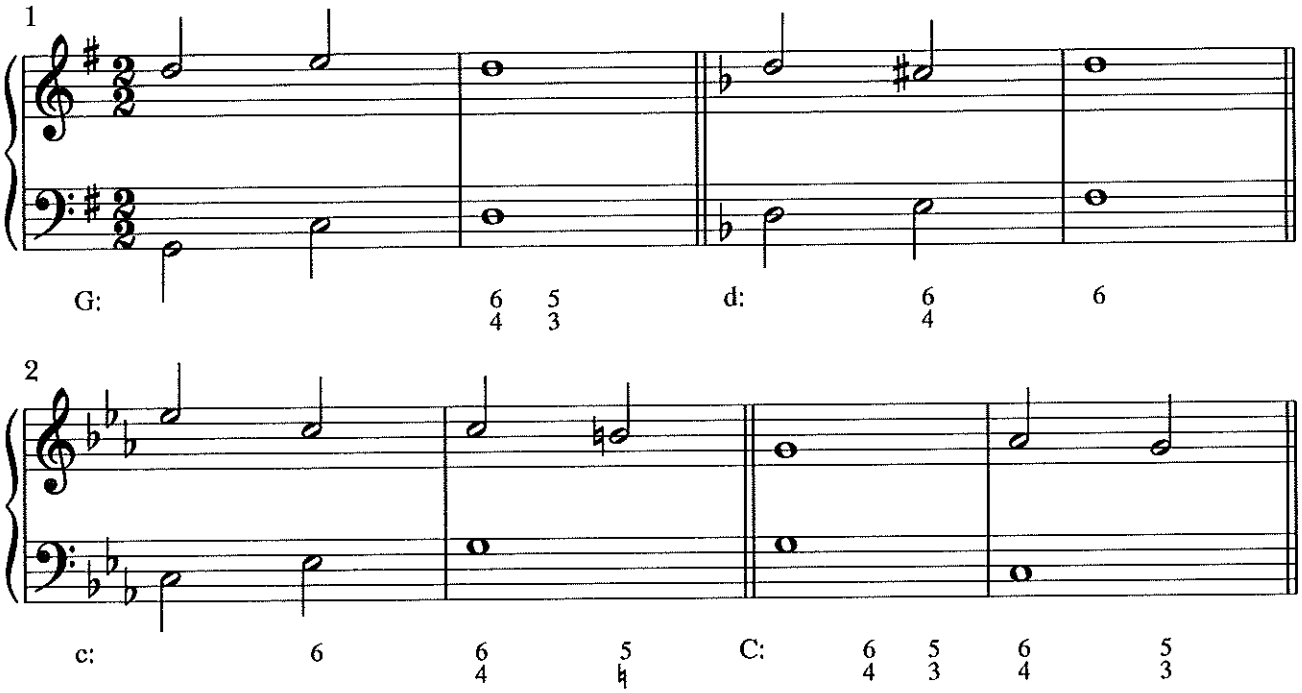


Note:

a. The bass is doubled.
b. The chord is unaccented.
c. The bass note remains stationary during the approach to, and departure from, the chord.

EXERCISES

A. Add parts for alto and tenor.



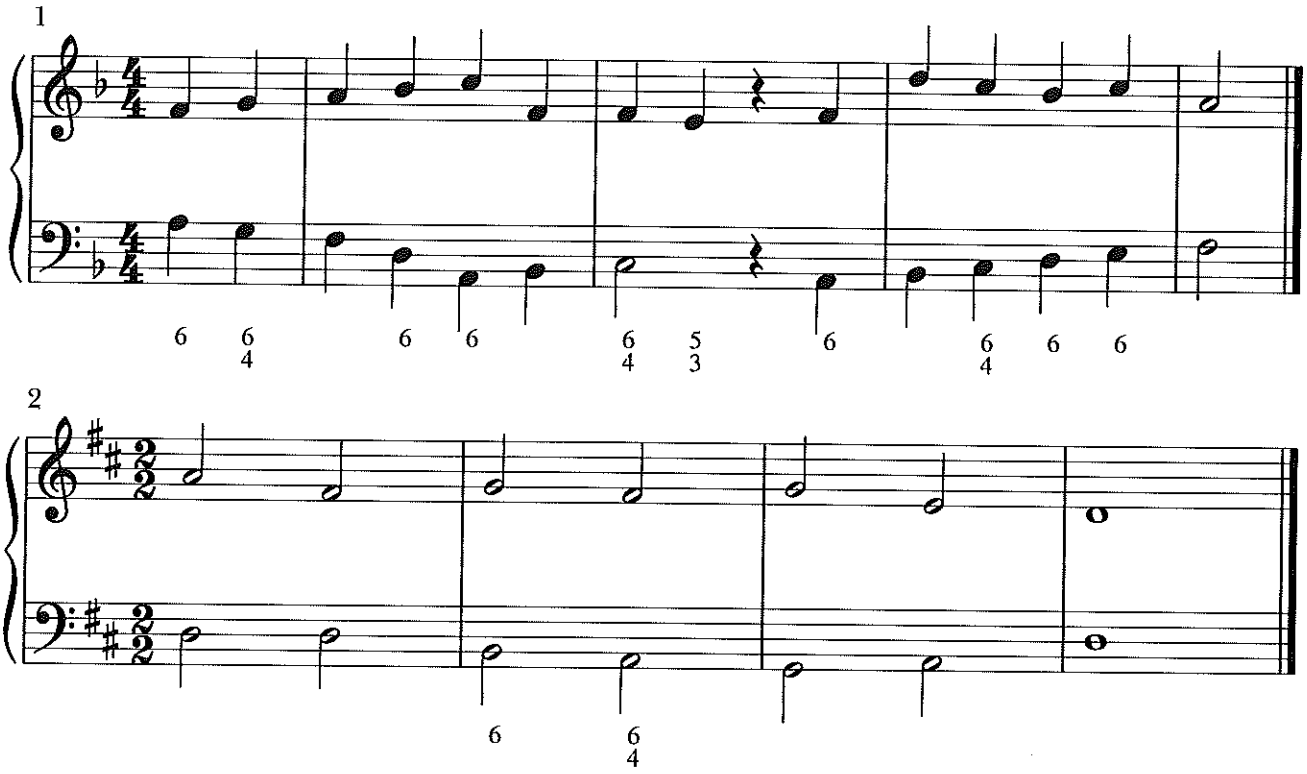
B. Add parts for soprano, alto, and tenor.



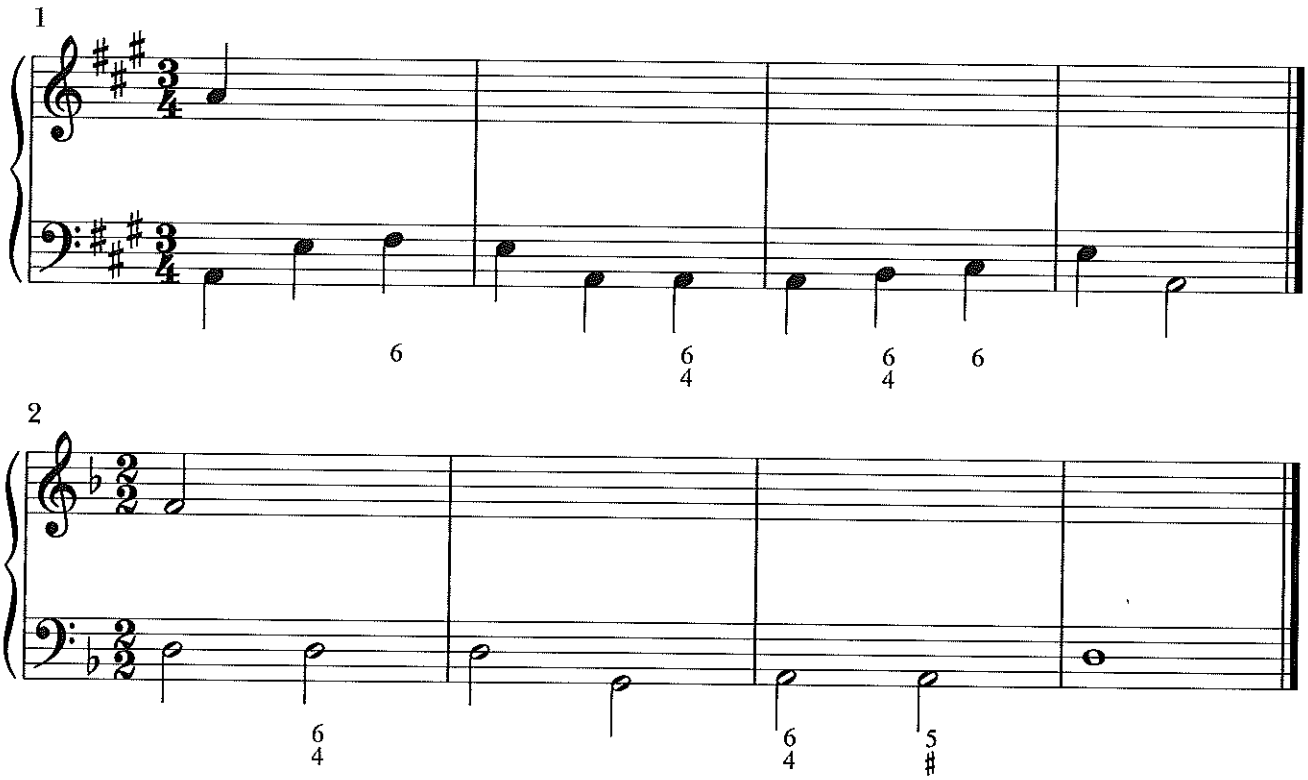
Note:

When the figured bass symbols indicate the use of a cadential $\frac{6}{4}$ chord, the figures $\frac{5}{3}$ (or $\frac{5}{\natural}$ or $\frac{5}{\sharp}$) follow them. These figures are included to show the linear motion of two of the upper parts.

C. Add parts for alto and tenor.



D. Add parts for soprano, alto, and tenor.



E. Add parts for alto, tenor, and bass. (Use first and second inversions.)

1



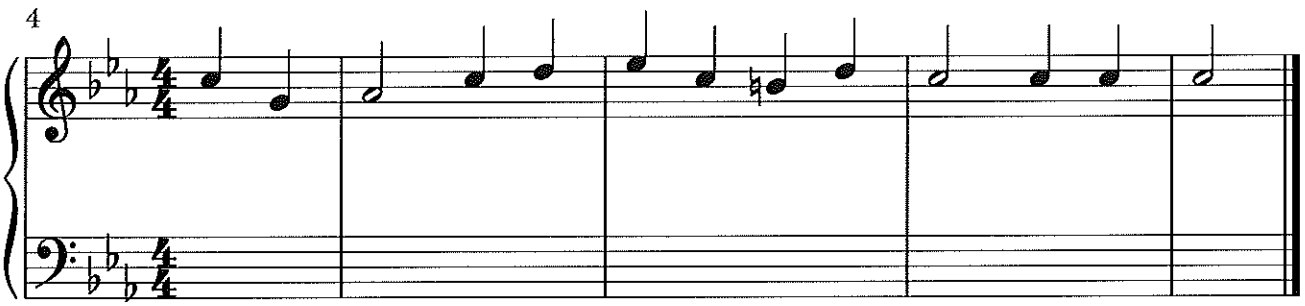
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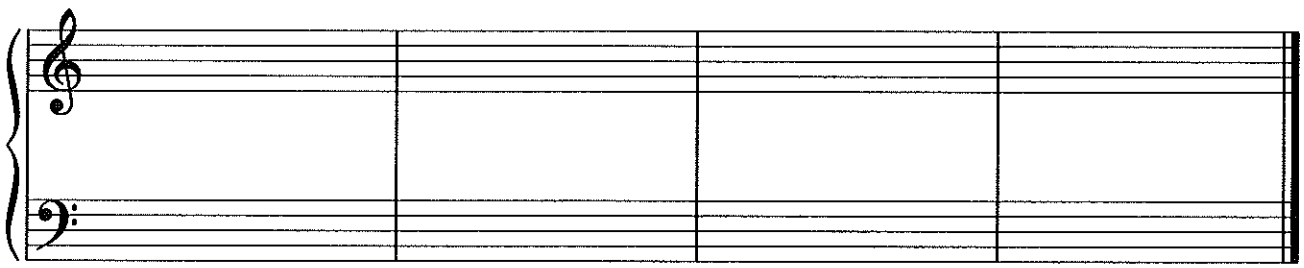
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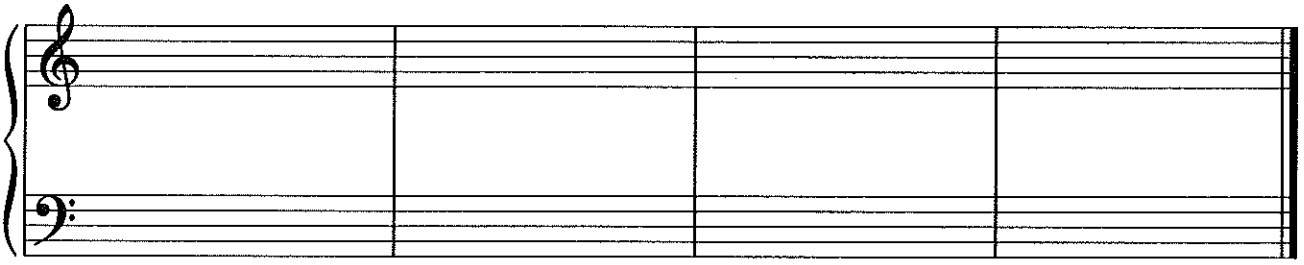
F. Using the given progressions, invent four-measure phrases for soprano, alto, tenor, and bass.

1. E♭: $\frac{3}{4}$ I–V–I–IV–I–IV–I $\frac{6}{4}$–V–I
2. a: $\frac{6}{8}$ i–V$\frac{6}{4}$–i6–iv–i$\frac{6}{4}$–iv6–iv–V–iv$\frac{6}{4}$–i
3. E: $\frac{3}{4}$ I–IV6–V–I–IV$\frac{6}{4}$–I–I6–V–I
4. g: $\frac{4}{4}$ i–V$\frac{6}{4}$–i6–iv–iv6–i$\frac{6}{4}$–V–i

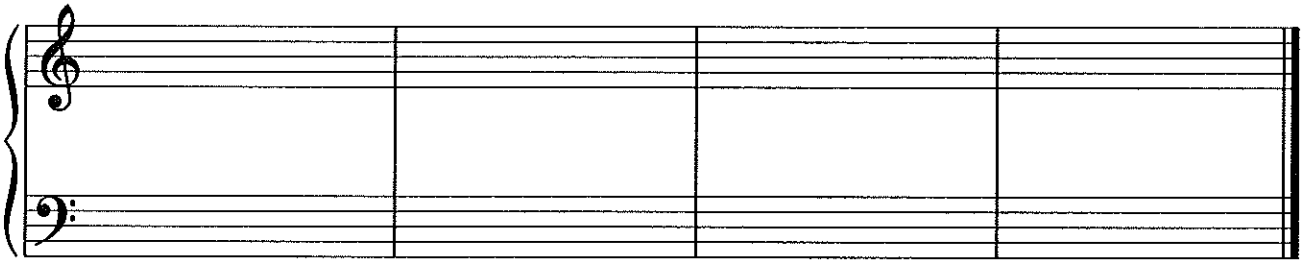
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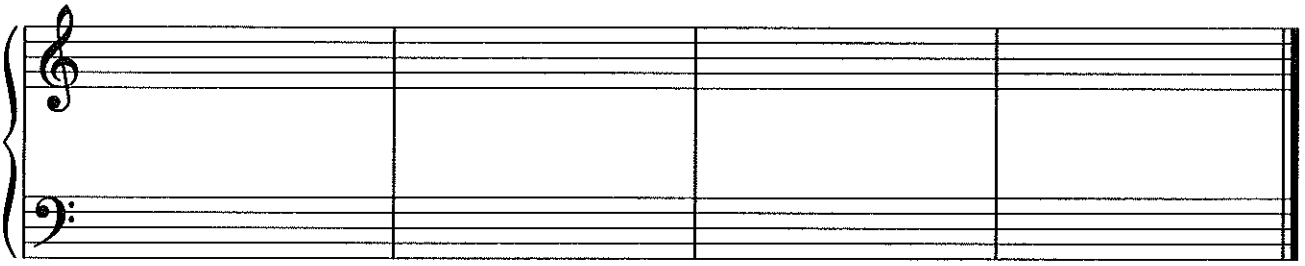
2



3



4



G. Make a harmonic analysis of the following phrases. Indicate all inversions with the appropriate symbols.

1



2



10 Secondary Triads

The *supertonic, mediant, submediant, subtonic,* and *leading tone triads* are the secondary triads.

Unlike the primary triads, which are either major or minor, the secondary triads include major, minor, augmented, and diminished triads, depending on the scale degree and scale type. Refer to page 67 for a listing of triad qualities in major and minor keys.

The table below shows the usage and principal doublings of the secondary triads.

	TRIAD	USAGE			DOUBLING
		ROOT POS.	FIRST INV.	SECOND INV.	
MAJOR KEYS	ii	yes	yes	no	root
	iii	yes	yes	no	root
	vi	yes	yes	no	root
	vii°	no	yes	no	third
MINOR KEYS	ii	no	rare	no	third
	III	yes	yes	no	root
	VI	yes	yes	no	root
	VII	yes	yes	no	root
	ii°	no	yes	no	third
	vi°	no	yes	no	third
	vii°	no	yes	no	third
	III+	no	yes	no	third

Note:

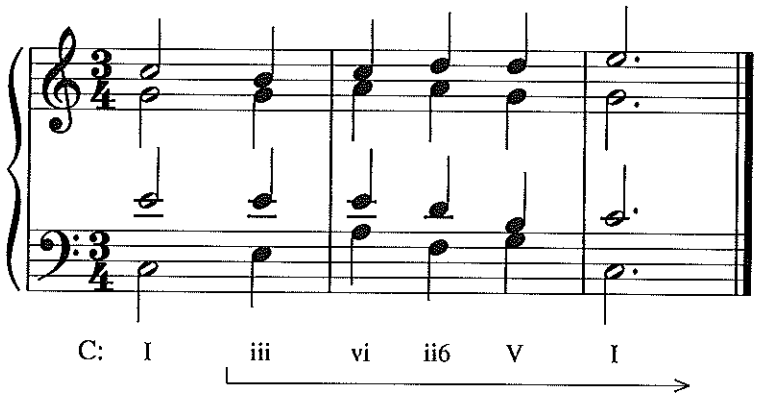
a. Secondary triads are not used in second inversion.
b. In minor keys the minor form of the supertonic triad is not used in root position, and is rarely used in first inversion.
c. Where the root is the principally doubled pitch (i.e., in major and minor triads), doubling procedures are the same as those for the primary triads.
d. The diminished and augmented triads are used in first inversion only.
e. Neither the root nor the fifth is doubled in the diminished or the augmented triad. In both cases the interval between the two pitches is unstable (dissonant), and thus the pitches are sensitive. In particular, the fifth of each triad has a strong tendency toward resolution—*downward* in the diminished triad, and *upward* in the augmented triad.

The addition of the secondary triads to the primary triads gives the student the potential to use a different chord on every scale degree. To do so indiscriminately, however, would certainly produce unsatisfactory music, and yet it is often difficult to exercise discrimination when such a wide variety of resources exists. The following observations will help the student in this respect.

CHORD RELATIONSHIPS

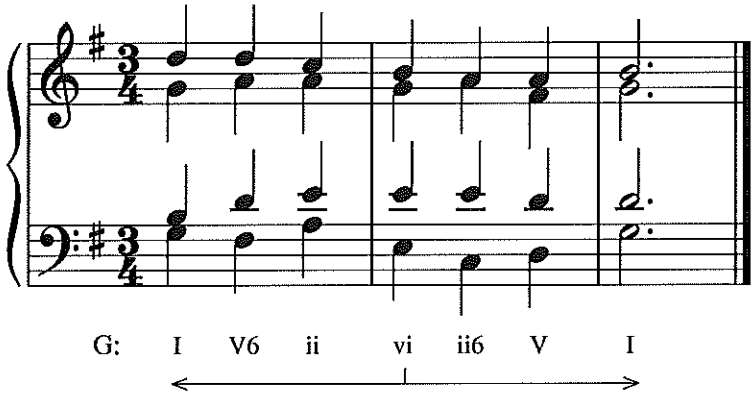
Harmonic movement may be considered strong, neutral, or weak, depending upon (1) *root* movement and (2) *bass* movement. For a given harmonic progression, movements in root position (assuming diminished and augmented triads are not employed) tend to be stronger than those that involve inversions. Since, however, inversions create melodic interest in the bass, a successful harmonization is the result of a proper balance between root and bass movement.

1. The strongest harmonic movement occurs when roots are separated by ascending fourths or descending fifths. Particularly strong is a progression that makes use of consecutive chords in any part of the following chord sequence: I–IV–VII–III–VI–II–V–I.

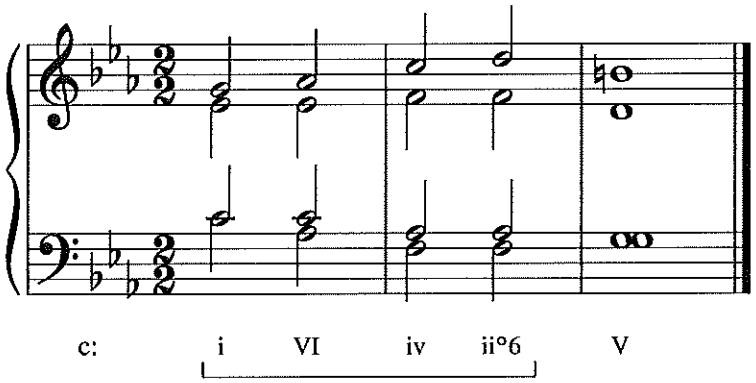


Movement in the opposite direction, descending fourths or ascending

fifths, also produces strong harmonic motion but tends to lead away from, rather than toward, cadence points. The term *retrogression* is often applied to this kind of motion. A sequence of retrogressive harmonies is commonly followed by movement in ascending fourths to terminate a phrase or period.

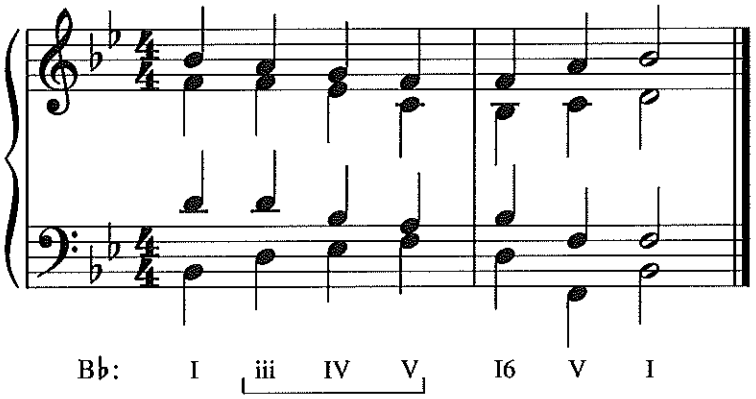


2. Chords a third apart progress weakly in a harmonic sense because there are always two common tones. They may be used effectively, however, in maintaining a smooth motion in preparation for a cadence.



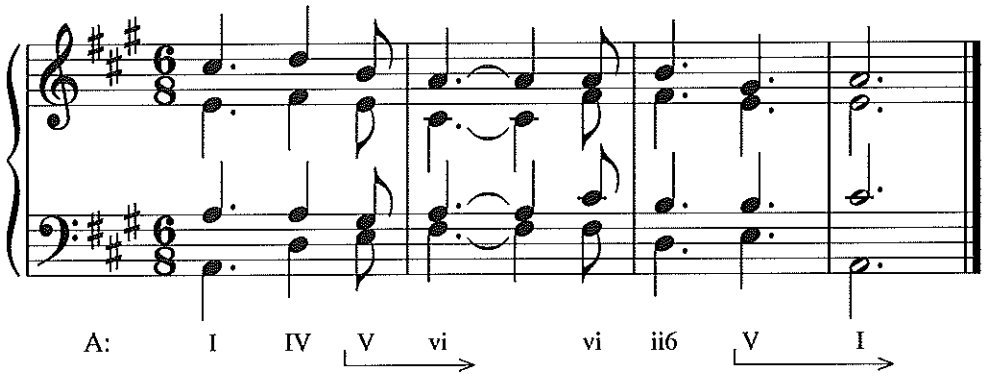
Harmonic strength (or lack thereof) differs little between chords in ascending or descending thirds.

3. Chords a second apart lack the harmonic drive of those a fourth or fifth apart, but because no common tones exist, they give a greater impression of harmonic motion than do those a third apart.



Harmonic movement in ascending seconds is generally considered to be stronger than that in descending seconds.

4. The deceptive cadence, V–vi or V–VI (when the dominant is followed by the submediant in a cadential situation), is said to be deceptive because the ear, which expects a tonic chord after the dominant, is deceived by the submediant chord. This cadence is used to prolong musical activity, since the composer must reach a cadence terminating with the tonic to conclude the phrase satisfactorily.



5. The leading tone triad, followed by the tonic triad, vii°6–i, is said to have a "dominant function," and so the progression below may be considered a weak form of an authentic cadence.

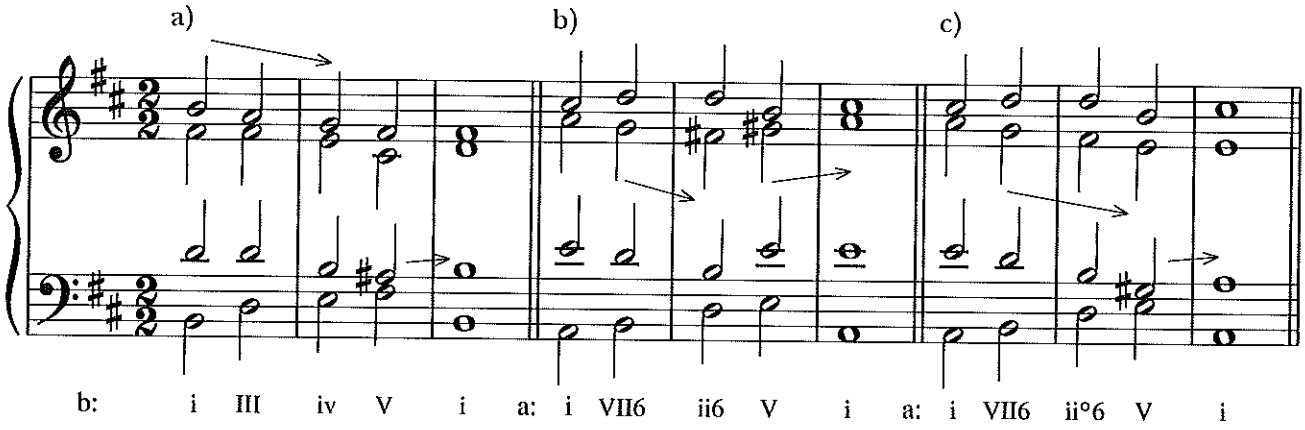


THE SIXTH AND SEVENTH SCALE DEGREES IN MINOR KEYS

In Chapter 2, the concept of a minor scale consisting of a minor tetrachord and a chromatic hexachord was introduced. Certain problems in part-writing are more easily solved by using this scale as a point of reference than by trying to determine which of the three forms of the traditional minor scale to employ. The fact is that under a wide variety of circumstances, a harmonization will contain elements of more than one of the forms of the minor scale—in other words, it will be based upon the synthetic minor scale.

Generally speaking, when the lower note of either of the chromatic scale degrees (sixth and seventh) is employed, if the voice which contains that note moves by step, it will do so *downward.* Conversely, when the higher note is employed, stepwise motion will be *upward.*

Study the following examples carefully:



Note:

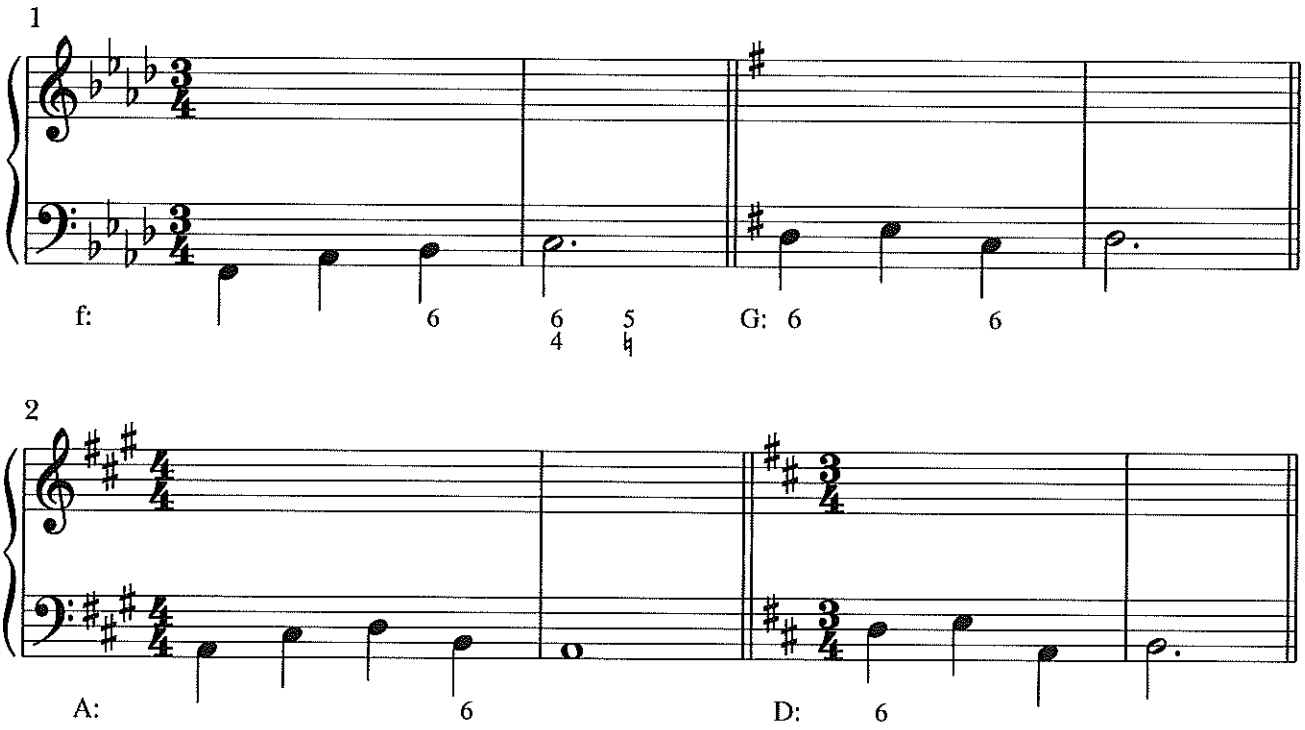
Harmonically, c) differs from b) only in the substitution of ii°6 for ii6. As a result of the altered pitch, however, the subsequent voice-leading in the middle voices is quite different.

EXERCISES

A. Add parts for alto and tenor.



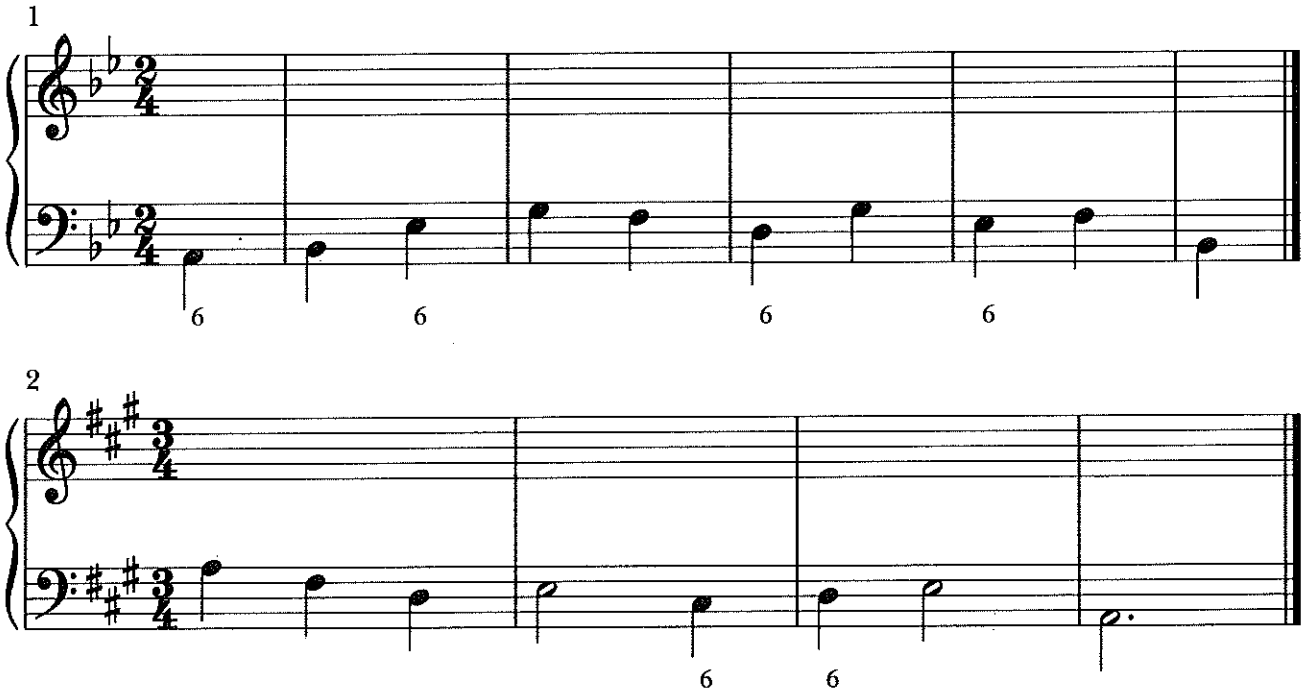
B. Add parts for soprano, alto, and tenor.



C. Add parts for alto and tenor.



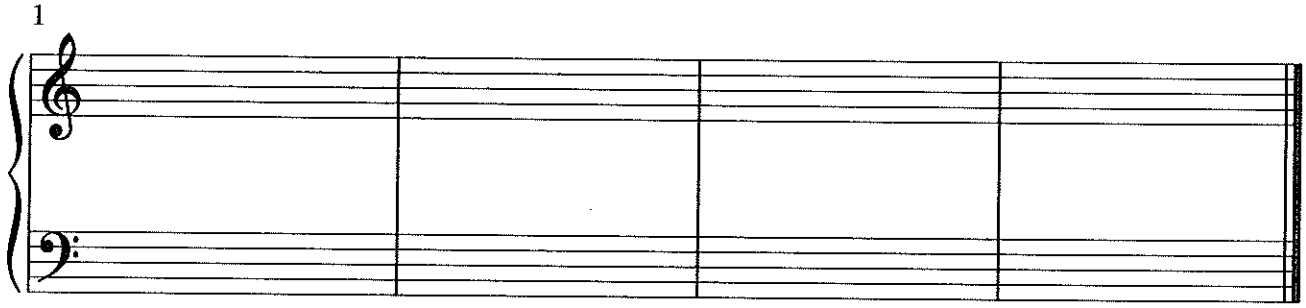
D. Add parts for soprano, alto, and tenor.



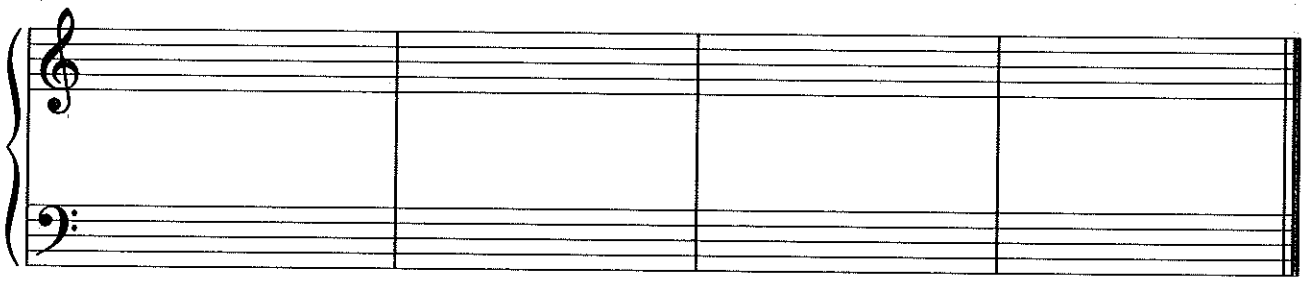
E. Using the given progressions, invent four-measure phrases for soprano, alto, tenor, and bass.

1. D: $\frac{4}{4}$ I–vii°6–I–I6–IV–iii–vi–ii6–V–I
2. b: $\frac{4}{4}$ i–III–iv–V–V6–VI6–ii°6–V
3. B♭: $\frac{9}{8}$ V6–I–V^6_4–I6–IV–vii°6–iii–vi6–ii–V–vi
4. a: $\frac{3}{4}$ i–V6–i–VI–ii°6–iv–V–VI

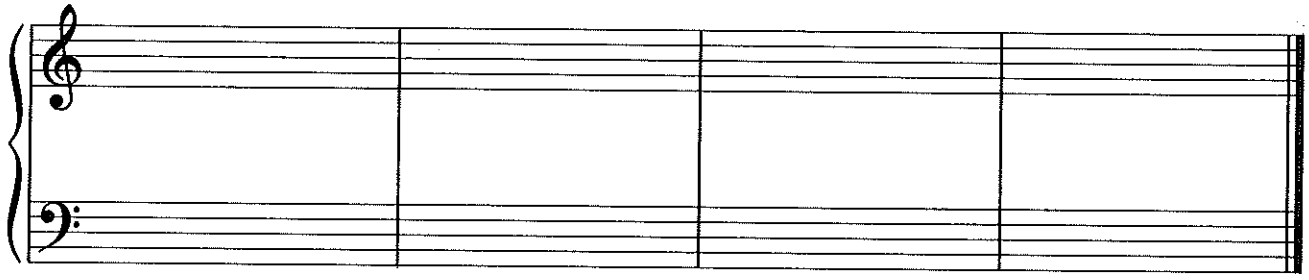
1



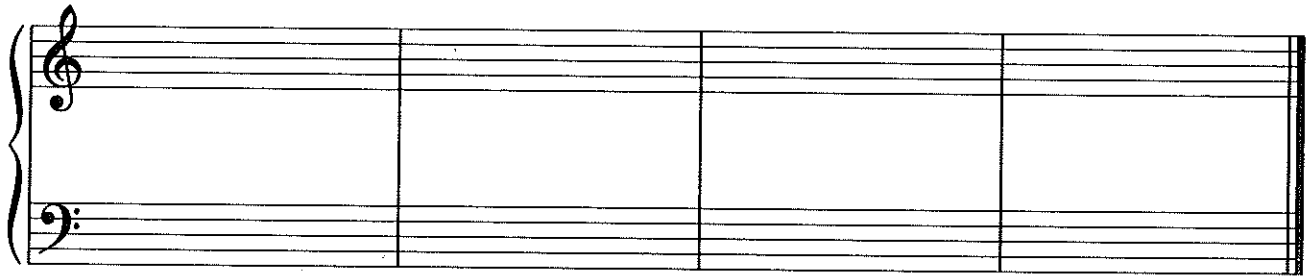
2



3



4



F. Make a complete harmonic analysis of the following phrases.

1



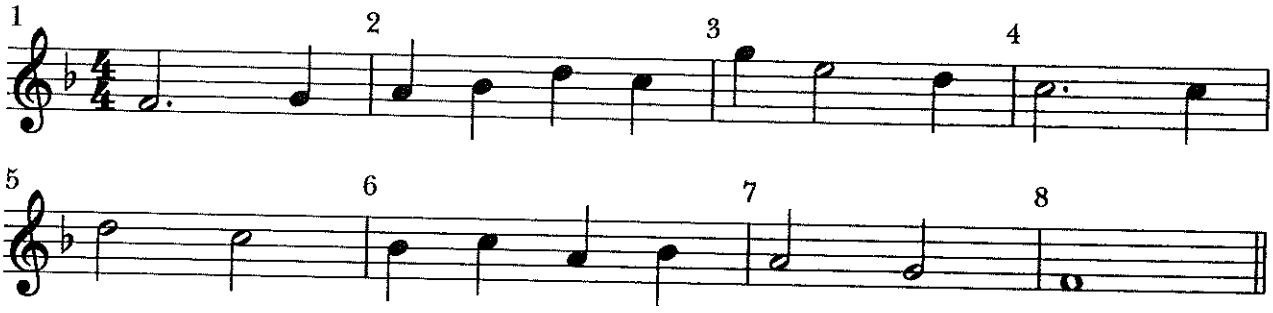
2



11 The Harmonization of Melodies I

In the last chapter it was emphasized that the introduction of secondary triads created the potential for a wide variety of harmonic progressions; as a result, musical problems may be encountered. These problems are brought into particularly sharp focus when students are confronted with a melody to harmonize; they face what seems to be a bewildering array of possibilities, and, consequently, they have difficulty in deciding where to begin.

Using the melody below as a model, the author describes a method that can be effectively applied to the harmonization of melodies in general. Students should follow the steps carefully and should routinely put the steps into practice when they begin their own harmonizations.



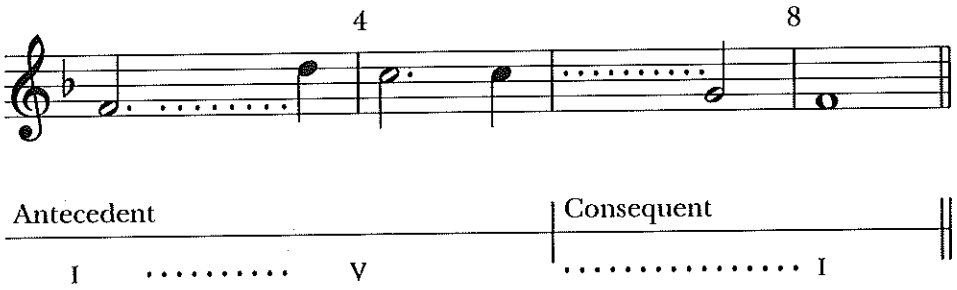
1. Determination of key.
   The possibilities are F major and d minor. The key is F major for at least two reasons:
   a. The melody begins and ends on f (although this fact does not automatically preclude d minor).
   b. C♮, the dominant in F, appears frequently. If the melody were in d, one would expect c♯ (the leading tone) rather than C♮ (the subtonic) to predominate.

2. Analysis of phrase structure, leading to identification of cadences. Cadences, which are points of relative repose, serve to terminate phrases (see Chapter 8). Clearly, the melody comes to rest in two places—on the dotted half note c^2 in measure 4, and on the final f^1. The arrival at c^2 strongly suggests a half cadence (conclusion on V); and one would expect an authentic cadence (V–I) to conclude on the final tonic note.

   This melody, in fact, exhibits two very common characteristics with respect to phrase structure:

   a. Each phrase is four measures long.

   b. The first phrase (antecedent) ends with a half cadence; the second phrase (consequent) ends with an authentic cadence.



Keep a mental picture of this diagram: it will be encountered frequently.

3. Working backward from cadences, assignation of chords whose relationships represent strong harmonic motion (descending fifths, ascending seconds)—this phase ends when such motion is no longer possible.

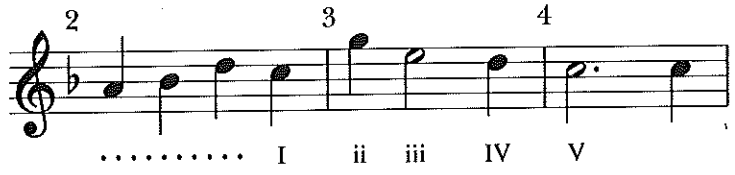
   Since the harmonic goal for each phrase is now known, it makes sense to try to reach the final cadence as clearly as possible.

   Phrase One: Strong half cadences: a) ii–V, b) IV–V] both fit the melody

   a. To continue to achieve strong harmonic motion, ii should be preceded by vi (descending fifths) or by I (ascending seconds). The E♮ does not belong to either harmony:

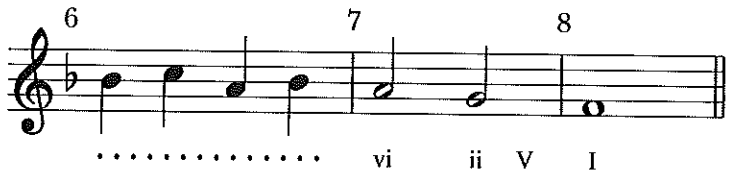


   b. To continue to achieve strong harmonic motion, IV should be preceded by I (descending fifths), or by iii (ascending seconds). The latter fits the melody. In fact, ascending seconds may be projected back to c^2 in the second measure before the pattern breaks down. Thus, the solution to the end of phrase one is:



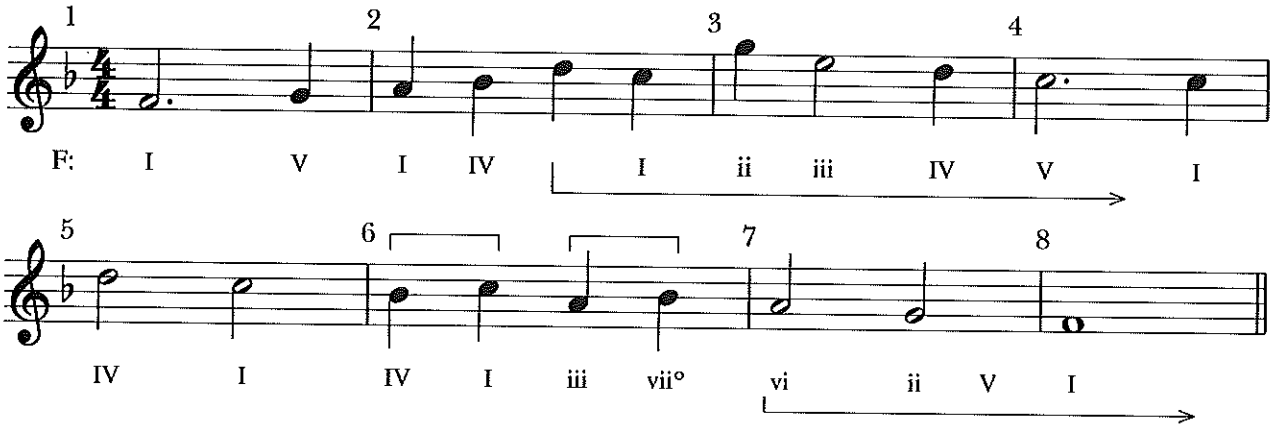
Phrase Two: Strong authentic cadence: V–I

To continue to achieve strong harmonic motion, V should be preceded by ii (descending fifths), or by IV (ascending seconds). In this case, descending fifths may be effectively projected back to the beginning of the seventh measure, and thus, the solution is:



4. Assignation of chords from the beginning of phrases to link with chords already assigned—strength of harmonic movement is less critical in this phase than establishment of tonality.

   The harmonic goals for the phrases cannot be effectively realized unless the tonality (key) is clear in the first place. Thus, the harmonization of the remaining pitches in the melody should reinforce the key, in this case F major. Primary triads serve this purpose very well:



Note:

a. In the beginning of each phrase, the rate of change of harmony is relatively slow compared with that in the end of each phrase. The rhythmic configuration that emerges from these changes of harmony is called the *harmonic rhythm.* Frequently, harmonic rhythm is faster at the end of phrases than at the beginning.

b. In measure 6, the first and second notes have been assigned a subdominant and tonic triad respectively; the third and fourth have been assigned a mediant and leading tone triad. This harmonization has been chosen to exploit the melodic pattern wherein the second pair of notes represents a transposition, by a step down, of the first pair. The repetition of a pattern at a different pitch level creates what is known as a *sequence.* Attention has been drawn to this melodic sequence by the harmonic sequence that accompanies it.

5. Construction of bass line with concentration on melodic interest, accomplished by appropriate use of inversions.



Several different bass lines could obviously be written. The previously established progressions to the cadences suggested largely conjunct (by step) motion in the first phrase and rather more disjunct (by leap) motion in the second. *Contrary motion* between the two voices is used throughout. Although such motion is not always manageable, it is desirable because it gives the outer voices melodic independence and usually facilitates the part-writing in the inner voices.

6. Addition of inner voices to complete harmonization.



Note:

In the complete harmonization, an effort has been made to give the inner voices, as well as the outer, an interesting melodic line. Although it is not always possible to achieve such a line (e.g., the alto voice in the second phrase), linear considerations should not be ignored. The combination of a logical harmonic progression and fluid linear activity typically leads to an effective musical setting.

EXERCISES

A. Harmonize the following one-phrase melodies.

1



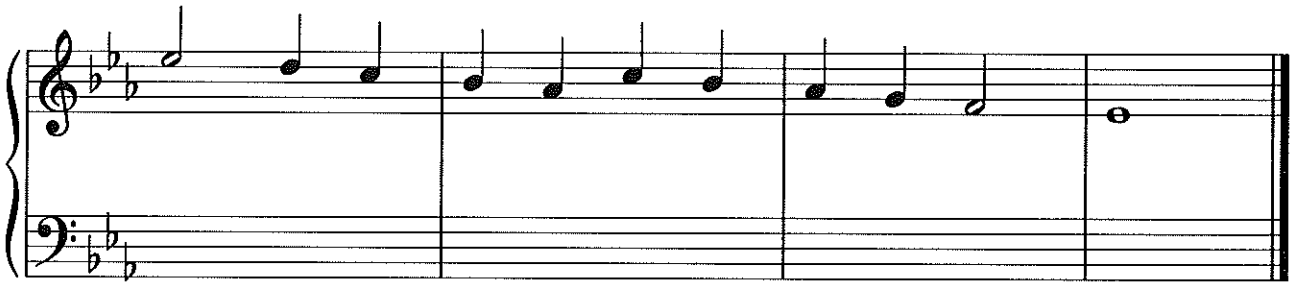
2



B. Harmonize the following two-phrase melodies.

1





2





3





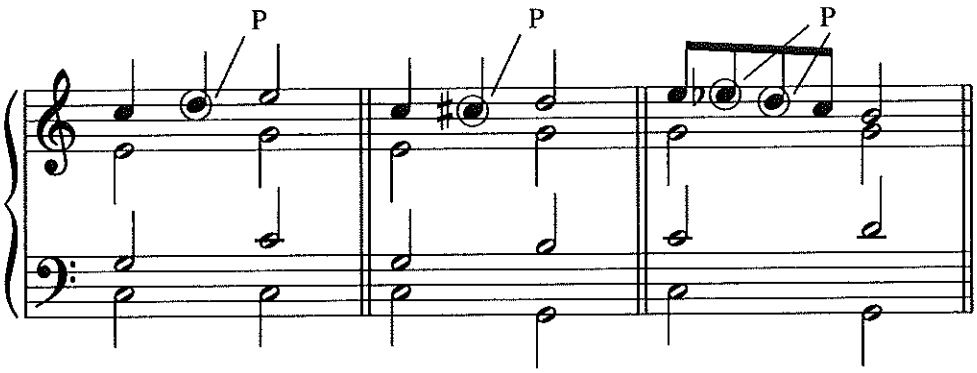
Nonchord Tones

12 Passing Tones, Neighboring Tones, Changing Tones, Appoggiaturas, Escape Tones, Anticipations

A nonchord tone is a tone that does not belong to the prevailing harmony. It may be diatonic (belonging to the key) or chromatic (not belonging to the key).

THE PASSING TONE

The passing tone (P) is a nonchord tone that is approached by step and resolved by step in the same direction. A *step* is typically an augmented unison, or a major or minor second.



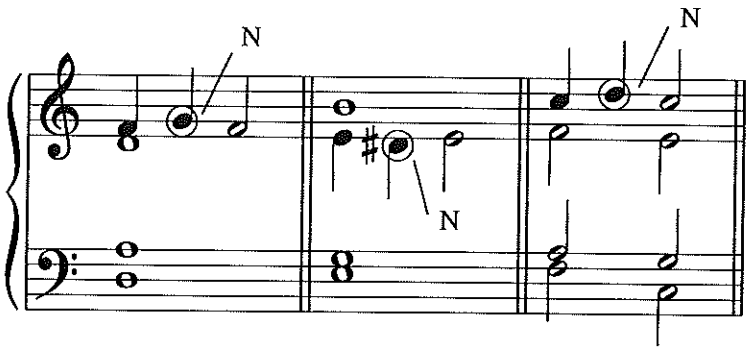
Note:

a. The C♯ and the E♭ are *chromatic* passing tones because they are not diatonic notes in the key of C.

b. The last example contains two adjacent passing tones. Such a procedure is common.

THE NEIGHBORING TONE

The neighboring tone (N) is a nonchord tone that is approached by step and resolved by step in the opposite direction.



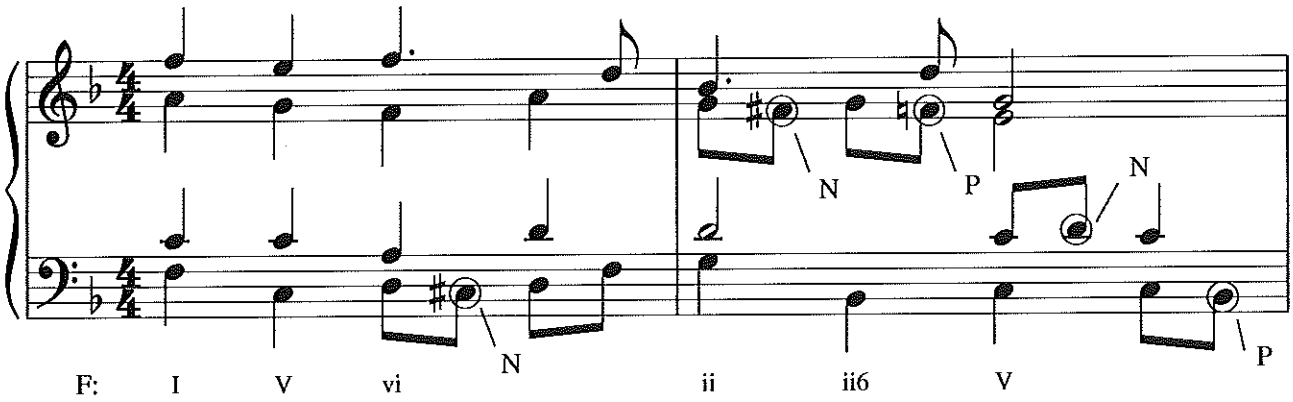
Note:

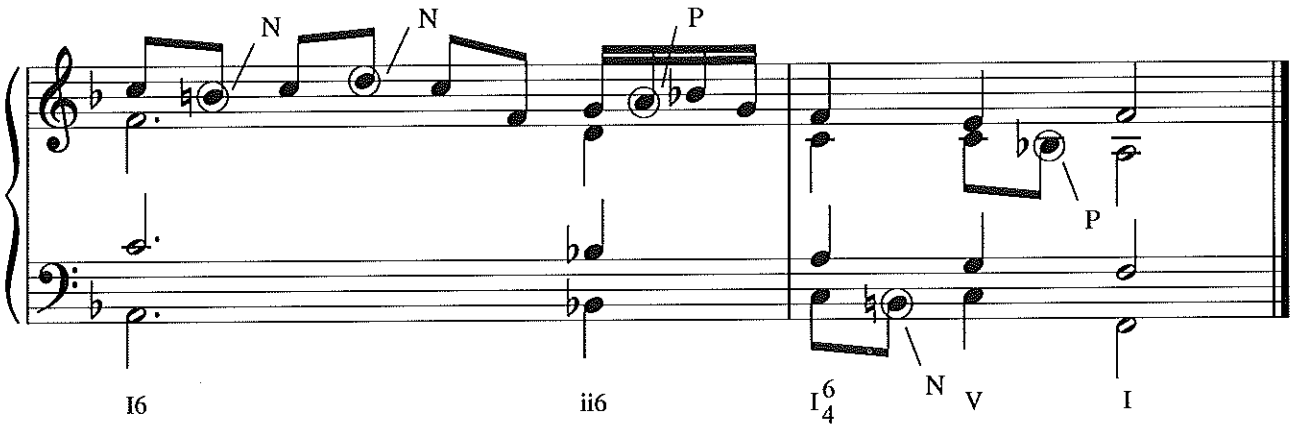
a. Upper neighboring tones are shown in the first and third examples; a lower neighboring tone is shown in the second.

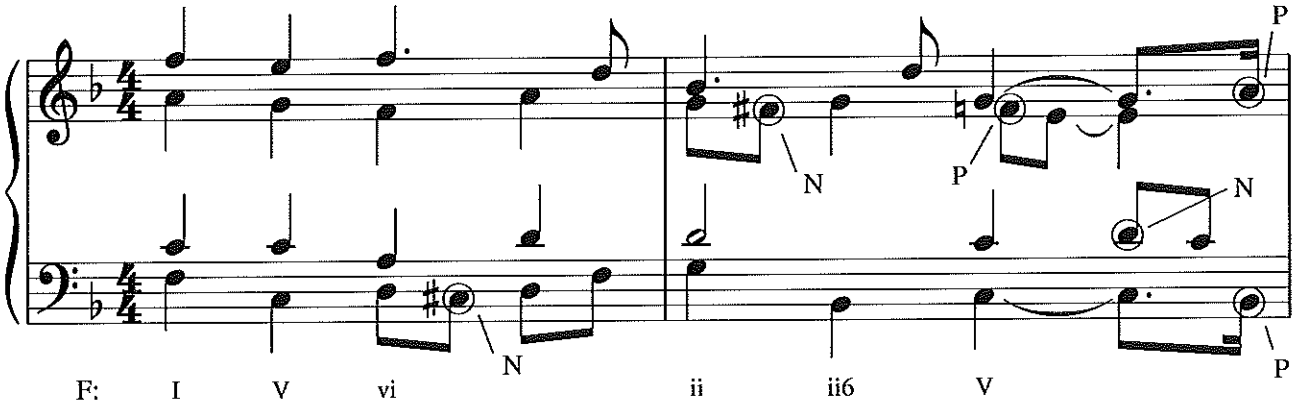
b. The D♯ represents a chromatic neighboring tone.

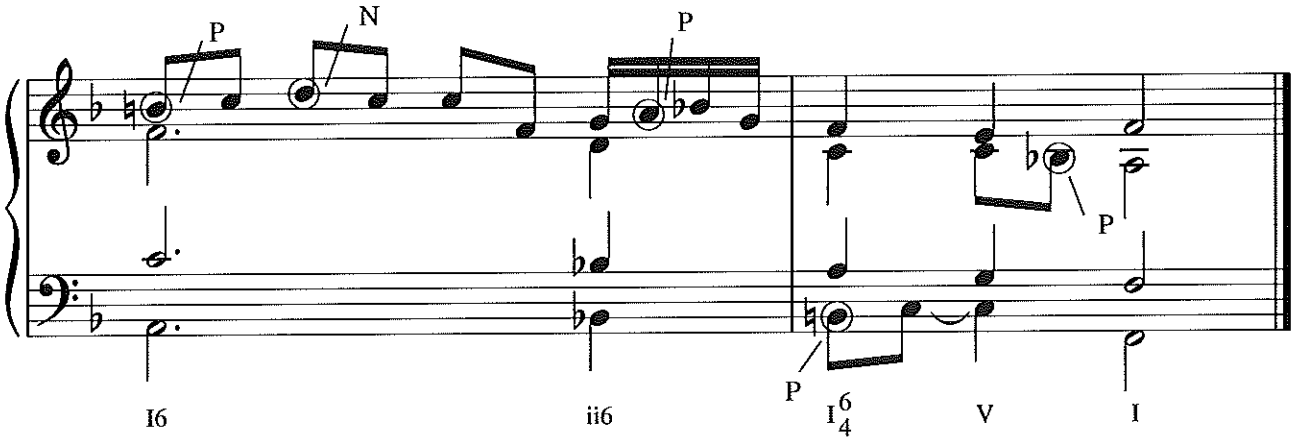
Passing tones and neighboring tones are usually unaccented, but they do not have to be. They occur in all voices.

The following examples show them both unaccented and accented.


N
P
N
N
P
F: I V vi ii ii6 V


N
N
P
P
N
I6 ii6 I6/4 V I

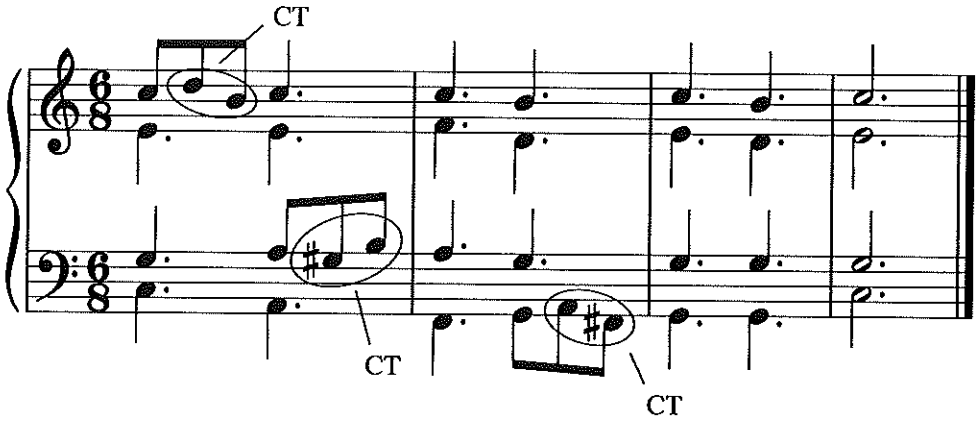

P
N
P
N
N
P
F: I V vi ii ii6 V


P
N
P
P
P
P
I6 ii6 I6/4 V I

CHANGING TONES

Changing tones (CT) are nonchord tones that occur in pairs and separate a chord tone from its repetition. The first is approached by step; the second is approached by leap and resolved by step. In each movement there is a change in direction.

Changing tones may be accented or unaccented and occur in all voices. They are more frequently used, however, in instrumental than vocal music.

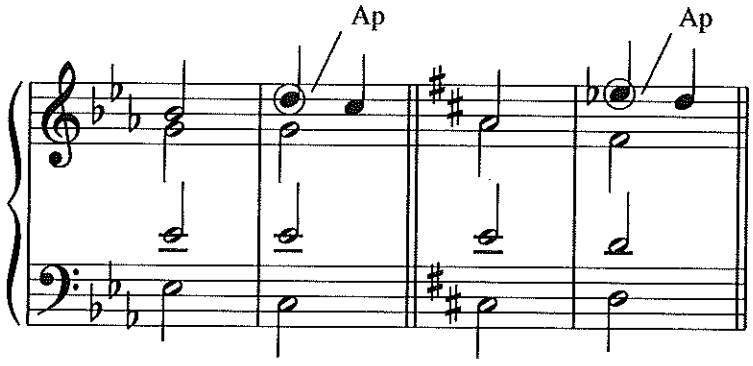


Note:

a. The first changing tone may be above or below the chord tone that precedes it.
b. A changing tone may be a diatonic or chromatic pitch.
c. A pair of changing tones may be regarded as the combination of an upper and lower neighboring tone.

THE APPOGGIATURA

The appoggiatura (Ap) is a nonchord tone that is approached by leap and resolved by step, usually in the opposite direction. It is typically accented and most frequently occurs in the soprano (or top) voice.

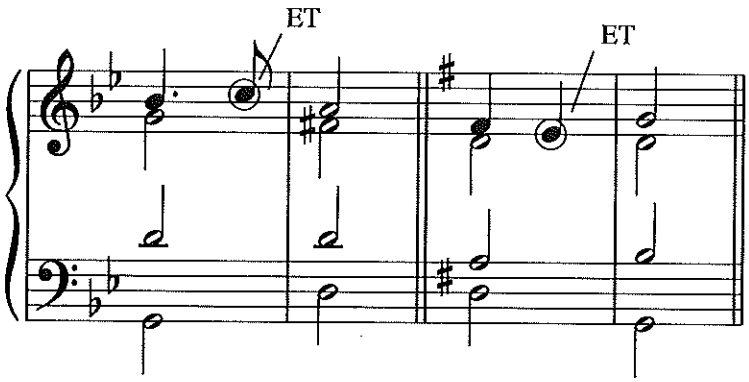


Note:

As the second example shows, the appoggiatura may be a chromatic pitch.

THE ESCAPE TONE

The escape tone (ET) is a nonchord tone that is approached by step and resolved by leap, usually in the opposite direction. It is typically unaccented and most frequently occurs in the soprano (or top) voice.

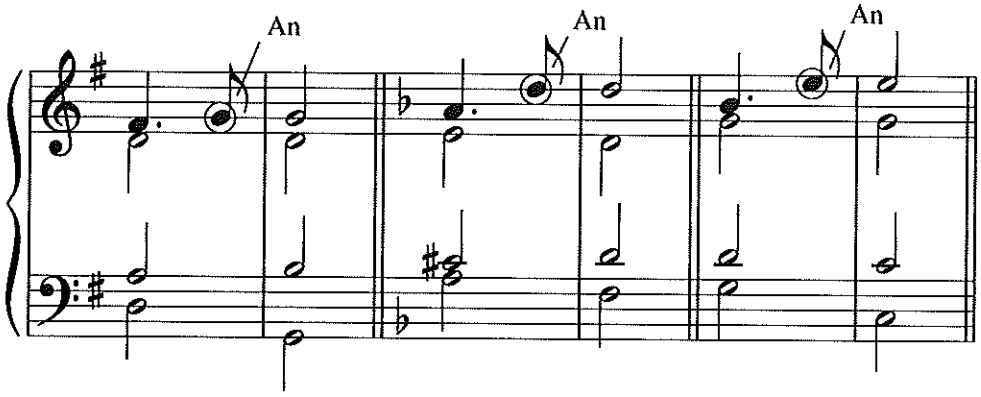


Note:

As the first example shows, the escape tone may be of shorter duration than the pitch that precedes it.

THE ANTICIPATION

The anticipation (An) is a nonchord tone that is approached by step or leap, and it becomes a chord tone with a subsequent change in harmony. It is unaccented and usually occurs in the soprano (or top) voice.



Note:

The anticipation is generally of shorter duration than the chord tone that precedes it.

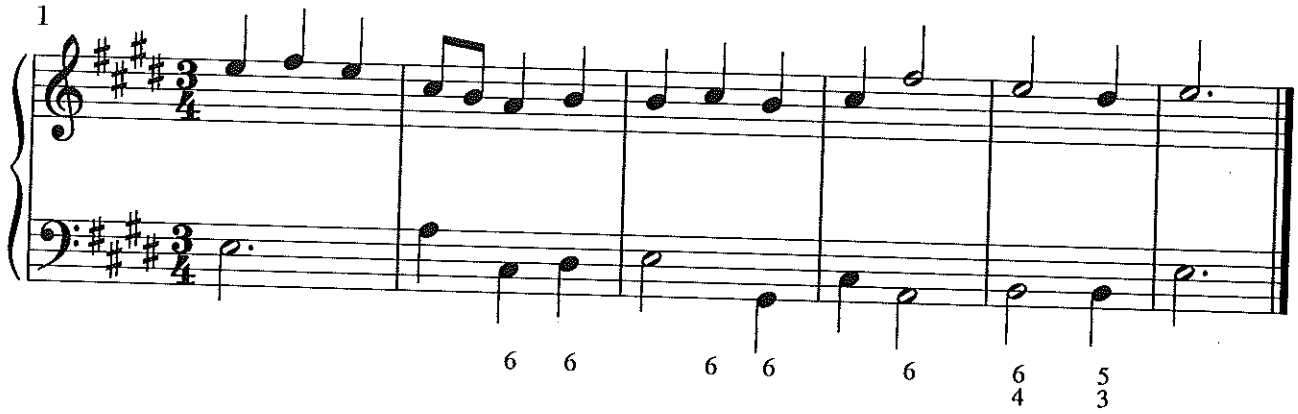
EXERCISES

A. Elaborate the following harmonizations by the addition of passing tones and neighboring tones. Label each one.

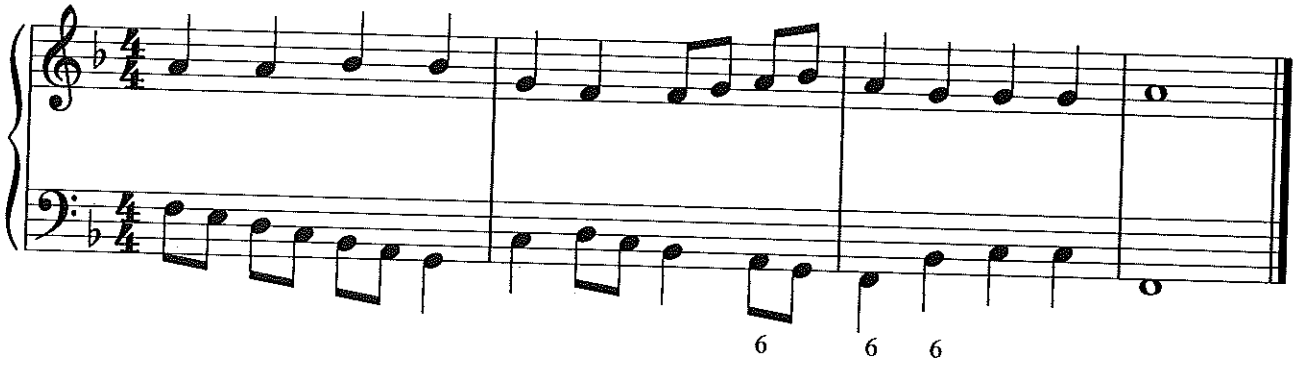


B. Add parts for alto and tenor, introducing passing tones and neighboring tones where appropriate.

1

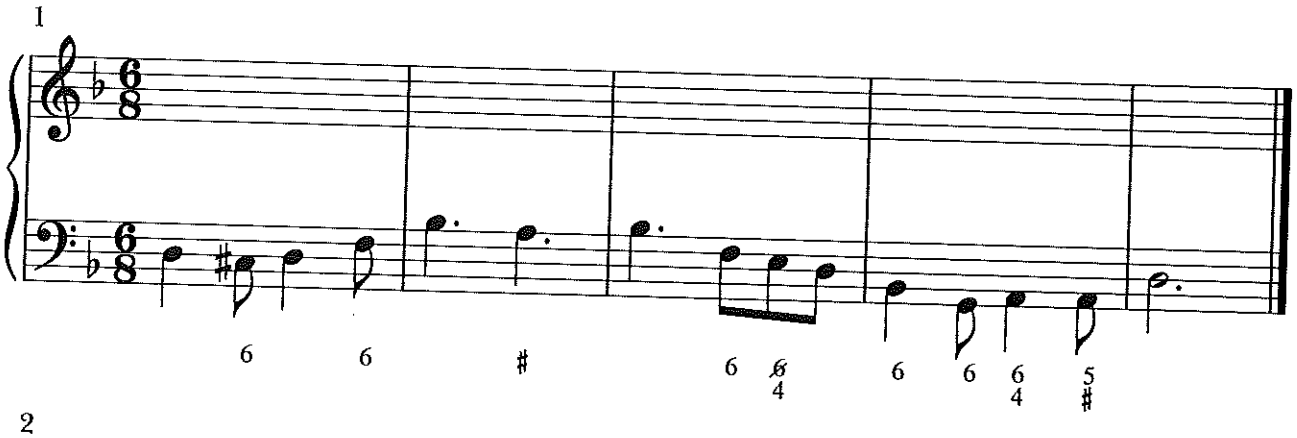


2



C. Add parts for soprano, alto, and tenor.

1



2



D. Harmonize the following melodies, introducing passing tones and neighboring tones where appropriate.

1



2



3





E. Add parts for alto and tenor. In the space provided, write an elaborated version introducing changing tones, appogiaturas, escape tones, and anticipations.




2
6
6
6
6
4

F. Make a complete harmonic analysis of the following phrases. Circle and label all nonchord tones.

1



2



3



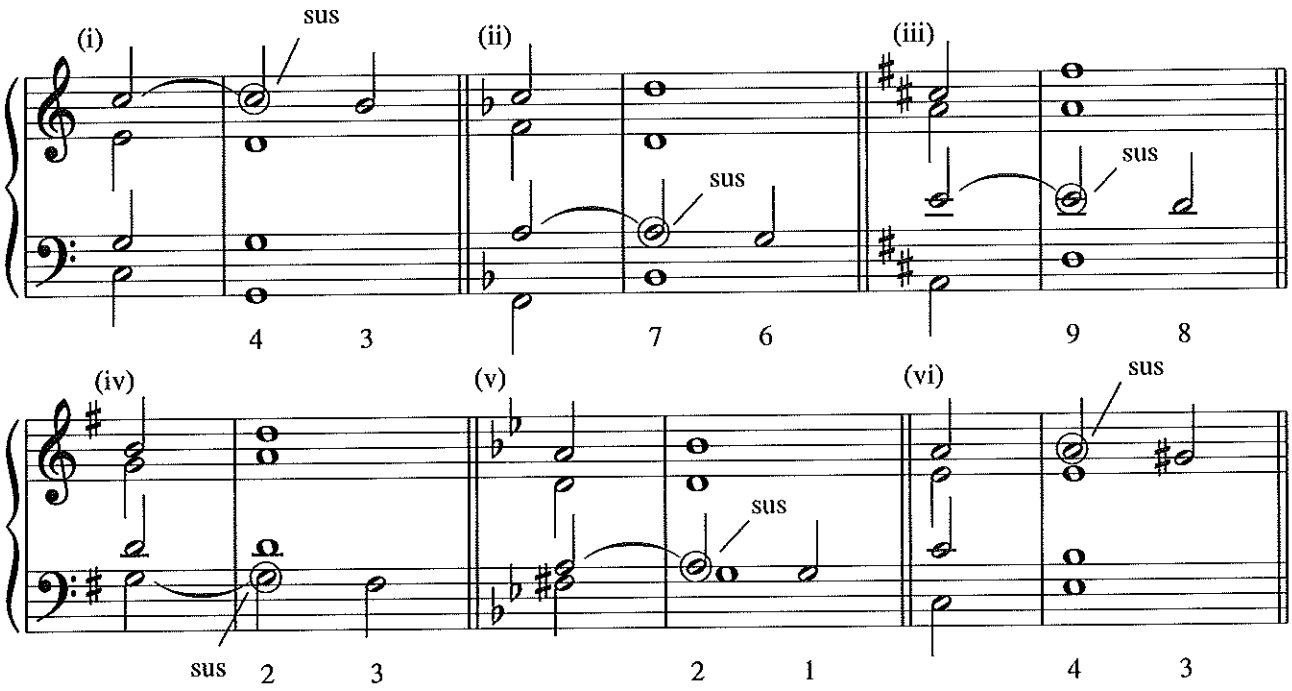
4



13 Nonchord Tones

Suspensions

The suspension is a nonchord tone that is the result of retaining a tone from a previous harmony and that resolves downward by step. It is accented.



Note:

a. A suspension is described by the intervals it and its resolution form with the bass. For example, (i) is said to be a 4–3 suspension, (iii), a 9–8 suspension, and so on.

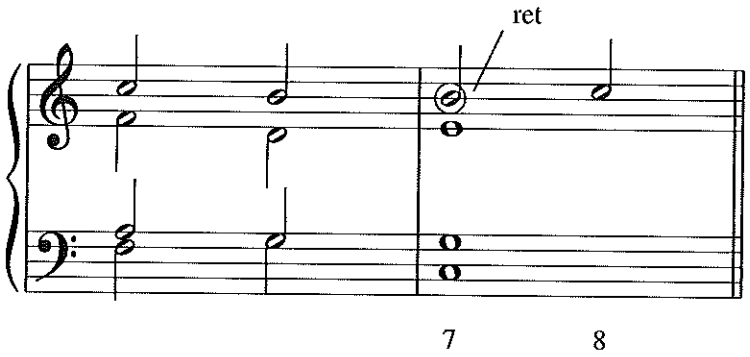
b. A suspension may occur in any voice. If it occurs in the bass, however, it is described by the interval that forms a dissonance with it, as shown in (iv).

c. Except in the cases of the 9–8 and 2–1 suspension, the pitch of resolution is not doubled.

d. The suspended pitch may be tied to the previous one or may be sounded again. If sounded again, it is known as an *articulated* suspension.

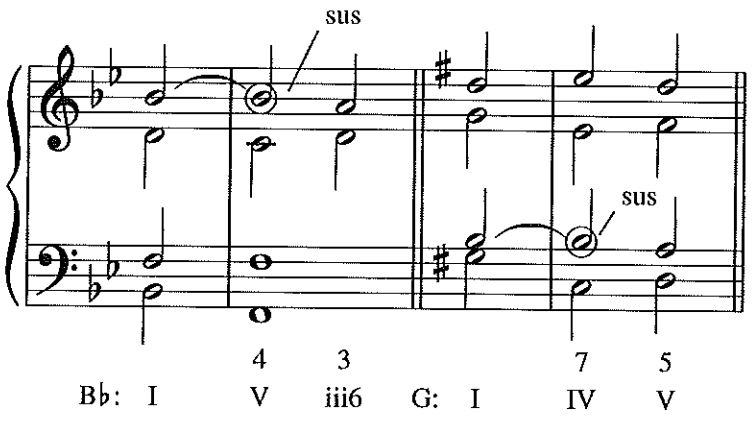
e. The suspension is in stronger position metrically than its resolution.

A suspension may be written to resolve upwards by step. Suspensions of this kind are usually called *retardations*.



When working with suspensions, the student should get used to thinking in terms of three events: preparation (consonance)–suspension (dissonance)–resolution (consonance).

Usually, only two harmonies are involved; the suspension delays the completion of the second one. It is possible (though rare), for the resolution to coincide with a change of harmony, thus:



Note:

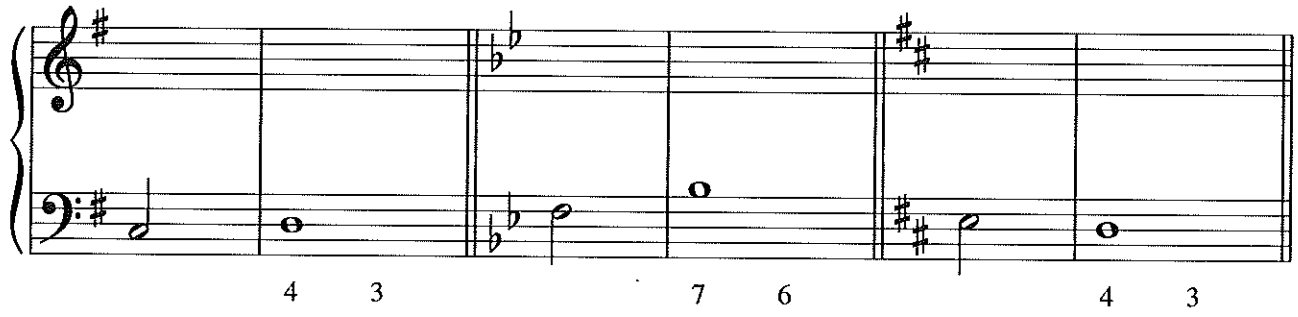
a. The change in harmony may or may not involve motion in the bass.

b. In the second example, the figured bass symbols (7–5) suggest a leap in the suspension's resolution. The unusual figures are produced by the ascending bass line; the suspended b resolves normally.

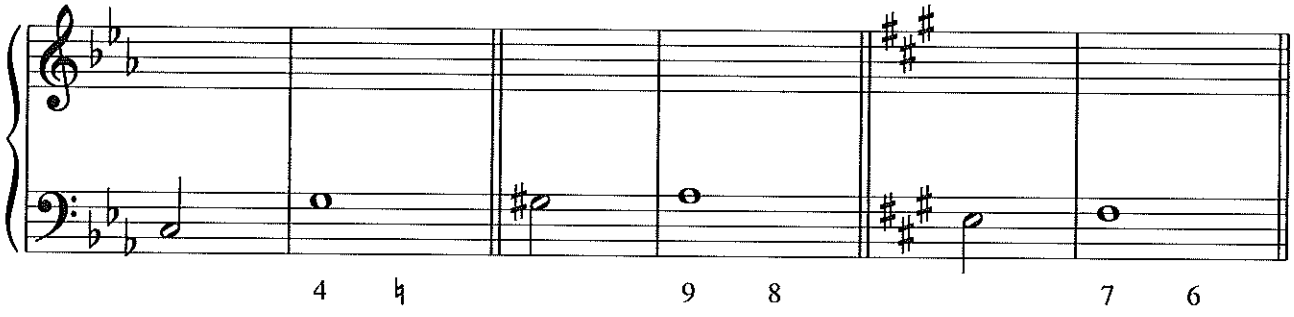
EXERCISES

A. Add to the given voice a second one so that the indicated suspensions are created. (Observe that the first interval in each example is consonant as a result of writing the suspension correctly.)

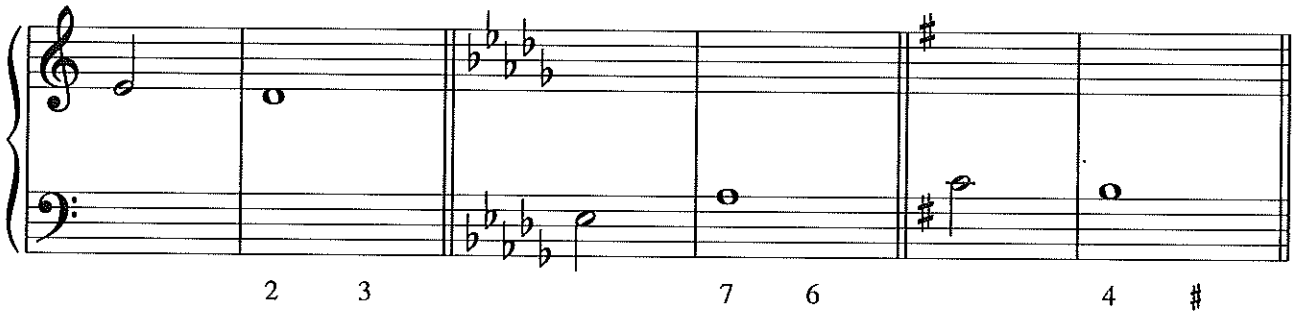
1



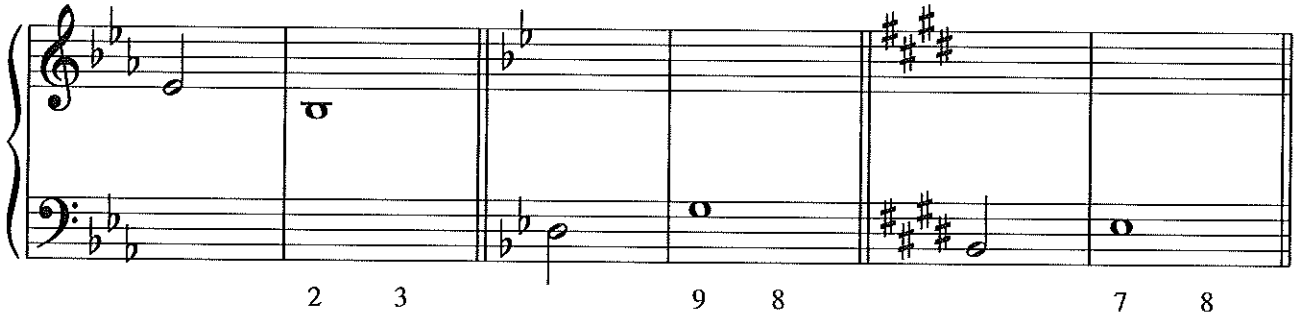
2



3

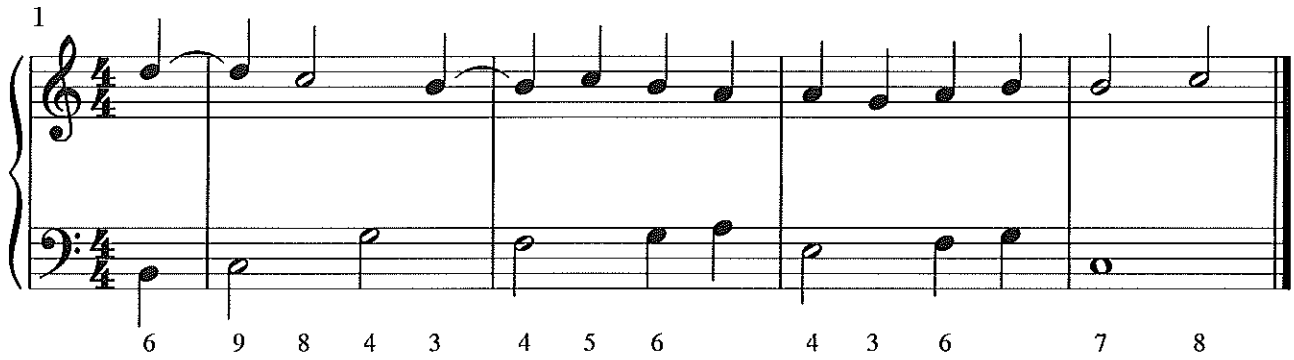


4

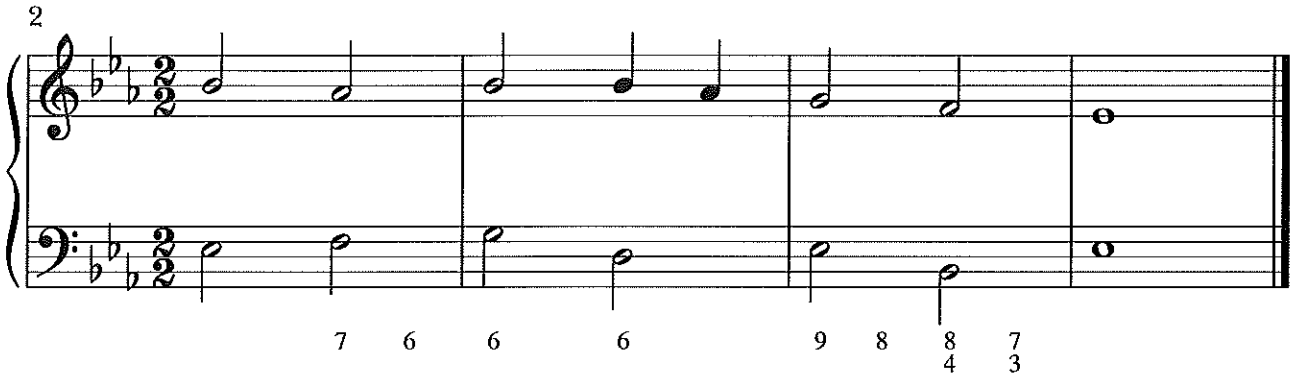


B. Add parts for alto and tenor.

1

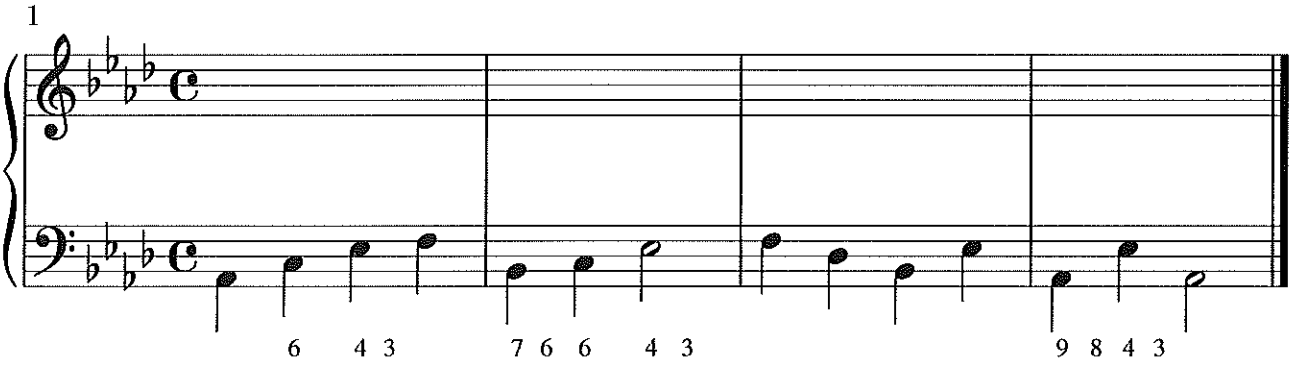


2

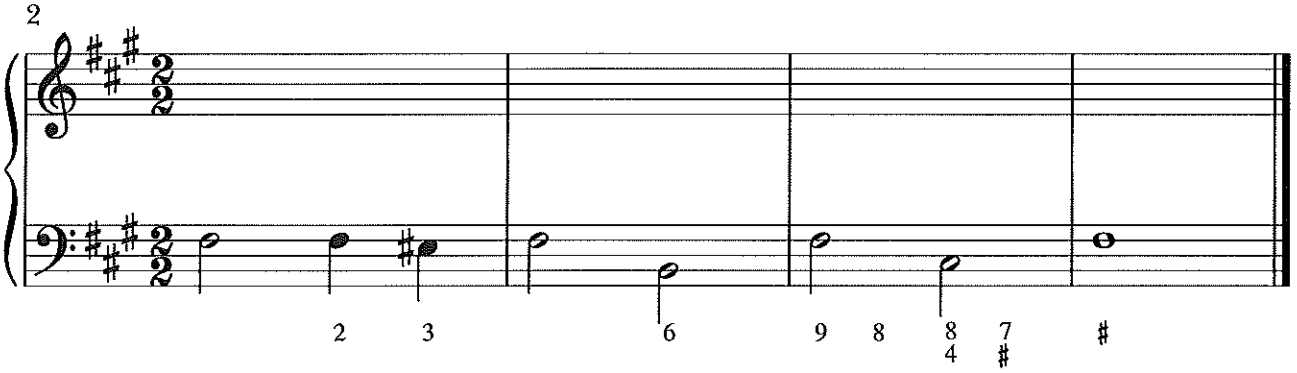


C. Add parts for soprano, alto, and tenor.

1

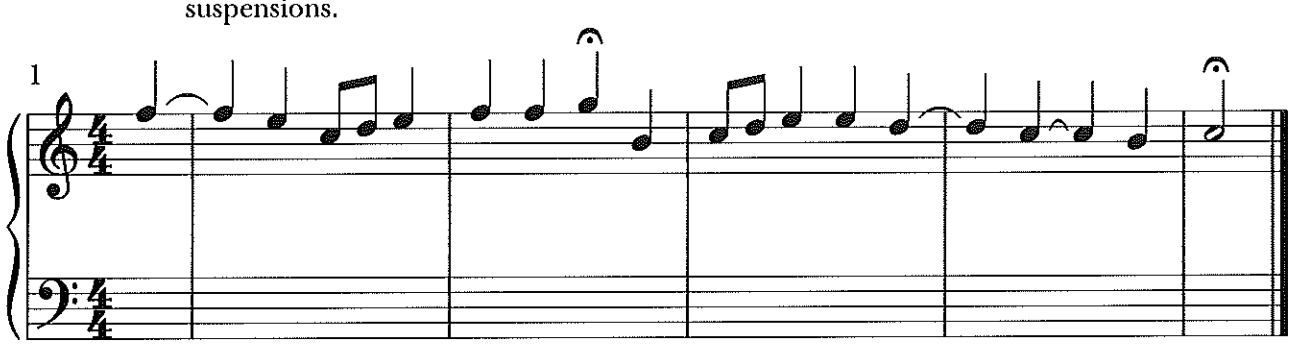


2

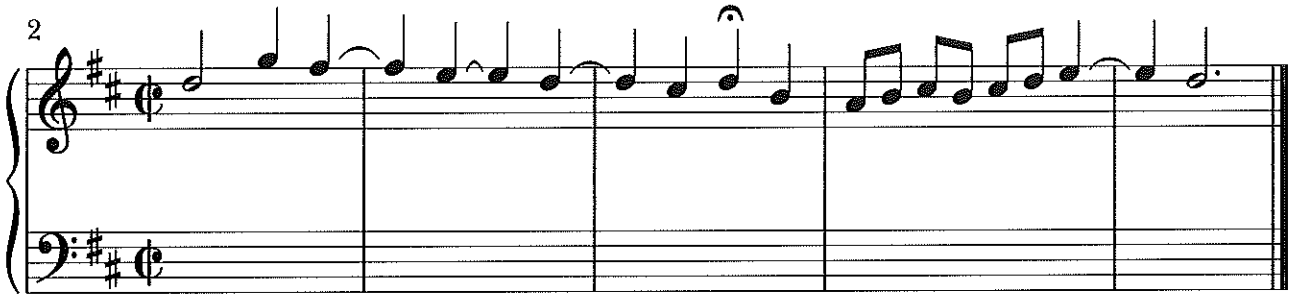


D. Add parts for alto, tenor, and bass. All tied notes should be treated as suspensions.

1

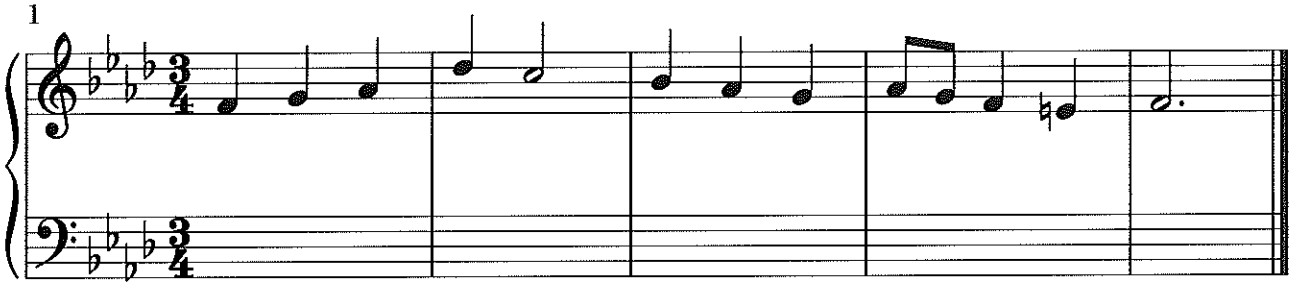


2

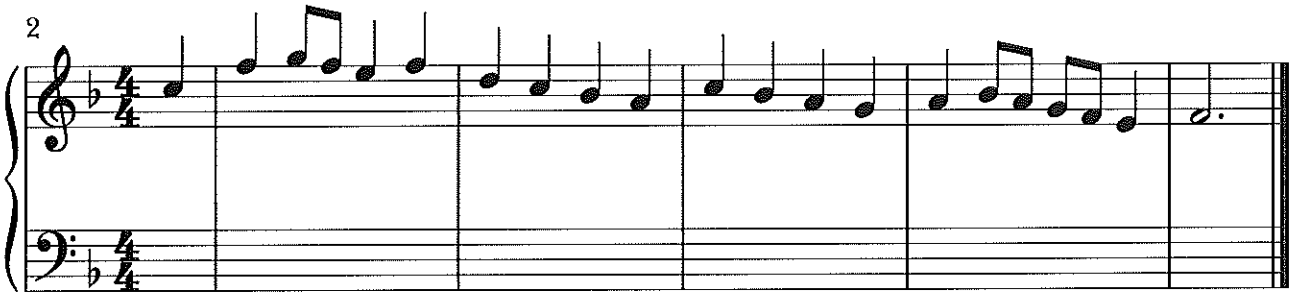


E. Add parts for alto, tenor, and bass. Use suspensions in the middle parts, where appropriate.

1



2



F. Make a complete harmonic analysis of the following phrases. Circle and label all nonchord tones.

1



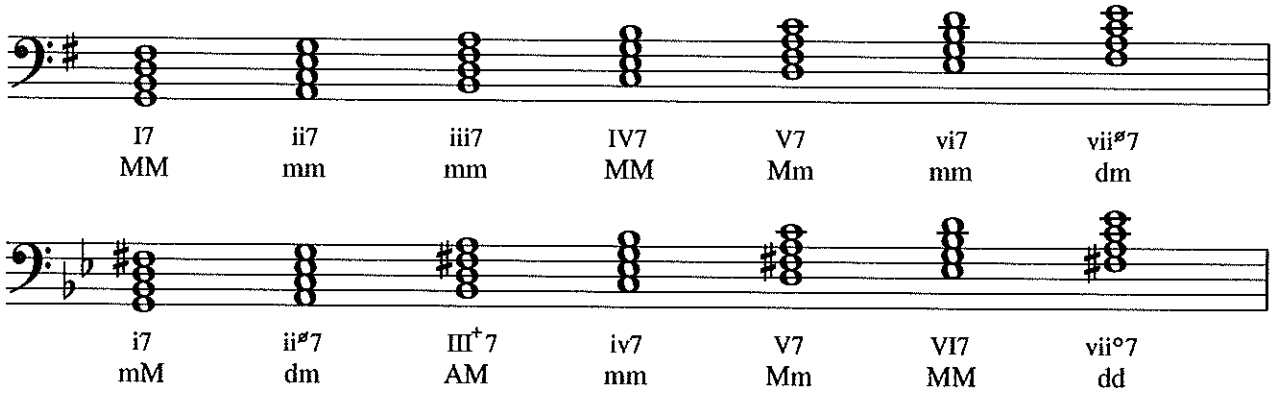
2



14 Diatonic Seventh Chords

A seventh chord is constructed by adding to a triad a pitch that is a seventh above its root.

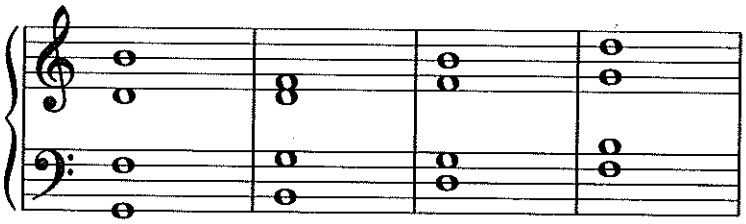
The result is a sonority (chord) with four pitches stacked in thirds. The quality of the seventh varies with the scale degree upon which it is built, as follows:



The diagram shows the seventh chords in the major and harmonic minor scales. Notice that under the Roman numerals, the qualities of the seventh chords are described. The first symbol shows the quality of the triad, and the second symbol shows the quality of the interval of the seventh.

FIGURED BASS SYMBOLS FOR SEVENTH CHORDS

The example below shows a seventh chord in root position and in its three inversions. For each one, the complete figured bass is given, and the most commonly used abbreviation.



7 5 3	6 5 3	6 4 3	6 4 2
7	6 5	4 3	4 2

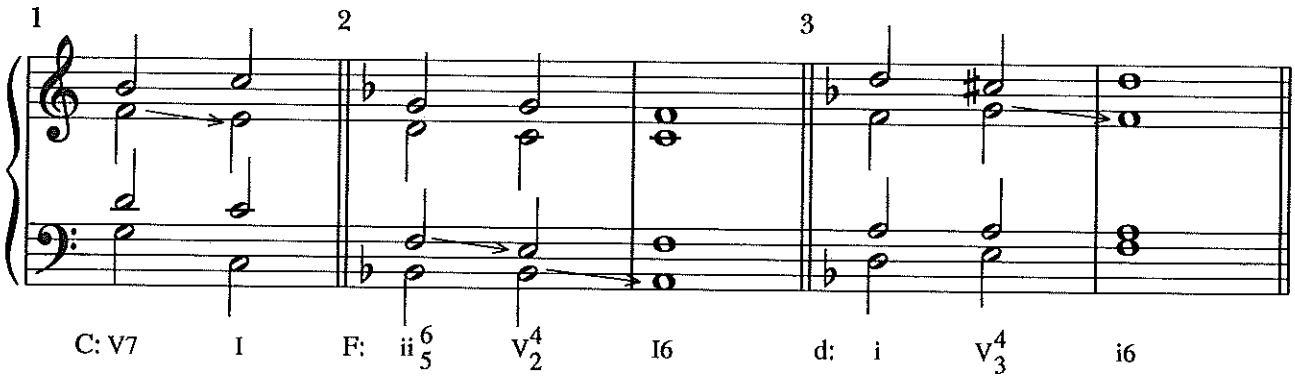
Seventh chords in root position and in inversion are described by their figured bass symbols. For example, a seventh chord in first inversion is called a "six-five chord."

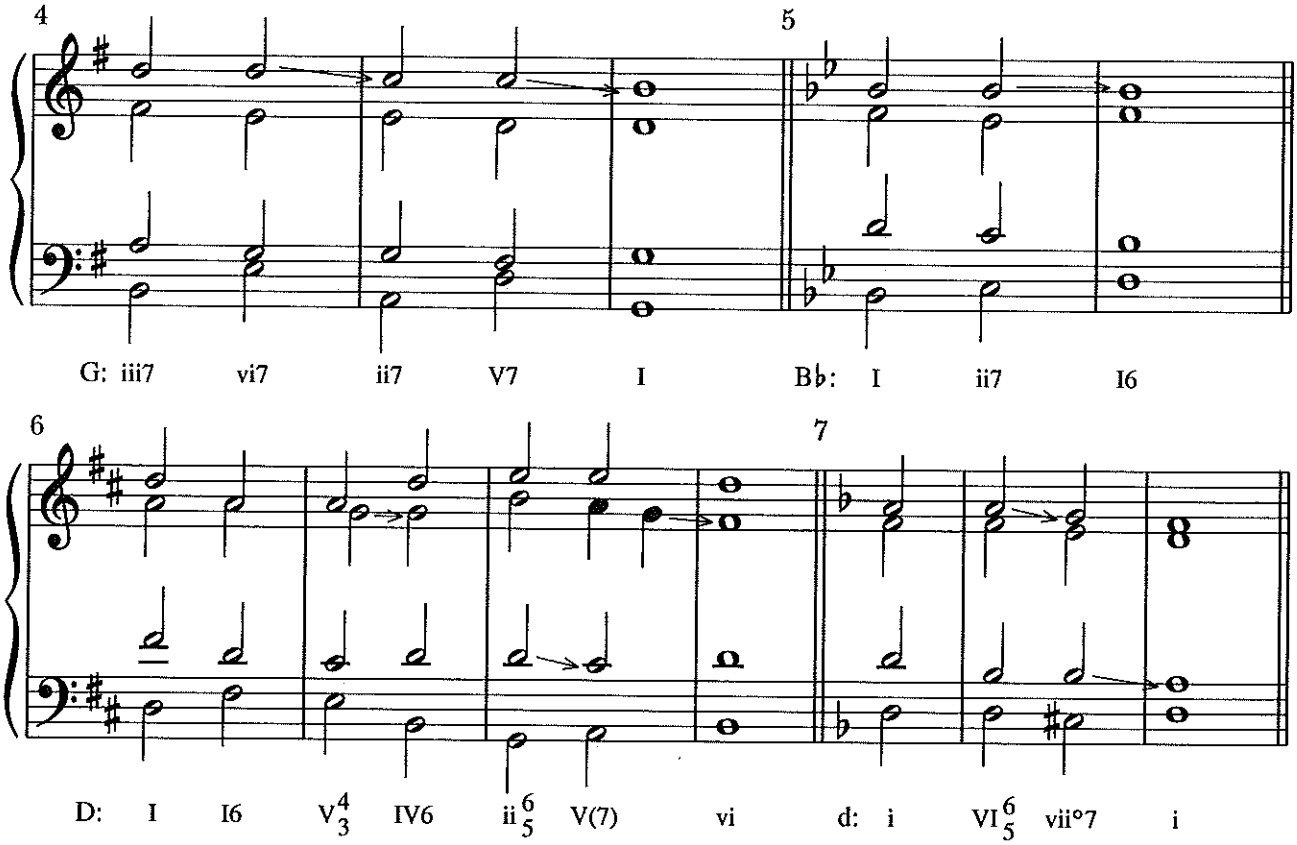
THE SEVENTH CHORD IN FOUR-PART WRITING

The seventh in the seventh chord is nearly always treated in one of two ways:

1. It resolves *downward* by step. This is its most natural resolution and the one most frequently encountered.
2. It remains as a common tone. Such a resolution is referred to as *passive.*

Study the following examples carefully:





Note:

a. The seventh is never doubled; it is a sensitive pitch.

b. Occasionally the fifth is omitted from the seventh chord, as in 4. By doing so, parallel fifths are prevented.

c. In some examples (see 2, 4, and 7), the seventh is "prepared" and treated in a manner similar to a suspension. This is a common procedure in part-writing the chord.

d. The seventh chord usually resolves by a root movement of a descending fifth. With the exception of vii°7–i, this is the chord's most natural resolution, and the one most frequently encountered.

e. The seventh chord is used freely in all inversions.

THE FUNCTION OF DIATONIC SEVENTH CHORDS

Because seventh chords are dissonant sonorities, they should be used sparingly. They have two important functions:

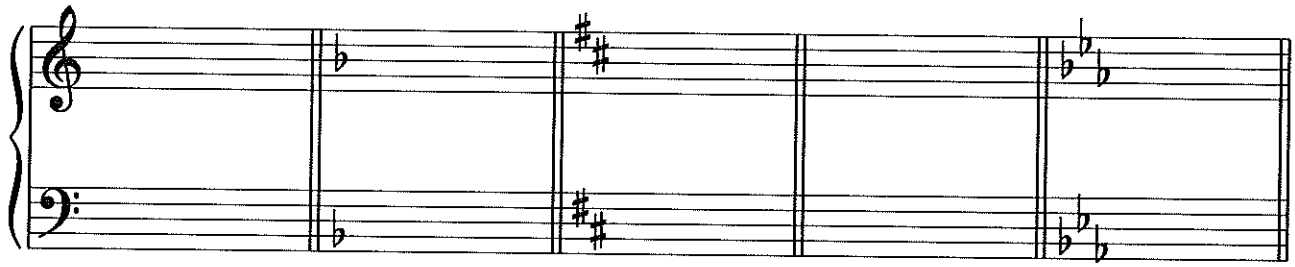
1. To add intensity to harmonic motion, particularly at cadence points.
2. To create harmonic color.

With reference to the first function, the dominant seventh (V7) and the supertonic seventh (ii7) are more widely used than other seventh chords. With reference to the second function, care must be taken that when seventh chords are used for harmonic color, they are used systematically. The presence of a seventh chord in a passage otherwise devoid of them is likely to be unsatisfactory.

EXERCISES

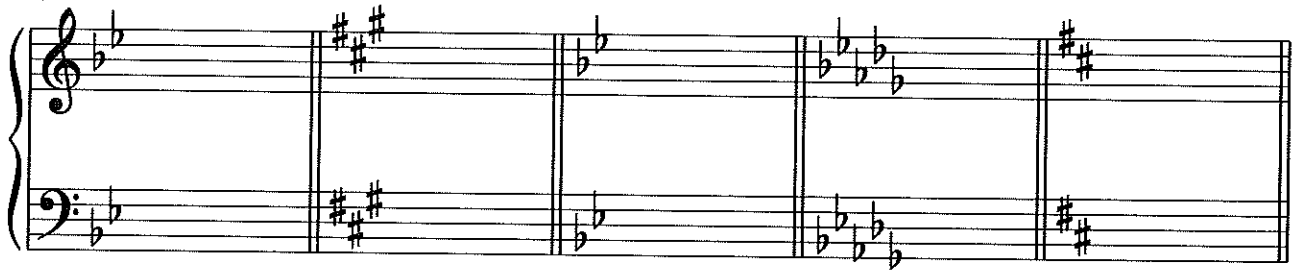
A. Write, in four voices, the following seventh chords.

1



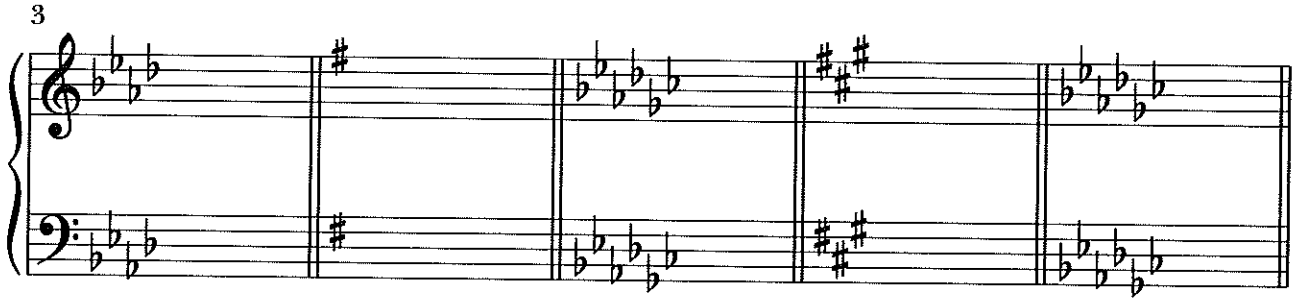
C: V7 F: V7 d: V7 a: V7 E♭: V^6_5

2



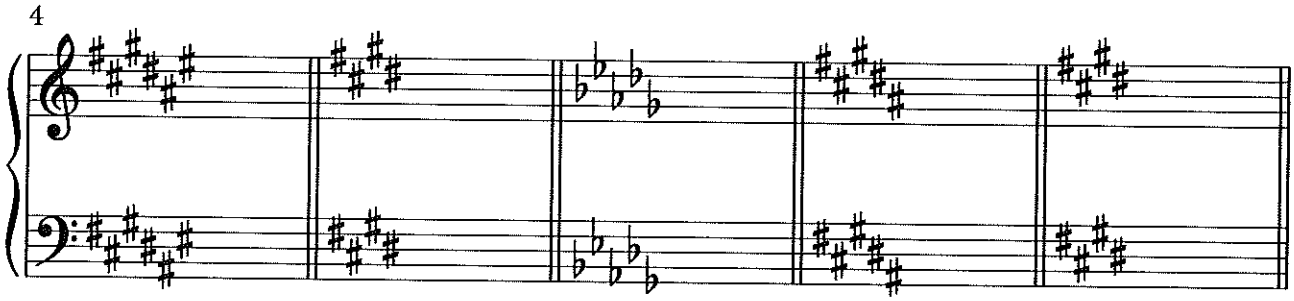
B♭: V^4_2 f♯: V^4_3 g: V^6_5 D♭: V^4_2 D: V^4_3

3



A♭: ii7 e: iv7 G♭: vi^6_5 A: vii^{ø}7 e♭: $ii^{ø6}_5$

4



F♯: IV^4_2 E: iii^4_3 b♭: VI^4_2 B: ii^4_2 c♯: vii°7

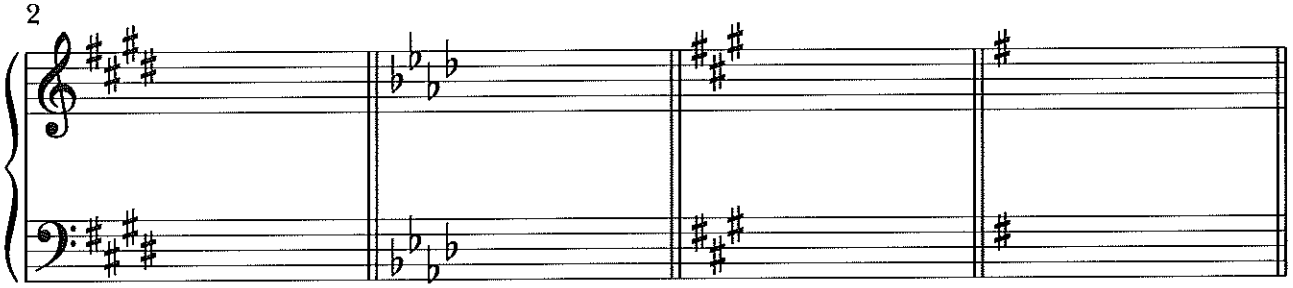
B. Write and resolve, in four voices, the following seventh chords.

1



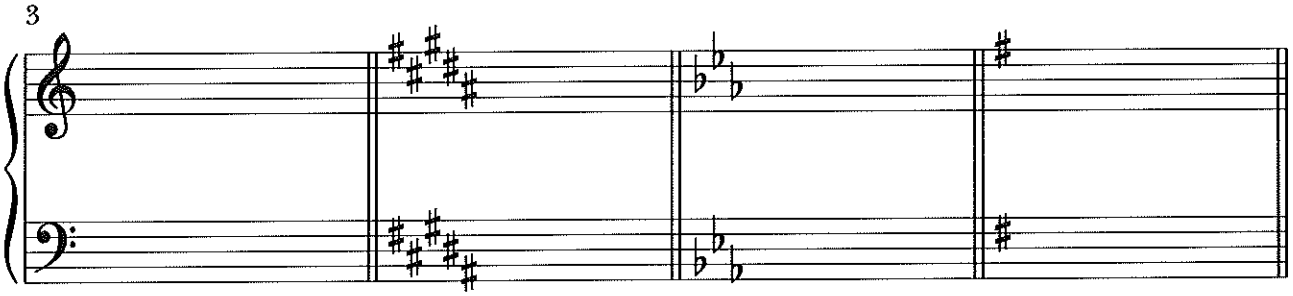
C: V^6_5 I d: V7 i E♭: V^4_2 I6 F: V^4_3 I6

2



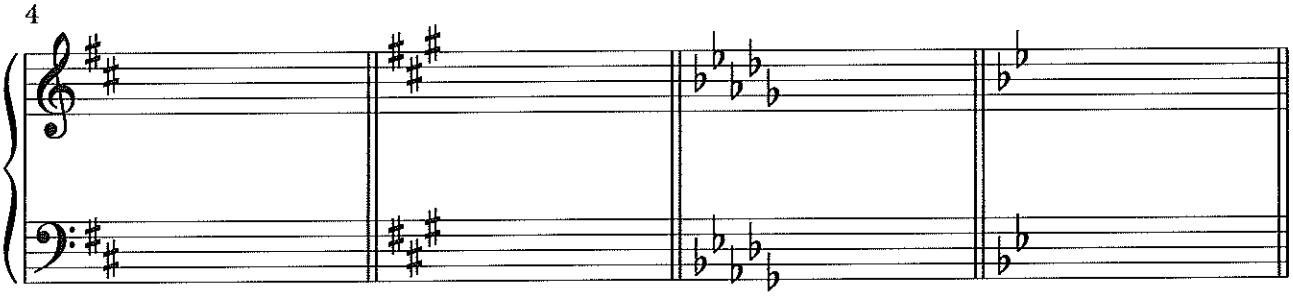
E: iii7 vi A♭: ii^6_5 V f♯: VI^4_2 ii°6 G: ii^4_2 V6

3



a: vii°7 i B: ii7 I6 E♭: IV7 ii6 e: $ii^{ø6}_5$ V

4

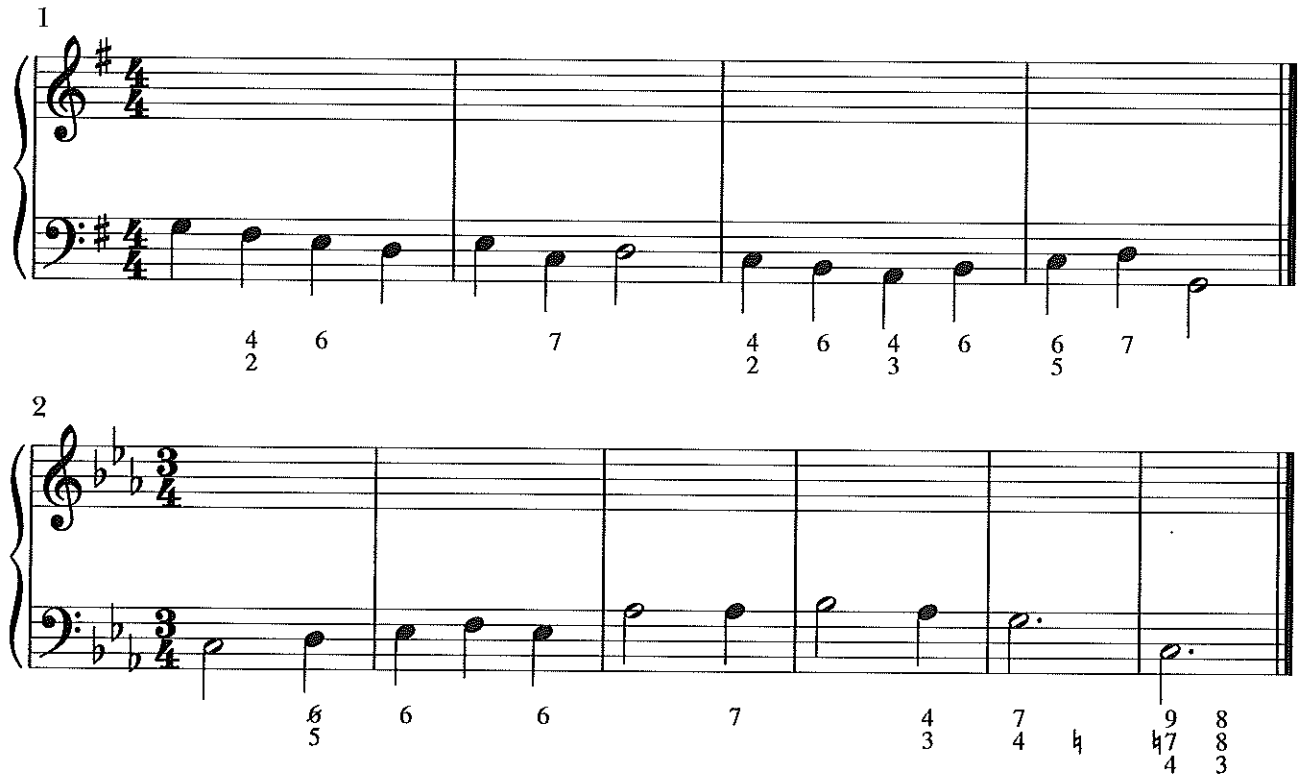


b: V^4_2 i6 A: V^4_3 IV6 D♭: vi7 ii g: V^6_5 i

C. Add parts for alto and tenor.



D. Add parts for soprano, alto, and tenor.



E. Harmonize the following melodies, using seventh chords.

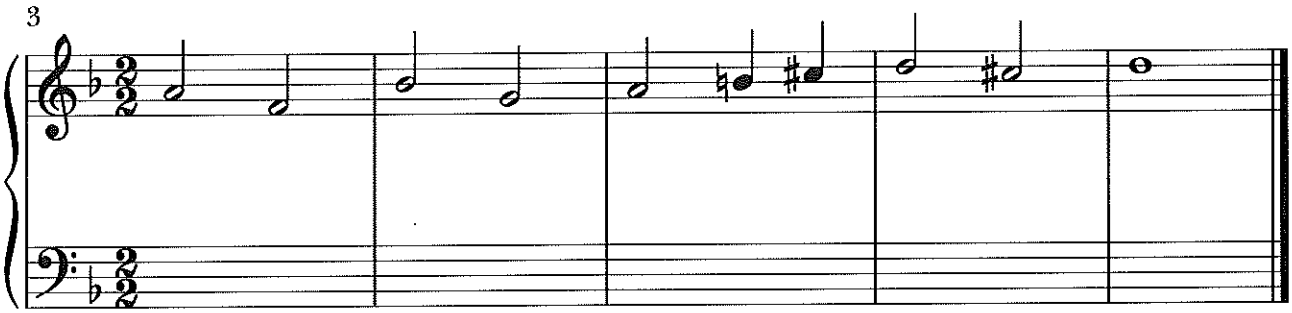
1



2



3



4

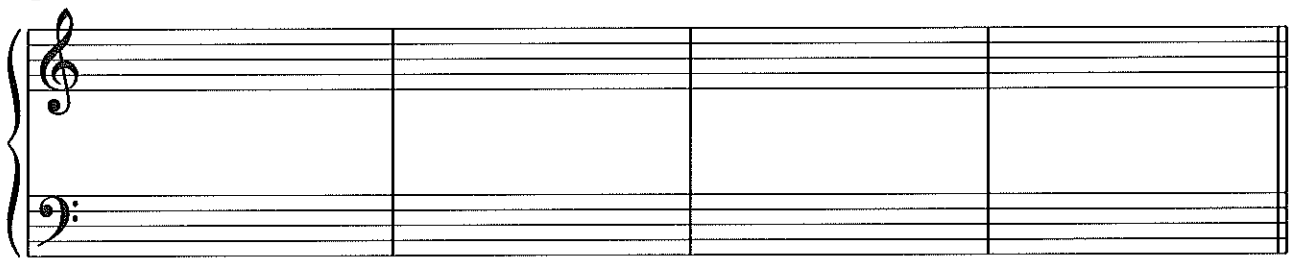



5
6

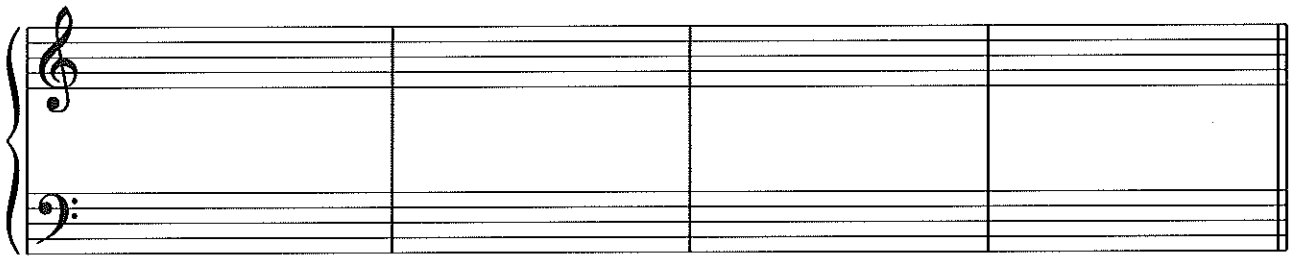
F. Using the given progressions, invent four-measure phrases for soprano, alto, tenor, and bass.

1. G: $\frac{3}{4}$ I–I^4_2–IV6–V^6_5–I–V^6_4–I6–ii^6_5–V7–I
2. c: $\frac{4}{4}$ i–V^6_5–i–iv7–VII–III7–VI–$ii^{\circ 6}_5$–V4–3–i
3. E: $\frac{2}{2}$ V^6_5–I–ii7–I6–IV–V7–I4–3
4. d: $\frac{2}{4}$ i–i^4_2–iv6–iv–V–V^6_5–i

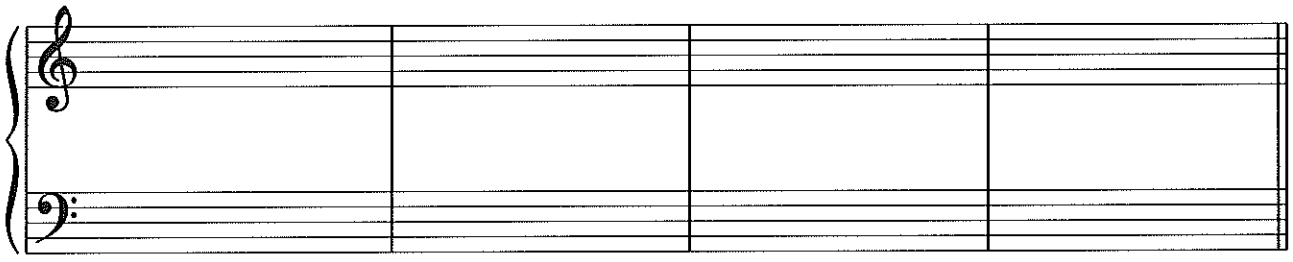
1



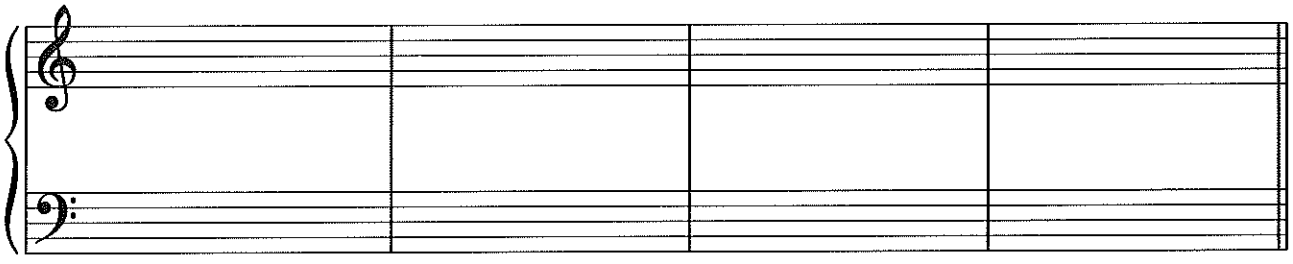
2



3

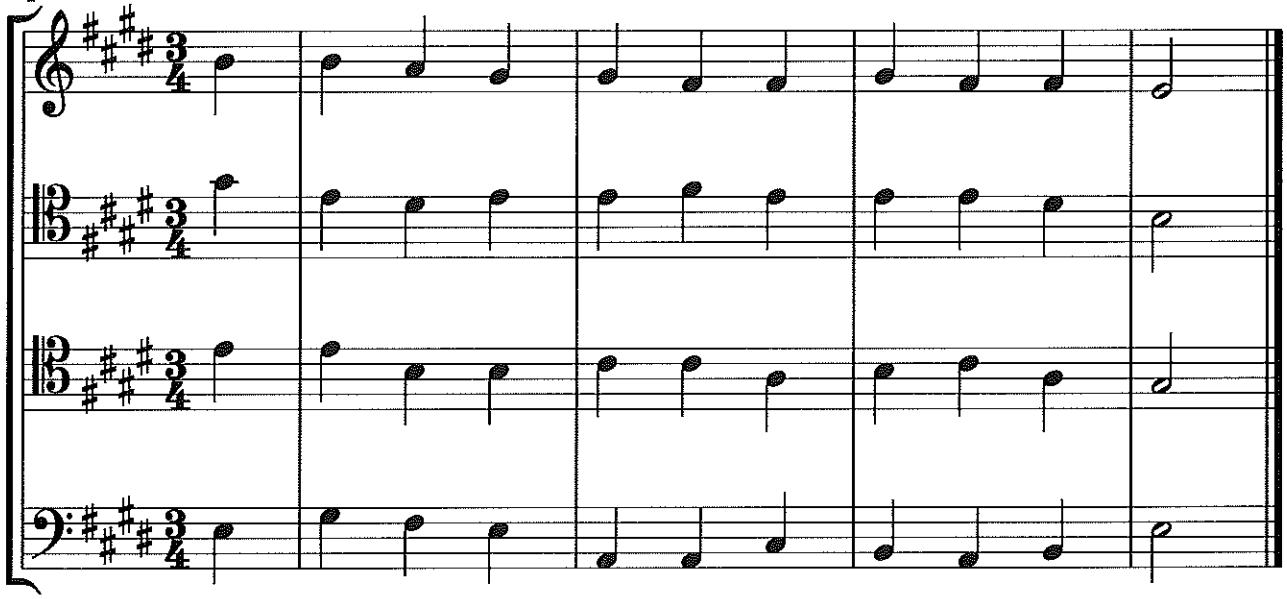


4



G. Make a complete harmonic analysis of the following phrases.

1



2



BURKHART, TUREK, AND WENNERSTROM EXERCISES FOR HARMONIC ANALYSIS

All excerpts for harmonic analysis are taken from Charles Burkhart's *Anthology for Musical Analysis*, fifth edition; Ralph Turek's *Analytical Anthology of Music*, second edition; and Mary Wennerstrom's *Anthology of Musical Structure and Style*.

For each exercise, the student should identify

1. The Key.
2. All chords in their appropriate inversions.
3. All nonchord tones.

Burkhart

p. 556, mm. 1–4.
p. 561, mm. 17–22 (setting a).
pp. 564–65, mm. 1–4 and mm. 13–16 (setting b).
p. 565, mm. 1–3 (setting b).
p. 567, mm. 1–6.
pp. 572–73, mm. 1–5 and mm. 11–15.
pp. 576–77, mm. 1–6.

Turek

p. 84, mm. 1–4.
p. 86, mm. 1–4.
p. 90, mm. 1–2 and mm. 11–12.
p. 91, mm. 15–18.
p. 92, mm. 1–6 (No. 2).
p. 93, mm. 1–8 and mm. 17–22 ("Ermuntre dich").
p. 94, mm. 31–38.
p. 94, mm. 1–9 ("Herr, Ich habe missgehandelt").
p. 95, mm. 1–5 ("Herr, Ich habe missgehandelt").
p. 96, mm. 1–2 ("Sei gregrüsset, Jesu gütig").
p. 96, mm. 1–4 ("Wach' auf, mein Herz").
p. 97, mm. 1–5 (setting by Nicolai).

Wennerstrom

pp. 146–47, mm. 1–4 and mm. 10–11 (settings A and B).
p. 148, mm. 1–6.
p. 149, mm. 1–6.
p. 152, mm. 1–4 and mm. 25–32.

15 The Harmonization of Melodies II

In Chapter 11, the first to address the harmonization of melodies, it was shown how a student could reach an effective solution by systematically applying six steps. These steps are, in essence,

1. The determination of the key.
2. The identification of phrases and cadences.
3. The identification of the harmonic activity leading to each cadence.
4. The identification of the harmonic activity that sets each phrase in motion.
5. The construction of an appropriate bass line.
6. The addition of the inner voices.

The application of these six steps is equally relevant in this chapter, with one important addition—the melodies to be harmonized contain nonchord tones. This added variable forces the student to consider, before completing steps 3 through 6, which of the pitches in a melody should or should not be harmonized. An awareness of typical melodic patterns, and the nonchord tones that may be suggested by them, goes a long way toward facilitating this process.

The following pages exemplify some of the more commonly encountered patterns and how they may be identified in diatonic melodies.

In the patterns exemplified below, the white notes represent chord tones, and the black notes, nonchord tones.

PASSING TONE PATTERNS [P]



Note:

a. Although single passing tones are the most commonly encountered, two or more are by no means rare.

b. The white notes are more likely to be accented than the black ones.

NEIGHBORING TONE PATTERNS [N]

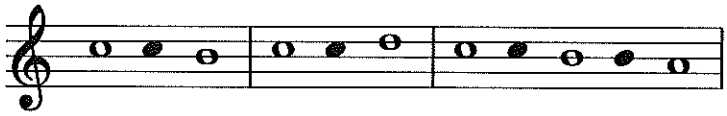


Note:

a. Changing tones are included in this category.

b. Again, the white notes are more typically accented than the black notes.

SUSPENSION PATTERNS [Sus]



Note:

a. The black notes are typically accented.

b. The last example shows the beginning of a chain of suspensions. This repetitive pattern may easily unfold over several measures of music.

APPOGGIATURA PATTERNS [Ap]



Note:

a. This group of patterns involves a leap followed by a step with or without a change of direction.
b. The black note is frequently (though not always) accented in each of these patterns.

ESCAPE TONE PATTERNS [ET]



Note:

a. This group of patterns involves a step followed by a leap with or without a change of direction.
b. The black note is frequently (though not always) unaccented in each of these patterns.

ANTICIPATION PATTERNS [An]



Note:

a. The black note is the result of a step or leap in either direction.
b. The black note is much more frequently unaccented than accented in this group of patterns.

Study the following examples carefully. The melody alone is first shown with the nonchord tone patterns bracketed above it and with a suggested harmonic progression below it. Then a complete harmonization is given.



Note:

a. The last beat of measure 3 is tied to the first of measure 4. This kind of suspension pattern is widely used at cadence points.

b. The harmonization of the first phrase is very simple, and the harmony changes only on the first beat of each measure.

c. In the second phrase, there are more frequent changes of harmony. It is quite common for the rate of change of harmony to accelerate as the final cadence is approached.

d. The term *harmonic rhythm* is used to describe changes in harmony. Thus, the harmonic rhythm is *slower* in the first phrase than in the second.



Note:

a. Throughout, the harmonic rhythm is faster in this example than in the previous one.

b. The tonality is clearly established in the first measure by the movement from the tonic to the dominant and back again.

c. The second half of measure 2 is harmonized with the supertonic triad. The c♯, therefore, is properly treated as an escape tone rather than a leading tone.

d. Because of the change in harmony on the third beat of measure 3, the chord tones are separated by adjacent passing tones.

e. The deceptive resolution of the dominant at the end of measure 6 allows for a very strong harmonic movement through the circle of fifths in measures 7 and 8.

In both examples, the procedures described in Chapter 11 and itemized at the beginning of this chapter were followed. The melodic phrase structure in each suggested a half cadence at the end of the first phrase and an authentic cadence at the end. The bass line in the first example is very simple, because the melody allows it to be so. In the second example, however, the sequential structure of the second and third measures of the melody implies a more elaborate harmonization. Finally, the complete harmonizations follow established voice-leading principles.

EXERCISES

A. Harmonize the following one-phrase melodies.

1

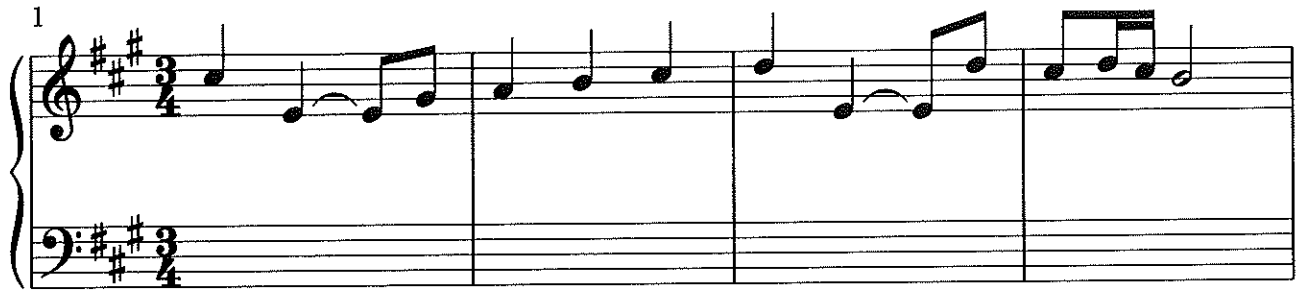


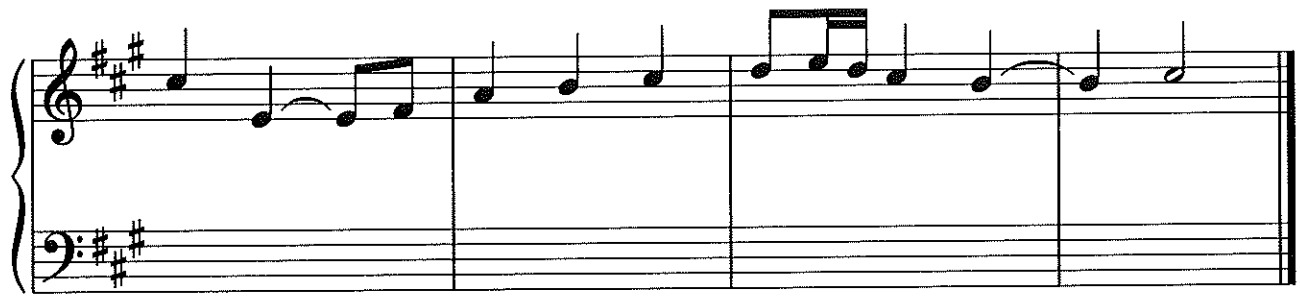
2



B. Harmonize the following two-phrase melodies.

1





2





3



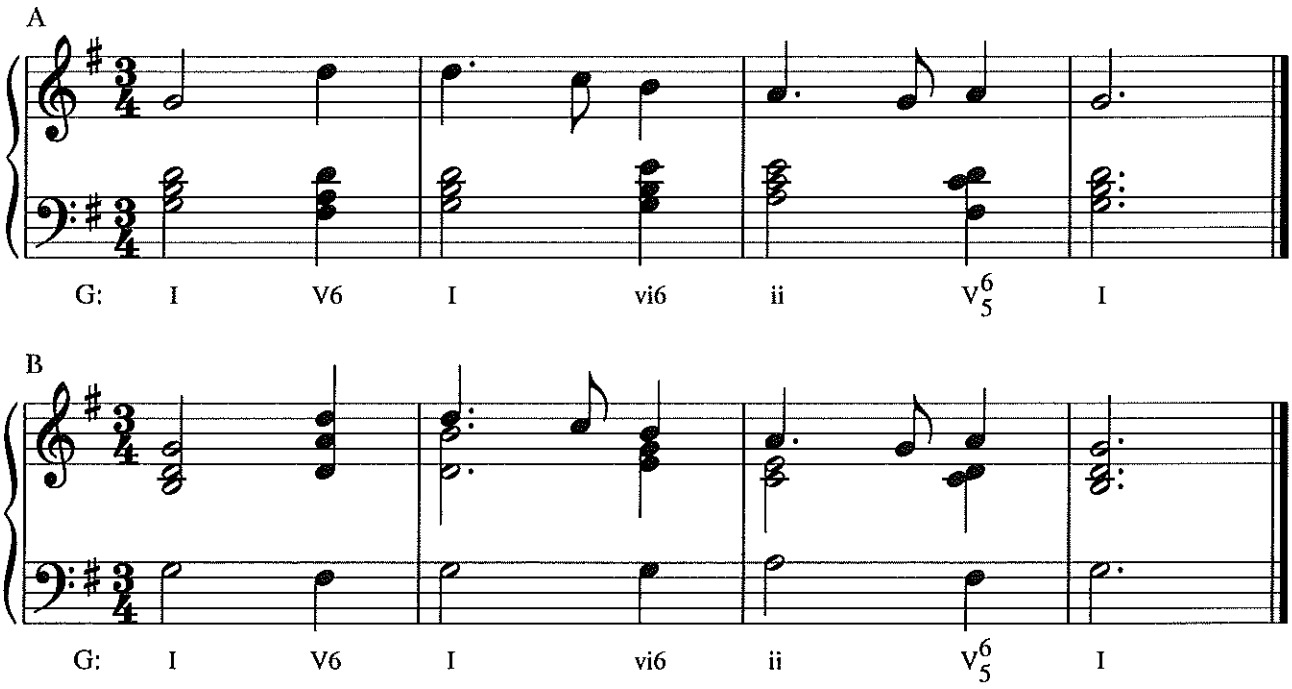


16 Writing for the Piano

FOUR-PART CHORDAL STYLES

Relatively little piano music is written in a four-part, quasi-choral style. Such writing is not particularly idiomatic for the instrument because, compared with a choir, the piano has a very limited capacity for sustaining the sound of a chord. Besides this fact, the piano's considerable range (over seven octaves) and the ability of the hands to execute a wide variety of configurations give composers for the piano an enormous number of ways to express their musical ideas.

Piano writing has two basic four-part chordal styles, from which a number of more elaborate configurations may be derived. The former involves the playing of a three-part chord in the left hand and a single note in the right hand (A), and the latter, precisely the reverse (B).



The following pages exemplify how the two examples above may be converted into more elaborate settings for the piano.

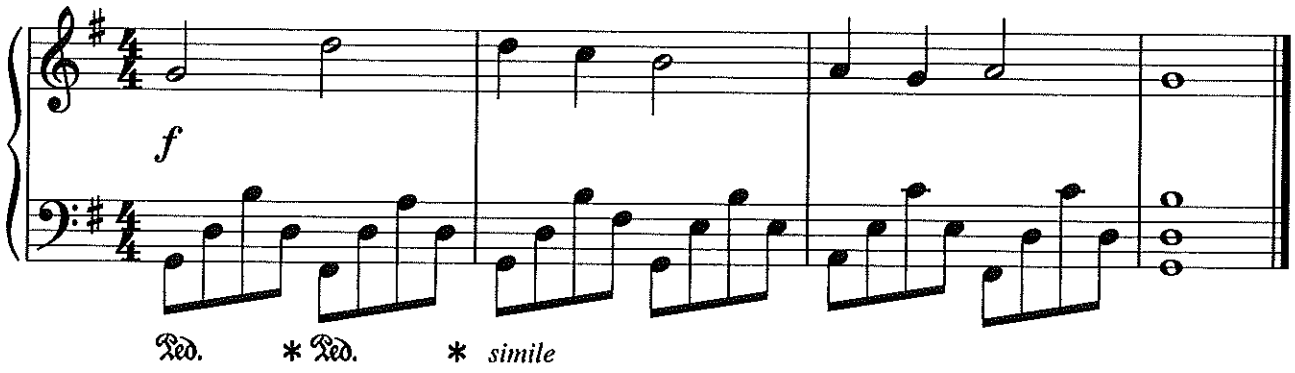
A1 **Andante**



A2 **Allegretto**



A3 **Moderato**



A4 **Semplice**




A5 Brilliante
ff
Ped. * Ped. * simile
B1 Andantino
p legato
B2 Alla marcia
f
B3 Presto
mf

Note:

a. Although all of the elaborations of the two basic chordal styles demonstrate considerable freedom in handling parts, the basic principles of voice leading are not violated.

b. Each configuration fits the hands well. This is an important consideration in writing piano music.

c. A general direction as to tempo and character (e.g, Brilliante) is placed above the meter signature, and the dynamic markings are placed between the staves.

d. Two examples make use of the sustaining pedal. When the hand cannot be in continuous contact with the keys (as in the case for the left hand in A3 and A5), the sustaining pedal is used if it is desirable to sustain the sound.

e. The figuration in the left hand in A2 is known as an *Alberti bass.* It is named after the composer Domenico Alberti, who used it excessively in his piano music.

The exercises that follow are related to the materials on pages 193–196. The piano accompaniment is covered on pages 201–202.

EXERCISES

Harmonize the following melodies, and set them for piano.

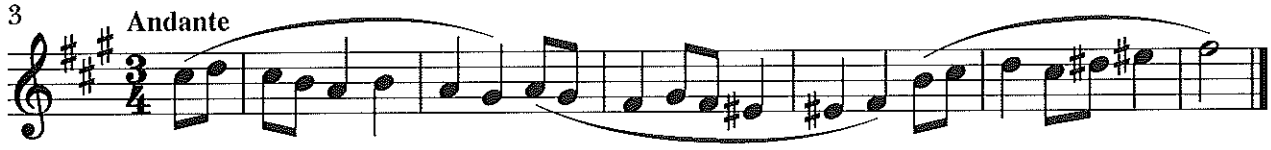
1 **Moderato**



2 **Allegro moderato**



3



THE PIANO ACCOMPANIMENT

When the piano is used to accompany another instrument, the composer is not compelled to write the melody in the piano part. As a result, piano accompaniments display a wide variety of configurations. Typically, the piano part will accomplish the following:

1. Supply the harmony through an established accompaniment pattern.
2. Supply the harmony and a countermelody.

The following examples are based upon some of the settings in the previous section. They should be studied carefully, with particular attention to the function of the right hand.







Note:

a. The bar lines for the solo instrument are separated from those for the piano to facilitate reading.
b. Dynamic markings are given for both the piano and solo instrument. They are placed beneath the staff for the latter.
c. In A3, the countermelody in the right hand violates no voice-leading principles either with reference to the melody or to the accompaniment.

Sometimes both the piano and solo instrument are given the melody, as the following example shows:



In all exercises that involve a solo instrument with piano accompaniment, students are advised to

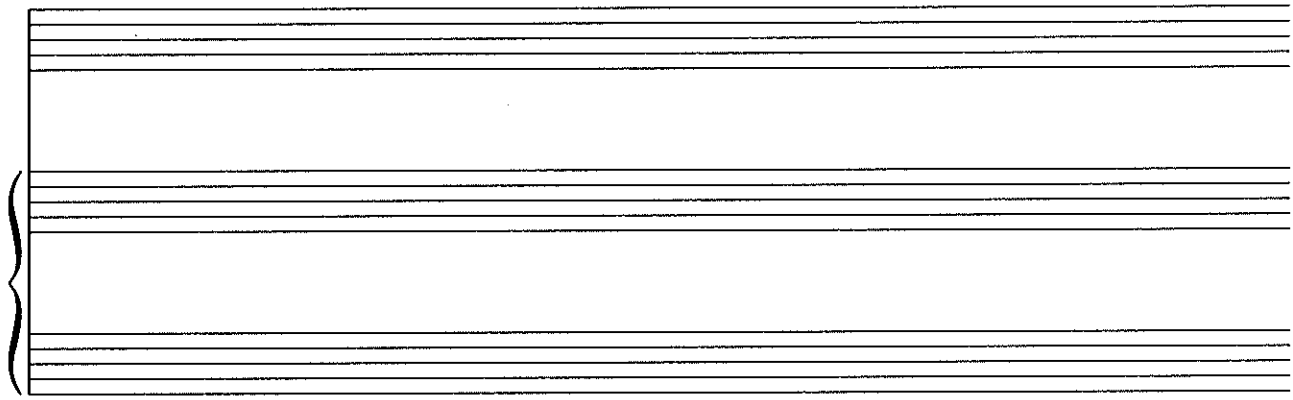
1. Equip themselves with manuscript paper.
2. Harmonize the melody according to the six procedures described in Chapters 11 and 15.
3. Decide upon an accompaniment that would seem to suit the character of the melody.
4. Write a countermelody when possible.
5. Copy the finished product in the space provided.

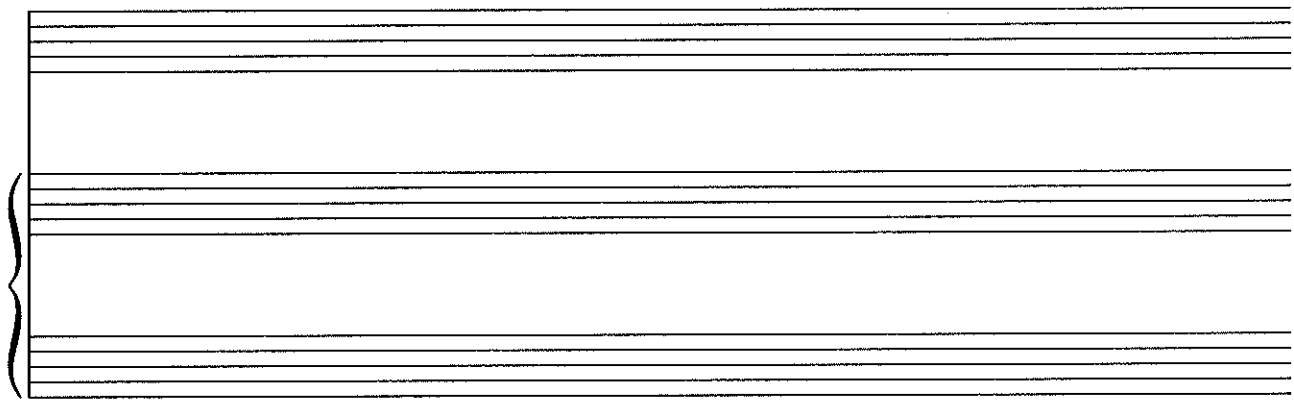
EXERCISES

Invent piano accompaniments for the following melodies.

1

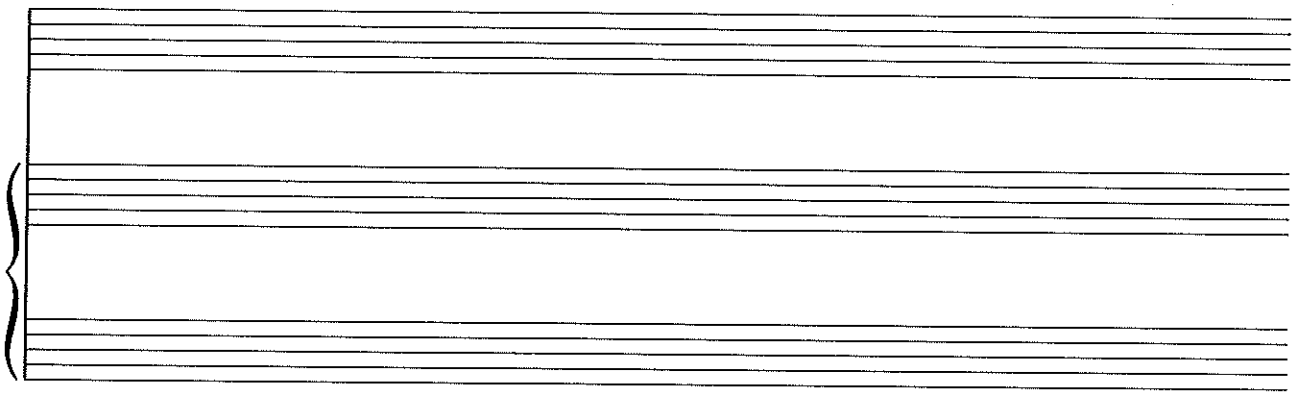


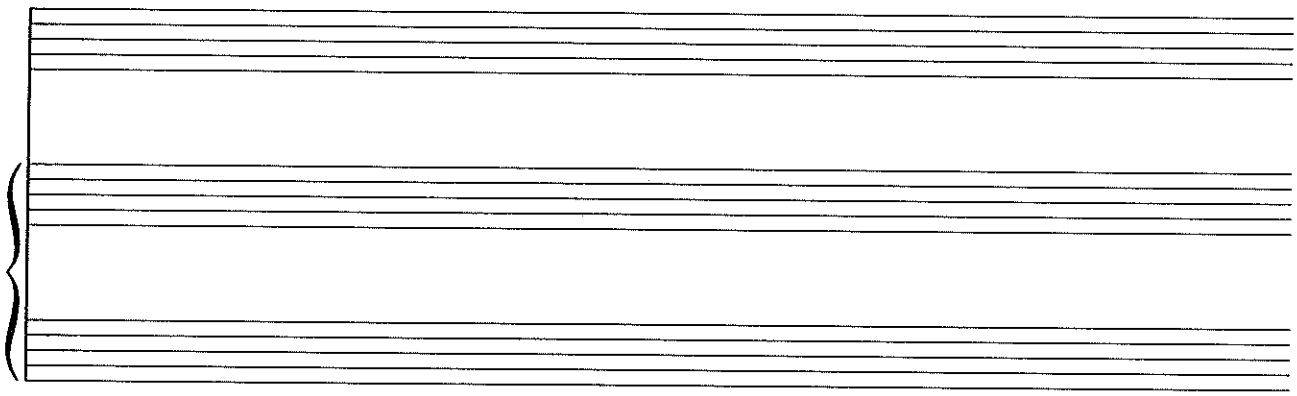




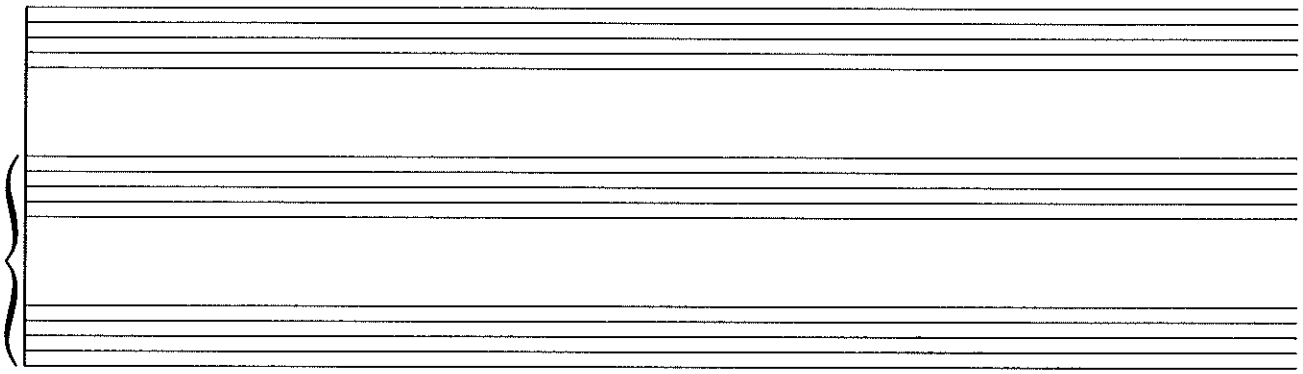
2 **Marcato** for viola

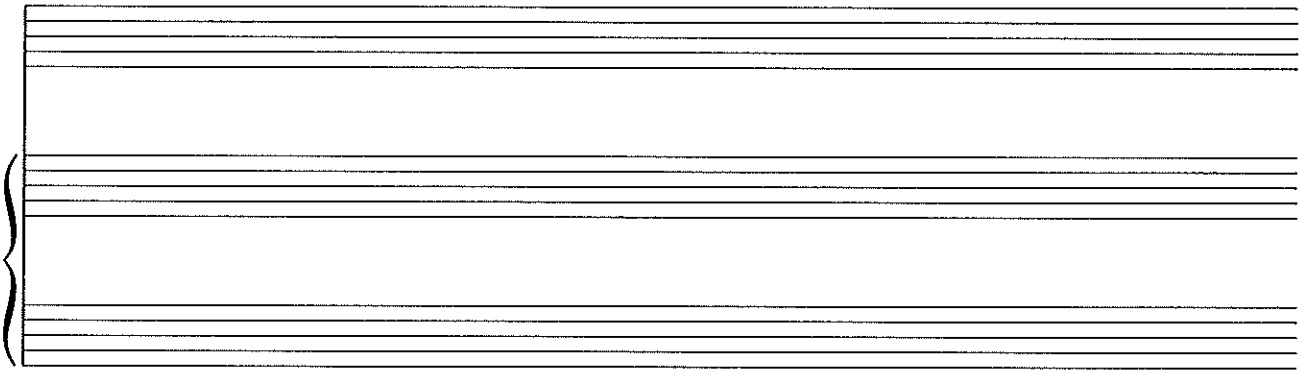







Molto moderato
3
for violin





BURKHART, TUREK, AND WENNERSTROM EXERCISES FOR HARMONIC ANALYSIS

Burkhart

p. 197, mm. 1–8.
p. 238, mm. 1–8.
p. 313, mm. 1–8.
p. 188, mm. 1–10.
p. 72, mm. 1–16.
p. 315, mm. 1–16.
p. 248, mm. 1–16.
pp. 314–15, mm. 1–20.

Turek

p. 255, mm. 1–8.
p. 309, mm. 1–8 (No. 3).
p. 309, mm. 1–8 (No. 23).
p. 318, mm. 1–8.
p. 334, mm. 17–32.
p. 187, mm. 1–16.
p. 193, mm. 61–80.

Wennerstrom

p. 176, mm. 1–8.
p. 204, mm. 1–8.
p. 235, mm. 1–8.
p. 317, mm. 1–8.
p. 206, mm. 1–9.
p. 172, mm. 1–13.
p. 316, mm. 1–16.
p. 375, mm. 1–16.

PART THREE

HARMONY IN COMMON PRACTICE

The Chromatic Vocabulary

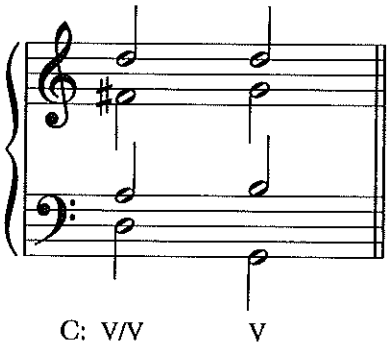
17 Secondary Dominants

It is always possible to precede a major or minor triad with a chord that can be analyzed as its dominant. For instance, if a G major triad is written, then a D major triad written before it could be considered a V chord, and indeed it is one, provided that G is a I chord.

If, however, G is not a tonic chord but, for example, the dominant chord



in the key of C, then D is no longer a V chord. Yet its relationship to G is still that of a dominant to a tonic. Under these circumstances, the D major triad is called a *secondary dominant,* and in the key of C is known as V of V (written V/V).



From the explanation above, it follows that a major or minor triad on each scale degree can be preceded by a chord that stands in a dominant relationship to it.

Study the following examples carefully:



Note:

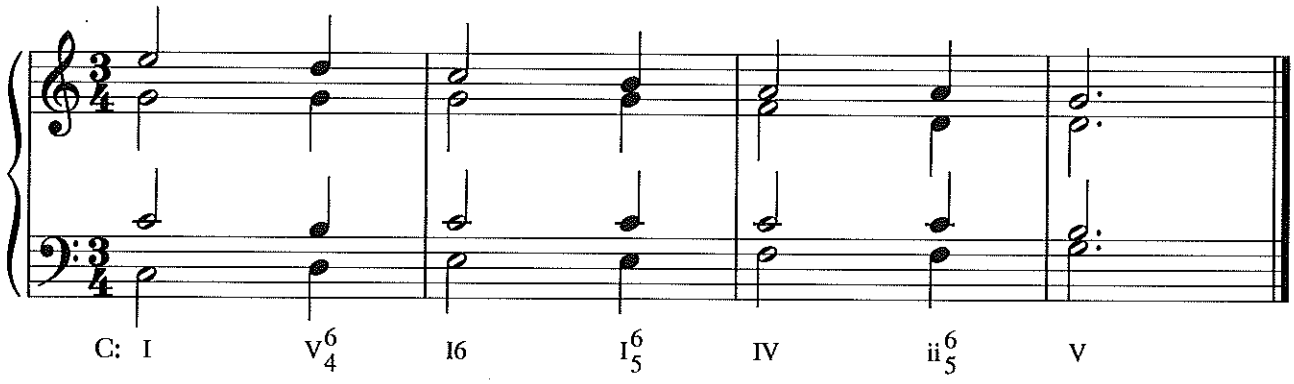
a. Secondary dominants are always chromatic chords; in other words, each contains at least one pitch that is not diatonic.

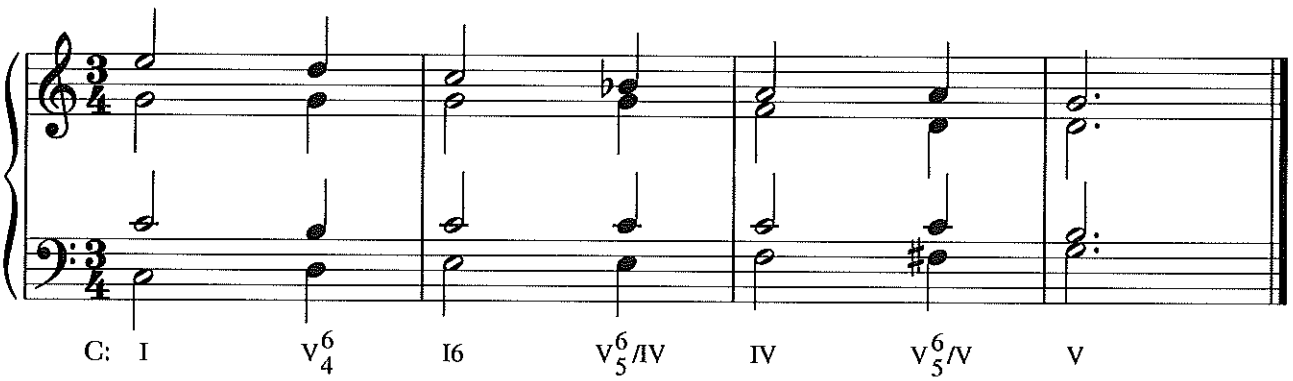
b. Secondary dominants are commonly used with the addition of the minor seventh and thus become major-minor seventh chords.
c. As seventh chords, secondary dominants are used in all inversions.
d. The resolution of a secondary dominant seventh is the same as that of a dominant seventh.
e. A secondary dominant may resolve to another secondary dominant rather than to a diatonic chord.

THE FUNCTION OF SECONDARY DOMINANTS

Secondary dominants have the effect of strengthening the harmonic movement to the chords that follow them. In addition to the root movement of a descending fifth, this strengthening effect is produced first by the resolution of the *secondary leading tone* to the root of the subsequent chord, and second, if the seventh is present, by its resolution to the third of that chord. The secondary dominant, then, allows the major or minor triad that follows it to sound like a temporary (or fleeting) tonic. The latter chord is said to be *tonicized*, and the process is known as *tonicization*. The function of the secondary dominant, therefore, is very similar to that of the dominant.

The following two examples represent phrases that terminate with a half cadence. Except for the chromatic pitches in measures 2 and 3, they are identical.





Note:

a. In the second example, the movement to IV is strengthened by the alteration of I^6_5 to V^6_5/IV, and the movement to V is similarly strengthened by changing ii $^6_5$ to V^6_5/V.

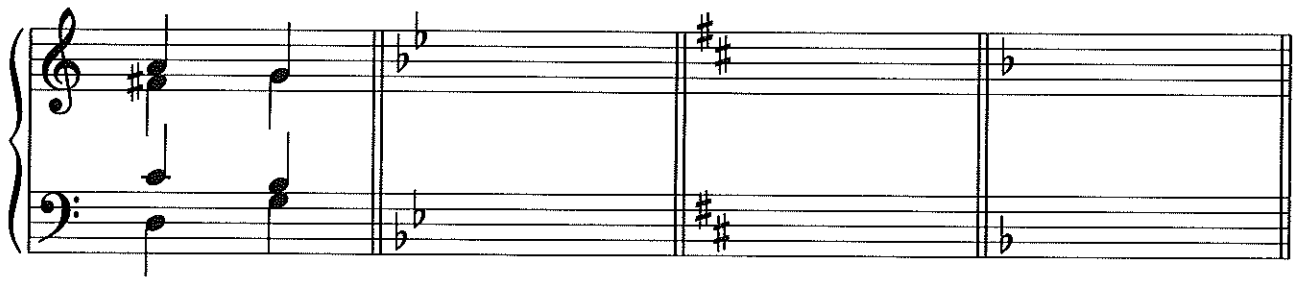
b. The half cadence in the second example creates a greater sense of repose than that in the first, because V is preceded by its dominant.

The exercises that follow are related to the materials introduced on pages 209–211. Deceptive resolutions of secondary dominants are covered on pages 219–220.

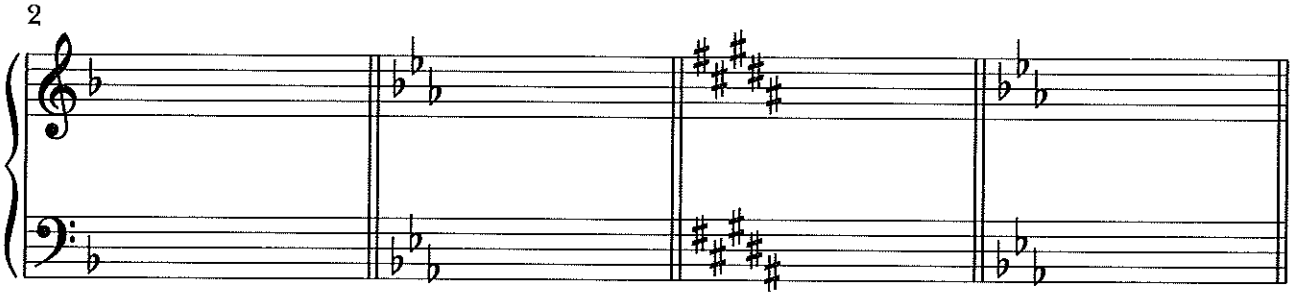
EXERCISES

A. Write, and resolve, the following secondary dominants.

1



2



3

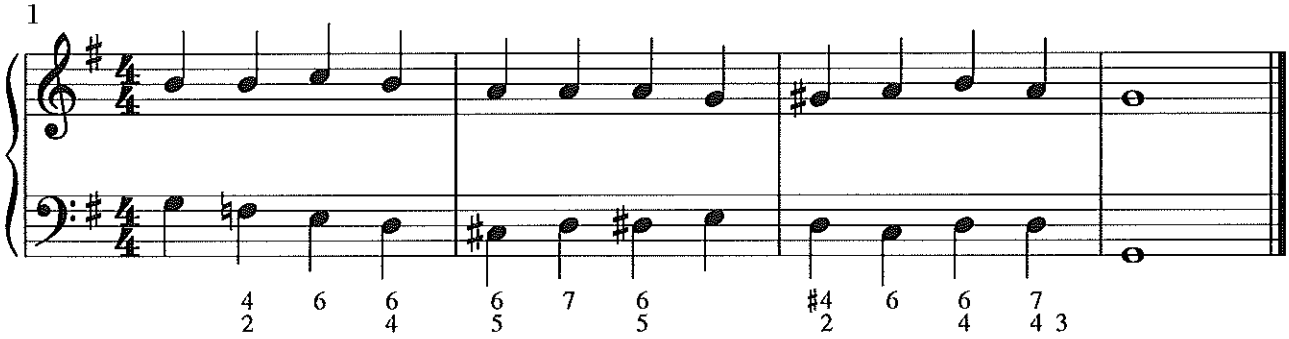


4

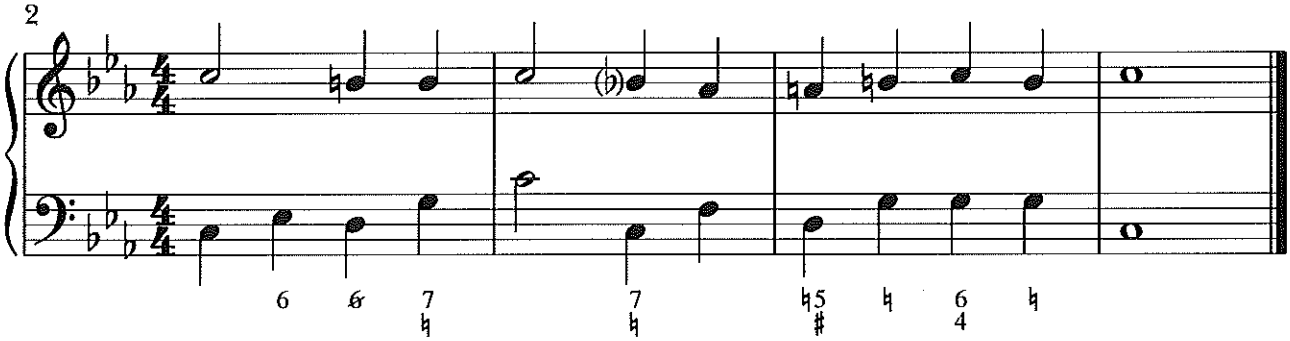


B. Add parts for alto and tenor.

1

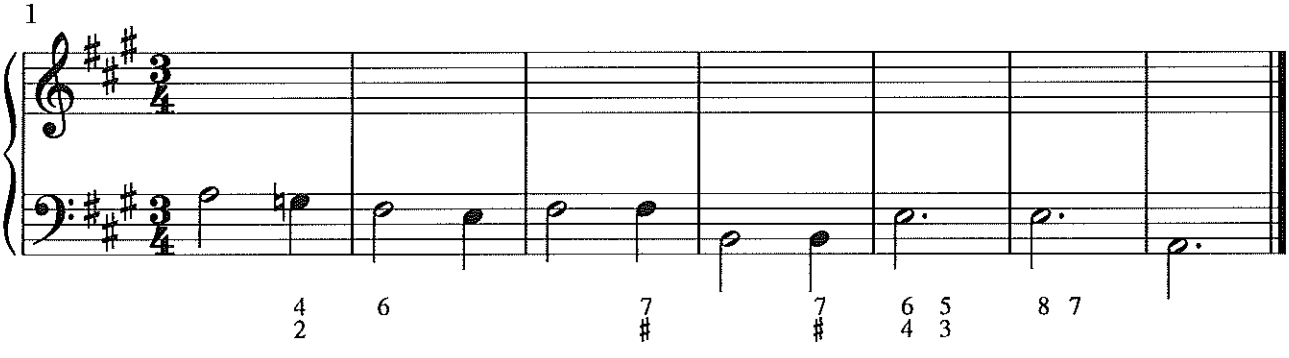


2

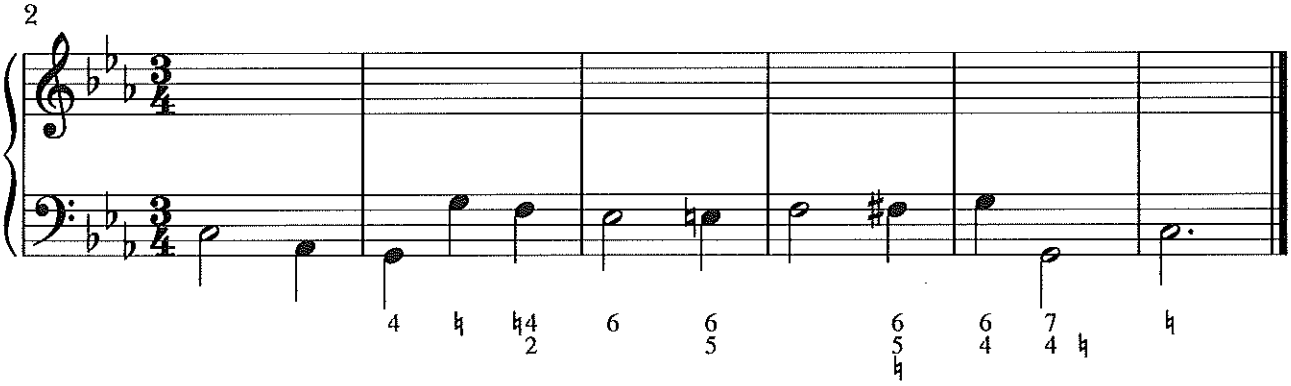


C. Add parts for soprano, alto, and tenor.

1

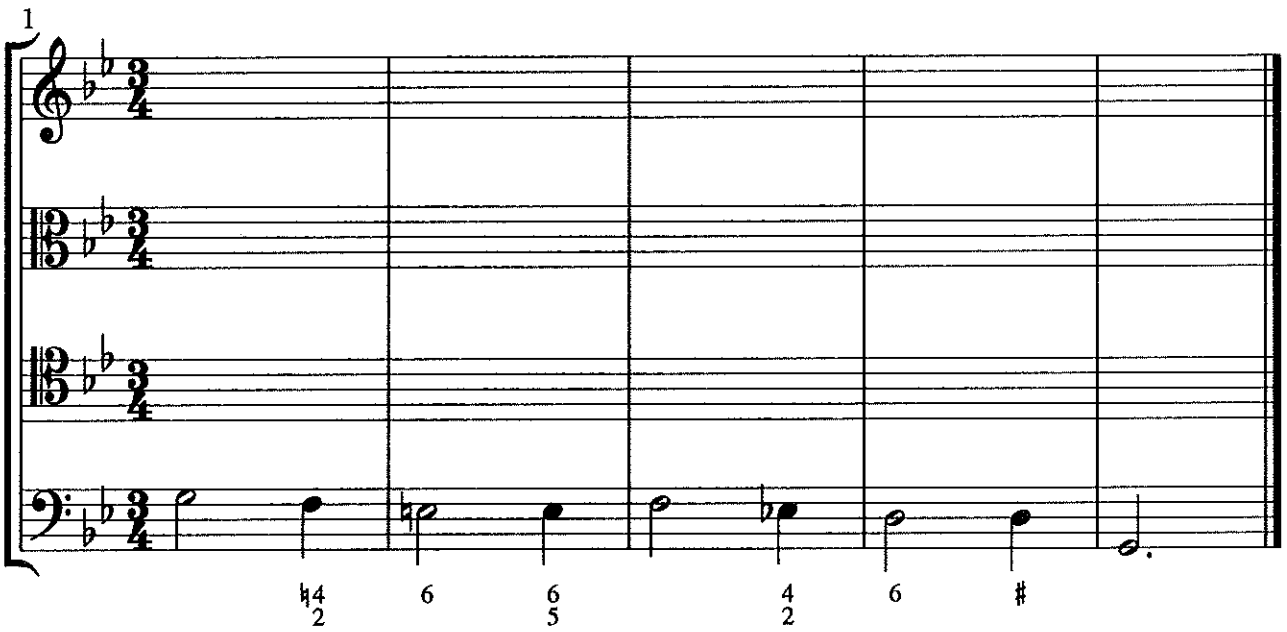


2

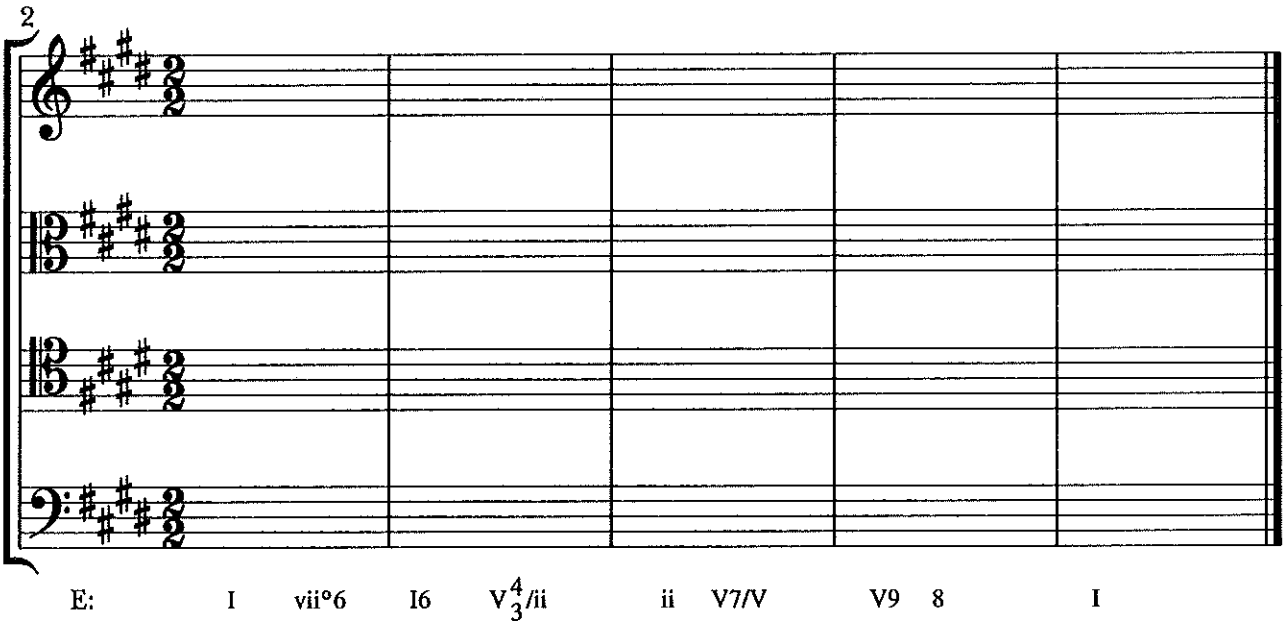


D. Harmonize the following in open score, making appropriate use of nonchord tones.

1



2



E. Add parts for alto, tenor, and bass.



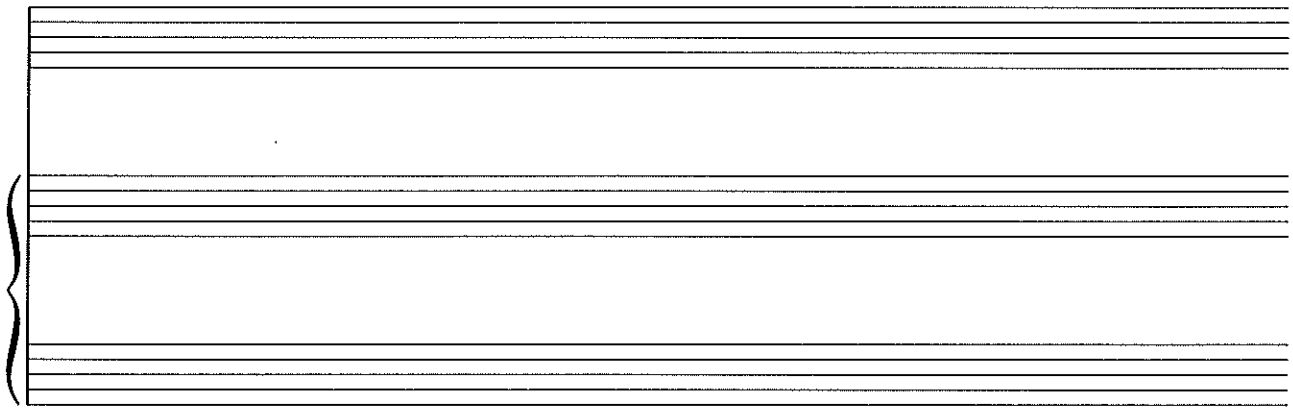
* indicates that a secondary dominant should be used

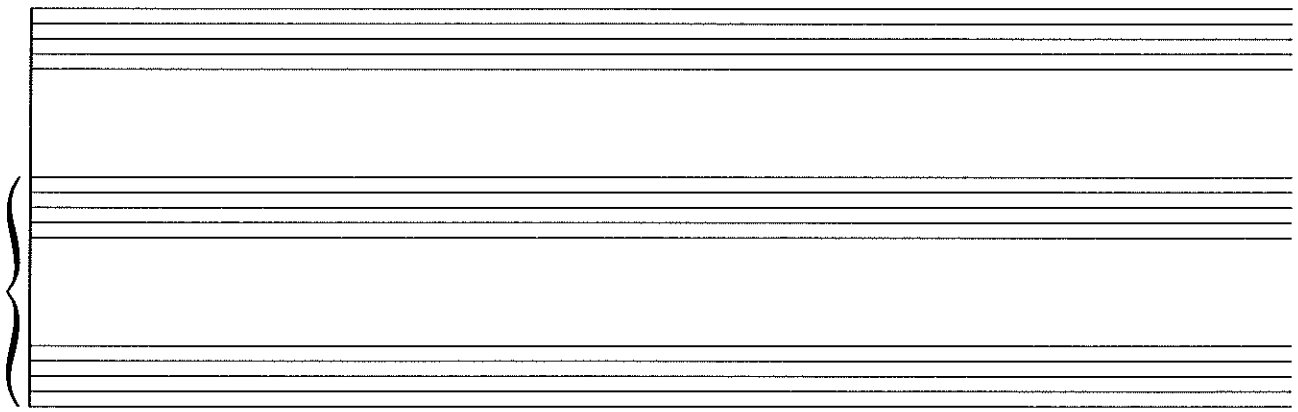
F. Harmonize and set for piano.



G. Invent a piano accompaniment for the following melody.







BURKHART, TUREK, AND WENNERSTROM EXERCISES FOR HARMONIC ANALYSIS

Burkhart

p. 311, mm. 1–4.
p. 318, mm. 1–4.
p. 304, mm. 60–65.
p. 305, mm. 87–96.

Turek

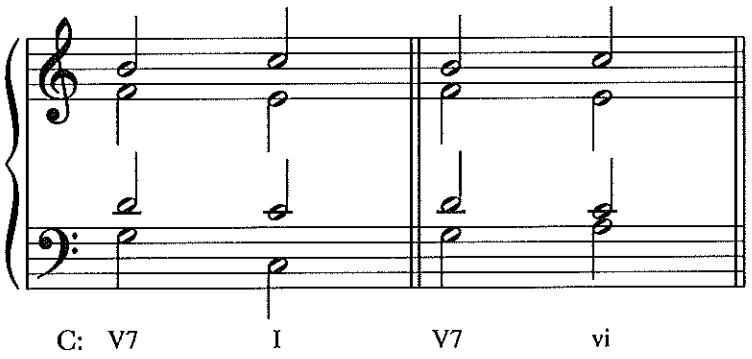
p. 131, mm. 1–8.
p. 263, mm. 1–8.
p. 309, mm. 1–16.
p. 338, mm. 73–80.
p. 214, mm. 1–12.

Wennerstrom

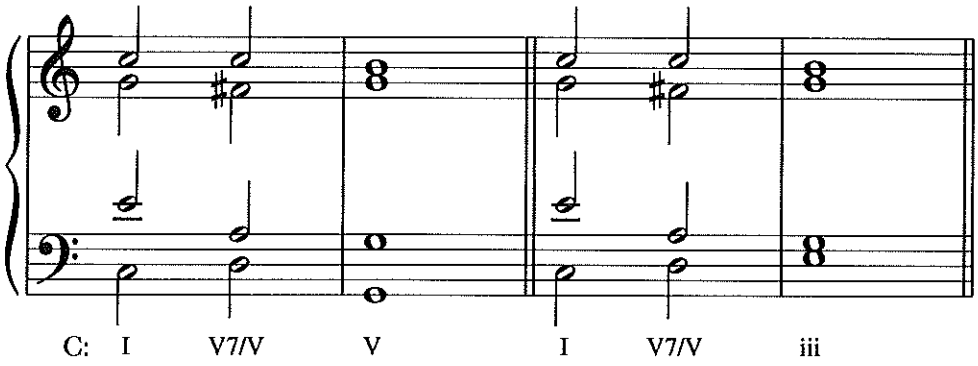
p. 176, mm. 9–12.
p. 120, mm. 1–8.
p. 284, mm. 1–8.

DECEPTIVE RESOLUTIONS OF SECONDARY DOMINANTS

The most natural resolution of V7 is to I, as in the authentic cadence. V7, however, frequently resolves to vi, a harmonic motion that occurs in the deceptive cadence. In the former progression, the root motion is a perfect fifth downward; in the latter, a second (major or minor, depending on the mode) upward.



Similarly, the secondary dominant naturally resolves to a chord whose root is a perfect fifth below its own. It may, however, resolve *deceptively* (in the manner of an ordinary dominant) by root motion ascending by step.



Notice that the third of the secondary dominant, which has a leading tone function, still resolves upward by step in the deceptive resolution.

EXERCISES

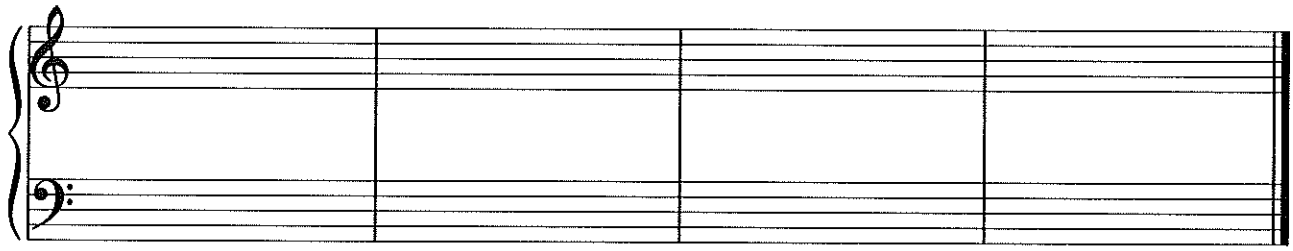
A. Harmonize and set for piano. At least one deceptive resolution of a secondary dominant should be employed.



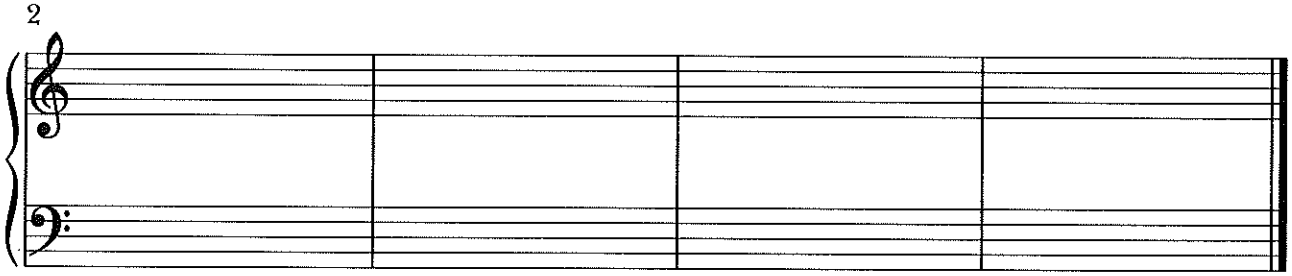
B. Using the given progressions, invent four-measure phrases for soprano, alto, tenor, and bass.

1. C: $\frac{3}{8}$ I–V^6_4–I6–V^6_5/V–iii6–vi7–V7/V–V
2. f: $\frac{2}{2}$ V^6_5–i–V^6_5/III–i6–iv–V–i
3. D: $\frac{2}{4}$ I–vii°6–V^6_5/IV–ii6–V^6_5/V–V–I
4. e: $\frac{4}{4}$ i–i6–V^6_5/iv–iv–V7/V–V–VI

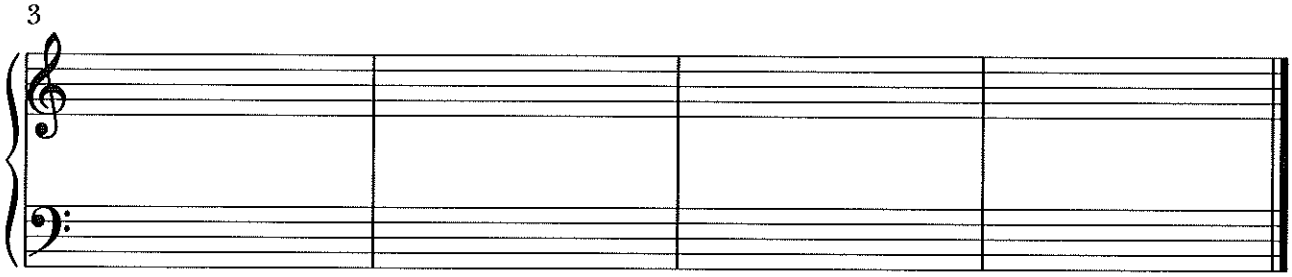
1



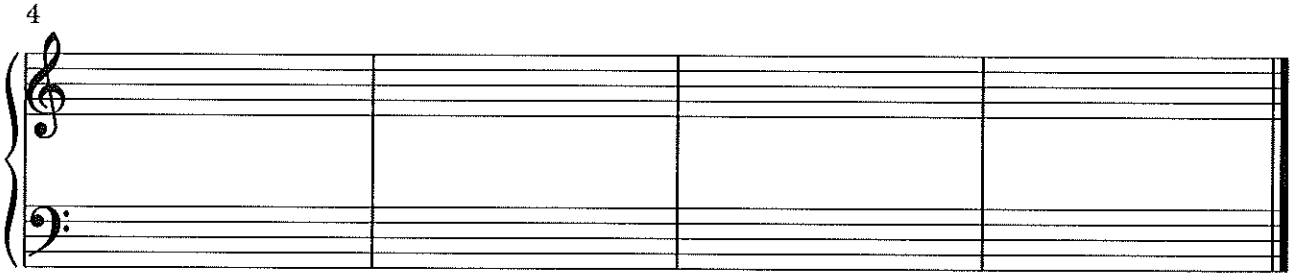
2



3

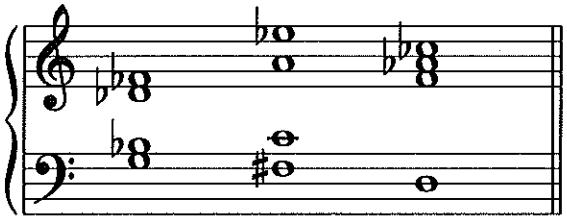


4



18 Secondary Diminished Seventh Chords

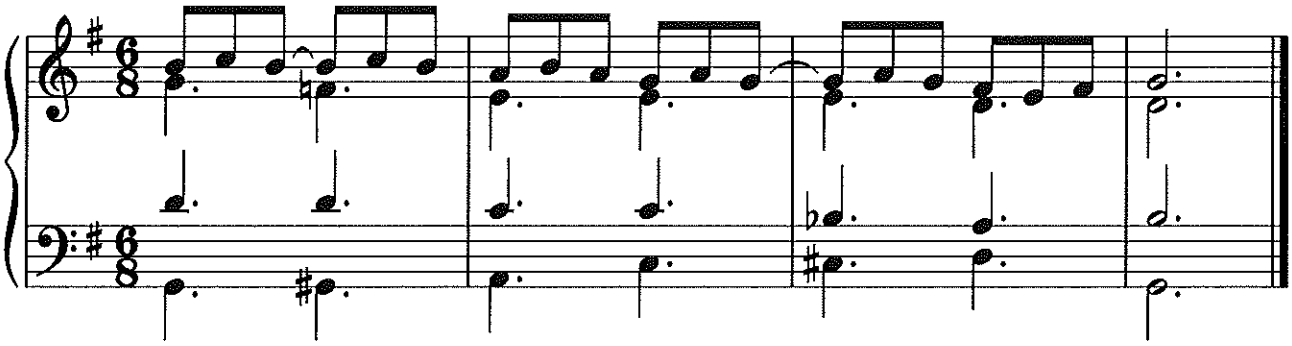
A diminished seventh chord is constructed by adding to a diminished triad a pitch that is a diminished seventh above its root (vii°7 in the minor key is a diatonic diminished seventh chord). The result is a sonority with four pitches stacked in minor thirds.



Example 7 (p. 173) in Chapter 14 shows a vii°7 resolving to i. In that resolution, the chord functioned as a dominant.

Since V7–i may be substituted by vii°7–i without causing a change in harmonic function, it follows that, for example, V7/V–V may similarly be substituted by vii°7/V–V. Thus, secondary diminished seventh chords may be used as secondary dominants.

Study the following examples carefully:





B♭: I vii°$^{4}_{3}$/IV IV6 iii6 vii°$^{6}_{5}$/vi vi vii°7/V V I

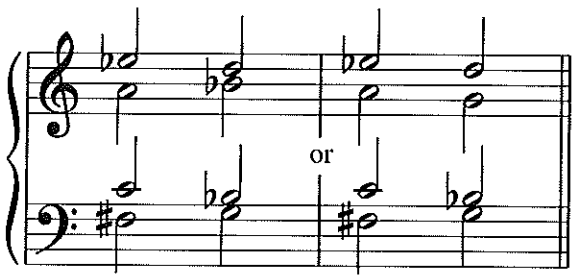


g: i vii°7/VII VII vii°7/VI VI vii°7/V V I

Note:

a. Each diminished seventh chord is built on a root that is a *minor second* below the root of its chord of resolution. Thus, each diminished seventh chord is in a leading tone relationship with its successor.

b. In resolution, the root of the diminished seventh chord moves *upward* by step, and its seventh moves *downward* by step. The remaining pitches resolve in either of the following ways:



Of the two resolutions, the first is stronger in terms of line, because the two diminished fifths are more naturally resolved. The second, however, gives a better doubling in the successive chord.

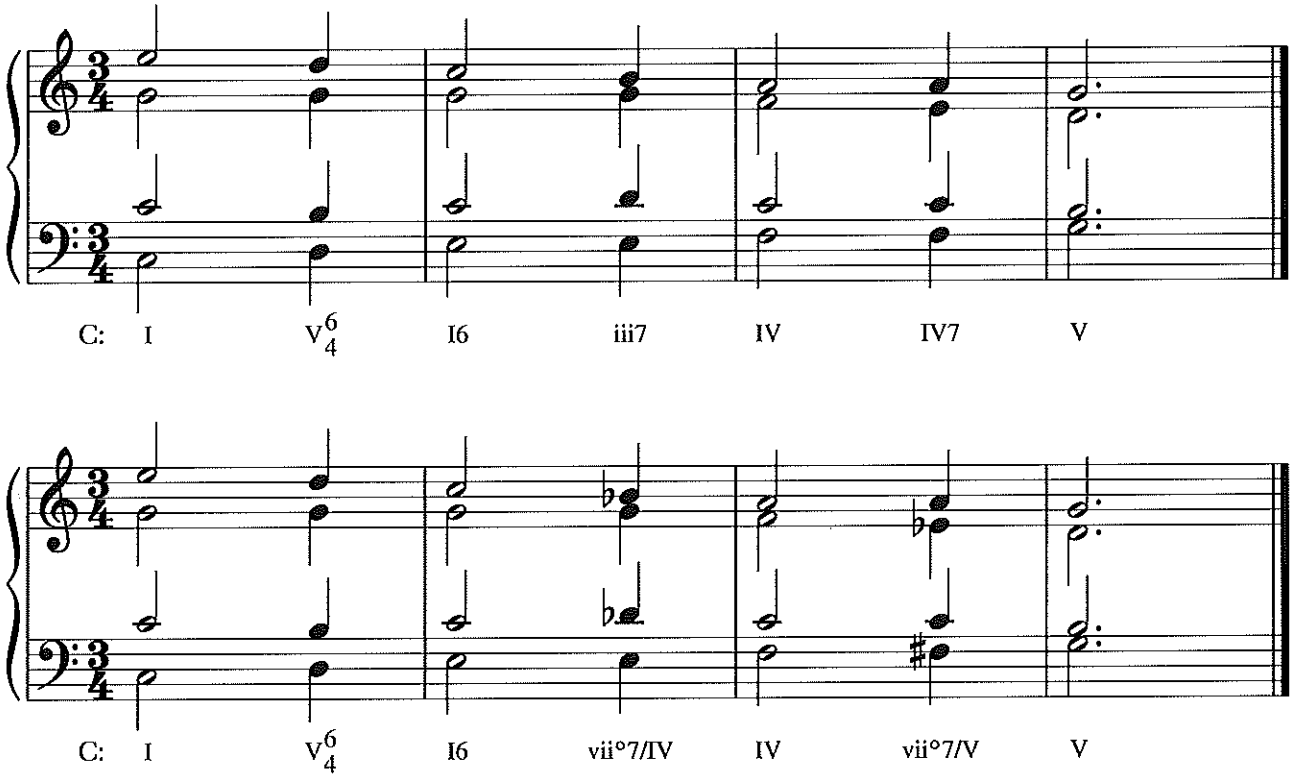
c. Diminished seventh chords are used freely in root position, first inversion, and second inversion. The third inversion is less frequently used because it normally resolves to a $^{6}_{4}$ chord. (As a means of modulation this resolution will be found to be valuable.)

d. These examples are variants of those on page 210 ("Secondary Dominants") and should be compared with them.

THE FUNCTION OF SECONDARY DIMINISHED SEVENTH CHORDS

Like secondary dominants, secondary diminished seventh chords have the effect of strengthening the harmonic movement to the chords that follow them. They produce this effect first by the resolution of the secondary leading tone to the root of the subsequent chord, and second, by the resolution of the seventh to the fifth of that chord. Thus, secondary diminished seventh chords also tonicize major or minor triads that follow them and may be said to have a dominant function.

The two examples below, variants of the examples on page 211 of the previous chapter, represent phrases that terminate with a half cadence. Once again, they are identical with the exception of the chromatic pitches in measures two and three in the second example.



Note:

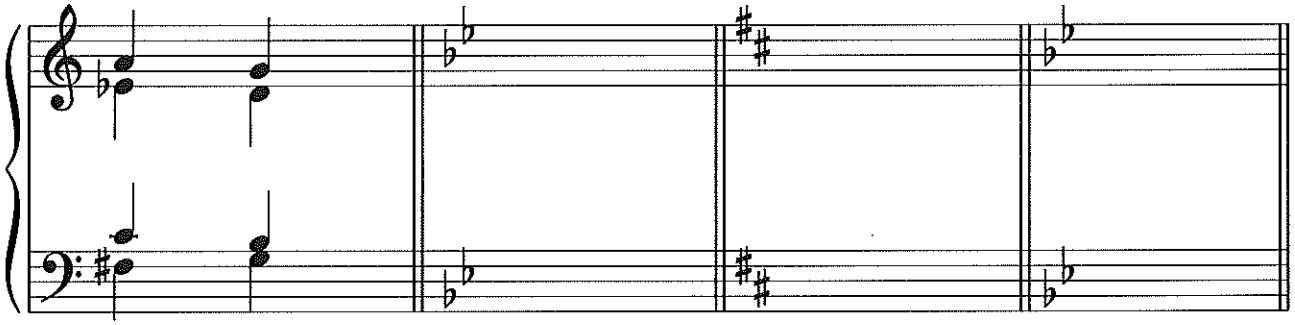
a. In the second example, the movement to IV is strengthened by the alteration of iii7 to vii°7/IV, and the movement to V is similarly strengthened by changing IV7 to vii°7/V.

b. The half cadence in the second example creates a greater sense of repose than that in the first, because V is preceded by its secondary diminished seventh, which has a dominant function.

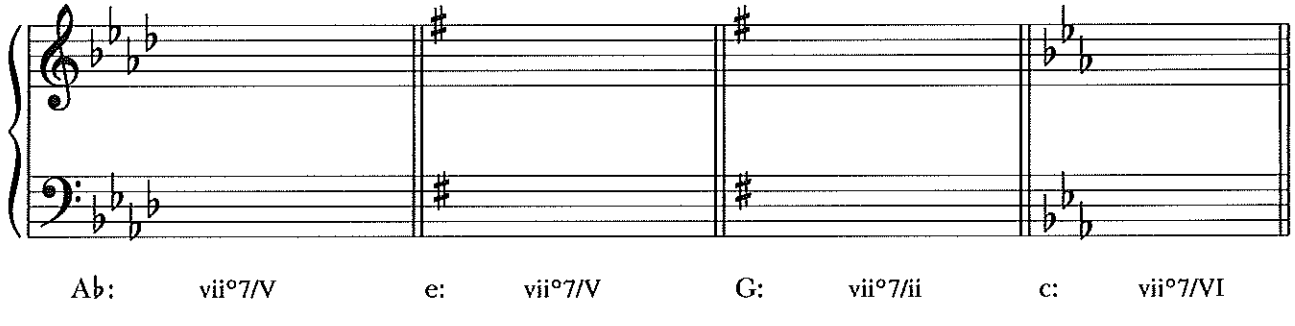
The exercises that follow are related to the materials introduced on pages 223–225. Irregular resolutions of secondary diminished seventh chords are covered on pages 233–234.

EXERCISES

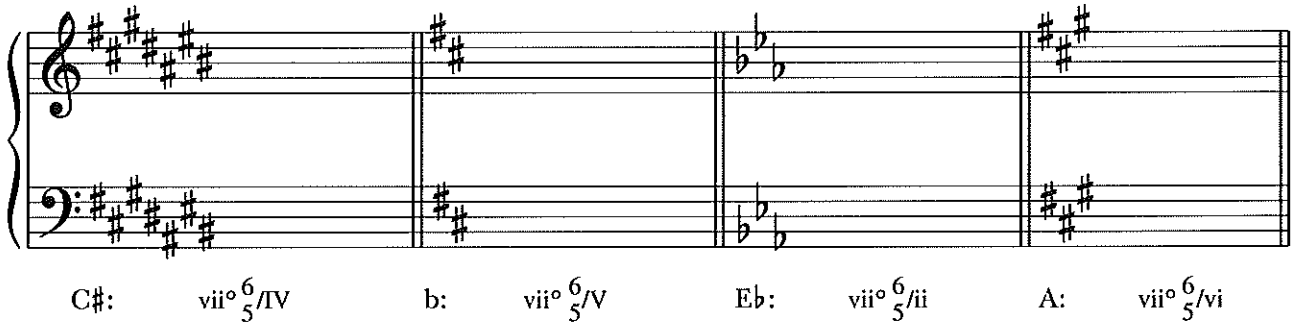
A. Write, and resolve, the following secondary diminished sevenths.



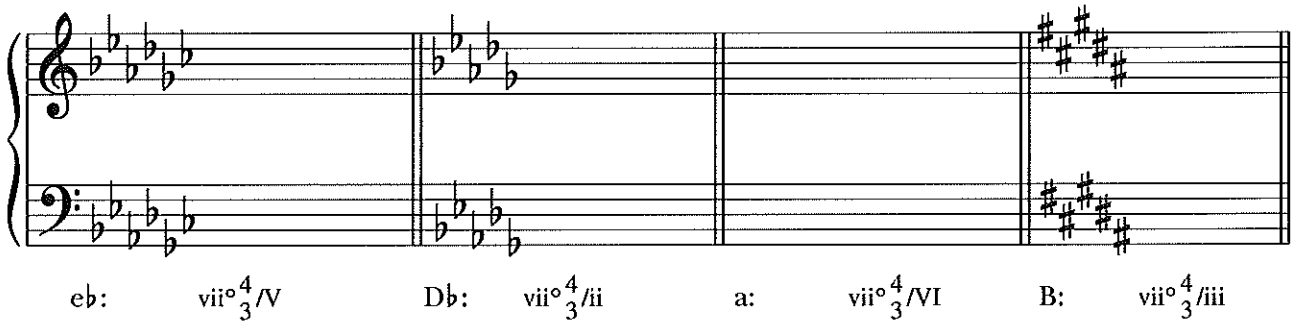
C: vii°7/V B♭: vii°7/iii D: vii°7/vi g: vii°7/iv



A♭: vii°7/V e: vii°7/V G: vii°7/ii c: vii°7/VI



C♯: vii°$^{6}_{5}$/IV b: vii°$^{6}_{5}$/V E♭: vii°$^{6}_{5}$/ii A: vii°$^{6}_{5}$/vi

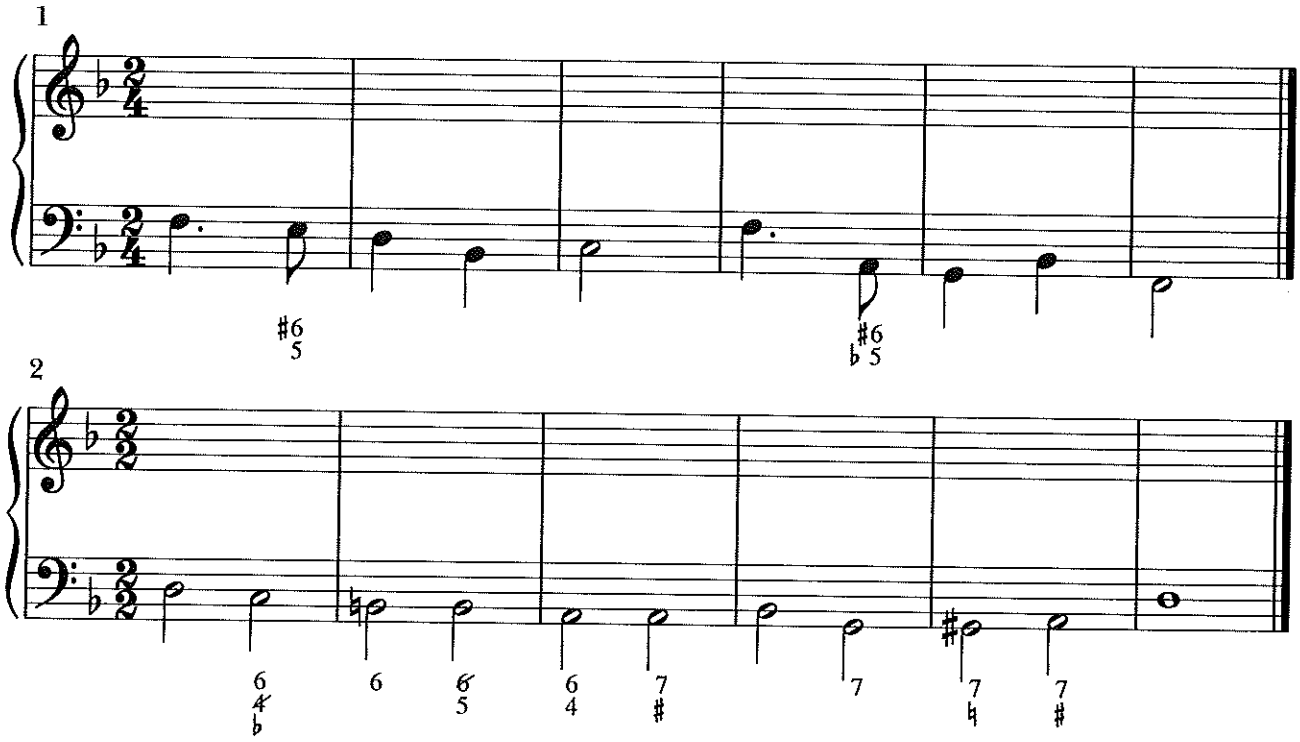


e♭: vii°$^{4}_{3}$/V D♭: vii°$^{4}_{3}$/ii a: vii°$^{4}_{3}$/VI B: vii°$^{4}_{3}$/iii

B. Add parts for alto and tenor.

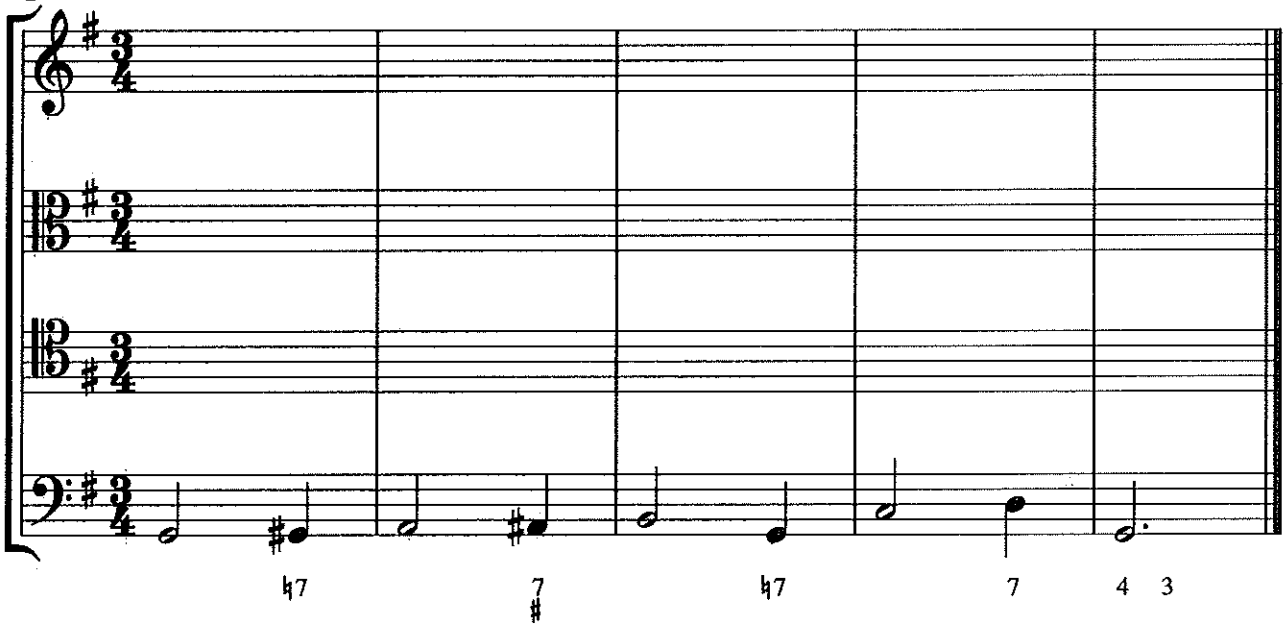


C. Add parts for soprano, alto, and tenor.

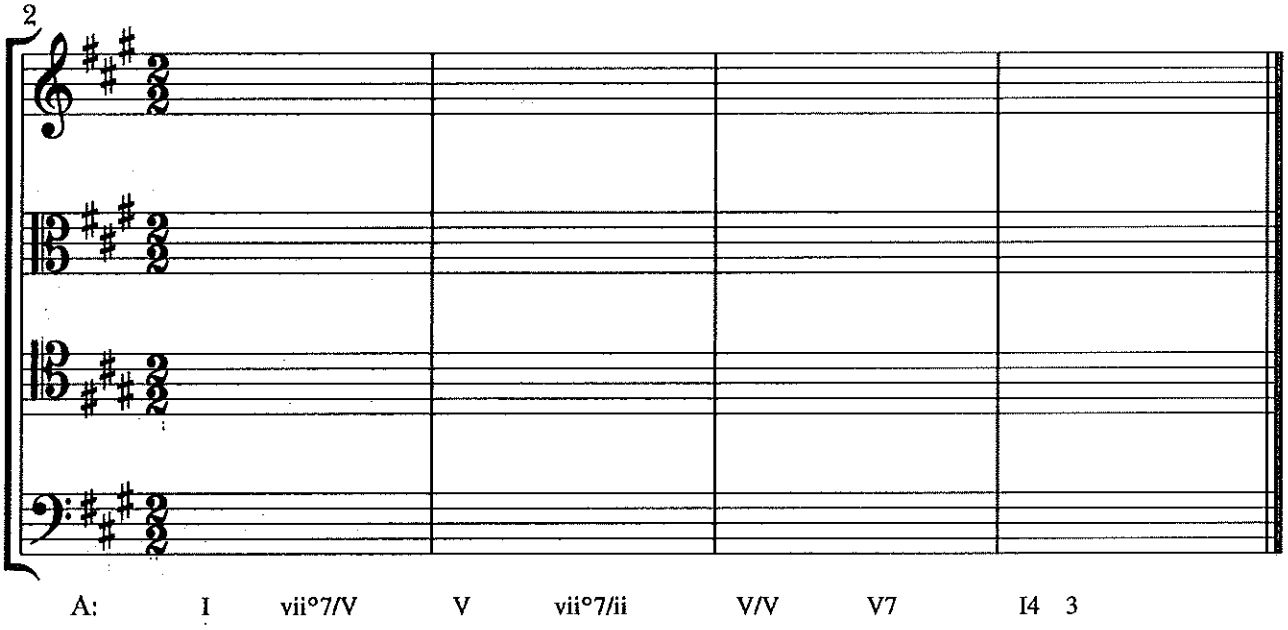


D. Harmonize the following phrases in open score, making appropriate use of nonchord tones.

1



2



E. Add parts for alto, tenor, and bass.



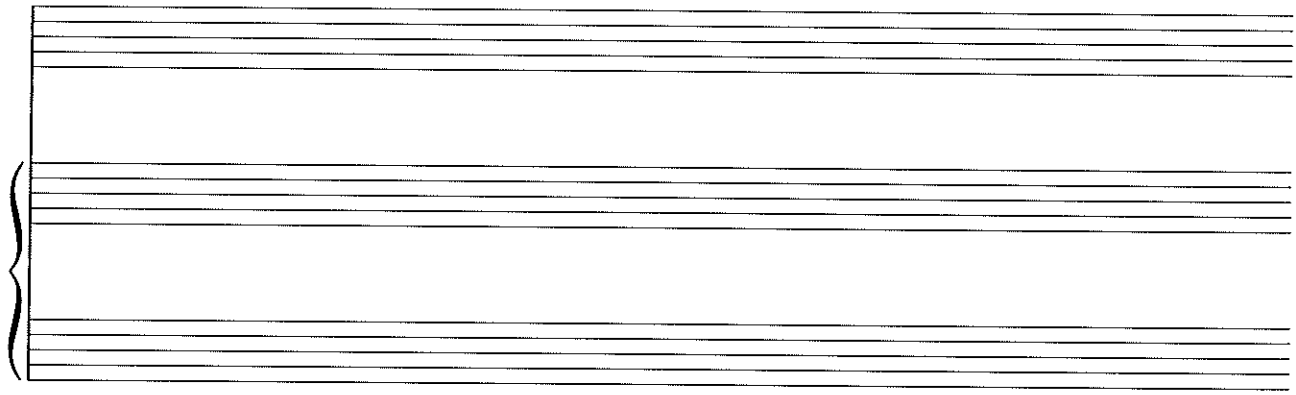
* indicates that a secondary diminished seventh should be used

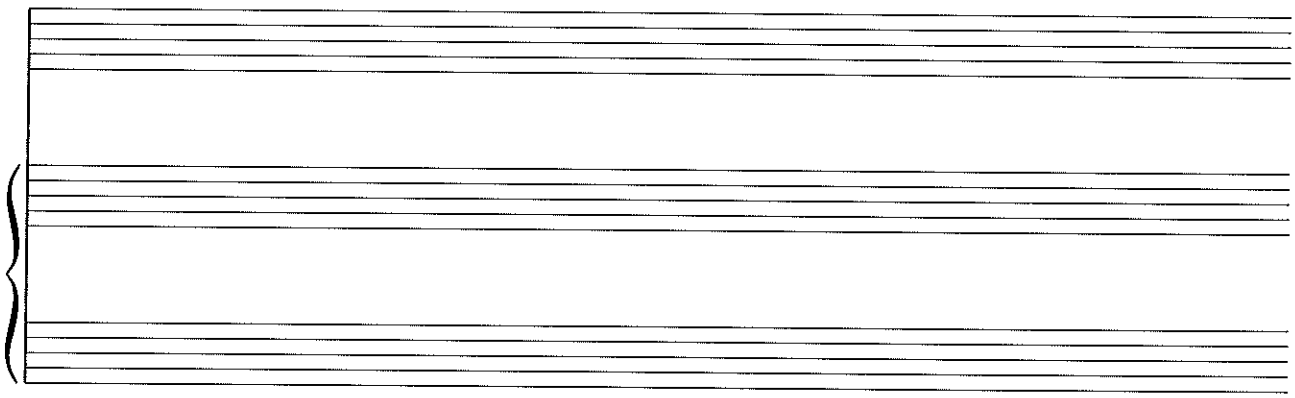
F. Harmonize and set for piano, making use of secondary diminished seventh chords.



G. Invent a piano accompaniment for the following melody.







BURKHART, TUREK, AND WENNERSTROM EXERCISES FOR HARMONIC ANALYSIS

Burkhart

p. 190, mm. 64–71.
p. 192, mm. 103–10.
pp. 316–17, mm. 1–15, 32–49.

Turek

p. 324, mm. 1–4.
p. 334, mm. 1–8.
p. 282, mm. 284–95.
pp. 312–13, mm. 9–26.

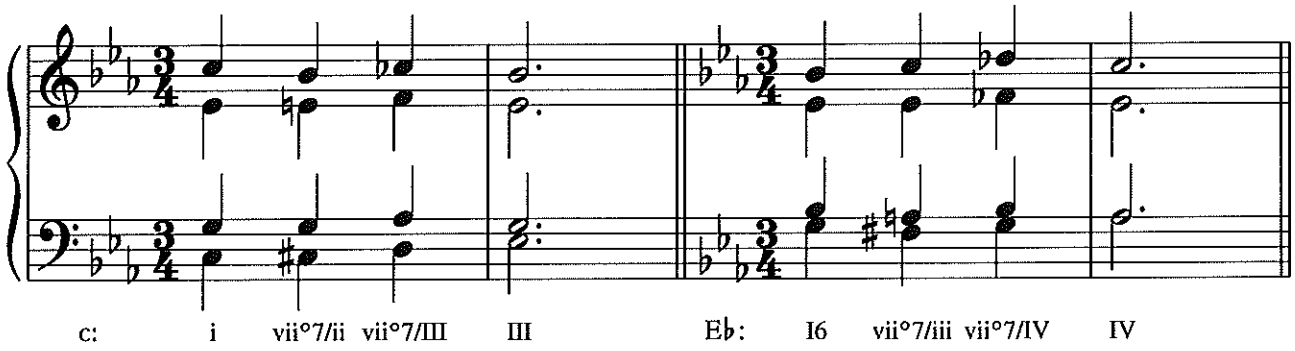
Wennerstrom

p. 312, mm. 1–8.
p. 237, mm. 68–76.
p. 316, mm. 17–29.

IRREGULAR RESOLUTIONS OF DIMINISHED SEVENTH CHORDS

Any resolution of the diminished seventh chord that does not exploit the chord's dominant function may be called irregular. The two most common uses of the chord in this respect are

1. The resolution of a diminished seventh chord to another diminished seventh chord.



Note:

The movement from one diminished seventh to another involves similar motion of all voices by half steps.

2. The diminished seventh chord as a "passing" chord.



Note:

In this example, the resolution of the seventh of the diminished seventh chord is passive.

EXERCISES

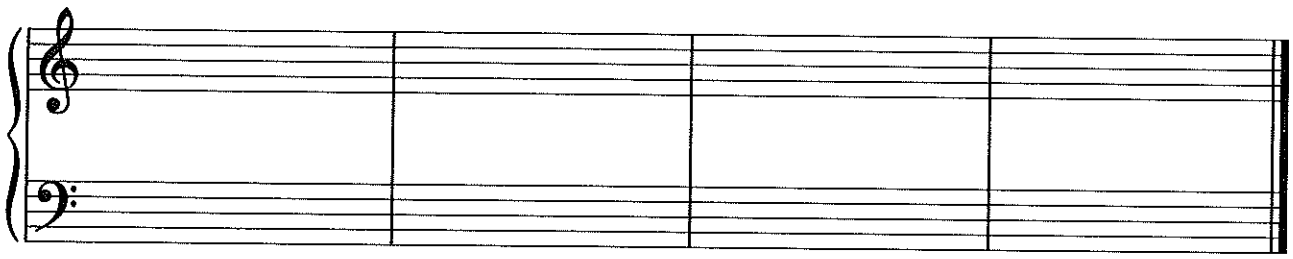
A. Add parts for soprano, alto, and tenor.



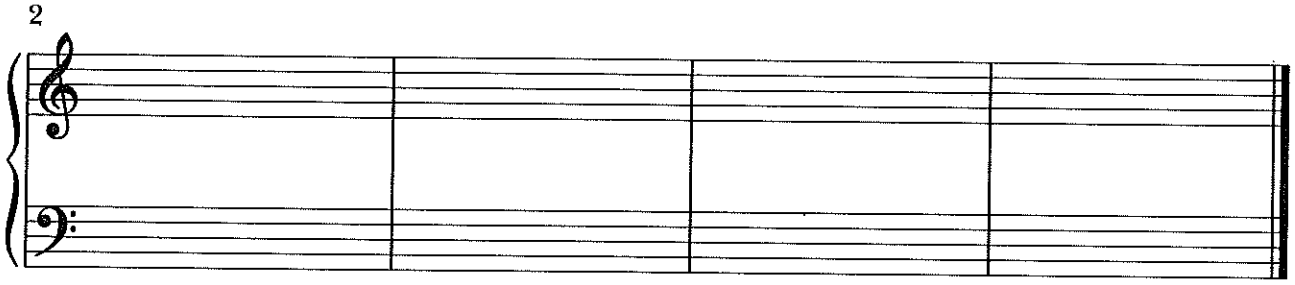
B. Using the given progressions, invent four-measure phrases for soprano, alto, tenor, and bass.

1. A♭: $\frac{3}{4}$ I–vii°$^{6}_{5}$/vi–vi–vii°$^{6}_{5}$/IV–IV–I6–vii°6–I
2. b: $\frac{2}{2}$ i–III–vii°/iv–iv–vii°7/V–V–iv6–i
3. A: $\frac{4}{4}$ I–ii7–vii°7/iii–I6–IV–V7–vii°7/vi–IV6–V$^{6}_{5}$/V–V–I
4. c: $\frac{3}{4}$ i–i6–vii°7/iv–iv–vii°7/V–V–VI

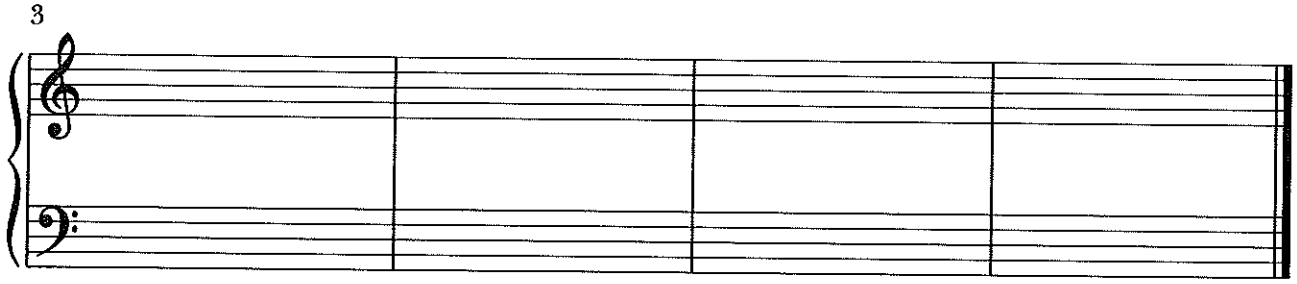
1



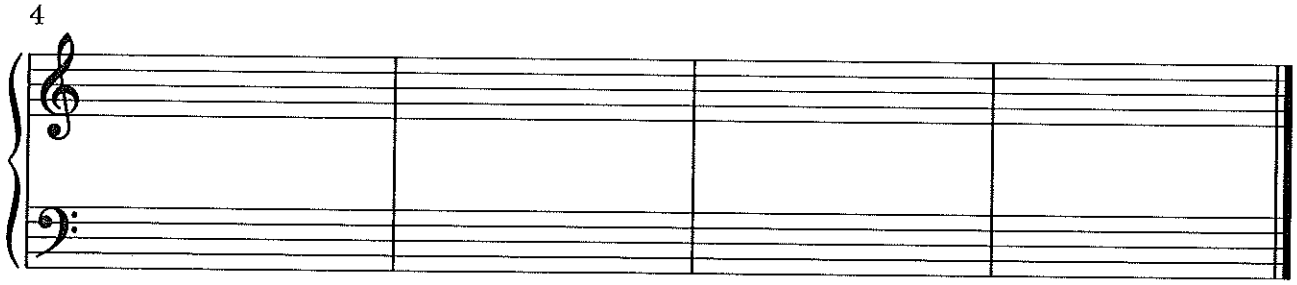
2



3

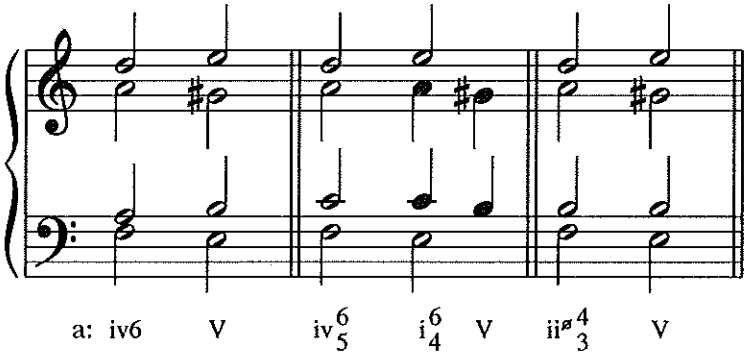


4

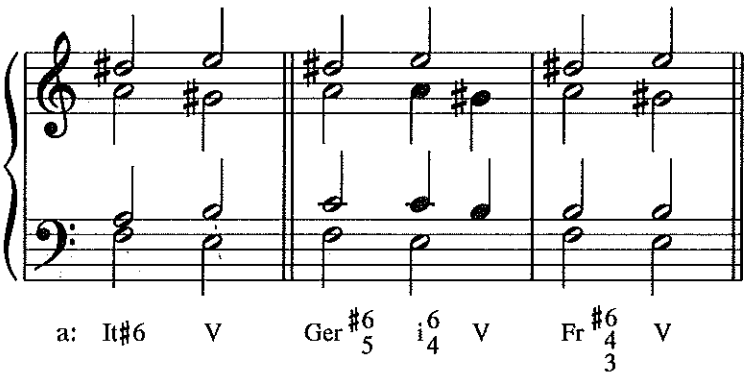


19 Augmented Sixth Chords

The progressions below show the three common approaches to the dominant in the key of a minor.



In each of the above progressions, the movement to V can be strengthened by rewriting d^2 as $d\sharp^2$, so that there is a melodic movement of only a half step to the dominant pitch. The progressions will now be



Each of the chords that precedes V is known as an augmented sixth chord, because in each case there exists an interval of an augmented sixth between the bass and soprano voice. The first is the *Italian sixth,* an altered iv6 chord; the second is the *German sixth,* an altered iv$^6_5$ chord; the third is the *French sixth,* an altered ii$^{\circ}{}^4_3$ chord.

Note:

a. The interval of the augmented sixth resolves outward to the octave in each progression.

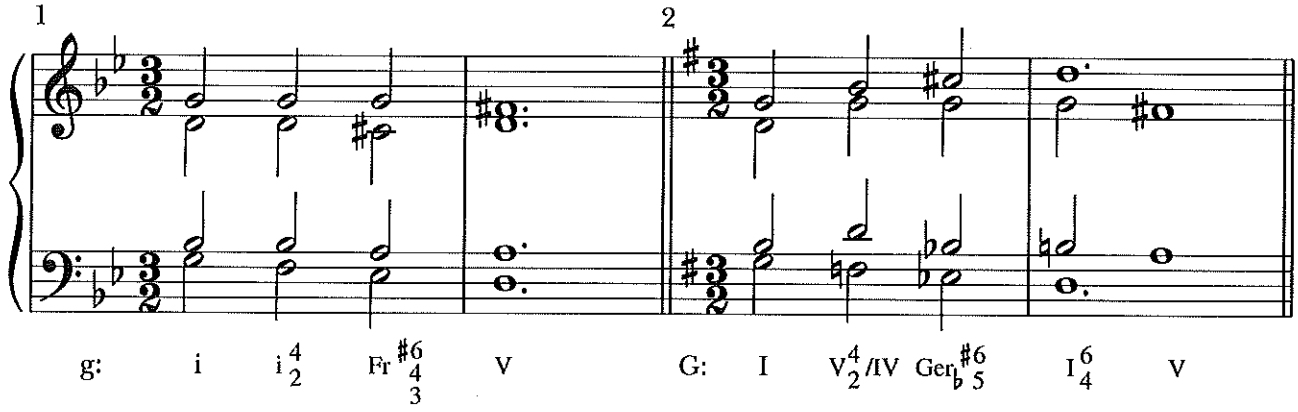
b. In the Italian sixth, the fifth is doubled.

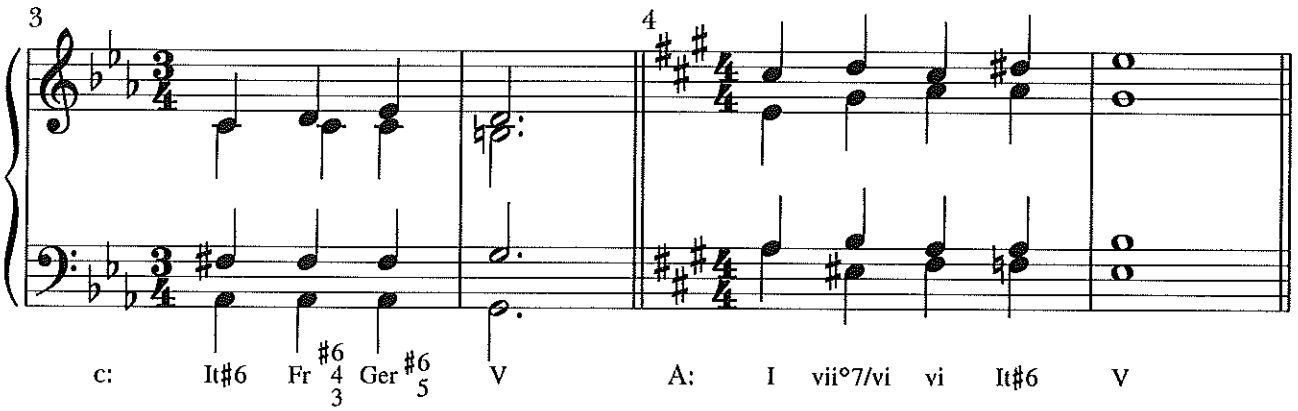
c. All three of the augmented sixth chords contain a pitch that is a *major third above the bass.*

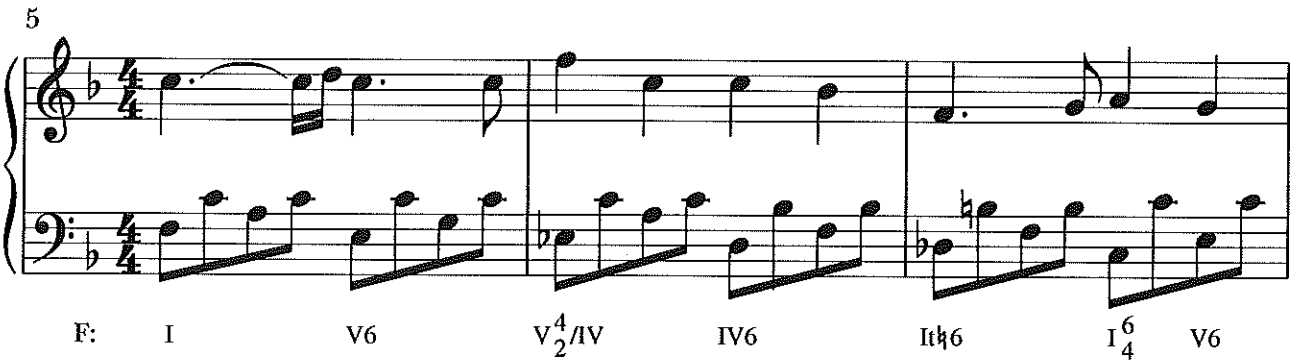
d. The German sixth also contains a pitch that is a *perfect fifth above the bass.*

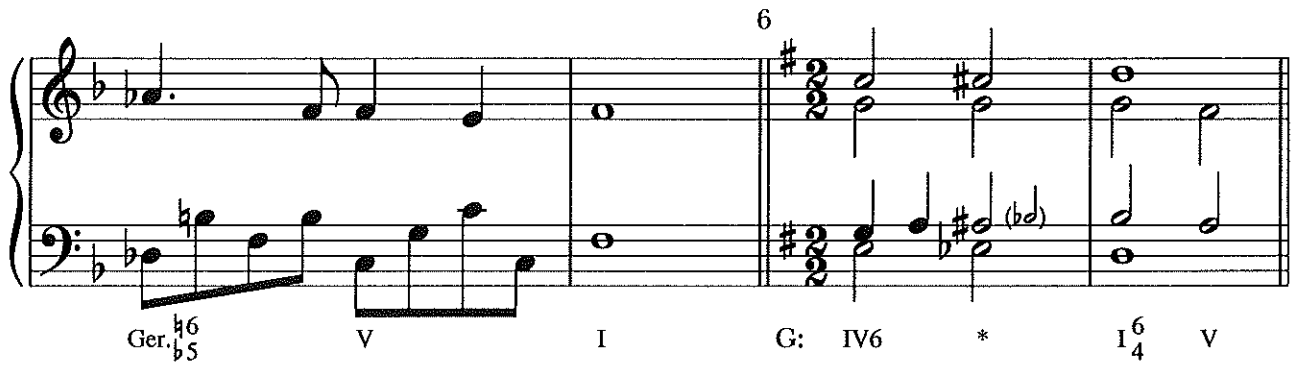
e. The French sixth also contains a pitch that is an *augmented fourth above the bass.*

Study the following examples carefully. They show the augmented sixth chords in different settings in both major and minor keys.









Note:

a. The interval of the augmented sixth may occur between the bass and any upper voice.

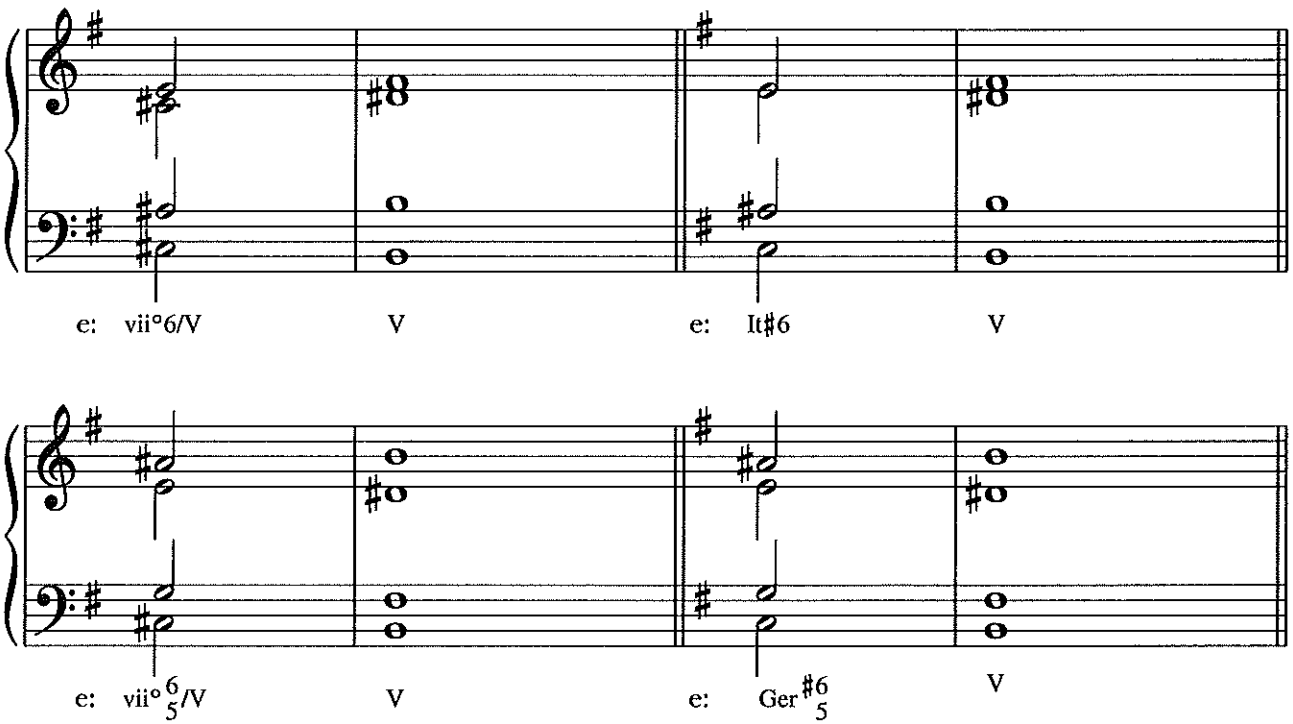
b. The direct resolution of the German sixth to V involves parallel fifths (see 3). This is an exceptional, but accepted, procedure.

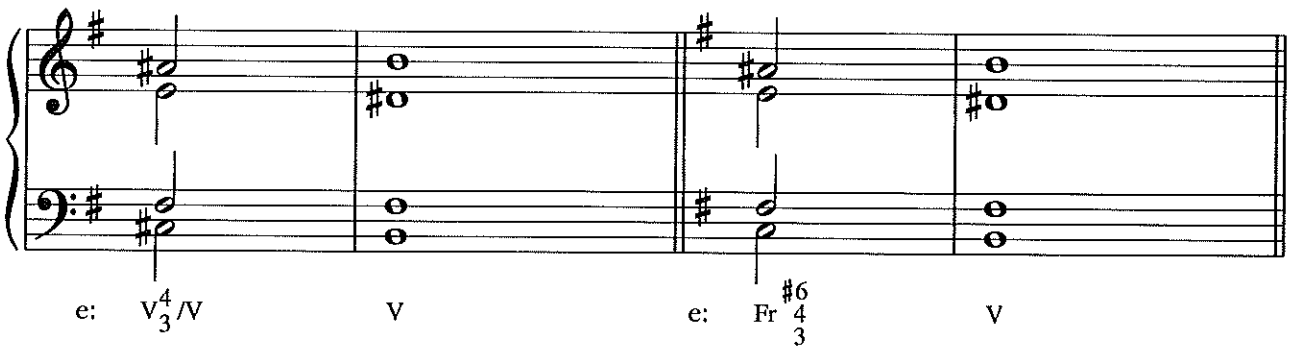
c. In example 6, the German sixth is respelled with an A♯ instead of a B♭ (see *) so that the tenor's resolution to b is more natural.

d. Augmented sixth chords are used in both major and minor keys.

THE FUNCTION OF AUGMENTED SIXTH CHORDS

The diatonic scale degrees from which the augmented sixth chords are derived give relatively little information about the chords' function. A comparison of these chords, however, with secondary dominants and diminished sevenths helps to reveal their true role in the chromatic vocabulary.





Note:

a. The Italian sixth may be viewed as vii°6/V with a lowered third.

b. The German sixth may be viewed as $vii^{\circ}{}_{5}^{6}/V$ with a lowered third.

c. The French sixth may be viewed as V_3^4/V with a lowered fifth.

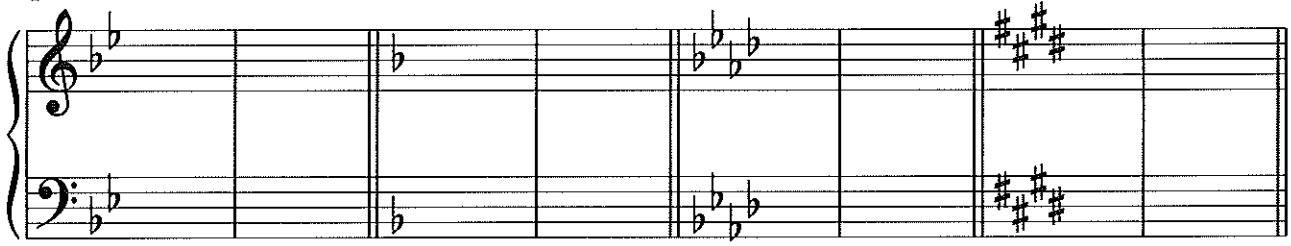
d. In each case, the augmented sixth chord contains those pitches that give the secondary dominants and secondary diminished sevenths their dominant function, that is, root, leading tone, and seventh. The augmented sixth chords, therefore, are functionally very similar to secondary dominants and secondary diminished sevenths.

Other uses of augmented sixth chords are covered on pages 247–248.

EXERCISES

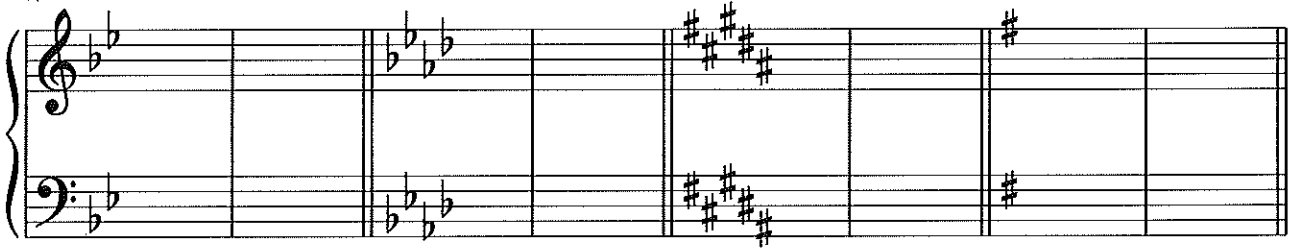
A. Write, and resolve to V, the following augmented sixth chords.

1



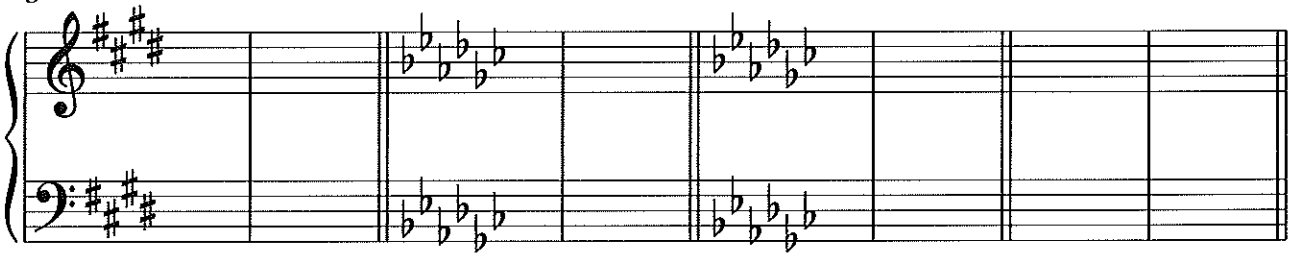
g: Ger V d: It V A♭: Fr V E: Fr V

2



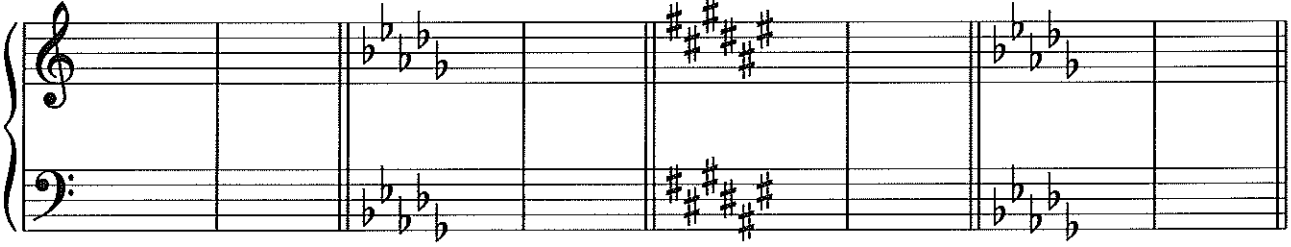
B♭: It V f: Ger V B: Fr V G: Ger V

3



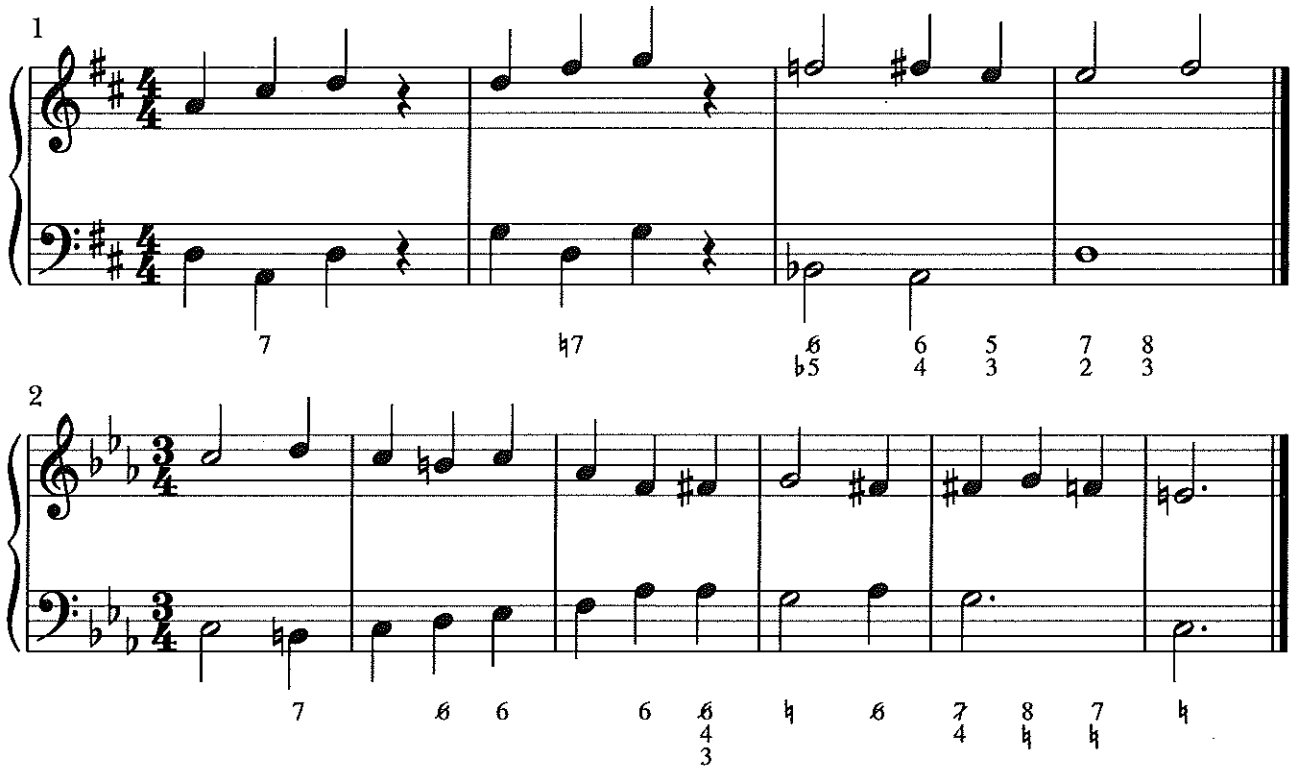
c♯: Fr V e♭: It V G♭: It V a: Ger V

4

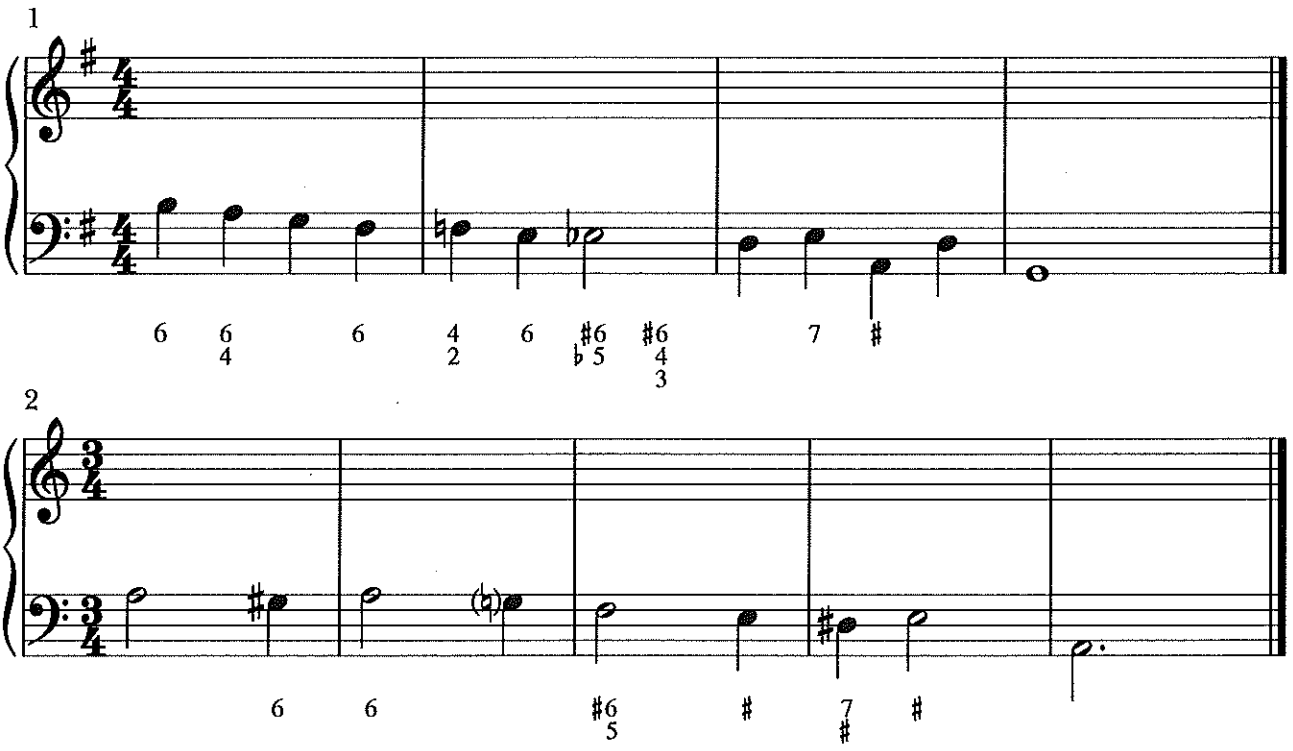


C: Fr V B♭: Ger V F♯: It V D♭: Fr V

B. Add parts for alto and tenor.

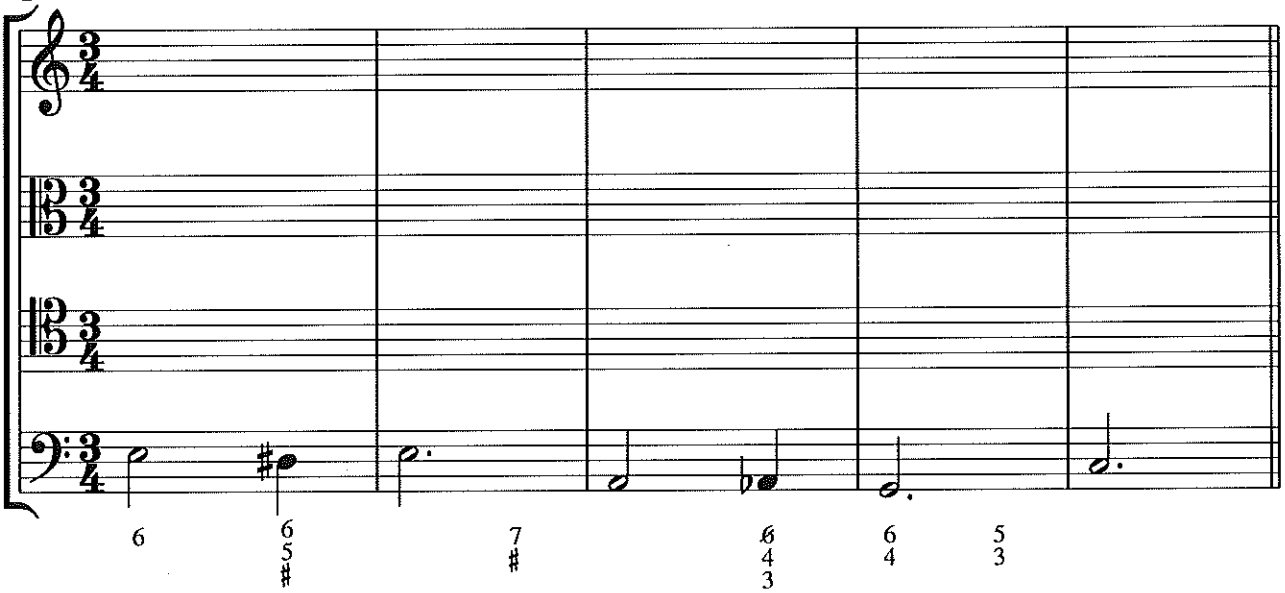


C. Add parts for soprano, alto, and tenor.

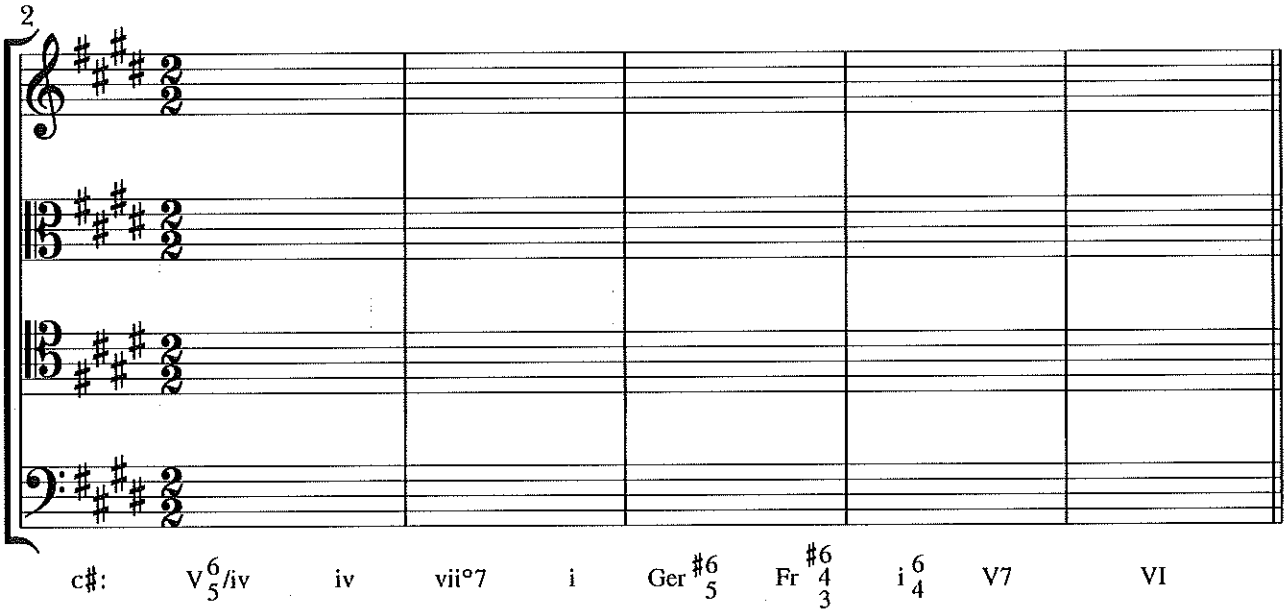


D. Harmonize the following phrases in open score, making appropriate use of nonchord tones.

1



2



E. Add parts for alto, tenor, and bass.



* indicates that an augmented sixth chord should be used

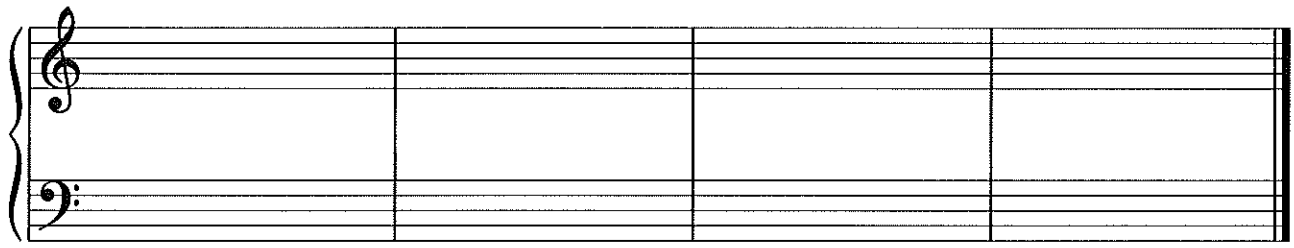
F. Harmonize and set for piano, making use of augmented sixth chords.



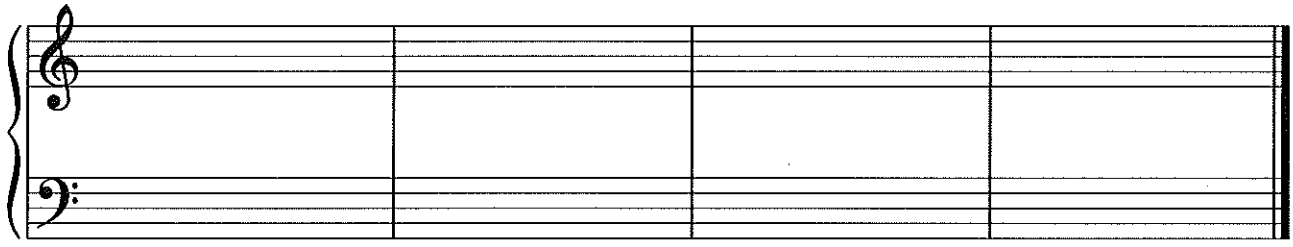
G. Using the given progressions, invent four-measure phrases for soprano, alto, tenor, and bass.

1. f: $\frac{4}{4}$ i–V6–v6–VI–Ger$^{\sharp 6}_{5}$–i$^{6}_{4}$–V–i
2. D: $\frac{3}{4}$ V6–I–V7/IV–IV–IV6–It$^{\sharp 6}$–V
3. a: $\frac{4}{4}$ i–vii°6–i6–iv–V$^{6}_{5}$/V–V–Fr$^{\sharp 6}_{4}$–V
4. C: $\frac{2}{2}$ I–vi7–ii7–V–It$^{\sharp 6}$–V–vi

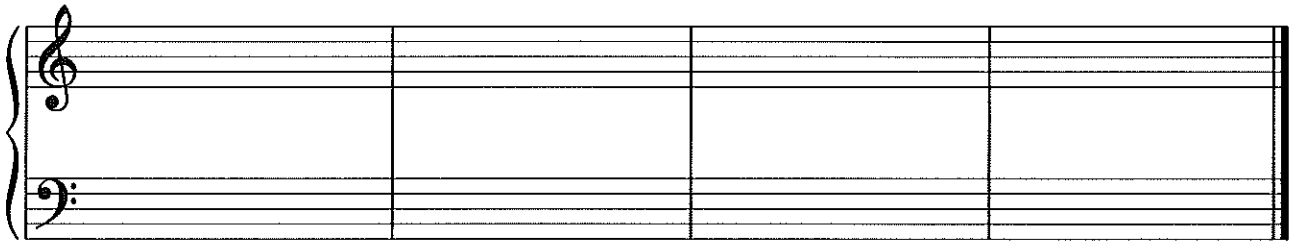
1



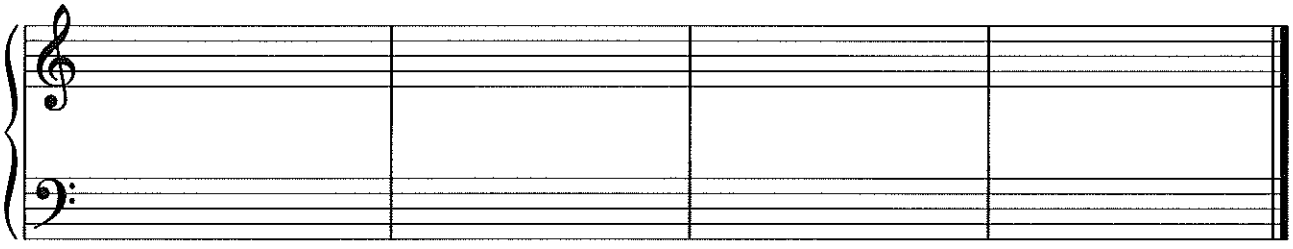
2



3



4



BURKHART, TUREK, AND WENNERSTROM EXERCISES FOR HARMONIC ANALYSIS

Burkhart

p. 182, mm. 1–4.
pp. 241–42, mm. 240–52.
p. 395, mm. 1–13.
p. 263, mm. 1–17.

Turek

p. 316, mm. 40–46.
p. 215, mm. 31–40.
p. 319, mm. 28–34.
p. 259, mm. 1–12.

Wennerstrom

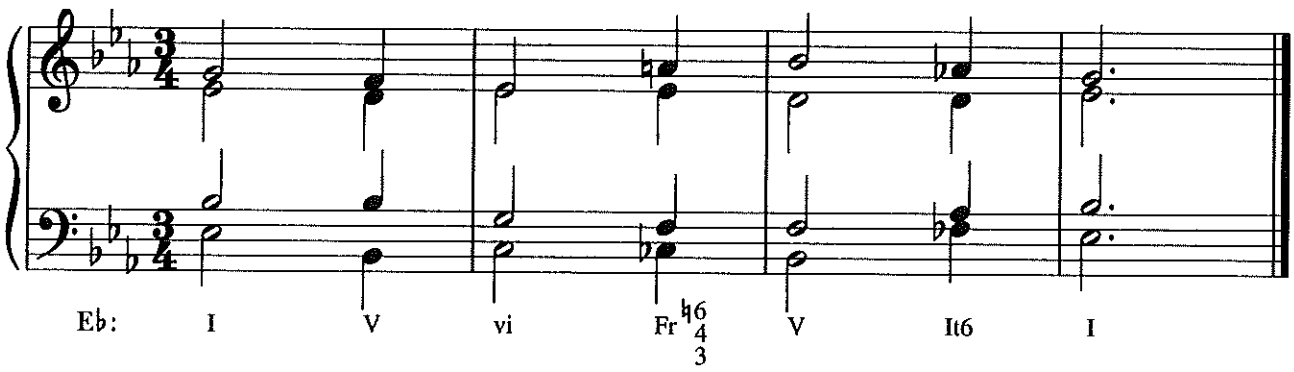
p. 286, mm. 1–4.
p. 330, mm. 1–8.
p. 402, mm. 26–33.
p. 327, mm. 1–8.

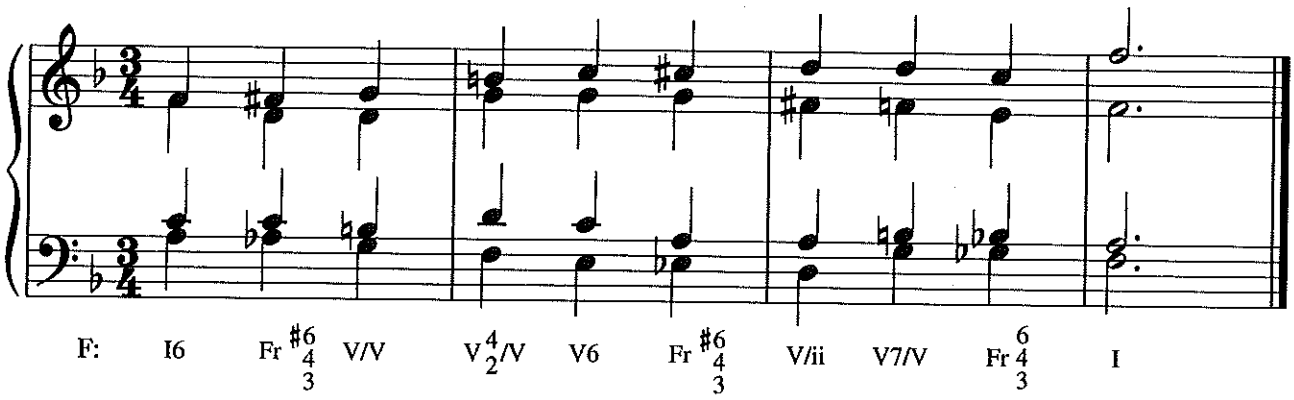
OTHER USES OF AUGMENTED SIXTH CHORDS

Although augmented sixth chords usually occur as pre-dominant sonorities, they may also be used to precede major triads on other degrees of the scale. (They very rarely resolve to minor triads.) Under these circumstances, the augmented sixth chords are obviously no longer altered forms of iv and ii; instead, their analysis depends upon the scale degree of the chord of resolution.

If an augmented sixth chord does not resolve to V, it is most commonly used in resolution to I as an altered chord with a dominant function, or possibly as a pre-secondary dominant sonority.

The following examples show some of these uses:





Note:

a. The Italian sixth in the first example is built on the seventh scale degree.

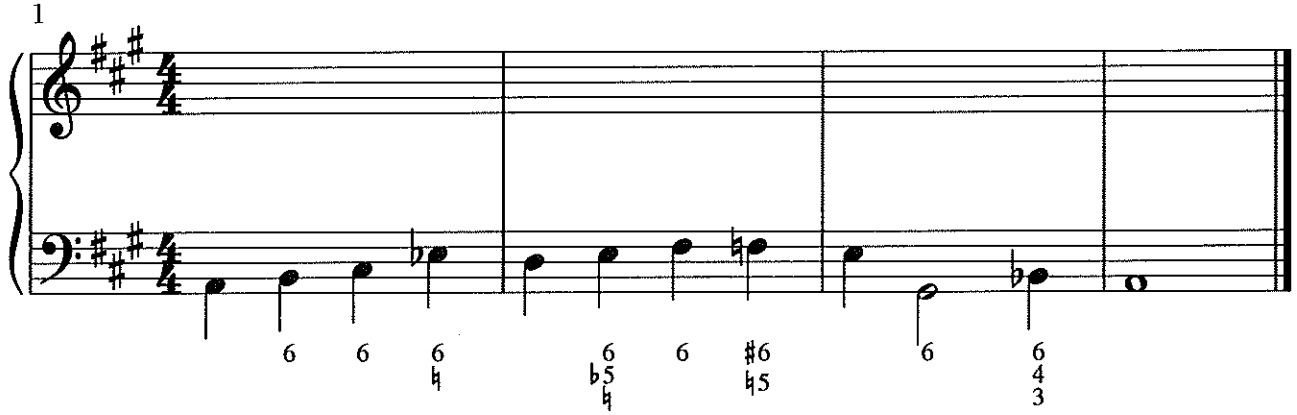
b. In the second example, the three French sixth chords are built on the sixth, second, and fifth scale degrees, respectively.

c. Regardless of the scale degree, the resolution of augmented sixth chords does not change.

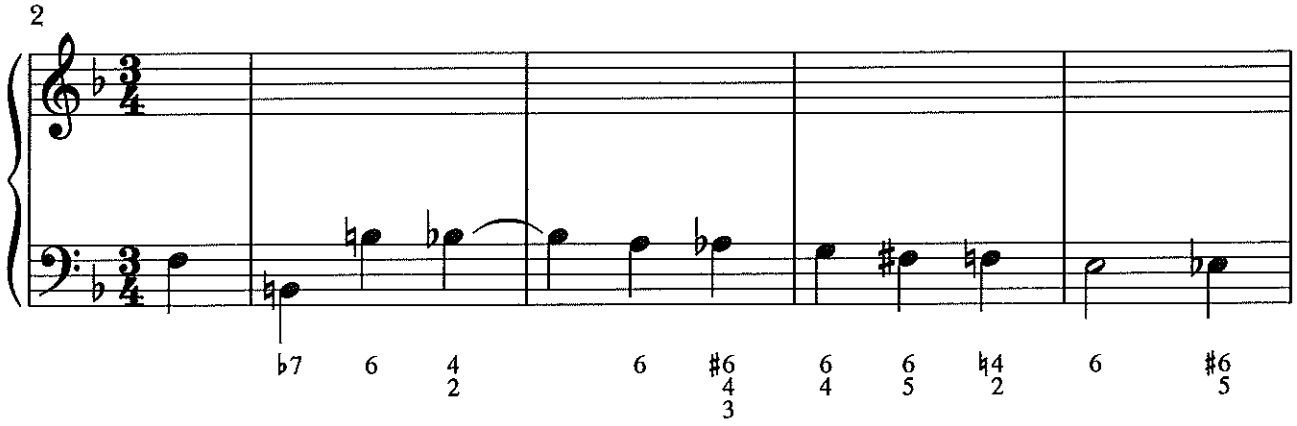
EXERCISES

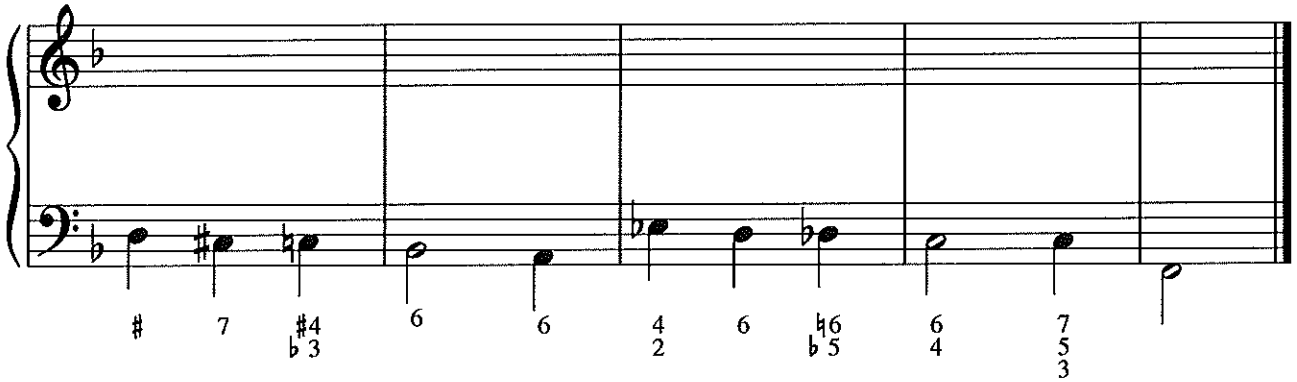
A. Add parts for soprano, alto, and tenor.

1



2





B. Harmonize and set for piano. + indicates that a chromatic chord should be used.



20 Borrowed Chords

All of the chromatic chords that have so far been encountered have been functional in nature. In other words, they have been used to complete a specific harmonic task, usually that of tonicizing a major or minor triad, or emphasizing a movement to the dominant.

Some chromatic chords, however, are used for the purpose of creating harmonic color and thus are said to be coloristic rather than functional. These chords nearly always appear in the major key and come from the parallel minor. They are referred to as *borrowed chords*.

Diatonic triads



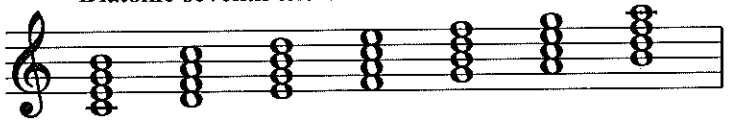
M m m M M m d

Borrowed triads



m d M m m M M

Diatonic seventh chords



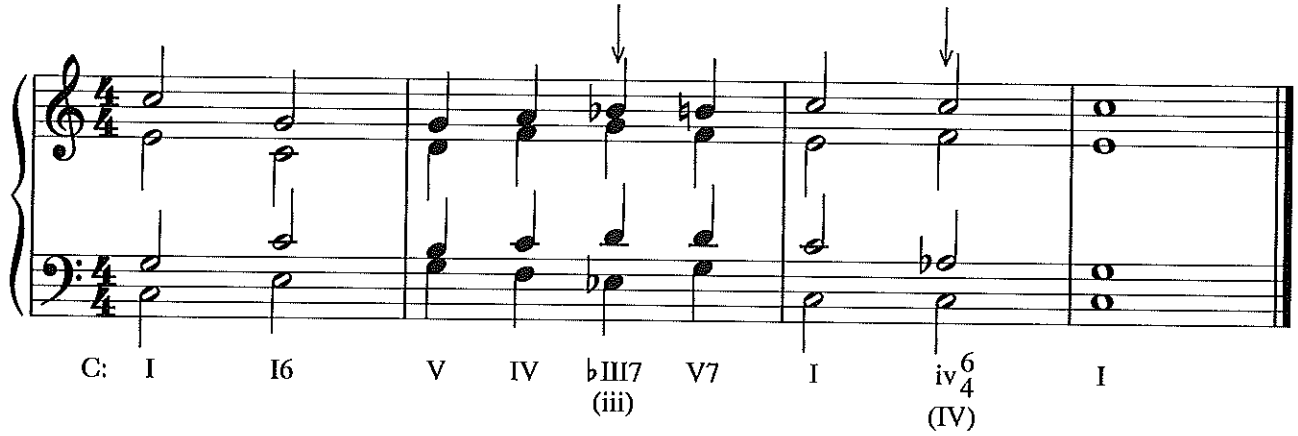
MM mm mm MM Mm mm dm

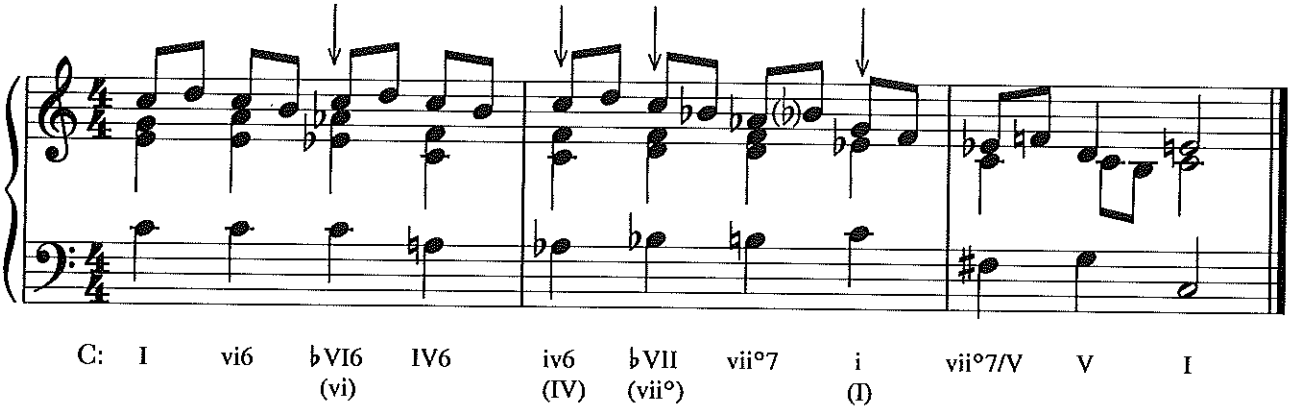
Borrowed seventh chords



mm dm MM mm mm MM Mm

The example above shows the triads and seventh chords on each of the degrees of the major scale, and their equivalents in the parallel minor. In a chromatic idiom, harmonic variety may be achieved by using chords from the minor mode in place of the diatonic ones, as the examples below show:

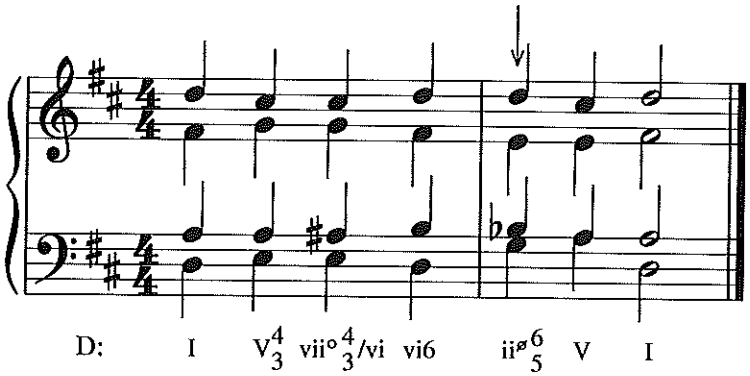




Note:

Each borrowed chord is either preceded by, or used directly in place of, its diatonic equivalent.

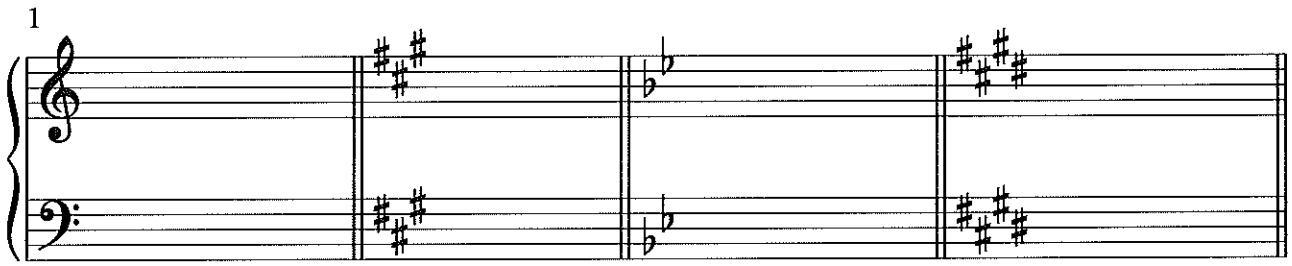
The borrowed supertonic, a diminished triad, contains a tritone and thus may be viewed in a functional as well as coloristic capacity. In the example below, the movement to the cadence is enhanced by the borrowed dissonant supertonic seventh. The dissonance of the augmented fourth between the tenor and the alto would, of course, be absent if the diatonic chord were employed.



EXERCISES

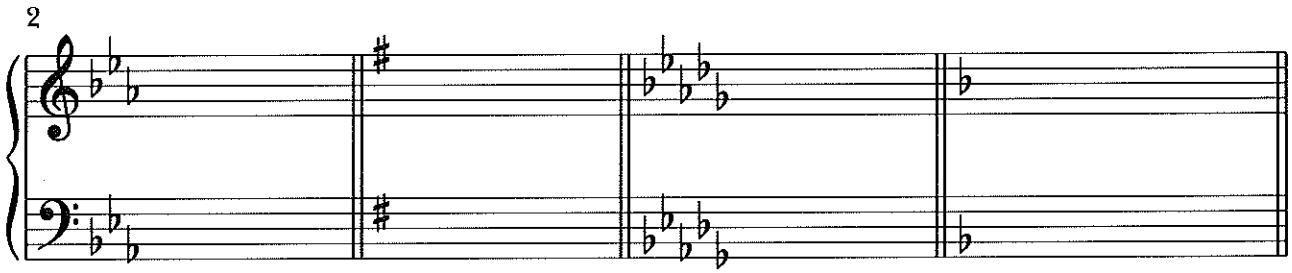
A. Write the following chords.

1



C: ♭VI A: ♭III B♭: iv E: ♭VII

2



E♭: v G: ♭VI D♭: ♭III F: ♭VI

3



♭6 ♭3 5 ♭3 6 ♭4 ♭3

4



♭5 3 ♭6 3 ♭6 ♭3 ♭

B. Add parts for alto and tenor.



C. Add parts for soprano, alto, and tenor.



D. Harmonize and set for piano, making use of borrowed chords.



BURKHART, TUREK, AND WENNERSTROM EXERCISES FOR HARMONIC ANALYSIS

Burkhart

p. 271, mm. 1–13.
p. 201, mm. 164–79.
p. 247, mm. 72–90.

Turek

p. 329, mm. 4–8.
pp. 340–41, mm. 7–12 (third beat).
p. 319, mm. 39–46.
p. 322, mm. 26–40.

Wennerstrom

p. 231, mm. 102–7.
p. 228, mm. 43–50.
p. 397, mm. 156–62.
pp. 313–14, mm. 13–25.

21 The Neapolitan

The major triad built on the lowered second degree of the scale is known as the *Neapolitan.* Because it is diatonic in neither the major nor the parallel minor mode, it may not be considered a borrowed chord. Although it is clearly derived from ii°, it is used in both major and minor keys.

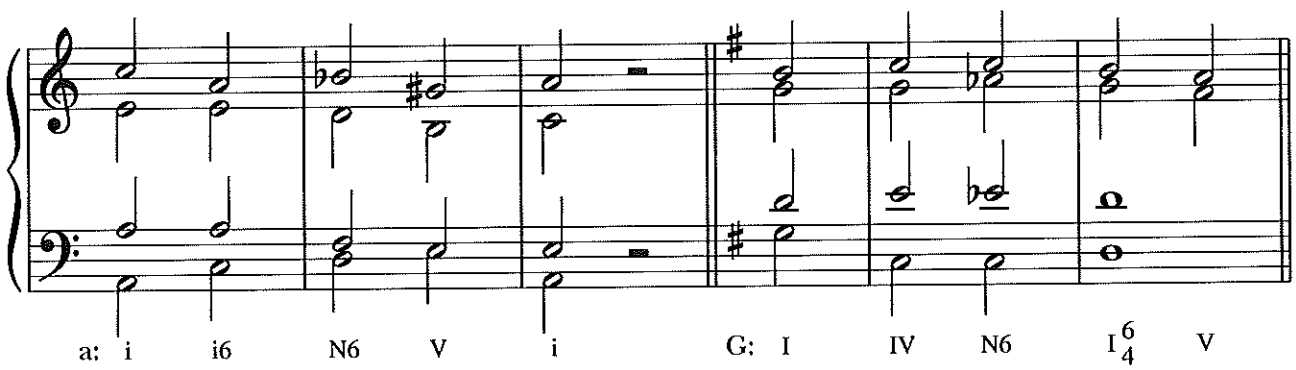
THE NEAPOLITAN SIXTH CHORD

The Neapolitan is used far more frequently in first inversion than in root position and is thus described as the *Neapolitan sixth chord.*

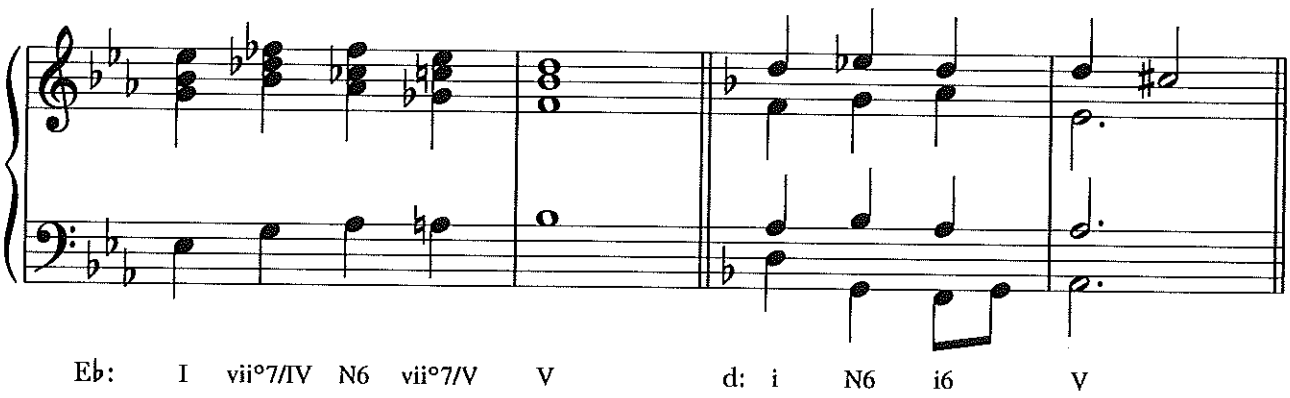


a: ii°6 N6 or ♭II6 E: ii6 N6 or ♭II6

Study the following examples carefully:



a: i i6 N6 V i G: I IV N6 I^{6}_{4} V



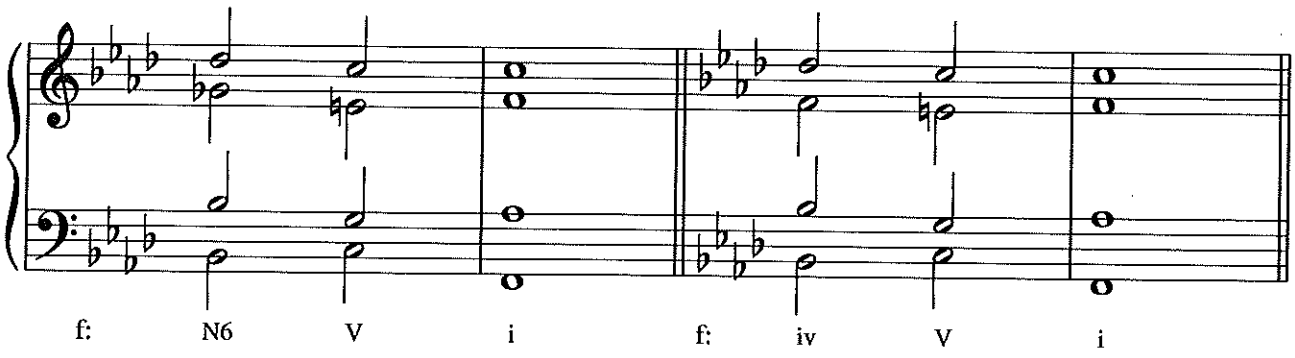
Note:

a. The *bass* of the Neapolitan sixth is doubled.

b. The root of the Neapolitan resolves *downward* to the closest pitch in the subsequent chord.

THE FUNCTION OF THE NEAPOLITAN SIXTH

In the four examples above, it may be observed that the Neapolitan sixth chord either directly precedes the dominant or is followed by a chord that precedes the dominant. The bass movement is from the subdominant note to the dominant note. In fact, as the example below shows, if the root of the Neapolitan sixth were lowered a half step, a subdominant triad (with its root doubled) would be formed.

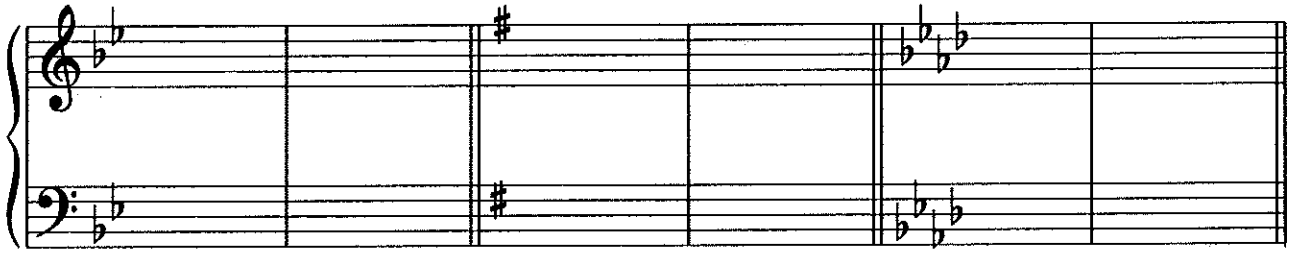


Because of the close similarity between the Neapolitan sixth and the subdominant, the chord is often used in place of the subdominant and is said to have a *subdominant function.*

EXERCISES

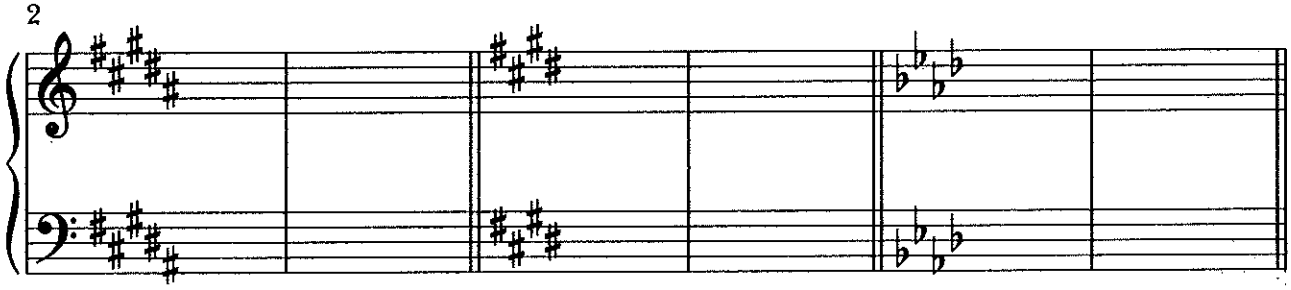
A. Write and resolve the following Neapolitan sixth chords as directed.

1



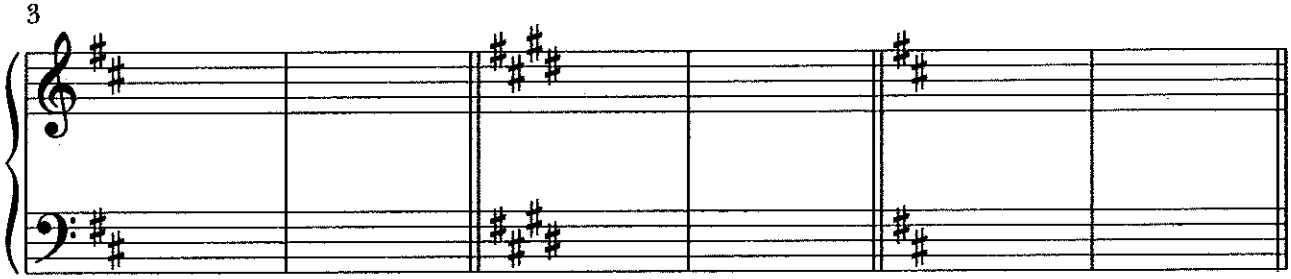
g: N6 V e: N6 vii°7/V V f: N6 i^{6}_{4} V

2



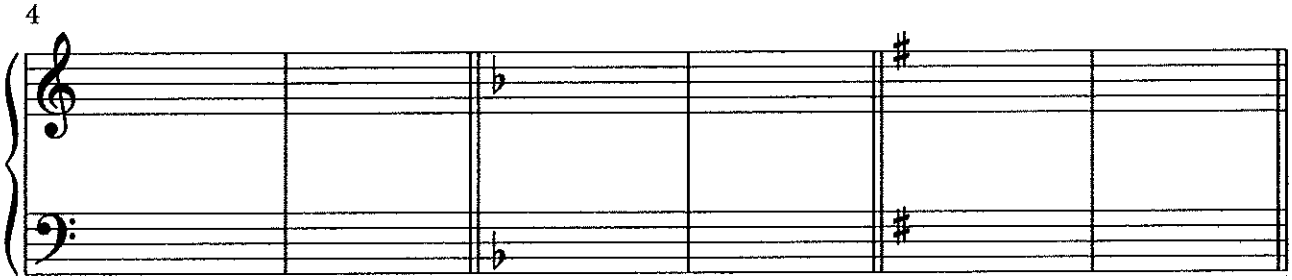
B: N6 V E: N6 V A♭: N6 I^{6}_{4} V

3



D: N6 V c♯: N6 i^{6}_{4} V b: N6 vii°7/V V

4



a: N6 vii°7/V V d: N6 vii°7/V V e: N6 i^{6}_{4} V

B. Add parts for alto and tenor.

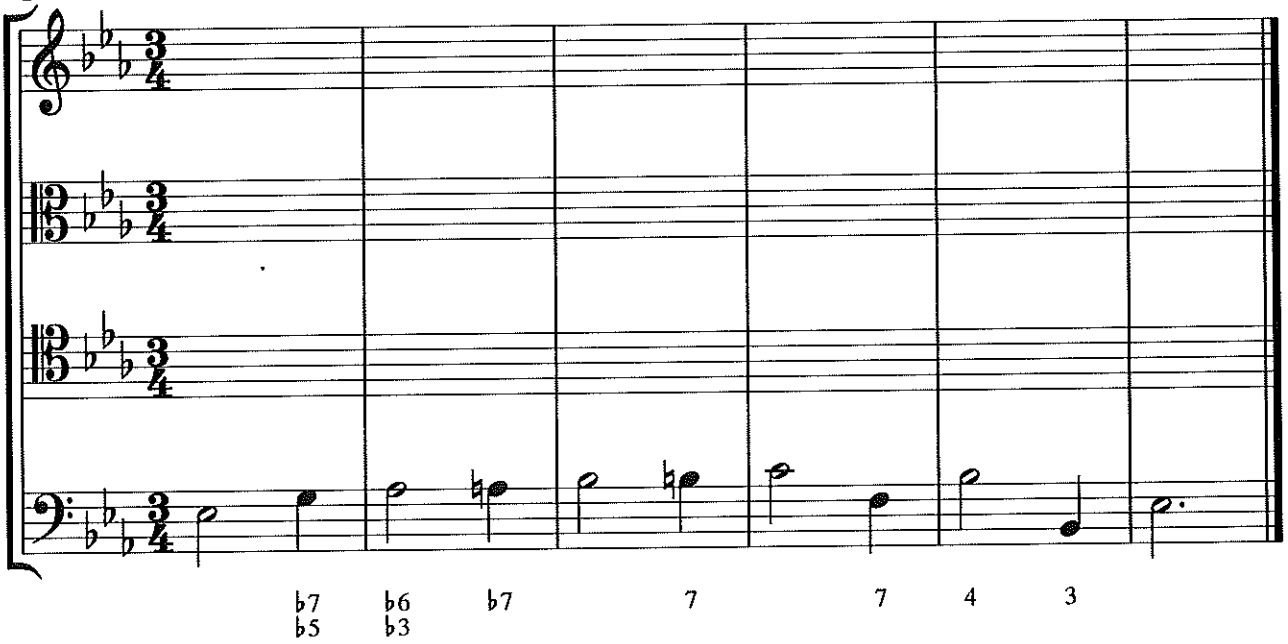


C. Add parts for soprano, alto, and tenor.

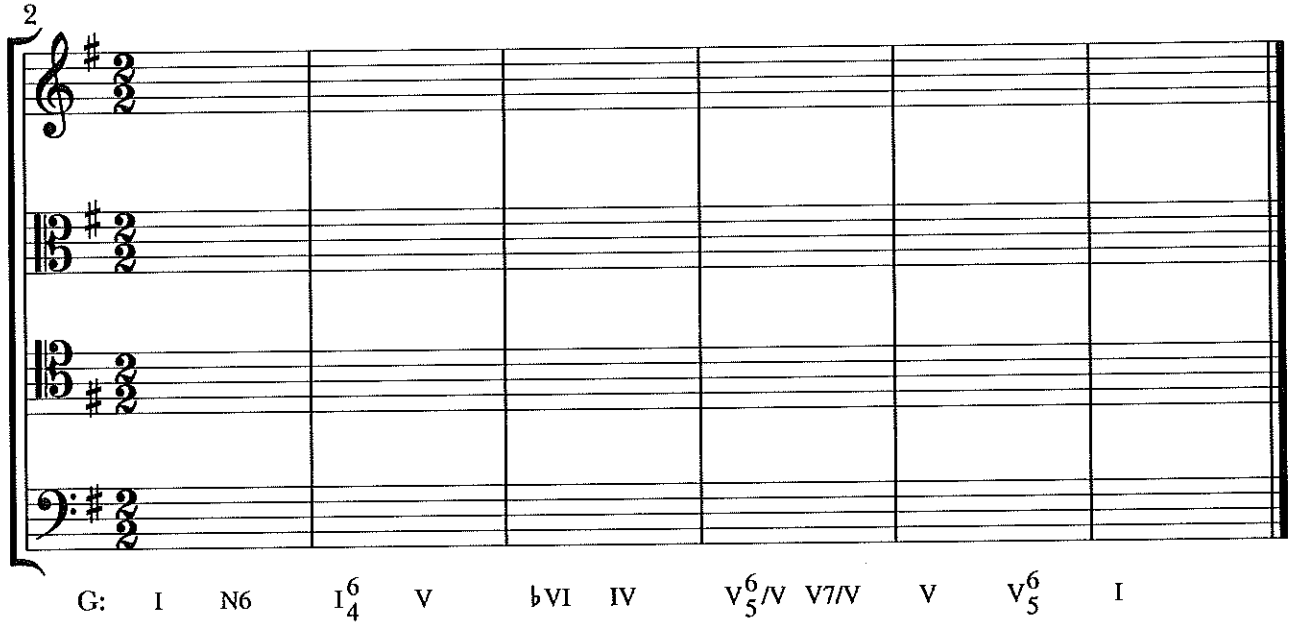


D. Harmonize the following phrases in open score, making appropriate use of nonchord tones.

1



2



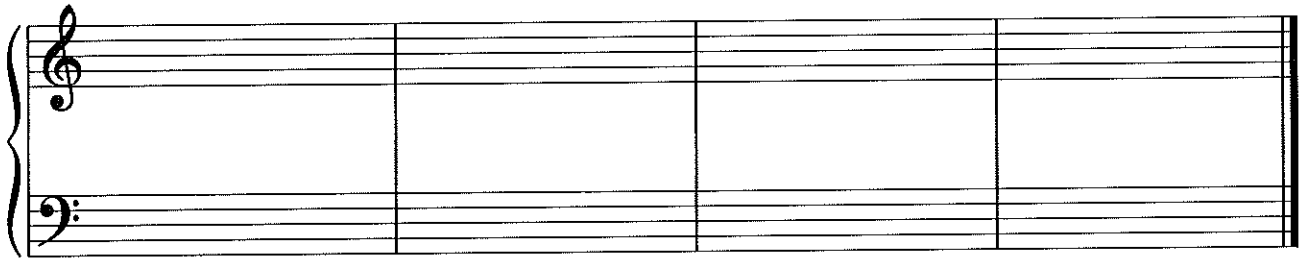
E. Harmonize and set for piano.



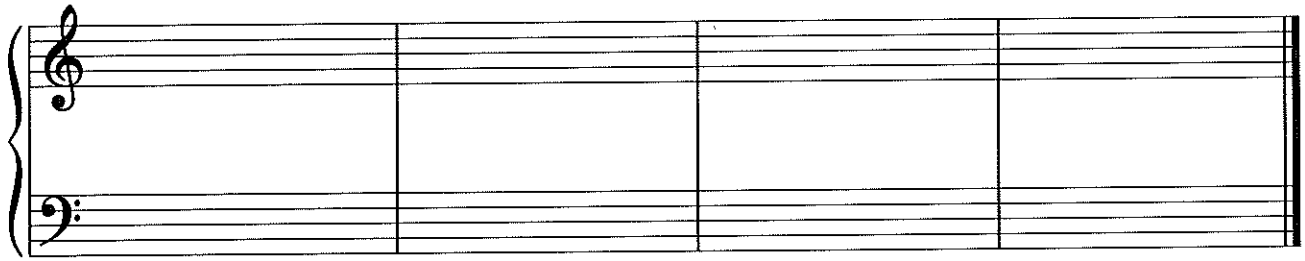
F. Using the given progressions, invent four-measure phrases for soprano, alto, tenor, and bass.

1. f♯: $\frac{4}{4}$ i–V^6_4–i6–V^6_5/iv–iv–N6–V–I
2. D: $\frac{3}{4}$ I–V7/vi–vi–ii–ii6–N6–vii°7/V–V
3. g: $\frac{4}{4}$ V^6_5–i–VI–It♯6–V–i–ii°6–N6–V–i
4. C: $\frac{3}{4}$ I–V^6_5/ii–ii–V^6_5/iii–iii–IV–N6–I^6_4–V–I

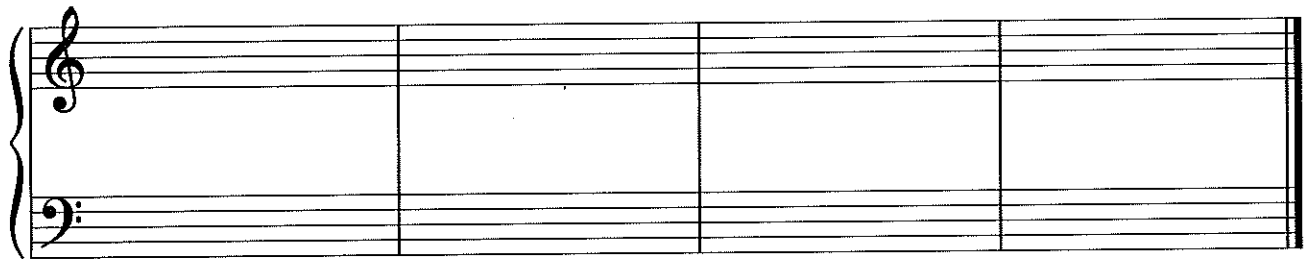
1



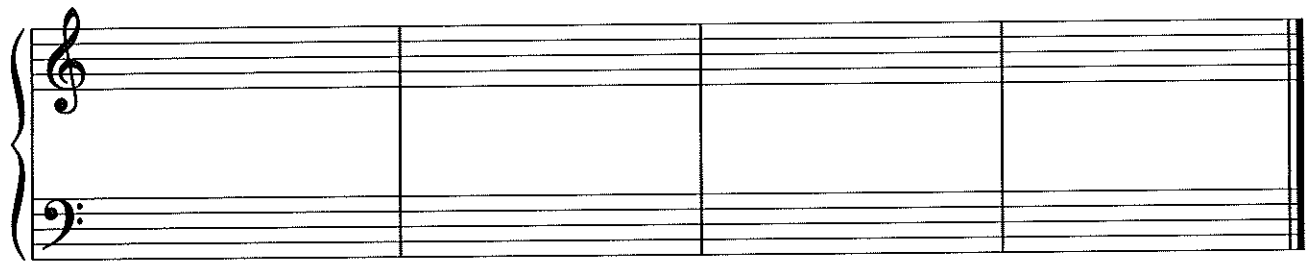
2



3



4



BURKHART, TUREK, AND WENNERSTROM EXERCISES FOR HARMONIC ANALYSIS

Burkhart

p. 282, mm. 235–39.
pp. 210–11, mm. 166–73.
p. 182, mm. 9–17.
p. 332, mm. 1–8.
p. 307, mm. 131–48.

Turek

p. 310, mm. 1–4.
p. 392, mm. 1–6.
p. 189, mm. 51–58.
p. 188, mm. 25–35.

Wennerstrom

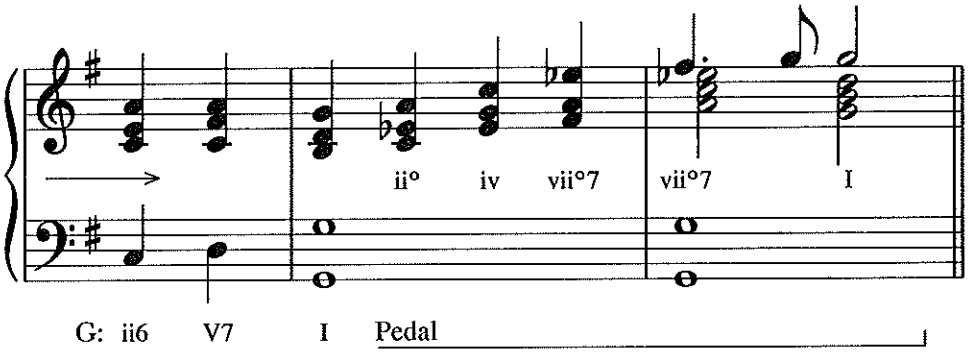
p. 281, mm. 13–16.
p. 336, mm. 54–58.
p. 337, mm. 1–8.
p. 291, mm. 8–20.
p. 206, mm. 45–60.

22 Pedal Point

A pedal point, or pedal, is a sustained pitch, usually in the bass, above which is juxtaposed a series of changing harmonies. Pedal points are named after the scale degrees upon which they are built, for example, "tonic pedal," "supertonic pedal," and so forth. In practice, however, they tend to occur most frequently on the tonic or dominant.

THE TONIC PEDAL

The tonic pedal is often used at the end of a piece of music to prolong the tonic harmony.



THE DOMINANT PEDAL

The main function of the dominant pedal is to create tension and thus to prepare the listener for the subsequent musical event, which is usually the resolution to the tonic.



Note:

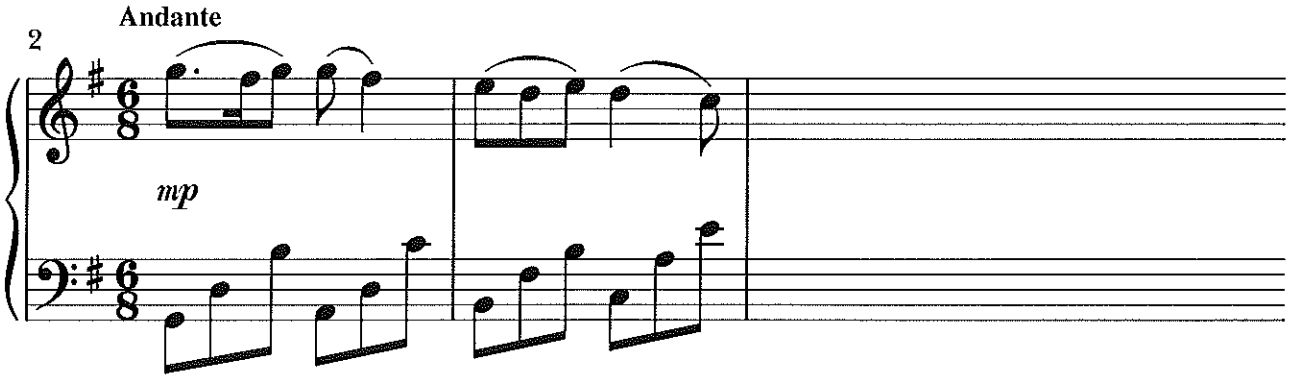
In the examples above, the pedals interact both consonantly and dissonantly with the chords above them. This interaction is the essence of the writing of pedal points, and their effectiveness is dependent upon the skill with which the composer controls the harmonic activity in the upper voices.

EXERCISES

A. Complete the following pieces for piano in the style suggested by the beginning of each. They should be between eight and sixteen measures in length.

The first piece has been worked as a model.




Andante
2
mp

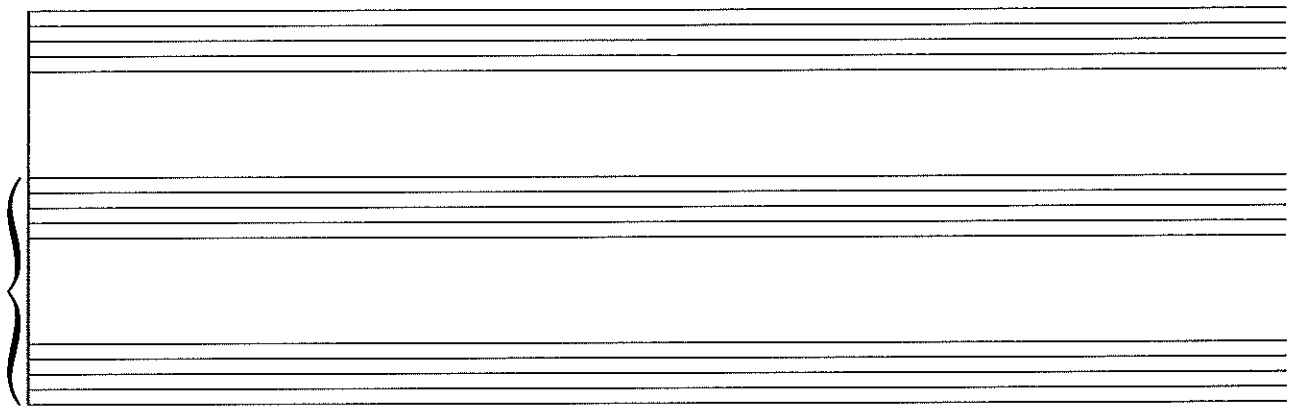

3
Allegro moderato
mp
mf

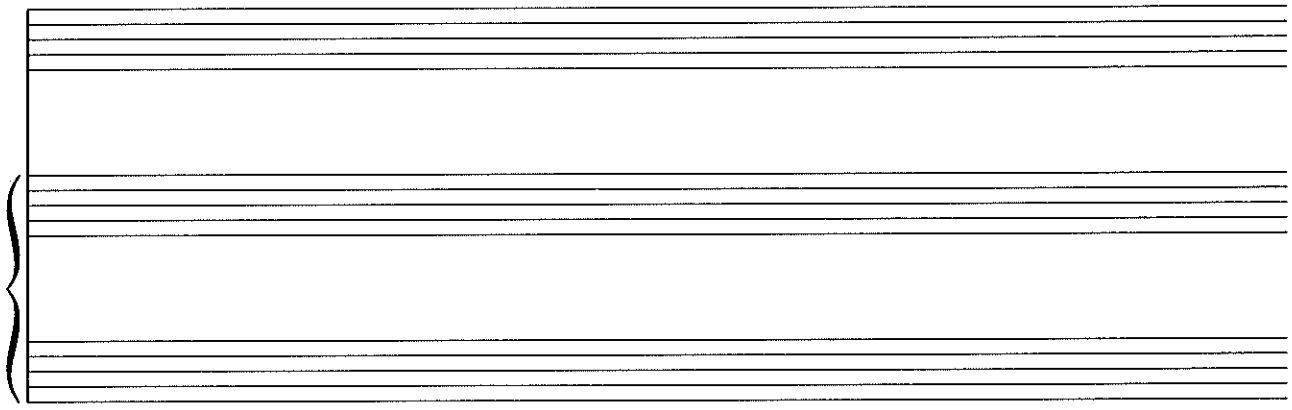

4
Andantino
p

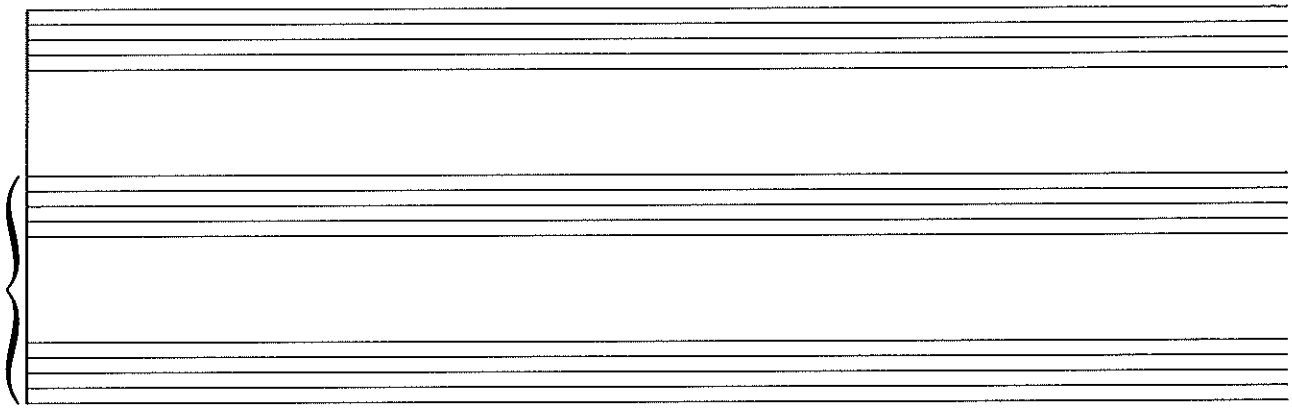

5
Tempo di valse
mf
Ped.
*
Ped.
*
sim.

B. Select two of the previously completed piano pieces and arrange each of them for a solo instrument with piano accompaniment.

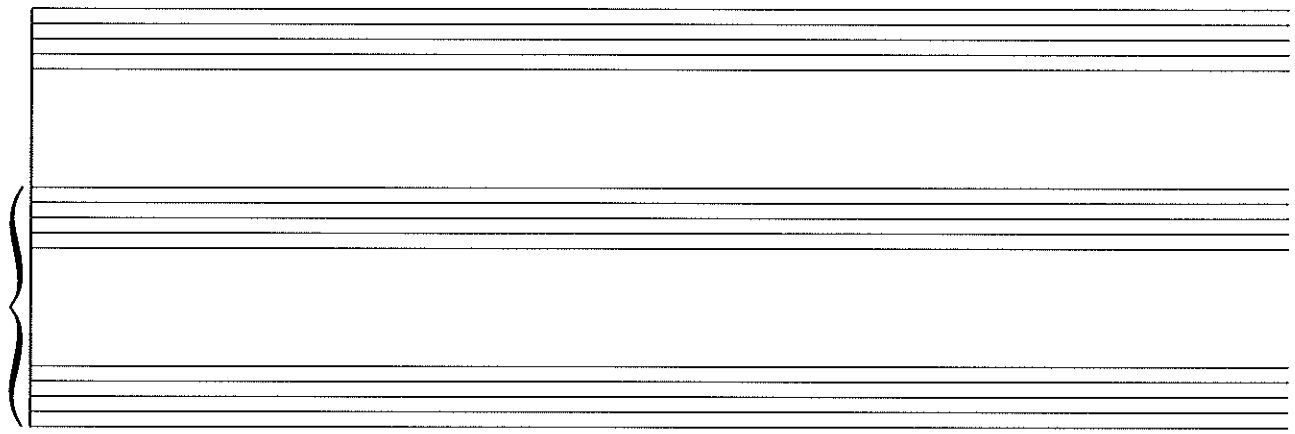
1

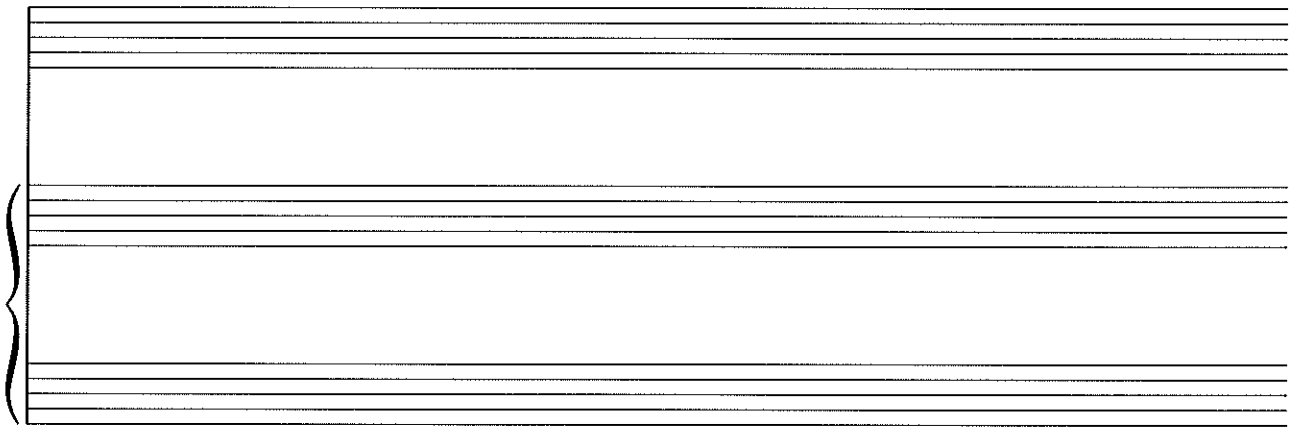


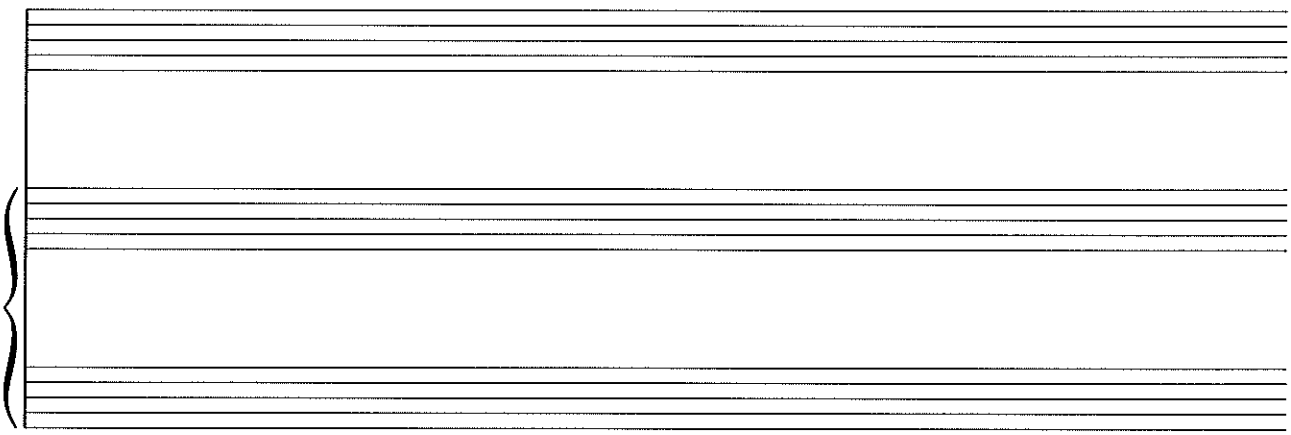




2







BURKHART, TUREK, AND WENNERSTROM EXERCISES FOR HARMONIC ANALYSIS

Burkhart

p. 318, mm. 9–29.
p. 114–15, mm. 1–35.
p. 316–17, mm. 1–64.

Turek

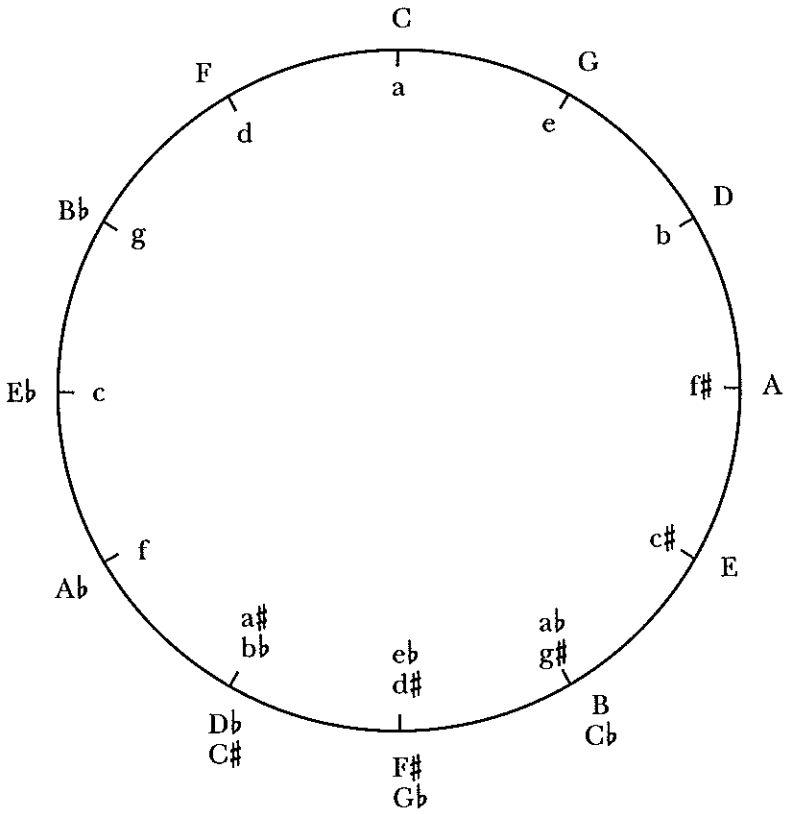
p. 331, mm. 1–8.
p. 268, mm. 68–76.
p. 347, mm. 61–69.
pp. 256–57, mm. 86–95.

Wennerstrom

p. 137, mm. 73–81.
p. 251, mm. 24–35.
pp. 372–74, mm. 17–45.

23 Modulation

Modulation may be defined as the process of moving from one key to another. Although all keys are related, the degree of relationship is a function of their relative positions in the circle of fifths.



The diagram shows that if C is the tonic, then, after a minor, the keys of G and F and their relative minors would be the most closely related to C because G and F respectively are a perfect fifth above and below C.

Similarly, D and B♭ and their relative minors would follow in terms of close relationship; and F♯/G♭ and their relative minors would be the most remote keys.

The essence of modulation lies in the appropriate writing of a chord that suggests a new key as a result either of the presence of a pitch (or pitches) foreign to the established key, or of the subsequent musical activity, or of both. In simple modulation, this chord, the *modulating chord,* almost always has a dominant function in the new key, for example, V, V7, vii°, vii°7, or I^6_4 in a cadential situation. In more complex modulation, however, the function of the modulating chord is less predictable. In any event, the new key is established by the harmonic progression that follows the modulating chord.

COMMON CHORD MODULATION

Common chord modulation may be effected only between keys that have diatonic chords in common. (The C major triad, for instance, is I in C, VI in e, IV in G, V in F, and so on.) This kind of modulation is usually employed to modulate from a given key to a *closely related key,* although it may be used for more distant modulations as well. The term *closely related* serves to describe keys that are no more than one step removed in the circle of fifths from a given key. Such keys have several chords in common with one another. For example, common chord modulation may easily be effected from A to f♯, E, c♯, D, and b.

Study the following examples carefully:





Note:

a. The common chord directly precedes the first appearance of a chord with dominant function in the new key.

b. The common chord is analyzed as a diatonic chord in both keys.

c. The common chord is the "pivot" of the modulation. At this point, the harmonic analysis expresses the last event in the established key and the first event in the new key.

The exercises that follow are related to the materials on pages 281–283. Abrupt and enharmonic modulation are covered on pages 297–300.

EXERCISES

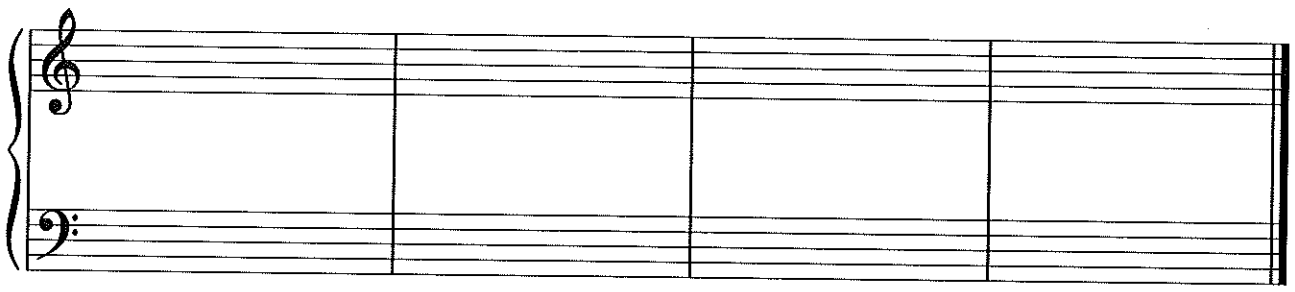
A. Write the diatonic function of each given chord in five keys.

1.	C	C: I	a: III	G: IV	F: V	e: VI
2.	A♭					
3.	E♭					
4.	g					
5.	f♯					
6.	D					
7.	B♭					
8.	a					
9.	e					
10.	F♯					
11.	F					
12.	c					

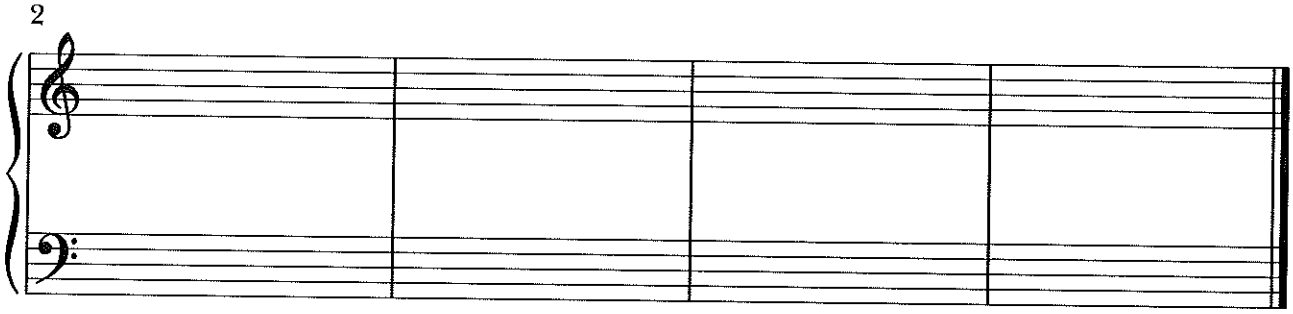
B. Using the given progressions, invent four-measure phrases for soprano, alto, tenor, and bass.

1. D: $\frac{3}{4}$ I–V–I–I6–IV
   G: I–V^6_5–vi6–ii6–V7–I
2. a: $\frac{4}{4}$ i–vii°6–i6–iv
   C: ii–ii^4_2–V6–I–V7–I
3. E♭: $\frac{6}{8}$ V^6_5–I–iii^6_5–vi
   c: i–$vii°^4_2$–i^6_4–V4–3–i

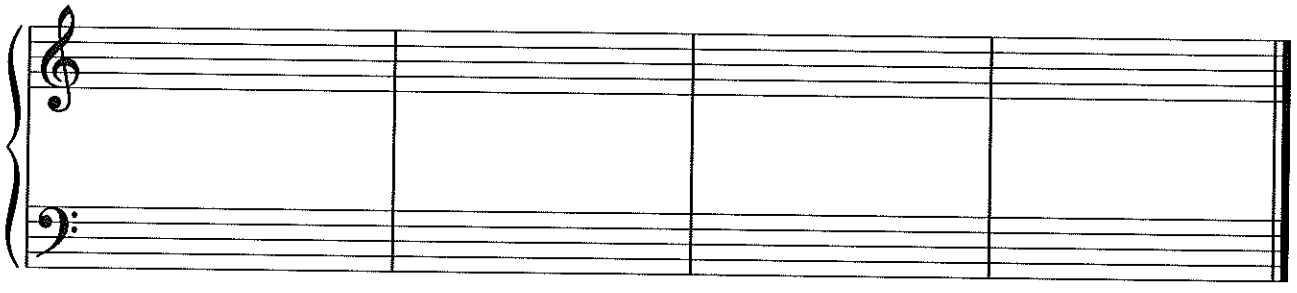
1



2



3

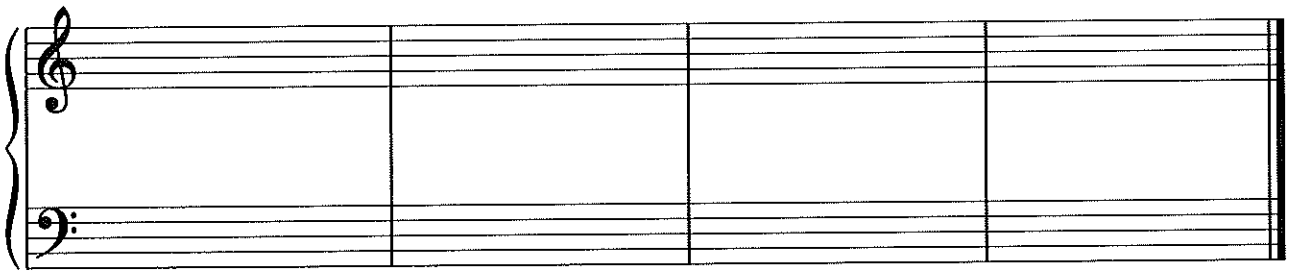


4. g: $\frac{4}{4}$ i–i6–V^6_5/iv–iv–vii$^{\circ 4}_{3}$/III–III6
d: VI6–i^6_4–V4–3–i

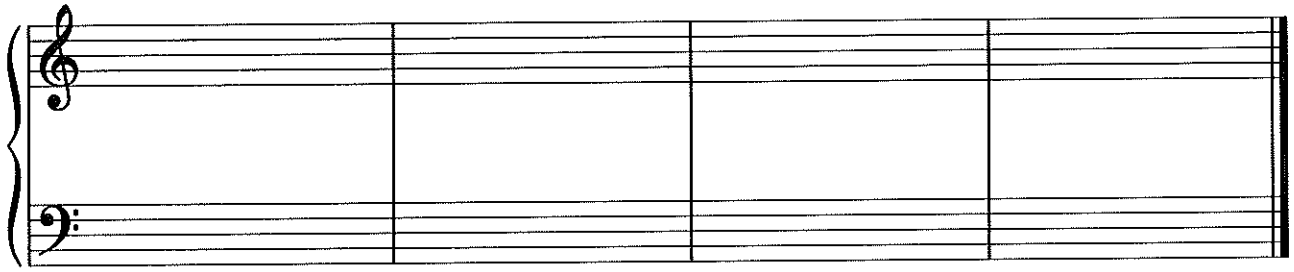
5. E: $\frac{3}{4}$ I–V^6_5/ii–ii–vii$^{\circ}$7/iii–iii
B: vi–V^6_5–vi6–ii7–V7–I

6. f♯: $\frac{4}{4}$ i–V^6_5–i^4_2–vii$^{\circ 6}_{5}$/VI–VI
b: III–vii$^{\circ 6}_{5}$–VI6–V^6_5–vii$^{\circ 6}_{5}$/V–V–i

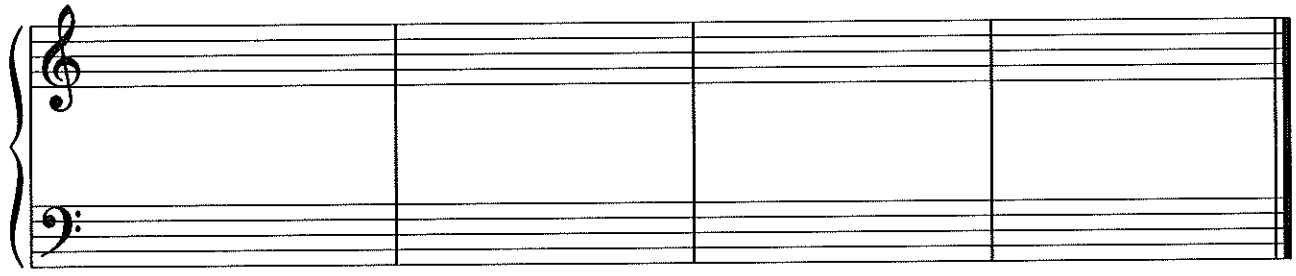
4



5



6



C. Harmonize the following melodies and set them for piano.

1 **Comodo**



2 **Alla breve**



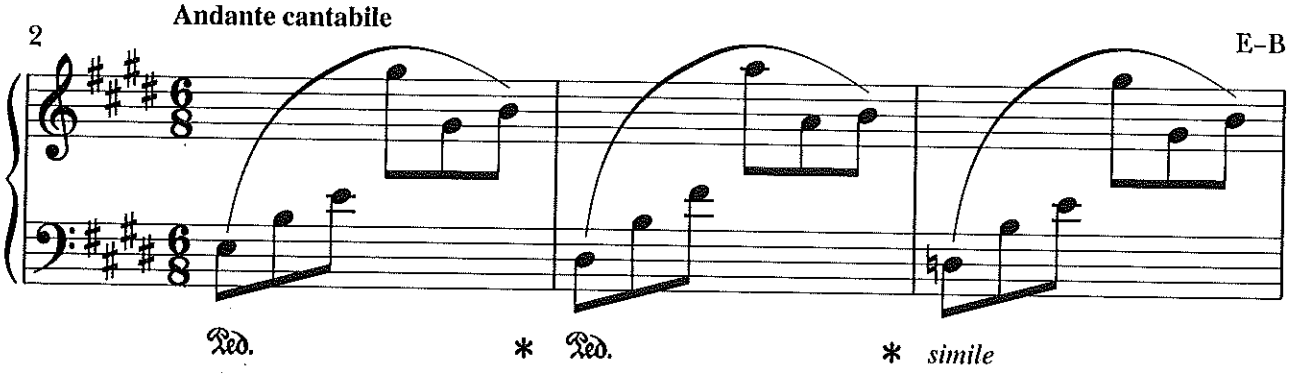
3

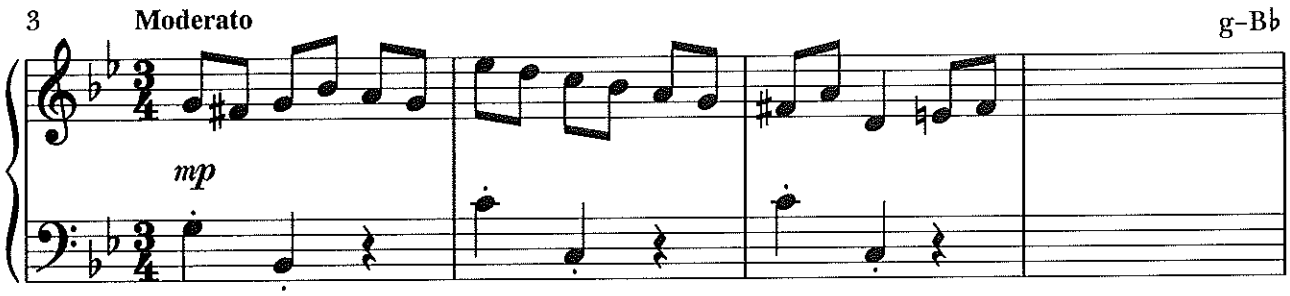
Andante pastorale



D. Complete the following pieces for piano in the style suggested by the beginning of each. They should be between eight and sixteen measures in length and should modulate as indicated.

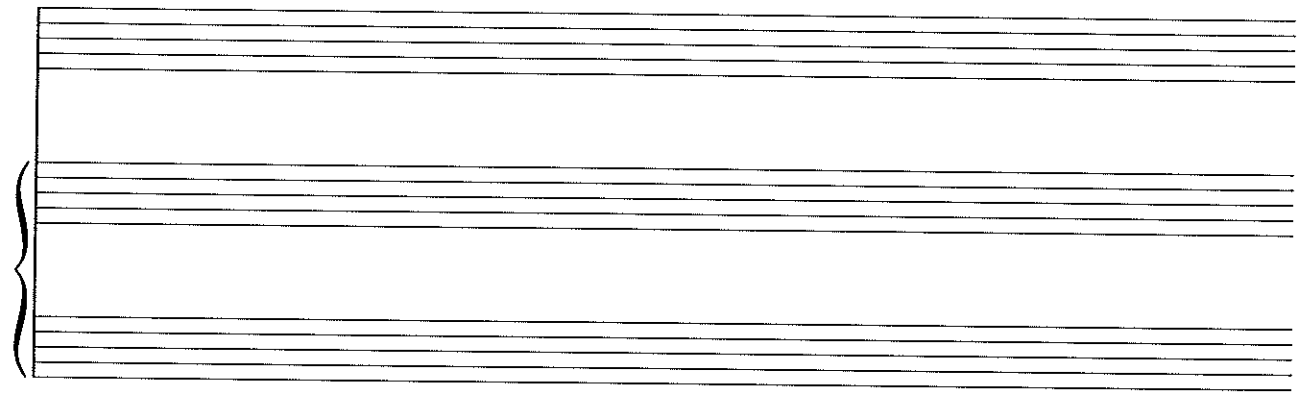


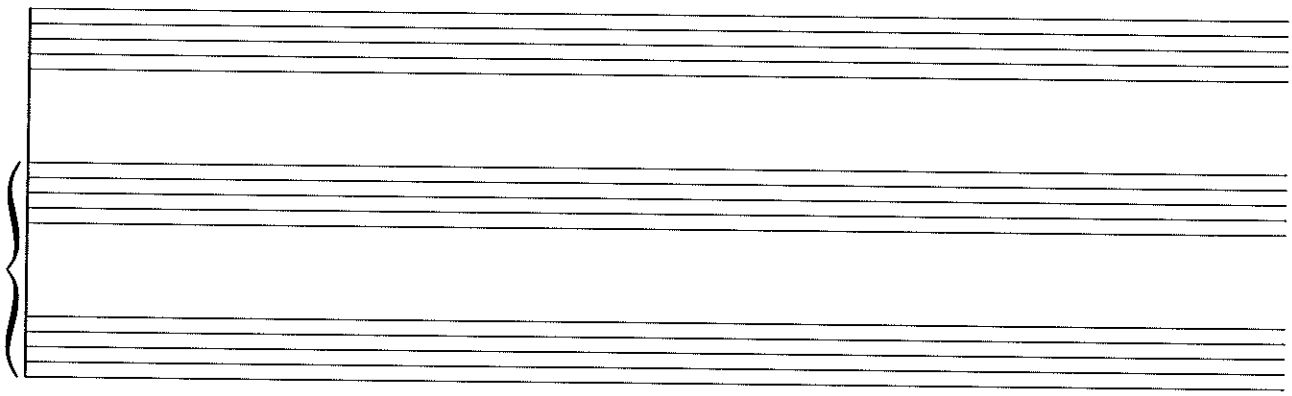

Andante cantabile
2
E–B
Ped.
*
Ped.
*
simile

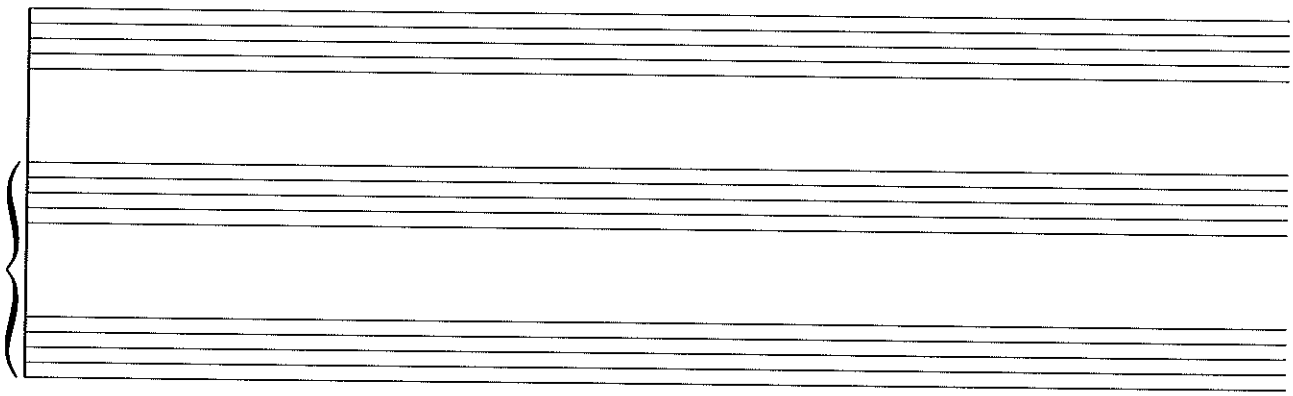

3
Moderato
g–B♭
mp

E. Select two of the previously completed piano pieces and arrange each of them for a solo instrument with piano accompaniment.

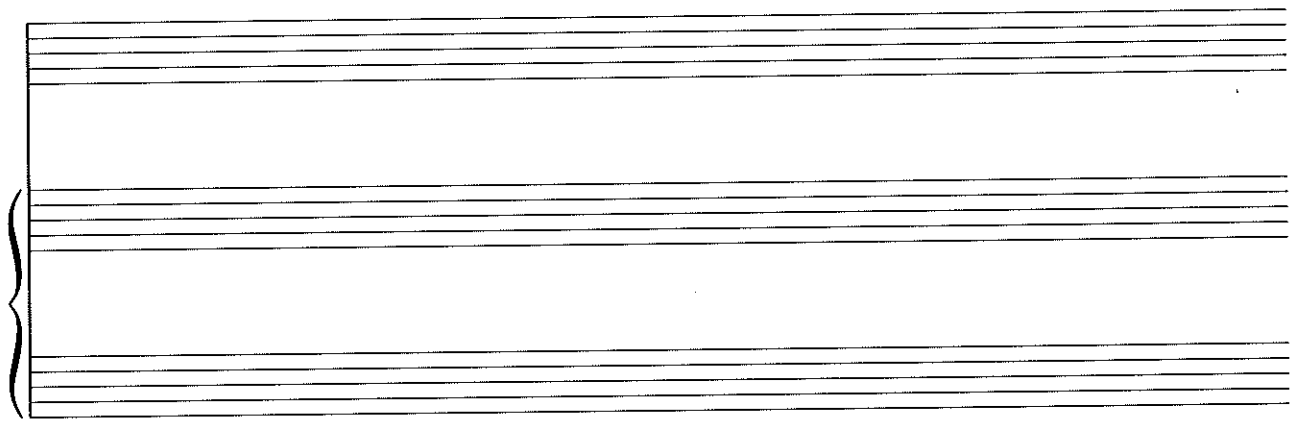
1

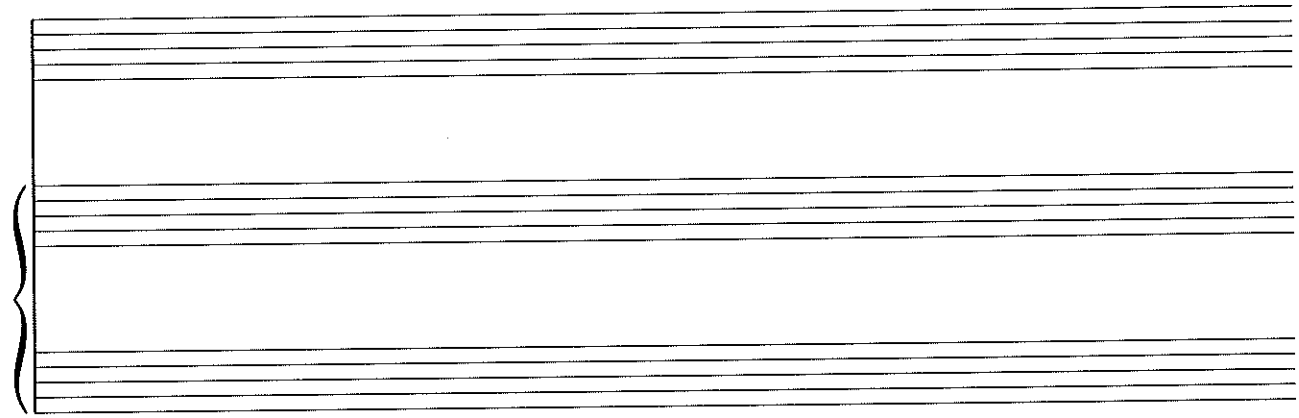


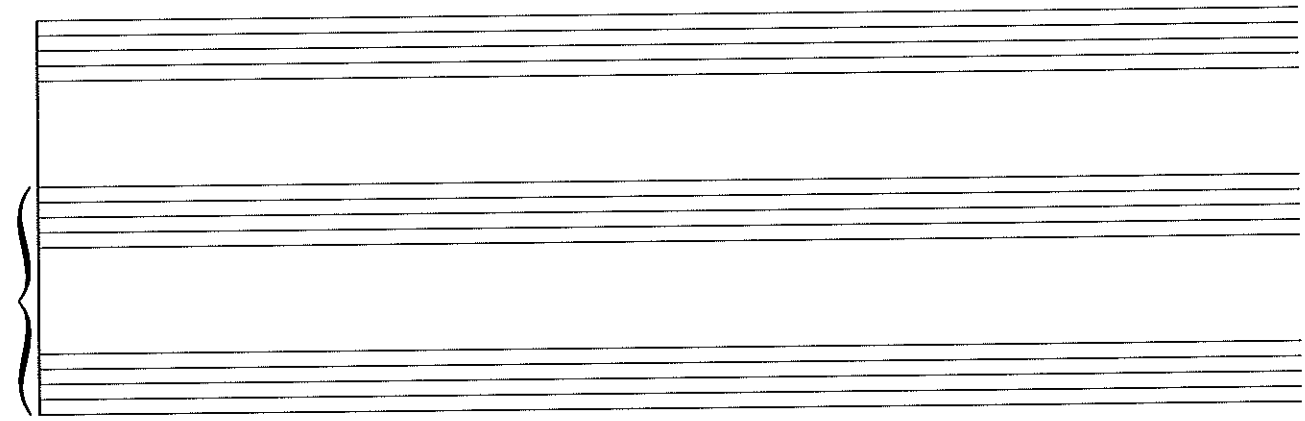




2







BURKHART, TUREK, AND WENNERSTROM EXERCISES FOR HARMONIC ANALYSIS

Burkhart

p. 178, mm. 1–8.
p. 125, mm. 1–12.
p. 139, mm. 1–18.
pp. 71–72, mm. 1–22.

Turek

p. 344, mm. 22–28.
p. 54, mm. 1–8.
p. 234, mm. 13–24.
pp. 337–38, mm. 25–38.
pp. 229–30, mm. 36–58.
p. 255, mm. 1–24.

Wennerstrom

p. 248, mm. 1–8 (var. VI).
p. 235, mm. 8–21.
p. 111, mm. 1–16.
p. 234, mm. 63–77.
pp. 172–73, mm. 1–34.

ABRUPT MODULATION

Abrupt modulation occurs when the chord that precedes the modulating chord is *not diatonic* in the new key. The emphasis, therefore, in this type of modulation is upon the modulating chord itself.

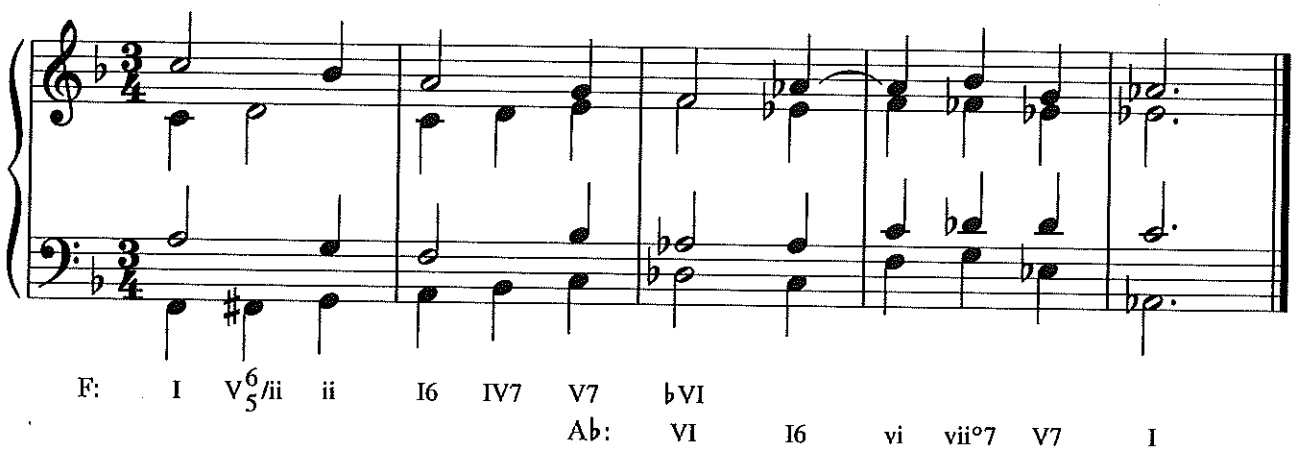
In the previous section, the modulating chord was identified as a chord with dominant function in the new key. The examples below, which should be studied carefully, show that in abrupt modulation the modulating chord may or may not be one with dominant function and may be either diatonic or chromatic in the new key.

1. Modulating chord *diatonic* in new key—dominant type:



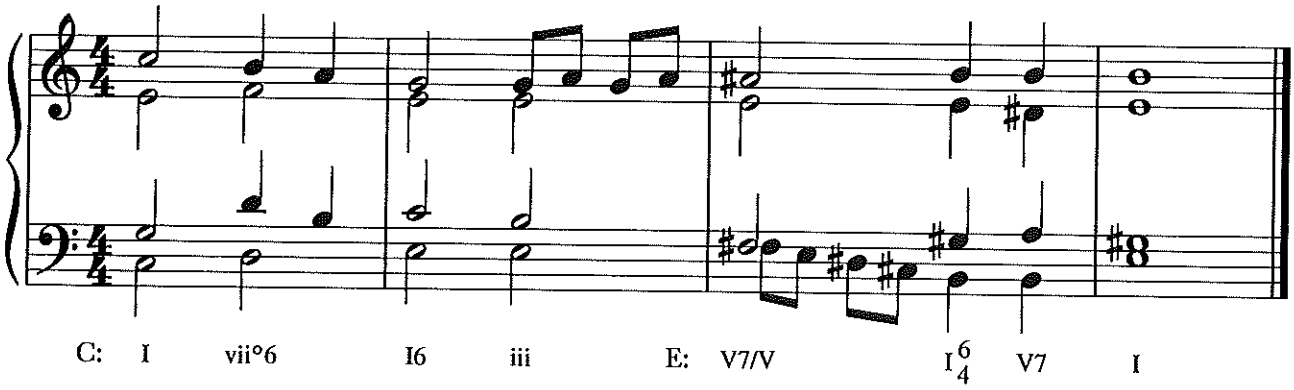
Note:
The modulating chord, the secondary dominant seventh of the submediant in the established key, becomes the dominant seventh in the new key.

2. Modulating chord *diatonic* in new key—nondominant type:



Note:
The modulating chord, the lowered submediant in the established key, becomes the subdominant in the new key.

3. Modulating chord *chromatic* in new key—dominant type:



Note:

The modulating chord, which has a very obscure function in the established key, is the secondary dominant seventh of the dominant in the new key.

4. Modulating chord *chromatic* in new key—nondominant type:



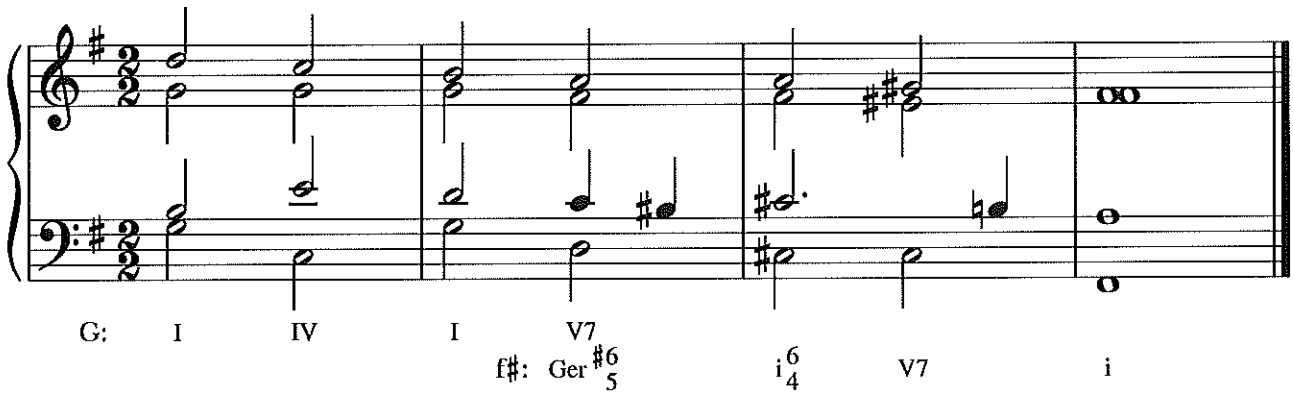
Note:

The modulating chord, which in the established key is borrowed from f minor, is the Neapolitan sixth chord in the new key.

ENHARMONIC MODULATION

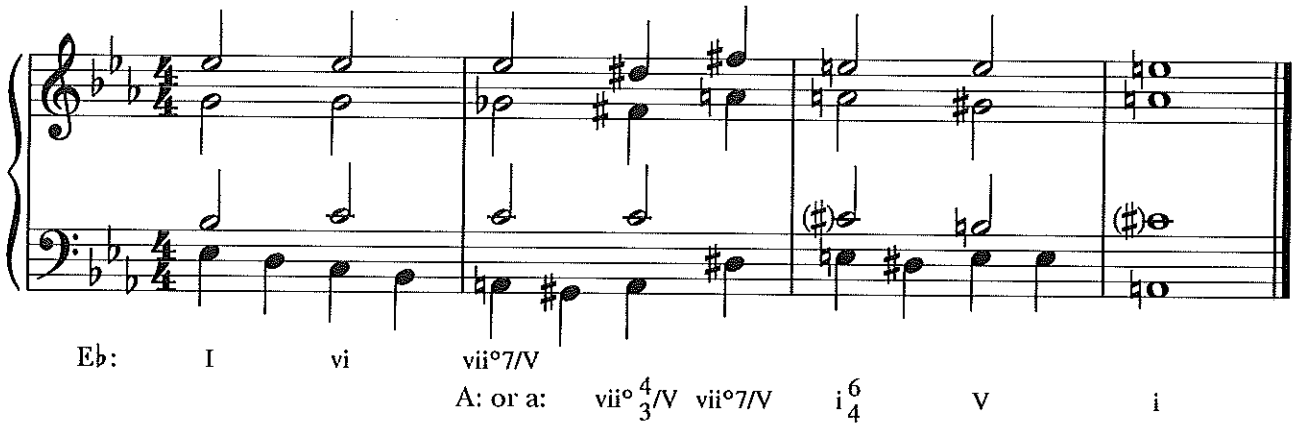
The word *enharmonicism* is used to describe the respelling of a pitch, or pitches, to produce a new harmonic function. In an enharmonic modulation, therefore, the modulating chord is respelled and, as a result, performs an appropriate function in the new key.

Enharmonic modulations by their very nature tend to be dramatically abrupt, as the following examples show.



Note:

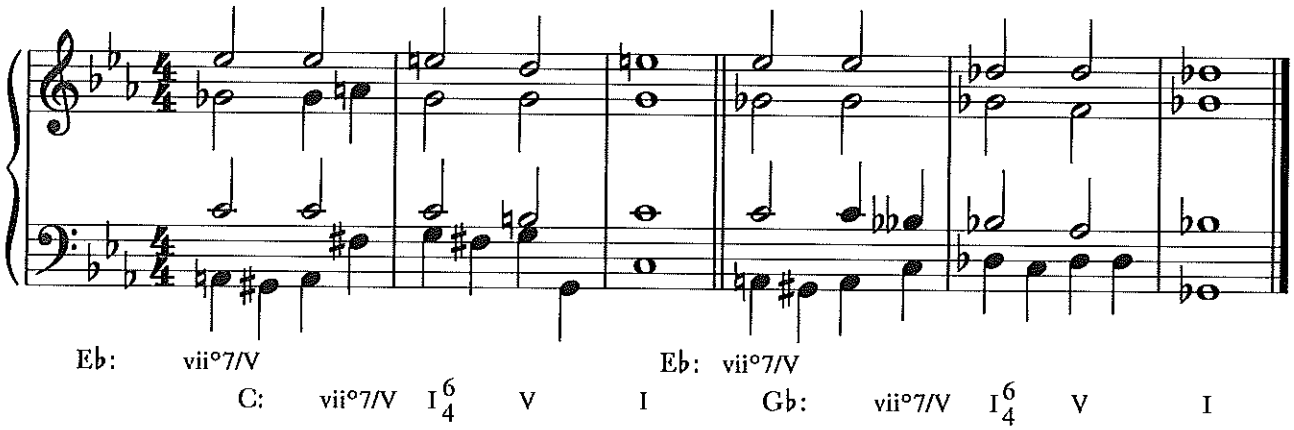
The modulating chord *sounds* as the dominant seventh in the established key, but by respelling the C♮, as a B♯, a German sixth chord is formed in the new key, and thus the modulation is effected.



Note:

In this example, the respelled diminished seventh chord resolves naturally to a cadential tonic six-four in the new key. Such a resolution is typically employed to establish tonality.

In fact, with respect to modulation, the diminished seventh chord is an extremely versatile sonority, since, with the exception of the augmented triad, it is the only tertian chord that retains its identity under inversion. In other words, it is impossible *by ear* to tell whether a diminished seventh chord is in root position. By way of illustration, the diminished seventh chord extracted from the example above is respelled and revoiced to produce another two modulations that are effective.



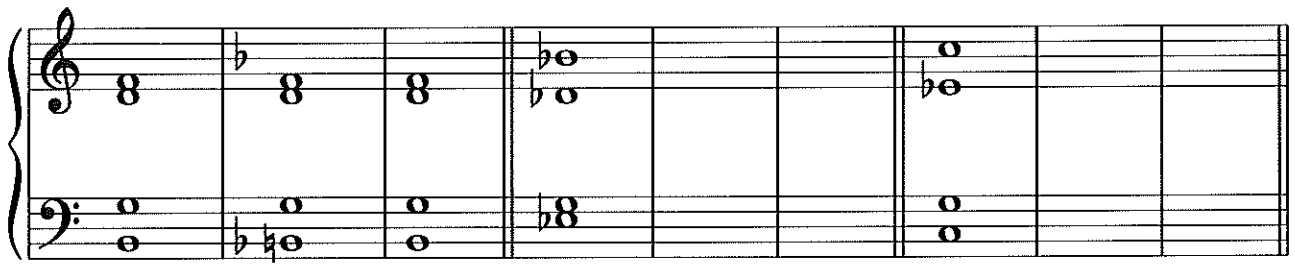
Note:

Modulation has been defined here as the process of moving from one key to another, and the mechanics of this process have been described and exemplified. Key change does occur, however, without the presence of a modulating chord. For example, a half or authentic cadence at the end of a phrase or period is frequently followed by musical activity in another key (see Burkhart, p. 173, mm. 67–68). Under these circumstances it makes no musical sense to designate any particular chord as a pivot between the two keys because the termination of activity in one key is not linked to the beginning of activity in another. Thus, key change occurs with the omission of the process of modulation.

EXERCISES

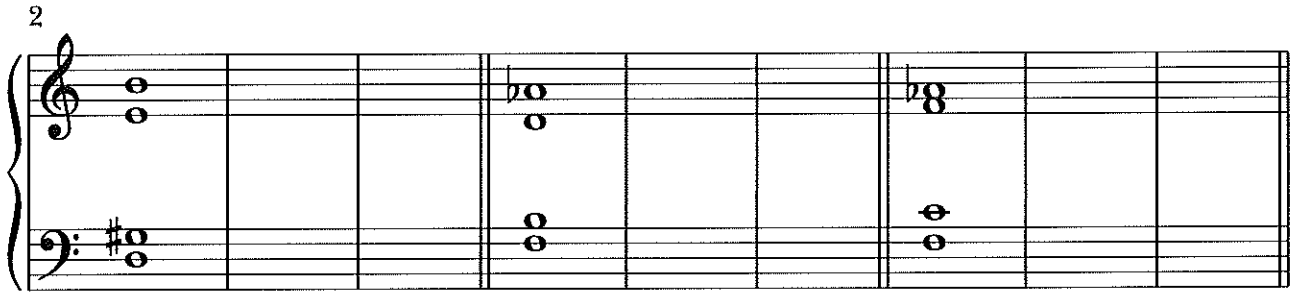
A. Analyze the given chords as chromatic in one key and diatonic in another.

1

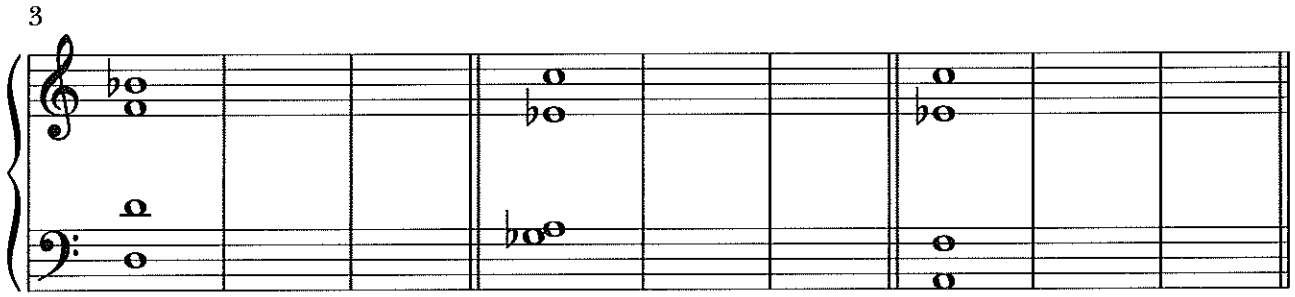


F: V_5^6/V C: V_5^6

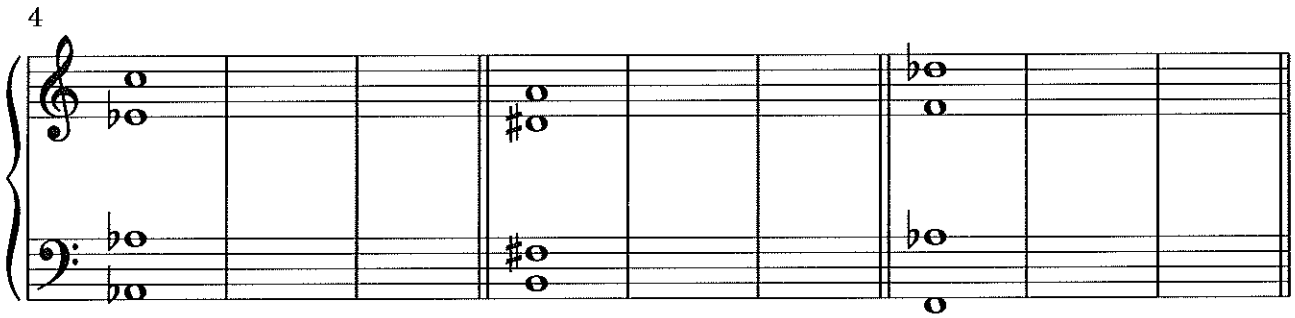
2



3

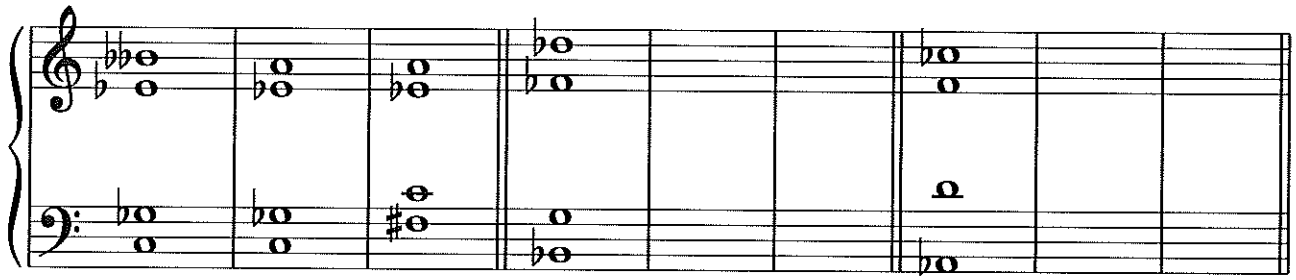


4



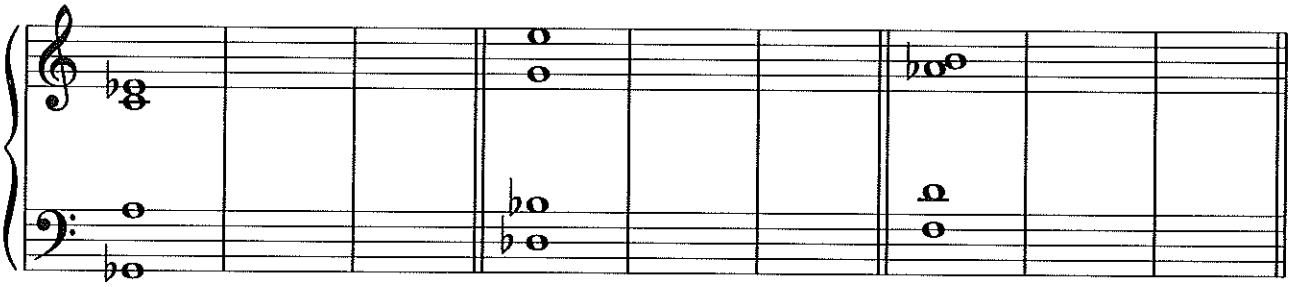
B. Respell and revoice the given diminished seventh chords in two keys, so that they would resolve naturally to $I(i)^6_4$ or to V.

1

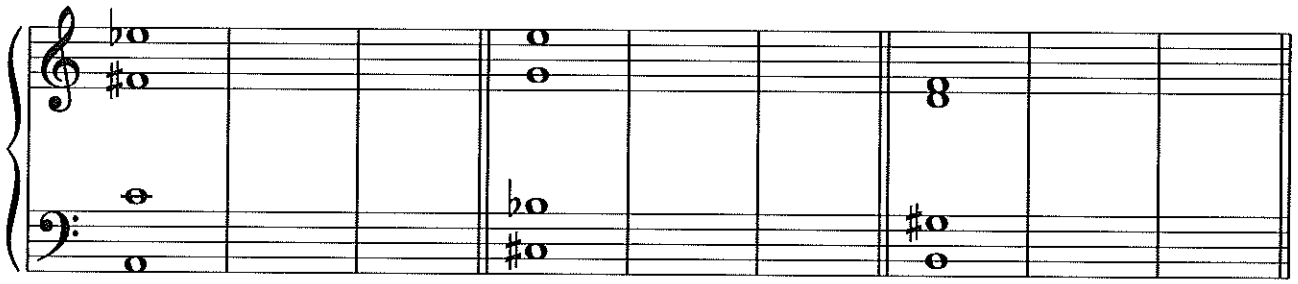


E♭: vii°$^6_5$/V c: vii°7/V

2



3



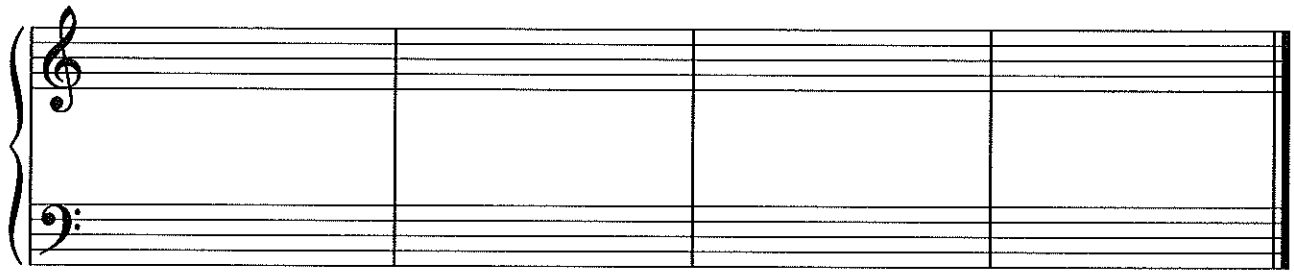
4



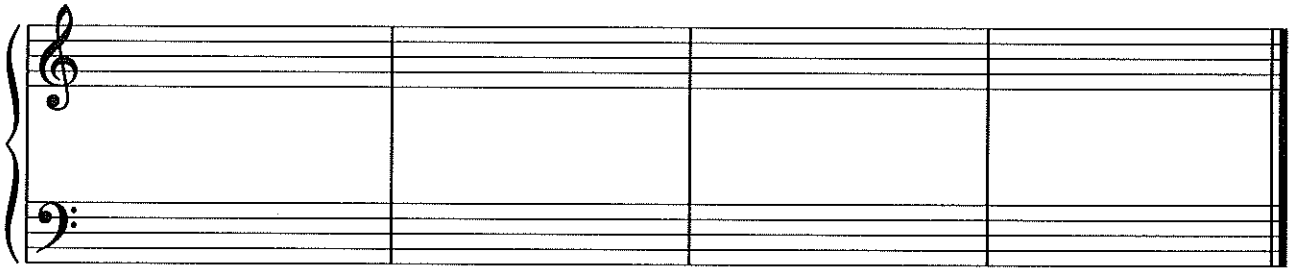
C. Using the given progressions, invent four-measure phrases for soprano, alto, tenor, and bass.

1. E: $\frac{3}{4}$ I–V$^6_4$–I6–IV–N6
   c: IV6–G$^6_5$–i$^6_4$–V–i
2. C: $\frac{6}{8}$ I–ii$^4_2$–V6–V7–♭VI
   D♭: V–V$^4_2$–I6–ii7–V4–3–I
3. B♭: $\frac{4}{4}$ I–I6–IV–I–V7
   a: G$^6_5$–i$^6_4$–V–iv$^6_4$–i

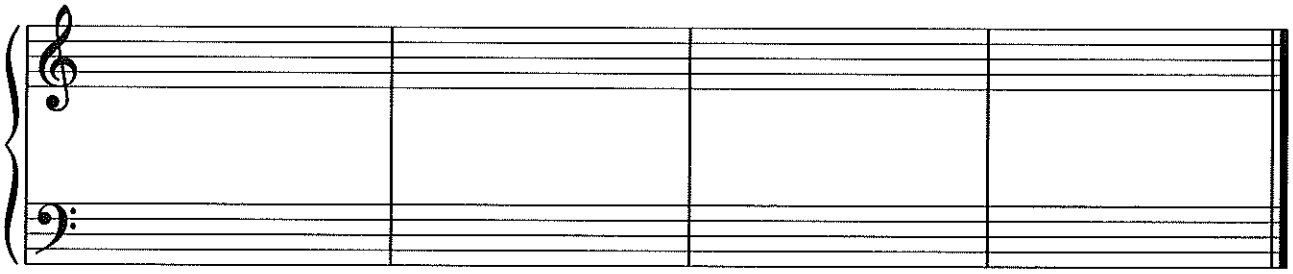
1



2



3

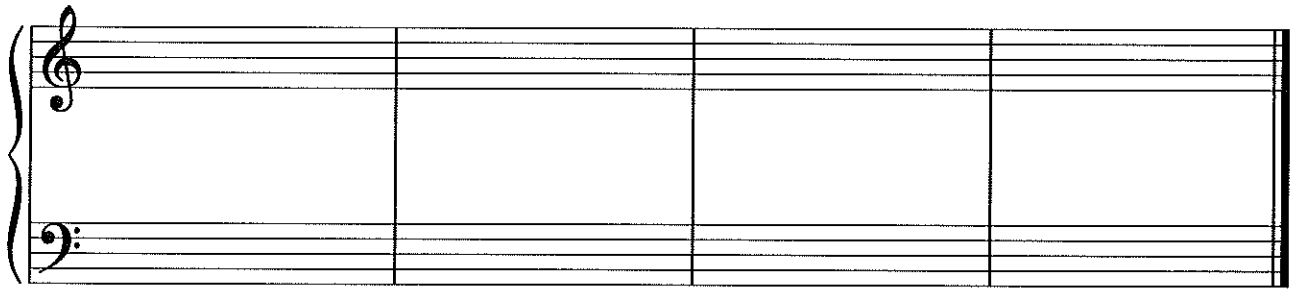


4. E♭: $\frac{4}{4}$ I–V^6_5–I–V^4_2/IV–IV6–iv6
G♭: ii6–vii°$^6_5$/iii–I6–vi^6_5–V7/V–V–I

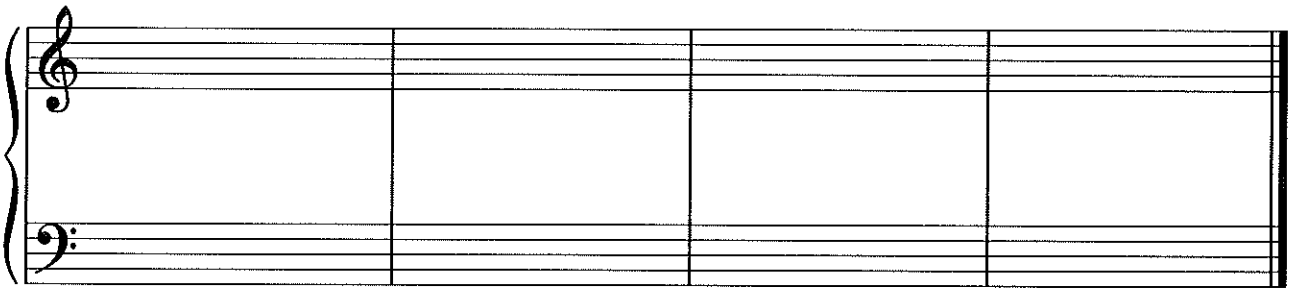
5. D: $\frac{6}{8}$ V7–I–V^6_5/ii–ii–V^6_5/iii–V/vi
E: V/V–V^4_2/V–V6–vi6–I^6_4–V–I

6. G: $\frac{3}{4}$ I–V–V^6_5/vi–vi–vii°7/V–vii°$^4_3$/V
B♭: vii°$^6_5$/V–I^6_4–V–I

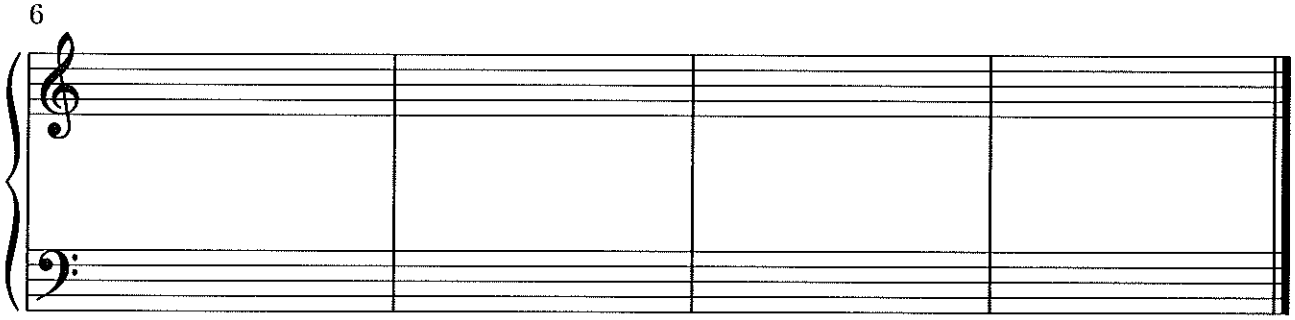
4



5



6

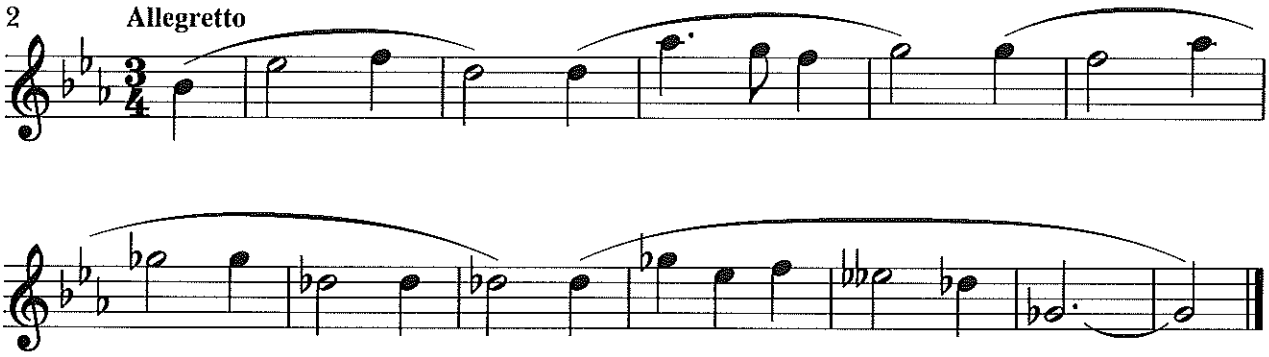


D. Harmonize the following melodies and set them for piano.

1 **Moderato**

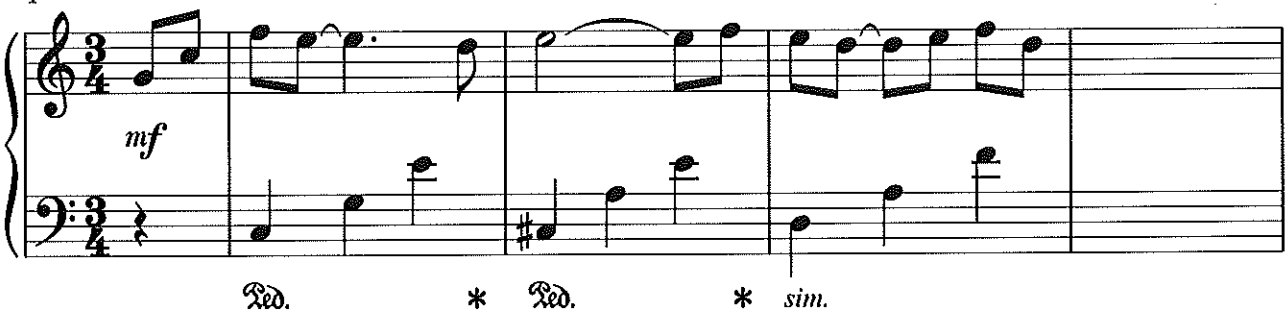


2 **Allegretto**



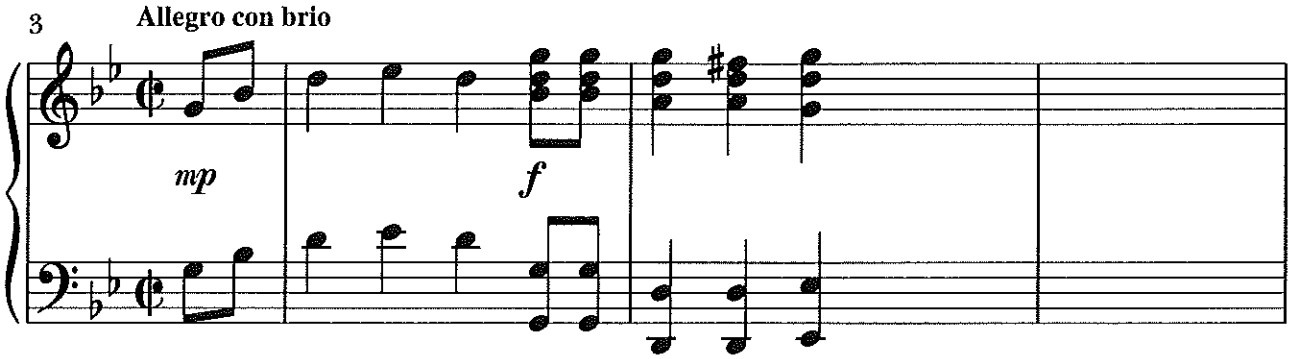
E. Complete the following pieces for piano in the style suggested by the beginning of each. They should be between twenty-four and thirty-two measures in length and should modulate at least twice before returning to the original key.

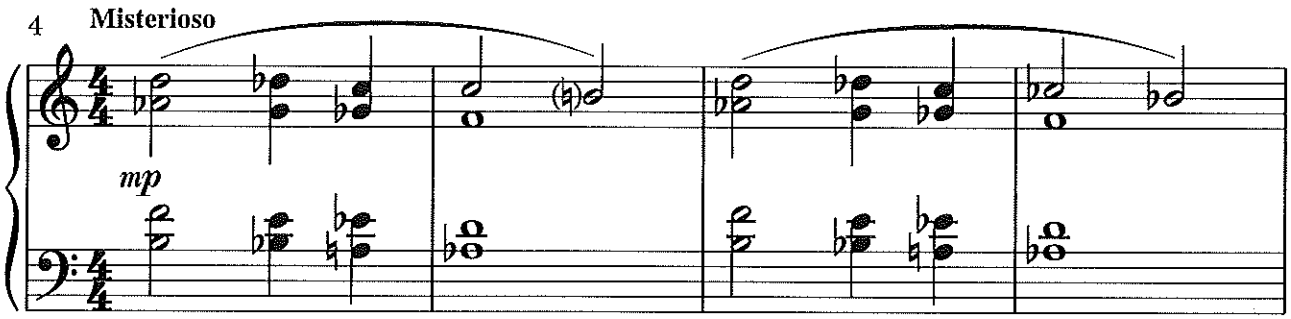
1 **Moderato**



2 **Moderately and simply**

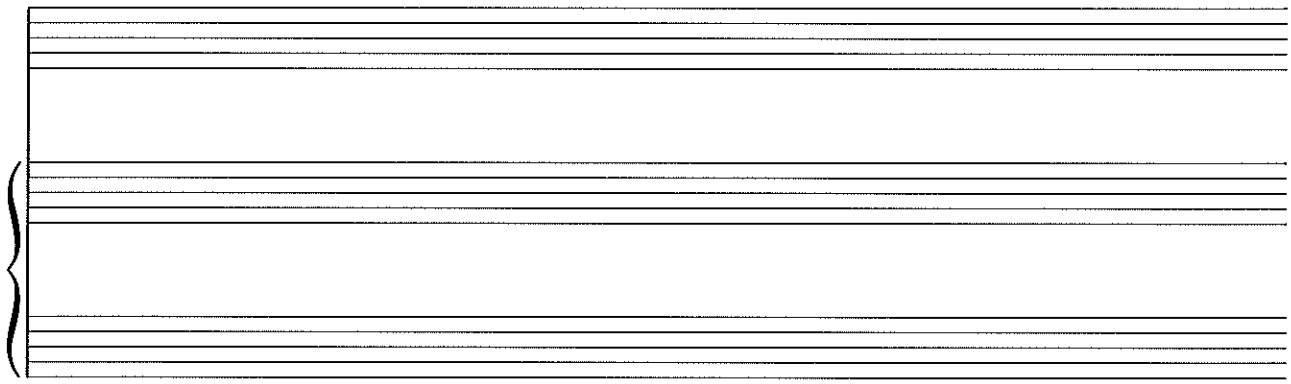


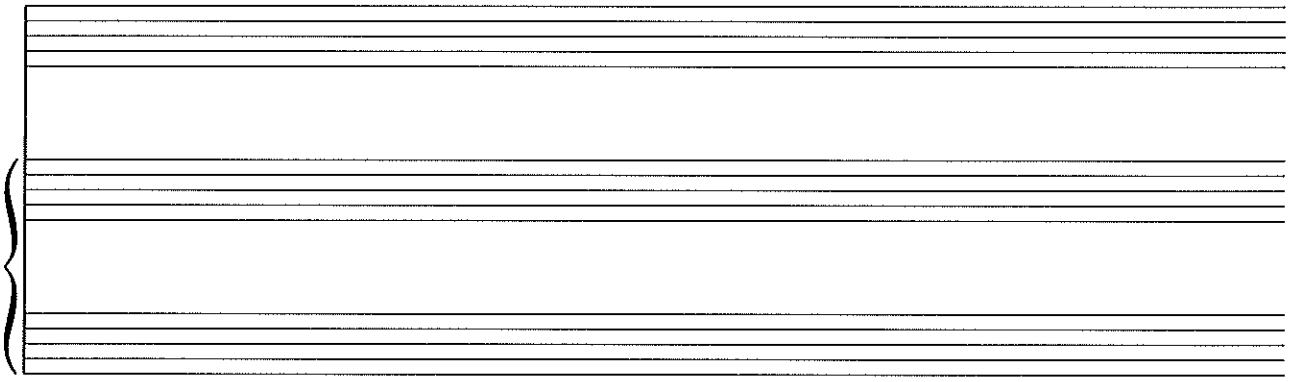

3
Allegro con brio
mp
f

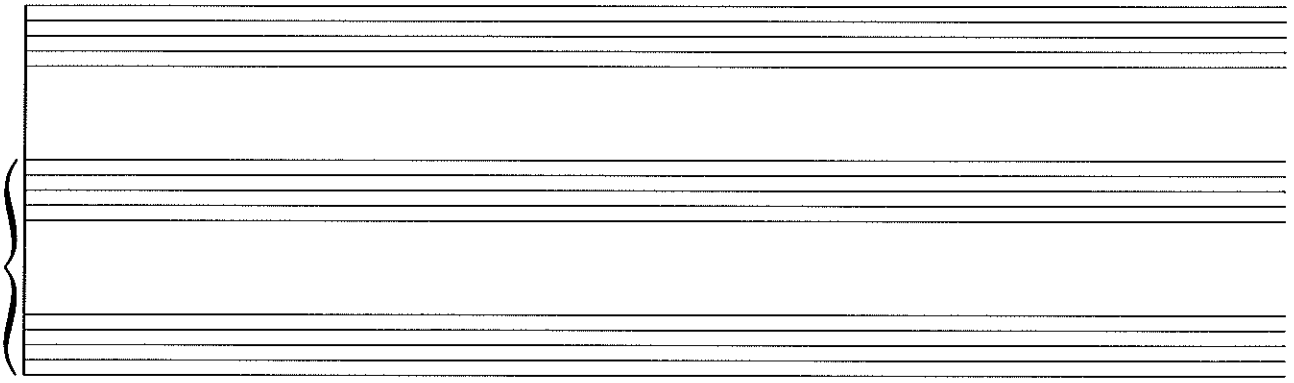

4
Misterioso
mp

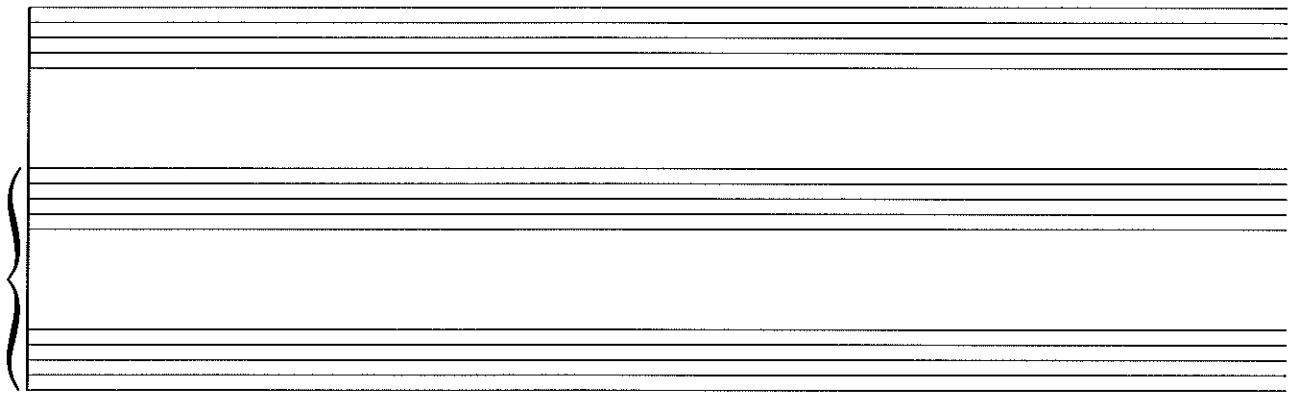
F. Select two of the previously completed piano pieces and arrange each of them for a solo instrument with piano accompaniment.

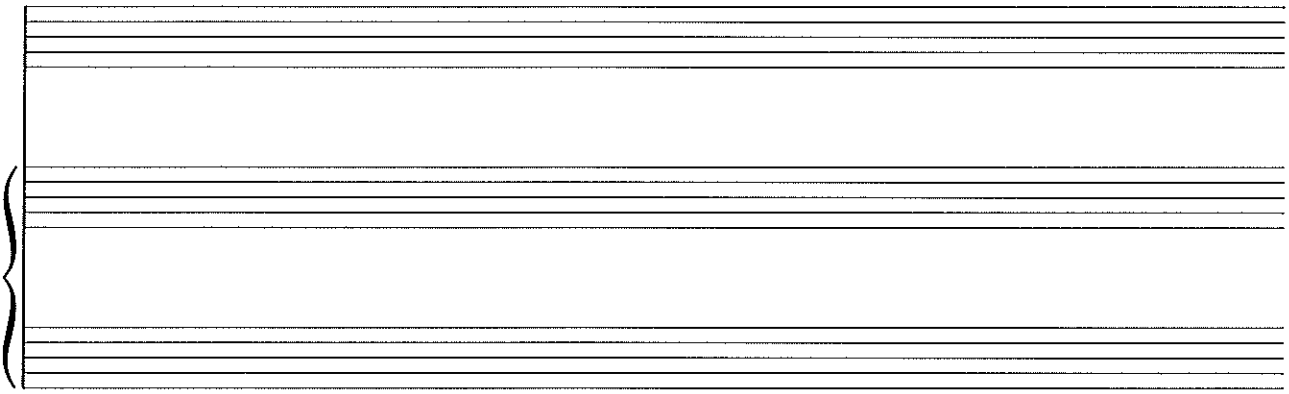
1

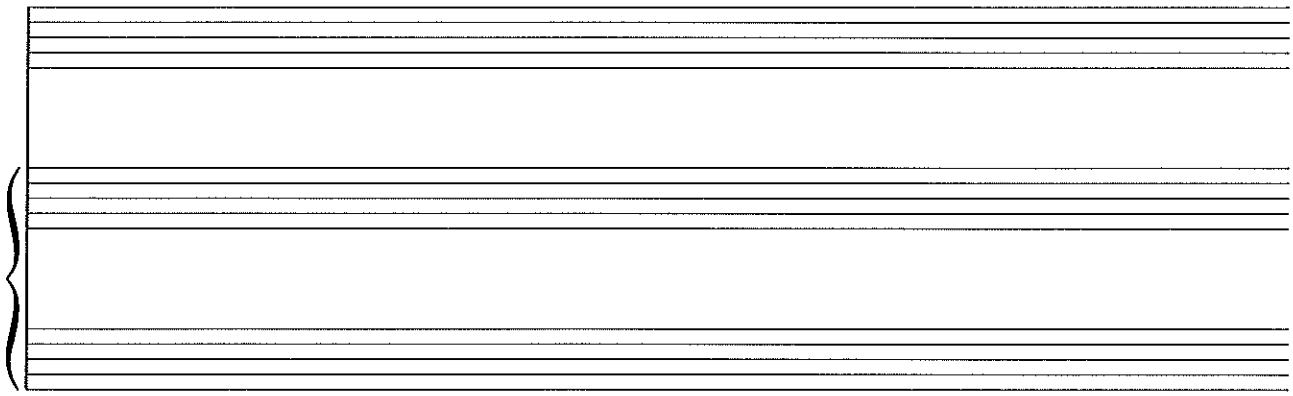




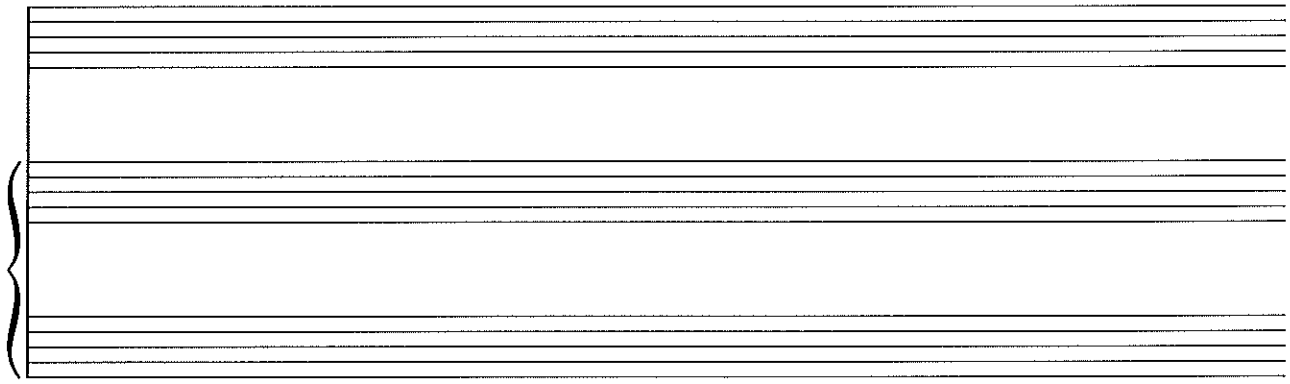


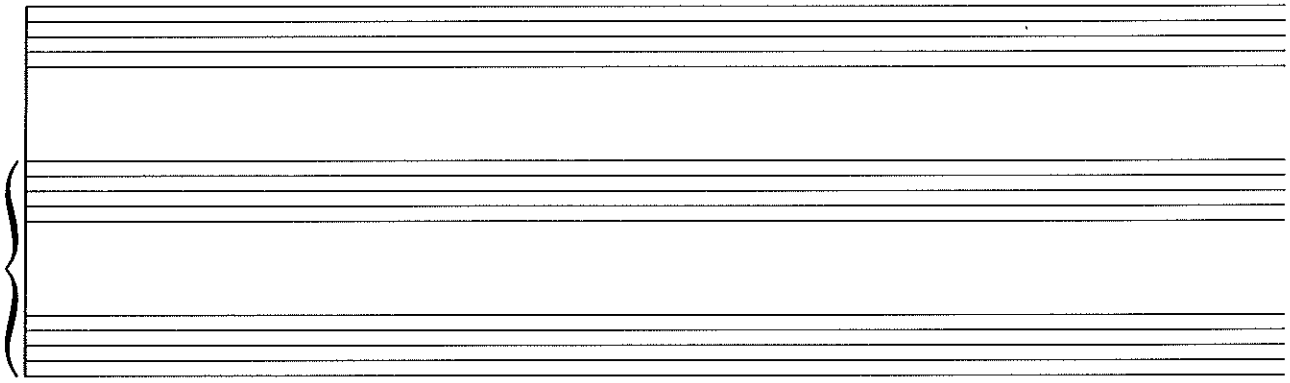


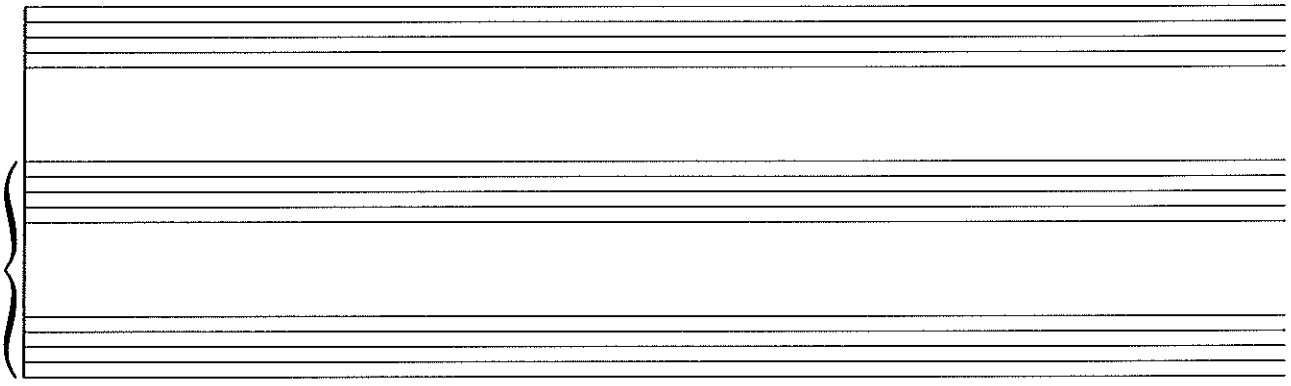


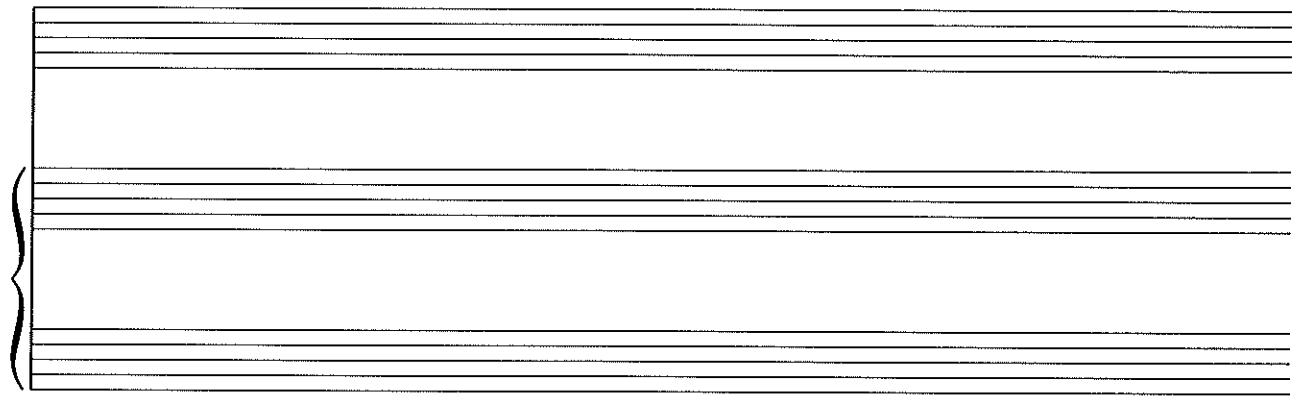


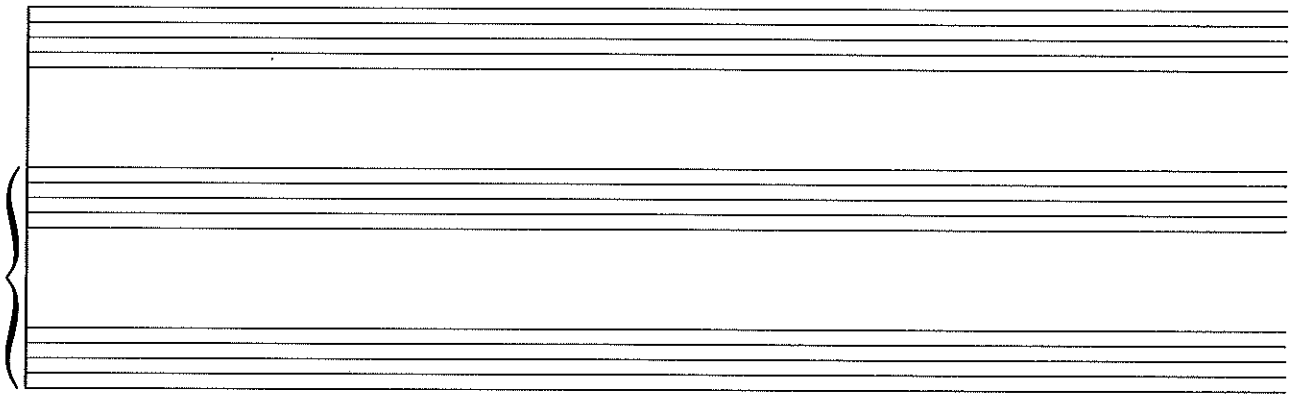
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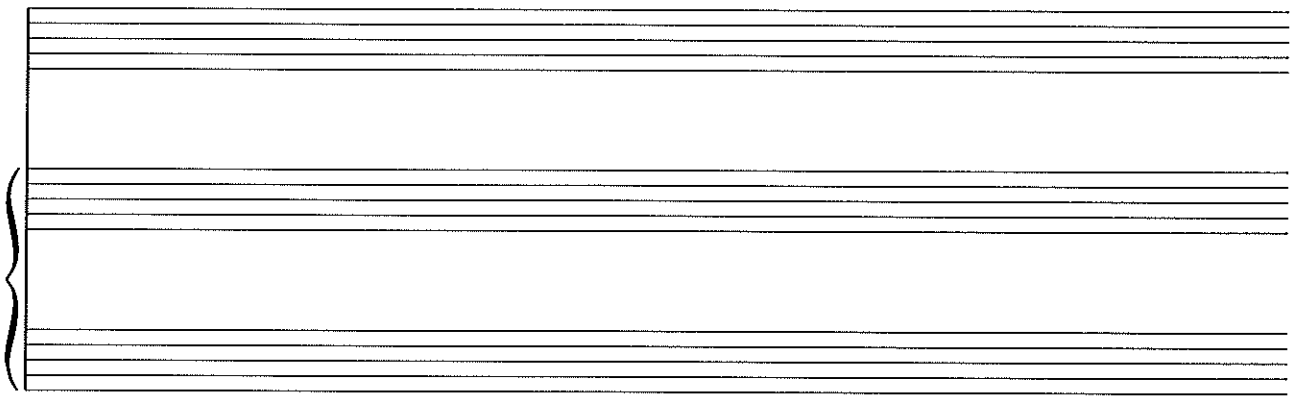












BURKHART, TUREK, AND WENNERSTROM EXERCISES FOR HARMONIC ANALYSIS

Burkhart

p. 182, mm. 5–8.
pp. 155–56, mm. 40–48.
p. 167, mm. 61–69.
pp. 190–91, mm. 71–81.
pp. 271–72, mm. 13–25.
pp. 309–10, mm. 1–77.

Turek

p. 264, mm. 37–44.
pp. 310–11, mm. 5–13.
p. 383, mm. 27–35.
p. 319, mm. 59–69.
pp. 259–60, mm. 12–24.

Wennerstrom

p. 374, mm. 44–48.
pp. 400–1, mm. 1–11.
pp. 398–99, mm. 1–18.
pp. 284–85, mm. 1–24.

COMPLETE PIECES FOR HARMONIC ANALYSIS

Burkhart

pp. 97–98. Bach, *Partita No. 1 in B Flat Major, Sarabande.*
pp. 114–15. Bach, *WTC, Book 1, Prelude in C Major.*
pp. 242–43. Beethoven, *Piano Sonata, Op. 2, No. 1, Menuetto.*
p. 308. Schubert, *Op. 9, No. 2, Waltz in A Flat Major.*
pp. 309–10. Schubert, *Six Moments Musical, No. 6.*
pp. 316–17. Schumann, *Album for the Young, Op. 68, Kleine Studie.*
p. 331. Chopin, *Preludes, Op. 28, No. 9 in E Major.*
p. 332. Chopin, *Preludes, Op. 28, No. 20 in C Minor.*
pp. 395–96. Wolf, *Two Songs, Das Verlassene Mägdlein.*

Turek

pp. 116–17. Bach, *WTC, Book 1, Prelude in C Major.*
pp. 131–32. Bach, *French Suite, No. 6, Sarabande.*
pp. 191–92. Haydn, *Piano Sonata in D Major, Largo e sostenuto.*
pp. 282–83. Beethoven, *Piano Sonata, Op. 53, Introducione.*
pp. 325–26. Schumann, *Scenes of Childhood, By the Fireside.*
pp. 327–28. Schumann, *Dichterliebe, Op. 48, Wenn ich in deine Augen seh'.*
p. 335. Schumann, *Album for the Young, Op. 68, Nordic Song.*
p. 351. Chopin, *Preludes, Op. 28, No. 20 in C Minor.*
pp. 390–91. Wolf, *Mörike Songs, No. 28.*

Wennerstrom

pp. 104–5. Bach, *WTC, Book 1, Prelude in G minor.*
pp. 119–20. Bach, *French Suite, No. 6, Sarabande.*
pp. 204–6. Mozart, *Piano Sonata, K.280, Adagio.*
pp. 233–34. Beethoven, *Piano Sonata, Op. 14, No.1, Allegretto.*
pp. 284–85. Schubert, *Waltzes, Op. 9, Nos. 14 and 29.*
pp. 313–14. Schumann, *Scenes from Childhood, Op. 15, No. 13.*
pp. 317–18. Schumann, *Carnaval, Op. 9, No. 4.*
p. 337. Chopin, *Preludes, Op. 28, No. 20 in C Minor.*
pp. 341–42. Chopin, *Mazurkas, Op. 67, No. 1 in G minor.*

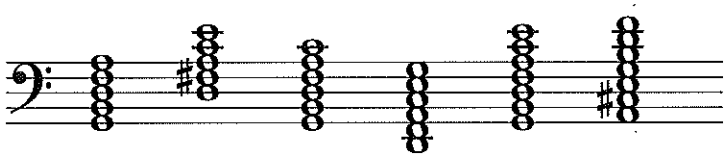
PART FOUR

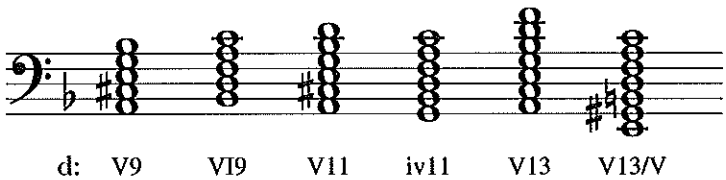
POST-COMMON PRACTICE HARMONY

24 Ninth, Eleventh, and Thirteenth Chords

Since the latter part of the nineteenth century, composers have expanded the tertian vocabulary by the use of multivoiced structures. Although these structures have been treated both conventionally and unconventionally with regard to part-writing and harmonic function, they enjoy wide usage largely as elaborations, or extensions, of seventh chords. Ninth, eleventh, and thirteenth chords may be employed on all scale degrees. In practice, however, they most frequently occur as extensions of the Mm7 chord and are exemplified below mainly in that role.

A seventh chord is constructed by adding to a triad a pitch that is a seventh above its root. Similarly, a ninth chord is constructed by adding to a seventh chord a pitch that is a ninth above its root. The logical extension of this procedure accounts for the eleventh and thirteenth chords, as shown by the examples below.



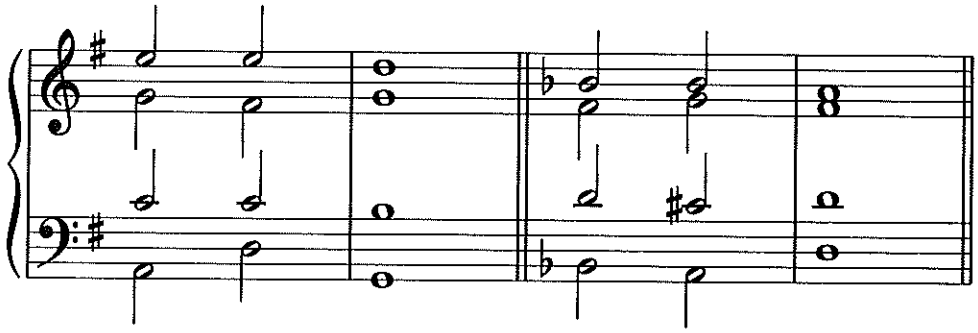


Note:

Each chord is designated by a single number.

NINTH CHORDS

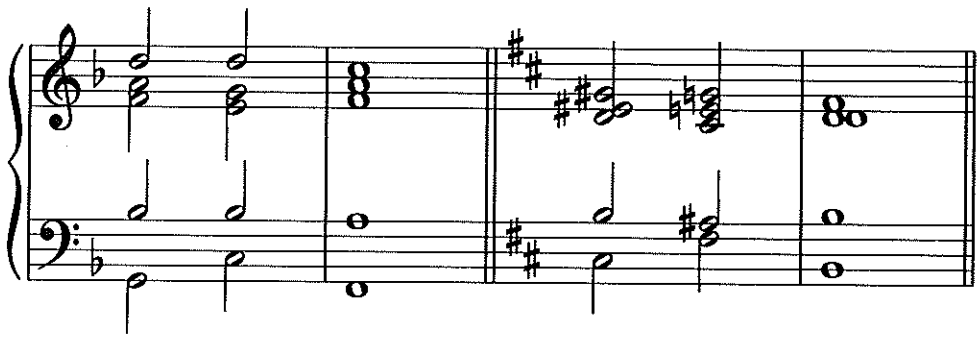
1. In four voices:



Note:

a. The fifth is omitted.
b. The ninth resolves downward by step.

2. In five voices:

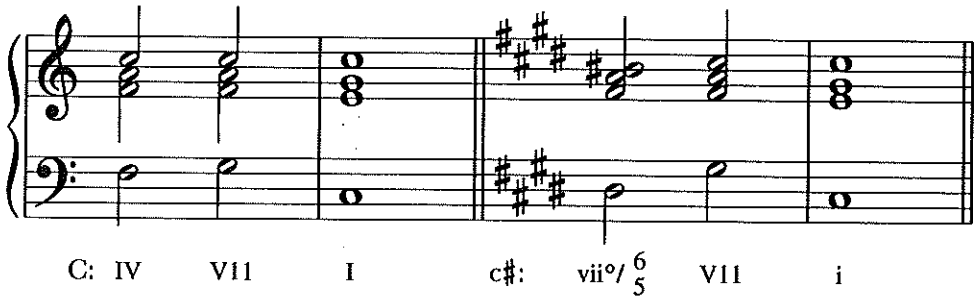


Note:

In the resolution of the dominant ninth to the tonic, the fifth is resolved upward by step to avoid parallel fifths.

PERFECT ELEVENTH CHORDS

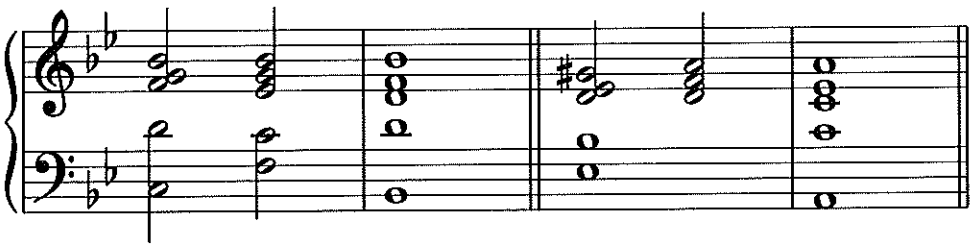
1. In four voices



Note:

a. The third and fifth are omitted.
b. The eleventh is resolved passively.
c. The sonority has characteristics of both dominant and subdominant harmony.

2. In five voices:

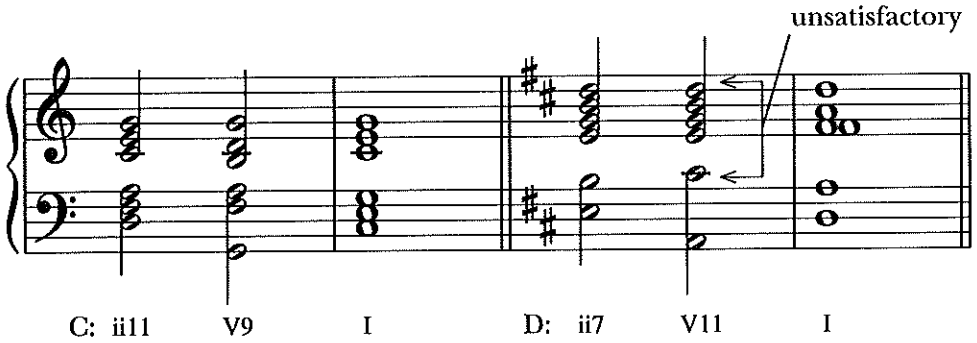


Note:

a. The third is omitted.

b. In the resolution of the dominant eleventh to the tonic, the fifth is resolved upward by step to avoid parallel fifths.

3. In six voices:

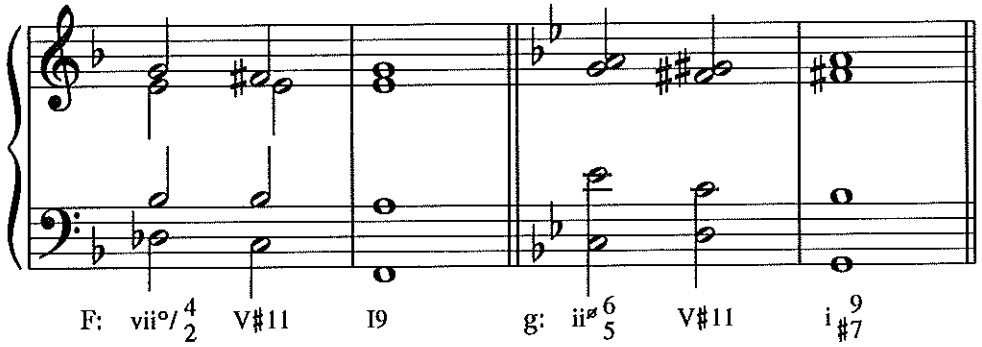


Note:

Because the complete dominant perfect eleventh includes both the leading tone and the tonic note, the chord is very rarely used.

AUGMENTED ELEVENTH CHORDS

1. In four voices:



Note:

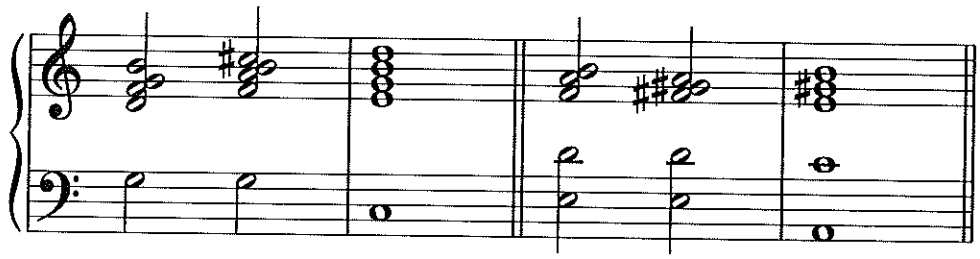
a. The fifth and ninth are omitted.

b. The eleventh resolves upward by a half step.

c. The third is resolved passively.

d. I9 is a dissonant sonority. The use, therefore, of the dominant augmented eleventh is restricted to musical styles in which this kind of activity is tolerated, for example, jazz and popular music, and in the works of many composers of the impressionist period through the present.

2. In five voices:

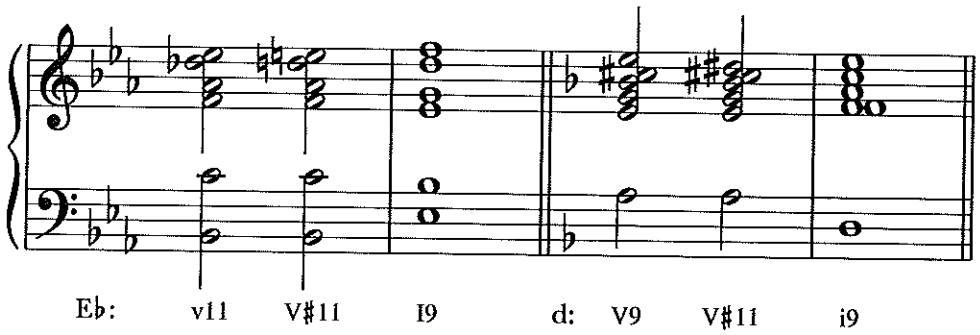


Note:

a. The fifth is omitted.

b. The resolution of the dominant augmented eleventh is a complete tonic ninth.

3. In six voices:

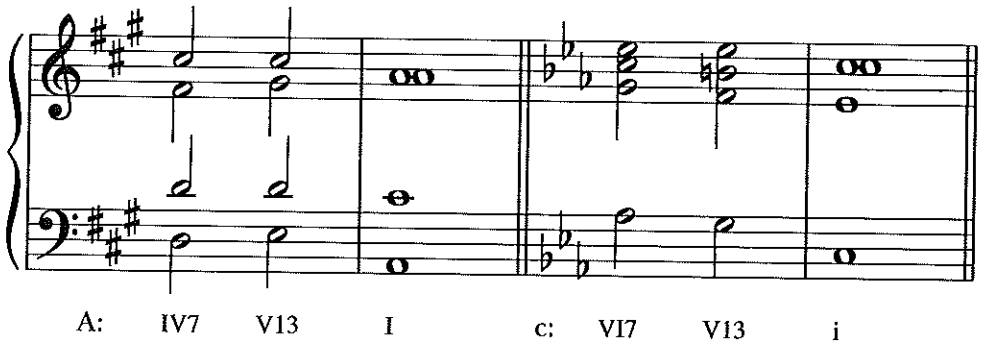


Note:

The fifth is resolved upward by step when doing so is necessary to avoid parallel fifths.

THIRTEENTH CHORDS

1. In four voices:

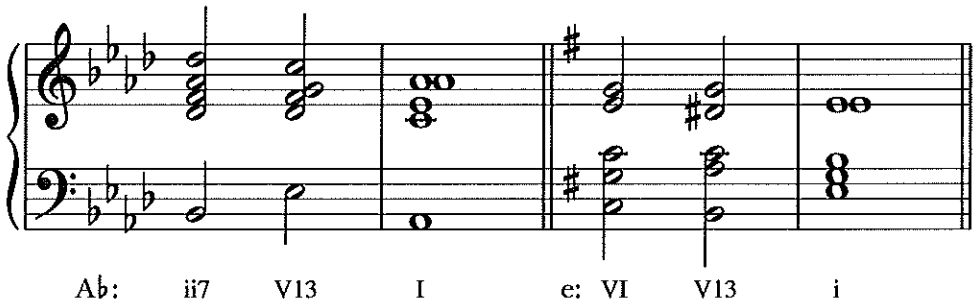


Note:

a. The fifth, ninth, and eleventh are omitted.

b. The thirteenth resolves downward by a third to the tonic note.

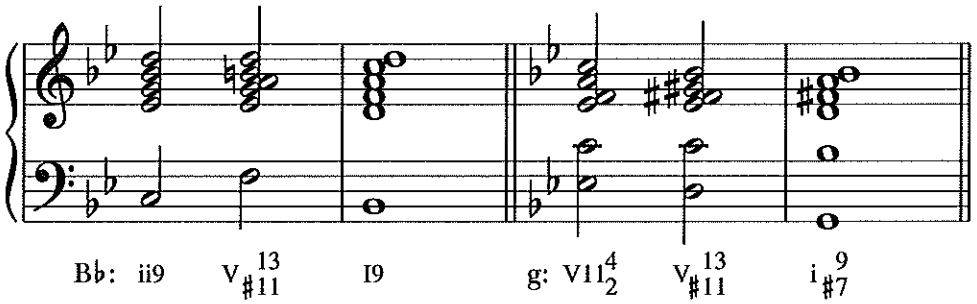
2. In five voices:



Note:

The fifth and the eleventh are omitted.

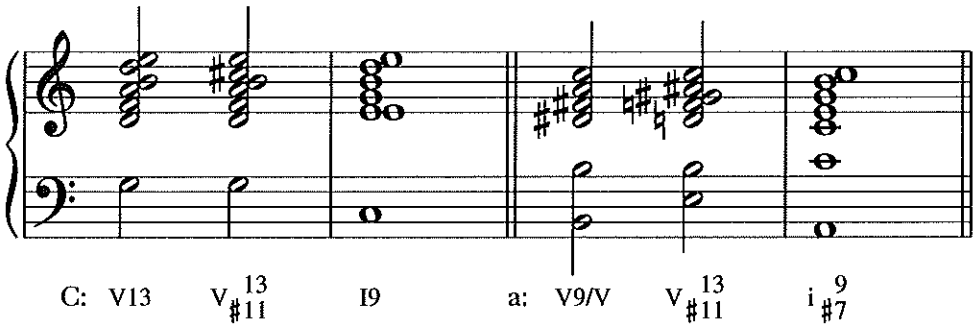
3. In six voices:



Note:

a. The fifth is omitted.
b. The augmented eleventh is used.
c. The thirteenth is usually resolved passively to avoid crossing voices.
d. The third is resolved passively.

4. In seven voices:



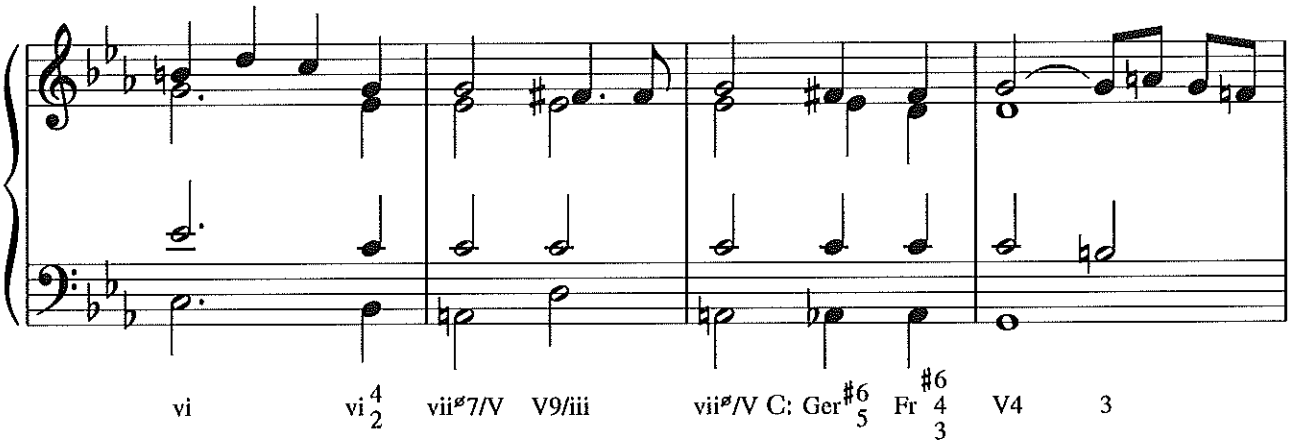
Note:

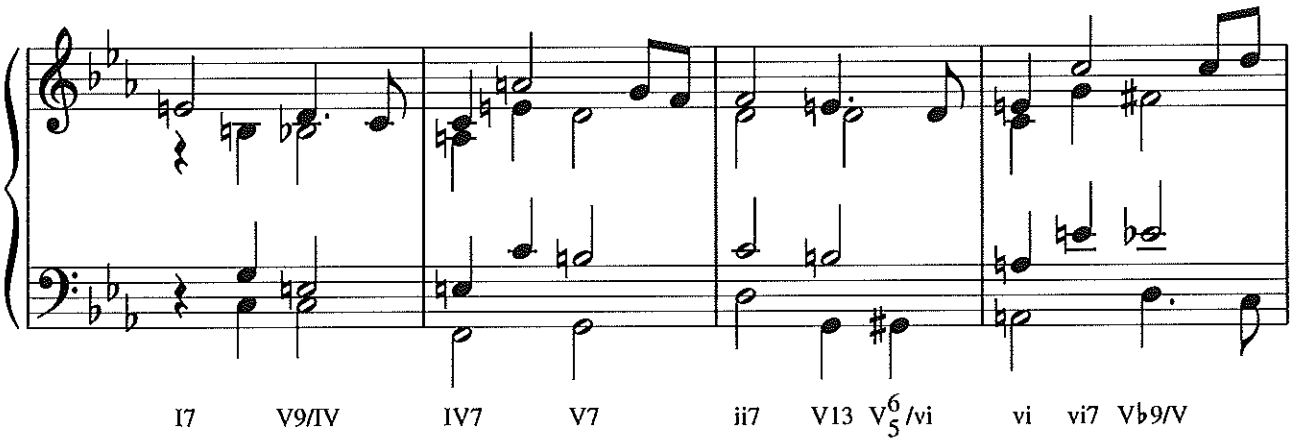
The fifth is resolved upward to avoid parallel fifths.

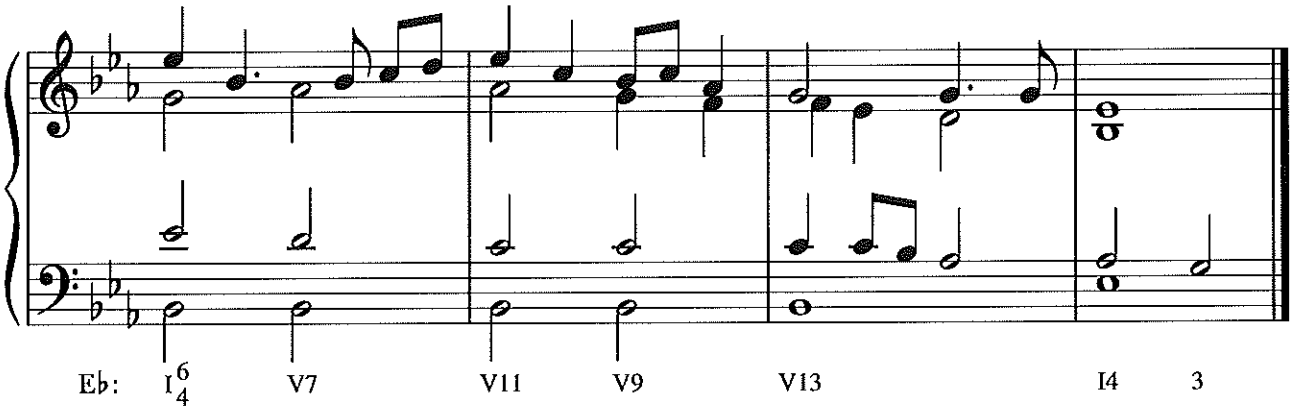
NINTH, ELEVENTH, AND THIRTEENTH CHORDS IN COMBINATION

The two musical examples that follow, one for piano and one for SATB, show how these extended chords may be used within the framework of functional tertian harmony. Careful study of the examples will reveal some minor irregularities in voice-leading. The addition of ninths, elevenths, and thirteenths to seventh chords frequently creates problems for which no "correct" solutions exist. Under such circumstances, one resolves the problems as smoothly as the situation allows.




vi
vi 4 2
vii⌀7/V
V9/iii
vii⌀/V
C: Ger #6 5
Fr #6 4 3
V4
3

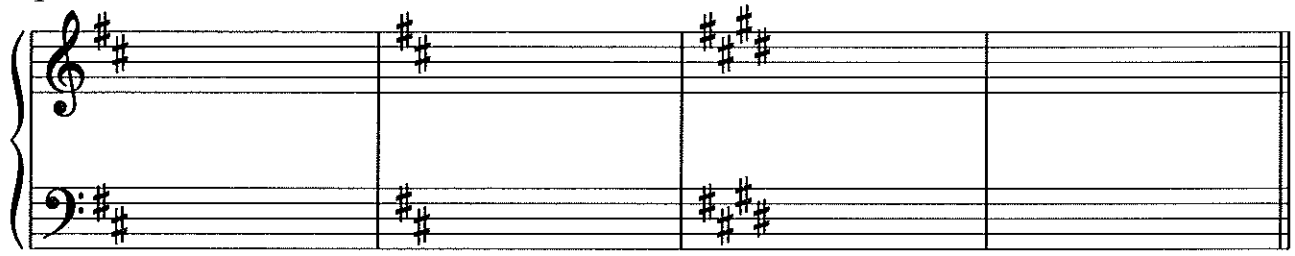

I7
V9/IV
IV7
V7
ii7
V13
V6 5 /vi
vi
vi7
V♭9/V


E♭:
I 6 4
V7
V11
V9
V13
I4
3

EXERCISES

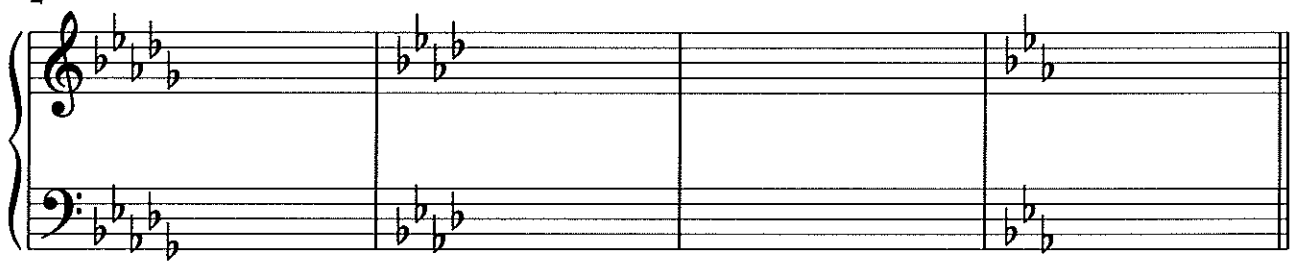
A. Write and resolve the following chords in four voices.

1



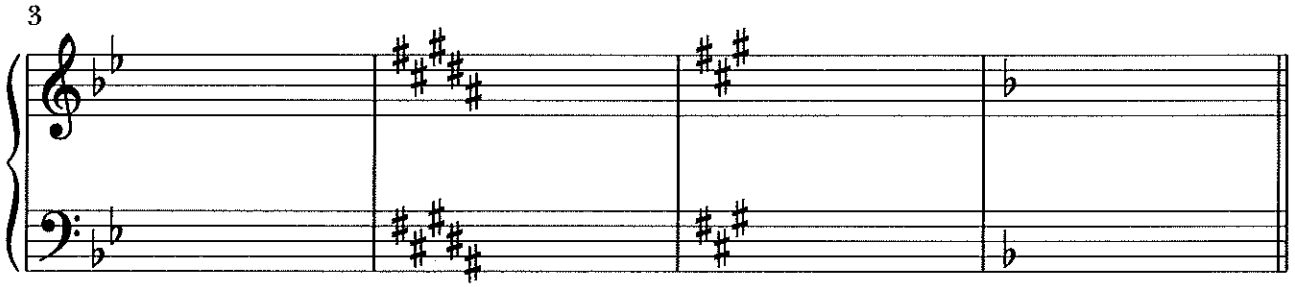
D: V9 b: V9 E: ii9 C: V9/V

2



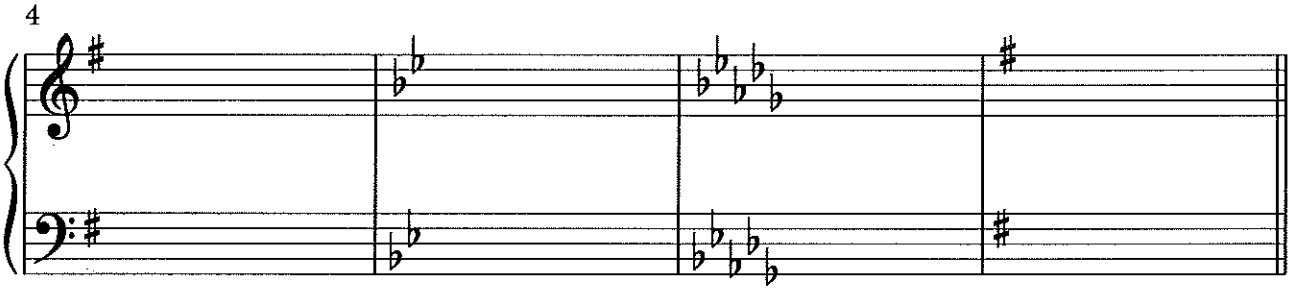
D♭: V11 A♭: ii11 a: V11 c: V11

3



g: V♯11 B: V♯11 f♯: V♯11 F: V♯11

4



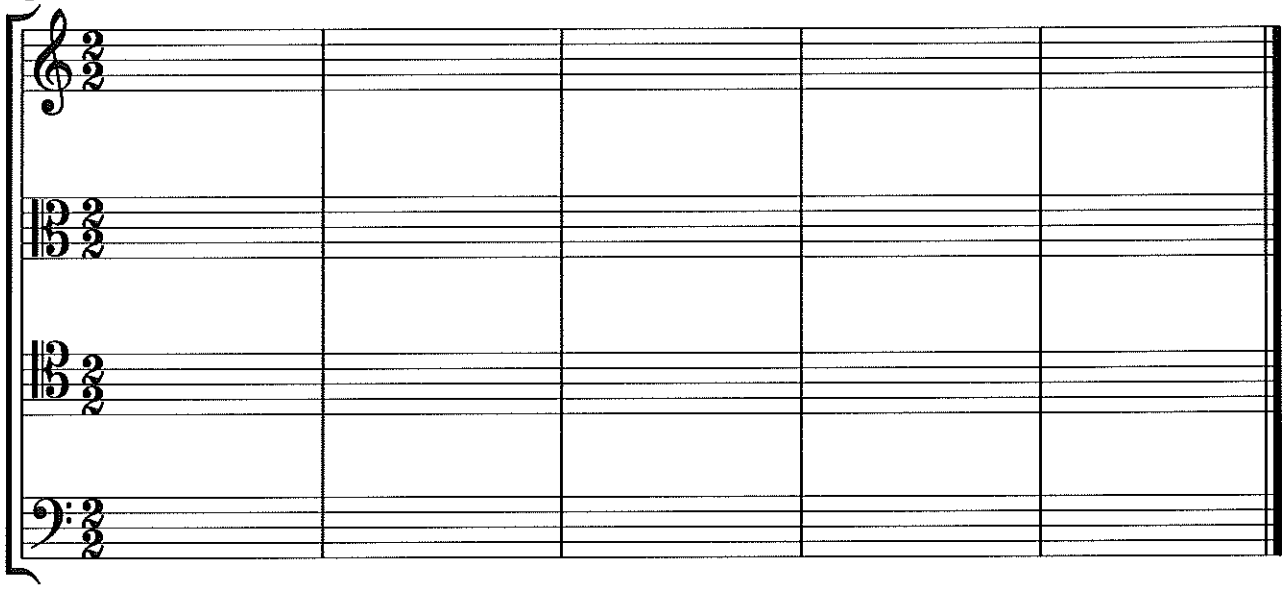
G: V13 B♭: V13/V b♭: V13 e: V13/VI

B. 1. Make a harmonic analysis of the piece below.
2. Reduce it to a four-voice structure by crossing out all the unessential pitches.



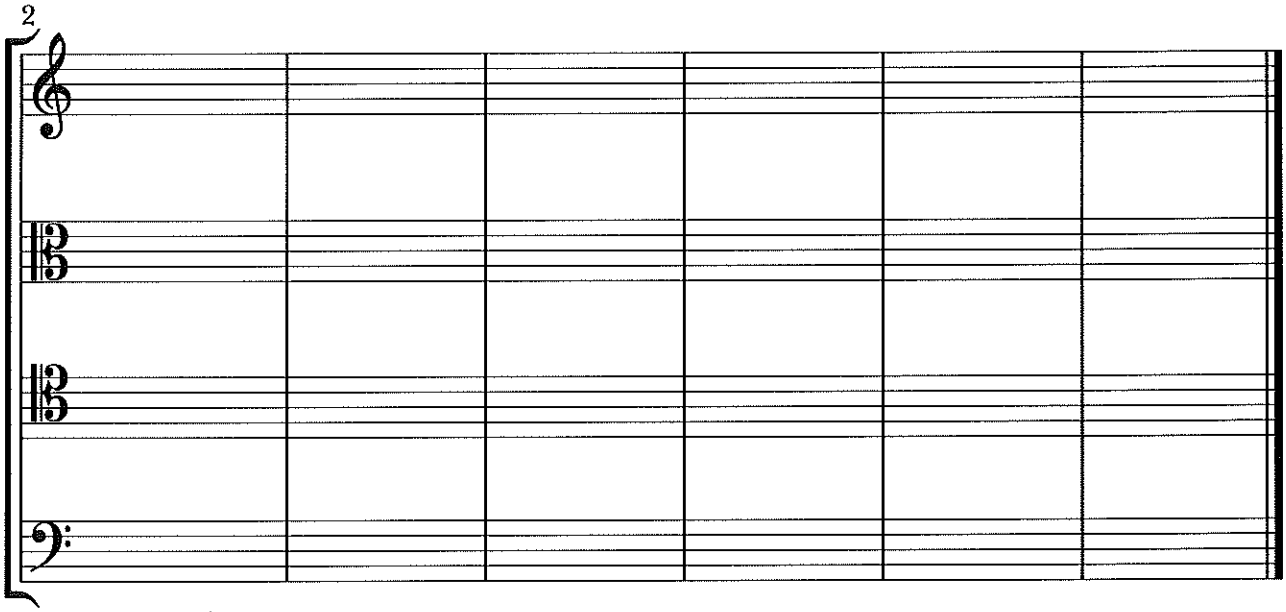
C. Harmonize the following in open score, making appropriate use of nonchord tones.

1



E♭: ii6 V13 I V11 I vi7 V♯11/V V9 I

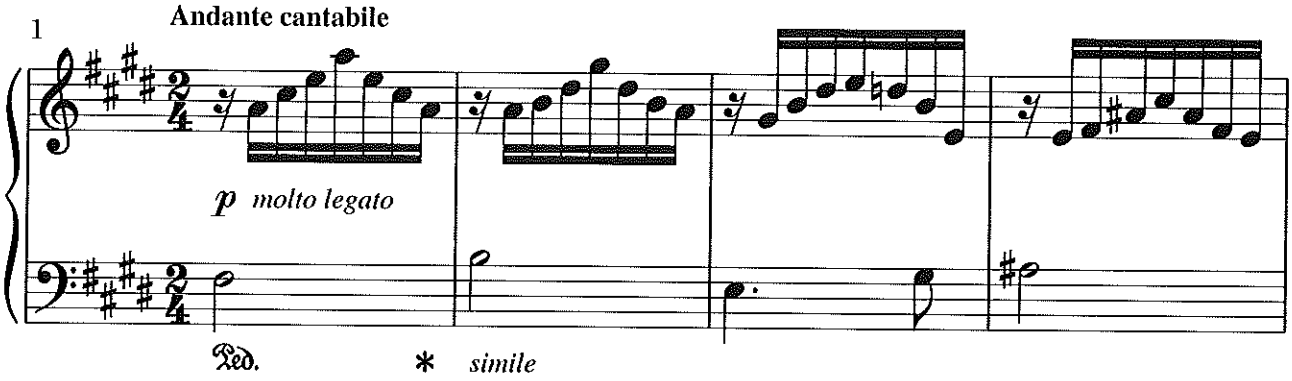
2

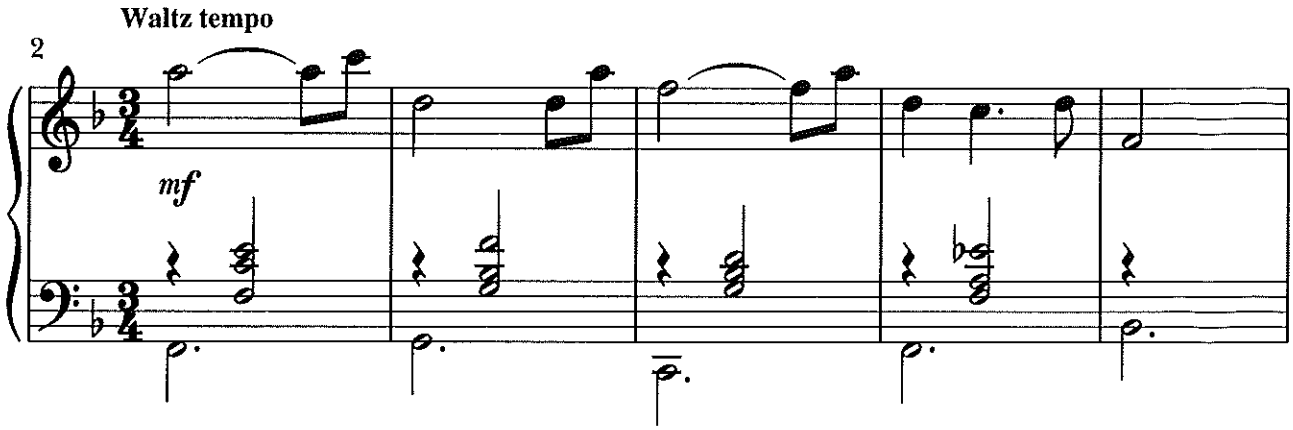


a: i V^4_2/iv iv6 V13 VI V9/N N N6 V11 V7 i4 3

D. Complete the following pieces for piano in the style suggested by the beginning of each. They should be no more than sixteen measures in length.

1




Waltz tempo
2
mf

BURKHART, TUREK, AND WENNERSTROM EXERCISES FOR HARMONIC ANALYSIS

Burkhart

pp. 349–55.

Turek

pp. 359–65.

Wennerstrom

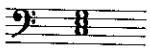
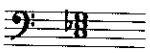
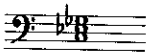
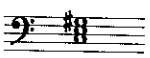
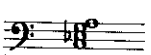
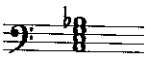
pp. 353–56.
pp. 357–58.

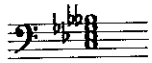
1. Analyze the key scheme of each example.
2. Identify all chords with dominant (or secondary dominant) function, and label the resolution of each one as normal (N) or deceptive (D).

25 Chord Symbols

In Chapter 9, figured bass was introduced and shown to be a musical shorthand to save composers' time; with the addition of Roman numerals representing the seven scale degrees, it has become a useful analytical tool. With a change in emphasis from bass to soprano (or melody), a similar shorthand has developed in this century. Commonly found in lead sheets of popular music is a melody line under (or above) which chord symbols are written. These symbols (which, unlike the combination of Roman numerals and figured bass, give no specific information about harmonic function and voice leading) allow performers both to harmonize the melody and, depending upon their expertise, to improvise upon it. In addition, chord symbols are printed above the vocal line in published sheet music and are used by performers who either cannot play the written-out piano part or prefer to invent their own accompaniment based upon the given harmonies.

The symbols below are those understood and used by most performers and composers of jazz and popular music.

SYMBOL	QUALITY	NOTATION
C	major triad	
Cm or C-	minor triad	
C° or Cdim	diminished triad	
C^{+} or Caug	augmented triad	
C6	major triad with added major sixth	
Cm6	minor triad with added major sixth	
C7	major-minor seventh (dominant seventh)	

CM7 or Cmaj7	major-major seventh	
Cm7 or C-7	minor-minor seventh	
C°7 or Cdim7 (or C° or Cdim)	diminished-diminished seventh	
C9	dominant ninth	
Cm9	minor-minor seventh with added ninth	
C11	dominant eleventh	
C13	dominant thirteenth	

Note:

a. The symbol for the diminished triad is also used when a diminished seventh chord is intended.

b. There is no simple symbol for a half-diminished seventh chord, but its equivalent is given below.

Chromatic alterations and more complex chords are indicated in the following manner:

C-9 or C♭9		
C+11 or C♯11		
Cm7-5 or Cm7♭5		(half-diminished seventh)

An inversion is indicated by writing the name of the bass note either below or to the right of the chord symbol, thus:

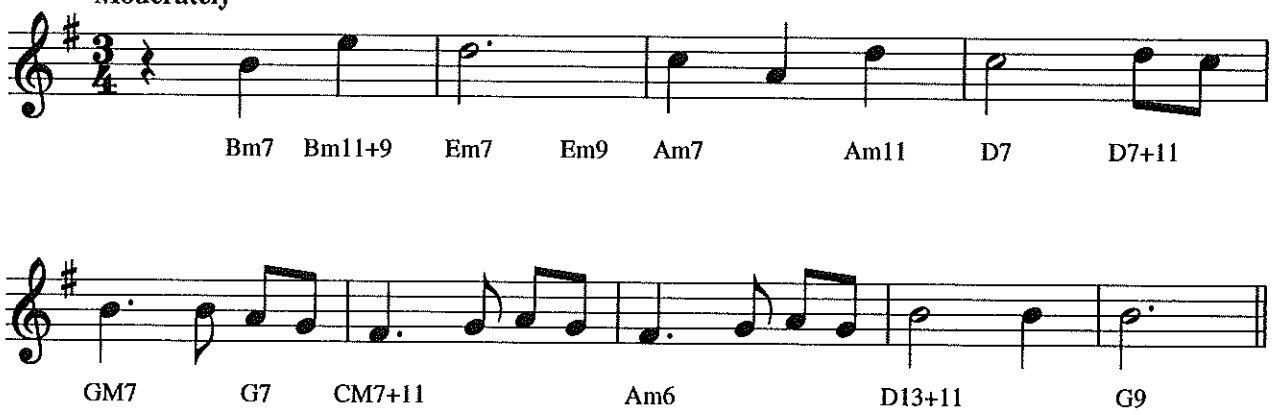
A pedal bass note may be written similarly:

$\frac{C7}{E}$ or C7/E 

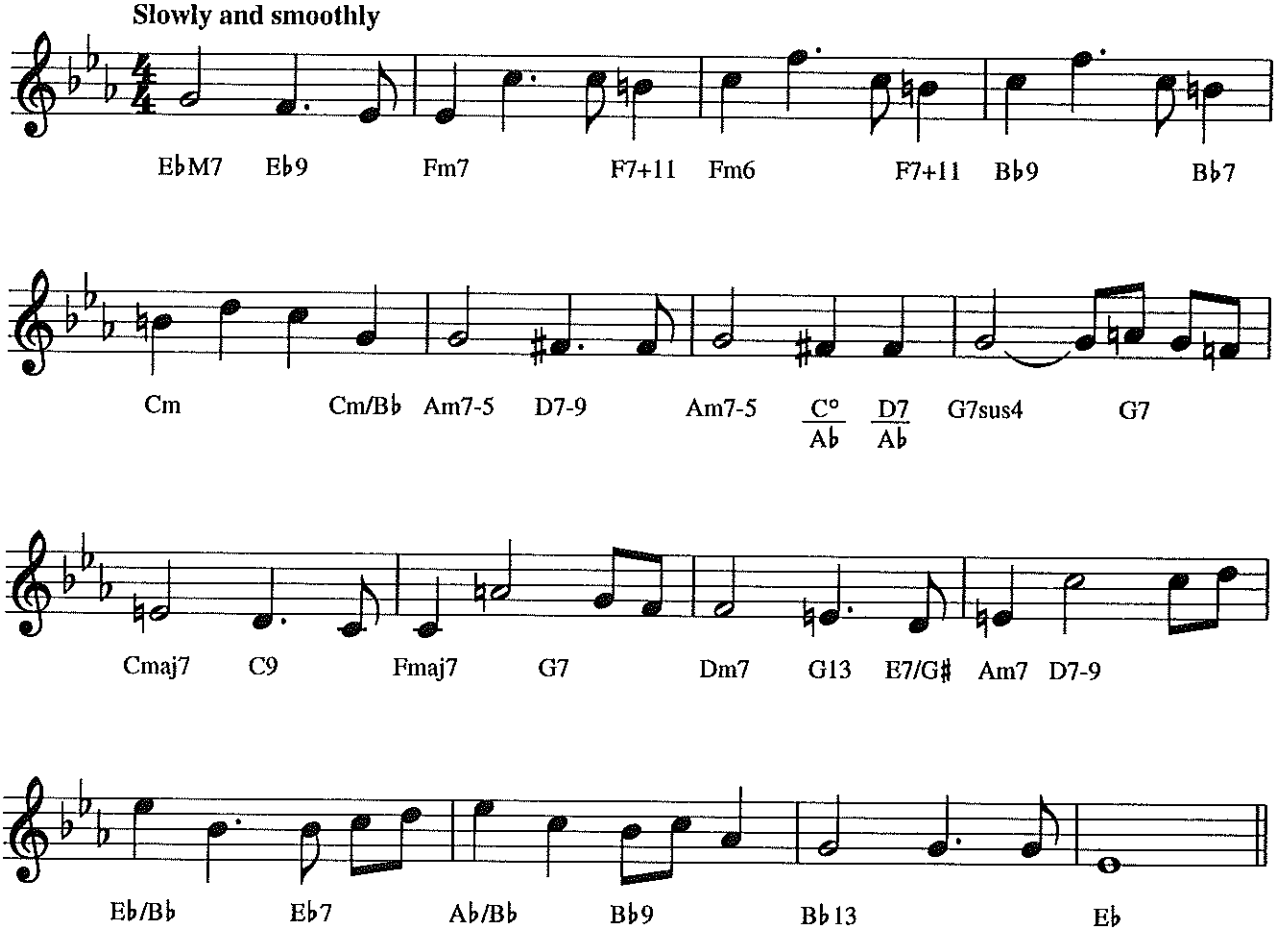
$\frac{Cm7}{F}$ or Cm7/F 

The two examples below are lead sheet, and therefore approximate, reductions of the examples of combined ninth, eleventh, and thirteenth chords in the previous chapter. Compare the two versions carefully, and make note of the translation of the Roman numeral analysis into chord symbols.

Moderately



Slowly and smoothly



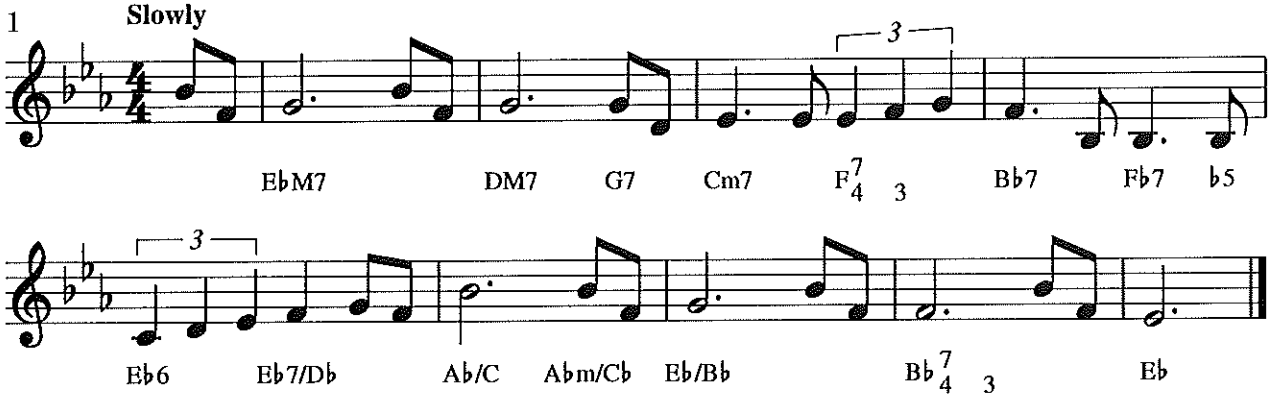
EXERCISES

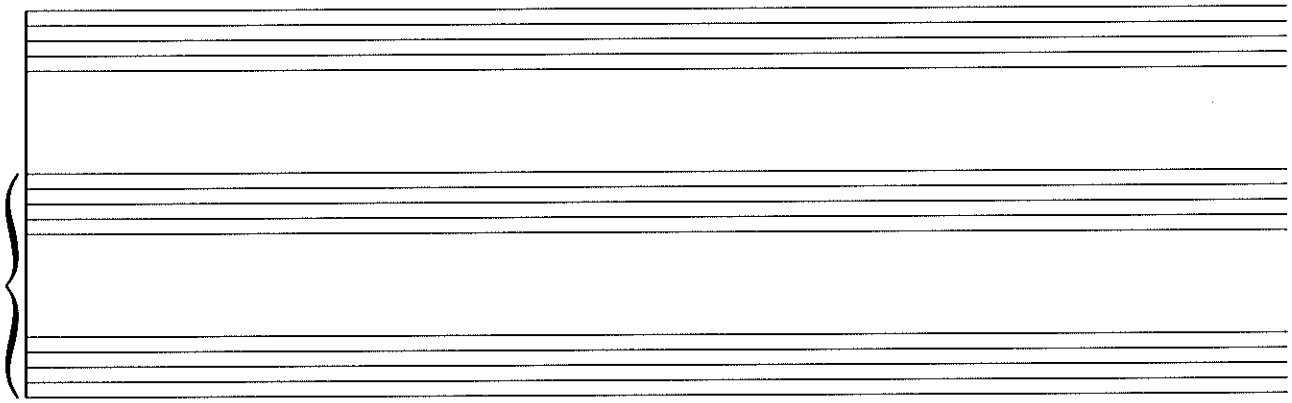
A. Assign symbols to the following chords.

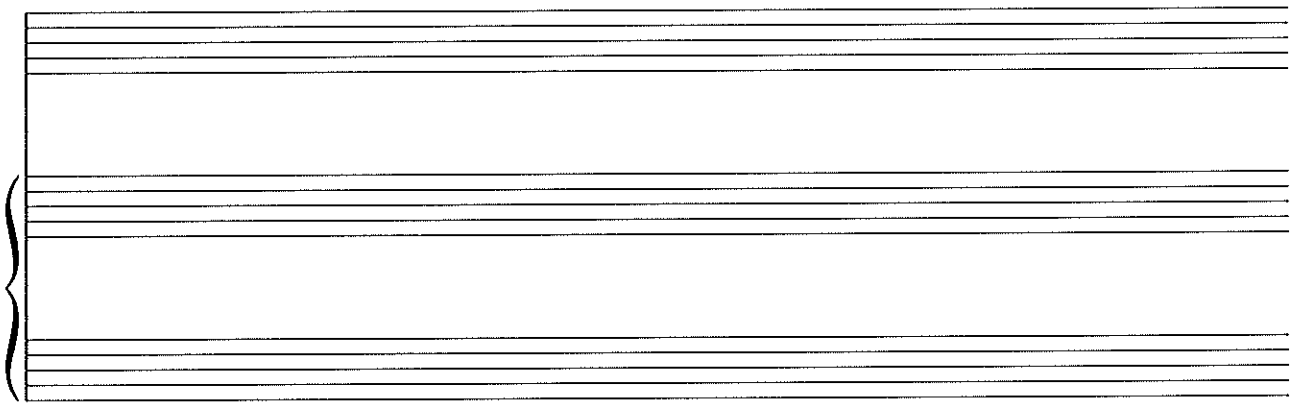


B. Using the given chords, invent piano accompaniments for the following melodies.

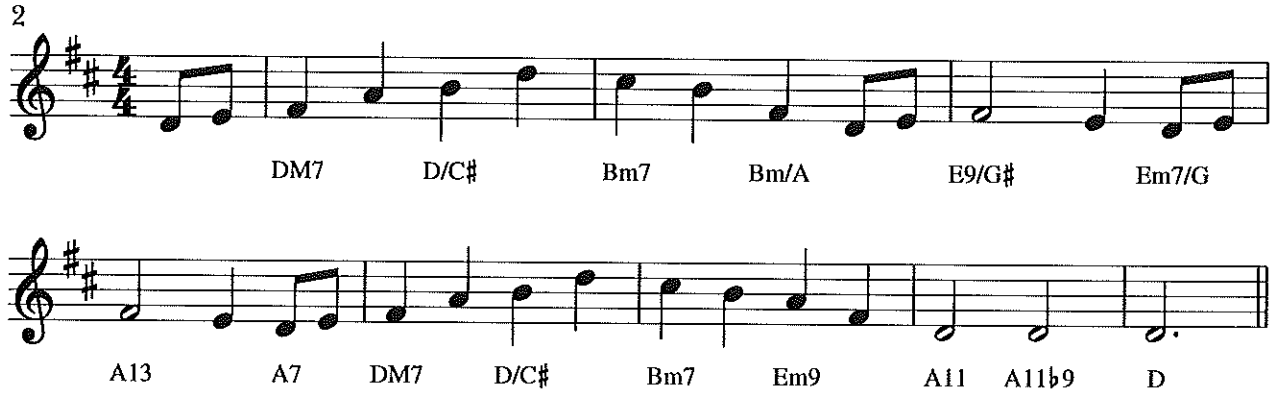
1 **Slowly**

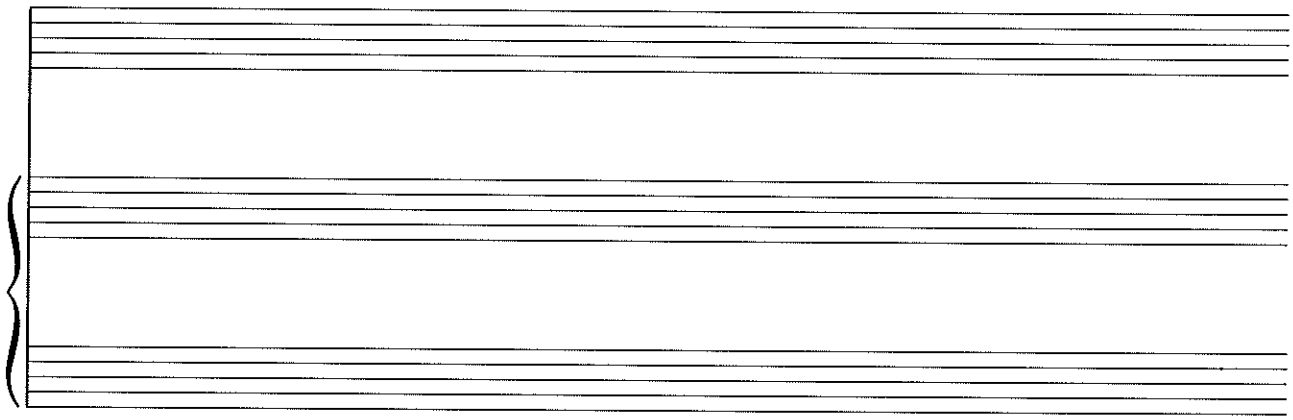


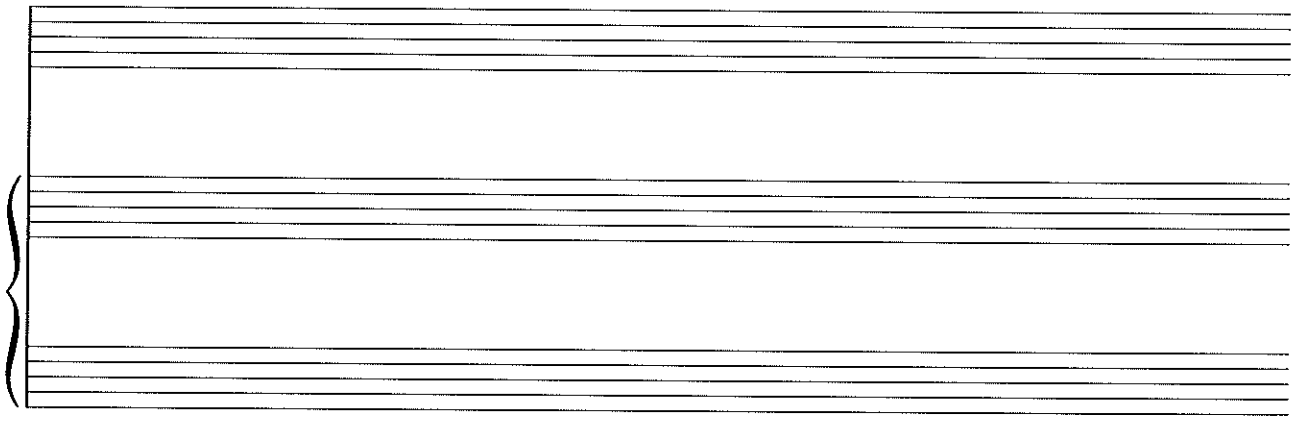




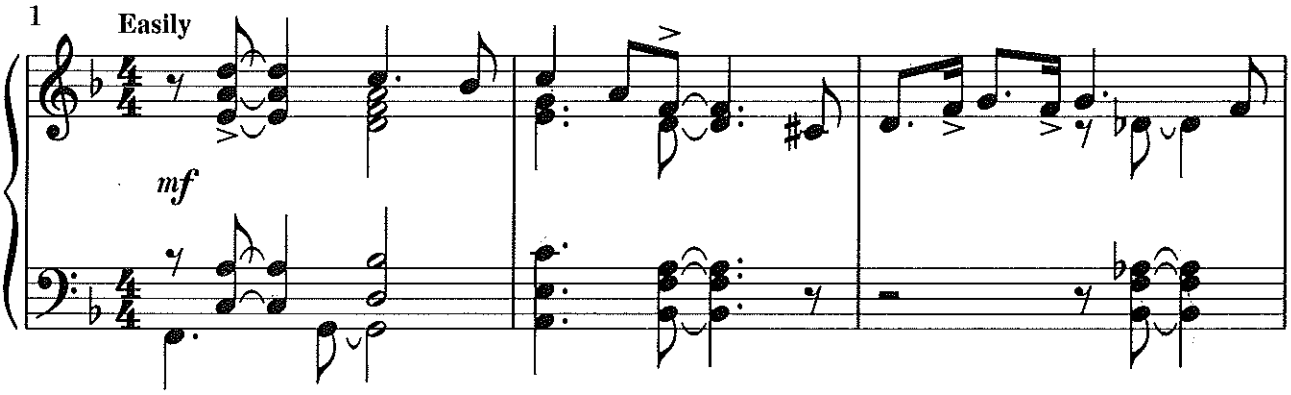
2







C. Complete the following pieces for piano in the style suggested by the beginning of each. They should be no more than sixteen measures in length.



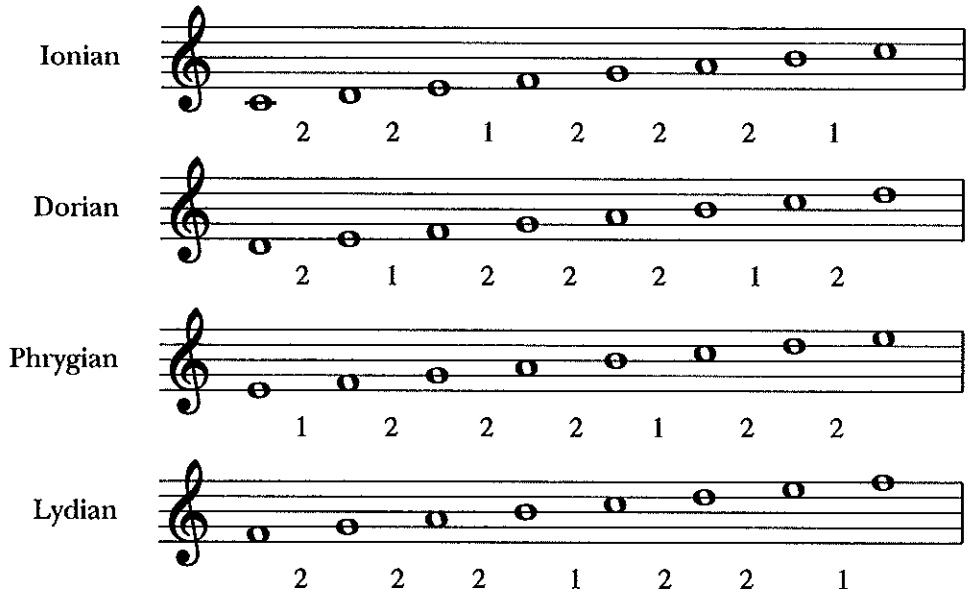

Slowly and rhythmically
p

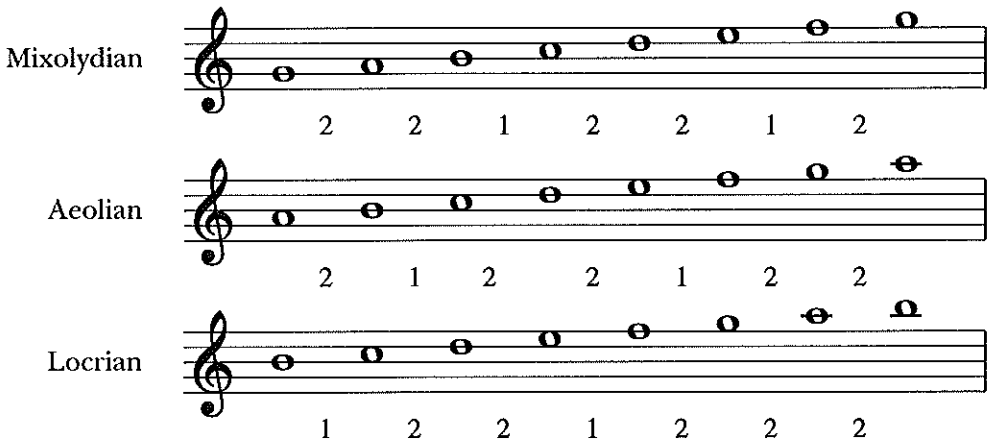
26 Modal Harmony

MODES

Despite the wealth of resources offered by the major-minor scale system, toward the end of the nineteenth century some composers began to seek out alternative scale patterns from which to build their compositions. As a result of these endeavors, the so-called Greek modes, which predate the major-minor scale system by several centuries, enjoyed a renaissance as the basis for both melodic and harmonic activity.

There are seven of these modes, each of which may be constructed by beginning upon a different degree of the major scale:





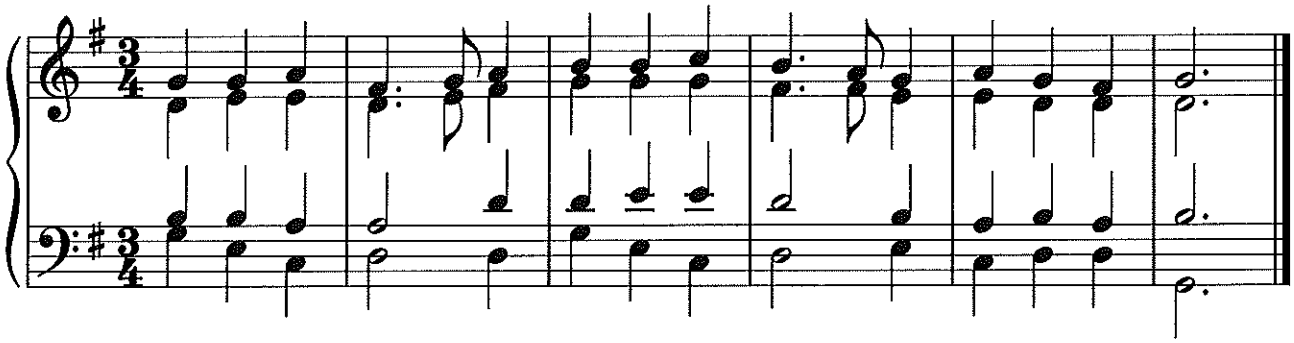
It can be seen that, because of its starting point, each mode has an individual intervallic organization which gives it its particular character.

There are, however, three logical groupings of the modes, based upon the qualities of the "tonic" triads.

1. *Major modes:*
   Ionian–The same as the major scale.
   Lydian–The major scale with a *raised fourth* degree.
   Mixolydian–The major scale with a *lowered seventh* degree
2. *Minor modes:*
   Dorian–The natural minor scale with a *raised sixth* degree.
   Phrygian–The natural minor scale with a *lowered second* degree.
   Aeolian–The same as the natural minor scale.
3. *Diminished mode:*
   Locrian–The natural minor scale with a *lowered second* and *lowered fifth* degree. The only mode without a dominant-tonic relationship.

The first six measures of "America" are set below in each of the seven modes. Study the examples carefully, and make note of how they differ one from another.

Ionian



Note:

Since the Ionian mode is the same as the major scale, this is a simple diatonic setting of the melody.

Lydian



Note:

a. The Lydian mode is built upon the fourth degree of the major scale. Since G is the fourth degree of the D major scale, G Lydian has the same key signature as D major.

b. The soprano $c\sharp^2$ at the end of the third measure has a strong tendency to resolve to d^2. For this reason, the melody is subsequently transferred to the alto voice.

c. Although G is emphasized throughout as a tonic, the presence of the leading tone to V creates a certain tonal ambiguity. This is a feature to exploit in harmonizing Lydian melodies.

Mixolydian

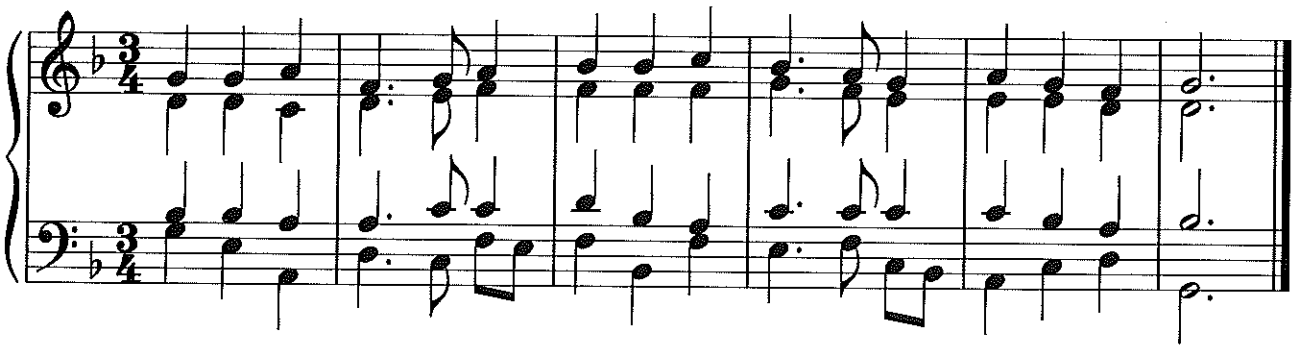


Note:

a. The Mixolydian mode is built upon the fifth degree of the major scale. Since G is the fifth degree of the C major scale, G Mixolydian has the same key signature as C major.

b. The tonal center of G is reinforced by a strong cadence and a paucity of C major chords.

Dorian

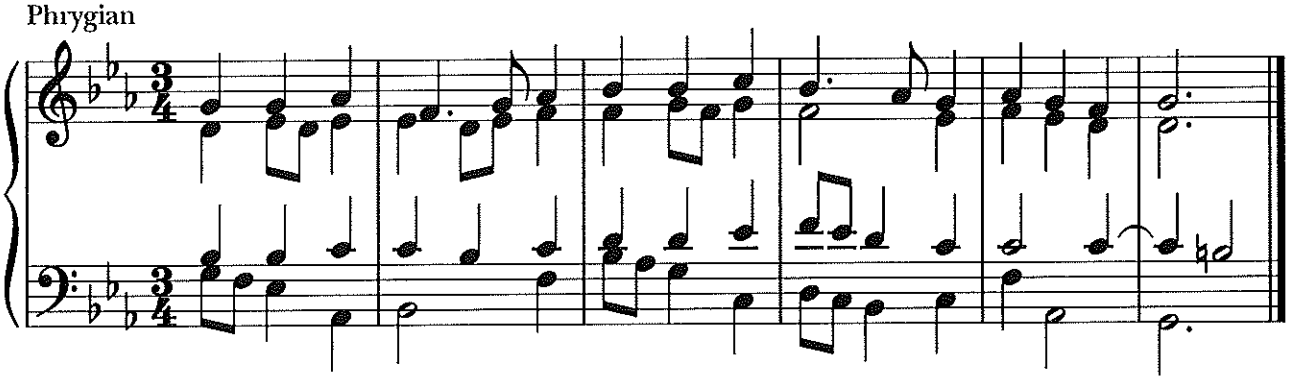


Note:

a. The Dorian mode is built upon the second degree of the major scale. Since G is the second degree of the F major scale, G Dorian has the same key signature as F major.

b. This setting tends to gravitate toward F major, largely due to the structure of the melody in this mode. The deceptive resolution of the C7 chord at the cadence aids in reaffirming G as tonic.

Phrygian

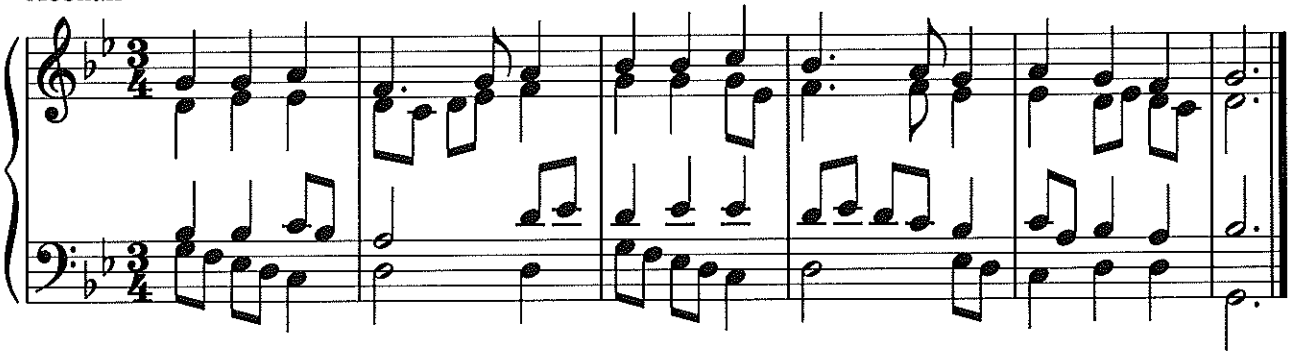


Note:

a. The Phrygian mode is built upon the third degree of the major scale. Since G is the third degree of the E♭ major scale, G Phrygian has the same key signature as E♭ major.

b. This setting typically concludes with the so-called Phrygian cadence, which, in this case, gives a strong impression of a half close in c minor.

Aeolian



Note:

a. The Aeolian mode is built upon the sixth degree of the major scale. Since G is the sixth degree of the B♭ major scale, G Aeolian has the same key signature as B♭ major.

b. This setting is an elaboration of the one in the Ionian mode but placed, of course, in the parallel natural minor. The two should be compared.

Locrian



Note:

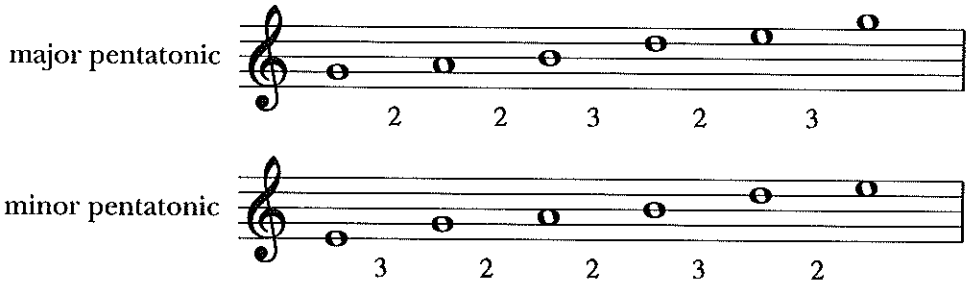
a. The Locrian mode is built on the seventh degree of the major scale. Since G is the seventh degree of the A♭ major scale, G Locrian has the same key signature as A♭ major.

b. The unstable diminished triad on G is no more than a theoretical tonic, and the setting concludes with a half cadence in A♭ major. The Locrian mode, in fact, is seldom used within the framework of functional harmony because a satisfactory conclusion on the tonic is almost impossible to achieve.

No chromatic pitches were used in any of the seven settings. Although such pitches are not prohibited, they should be handled with great care so that the flavor of the mode is not destroyed. For example, the excessive use of a raised seventh degree in a Mixolydian setting would negate the reason for using the Mixolydian mode in the first place.

THE PENTATONIC SCALE

An ancient scale with a number of modal connotations is the *pentatonic* scale. It contains five different pitches whose intervallic organization corresponds to the black keys on the piano. Although, technically, any one of its notes may be considered a tonic, the two forms given below are the most common since they alone allow the construction of a tonic triad.



Note:

a. The scales are made up entirely of major seconds and minor thirds.

b. Neither scale contains a dominant triad. As a result, pentatonic harmonizations tend to have a modal flavor.

EXERCISES

A. The key signature and the tonic are given. Identify the mode.

1



Mixolydian

2



3



4



5



6



7

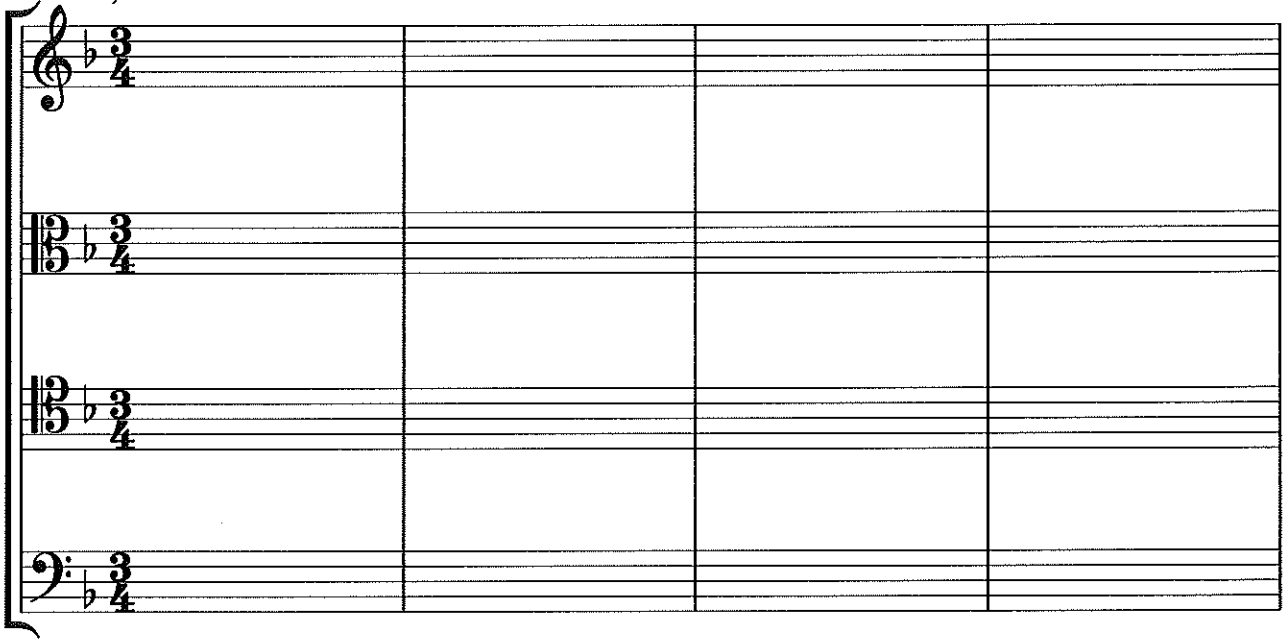


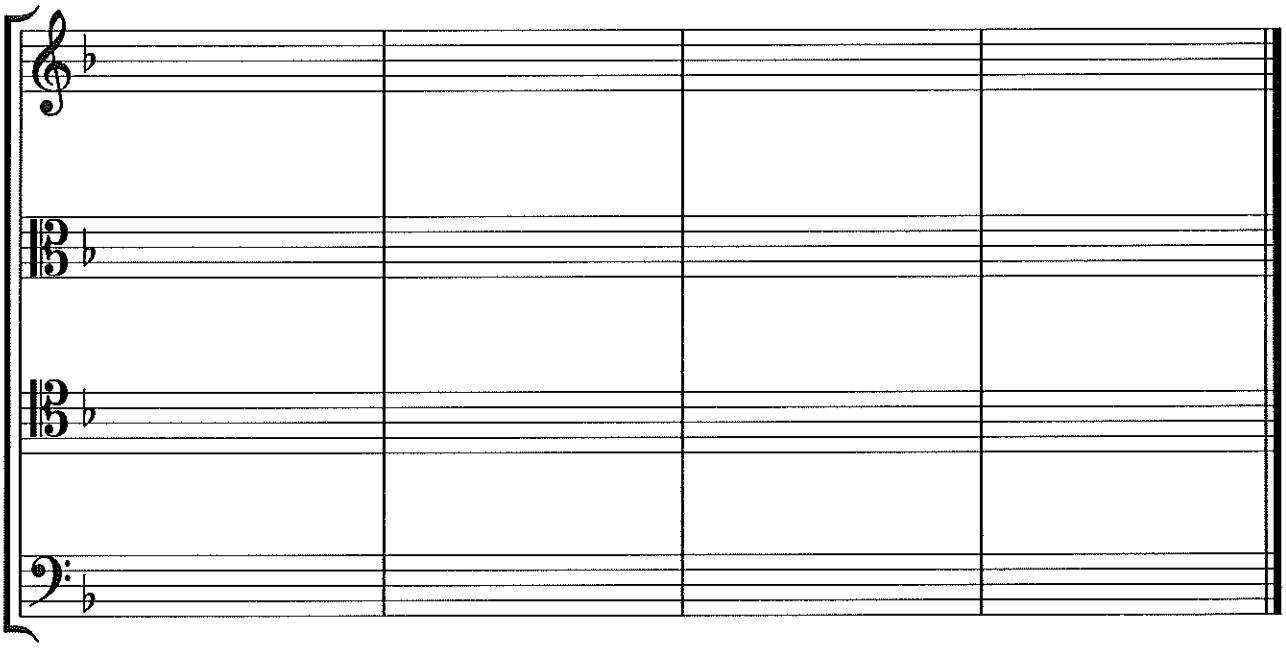
8



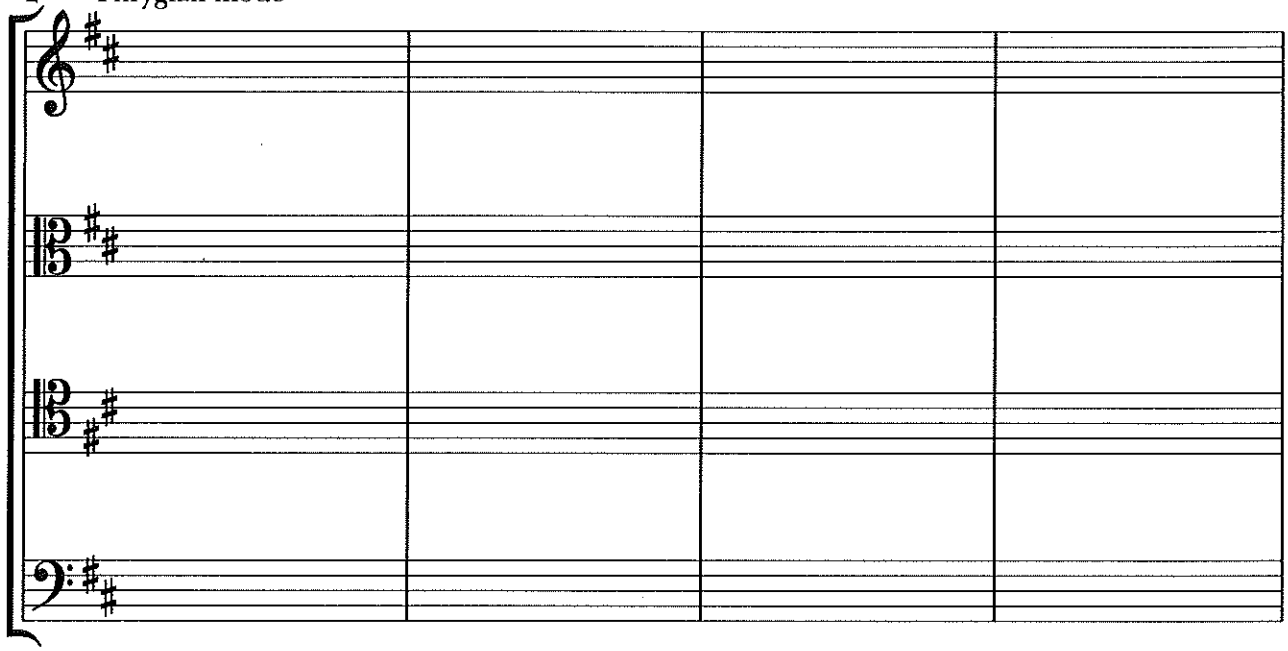
B. Write a melody of two four-measure phrases in each of the given modes, and harmonize it in open score.

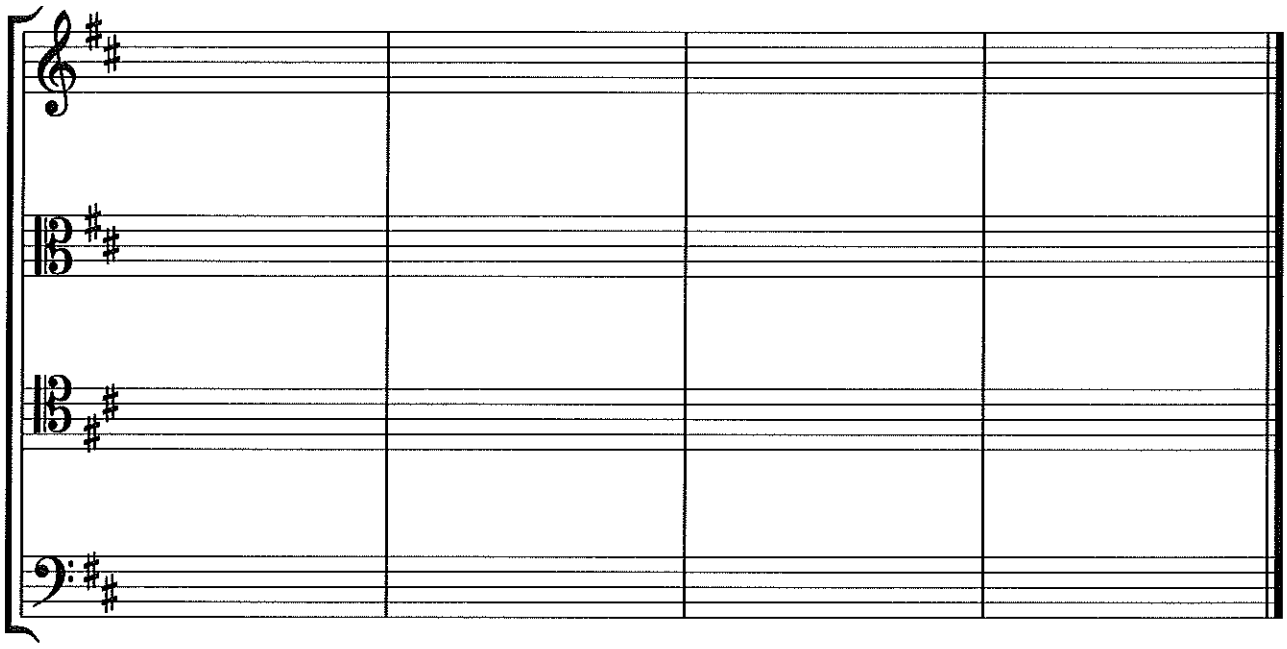
1 Lydian mode





2 Phrygian mode





BURKHART, TUREK, AND WENNERSTROM EXERCISES FOR HARMONIC ANALYSIS

Burkhart

pp. 421–25.

Turek

pp. 408–11.

Parts of both movements lend themselves to a modal analysis.

1. Identify the sections that seem to be modal.
2. To each modal section, assign a tonal center.
3. To each tonal center, assign a mode.

27 Nonfunctional Harmony

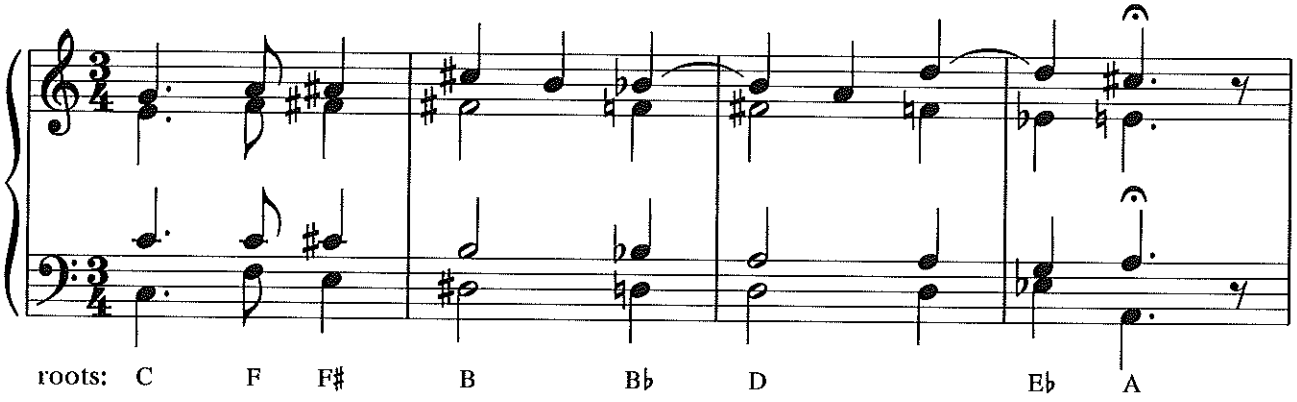
Tertian harmony may be described as nonfunctional under one, or both, of the following conditions:

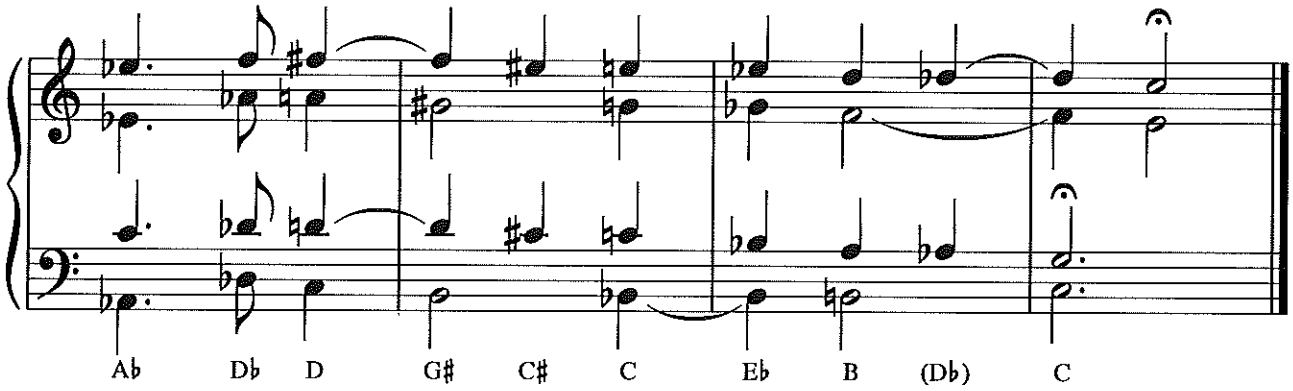
1. The abnegation of the hierarchy of root relationships associated with the major-minor key system.
2. The violation of the principles governing voice leading, with particular reference to sensitive pitches and intervals.

Music that demonstrates a conscious exploitation of either of these two conditions is not necessarily extremely complex from an analytical standpoint, nor even modern anymore, but nonetheless very much at odds with the "common practice" tradition.

ROOT MOVEMENTS BASED ON THE CHROMATIC SCALE

The following example, constructed in two four-measure phrases, begins and ends with a C major triad and may reasonably be said to have C as a tonal center. It exhibits very few harmonic progressions, however, that one would normally associate with the key of C.





Note:

a. The dominant of C is avoided.

b. With the exception of E and G, chord roots occur on every degree of the chromatic scale.

c. The example consists largely of chromatic root movements, including those of the tritone.

d. In spite of the chromaticism, the voice leading is quite logical.

PARALLELISM

Parallelism describes the activity of a chain of chords whose voices proceed in parallel motion. The result usually contributes to a complete breakdown of traditional voice-leading procedures, and instead the emphasis shifts to the production of a "thickened" melodic line. "Real" parallelism involves a series of exact transpositions of a given chord; "tonal" parallelism involves the rewriting of a chord type (for example, triad, seventh chord) upon different degrees of a scale so that changes in quality may occur.

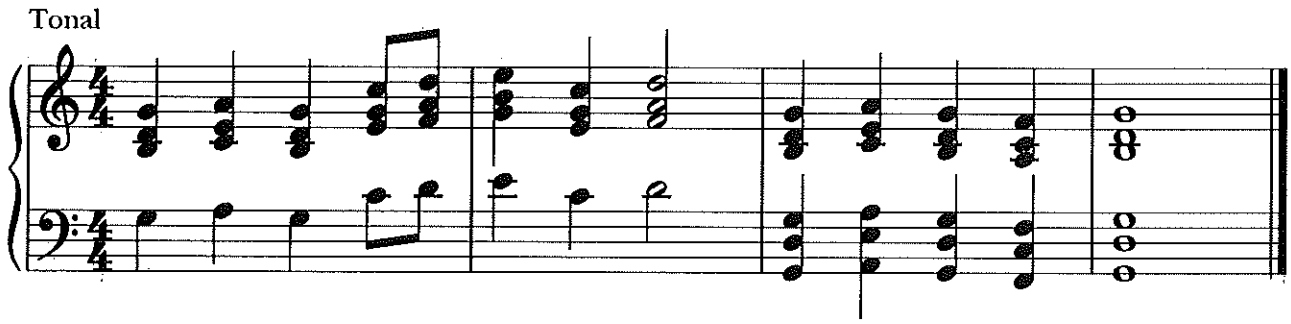
Real



Note:

a. The example is based upon a chain of complete dominant ninth chords.

b. Neither the voice-leading nor the root relationships are treated functionally.



Note:

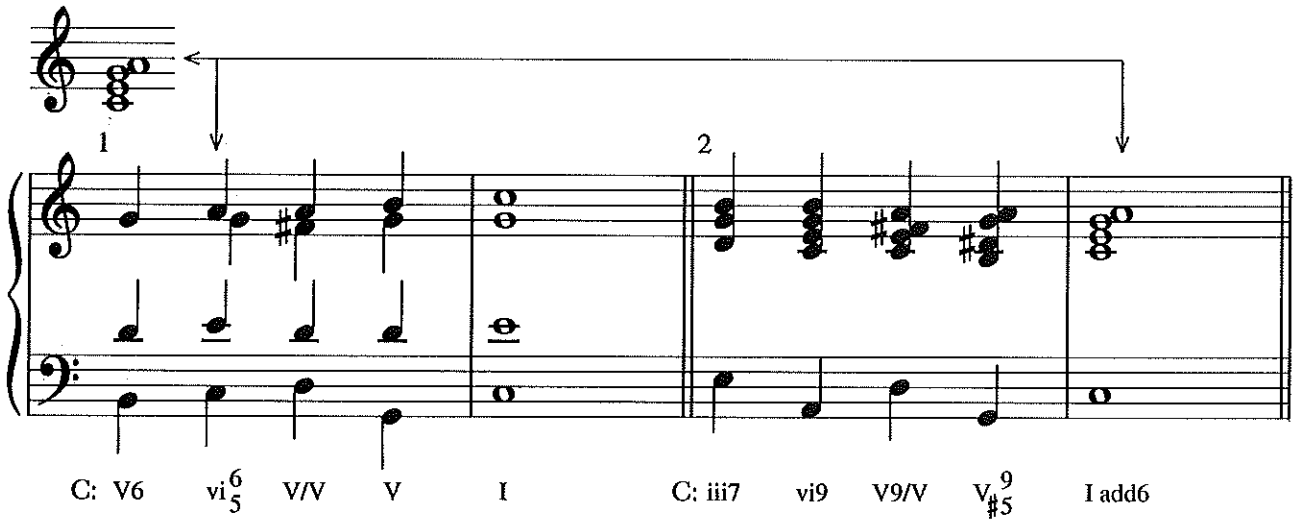
a. The example is written in the Mixolydian mode on G.

b. Unlike the previous example, the quality of each chord is determined by the scale degree upon which it is built. The motion is, nonetheless, parallel throughout.

CHORDS OF ADDITION

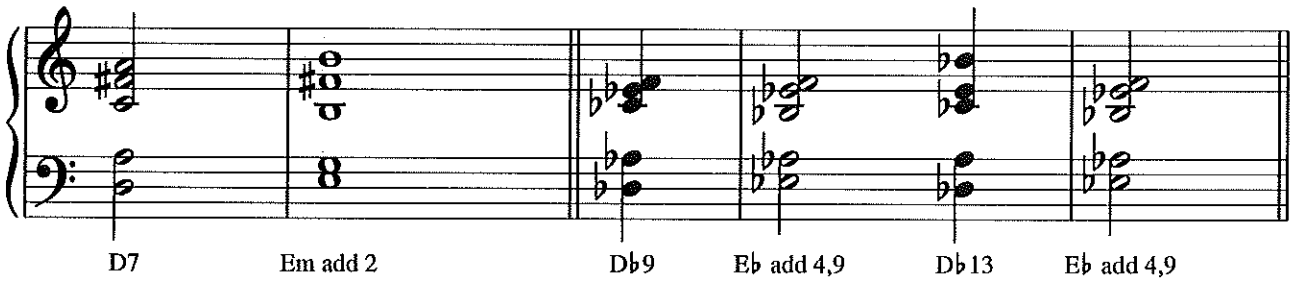
A chord of addition is formed by adding a nonchord tone (or tones) to a tertian structure. This tone is usually a second, fourth, or sixth above the root of the chord.

Consider the following sonority and its use in the examples that follow:



In example 1, a^1 is the root of the submediant $\frac{6}{5}$ chord. In example 2, by contrast, there is no doubt that C is the root and that a^1 is simply "added" to the chord.

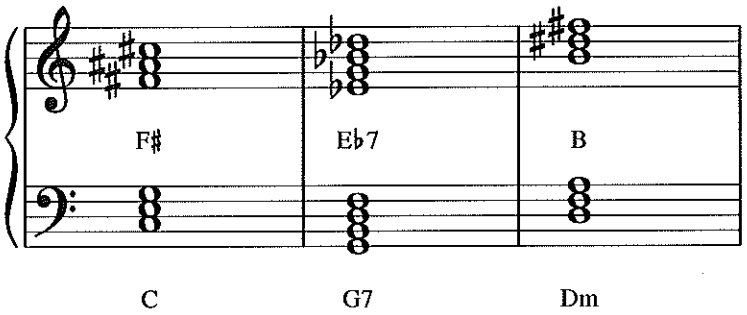
Thus, chords of addition tend to be used to create harmonic color:



POLYCHORDS

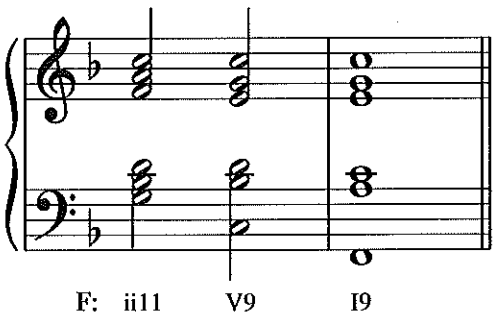
A polychord is formed by superimposing one tertian structure upon another with a different root.

Thus:



are polychords because in each case clearly distinguishable pairs of roots can be identified.

Care must be taken not to mistake polychords for extended single-root tertian sonorities.



In the example above, for instance, the first chord is obviously a supertonic eleventh in F major, even though the bottom and top three notes form g minor and F major triads respectively.

The following examples represent typical uses of polychords. Study them carefully.



Note:

a. The right hand part consists of alternating major and minor triads.

b. The left hand part consists of only major triads. Each triad, however, contains a pitch that is related to the top line by a whole step, i.e., D–E, E–F♯, C–D, etc.



Note:

The example is constructed entirely from pairs of Mm7 chords that move in contrary motion.

BITONALITY

Music in which two tonal centers simultaneously coexist is said to be *bitonal.* For such music to be comprehensible to the listener, the two tonalities are usually set in different registers or, in the case of instrumental music, allotted to instruments of contrasting timbres.

The term *polytonality*, which in theory refers to an unspecified number of simultaneous tonalities, is used synonymously with bitonality. In practice, music with more than two concurrent tonal centers is rarely encountered.

In the examples that follow, within each tonality the harmonic and melodic treatment is functional. The sense of function, however, is largely destroyed by the choice of tonal combinations.



PANDIATONICISM

Music that is the result of the free use of the diatonic rather than the chromatic scale is said to be *pandiatonic.* Implicit in the term is the organization of diatonic materials (major or minor) in a nonfunctional manner with respect both to chord progression and voice leading.

Although pandiatonic compositions are usually based upon tertian harmony, the linear movement "stretches" the chords so that conventional harmonic analysis is impossible. Chords tend to be implied, not stated, as the following example shows:



Note:

a. There are no chromatic pitches. The piece consists entirely of notes played on the white keys of the piano.

b. The harmonic motion evolves from the linear motion.

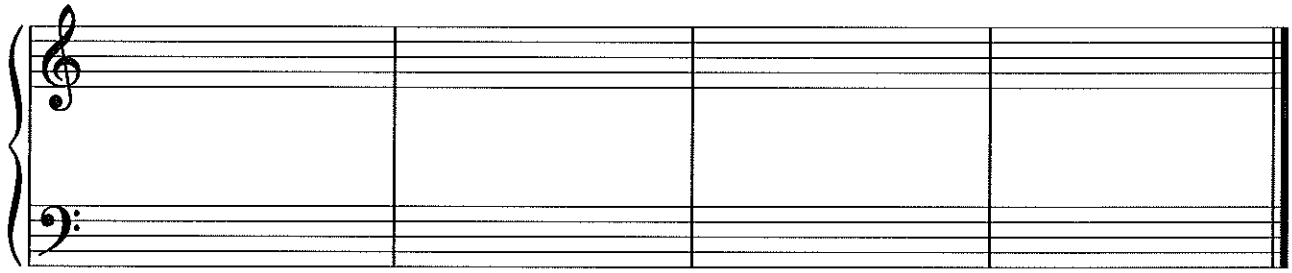
c. The final cadence on to the tonic, C, is voiced traditionally to achieve a sense of conclusion.

EXERCISES

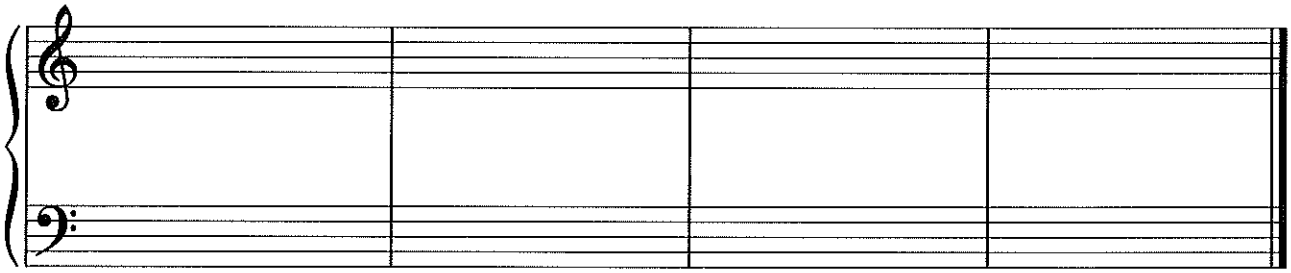
A. Using the given chord roots, invent four-measure phrases for soprano, alto, tenor, and bass. The chords may be major or minor, and freely inverted.

1. $\frac{3}{4}$ D–A♭–B♭–E–C–F♯–B–E♭–D.
2. $\frac{2}{2}$ A–B♭–C–D♭–G♭–F–E–A.
3. $\frac{6}{8}$ E♭–A♭–F–B♭–G–G♭–C–C♭–B♭–F♭–E.

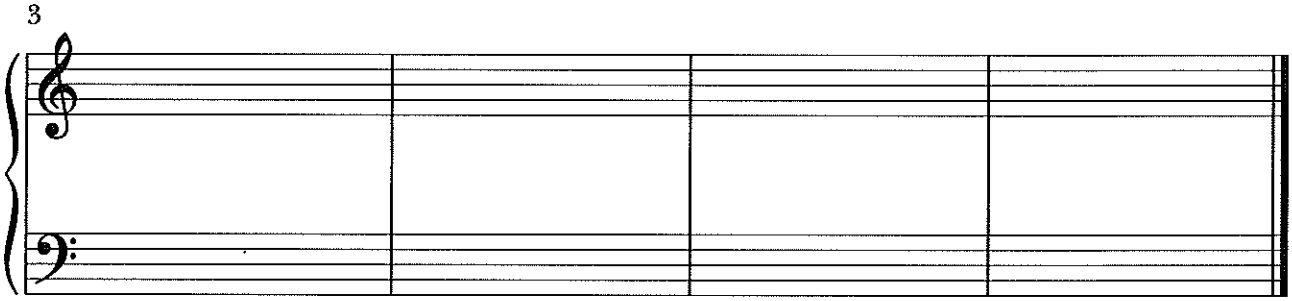
1



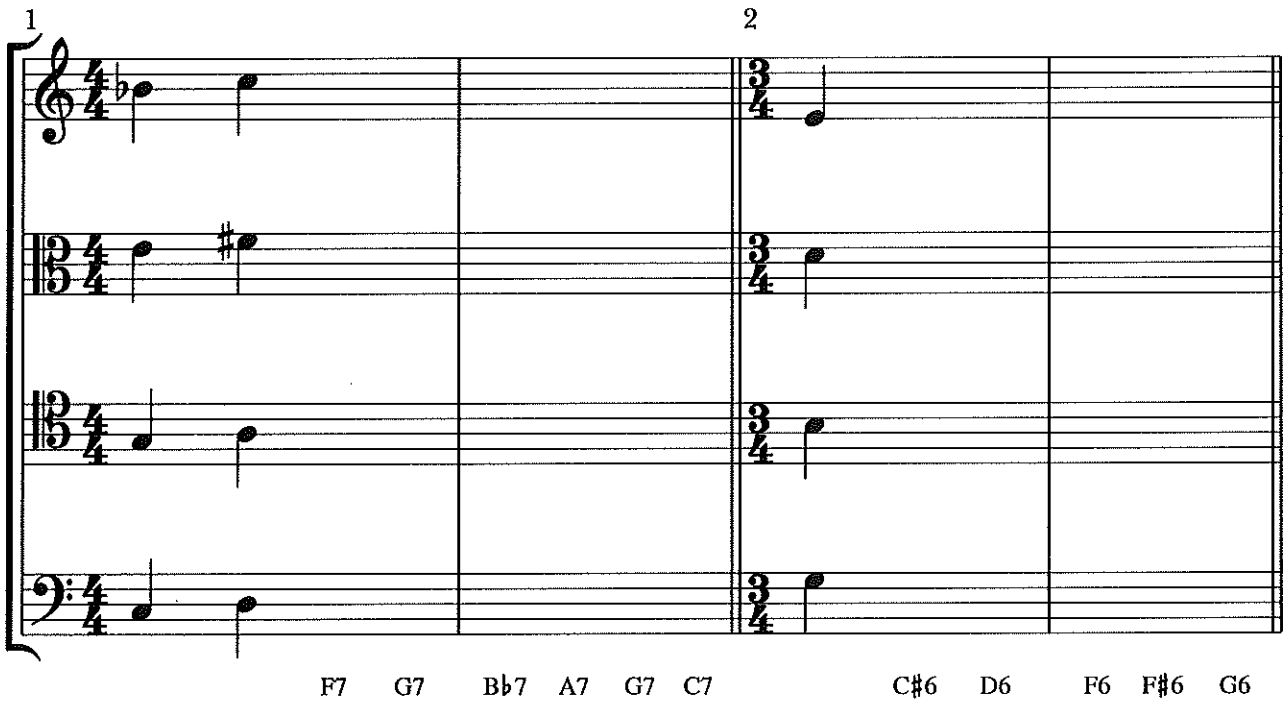
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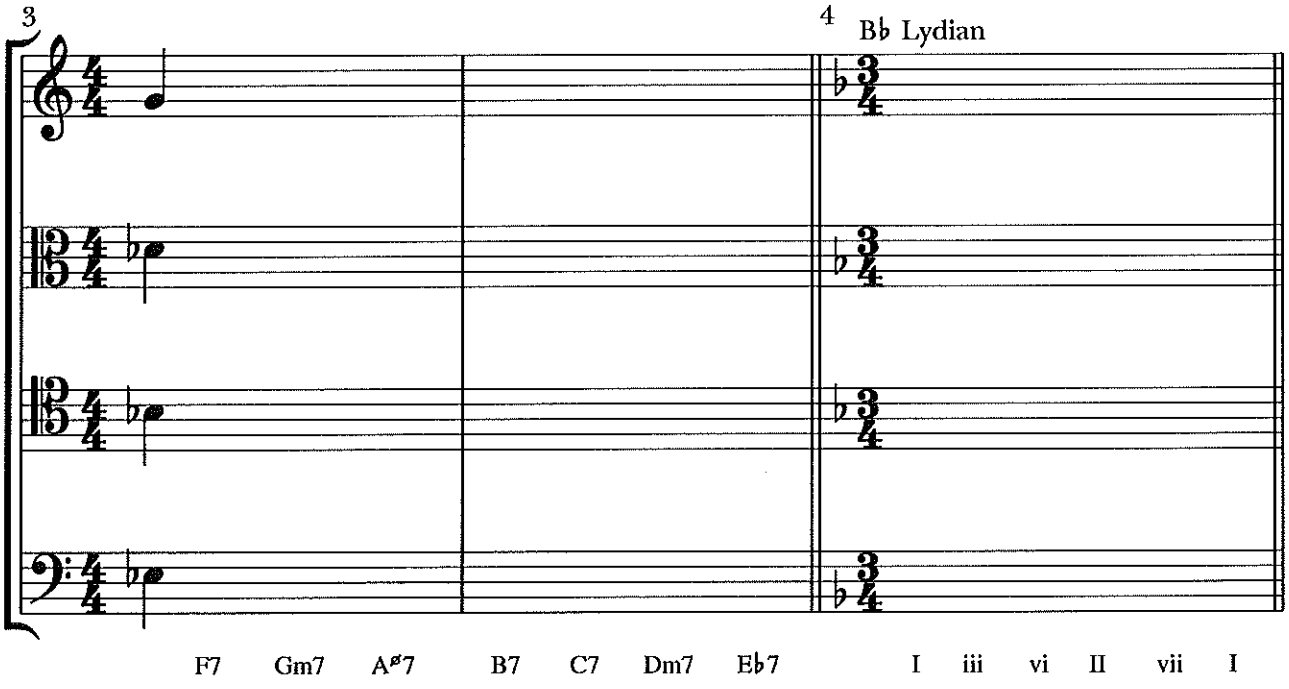


3



B. Continue the following in parallel motion as indicated.





C. Beginning with the four given polychords, write a twelve-measure piece for piano made up of three four-measure phrases.

Phrase 1: Minor triads in the right hand, major triads in the left.
Phrase 2: Major triads in the right hand, minor triads in the left.
Phrase 3: Major triads in both hands.

Each polychord should have one, but not more than one, common note.



D. Invent a piano accompaniment for the following melody in a freely diatonic style.



BURKHART, TUREK, AND WENNERSTROM EXERCISES FOR HARMONIC ANALYSIS

Burkhart

pp. 421–25;
pp. 441–42.
Identify and describe all uses of parallelism.

p. 517.

a. Identify the roots and qualities of the triads of which the polychords are constructed.
b. Try to determine the logic behind the construction of the polychords.

Turek

pp. 408–9, mm. 1–4 and mm. 35–38;
pp. 414–16, mm. 21–30.
Identify and describe all uses of parallelism.

pp. 458–62.
This movement contains some of the compositional techniques discussed in this chapter. Identify and describe as many of them as you can.

Wennerstrom

pp. 505–6, mm. 1–15.
This *Tema* may be described as pandiatonic.

a. Identify all vertical structures of three or four pitches.
b. Divide the structures into those that are obviously tertian sonorities and those that are not so clearly tertian sonorities.
c. Determine the extent to which the clear tertian sonorities play a functional role in the music.

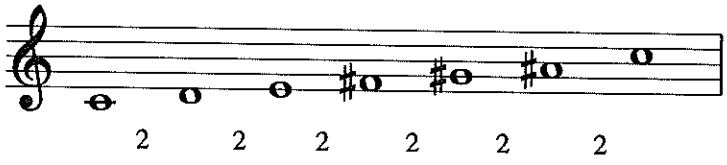
28 Artificial Scales

A scale that does not conform to a traditional pattern is said to be *artificial.* Although the possibilities for the construction of artificial scales are enormous, composers have tended to use those scales that possess an intrinsic mathematical logic, or scales from which unusual harmonic relationships may be derived.

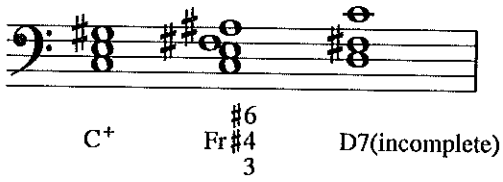
Two artificial scales are in common use.

THE WHOLE TONE SCALE

The whole tone scale is constructed entirely of whole steps:



Derived tertian sonorities:



Note:

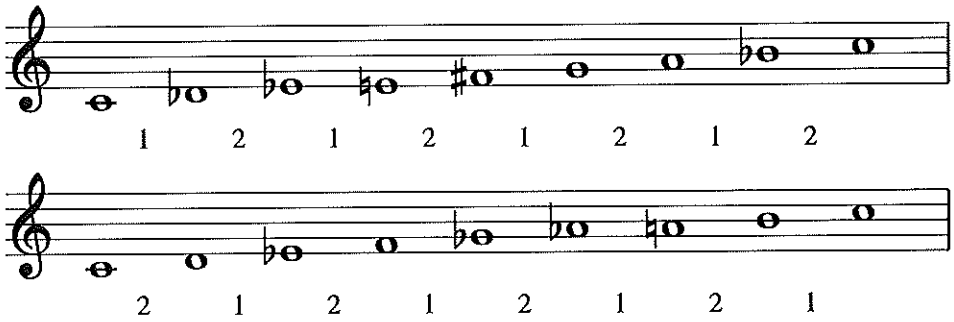
a. As a result of its construction, any note of the scale may be chosen as a tonic; the intervallic organization is unchanged.

b. The only transposition of the scale that allows a different pitch collection is that of a half step up or down. Thus, with reference to pitch, there are two distinct whole tone scales.

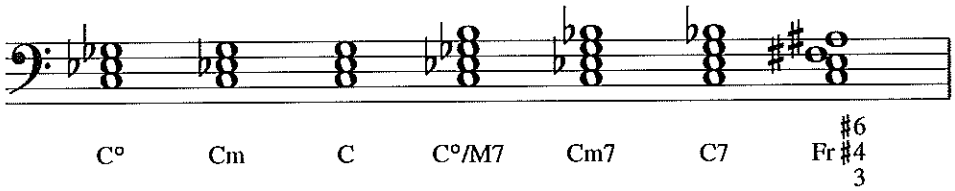
c. No stable tertian structures can be derived from the whole tone scale.

THE OCTATONIC SCALE

The octatonic scale, made up of alternating half steps and whole steps, exists in two forms:



Derived tertian sonorities:

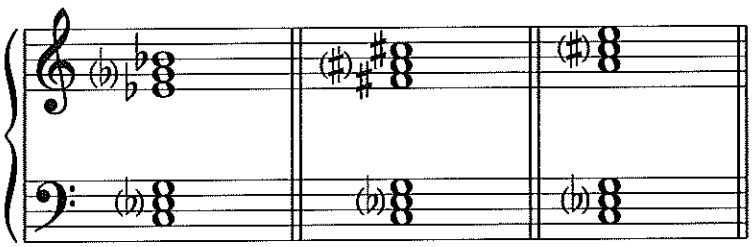


Note:

a. As a result of its construction, the first, third, fifth, and seventh note of either scale form may be chosen as tonics; the intervallic organization is unchanged.

b. Each form may be transposed up or down a half step *twice* before the pitch content and the intervallic organization are duplicated. Thus, for each form, three distinct octatonic scales exist. (It is not accidental that each of the forms consists of two interlocking diminished seventh chords, for it is the symmetrical property of these sonorities that explains the limited number of distinct transpositions of the scale.)

c. The scale abounds with bitonal implications because major and minor triads that are obscurely related to one another can be formed on four of the eight scale degrees:

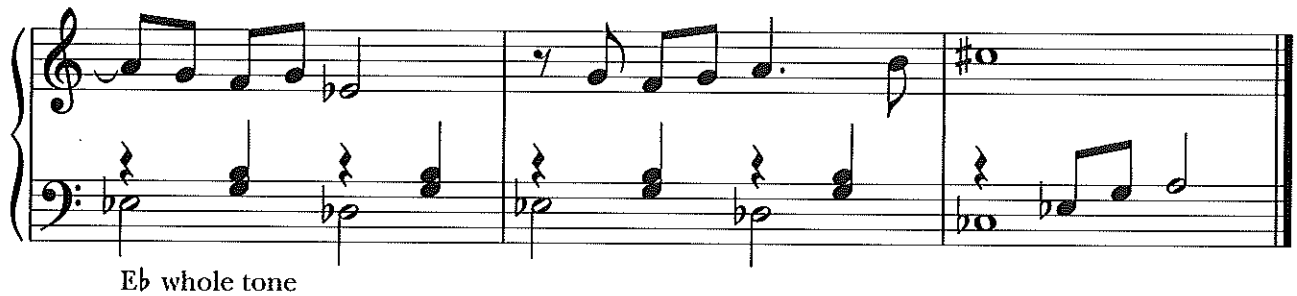


The two examples that follow are based upon the whole tone and octatonic scales, respectively. Although the tonal resources of the latter are richer, notice how in both

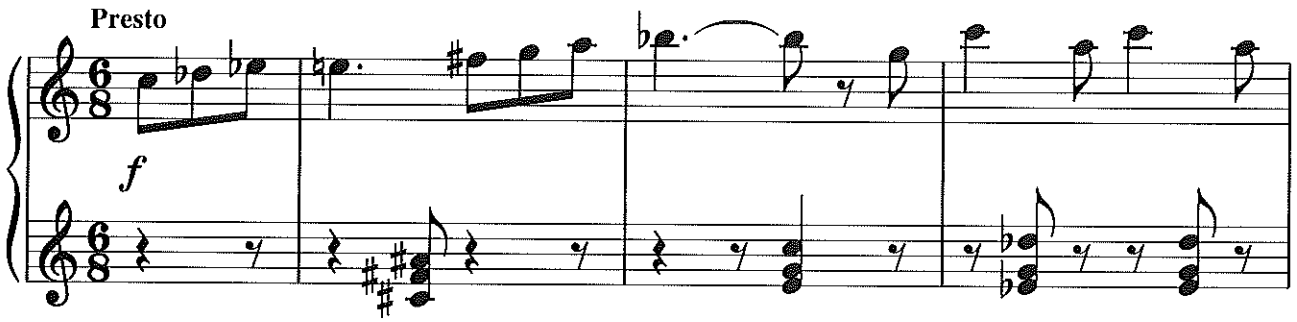
1. The interval of the tritone predominates, particularly melodically.
2. The tertian harmony is treated nonfunctionally.



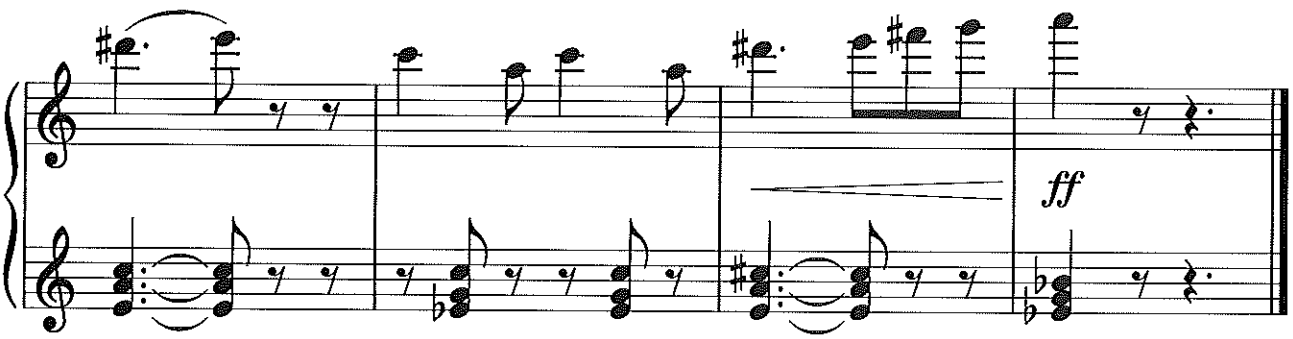
C whole tone



Eb whole tone

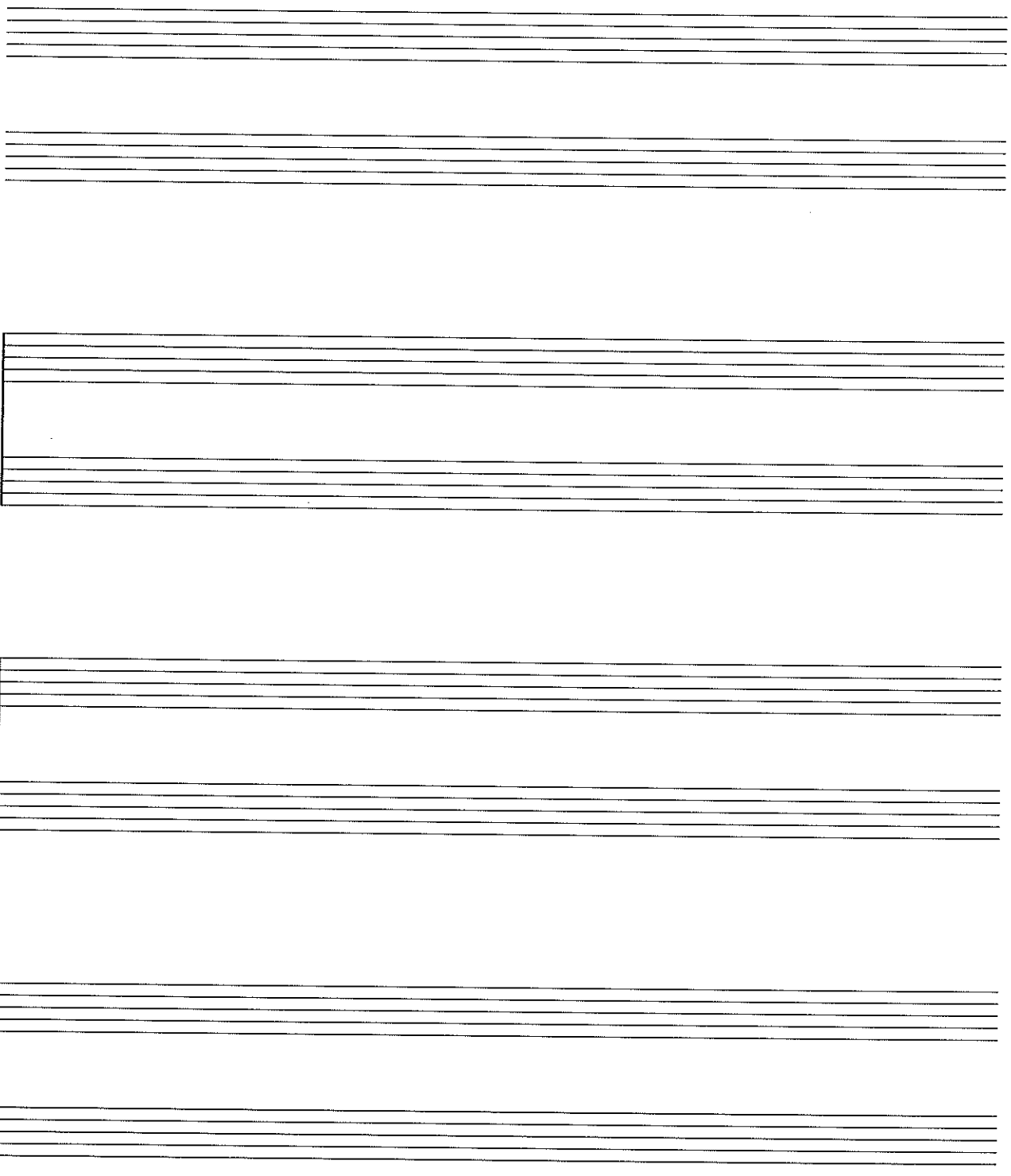


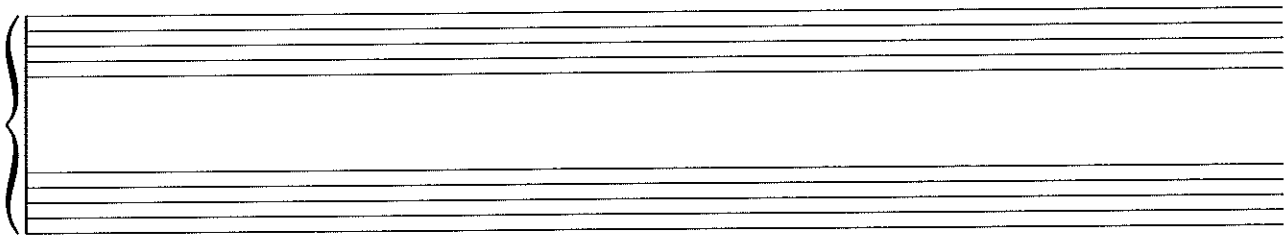
C octatonic (1 2 1 2 1 2 1 2) throughout

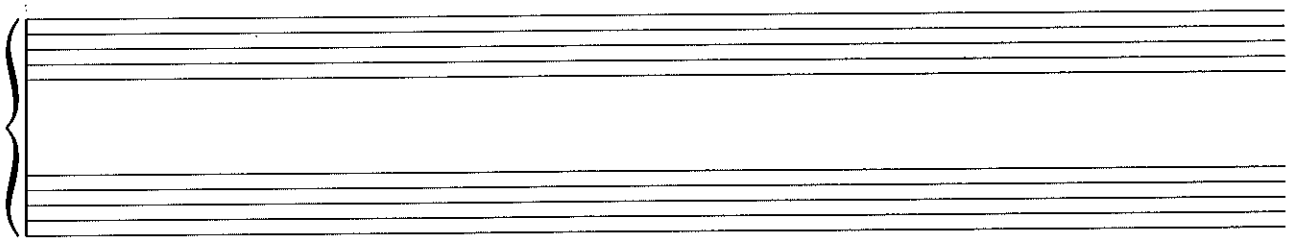


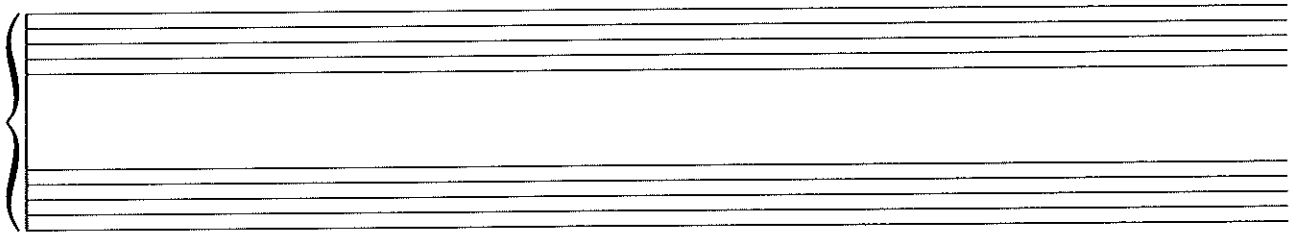
EXERCISES

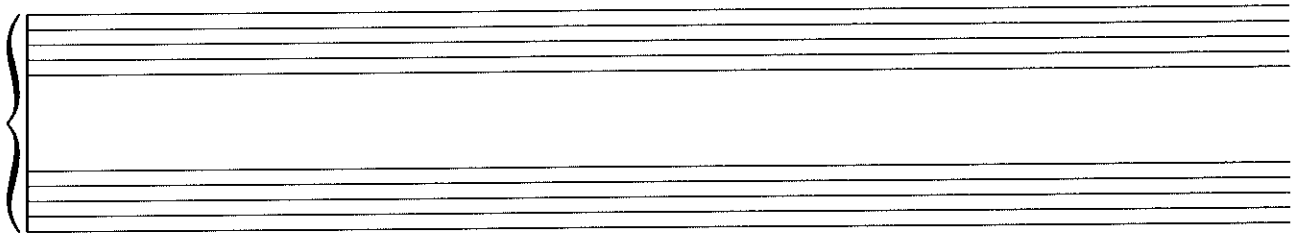
1. Invent a symmetrical artificial scale of not more than eight notes.
2. From the scale, derive all the possible tertian structures.
3. Using the untransposed scale and tertian structures, write a piece for piano of not more than sixteen measures.











BURKHART, TUREK, AND WENNERSTROM EXERCISES FOR HARMONIC ANALYSIS

Burkhart

pp.441–42 (voice part only).
p.458.

Turek

pp. 430–32, mm. 21–23 and mm. 46–48.
pp. 453–54, mm.1–4.
p. 516, mm. 36–38 (clarinet part).

Wennerstrom

pp. 479–80.

Analyze the scale structure of the given pieces.

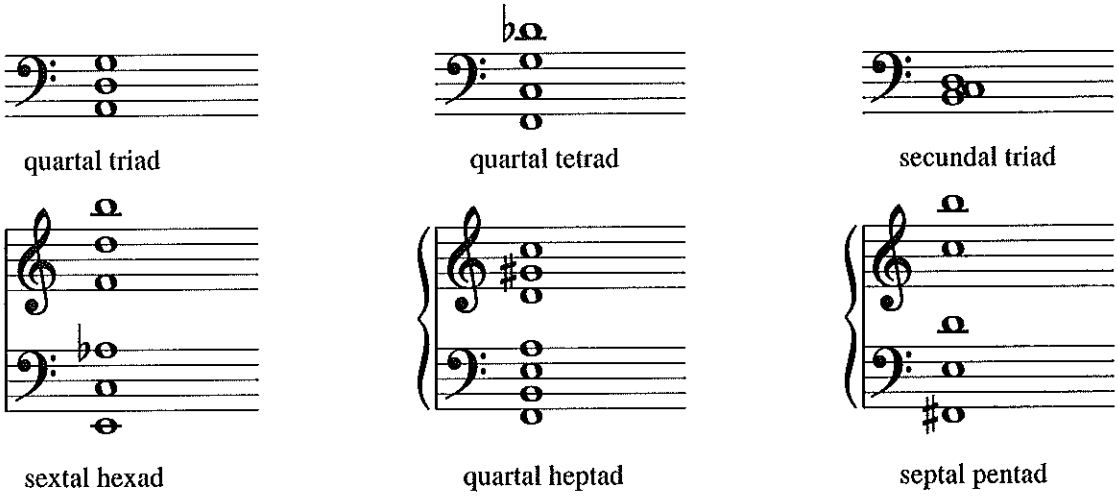
29 Nontertian Harmony

This chapter will serve to introduce the student to a number of harmonic procedures of the twentieth century that are not based on the interval of the third. The contents of Chapters 27 and 28 exemplified how tertian structures could be manipulated despite a departure from functional root relationships. With the abandonment of the third as the governing interval, harmonic procedures are obviously perceived as even further removed from common practice in the traditional sense. Composers have been experimenting with nontertian harmony since the first decade of this century. It is interesting to observe, however, that after three generations no single harmonic system has evolved that can reasonably be said to represent a new common practice.

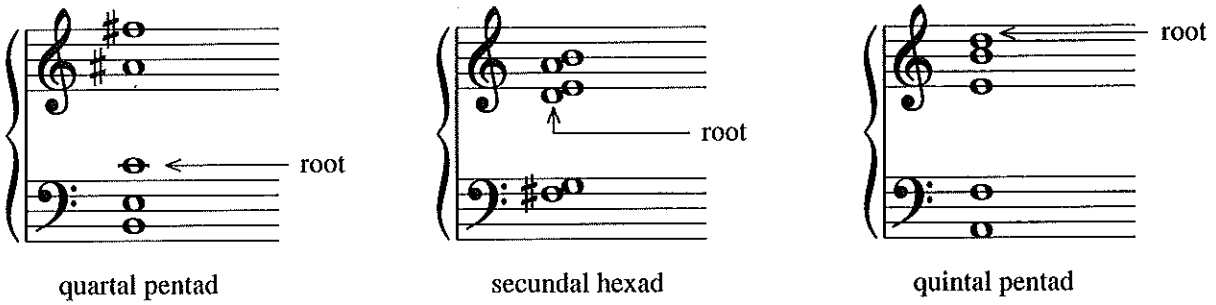
NONTERTIAN PROJECTIONS

The tertian harmonic system is, by implication, founded on the projection of the interval of the third from a given series of pitches. On the basis of this implication one may always determine the root of a chord, however the pitches are arranged, by discovering which member of the chord allows the remaining pitches to be stacked in thirds upon it. Thus, a nonchord tone is definable, in a general sense, as a pitch that is not a member of a projection from a given chord root.

Similar systems may be developed by projecting intervals other than the third. The resulting sonorities of the systems are commonly described by the organizing interval (secundal, quartal, quintal, etc.) and by the number of pitches (triad, tetrad, pentad, etc.). A three-note chord built in fourths, therefore, is said to be a *quartal triad*; a five-note chord built in seconds is a *secundal pentad*, and so on.



Provided that the generating interval is known, chord roots may be identified when the sonorities are inverted.



Unlike the tertian system, there is no universally accepted common practice involving chords built upon other intervals. In fact, relatively few well-known works demonstrate the *systematic* projection of any single interval other than the third. Thus, voice-leading procedures and root relationships are assumed to be the prerogatives of the individual composer.

The following chorale exemplifies the use of *quintal tetrads.* The student should analyze each chord with a view to determining its root.

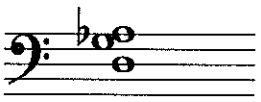
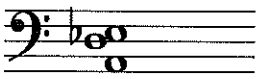


FREELY FORMED HARMONIC STRUCTURES

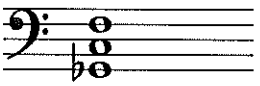
The final break with the common practice harmonic tradition occurs when one conceives of a chord as a set of pitches constructed not by intervallic projection but by free intervallic selection. Thus, any set of pitches sounded simultaneously can be the harmonic basis for all or part of a composition. With the elimination of a single interval as an organizing agent, the concept of inversion ceases to have any real meaning. A chord, therefore, may be more conveniently identified by its *interval content* than by its pitch structure.



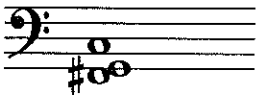
The three-note chord above has no clear root, and so it makes little sense to single out one of its pitches by which to designate it. The chord does, however, contain a perfect fourth (C–F), a diminished fifth (C–G♭), and a minor second (F–G♭). Furthermore, when the chord is transposed,

 or  etc.,

or rearranged,

 or  etc.,

or mirror inverted,



the same interval content prevails.

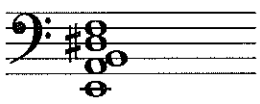
This interval content is effectively described by the *interval vector*, a structure that allows one to determine how many and what intervals a chord contains. The interval vector is based upon two assumptions. First, all intervals can be reduced to a span of a tritone or less by the process of inversion. Thus, the major seventh is equivalent to the minor second, the minor seventh to the major second, the major sixth to the minor third, and so on. Second, a note name (C, D♭, D, E♭, etc.) or its enharmonic equivalent is counted only once in a chord. A major triad, for example, written in four voices, is considered to be a three-note chord.

The chord C–F–G♭, and its various transpositions, rearrangements, and mirror inversions, therefore, is classified by its interval vector as 100011. The first entry gives the number of minor seconds, the second gives the number of major seconds, the third gives the number of minor thirds, the fourth gives the number of major thirds, the fifth gives the number of perfect fourths, and the sixth gives the number of tritones.

Further examples follow:



I.V. 012120



I.V. 220222



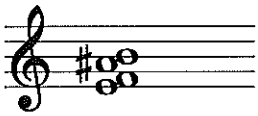
I.V. 010020

Whereas there are four tertian triads, there are twelve freely constructed three-note chords that have different interval contents; compared with the five or six seventh chords in common usage, the complete set of four-note chords is represented by twenty-eight different interval vectors. The composer's potential harmonic resources, therefore, are dramatically expanded. Typically, these resources are kept under control by:

1. The exploitation of a small number of predetermined chord types in a given movement or work.
2. The derivation of vertical (chordal) from horizontal (melodic) structures.

The two examples for piano that follow should be studied carefully.

1. This example is based upon the following four-note chord:



I.V. 200022



Note:

a. Eleven four-note chords have the same interval content.

b. The first chord is never restated at its original pitch level. Instead, it is transposed and revoiced to create variety.

2. This example is based upon the following two melodic fragments:





Note:

a. The accompaniment in measures 2, 3, and 8 is derived from the melody in measure 1.

b. The accompaniment in measures 4, 5, 6, and 7 is derived from the melody in measure 2.

c. The entire melody can be derived from the two four-note fragments.

EXERCISES

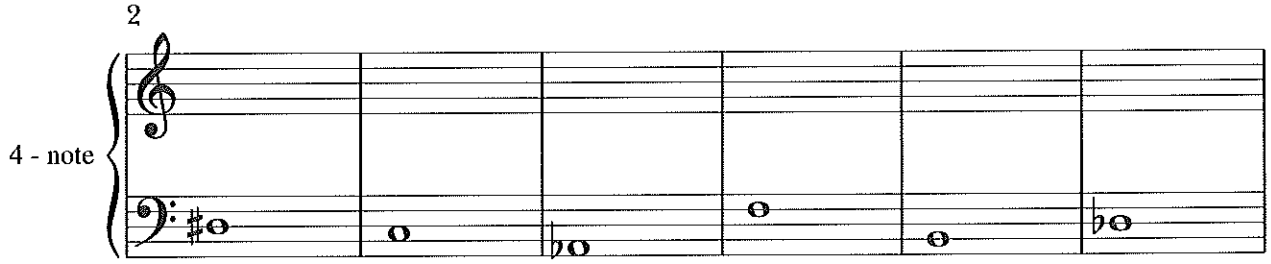
A. Project intervals from the given pitches as indicated.

1



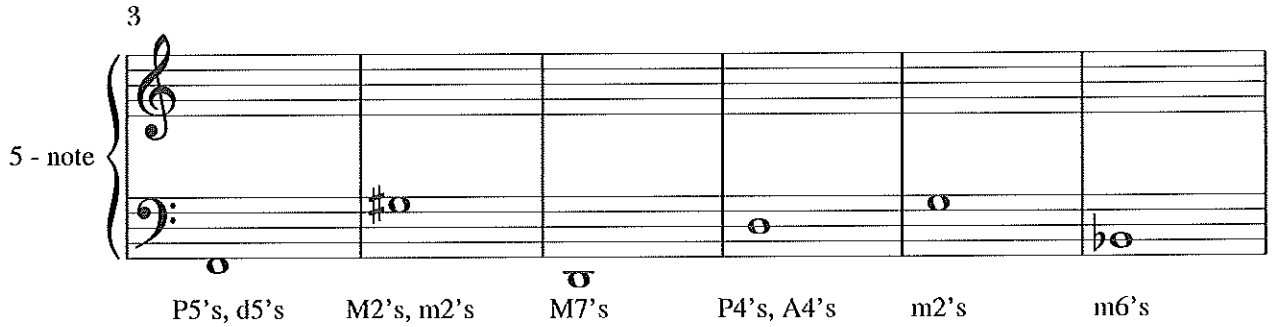
P4's, A4's m6's P5's M2's M6's, m6's M7's

2



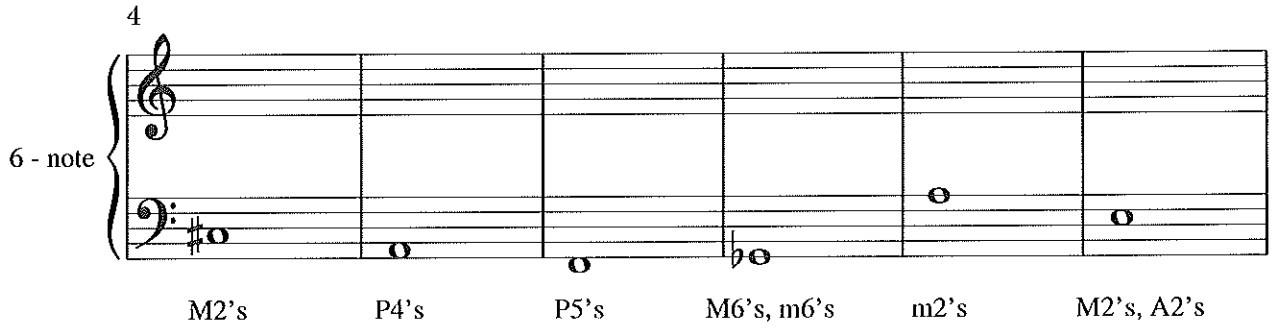
P4's P5's, d5's m7's M2's m7's, d7's P5's

3



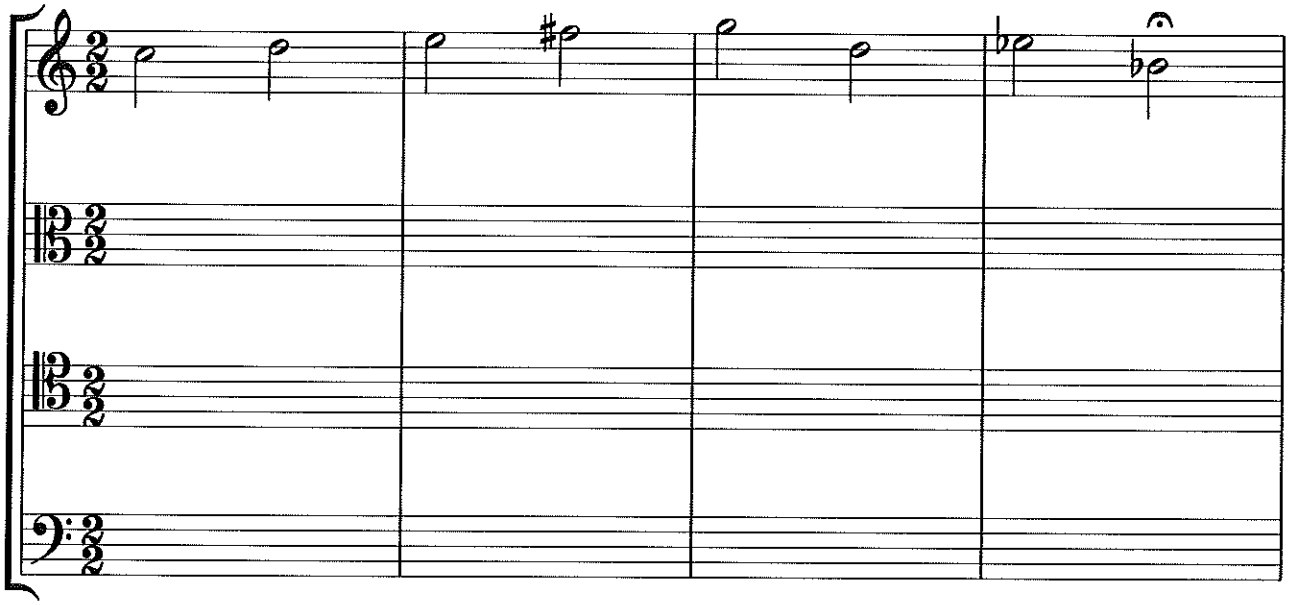
P5's, d5's M2's, m2's M7's P4's, A4's m2's m6's

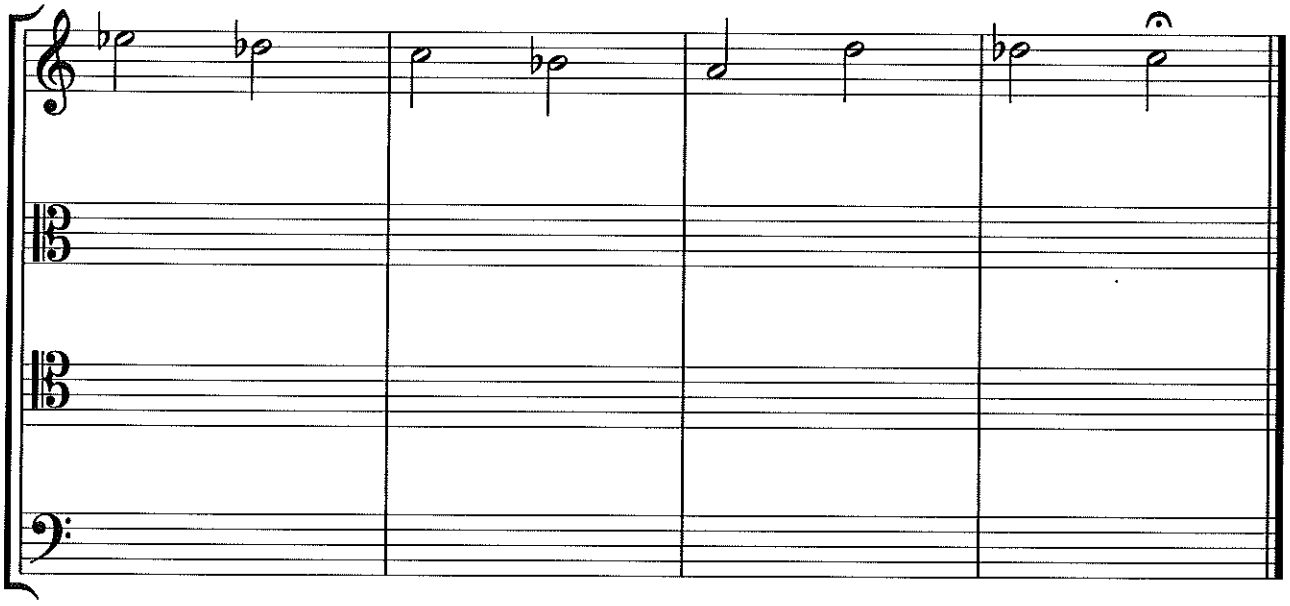
4



M2's P4's P5's M6's, m6's m2's M2's, A2's

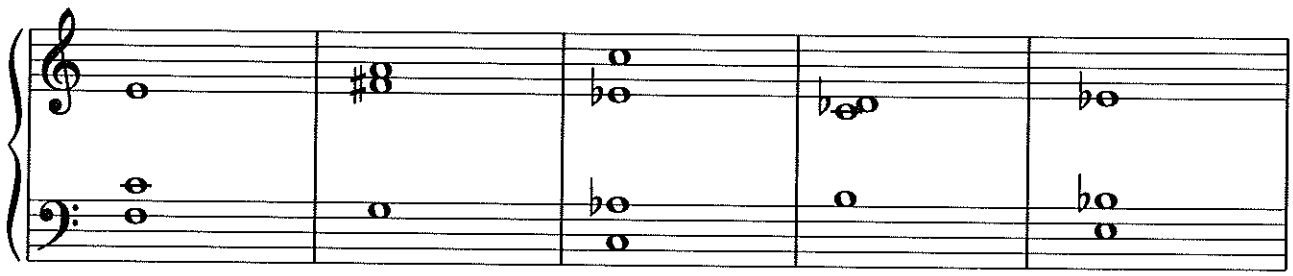
B. Harmonize the given chorale melody in quartal tetrads.





C. Find the interval vector of each of the following chords.

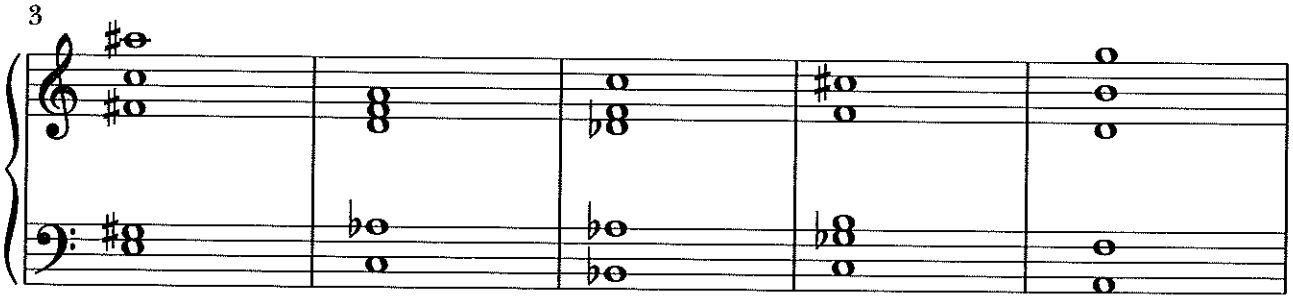
1



2



3



4



D. Choose two of the four-note chords from C2, and write a short piece for piano based upon some of the transpositions and inversions of these chords.

BURKHART, TUREK, AND WENNERSTROM EXERCISES FOR HARMONIC ANALYSIS

Burkhart

pp. 437–40.

Turek

p. 437.

Wennerstrom

pp. 428–32.

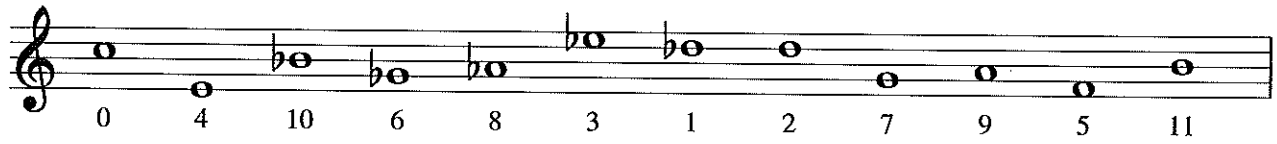
1. Identify all *clearly stated* three- and four-note chords.
2. Make an interval vector analysis of all of them.
3. As a result of this analysis, does it seem to be possible to come to some conclusions about the nature of the sonorities the composer used?

30 Harmonic Procedures in Twelve-Tone Serialism

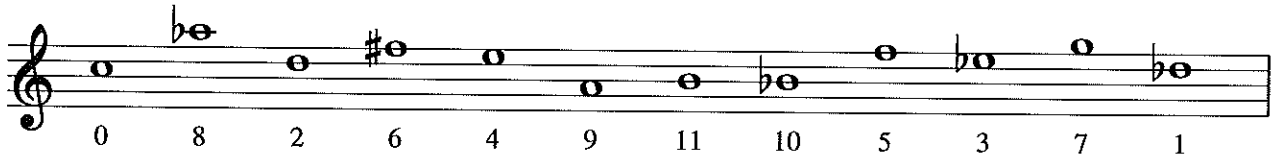
INTRODUCTION

Serial composition is based on the construction of a series of pitch classes (see page 2) that is designated as the *basic set*. The basic set may be expressed in four different forms—the *prime*, the *inversion*, the *retrograde*, and the *retrograde-inversion*. Furthermore, the notation may be expressed conventionally or through *pitch class integers*, as the example below shows:

Prime



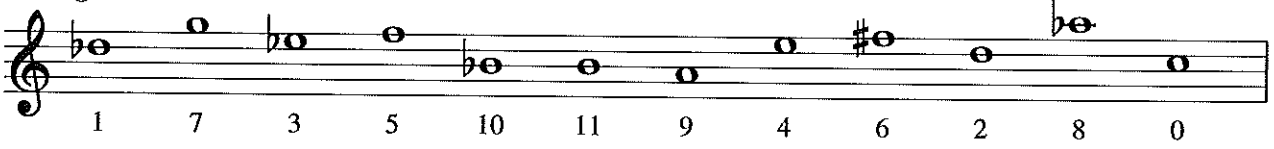
Inversion



Retrograde



Retrograde-inversion



Note:

a. The prime (P) is constructed of all twelve of the pitch classes of the chromatic scale. There is no duplication. For the notation in pitch class integers, 0 is assigned to the first pitch class (C), 1 to D♭, 2 to D, 3 to E♭, 4 to E, and so on.

b. The inversion (I) begins with the same pitch class as the prime; subsequently, all of the intervals are inverted. The notation in pitch class integers clearly shows the process of *complementation*. 0 to 8 is the inversion of 0 to 4 (8 + 4 = 12); 8 to 2 is the inversion of 4 to 10 (6 + 6 = 12); 2 to 6 is the inversion of 10 to 6 (4 + 8 = 12), and so on.

c. The retrograde (R) is derived from the writing of P in reverse order.

d. The retrograde-inversion (RI) is derived from the writing of I in reverse order.

Because the process of transposition allows each form of the basic set to begin on any one of the twelve pitch classes in the chromatic scale, forty-eight different permutations are possible. They may be conveniently tabulated thus:

	I0	I4	I10	I6	I8	I3	I1	I2	I7	I9	I5	I11	
P0	C	E	B♭	G♭	A♭	E♭	D♭	D	G	A	F	B	R0
P8	A♭	C	G♭	D	E	B	A	B♭	E♭	F	D♭	G	R8
P2	D	F♯	C	A♭	B♭	F	D♯	E	A	B	G	C♯	R2
P6	F♯	A♯	E	C	D	A	G	G♯	C♯	D♯	B	F	R6
P4	E	G♯	D	B♭	C	G	F	F♯	B	C♯	A	E♭	R4
P9	A	C♯	G	E♭	F	C	B♭	B	E	F♯	D	A♭	R9
P11	B	D♯	A	F	G	D	C	C♯	F♯	G♯	E	B♭	R11
P10	B♭	D	A♭	E	F♯	C♯	B	C	F	G	E♭	A	R10
P5	F	A	E♭	B	C♯	G♯	F♯	G	C	D	B♭	E	R5
P3	E♭	G	D♭	A	B	F♯	E	F	B♭	C	A♭	D	R3
P7	G	B	F	C♯	D♯	A♯	G♯	A	D	E	C	F♯	R7
P1	D♭	F	B	G	A	E	D	D♯	A♭	B♭	G♭	C	R1
	RI0	RI4	RI10	RI6	RI8	RI3	RI1	RI2	RI7	RI9	RI5	RI11	

or, using numbers to represent pitch classes, where C = 0:

	I0	I4	I10	I6	I8	I3	I1	I2	I7	I9	I5	I11	
→ P0	0	4	10	6	8	3	1	2	7	9	5	11	R0 ←
P8	8	0	6	2	4	11	9	10	3	5	1	7	R8
P2	2	6	0	8	10	5	3	4	9	11	7	1	R2
P6	6	10	4	0	2	9	7	8	1	3	11	5	R6
P4	4	8	2	10	0	7	5	6	11	1	9	3	R4
P9	9	1	7	3	5	0	10	11	4	6	2	8	R9
P11	11	3	9	5	7	2	0	1	6	8	4	10	R11
P10	10	2	8	4	6	1	11	0	5	7	3	9	R10
P5	5	9	3	11	1	8	6	7	0	2	10	4	R5
P3	3	7	1	9	11	6	4	5	10	0	8	2	R3
P7	7	11	5	1	3	10	8	9	2	4	0	6	R7
P1	1	5	11	7	9	4	2	3	8	10	6	0	R1
	RI0	RI4	RI10	RI6	RI8	RI3	RI1	RI2	RI7	RI9	RI5	RI11	

Note:

a. The prime is read horizontally from left to right.

b. The inversion is read vertically from top to bottom.

c. The retrograde is read horizontally from right to left.

d. The retrograde-inversion is read vertically from bottom to top.

e. Each transposition is designated by its form (P, I, R, or RI) and by the number of half steps it has been *raised* above the 0 transposition. For example, P7 begins with the pitch class G (=7) because it is a perfect fifth (or interval 7) above P0. Similarly, I9 begins with the pitch class A (=9) because it is a major sixth (or interval 9) above I0.

f. The tables show only pitch classes and pitch class integers; the specific octave is not indicated. In serial composition, it is important to understand that, after the basic set has been established, the pitches may be written *in any octave.* Thus, as an example, I8 may be notated as



or, indeed, in thousands of other ways.

g. Enharmonic spellings are routinely employed with a view to producing the most logical and readable notation.

THE CONSTRUCTION OF THE BASIC SET

The intervallic organization of the basic set plays a significant part in determining the tonal character of the subsequent composition. Although the number of possible twelve-tone series exceeds 479 million, basic sets tend to fall into two types—those with and those without obvious tonal implications. For instance, the series



suggests that the pitches B and F are tonal centers around which the other pitches are organized. By contrast,



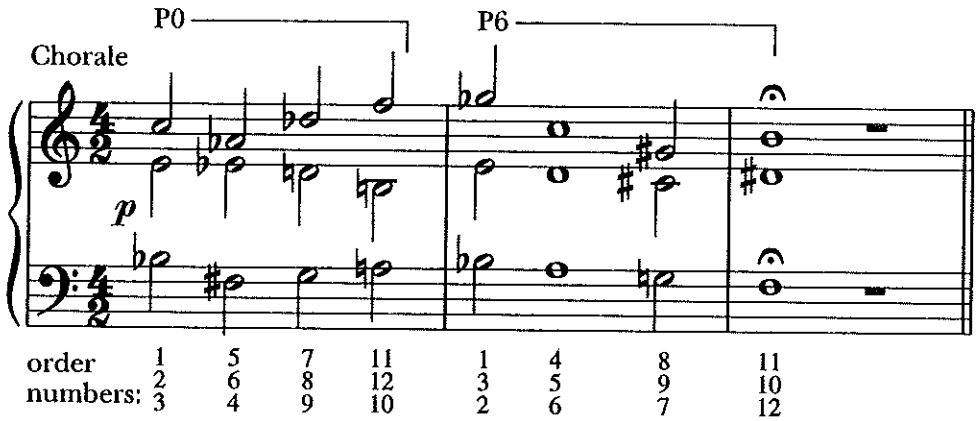
gives no clear indication of a tonal center.

The idea of basing a composition upon a twelve-tone series originated (with Arnold Schoenberg) in an attempt to write extended movements without tonal centers. Composers, however, have since used the twelve-tone technique in such a wide variety of ways that the treatment of tonality, or the absence thereof, is only one of several considerations that have to be made in the construction of a serial work.

THE HARMONIC BASIS

Vertical structures in twelve-tone music are commonly formed by the segmentation of the basic set into three, four, or six parts and by the consequent notation of the segments as chords.

The musical examples below, based on the basic set presented at the beginning of this chapter, clearly illustrate the principles of segmentation.

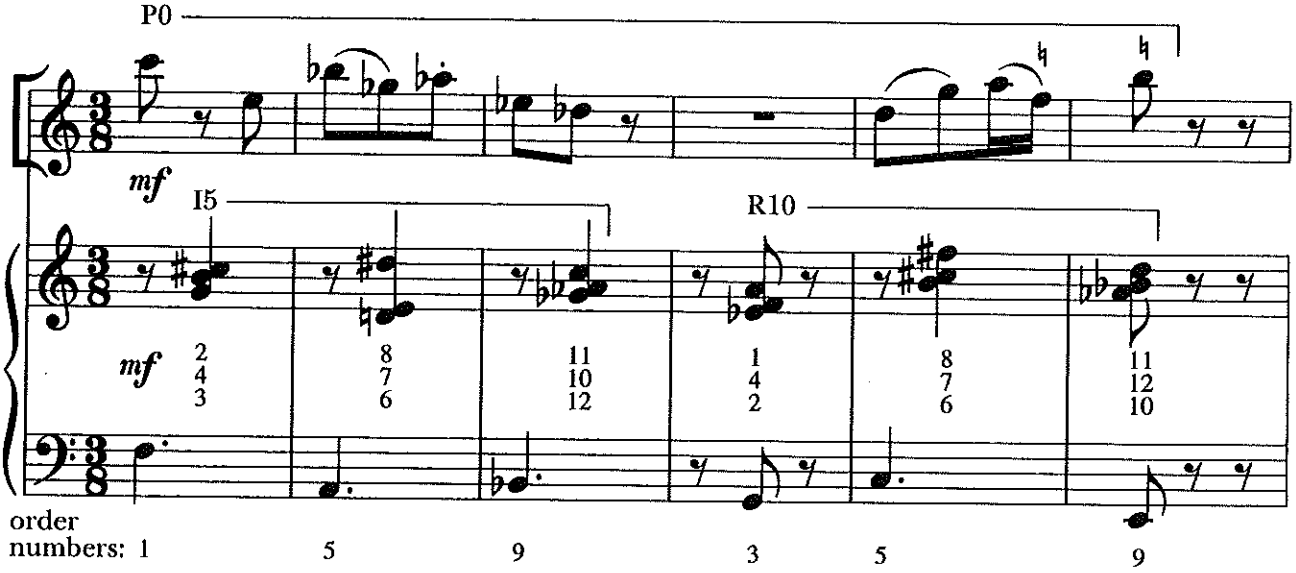


Note:

a. The chorale phrase is constructed exclusively from P0 and P6.

b. The position (order number) of each pitch in P0 and P6 is indicated beneath the example. The sets are thus segmented (or partitioned) into four three-note groups. Furthermore, the pitches are voiced freely within each group, so that the precise order of the basic set is not evident.

c. The individual lines are notated as far as possible to conform with traditional voice-leading procedures.



Note:

a. The solo part (P0) is constructed from a different form of the set than the accompaniment.
b. The accompaniment is based upon I5 and R10, which are both segmented into four-note groups.
c. A certain harmonic homogeneity is achieved by the presence of a major second in each chord.
d. At no time does a pitch in the accompaniment sound simultaneously with the same pitch, or its transposition by one or more octaves, in the solo part. Generally speaking, octave duplication in serial works is avoided; unison duplication is more frequently encountered.

The approach to harmony in twelve-tone writing varies dramatically from composer to composer, so that the realizations of vertical events occur as a result of a large number of techniques, many of which are highly complex.

As a final illustration, the series used in the examples above is manipulated so that the projection of the interval of the perfect fifth (see Chapter 29) is employed for all vertical structures.



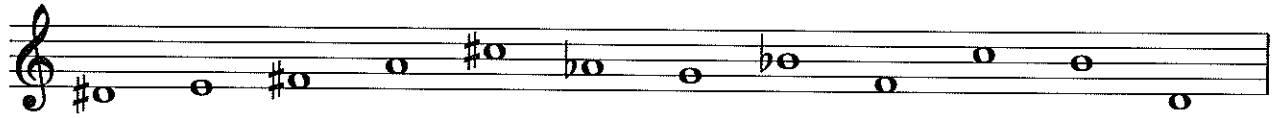
Note:

The three voices are in *canon*; they display the same melodic and rhythmic organization.

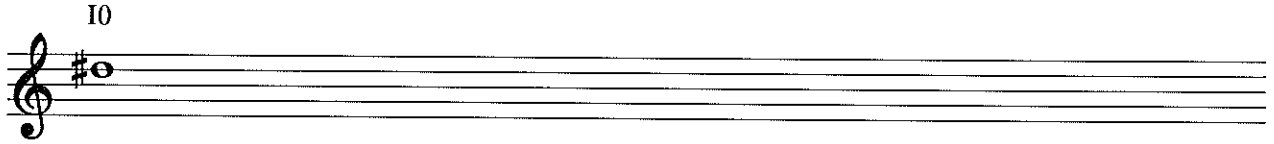
EXERCISES

A. Write the inversions of the given prime forms.

1 P0



I0



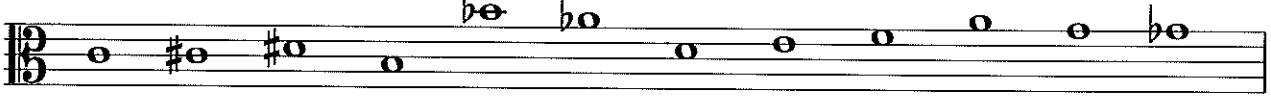
2 P0



I0



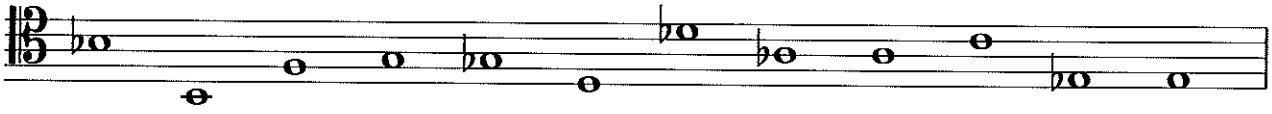
3 P0



I0



4 P0

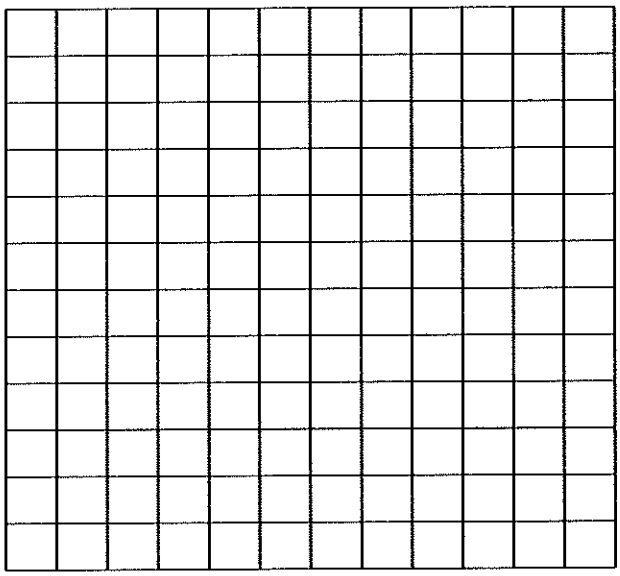


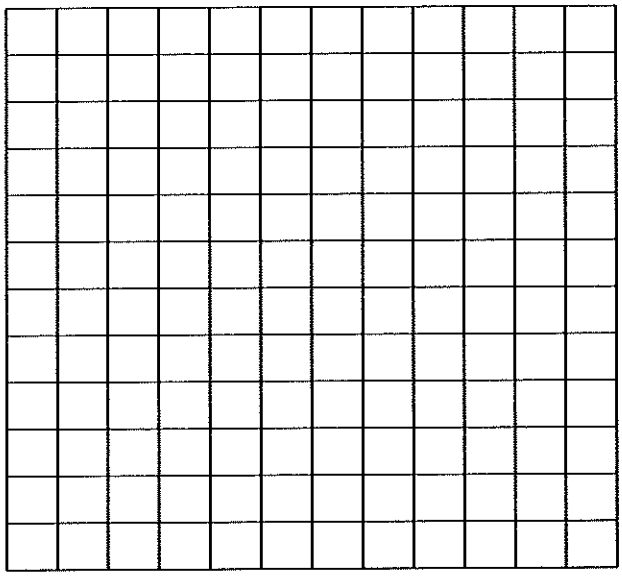
I0



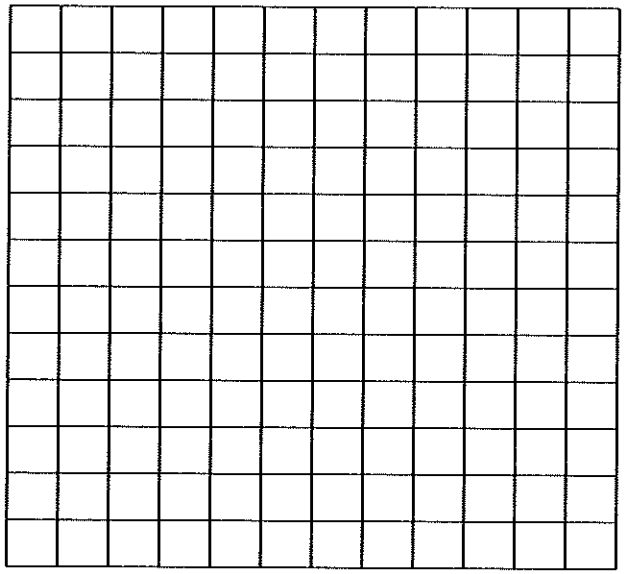
B. Invent three basic sets, and, for each, notate P0 and I0. In the space provided, tabulate all the transpositions of the series in a manner similar to that described on page 388.

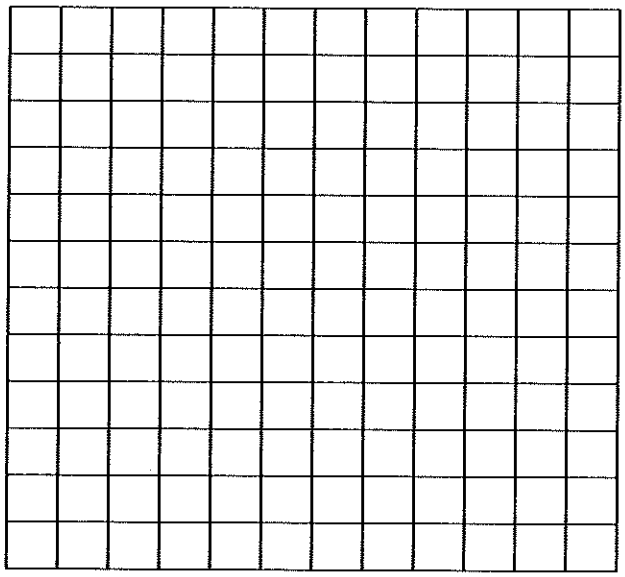
1



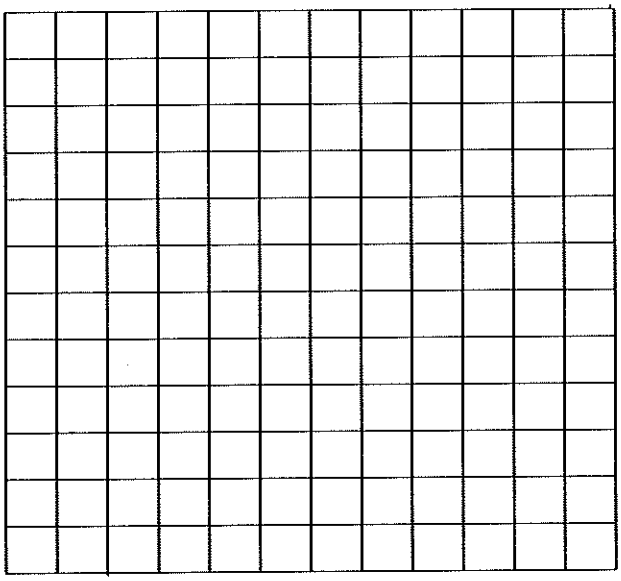


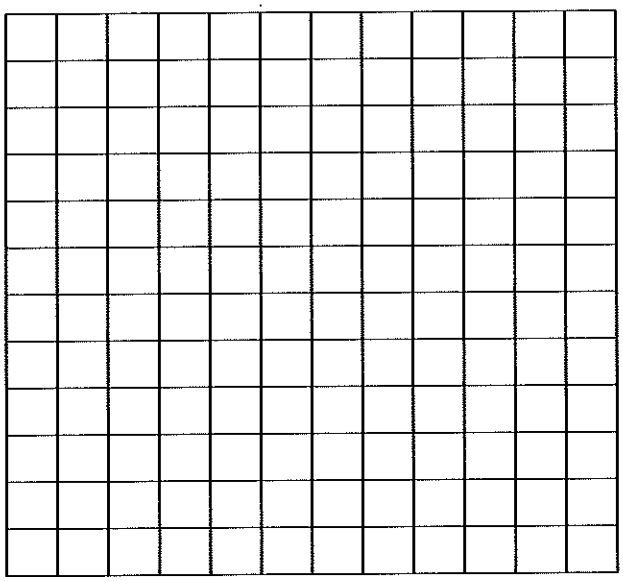
2



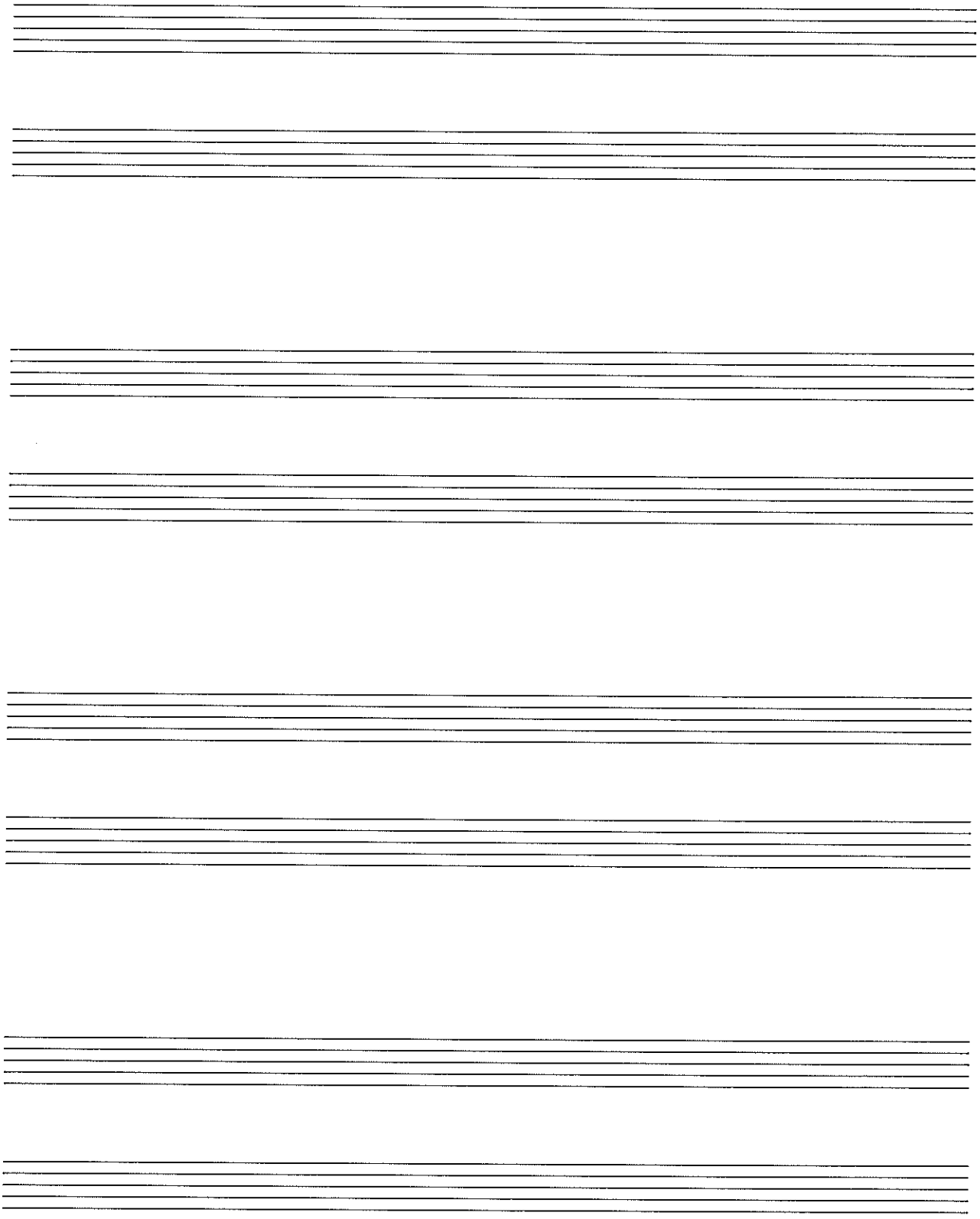


3



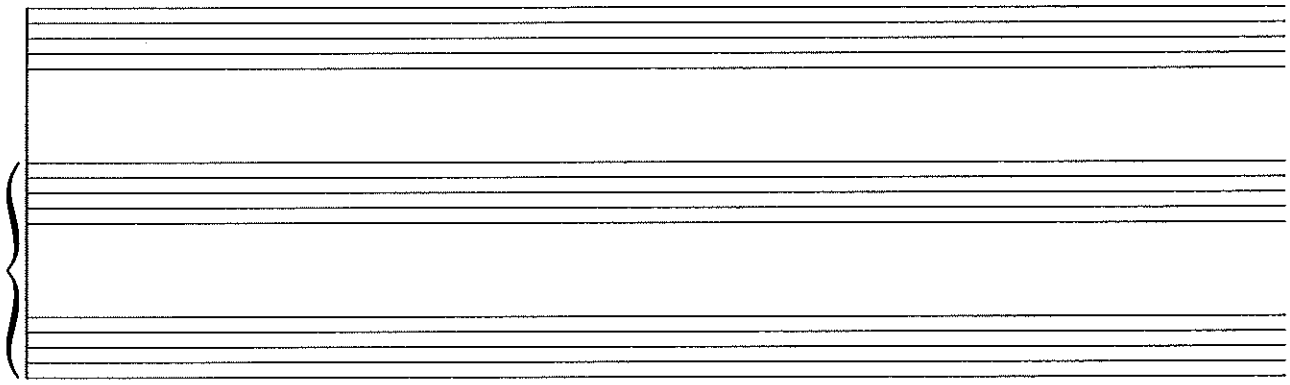


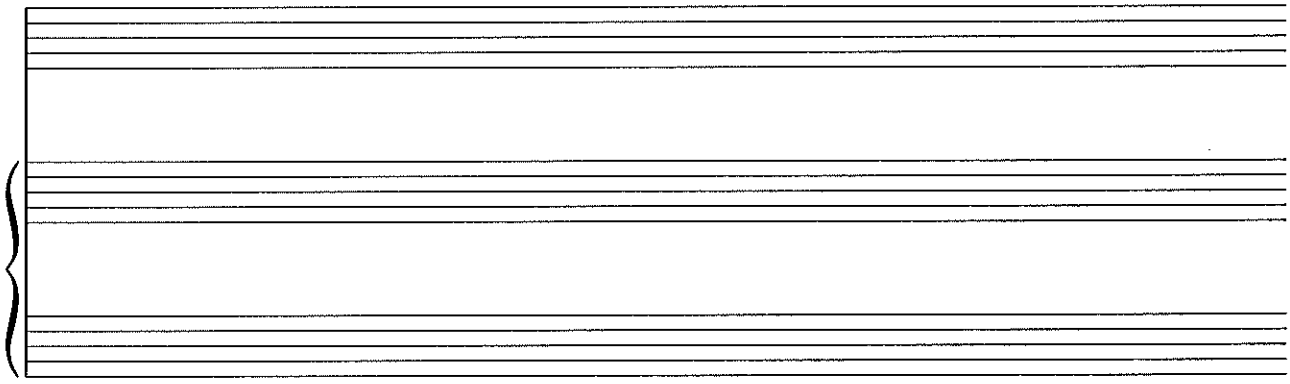
C. Choose one of the series from B, and, exploiting four-note segmentation, write a short piece for piano.

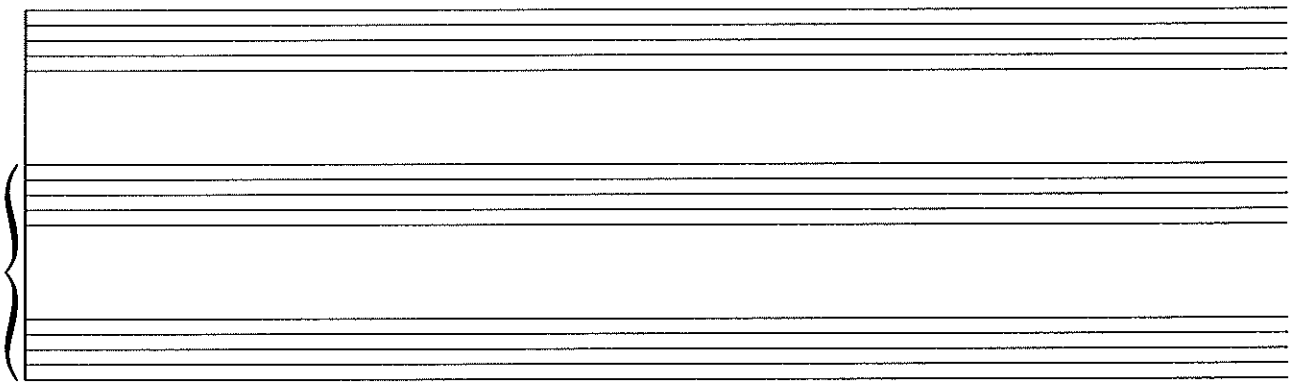


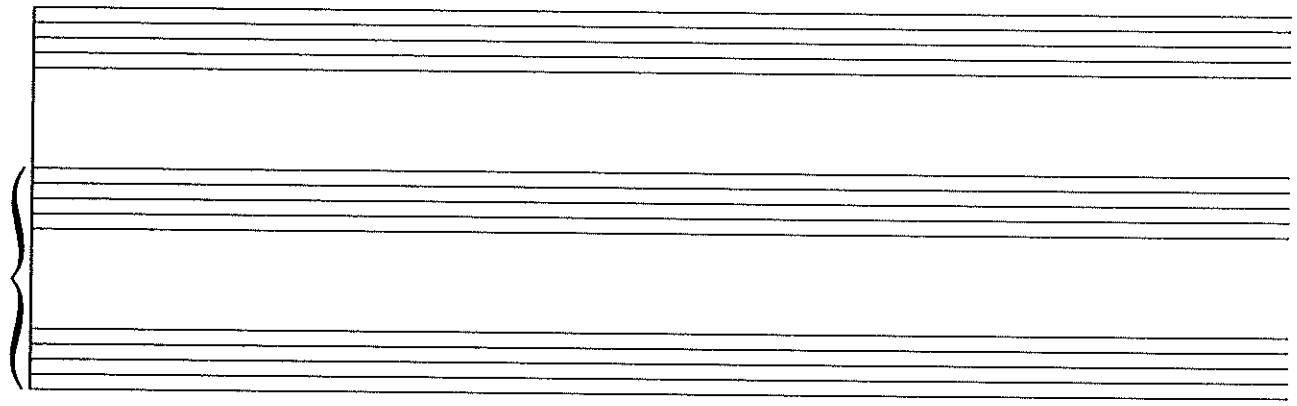
D. Write two short pieces for piano and a solo instrument of your choice. Make use of the two series you have not already set.

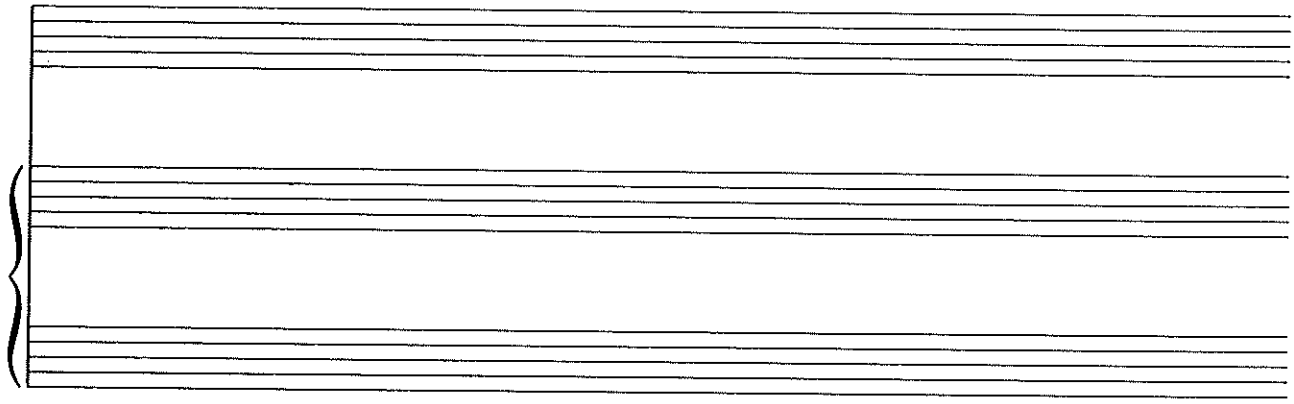
1

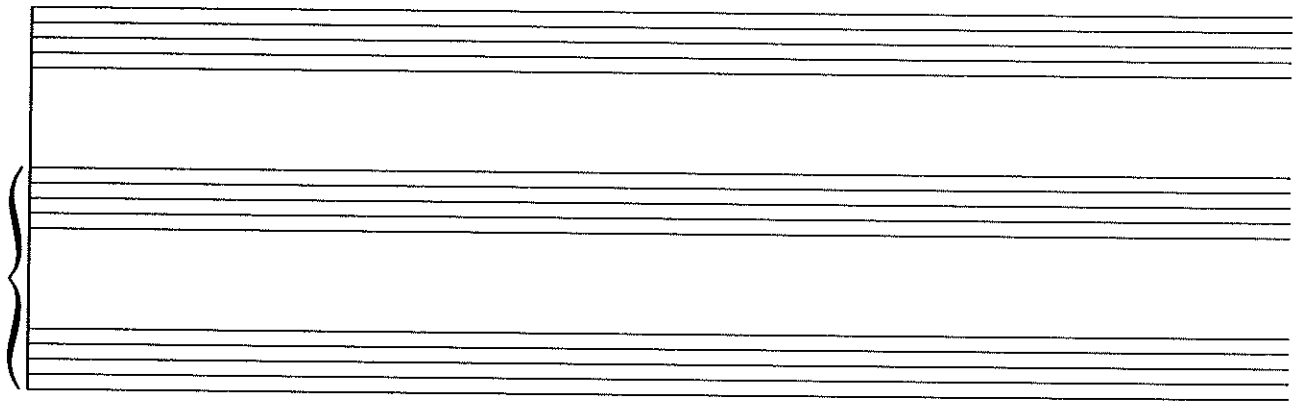




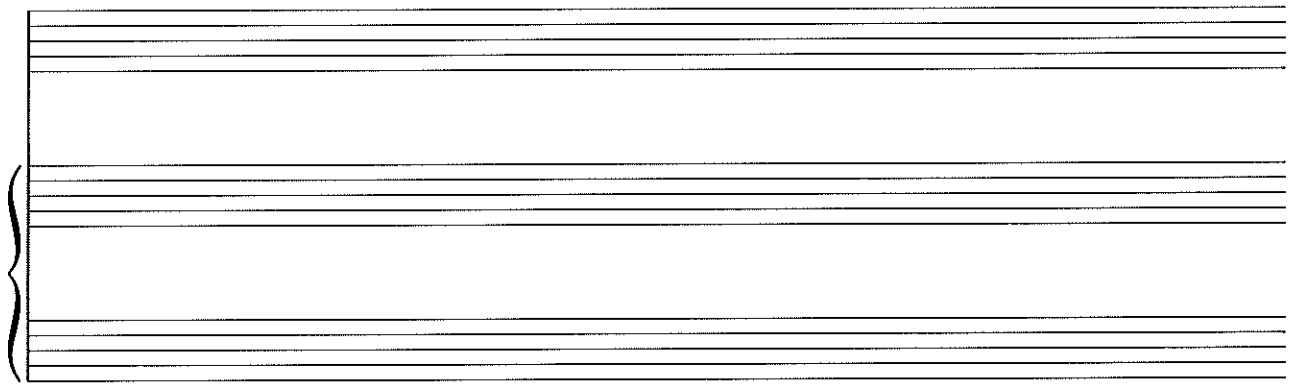


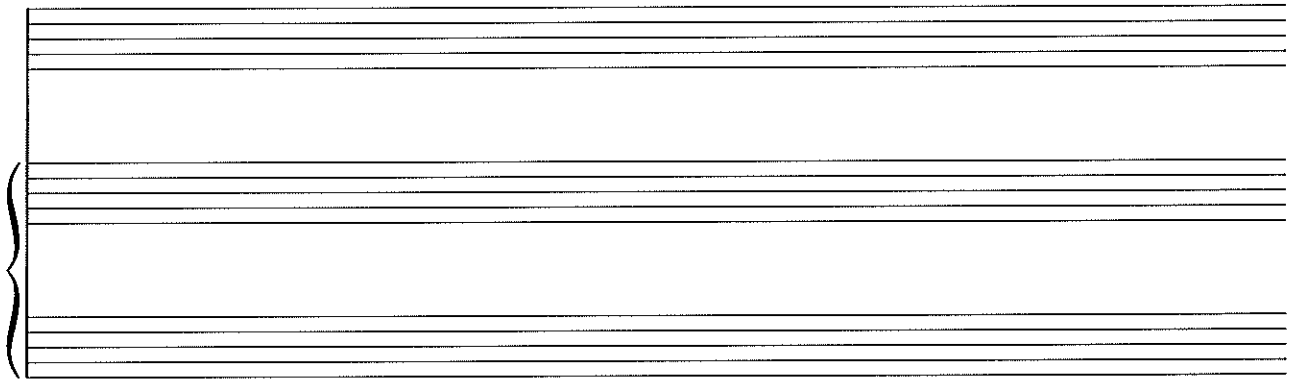


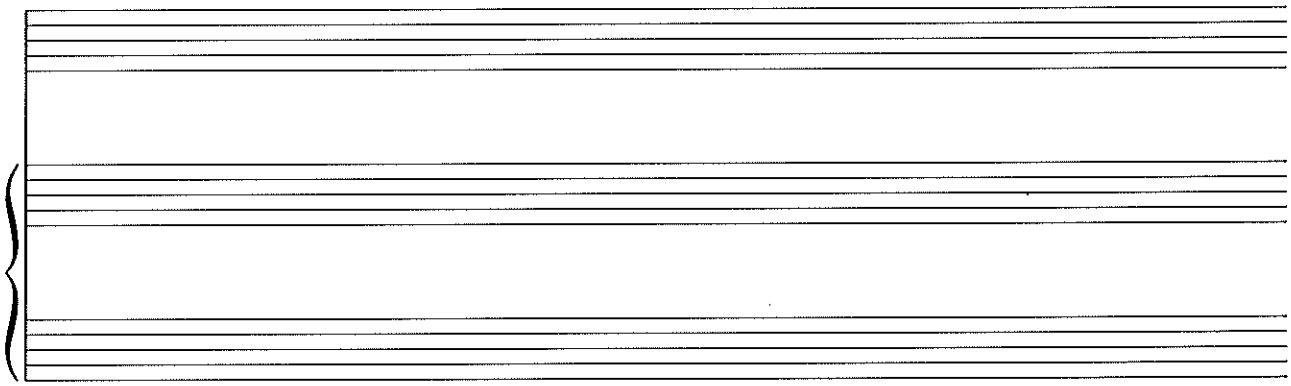


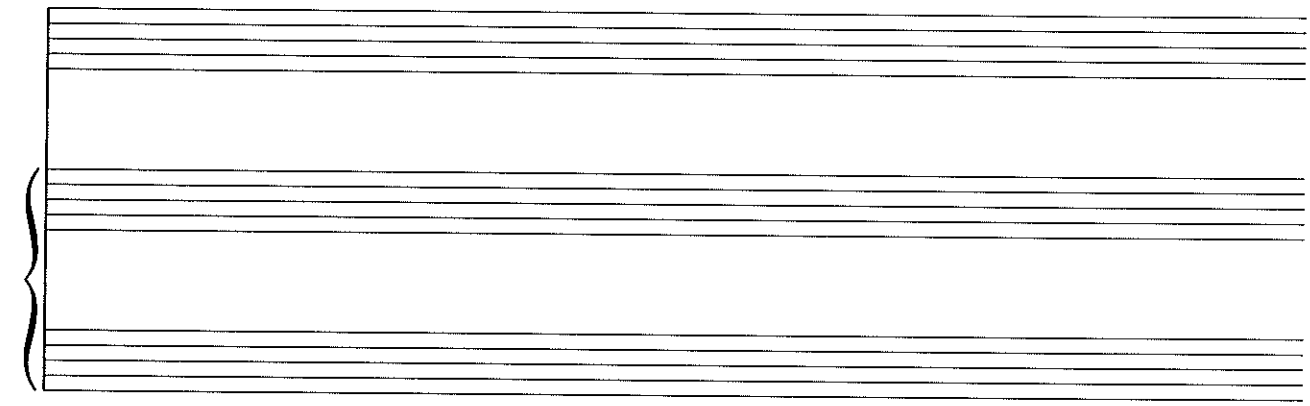


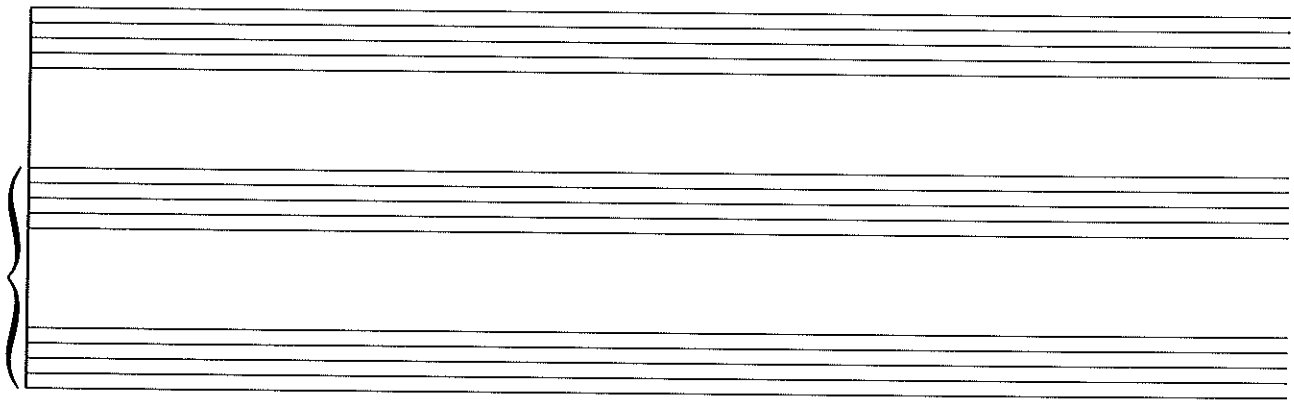
2

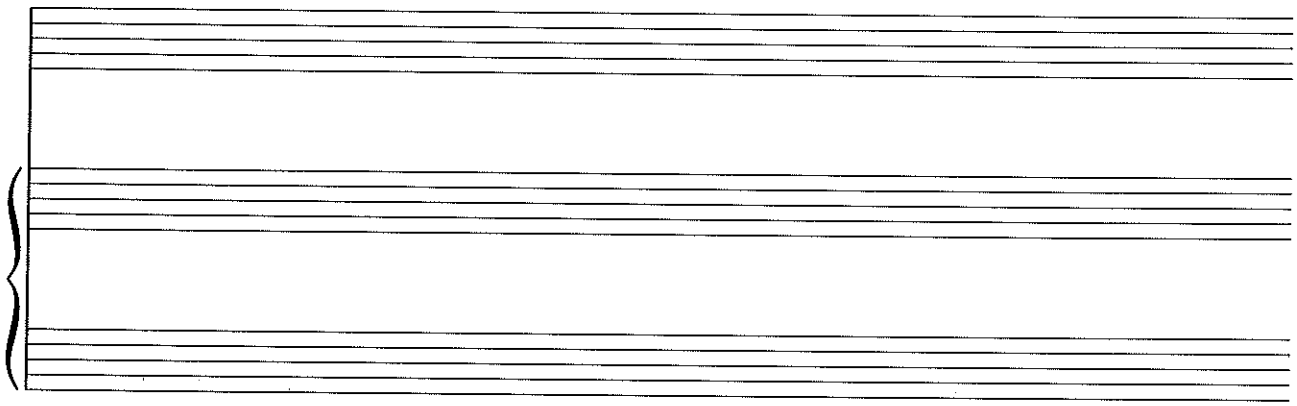












BURKHART, TUREK, AND WENNERSTROM EXERCISES FOR HARMONIC ANALYSIS

Burkhart

p. 504.
pp. 512–13.
pp. 485–87.
pp. 488–89.
pp. 525–28.

Turek

pp. 442–43.
pp. 494–500.
pp. 527–28.

Wennerstrom

pp. 518–21.
pp. 435–36.
pp. 440–43.
For each movement:

1. Read the author's introduction.
2. Determine precisely how each statement of the series is constructed.
3. Where applicable, investigate the degree to which there is a systematic approach to harmony.

Appendix

MUSICAL CALLIGRAPHY

Musical calligraphy is musical handwriting. Just as legible and fluid handwriting is desirable in the communication of the written word, so it is in the communication of written music. Since students will notate a great deal of music in the course of their studies, it will be well worth their while to pay attention to basic calligraphic procedures and to allow their calligraphy to develop until their notation is both legible and mature.

Possibly the most effective way of improving one's calligraphy is by copying printed music and taking note of the construction of the various musical characters. The following example is written for viola and piano. Examine both the format and the formation of the musical symbols. The notes below the example point out some of the basic elements of the notation of music.

Chanson Triste

Heinrich Nicht-Hören
(1799–1902)





Note:

a. The title is centered; the composer's name and dates are printed on the right of the page; the tempo mark is printed on the left.

b. Following the clef symbol, the key signature is written *before* the meter signature.

c. The bar lines are perpendicular to the staff and are arranged so that the entire line is embraced.

d. The note heads are lined up vertically according to the metrical organization of the music. Since the meter here is $\frac{4}{4}$, the quarter note beat occupies approximately one-quarter of the space taken up to notate the whole measure.

e. The note head is *oval* and occupies a space equivalent to the distance between two adjacent staff lines.

f. The stems ascend from the right or descend from the left of the note heads, depending upon the position of the note heads on the staff. Common sense largely dictates in which direction the stems go.

g. The dynamic marks are placed below the solo instrument's part, and between the staves in the piano part.

h. The braces, clef symbols, accidentals, and rests are always formed in the same manner. Thus, the student should practice writing them until they can be committed to paper accurately and automatically.

The student is advised to use a medium-density (No. 2 or HB) pencil initially. An automatic 0.5 mm pencil is most useful because it requires no sharpening. A good eraser and transparent ruler should always be at hand.

When the student feels sufficiently confident, work in ink may be attempted. Because such a wide variety of materials is available it is advised that, before making a purchase, he or she consult a book on musical calligraphy with a view to selecting the most suitable equipment.

INSTRUMENTS

Following is a list of instruments, their common abbreviations, their ranges, and their notation relative to concert pitch.

With respect to the last item, instruments fall into two categories: a) nontransposing (or concert pitch) instruments, which sound as written; and b) transposing instruments, which do not sound as written. The B flat clarinet, for example, sounds a major second lower than written. To compensate, the part for this instrument must be written a major second higher than that for a concert pitch instrument.

This legend is used in the following list to indicate in which part of the instrument's range the performer may have difficulty in producing accurate tones.

(U)—upper extreme
(L)—lower extreme
(UL)—both the upper and lower extremes

The student is advised either to avoid the use of problematic extremes of range or to consult an orchestration text to discover the nature of the problems.

INSTRUMENT	ABBREVIATION	WRITTEN RANGE	ACTUAL SOUND
Piccolo (UL)	Picc.	d^1–c^4	P8 higher
Flute (UL)	Fl.	c^1–d^4	as written
Oboe (UL)	Ob.	$b\flat$–a^3	as written
English horn (U)	Eng. Hrn., E.H.	b–g^3	P5 lower
Clarinet in B♭ (U)	Cl. in B♭	e–c^4	M2 lower
Clarinet in A (U)	Cl. in A	e–c^4	m3 lower
Bassoon (U)	Fag., Bsn.	BB♭–$e\flat^2$	as written
Contrabassoon (U)	C. Fag., C. Bsn.	BB♭–$b\flat^1$	as written
Alto saxophone in E♭ (L)	Alt. Sax., A. Sax	$b\flat$–f^3	M6 lower
Tenor saxophone in B♭ (L)	Ten. Sax., T. Sax.	$b\flat$–f^3	M9 lower
Baritone saxophone in E♭	Bar. Sax., B. Sax.	$b\flat$–e^3	M13 lower
Horn in F, French horn (UL)	Cor. in F, Hn. in F	F♯–c^3	P5 lower
Trumpet in B♭ (UL)	Tpt. in B♭	F♯–d^3	M2 lower
Trumpet in C (UL)	Tpt. in C	F♯–d^3	as written
Tenor trombone (UL)	Trb.	E–f^2	as written
Bass trombone (U)	B. Trb.	C–b^1	as written
Tuba (UL)	Tba.	DD–f^1	as written
Violin (U)	Vl.	g–g^4	as written
Viola (U)	Vla.	c–e^3	as written

Cello (violoncello) (U)	Vc.	C–g^2	as written
Double bass (U)	D.B., Bs.	E–d^2	P8 lower

TEMPO AND EXPRESSION MARKS

Following is a list of basic terms associated with tempo and expression. As a general rule, tempo marks are placed above the staff. Expression marks are found both above and below the staff.

accelerando	becoming faster	*marcato* (marc.)	marked
agitato	excited	*mezzo forte* (mf)	moderately loud
allargando	becoming broader	*mezzo piano* (mp)	moderately soft
allegretto	moderately fast	*moderato*	at moderate tempo
allegro	fast	*pianissimo* (pp)	very soft
andante	moderately slow	*piano* (p)	soft
animato	animated	*presto*	very fast
appassionato	impassioned	*rallentando* (rall.)	becoming slower
a tempo	at normal tempo	*risoluto*	determined
brilliante	brilliant	*ritardando* (rit.)	becoming slower
cantabile	singing	*schnell*	fast
crescendo (cresc)	increasing in volume	*sostenuto*	sustained
decrescendo (decresc.)	decreasing in volume	*spiritoso*	spirited
diminuendo (dim.)	decreasing in volume	*staccato* (stacc.)	detached
dolce	sweet, soft	*stringendo* (string.)	quickening
energico	energetic	*tempo primo*	at the original tempo
espressivo (espr.)	expressive	*tenuto* (ten.)	held, sustained
forte (f)	loud	*tranquillo*	quiet, calm
fortissimo (ff)	very loud	*vif*	lively
furioso	furious	*vivace*	quick
giocoso	humorous	*vivo*	lively
grazioso	graceful	*zart*	tender, delicate
langsam	slow		
larghetto	slow		
largo	very slow		
lento	slow		
maestoso	majestic		

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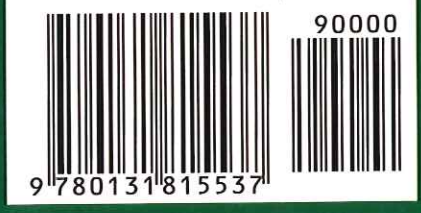
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